BIP GROUP

PEROLO

ISO & ROAD

2022

MANLID & INSPECTION HATCHES

FUNCTION

A manlid allows access to the tank or tanker to carry out inspections, maintenance or repair. Perolo offer a wide range of manlids to suit tank containers, road tankers, rail tank cars and storage tanks intended for chemical and foodstuff liquid products. Most common options are low-profile for ISO tanks or standard height straight neckring for Road Tankers.

- Various internal diameters: 150, 300, 450, 460, 500, 600, 900
- Standard MAWP: 0-4 Bar
  Special MAWP > 4 Bar
- Fixing points: 3, 4, 6, 8
  Special service : 10, 12, 20
- Large choice of seals and accessories
- Special Uses: H2O2, Drybulk, Bolted Manlid, Rail Tank Car
- Necking: low-profile or standard height
  Profile: straight or radiused
  Thickness: 6 or 8mm
  Compensated neckring: with or without
- St st eyebolts with brass wing nuts
  St st eyebolts with st st wing nuts
  Hexagonal nuts
  Protective washer for coated coverlid
  Safebolt (1 per manlid)
  TIR sealing
- Contact Materials:
  Stainless steel: 316, 304, 904
  Duplex, Carbon steel; Aluminium
- Many customisations possible eg fittings, flanges, logo
- Working Temperature: Standard or High (+250°C)
- Surface Finish: Acid etched, raw, electropolished, Halar lined coverlid
- Horizontal or Vertical (reinforced hinge bracket)
- ADR, RID, IMDG
  PED on request

ND150 Inspection Hatch

ND300 Cleaning Hatch

ND460 Manlid

ND500 Low-profile Manlid

ND600 Straight Manlid

ND500 Manlid 10 points

ND500 Bolted Manlid

ND500 H2O2 Manlid

Customised ND460 Manlid

ND450 Drybulk Manlid

ND500 Manlid for dished end

ND900 Vertical Manlid

MANLID

ND150 CLEANING HATCH
INSPECTION HATCH
11 18 94 00 00

ND150 Cleaning Hatch is designed to be installed on IBC's and static tanks intended for chemical and foodstuff liquid products.

SPECIFICATION	
Nominal diameter	154 mm (6")
Weight	8 Kg
Fastening points	3
Working Pressure	4 Bar
Test Pressure	6 Bar
Working Temperature	-40° to +200°C

Temperature range may vary according to the seal material

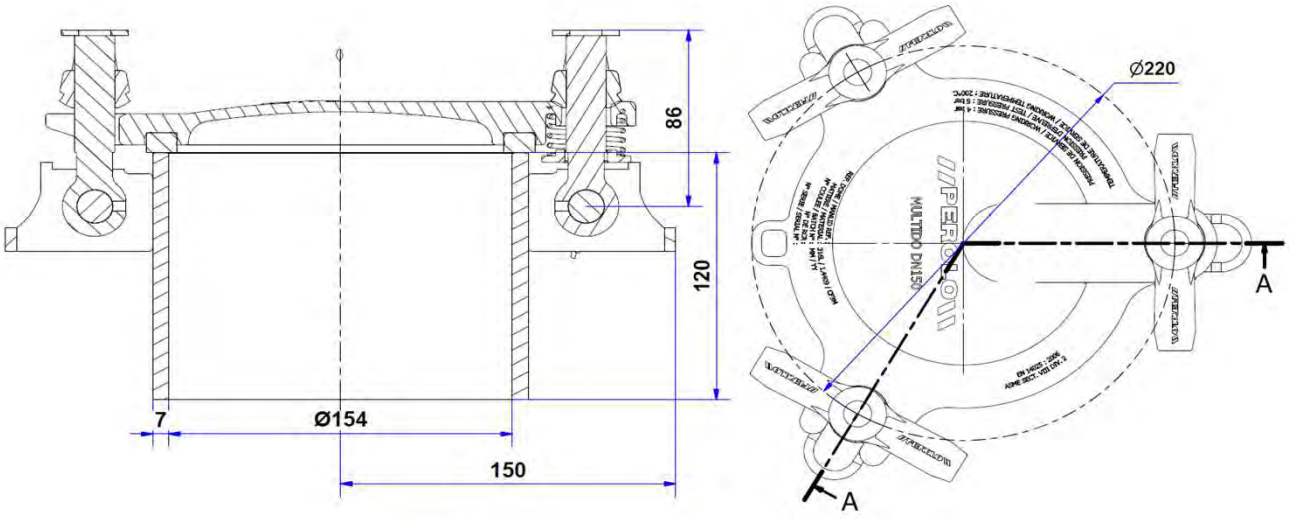
CONFORMITY, CERTIFICATION & APPROVAL
ADR / RID / IMDG / ASME / EN14025

ACCESSORIES	
Coverlid	11 18 94 10 02
Coverlid kit (coverlid, seal, pins & bolts)	11 18 94 90 00
Coverlid seal	11 13 91 10 02
Neckring	11 13 91 10 00

STANDARD FEATURES	
Made of (material)	316L / 1.4404/1.4409 st. steel manlid
Seal material	Black Nitrile / EPDM-PTFE
Neckring height	120 mm
Neckring thickness	7 mm
Fastening	3 Swingbolts with captive wing nuts
Perfect tightness	Trapezoïdal seal groove - Seal 14x10

CODIFICATION	
Full ND150 Cleaning Hatch - Design pressure 4 Bar - 316L/1.4404 stainless steel - black nitrile seal	11 18 94 00 00

DIMENSIONAL DRAWING



MANLID

ND150 CLEANING HATCH
INSPECTION HATCH
11 18 94 00 00

IBC

INSTALLATION

1. Disassemble the bolts with the screw
2. Weld the neckring on the tank.

TIG, MIG or MAG weld process are allowed. Check the material according to your tank and welding process.

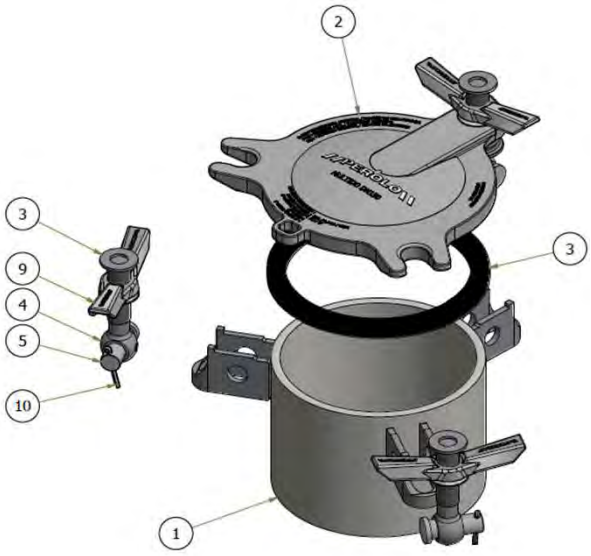
3. **Check that the sealing surface has no more than 1 mm flatness default.**
4. Assemble the cover and the security bolt. Assemble the 2 standard bolts.

NB: the tank radius should be +/- 50mm compared to neckring radius in order to allow correct neckring welding on the tank.

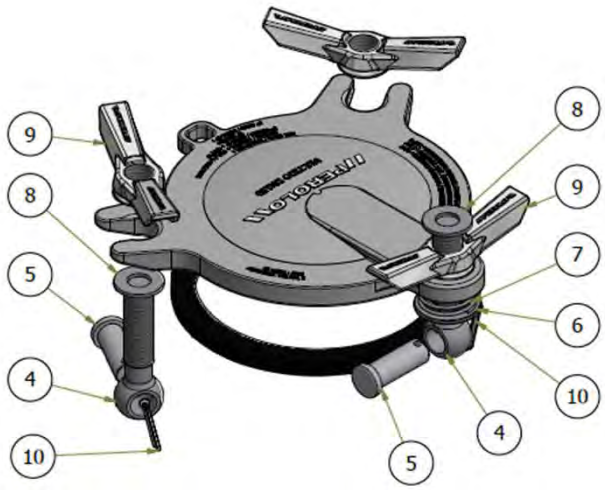
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
Coverlid, Neckring & seal*			
2	1	ND150 Coverlid, st st	11 18 94 10 02
1	1	Neckring, H 120mm, st st	11 13 91 10 00
3	1	Coverlid seal, black nitrile	11 13 91 10 02
Swingbolt assembly & spring			
7	1	Spring, st st	17 10 06 00 08
3,9,4 ,5,10	1	Swingbolt assembly (1x eye bolt 83mm, 1x wing nut, 1x pivot pin, 1x split pin 1x washer)	11 18 94 92 00

*for other seal material, please conctact our sales department



ITEM	QTY	DESIGNATION	PART N°
Coverlid kit			
3	1	1 x 10x14, Black nitrile seal	11 18 94 91 00
2	1	1 x ND150 coverlid, st st	
4	3	3 x eye bolts, M18 Ø16 L83, st st	
5	3	3 x pivot pins, st st	
9	3	3 x wing nuts, copper alloy	
6	1	1 x washer	
7	1	1 x spring	
8	3	3 x washer st st	
10	3	3 x split pin st st	



MANLID

ND300 MULTIDO
MANLIDS
MS3

ND300 MULTIDO manlid : A wide range to suit tank containers, road tankers and storage tanks intended for chemical and foodstuff liquid products.

SPECIFICATION	
Nominal diameter	300 mm (12")
Weight	15 Kg for 21mm Neckring
Fastening points	4
Working Pressure	3 Bar or 4 Bar
Test Pressure	4,5 Bar or 6 Bar
Working Temperature	-40° to + 200°C

Temperature range may vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
ADR / RID / IMDG / ASME / EN14025/ PED compliant on request
Lloyd's Register and Bureau Veritas

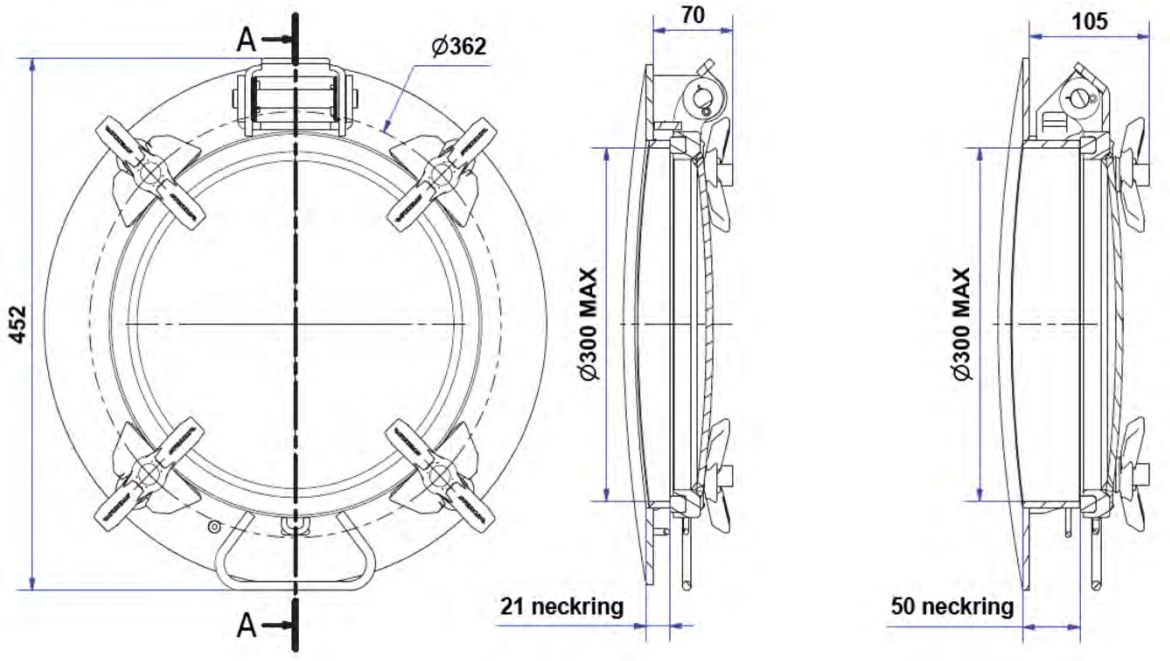
ACCESSORIES	
Coverlid seal EPDM/PTFE	11 13 75 00 20
Swingbolt ass. (58mm)	12 93 86 92 00
Swingbolt ass. (83mm)	12 93 86 91 00
Hinge kit	12 13 68 91 00
4 points low profile coverlid no TIR (316L)	CMS354D4**S**S
4 points standard coverlid no TIR (316L)	CMS354D4**D**S

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4404 stainless steel manlid
Seal material	seal not included
Neckring height	from 21 to 200 mm - Other on request
Neckring thickness	6 mm - Other on request
Fastening	Swingbolts with captive wing nuts
Perfect tightness	Trapezoïdal seal groove - Seal 14x14

DIMENSIONAL DRAWING

Low profile : neckring ≤ 40 mm

Standard profile : neckring > 40 mm



MANLID

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 15 27 20 00 E
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND300 MULTIDO
MANLIDS
MS3

CODIFICATION

STANDARD PART NUMBER	M	S	3	4	D	4	6	F	0	F	20	T	B	5

YOUR PART NUMBER	M		3			4			0					

MATERIAL			
St.st 904L	U	St.st 316L	S
St.st Duplex	D	St.st 304L	Y
		C.st P355NL	P

INTERNAL DIAMETER	
300mm internal diameter	3

MAWP	
4 Bar (test pressure 6 bar)	4
3 Bar (test pressure 4,5 bar)	3

FINISHING - other on request	
No surface treatment	B
Acid etched	D
Electropolished	P

FASTENING POINTS	
4 fastening points	4

NECKRING THICKNESS			
8 mm	8	6 mm	6

NECKRING HEIGHT - other on request	
21 mm (low profile)	F
24 mm (low profile)	A
58 mm (standard)	E
97 mm (standard)	D
150 mm (standard)	I

PROTUSION HEIGHT	
No protusion	0

COMPENSATED NECKRING-oth. on request	
No compensated neckring	E
70 mm width & 6 mm thick	F
85 mm width & 8 mm thick	G

COMP. NECKRING RADIUS-oth. on request	
2nd & 3rd digit eg: radius 1200mm	20
No radius	XX

SEALING			
1 sealing	S	T.I.R. sealing	T

SWINGBOLTS - other on request	
Standard swingbolts in 304 SS - wing nut in Brass	B
Swingbolts in SS with hexagonal nuts in SS	X
Safebolt in SS - wing nut in SS	S

ASSSEMBLY OPTIONS			
fully assembl.	1	not assembled	5

MANLID

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ND300 MULTIDO
MANLIDS
MS3

INSTALLATION

1. Disassemble the 4 bolts with the screw
2. Weld the neckring on the tank.

TIG, MIG or MAG weld process are allowed. Check the material according to your tank and welding process.

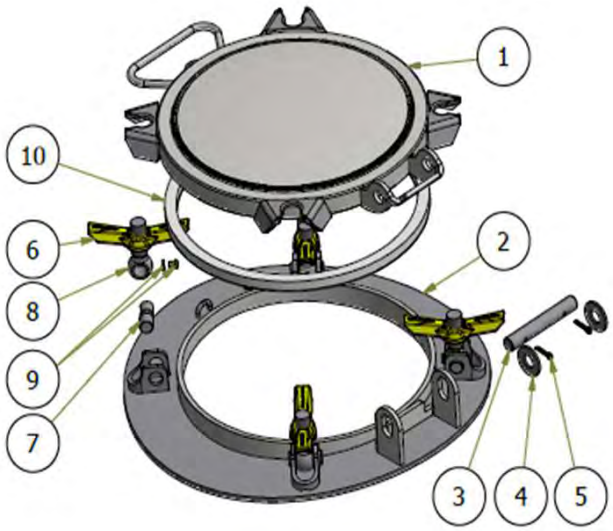
3. **Check that the sealing surface has no more than 1 mm flatness default.**
4. Assemble the cover and the 4 bolts (Tightening Torque 60 N.m +/-5)

NB: the tank radius should be +/- 50mm compared to neckring radius in order to allow correct neckring welding on the tank.

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
Coverlid, neckring & seal*			
1	1	4 points low profile coverlid with TIR	CMS354D4ST
1	1	4 points standard coverlid without TIR	CMS354D4DS
10	1	Seal, PTFE/EPDM 14x14	11 13 75 00 20
10	1	Seal, Nitrile/PTFE 14x14	11 13 75 00 13
10	1	Seal, braided PTFE 15x14	11 13 75 00 12
10	1	Seal, Sweet White Rubber 14x14	11 13 75 00 26
2	1	4 points low profile neckring with TIR	VS34D46F0F1180T
2	1	4 points standard neckring without TIR	VS34D46L0E0000S
Hinge pin kit			
3	1	Hinge pin 304 st st	12 13 68 91 00
4	2	Washers	
5	2	V4-28 pins	
58 mm swingbolt assembly			
8	1	58 mm eye bolt 304L st st	12 93 86 92 00 (for low profile manlid)
6	1	Wing nut copper alloy	
9	1	M6x8 TBHC screw and washer	
7	1	Pivot pin	
Bolting Kit			
3	1	Hinge pin 304 st st	11 15 27 91 00 (for R>1000)
5	2	V4-28 pins	
4	2	Washers	
8	4	58 mm eye bolt 304L st st	
6	4	Wing nut copper alloy	
9	4	M6x8 TBHC screw and 1x washer	
7	4	Pivot pin	

*for other seal material, please check the 'manlid gasket' sheet in accessories section



Low profile : neckring ≤ 40 mm

Standard profile : neckring > 40 mm

MANLID

ND460 MULTIDO
MANLIDS
MSA

ND460 Multido manlid - a wide range to suit tank containers and storage tanks intended for chemical and foodstuff liquid products.

SPECIFICATION	
Nominal diameter	460mm
Weight	22KG for 50mm neckring
Fastening points	4 or 6
Working Pressure	2 Bar to 4 Bar - Other on request
Test Pressure	3 Bar to 6 Bar
Working Temperature	-40° to + 200°C

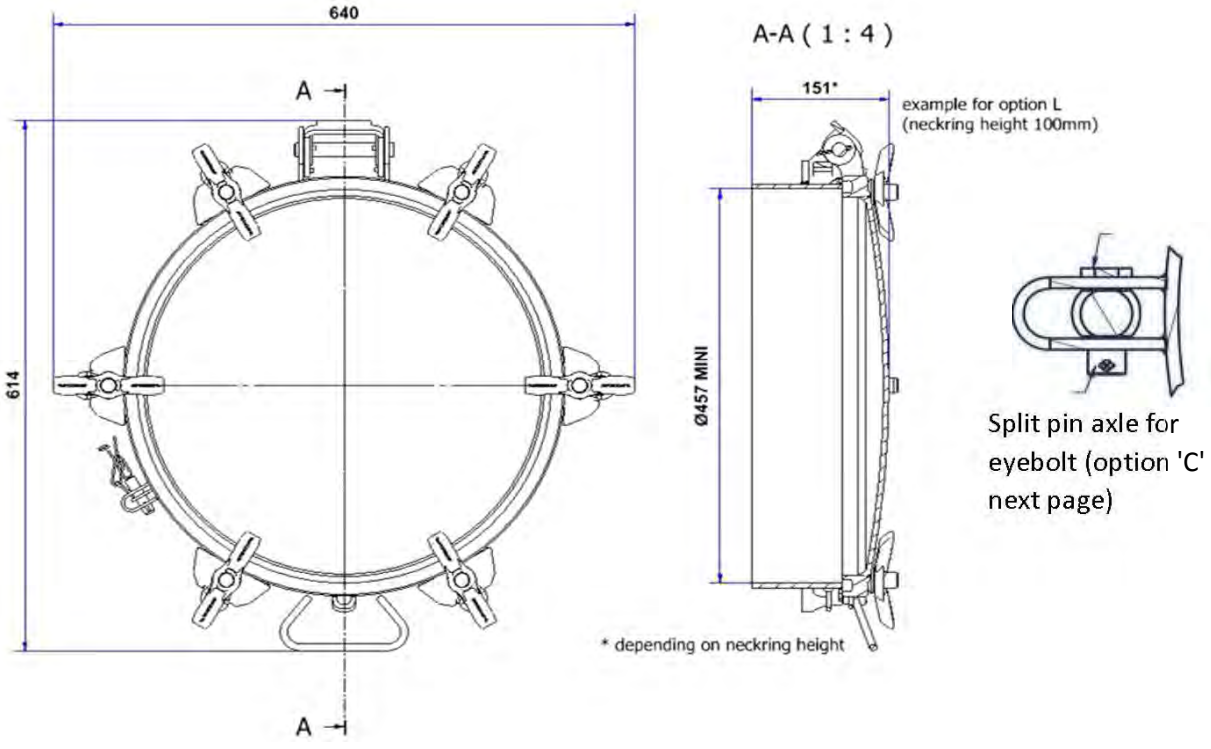
Temperature range may vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
ASME / EN14025/ PED compliant on request
Lloyd's Register

ACCESSORIES	
Coverlid seal PTFE/EPDM	11 13 75 00 44
Swingbolt ass. (83mm)	12 93 86 91 00
Swingbolt ass. (83mm) with split pin	On request
6 points standard coverlid with TIR (316L)	CMSA54D6D**T**
6 points standard coverlid no TIR (316L)	CMSA54D6D**S**

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4404 stainless steel manlid
Seal material	seal not included
Neckring height	from 21 to 200 mm - Other on request
Neckring thickness	6mm - Other on request
Fastening	Swingbolts with captive wing nuts
Perfect tightness	Trapezoïdal seal groove - Seal 14x14

DIMENSIONAL DRAWING



MANLID

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It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND460 MULTIDO
MANLIDS
MSA

CODIFICATION

STANDARD PART NUMBER	M	S	A	4	D	6	6	D	0	E	XX	T	C	5

YOUR PART NUMBER	M		A						0					

MATERIAL			
St.st 904L	U	St.st 316L	S
St.st Duplex	D	St.st 304L	Y
		C.st P355NL	P

INTERNAL DIAMETER	
460 mm internal diameter	A

MAWP - other on request	
4 Bar (test pressure 6 bar)	4
3 Bar (test pressure 4,5 bar)	3

FINISHING - other on request	
No surface treatment	B
Acid etched	D
Electropolished	P

FASTENING POINTS			
4 points	4	6 points	6

NECKRING THICKNESS			
8 mm	8	6 mm	6

NECKRING HEIGHT - other on request	
21 mm (low profile)	F
24 mm (low profile)	A
58 mm (standard)	E
97 mm (standard)	D
150 mm (standard)	I

PROTUSION HEIGHT	
No protusion	0

COMPENSATED NECKRING-oth. on request	
No compensated neckring	E
70 mm width & 6 mm thick	F
85 mm width & 8 mm thick	G

COMP. NECKRING RADIUS-oth. on request	
2nd & 3rd digit eg: radius 1200mm	20
No radius	XX

SEALING			
1 sealing	S	T.I.R. sealing	T

SWINGBOLTS - other on request	
Standard swingbolts in 304 SS - wing nut in Brass	B
Swingbolts in 304 SS - wing nut in Brass, split pin	C
Safebolt in SS - wing nut in SS	S

ASSSEMBLY OPTIONS			
fully assembl.	1	not assembled	5

MANLID

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ND500 MULTIDO
MANLIDS
MS5

ND500 Multido manlid - a wide range to suit tank containers, road tankers and storage tanks intended for chemical and foodstuff liquid products.

SPECIFICATION	
Nominal diameter	500 mm (20")
Weight	35.50 Kg for 21mm Neckring
Fastening points	6 or 8
Working Pressure	2 Bar to 4 Bar - other on request
Test Pressure	3 Bar to 6 Bar
Working Temperature	-40° to + 200°C

Temperature range may vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
ADR / RID / IMDG / ASME / EN14025/ PED compliant on request
Lloyd's Register

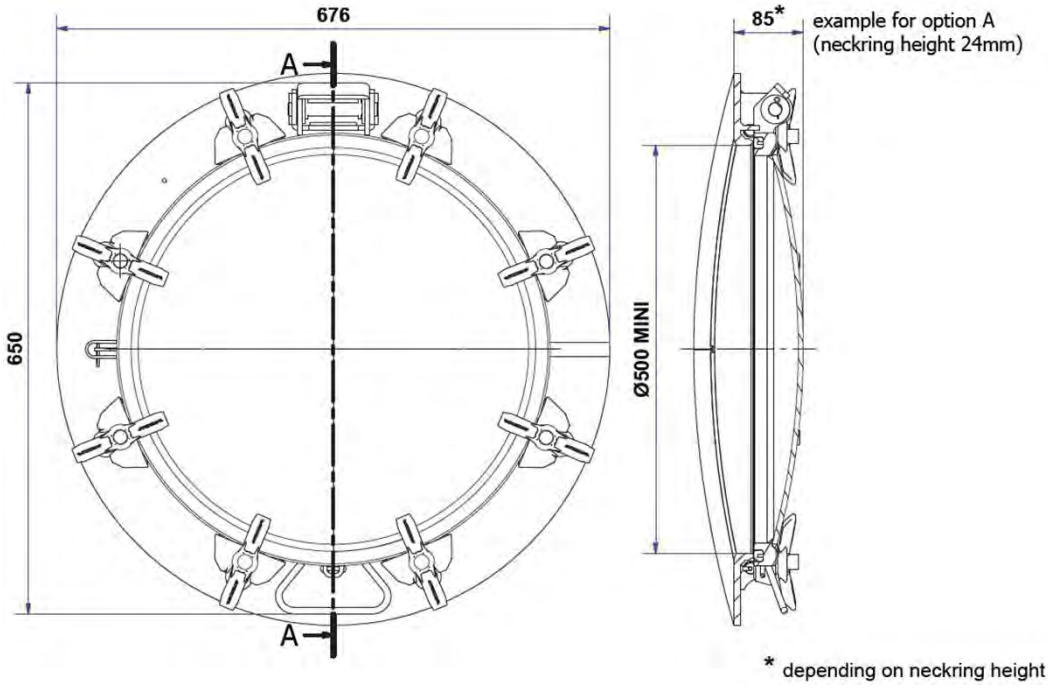
ACCESSORIES	
Coverlid seal PTFE/EPDM	11 13 75 00 19
Swingbolt ass. (58mm)	12 93 86 92 00
Swingbolt ass. (83mm)	12 93 86 91 00
Hinge kit	12 13 68 91 00
8 points low profile coverlid with TIR (316L)	CMS554D8**S**T
6 points standard coverlid no TIR (316L)	CMS554D6**D**S

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4404 stainless steel manlid
Seal material	seal not included
Neckring height	from 21 to 200 mm - Other on request
Neckring thickness	6 or 8 mm - Other on request
Fastening	Swingbolts with captive wing nuts
Perfect tightness	Trapezoïdal seal groove - Seal 14x14

DIMENSIONAL DRAWING

Low profile : neckring ≤ 40 mm

Standard profile : neckring > 40 mm



MANLID

ND500 MULTIDO
MANLIDS
MS5

CODIFICATION

STANDARD PART NUMBER	M	S	5	4	D	8	8	A	0	G	20	T	B	5
YOUR PART NUMBER	M		5						0					

MATERIAL			
St.st 904L	U	St.st 316L	S
St.st Duplex	D	St.st 304L	Y
		C.st P355NL	P
INTERNAL DIAMETER			
500 mm internal diameter			5
MAWP - other on request			
4 Bar (test pressure 6 bar)			4
3 Bar (test pressure 4,5 bar)			3
FINISHING - other on request			
No surface treatment			B
Acid etched			D
Electropolished			P
FASTENING POINTS			
8 points	8	6 points	6
NECKRING THICKNESS			
8 mm	8	6 mm	6
NECKRING HEIGHT - other on request			
21 mm (low profile)			F
24 mm (low profile)			A
58 mm (standard)			E
97 mm (standard)			D
150 mm (standard)			I
PROTUSION HEIGHT			
No protusion			0
COMPENSATED NECKRING-oth. on request			
No compensated neckring			E
70 mm width & 6 mm thick			F
85 mm width & 8 mm thick			G
COMP. NECKRING RADIUS-oth. on request			
2nd & 3rd digit eg: radius 1200mm			20
No radius			XX
SEALING			
1 sealing	S	T.I.R. sealing	T
SWINGBOLTS - other on request			
Standard swingbolts in 304 SS - wing nut in Brass			B
Swingbolts in SS with hexagonal nuts in SS			X
Safebolt in SS - wing nut in SS			S
ASSSEMBLY OPTIONS			
fully assembl.	1	not assembled	5

MANLID

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 15 18 20 00 F
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ND500 MULTIDO
MANLIDS
MS5

INSTALLATION

1. Disassemble the bolts with the screw
2. Weld the neckring on the tank.

TIG, MIG or MAG weld process are allowed. Check the material according to your tank and welding process.

3. **Check that the sealing surface has no more than 1 mm flatness default.**
4. Assemble the cover, hinge pin and all bolts. (Tightening Torque 60 N.m +/-5)

NB: the tank radius should be +/- 50mm compared to neckring radius in order to allow correct neckring welding on the tank.

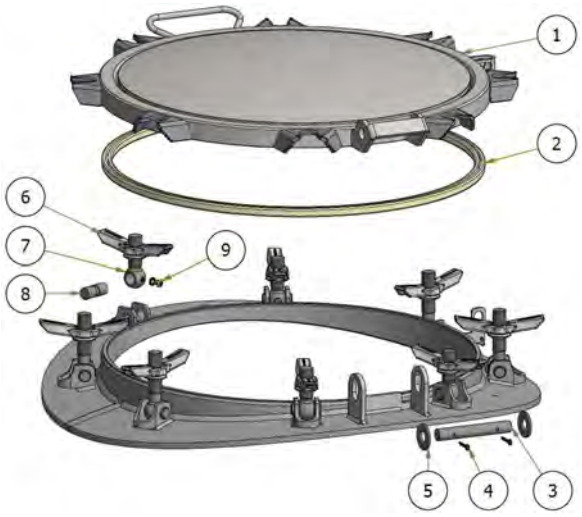
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
Coverlid & seal*			
1	1	8 points **low profile** standard sealing	CMS554D8SS
1	1	8 points **low profile** T.I.R. sealing	CMS554D8ST
1	1	6 points **low profile** standard sealing	CMS554D6SS
1	1	6 points **low profile** T.I.R. sealing	CMS554D6ST
1	1	8 points **standard** standard sealing	CMS554D8DS
1	1	6 points **standard** standard sealing	CMS554D6DS
2	1	Seal, PTFE/EPDM 14x14	11 13 75 00 19
2	1	Seal, Nitrile/PTFE 14x14	11 13 75 00 14
2	1	Seal, braided PTFE 15x14	11 13 75 00 15
2	1	Seal, Sweet White Rubber 14x14	11 13 75 00 18
Hinge pin kit			
3	1	Hinge pin 304 st st	12 13 68 91 00
5	2	Washers	
4	2	V4-28 pins	
58 mm swingbolt assembly (low profile)			
7	1	58 mm eye bolt 304L st st	12 93 86 92 00
6	1	Wing nut copper alloy	
9	1	M6x8 TBHC screw and washer	
8	1	Pivot pin	
83 mm swingbolt assembly (low profile & standard)			
7	1	83mm eye bolt 304L st st	12 93 86 91 00
6	1	Wing nut copper alloy	
9	1	M6x8 TBHC screw and washer	
8	1	Pivot pin	
Complete 8 points bolting kit low profile R>1000			
-	1	Hinge pin kit	11 18 53 92 00
-	4	58 mm Eye bolt assembly	
-	4	83 mm Eye bolt assembly	
Complete 6 points bolting kit standard			
-	1	Hinge pin kit	11 15 20 93 00
-	6	83 mm Eye bolt assembly	

*for other seal material, please check the 'manlid gasket' sheet in accessories section

Low profile : neckring ≤ 40 mm

Standard profile : neckring > 40 mm



MANLID

ND500 MULTIDO COVERLIDS
MANLIDS
CMS5

ND500 Multido Coverlid - designed to suit Perolo neckrings welded on tank containers and road tankers intended for chemical and foodstuff liquid product - for standard or low profile Perolo Multido Manlids.

SPECIFICATION	
Nominal diameter	500 mm (20")
Weight	17.0 Kg
Fastening points	6, 8 (others on request)
Working Pressure	4 Bar
Test Pressure	6 Bar
Working Temperature	-40° to + 200°C

Temperature range may vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
ADR / RID / IMDG / ASME / EN14025
Lloyd's Register, Bureau Veritas

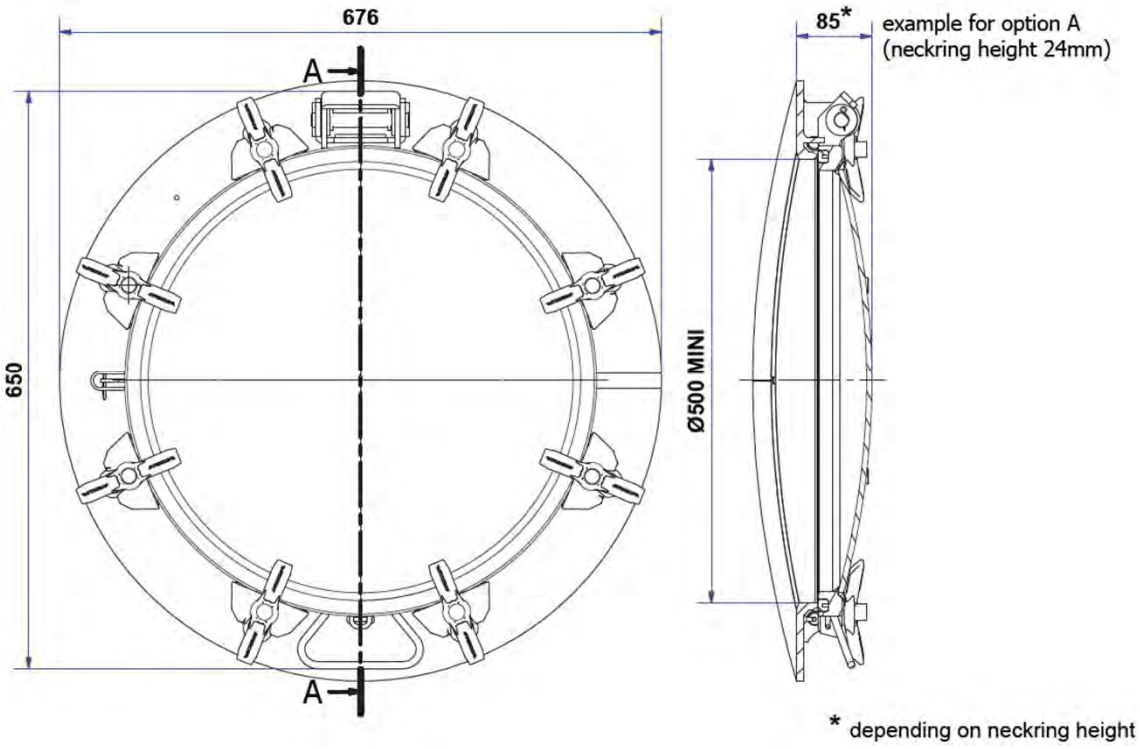
ACCESSORIES	
Coverlid seal PTFE/EPDM	11 13 75 00 19
Swingbolt ass. (58mm)	12 93 86 92 00
Swingbolt ass. (83mm)	12 93 86 91 00
Hinge kit	12 13 68 91 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4404 stainless steel manlid
Cover thickness	6 mm
Perfect tightness	Trapezoïdal seal groove - Seal 14x14
Special product	Double Skin Insulated cover on request

DIMENSIONAL DRAWING

Low profile : neckring ≤ 40 mm

Standard profile : neckring > 40 mm



MANLID

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 18 43 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND500 MULTIDO COVERLIDS
MANLIDS
CMS5

CODIFICATION

STANDARD PART NUMBER	CM	S	5	5	4	D	8	S	T

YOUR PART NUMBER CM _ _ _ _ _ _ _ _ _

MATERIAL - Other on request	
316L stainless steel	S

NOMINAL DIAMETER	
ND 500	5

DISHED BOTTOM THICKNESS	
5 mm (others on request)	5

DESIGN PRESSURE	
3 Bar (test pressure 4.5 bar)	3
4 Bar (test pressure 6 bar)	4

FINISHING	
No surface treatment	B
Acid Etched	D
HALAR Coated	H
Electropolished	E
Sand blasted/protected (carbon st)	B

FASTENING POINTS	
6 fastening points	6
8 fastening points	8

TYPE OF MANLID	
Low profile (neckring < 58mm height)	S
Standard (neckring ≥ 58mm height)	D

SEALING	
Standard sealing	S
T.I.R. sealing	T
no sealing	W

MANLID

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ND600 MULTIDO
MANLIDS
MS6

ND600 Multido manlid - a wide range to suit tank containers, road tankers and storage tanks intended for chemical and foodstuff liquid products.

SPECIFICATION	
Nominal diameter	600 mm (24")
Weight	34 Kg For 100mm Neckring
Fastening points	6, 8, 10 or 12 (depending on MAWP)
Working Pressure	0,5 to 4 Bar
Test Pressure	0,75 to 6 Bar
Working Temperature	-40° to + 200°C

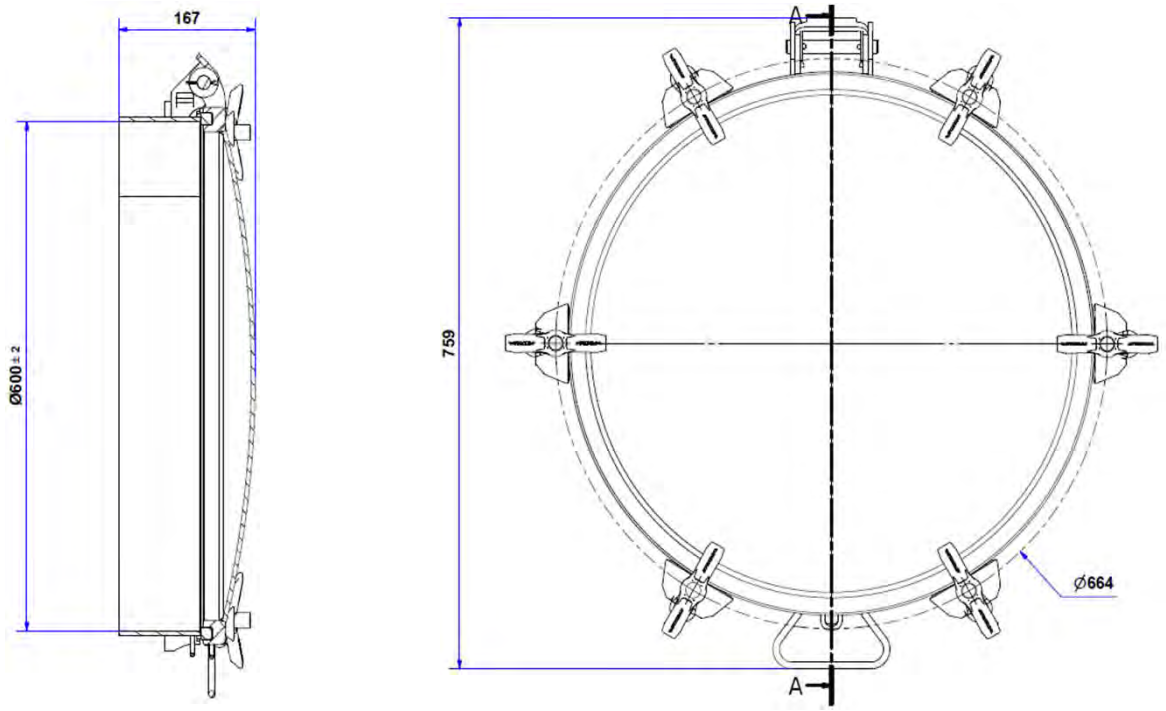
Temperature range may vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
ADR / RID / IMDG / ASME / EN14025/ PED compliant on request
Lloyd's Register

ACCESSORIES	
Coverlid seal Neoprene	11 13 75 00 28
Swingbolt ass. (58mm)	12 93 86 92 00
Swingbolt ass. (83mm)	12 93 86 91 00
Hinge kit	12 13 68 91 00
8 points standard coverlid no TIR (316L)	CMS654D8DS
6 points standard coverlid no TIR (316L)	CMS654D6DS

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4404 stainless steel manlid
Seal material	seal not included
Neckring height	from 50 to 200 mm - Other on request
Neckring thickness	6, 8 or 10 mm
Fastening	Swingbolts with captive wing nuts
Perfect tightness	Trapezoïdal seal groove - Seal 14x14

DIMENSIONAL DRAWING



MANLID

BIP GROUP VALVES SOLUTIONS FOR YOUR PEROLO ISO RAIL IBC ROAD

ND600 MULTIDO
MANLIDS
MS6

CODIFICATION

STANDARD PART NUMBER	M	S	6	3	D	6	6	L	0	E	XX	W	B	5

YOUR PART NUMBER	M		6						0					

MATERIAL			
St.st 904L	U	St.st 316L	S
C.st P355NL	P	St.st 304L	Y

INTERNAL DIAMETER	
600 mm internal diameter	6

MAWP - other on request	
4 Bar (test pressure 6 bar)	4
3 Bar (test pressure 4,5 bar)	3

FINISHING - other on request	
No surface treatment	B
Acid etched	D
Electropolished	P

FASTENING POINTS			
8 points	8	6 points	6
10 points	10		

NECKRING THICKNESS			
8 mm	8	6 mm	6
10 mm	10		

NECKRING HEIGHT - other on request	
100 mm	L
150 mm	I
200 mm	H

PROTUSION HEIGHT	
No protusion	0

COMPENSATED NECKRING-oth. on request	
No compensated neckring	E
70 mm width & 6 mm thick	F
85 mm width & 8 mm thick	G

COMP. NECKRING RADIUS-oth. on request	
2nd & 3rd digit eg: radius 1200mm	20
No radius	XX

SEALING			
1 sealing	S	T.I.R. sealing	T
No sealing	W		

SWINGBOLTS - other on request	
Standard swingbolts in 304 SS - wing nut in Brass	B
Swingbolts in SS with hexagonal nuts in SS	X
Safebolt in SS - wing nut in SS	S

ASSSEMBLY OPTIONS			
fully assembl.	1	not assembled	5

ISO ROAD IBC

MANLID

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 18 33 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND500 FLAT BOLTED MANLID

MANLIDS

11 15 07 00 00

ND500 bolted manlid is designed to suit tank containers intended for chemical and foodstuff liquid products.

SPECIFICATION	
Nominal diameter	500 mm (20")
Weight	110 Kg
Fastening points	20 on a 586 PCD
Working Pressure	4 to 10 Bar
Test Pressure	6 to 15 bar
Working Temperature	-40° to + 190°C

Temperature range may vary according to the seal material

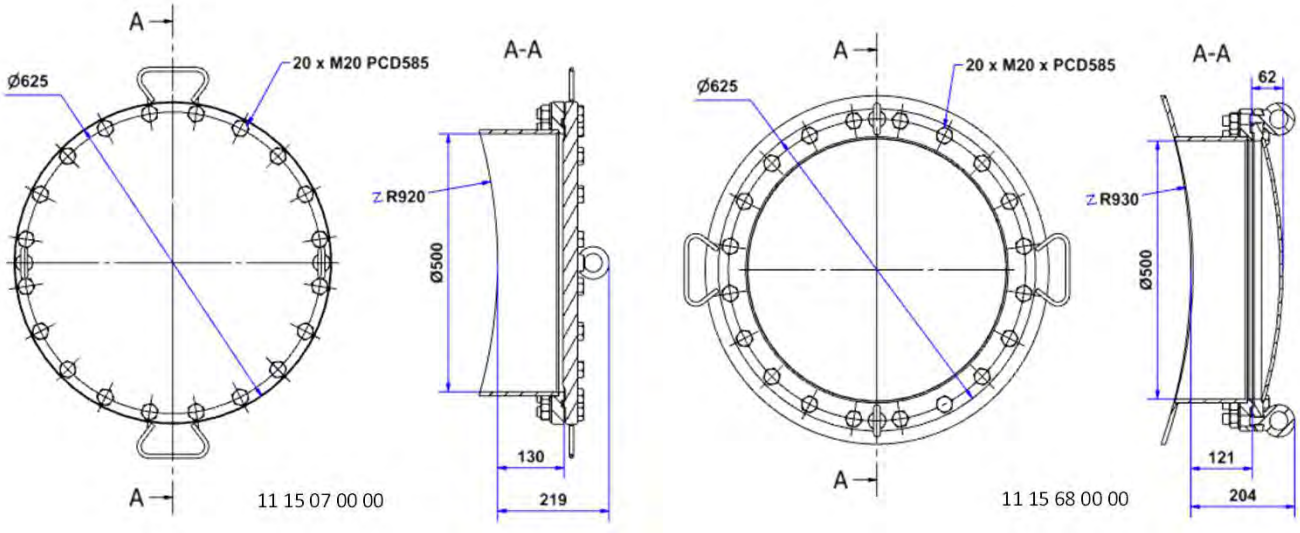
CONFORMITY, CERTIFICATION & APPROVAL
ADR/ RID/ IMDG/ ASME/ EN14025 (U STAMP on request)

STANDARD FEATURES	
Made of (material)	316L / 1.4404 stainless steel manlid
Seal material	EPDM/PTFE -NBR/PTFE -PTFE
Neckring height	100 mm - other on request
Neckring thickness	8 mm - other on request
Cover thickness	29 mm or 53mm domed dish end
Reinforcement ring thick.	8mm R900 to R1200
Handling	100 mm - other on request

ACCESSORIES	
Coverlid	11 15 07 20 00
Coverlid seal	11 15 07 00 03
Bolting kit	11 19 13 00 91
Neckring	11 15 07 10 00

CODIFICATION	
Complete ND500 bolted manlid - working pressure 6.67 Bar - made of 316L/1.4404 stainless steel with tongue and groove, PTFE seal	11 15 07 00 00
Complete ND500 bolted manlid - working pressure 6.67 Bar - made of 316L/1.4404 stainless steel -with reinforcement ring, tongue and groove, PTFE seal	11 15 68 00 00

DIMENSIONAL DRAWING



available with or without tongue & groove

ND500 Hydrogen Peroxide Manlid

MANLIDS

11 15 77 00 00

ND500 Multido - custom protusion with or without compensated neckring - designed exclusively for hydrogen peroxide carriage.

SPECIFICATION	
Nominal diameter	500 mm (20")
Weight	47 Kg
Fastening points	8
Working Pressure	4 Bar
Test Pressure	6 Bar
Working Temperature	-40° to + 200°C

Temperature range may vary according to the seal material

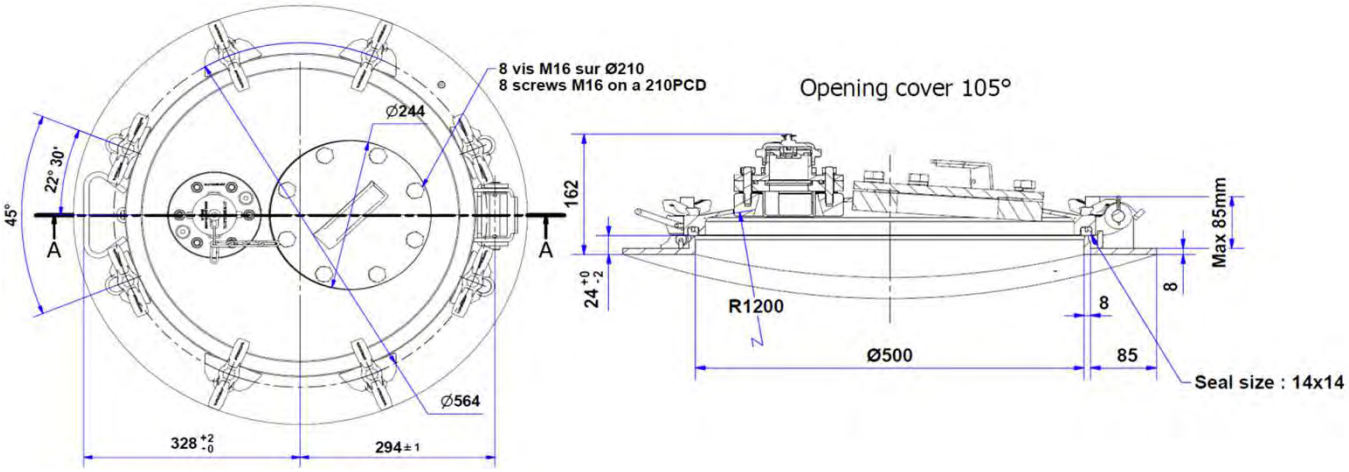
CONFORMITY, CERTIFICATION & APPROVAL
ADR / RID / IMDG / ASME / EN14025
Lloyd's Register and Bureau Veritas

ACCESSORIES	
Coverlid	11 15 77 10 00
Neckring	VS54D88A0G1100S
H2O2 Breather	11 15 13 00 00
Coverlid seal PTFE/EPDM	11 13 75 00 19
Swingbolt ass. (58mm)	12 93 86 92 00
Swingbolt ass. (83mm)	12 93 86 91 00
Hinge kit	12 13 68 91 00

STANDARD FEATURES	
Made of (material)	316L / 1.4404 stainless steel manlid
Seal material	seal not included
Neckring height	24 mm - low profile design
Neckring thickness	8 mm
Fastening	Swingbolts with captive wing nuts
Perfect tightness	Trapezoïdal seal groove - Seal 14x14

CODIFICATION	
Complete ND500 Multido manlid - stainless steel - working pressure 4 Bar - 8 fastening points - low profile - R= 1100	11 15 77 00 00

DIMENSIONAL DRAWING



MANLID

ND500 Hydrogen Peroxide Manlid

MANLIDS

11 15 77 00 00

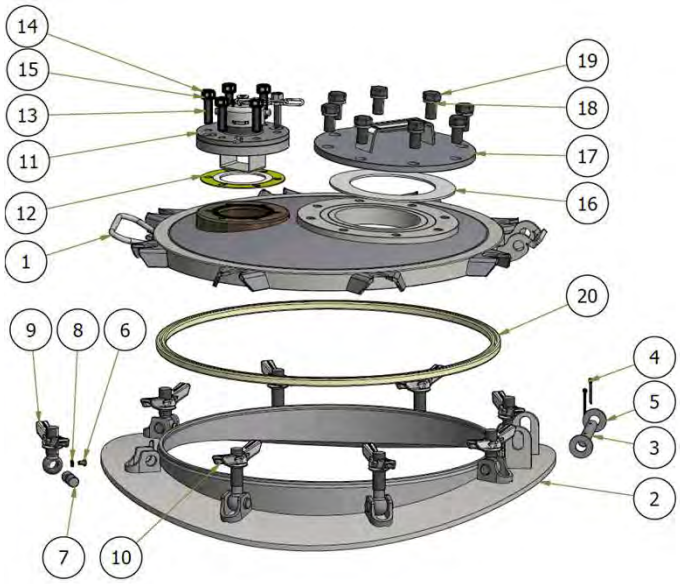
INSTALLATION

1. Carrefully weld the neckring on the tank avoiding neckring seal to face distorsion (max allowed flatness distorsion 1mm)
2. Check the neckring is clean
3. When tightening bolts are delivered separately check the stop washer is installed under the screw head on every bolt.

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
Coverlid, Neckring & seal*			
2	1	ND500 8 points Neckring R1100mm	VS54D88A0G1100S
1	1	ND500 Peroxide Coverlid	11 15 77 10 00
20	1	Coverlid Seal 14x14 (PTFE/EPDM)	11 13 75 00 19
Hinge pin kit			
3	1	Hinge pin axis, AISI 303	12 13 68 91 00
5	2	M16 washer, AISI 303	
4	2	V4-28 split pin, AISI 303	
58mm Swingbolt assembly			
9	1	58 mm eye bolt & wing nut st st	11 13 92 94 00
7	2	Pivot pin, 304L	
8,6	2	M6x8 TBHC screw &M6 washer 304	
83mm Swingbolt assembly			
10	1	83 mm eye bolt & wing nut st st	12 13 68 93 00
7	2	Pivot pin, 304L	
8,6	2	M6x8 TBHC screw &M6 washer 304	
Complete 8 points bolting kit			
-	1	hinge pin kit	11 15 18 98 00
-	4	58 mm eye bolt assembly	
-	4	83 mm eye bolt assembly	
ND50 H2O2 breather assembly			
11	1	H2O2 breather	11 15 13 00 00
14 15 13	1	Stud bolts, nuts & W10grower washers	11 91 35 91 00
12	1	H2O2 breather gasket PTFE	11 15 13 00 04
Blind flange assembly			
17	1	Blind flange / Cover, 316L	12 13 80 40 00
18,19	8	M16-30 screw & washers, 304	12 91 72 00 00
16	1	Blind flange gasket, PTFE	12 13 80 00 09

*for other seal material, please conctact our sales department



MANLID

ND450 Dry Bulk Powder Manlid

MANLIDS

13 15 24 00 00

ND450 Dry Bulk Powder manlid - combinating handle and swingbolts - offers an innovativing closure system which allows the cover to swing on the side and to have a 180° opening with a Ø 450 mm hole.

SPECIFICATION	
Nominal diameter	450 mm (16")
Weight	19 Kg
Fastening points	4
Working Pressure	2 Bar
Test Pressure	3 Bar
Working Temperature	-40° to + 80°C

Temperature range may vary according to the seal material

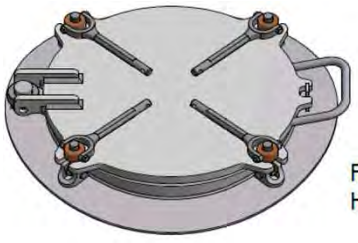
CONFORMITY, CERTIFICATION & APPROVAL
PED 2014/68/UE / EN14025 / (AD2000 on request)
Lloyd's Register

ACCESSORIES	
Nitrile seal 14x10	13 15 01 00 04
EPDM seal 14x10	13 15 24 00 04
Coverlid	13 15 11 10 00
Neckring	13 15 12 10 00
Extended wing nut ass.	13 15 12 91 00
Folding hand nut ass.	13 15 02 91 00
4 points coverlid	13 15 11 10 00

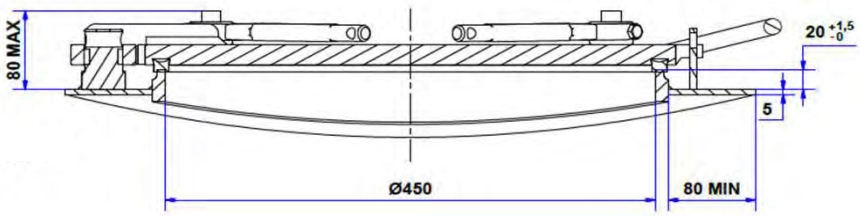
STANDARD FEATURES (See Codification for option)	
Made of (material)	5083 Aluminium
Seal material	seal not included
Neckring height	22 mm - low profile design
Neckring thickness	12 mm
Fastening	Folding Handnut or Extended Wing nuts
Perfect tightness	Trapezoïdal seal groove
Reinforcement ring thick.	5 or 8 mm

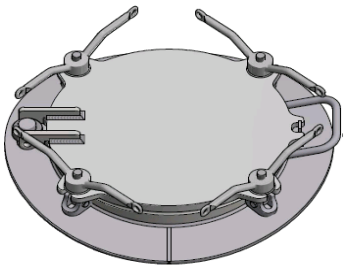
CODIFICATION	
Full ND450 Dry Bulk Powder manlid - design pressure 2 Bar - made of 5083 aluminium - Without gasket - Folding Handnut - R=1260 ±50	13 15 02 00 00
Full ND450 Dry Bulk Powder manlid - design pressure 2 Bar - made of 5083 aluminium - Without gasket - Extended WingNuts - R=1260 ±50	13 15 24 00 00

DIMENSIONAL DRAWING

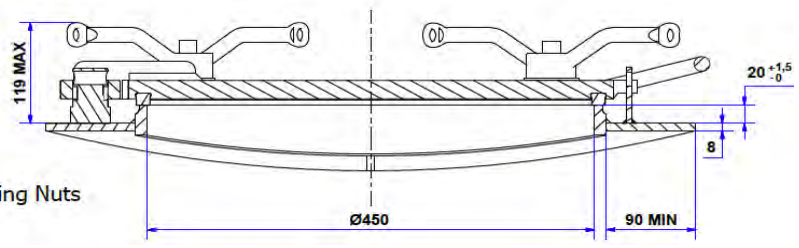


Folding Handnut





Extended Wing Nuts



MANLID

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 13 15 24 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND450 Dry Bulk Powder Manlid

MANLIDS

13 15 24 00 00

INSTALLATION

1. Service conditions

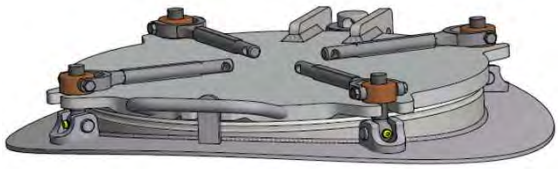
The manlid is designed for aluminium tankers carrying powders such as plastics, cements, granular…

Check the material compatibility with the transporter product

The tank must be fitted with a safety device to limit operating pressure to 3 bar maximum.

2. Installation on the tank

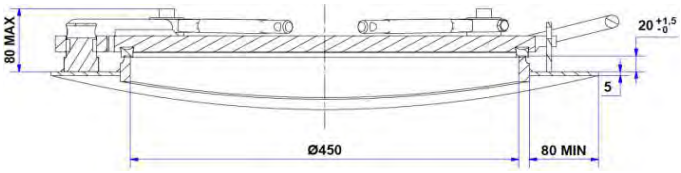
The neckring subassembly is welded on the tank by the customer.
The manlid material can be welded by TIG or MIG welding process, the customer is fully responsible of this final welding. **Max allowed flatness distorsion 1mm**

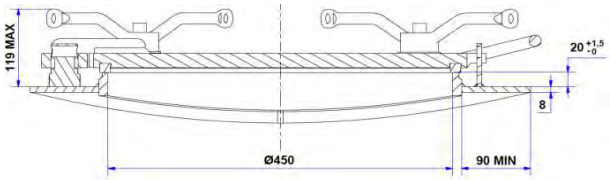


We suggest to use TIG process which needs less energy and limits the neckring seal face distortion after welding (welding material per ISO EN 18273).

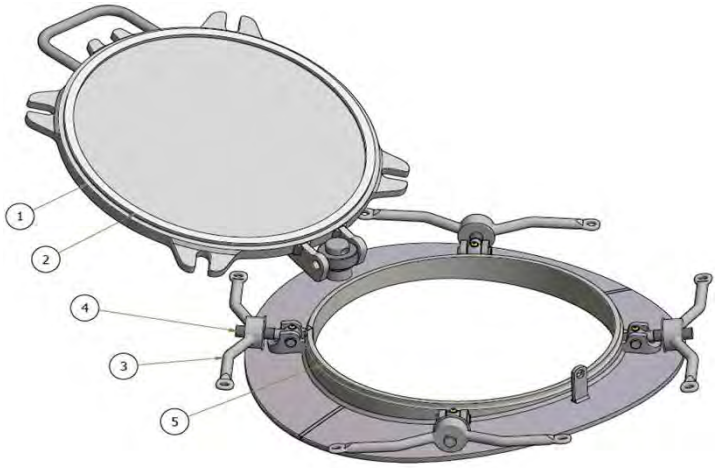
Before welding the neckring on the tank, the cover and the 4 bolts must be removed. They should be reassembled after welding.

The 4 M6 x 8 screws should be secured with glue ('Loctite' or equivalent)





	Description	Part Number
1	Coverlid	13 15 11 10 00
2	Coverlid seal	13 15 01 00 04
3	Folding hand nut	13 15 01 50 00
3	Extended wing nut	13 15 11 35 00
4	Eye bolt	13 15 01 00 02
5	Neckring	13 15 12 10 00



3. Check before use

Check the neckring seal flatness (maximum defect value 1 mm)
When the flatness defect is over this value a seal with higher dimensions should be used to match with the flatness defect.
The seal tightness should be checked during tank pressure test, this test can be done with water or with air.

ND450 Dry Bulk Powder Manlid

MANLIDS

13 15 24 00 00

4. Operating

Cover Opening

Unscrew the four nuts using the handle attached to every one , the torque given by the handle in horizontal position is enough to get the torque needed to unscrew.

If a residual pressure exists into the tank , the cover should slightly rotate around his articulation axis to evacuate the pressure before the 4 swing bolts have been rotated a stop exist on the cover near each swing bolt to secure the rotation before the pressure is evacuated.

Before rotate completely the 4 swing bolts, pull the cover to check there is no more pressure, then rotate finish the unscrewing and rotate the swing bolts .

The cover can be open in two ways : rotation around the horizontal axis , or rotation around the vertical axis , comfortable when the space is limited upper side .

Cover closing

Position the seal cover in front of the neckring sealing surface

Rotate the 4 swing bolts, pass over the cover stop to finish the tightening

A strength value of 20 daN on the handle is enough to get the seal tightness

Sealing

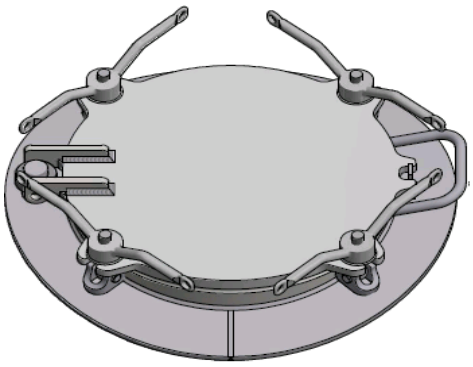
The cover vertical articulation spindle and the handles are drilled to allow installing a cable or a bar diameter 10 mm to secure the manlid against undesirable openings

Special precautions

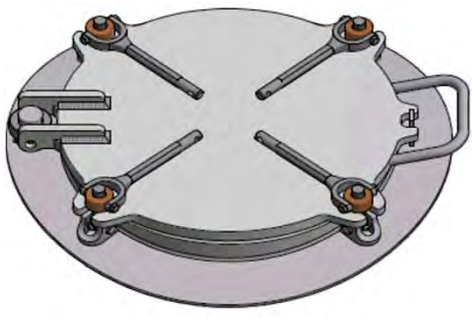
The neckring pressure test at 3 bar should be run by the customer after the neckring is welded on the tank.

(This test is not done during our final inspection.)

The cover should not be open when the tank is pressurized



Extended Wing Nuts



Folding Straight Handnut

MANLID

ND450 Dry Bulk Powder Manlid

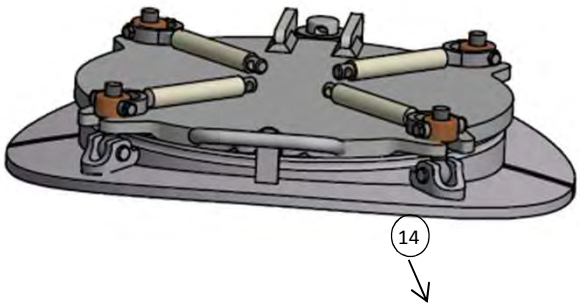
MANLIDS

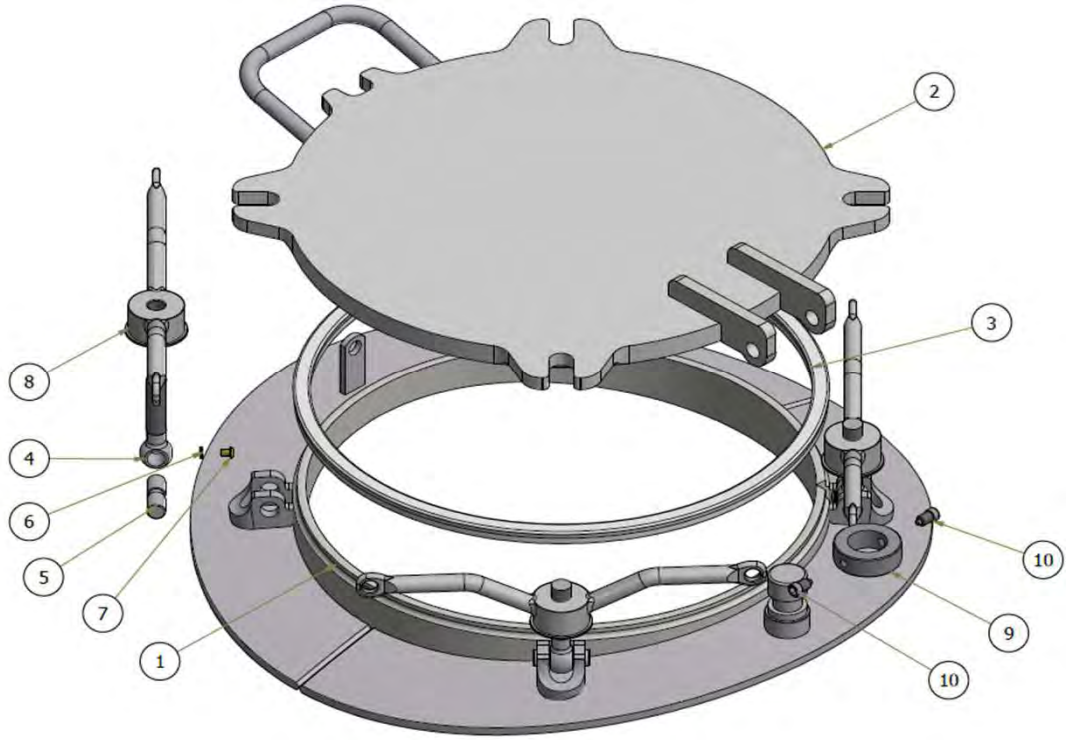
13 15 24 00 00

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Coverlid, Neckring & seal*	
2	1	ND450 Coverlid	13 15 11 10 00
3	1	Seal Nitrile 10 x 14 rectangular	13 15 01 00 04
3	1	Seal EPDM 10 x 14 rectangular	13 15 24 00 04
1	1	Neckring	13 15 12 10 00
		Articulation ring kit	
9	1	Articulation ring	13 15 12 94 00
10	2	Articulation pin	
		Swingbolt assembly - Extended Wing Nuts	
5	1	Pivot pin	13 15 12 91 00
4	1	Eye bolt	
8	1	Wing nut (Carbon St)	
7	1	6x8 TBHC Screw	
6	1	M6 Washer	
		Swingbolt assembly - Folding Hand Nuts	
5	1	Pivot pin	13 15 02 91 00
4	1	Eye bolt	
14	1	Wing nut	
7	1	6x8 TBHC Screw	
6	1	M6 Washer	

*for other seal material, please conctact our sales department





MANLID

ND900 MULTIDO
MANLIDS
17 13 01 00 00

ND900 Multido manlid - to suit tank containers, road tankers and storage tanks intended for chemical waste and foodstuff liquid products.
Can be installed in horizontal or vertical position.
2 Hinge System for easier opening and perfect thightness

SPECIFICATION	
Nominal diameter	900 mm (36")
Weight	112 Kg
Fastening points	12
Working Pressure	1 Bar
Test Pressure	1.5 Bar
Working Temperature	-40° to + 120°C

Temperature range may vary according to the seal material

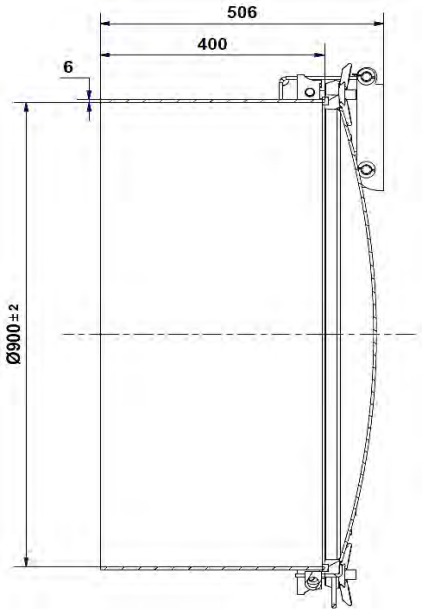
CONFORMITY, CERTIFICATION & APPROVAL
ADR / RID / EN14025

STANDARD FEATURES	
Made of (material)	304L/1.4307-316L/1.4404-Carbon Steel
Seal material	seal not included
Neckring height	from 100 mm to 400 mm
Neckring thickness	6 mm
Fastening	Swingbolts with captive wing nuts
Perfect tightness	Trapezoïdal seal groove 10x14

ACCESSORIES	
Coverlid	17 13 01 20 00
Coverlid seal neoprene	17 13 01 00 01

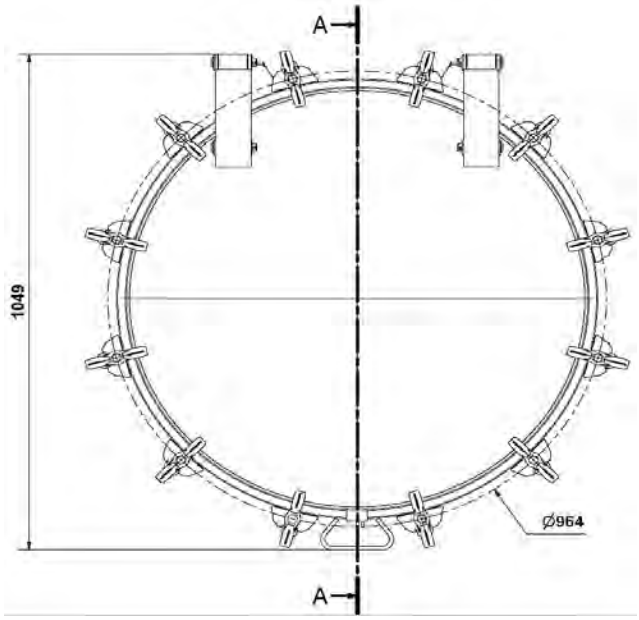
CODIFICATION	
Complete ND900 Multido manlid - 304L st st - 12 points - neckring 400mm high - with safebolt	17 13 01 00 00

DIMENSIONAL DRAWING



Vertical or horizontal

Installation



MANLID

ND900 MULTIDO
MANLIDS
17 13 01 00 00

INSTALLATION

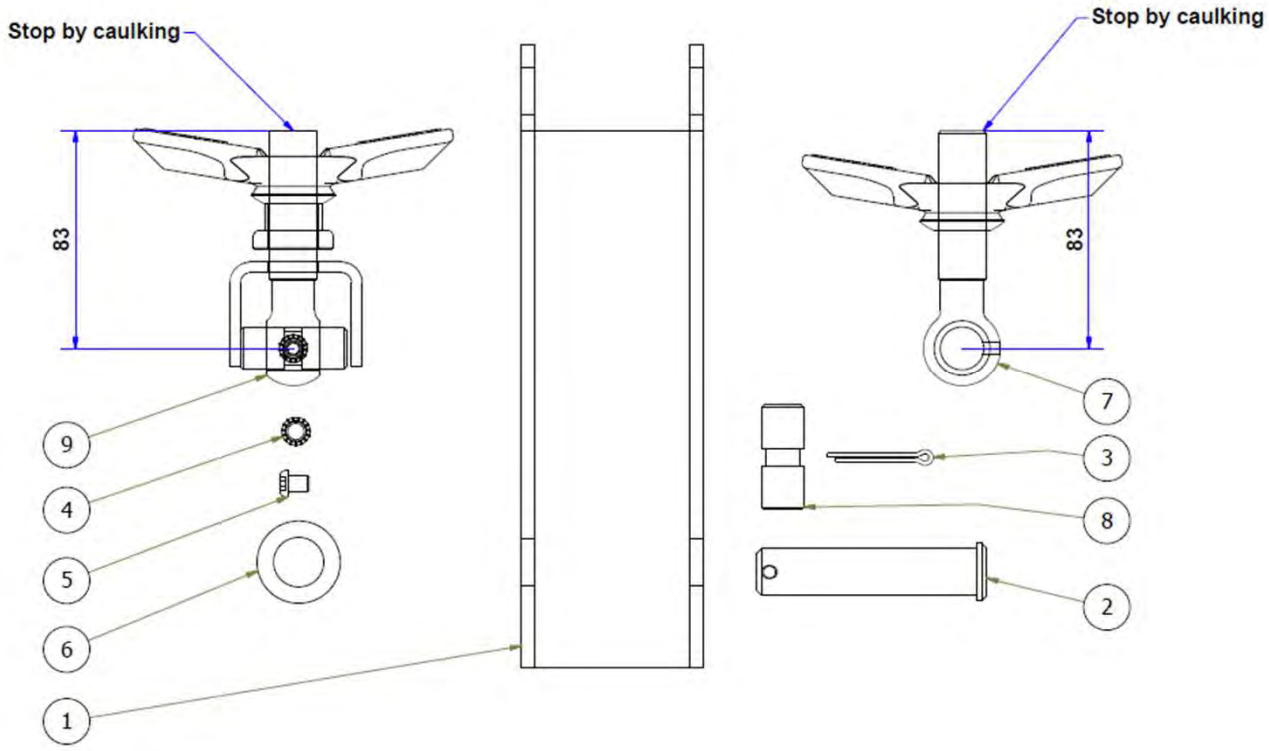
1. Disassemble the 4 bolts with the screw
2. Weld the neckring on the tank.

TIG, MIG or MAG weld process are allowed. Check the material according to your tank and welding process.

3. Check that the sealing surface has no more than1 mm flatness default.
4. Assemble the cover and the bolts

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
Coverlid, Neckring & seal*			
1	2	Rod st st	17 13 11 90 00
2	4	Hinge Pin st st	
3	4	Split Pin st st	
4	11	Washer st st	
5	11	Screw st st	
6	8	Washer st st	
7	11	83 mm EyeBolt (304L) with brass wing nut	
8	11	Pivot Pin st st	
9	1	Safebolt st st ass. (screw, washer, bolt, nut)	



MANLID

AIRLINE VALVES: Ball & Butterfly valves

FUNCTION

An Airline prevents the tank or tanker from implosion or explosion during loading and unloading operations and can be connected to a vapour recovery line if necessary. The most common airline is a 1 ½" BSP straight ball valve, but Perolo offers many different models (ball or butterfly valves) with several options for outlet connections (BSP, flanged, dual).

- Various internal diameters:
  1½": 32mm & 40mm
  2": 50mm
- Ball Valve or Butterfly valve
- Inlet connection:
  Various drilling patterns
  Open or closed slots
- Outlet connection:
  Flanged, Threaded (BSP or NPT)
  Dual outlet
- Operating mechanism:
  Manual: handle or remote lever
  Pneumatic actuator
- Contact Material:
  316L stainless steel
- Working Temperature:
  Standard or High (+250°C)
- Surface Finish:
  Acid etched, Halar lined
- Large choice of accessories:
  Gauges, Gauge guard, Filter
- ADR, RID, IMDG

1½" Ball Valve AD3	1½" Ball Valve AF3	1½" DUAL Ball Valve AT4	2" DUAL Ball Valve ATC5
1½" Ball Valve ADN4	2"-1½" Ball Valve ADC5	2" Ball Valve AD5X7	2" self-contained BV AD5X16
2" Butterfly Valve TF5	2" Butterfly Valve TT5	2" Butterfly Valve TP5	1½" Pneumatic BV AP3
1½" Pneumatic BV AG3	Remote Control *Section Accessories*		

1 ½" BSP Threaded ATCO™
ND32 Ball Valve
AD3X

The Perolo 1 ½" ATCO™ ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. With a super compact design, the 1 ½" ATCO™ is light & economic.

SPECIFICATION	
Nominal diameter	32mm (outlet 1 ½")
Weight	1.2 Kg
Inlet Connection	4 x Ø11 x 103.5 PCD
Outlet Connection	1 1/2" BSP
Working Pressure	6.7 Bar
Test Pressure	10 Bar
Working Temperature	- 40° to +150°C (+250°C for HT)
Design Temperature	- 40° to +190°C

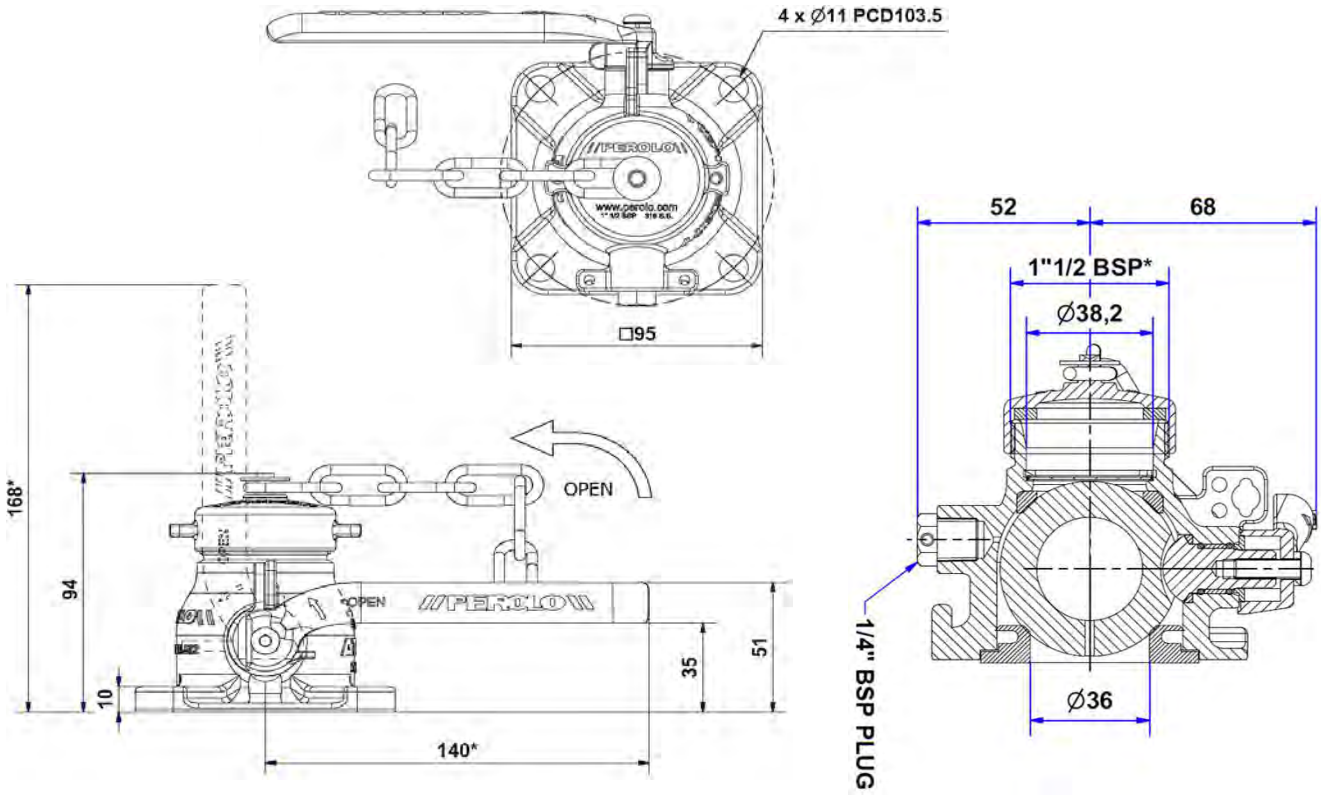
Temperature range vary according to the seal material

ACCESSORIES	
Weld-in flange	11 90 20 00 00
Recess Weld In Flange	11 91 12 00 00
Screw Kit	12 91 99 00 00
Filter Kit	11 14 01 95 00
Remote Control	RMT 33
Remote control Handle	11 14 15 91 00
Pressure gauge	70 18 06 00 95
1 1/2" BSP Cap	11 14 01 30 00

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas - RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face To Face 94mm with Cap
Operation	Handle - 1/4 Turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

1 ½" BSP Threaded ATCO™
ND32 Ball Valve
AD3X

CODIFICATION

STANDARD PART NUMBER	AD3	X	7	T	2	1C	S	2	1	.
YOUR PART NUMBER	AD3	__	7	__	__	__	__	__	__	__

MATERIAL - Other on request	
316L / 1.4409 stainless steel	X

MAWP	
6.7 Bar	7

WORKING TEMPERATURE	
Ball Seals PTFE +150°C	T
High temp +250°C ball seals PEEK	H

INLET FLANGE	
4 x Ø11 x 103.5 PCD	2

CONNECTION & CAP	
BSP without Cap	1
BSP with Standard Cap/Chain	1C
BSP with Reinforced Chain	1K
NPT with Standard Cap	2C

FILTER	
Without	S
With	P

GAUGE CONNECTION & PLUG	
1/4" BSP Connection & Plug	2

GAUGE PROTECTION	
Without	1
With	2

REMOTE CONTROL HANDLE	
Without remote control handle	.
With remote control handle	W

AIR LINE

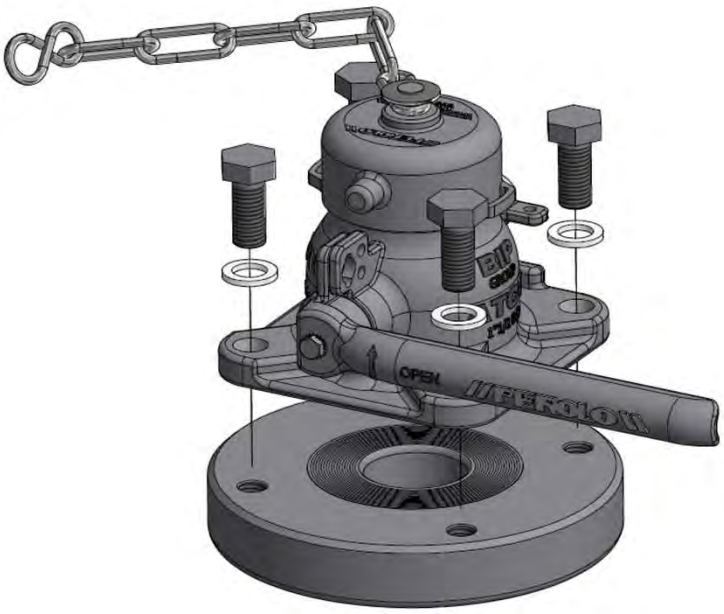
Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 01 20 00 J
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 ½" BSP Threaded ATCO™
ND32 Ball Valve
AD3X

INSTALLATION

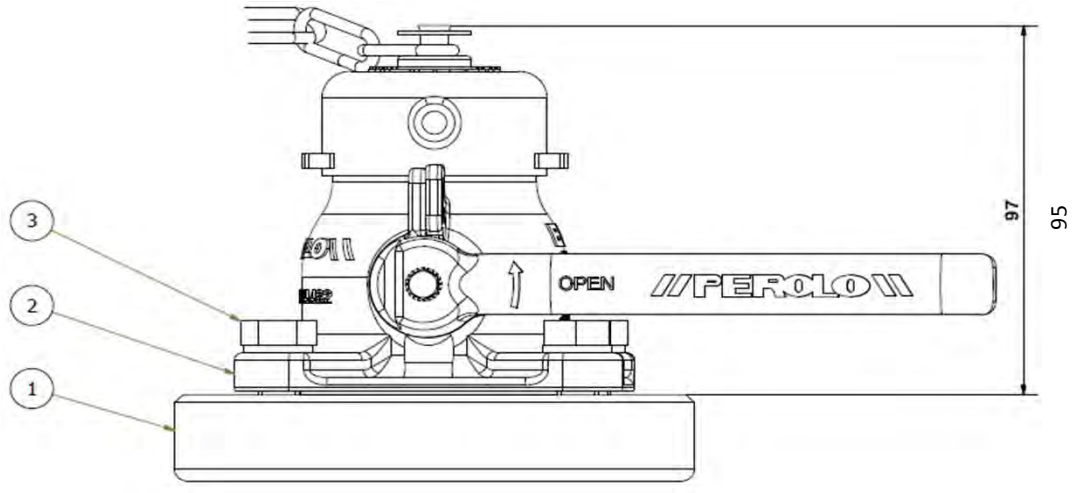
1. check Inlet (tank side) flange is clean, free from dirt and grit.
2. Screw the studs into the weldin flange, lubricate with PTFE added grease before installing the studs
3. Install the valve on the flange, secure with washer and nuts *(refer to PWAGEN001 fort bolt tightening sequence)*
4. Tighten the nuts torque : 30 N.m +/- 5
5. Open and close the valve 3 times to check the operational torque.

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs



NO GASKET

ITEM	DESCRIPTION	SPECIFICATION	PART N°
2	ND32 ATCO Ball Valve	ND32 Ball Valve 4 x Ø11 x 103.5 PCD	AD3X7T21CS21
3	Screw Kit	4 x M10-25 screws, 4 x grower washers	12 91 99 00 00
1	Weld-in Flange	int Ø32mm, ext Ø127mm, thickness 22mm 4 x M10 x 103.5 PCD 316L Stainless Steel	11 90 20 00 00



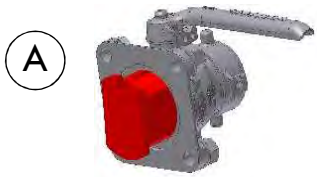
AIR LINE

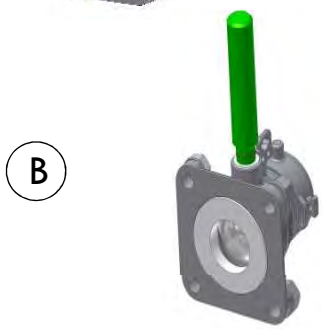
1 ½" BSP Threaded ATCO™
ND32 Ball Valve
AD3X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°	
			Standard	High Temp
		Ball seals kit:	**11 14 01 93 00**	**11 14 01 96 00**
10	1	Ball seal (PTFE / PEEK)	11 14 01 00 01	11 14 22 00 01
12	1	Main seal (PTFE / PEEK)	11 14 01 00 04	11 14 22 00 02
		Ball kit:	**11 14 01 94 00**	
11	1	Ball (316)	12 96 06 00 09	
		Handle kit:	**11 14 01 91 00**	**on request**
1	1	Screw M6x20 (st st)	-	
2	1	Washer W6 (st st)	-	
3	1	Handle (st st)	11 14 01 10 03	
4	1	Bearing (PTFE / PEEK)	12 96 06 00 13	11 14 22 00 04
5	1	O'ring (Viton)	70 10 03 02 08	
6	1	Spacer (PTFE / PEEK)	11 14 02 00 03	11 14 22 00 03
7	1 / 2	O'ring (FEP+Silicon / PFA+Silicon)	70 10 15 00 66	70 10 17 00 38
8	1	Spindle Gasket (PTFE / PEEK)	11 14 02 00 01	11 14 22 00 05
9	1	Spindle (st st)	11 14 02 00 02	
		Spindle seals kit:	**11 14 01 92 00**	**on request**
4	1	Bearing (PTFE / PEEK)	12 96 06 00 13	11 14 22 00 04
5	1	O'ring (Viton)	70 10 03 02 08	
6	1	Spacer (PTFE / PEEK)	11 14 02 00 03	11 14 22 00 03
7	1 / 2	O'ring (FEP+Silicon / PFA+Silicon)	70 10 15 00 66	70 10 17 00 38
8	1	Spindle Gasket (PTFE / PEEK)	11 14 02 00 01	11 14 22 00 05
		Filter kit:	**11 14 01 95 00**	
13	1	Filter (st st)	-	
14	1	Lock Ring (st st)	-	
		1 1/2" BSP Cap kit:	**11 14 01 30 00**	
15	1	Cap Seal	12 85 97 00 10	
16	1	BSP Cap & Chain	-	
		Tools kit:	**11 14 01 90 00**	
A	1	Spindle stop tool (Alu 2017T4)	11 14 01 90 01	
B	1	Spindle seal tool (st st)	11 14 01 90 02	

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed simultaneously.
- Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details





AIR LINE

1 ½" Flanged ATCO™
ND32 Ball Valve
AF3X

The Perolo 1 ½" ATCO™ ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. With a super compact design, the 1 ½" ATCO™ is light & economic.

SPECIFICATION	
Nominal diameter	32mm (outlet 1 ½")
Weight	3.5 Kg
Inlet Connection	4 x Ø11 x 103.5 PCD
Outlet Connection	4 x M16 x 98.4 PCD
Working Pressure	6.7 Bar
Test Pressure	10 Bar
Working Temperature	- 40° to +150°C (+250°C for HT)
Design Temperature	- 40° to +190°C

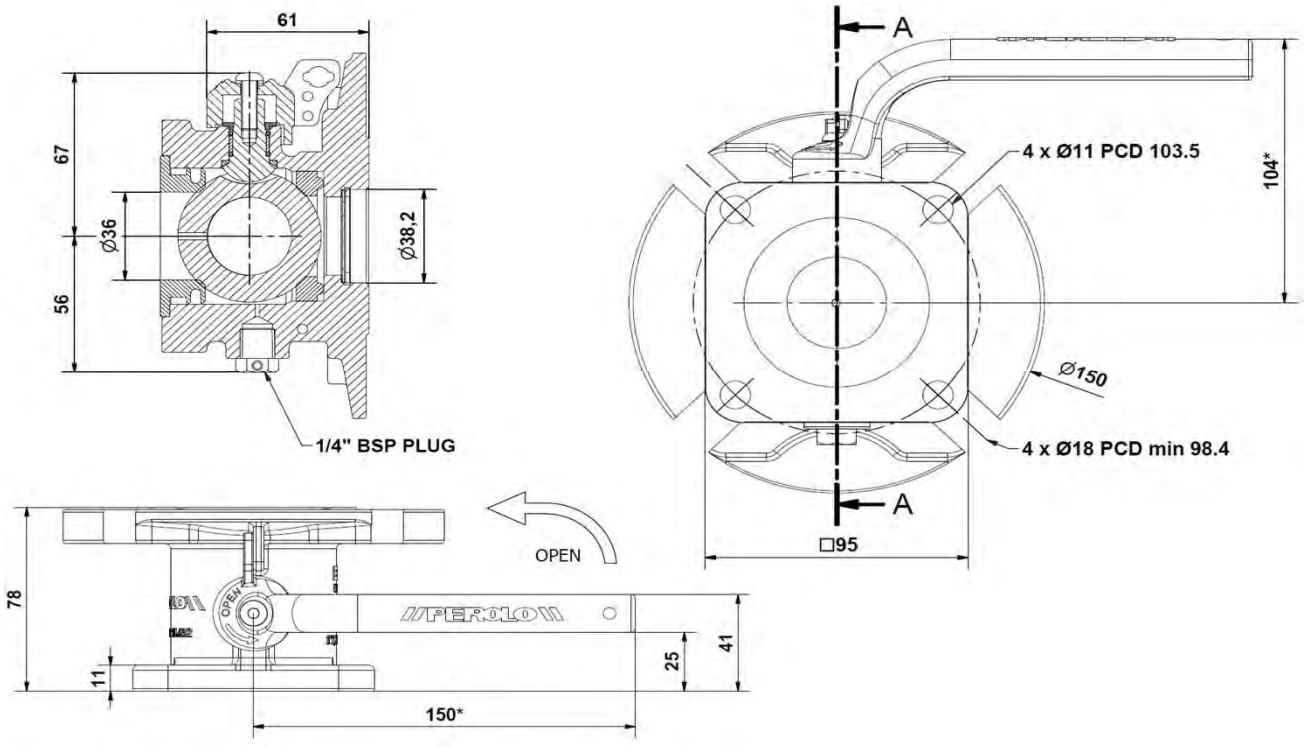
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas - RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face To Face 78mm
Operation	Handle - 1/4 Turn or Remote Control
Ball seals material	PTFE

ACCESSORIES	
Weld-in flange	11 90 20 00 00
Recess Weld In Flange	11 91 12 00 00
Blind Flange	11 95 87 00 00
Gasket	12 91 90 00 00
Bolting Kit	17 12 10 95 00
Screw Kit	12 91 99 00 00
Gasket	12 91 90 00 00
Filter Kit	11 14 01 95 00
Remote Control	RMT 33
Remote control Handle	11 14 02 21 00
Pressure gauge	70 18 06 00 95

DIMENSIONAL DRAWING



AIR LINE

1 ½" Flanged ATCO™
ND32 Ball Valve
AF3X

CODIFICATION

STANDARD PART NUMBER	AF3	X	7	T	2	5	S	2	1	.
YOUR PART NUMBER	AF3	__	7	__	2	__	__	2	__	__

MATERIAL - Other on request	
316L / 1.4409 stainless steel	X

MAWP	
6.7 Bar	7

WORKING TEMPERATURE	
Ball Seals PTFE +150°C	T
High temp +250°C ball seals PEEK	H

INLET FLANGE	
4 x Ø10.5 x 103.5 PCD	2

OUTLET FLANGE - other on request	
4 x M16 x 98.4 PCD	4
4 x slots Ø18 x min98.4 PCD	5
4 x M16 x 125 PCD	6
4 x M16 x 110 PCD	8
4 x Ø18 x 110 PCD	9

FILTER	
Without	S
With	P

GAUGE CONNECTION & PLUG	
1/4" BSP Connection & Plug	2

GAUGE PROTECTION	
Without	1
With	2

REMOTE CONTROL HANDLE	
Without remote control handle	.
With remote control handle	W

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 02 20 00 J
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 ½" Flanged ATCO™
ND32 Ball Valve
AF3X

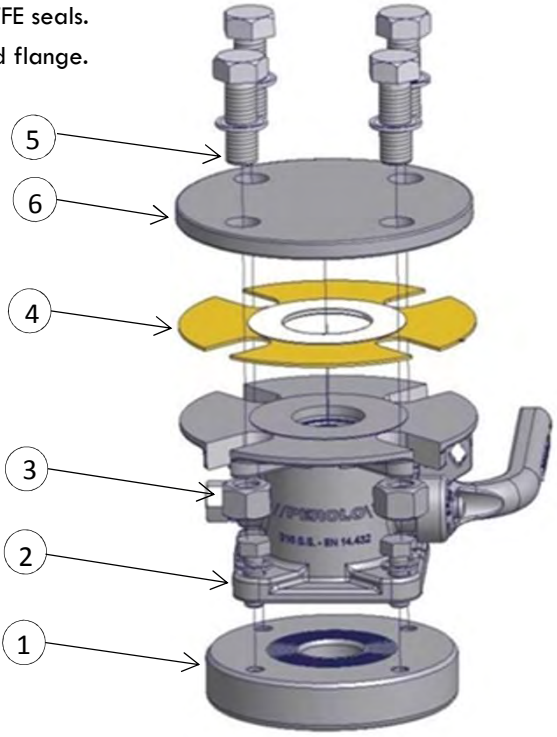
INSTALLATION

To install the ND32 ATCO™, the weldin flange must be machined with stock finish (serrated surface).

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Install the valve on the welding flange, **handle in closed position**. NO GASKET NEEDED
3. Install the 4 screws and washers.

Tightening torque required : 30 N.m +/- 5

4. Open and close the valve 3 times before operating to install the PTFE seals.
5. Install the gasket between the 1 ½" Flanged ATCO™ and the blind flange.
6. Secure with the M16 screws, washers and nuts.
7. Refer to PWAGENE001 for bolt tightening sequence.



NO GASKET

ITEM	DESCRIPTION	SPECIFICATION	PART N°
6	Blind flange	ext ø 150 mm - 4x18x110 PCD thickness 14 mm 316L stainless steel	11 95 87 00 00
5	Screw kit	4xM16-60 screws, 4 x lock washers 4xM16 nuts	17 12 10 98 00
4	Seal	Seal, flat, round, CNAF/PTFE, internal Ø 45mm, external Ø 150mm, 4 slots x 19 x min. 101.4 PCD	12 91 90 00 00
3	Screw kit	4 x M10 -25 screws, 4 x grower washers	12 91 99 00 00
2	ND32 ATCO ball valve	ND32 Flanged Ball valve 4xø11x103.5PCD	AF3X7T25S21
1	Weld-in flange	int ø 32 mm, ext ø 127 mm thickness 22mm 4xM10x103.5PCD 316L stainless steel	11 90 20 00 00

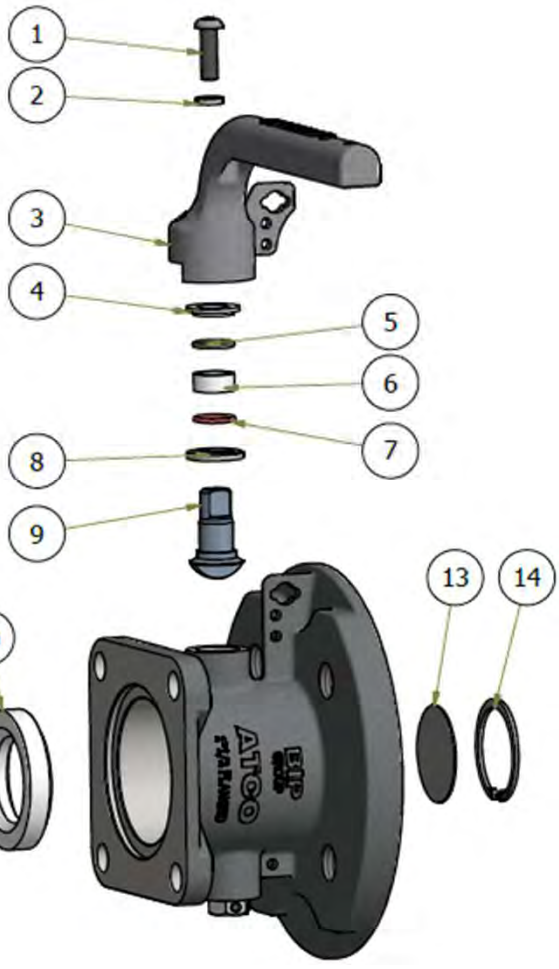
if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

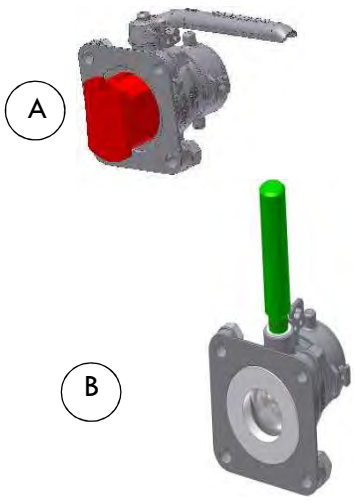
1 ½" Flanged ATCO™
ND32 Ball Valve
AF3X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 91 09 00 00**
10	1	Ball seal (PTFE)	12 96 06 00 10
12	1	Main seal (PTFE)	11 14 02 00 06
		Ball kit:	**11 14 01 94 00**
11	1	Ball (316)	12 96 06 00 09
		Handle kit:	**11 14 02 91 00**
1	1	Screw M6x16 (st st)	-
2	1	Washer W6 (st st)	-
3	1	Handle (st st)	11 14 02 10 01
4	1	Bearing (PTFE)	12 96 06 00 13
5	1	O'ring (Viton)	70 10 03 02 08
6	1	Spacer (PTFE)	11 14 02 00 03
7	1	O'ring (FEP/Silicon)	70 10 15 00 66
8	1	Spindle Gasket (PTFE)	11 14 02 00 01
9	1	Spindle (st st)	11 14 02 00 02
		Spindle seals kit:	**11 14 01 92 00**
4	1	Bearing (PTFE)	12 96 06 00 13
5	1	O'ring (Viton)	70 10 03 02 08
6	1	Spacer (PTFE)	11 14 02 00 03
7	1	O'ring (FEP+Silicon)	70 10 15 00 66
8	1	Spindle Gasket (PTFE)	11 14 02 00 01
		Filter kit:	**11 14 01 95 00**
13	1	Filter (st st)	-
14	1	Lock Ring (st st)	-
		Tools kit:	**11 14 01 90 00**
A	1	Spindle stop tool (Alu 2017T4)	11 14 01 90 01
B	1	Spindle seal tool (st st)	11 14 01 90 02

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed simultaneously.
- Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details





AIR LINE

1 ½" Threaded ATCO™
Dual Outlet ND32 Ball Valve
AT4X

The Perolo 1 ½" ATCO™ ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	32mm (outlet 1 ½")
Weight	2.3 Kg
Inlet Connection	4 holes x Ø10.5 x 103.5 PCD
Outlet Connection	1 1/2" BSP &/or 1 1/2" NPT
Working Pressure	6 Bar
Test Pressure	9 Bar
Working Temperature	- 40° to +150°C (+180° for HT)
Design Temperature	- 40° to +190°C

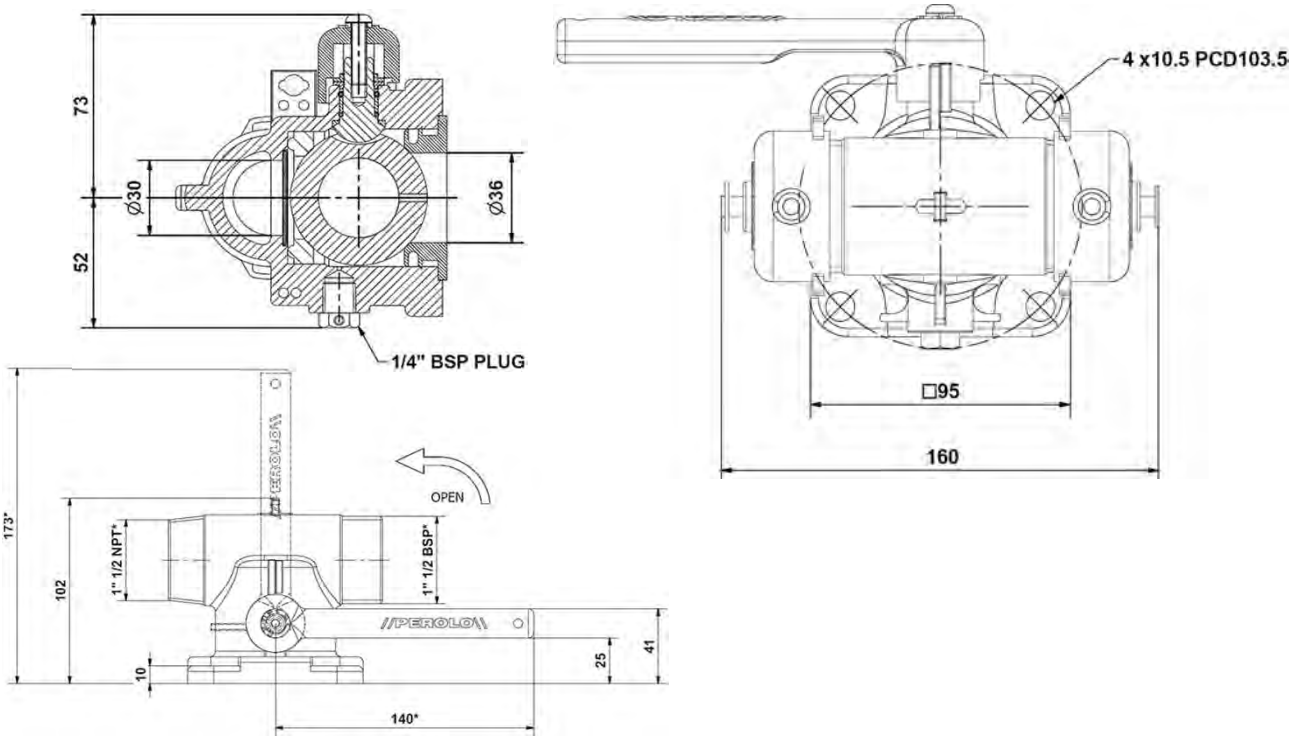
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

ACCESSORIES	
Weld-in flange	11 90 20 00 00
Screw kit	12 91 99 00 00
Remote control	RMT 33
Remote control handle	12 96 07 16 00
Pressure gauge	70 18 06 00 95
1 1/2" BSP Cap	11 14 01 30 00
1 1/2" NPT Cap	12 96 03 00 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Height 102mm
Operation	Handle - 1/4 Turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

1 ½" Threaded ATCO™
Dual Outlet ND32 Ball Valve
AT4X

CODIFICATION

STANDARD PART NUMBER	AT4	X	6	T	2	3C	S	2	1	.	.
YOUR PART NUMBER	AT4		6	T							

MATERIAL - Other on request	
316L / 1.4409 stainless steel	X

MAWP	
6.7 Bar	6

WORKING TEMPERATURE	
Ball Seals PTFE +150°C	T
High temp +180°C ball seals PTFE/Gr	H

INLET FLANGE - other on request	
4 x Ø10.5 x 103.5 PCD	2

CAP	
With Standard Cap (BSP & NPT)	3C
Without Cap	3

FILTER	
Without	S
With	P

GAUGE CONNECTION & PLUG	
With 1/4" BSP Connect. without Plug	0
With 1/4" NPT Connection & Plug	1
With 1/4" BSP Connection & Plug	2

GAUGE PROTECTION	
Without	1
With	2

CONNECTION	
2 x 1 1/2" BSP	B
1 x 1 1/2" BSP & 1 x 1 1/2" NPT	.

REMOTE CONTROL HANDLE	
Without remote control handle	.
With remote control handle	W

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 12 96 08 20 00 J
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 ½" Threaded ATCO™
Dual Outlet ND32 Ball Valve
AT4X

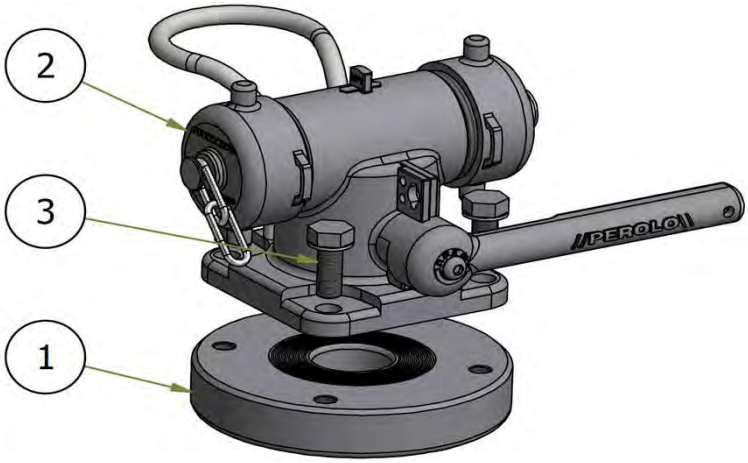
INSTALLATION

To install the ND32 ATCO™, the weldin flange must be machined with stock finish (serrated surface).

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Install the valve on the weldin flange, handle in closed position. NO GASKET NEEDED
3. Install the 4 screws and washers.

Tightening torque required : 30 N.m +/- 5

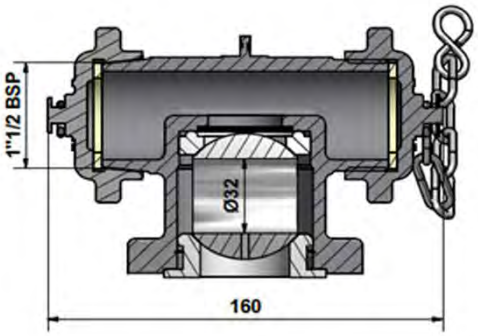
4. Open and close the valve 3 times before operating to install the PTFE seals.
5. Refer to PWAGENE001 for bolt tightening sequence.

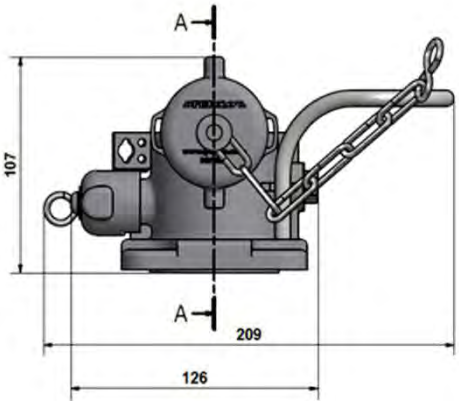


NO GASKET

ITEM	DESCRIPTION	SPECIFICATION	PART N°
2	ND32 ATCO Ball Valve	ND32 Ball Valve 4 x Ø11 x 103.5 PCD, with gauge protection	AT4X6T23CS22
3	Screw Kit	4 x M10-25 screws, 4 x grower washers	12 91 99 00 00
1	Weld-in Flange	int Ø32mm, ext Ø127mm, thickness 22mm 4 x M10 x 103.5 PCD 316L Stainless Steel	11 90 20 00 00

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs





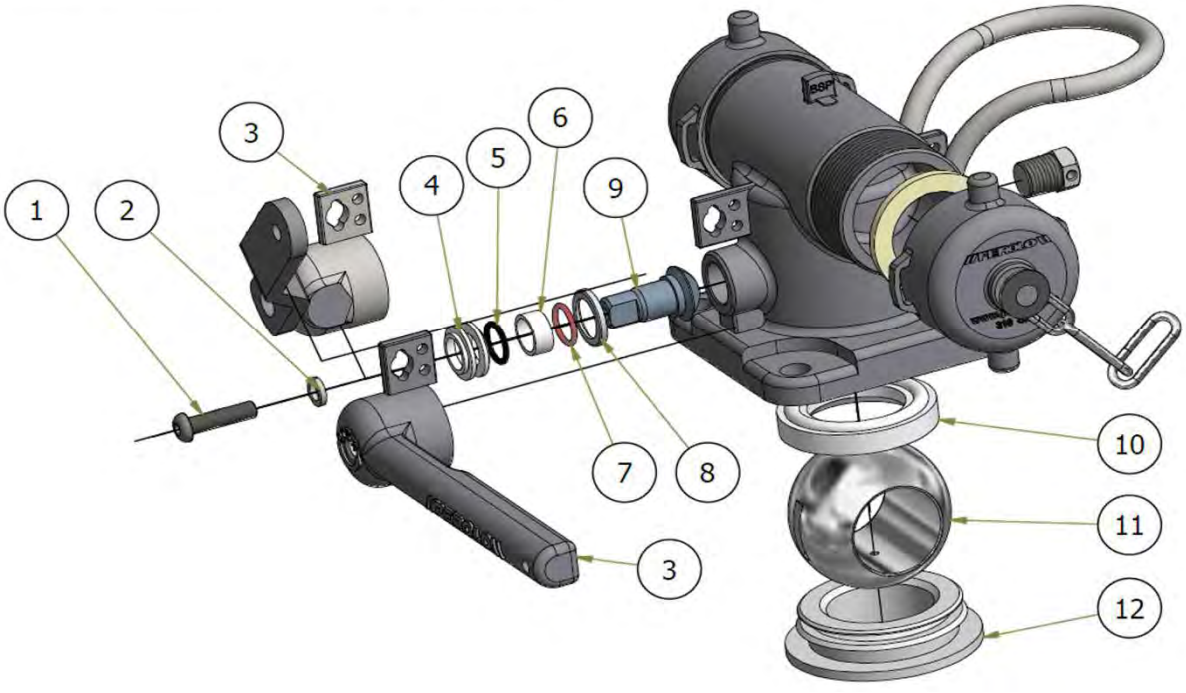
AIR LINE

1 ½" Threaded ATCO™
Dual Outlet ND32 Ball Valve
AT4X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 91 09 00 00**
10	1	Ball seal (PTFE)	12 96 06 00 10
12	1	Main seal (PTFE)	11 14 02 00 06
		Ball kit:	**11 14 01 94 00**
11	1	Ball (316)	12 96 06 00 09
		Handle kit:	**12 96 06 92 50**
1	1	Screw (st st)	-
2	1	Washer (st st)	-
3	1	Handle (st st)	12 96 08 00 18
4	2	Bearing (PTFE)	12 96 06 00 13
5	1	O'ring (Viton)	70 10 03 02 08
6	1	Spacer (PTFE)	11 14 02 00 03
7	1	O'ring (FEP/Silicon)	70 10 15 00 66
8	1	Spindle gasket (PTFE)	11 14 02 00 01
9	1	Spindle (st st)	11 14 02 00 02
		Spindle seals kit:	**11 14 01 92 00**
4	1	Bearing (PTFE)	12 96 06 00 13
5	1	O'ring (Viton)	70 10 03 02 08
6	1	Spacer (PTFE)	11 14 02 00 03
7	1	O'ring (FEP+Silicon)	70 10 15 00 66
8	1	Spindle Gasket (PTFE)	11 14 02 00 01
		Caps:	
-	1	Cap Seal	12 85 97 00 10
-	1	BSP Cap + Seal + Chain	11 14 01 30 00
-	1	NPT Cap + Seal + Chain	12 96 03 00 00

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed simultaneously.
- Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details



AIR LINE

2 x 2" BSP or NPT Threaded ATCO™
Dual Outlet ND50 Ball Valve
ATC5X7T

The Perolo ND50 ATCO FULL BORE DUAL 2" BSP or NPT threaded ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. With a super compact design, the DN50 ATCO is light & economic.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	5,5 KG
Inlet Connection	4 holes x Ø18 x 125 PCD
Outlet Connection	2 X 2" BSP or 2" NPT
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	- 40° to +150°C
Design Temperature	- 40° to +190°C

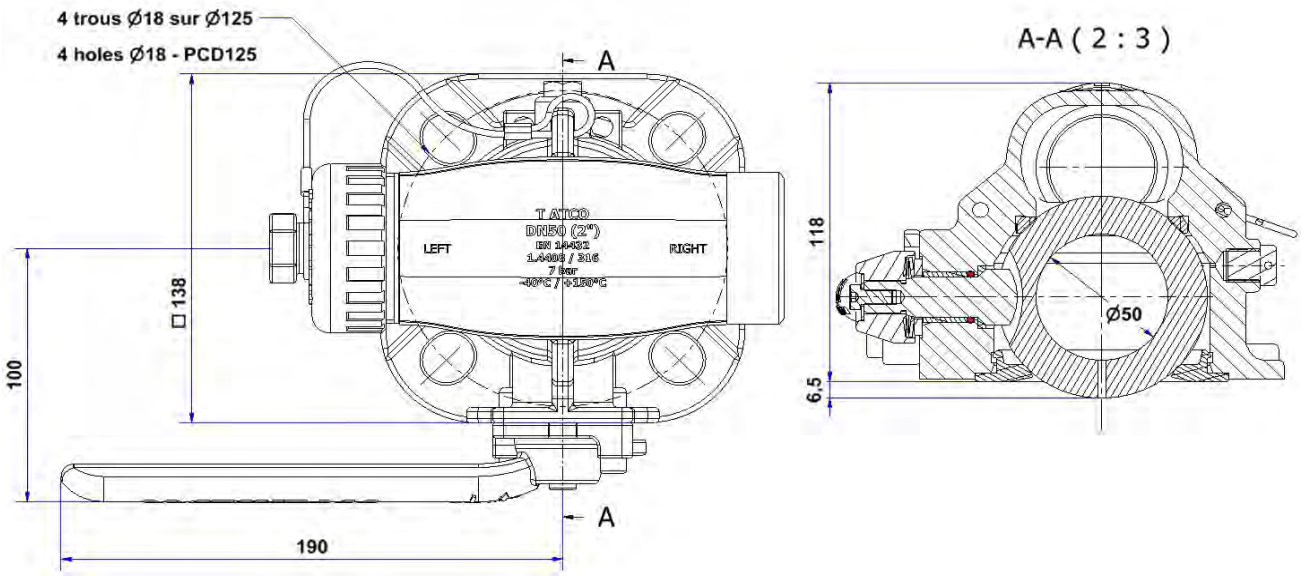
Temperature range vary according to the seal material

ACCESSORIES	
Weld in flange	11 90 25 00 00
Recess flange	On request
Stud kit	11 91 13 91 00
Remote Control	RMT 53
Remote control handle	11 24 06 00 01
Pressure gauge	70 18 06 00 95
2" BSP Cap kit	11 24 06 30 00
2" NPT Cap kit	On request

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Compact Design	Face to face 118 mm
Operation	Handle 1/4 Turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

2 x 2" BSP or NPT Threaded ATCO™
Dual Outlet ND50 Ball Valve
ATC5X7T

ISO ROAD IBC

AIR LINE

CODIFICATION

STANDARD PART NUMBER	ATC5	X	7	T	1	B	B	0	0	0
YOUR PART NUMBER	ATC5	__	7	T	1	__	__	__	__	__

MATERIAL - Other on request	
316 / 1.4408 stainless steel	X

MAWP	
7 Bar	7

WORKING TEMPERATURE	
Ball Seals PTFE -40°C +150°C	T

INLET FLANGE DRILLING	
4 x Ø18 x 125 PCD	1

RIGHT OUTLET CONNECTION*	
2" Buttwelding	0
2" BSP with cap	B
2" BSP without cap	D

*other (eg NPT) on request

LEFT OUTLET CONNECTION*	
2" Buttwelding	0
2" BSP with cap	B
2" BSP without cap	D

*other (eg NPT) on request

HANDLE	
Reverse handle	0
Standard handle	1
Right remote control handle	2
Left remote control handle	3

GAUGE CONNECTION	
Without	0
With	1

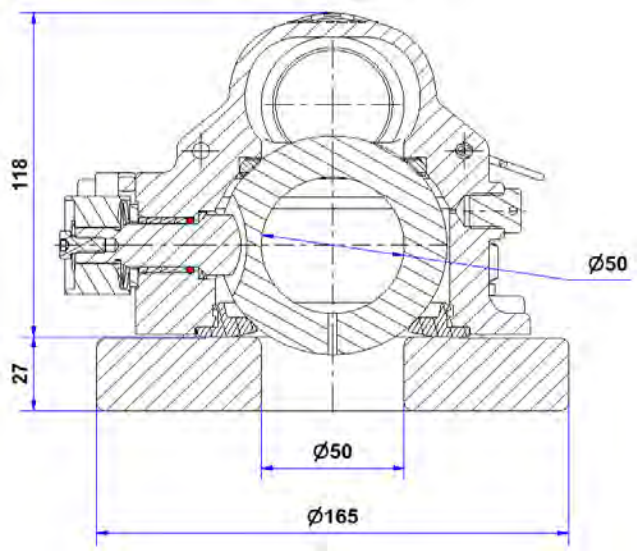
GAUGE PROTECTION	
Without	0
With	1

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 24 06 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 x 2" BSP or NPT Threaded ATCO™
Dual Outlet ND50 Ball Valve
ATC5X7T

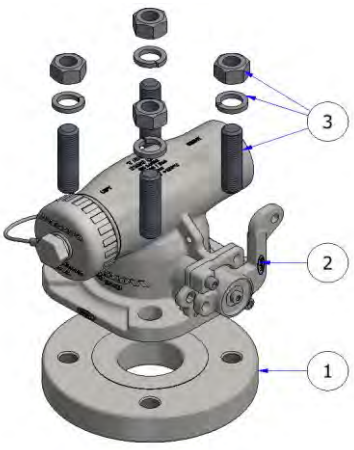
INSTALLATION

1. Check Inlet (tank side) flange is clean, free from dirt and grit.
2. Screw the studs into the weldin flange, lubricate with PTFE added grease before installing the studs
3. Install the valve on the flange, secure with washer and nuts *(refer to PWAGEN001 fort bolt tightening sequence)*
4. Tighten the nuts torque : 30 N.m -/+5
5. Open and close the valve 3 times to check the operational torque.



ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	Bolting Kit	4 x M16 33 studs 4 x nuts and 4 x washers	11 91 13 91 00
2	T ND 50 ATCO "Compact"	PN 7 Full Bore, 316 stainless steel Inlet flange 4xø17x125PCD 2 x 2" BSP with cap	ATC5X7T1BB000
1	Weld-in flange	ext ø 165mm, int ø 50mm thickness 26,5mm 4xM16x125PCD 316L stainless steel	11 90 25 00 00

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs



AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 24 06 30 00 A
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PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 x 2" BSP or NPT Threaded ATCO™
Dual Outlet ND50 Ball Valve
ATC5X7T

AIR LINE

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PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 x 2" BSP or NPT Threaded ATCO™
Dual Outlet ND50 Ball Valve
ATC5X7T

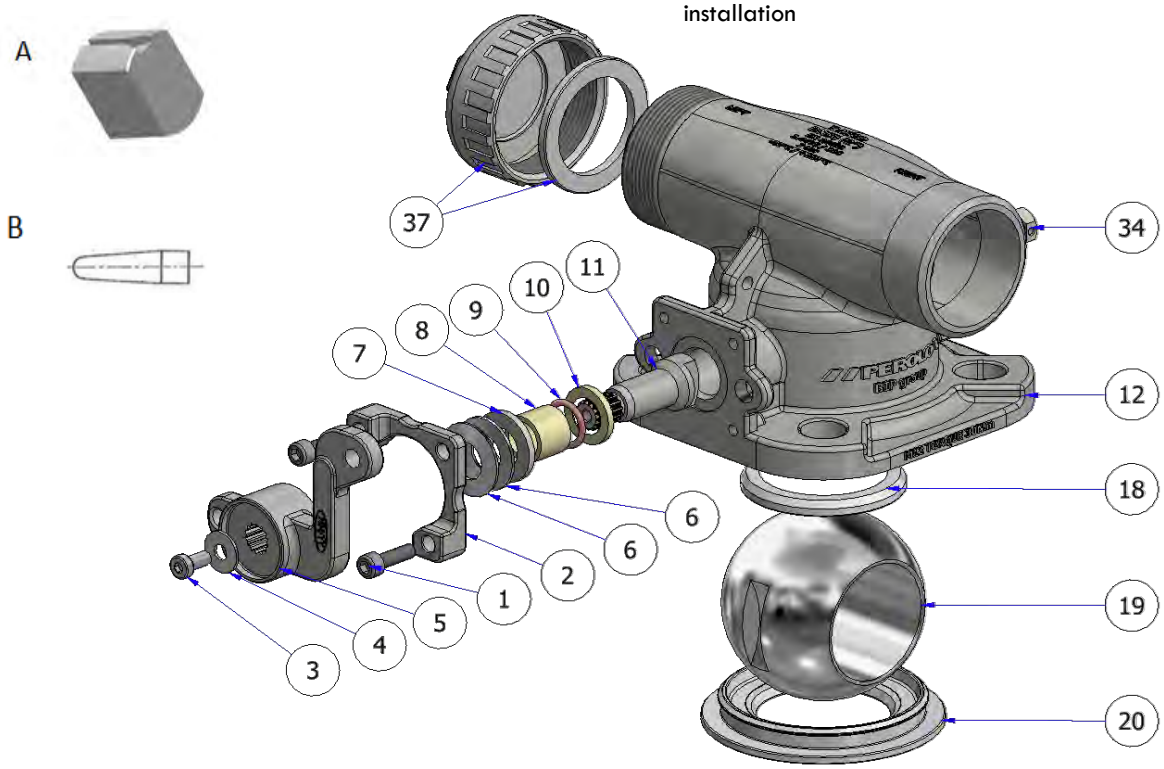
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 14 99 90 00**
18	1	Upper ball seal (PTFE)	11 12 39 00 09
20	1	Lower gasket (PTFE)	11 14 99 10 04
		Ball kit:	
19	1	Ball (316)	11 12 42 00 11
		Drilled Ball kit:	
19	1	Drilled Ball (316)	11 12 39 00 11
		Handle kit:	
5	1	Kit Right handle (11 14 99 10 01L)	**11 14 99 95 00**
	1	Kit Left handle (11 14 99 10 01)	**11 14 99 92 00**
	1	Kit Lever (11 24 06 00 01)	**11 24 06 92 00**
3	1	Screw M6x12	-
4	1	Washer Ø6	-
6	2	Elastic washer Ø16	-
7	1	Washer Ø16 wide series	-
		Stop handle kit:	**11 14 99 30 00**
2	1	Stop handle	11 14 99 1A 02
1	2	Screw M6x20	-
		Spindle seals kit:	**11 14 99 93 00**
8	1	Bearing Bush (PTFE)	11 12 39 00 14
9	1	O'Ring (FEP/silicone)	70 10 15 00 12
10	1	Spindle Gasket	11 12 39 00 21

ITEM	QTY	DESIGNATION	PART N°
		Spindle kit:	**11 14 99 94 00**
11	1	Spindle (Duplex)	11 14 99 10 05
6	2	Elastic washer Ø16	-
7	1	Washer Ø16 wide series	-
		2"BSP Cap kit:	
37	1	2" std BSP Cap + Seal + cable	11 24 06 30 00
-	1	Cap seal (PTFE)	11 12 03 40 02
		Pressure gauge and protection:	
-	1	Plug 1/4"	70 12 39 37 07
-	1	Gauge (many models on request)	70 18 07 00 34
-	1	Gauge protection	11 14 99 00 01
-	2	Screw HC M5-5	70 13 08 37 10
-	1	Union Fitting 1/4" BSP-M/F	70 12 13 37 41
		Tools kit:	**11 12 69 94 00**
A	1	Spindle Assembly tool	11 12 69 92 01
B	1	O'Ring Assembly tool	11 12 69 92 02

**When the ATCO is installed on recess flange*

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed
-Valve should be in closed position during installation



AIR LINE

1 ½" BSP or NPT Threaded ATCO™
ND40 Ball Valve
ADN4X

The Perolo 1 ½" ATCO™ ND40 FULL Bore self-contained ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. With a super compact design, the 1 ½" ATCO™ is light & economic. It has a full bore ND40 opening and can withstand 16 MAWP.

SPECIFICATION	
Nominal diameter	40 mm (outlet 1 ½")
Weight	3,63 Kg
Inlet Connection	4 x Ø18 x 110 PCD
Outlet Connection	1 1/2" BSP or NPT
Working Pressure	16 Bar
Test Pressure	24 Bar
Working Temperature	-40°C to 150°C
Design Temperature	-40°C to 200°C

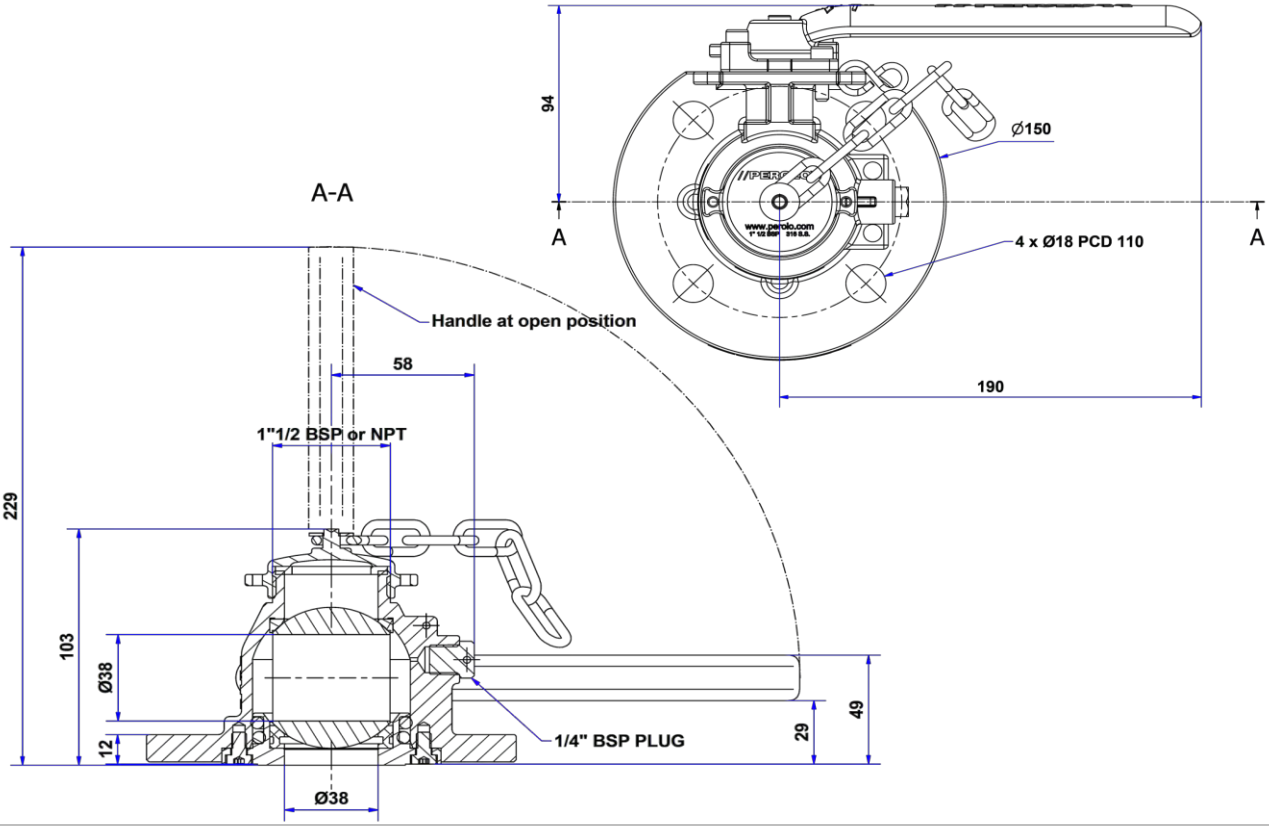
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

ACCESSORIES	
Weld-in flange	11 96 63 00 00
Recess Weld In Flange	On request
Screw Kit	11 12 69 91 00
Gasket	12 91 90 00 00
Filter Kit	11 24 01 95 00
Remote Control	RMT 33
Remote control Handle	11 14 99 30 01
Pressure gauge	70 18 06 00 95
1 1/2" BSP Cap	11 14 01 30 00
1 1/2" NPT Cap	On request

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 /1.4408 Stainless steel
Compact Design	Face to face 103mm with cap
Operation	Handle - 1/4 Turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 24 01 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 ½" BSP or NPT Threaded ATCO™
ND40 Ball Valve
ADN4X

CODIFICATION

STANDARD PART NUMBER	ADN4	X	16	T	1	BC	S	0	0	0	0
YOUR PART NUMBER	ADN4	__	16	T	1	__	__	__	__	__	__

MATERIAL - Other on request	
316 / 1.4408 stainless steel	X

MAWP	
16 Bar	16

WORKING TEMPERATURE	
Ball Seals PTFE +150°C	T

INLET FLANGE	
4 x Ø18 x 110 PCD	1

CONNECTION & CAP	
BSP with Cap	BC
BSP without Cap	B0
NPT with Cap	NC
NPT without Cap	N0

REMOTE CONTROL HANDLE	
With standard handle	S
With reverse handle	L
With remote control handle	W

BALL	
Undrilled ball	0
Drilled ball	1

GAUGE CONNECTION	
Without 1/4" BSP Connection	0
With 1/4" BSP Connection	1

GAUGE PROTECTION	
Without Gauge Protection	0
WithGauge Protection	1

FILTER	
Without	0
With	1

AIR LINE

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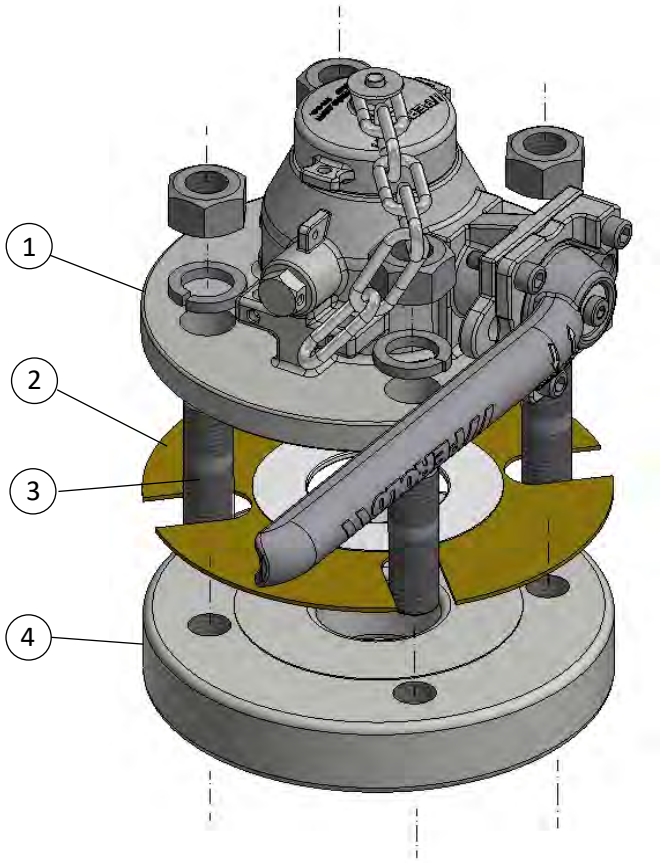
1 ½" BSP or NPT Threaded ATCO™
ND40 Ball Valve
ADN4X

INSTALLATION

1. check Inlet (tank side) flange is clean, free from dirt and grit.
2. Screw the studs into the weldin flange, lubricate with PTFE added grease before installing the studs
3. Install the valve on the flange, secure with washer and nuts *(refer to PWAGEN001 fort bolt tightening sequence)*
4. Tighten the nuts torque : 30 N.m +/- 5
5. Open and close the valve 3 times to check the operational torque.

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs



ITEM	DESCRIPTION	SPECIFICATION	PART N°
1	ND40 ATCO Ball Valve	ND40 Ball Valve 4 x Ø18 x 110 PCD	ADN4X16T1BCS0000
2	Gasket	ext Ø 150mm, int Ø 45mm, 4x slotted Ø19x minimum 101,4 PCD, CNAF/PTFE	12 91 90 00 00
3	Screw Kit	4 x M16 36/28 studs 4 x nuts and 4 x washers	11 12 69 91 00
4	Weld-in Flange	int Ø40mm, ext Ø150mm, thickness 27mm 4 x M16 x 110 PCD 316L Stainless Steel	11 96 63 00 00

AIR LINE

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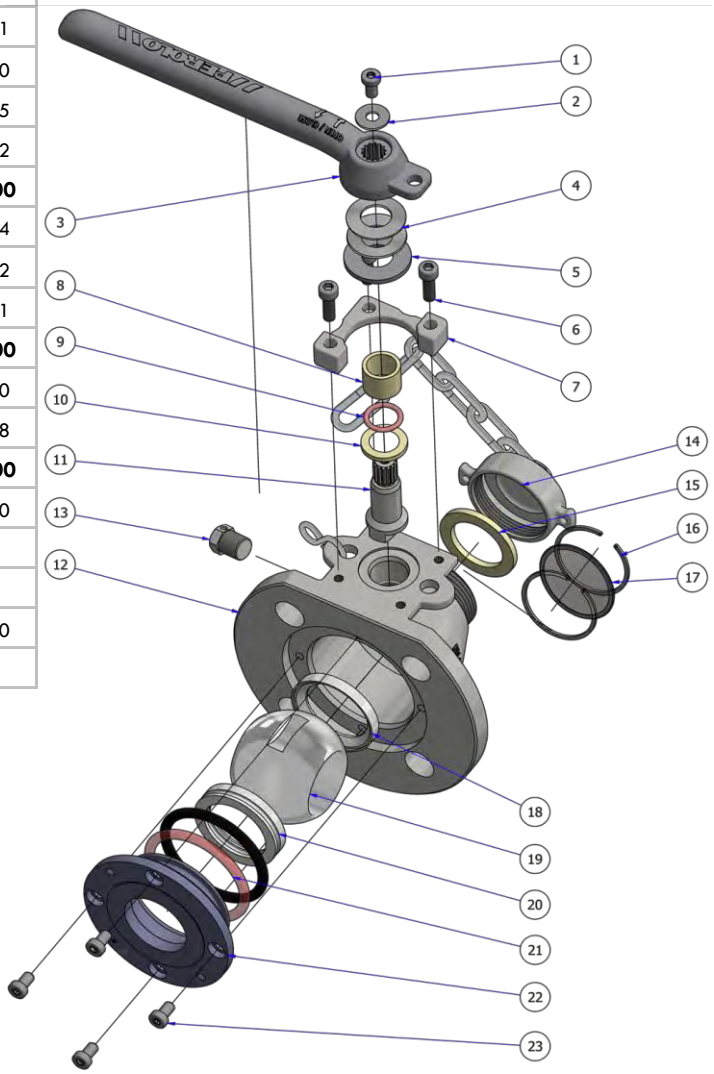
1 ½" BSP or NPT Threaded ATCO™
ND40 Ball Valve
ADN4X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 24 01 93 00**
18	1	Upper Gasket (PTFE)	11 12 82 00 03
20	1	Lower Gasket (PTFE)	11 24 01 30 00
21		O'ring (FEP+Silicon)	70 10 15 00 95
-		O'ring (Viton/FPM)	70 10 01 02 32
		Ball kit:	**11 24 01 94 00**
19	1	Ball undrilled (316)	11 12 82 00 05
		Handle kit:	
3	1	Kit Right handle (11 14 99 10 01L)	11 14 99 95 00
	1	Kit Left handle (11 14 99 10 01)	11 14 99 92 00
	1	Kit Remote Lever (11 24 06 00 01)	11 24 06 92 00
1	1	Screw M6x12	70 13 10 37 31
2	1	Washer W6	70 13 25 37 20
5	1	Washer W16	70 13 25 37 15
4	2	Spring washer	70 13 26 37 22
		Spindle seals kit:	**11 24 01 92 00**
8	1	Spacer (PTFE)	11 12 39 00 14
9	1	O'ring (FEP+Silicon)	70 10 15 00 12
10	1	Spindle Gasket (PTFE)	11 12 39 00 21
		Filter kit:	**11 24 01 95 00**
17	1	Filter	11 14 01 20 00
16	2	Circlip	70 13 14 37 08
		1 1/2" BSP Cap kit:	**11 14 01 30 00**
15	1	Cap Seal	12 85 97 00 10
14	1	BSP Cap & Chain	-
		1 1/2" NPT Cap kit:	**on request**
15	1	Cap Seal	12 85 97 00 10
14	1	BSP Cap & Chain	-

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed simultaneously.
- Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details



AIR LINE

2" - 1 1/2" BSP Threaded ATCO™
ND50 Ball Valve - COMPACT
ADC5X7T

The Perolo ND50 ATCO FULL BORE 1 1/2" BSP threaded ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. With a super compact design, the DN50 ATCO is light & economic.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	4,2 Kg
Inlet Connection	4 holes x Ø18 x 125 PCD
Outlet Connection	1 1/2" BSP
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	- 40° to +150°C
Design Temperature	- 40° to +200°C

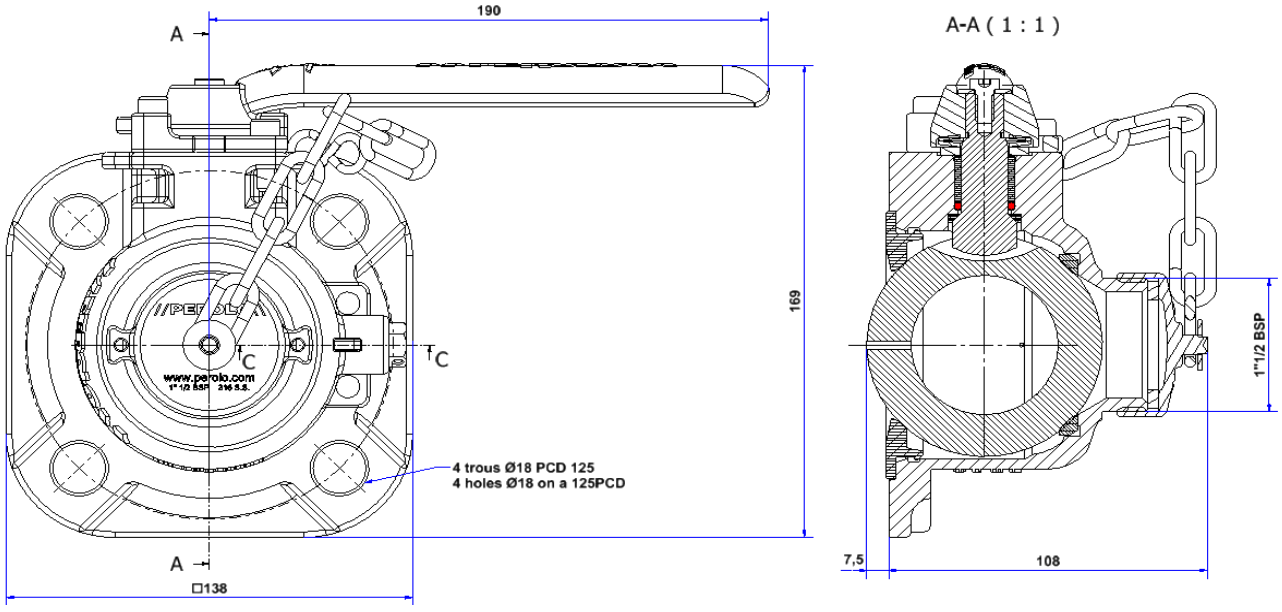
Temperature range vary according to the seal material

ACCESSORIES	
Weld in flange	11 90 25 00 00
Recess flange	11 92 44 00 00
Stud kit	11 91 13 91 00
Remote Control	RMT 53
Remote control Handle	11 14 99 30 01
Pressure gauge	70 18 06 00 95
1 1/2" BSP Cap kit	11 14 01 31 00

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Compact Design	Face to face 108 mm with Cap
Operation	Handle 1/4 Turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

2" - 1 1/2" BSP Threaded ATCO™
ND50 Ball Valve - COMPACT
ADC5X7T

CODIFICATION

STANDARD PART NUMBER	ADC5	X	7	T	1	BC	S	0	0	0
YOUR PART NUMBER	ADC5	__	7	T	1	__	__	__	__	__

MATERIAL - Other on request	
316 / 1.4408 stainless steel	X

MAWP	
7 Bar	7

WORKING TEMPERATURE	
Ball Seals PTFE +150°C	T

INLET FLANGE DRILLING	
4 x Ø18 x 125 PCD	1

CONNECTION & CAP	
1.5" BSP with cap	BC
1.5" BSP without cap	B0

HANDLE	
Standard handle	S
Reverse handle	L
Remote controle handle	W

GAUGE CONNECTION	
Without	0
With	1

GAUGE PROTECTION	
Without	0
With	1

FILTER	
Without	0
With	1

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 99 20 00 A
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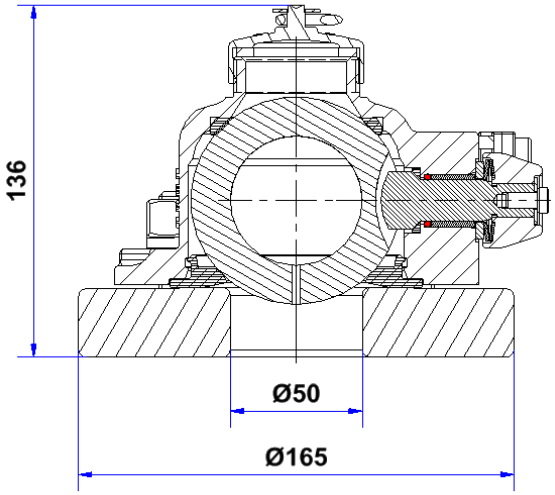
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" - 1 1/2" BSP Threaded ATCO™
ND50 Ball Valve - COMPACT
ADC5X7T

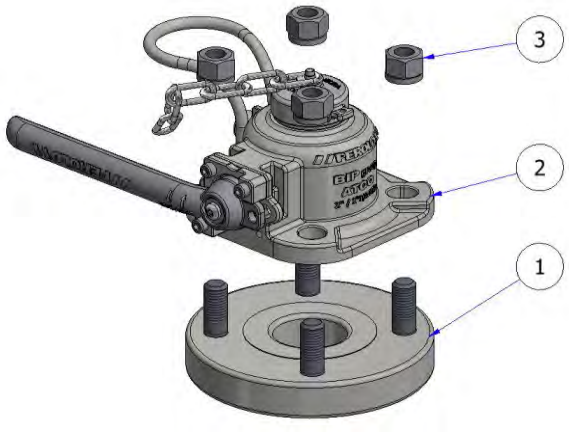
INSTALLATION

1. Check Inlet (tank side) flange is clean, free from dirt and grit.
2. Screw the studs into the weldin flange, lubricate with PTFE added grease before installing the studs
3. Install the valve on the flange, secure with washer and nuts *(refer to PWAGEN001 fort bolt tightening sequence)*
4. Tighten the nuts torque : 30 N.m -0/+10
5. Open and close the valve 3 times to check the operational torque.



ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	Bolting Kit	4 x M16 33 studs 4 x nuts and 4 x washers	11 91 13 91 00
2	ND 50 ATCO "Compact"	PN 7 Full Bore, 316 stainless steel Inlet flange 4xø17x125PCD 2" BSP	ADC5X7T1BCS0000
1	Weld-in flange	ext ø 165mm, int ø 50mm thickness 26,5mm 4xM16x125PCD 316L stainless steel	11 90 25 00 00

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs



AIR LINE

2" - 1 1/2" BSP Threaded ATCO™
ND50 Ball Valve - COMPACT
ADC5X7T

MAINTENANCE

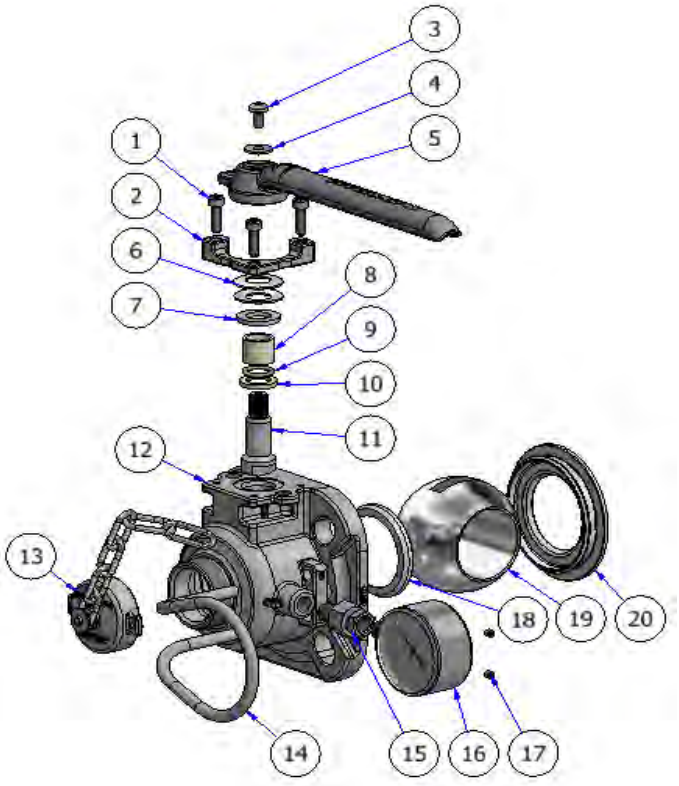
ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 14 99 90 00**
18	1	Uper ball seal (PTFE)	11 14 99 10 03
20	1	Lower gasket (PTFE)	11 14 99 10 04
		Ball kit:	
19	1	Ball (316)	11 12 42 00 11
		Drilled Ball kit:	
19	1	Drilled Ball (316)	11 12 39 00 11
		Handle kit:	**11 14 99 92 00**
3	1	Screw M6x12	-
4	1	Washer Ø6	-
5	1	Handle	11 14 99 10 01
6	2	Elastic washer Ø16	-
7	1	Washer Ø16 wide series	-
		Stop handle kit:	**11 14 99 30 00**
2	1	Stop handle	11 14 99 1A 02
1	3	Screw M6x20	-
		Spindle seals kit:	**11 14 99 93 00**
8	1	Bearing Bush (PTFE)	11 12 39 00 14
9	1	O'Ring (FEP/silicone)	70 10 15 00 12
10	1	Spindle Gasket	11 12 39 00 21
		Spindle kit:	**11 14 99 94 00**
11	1	Spindle (Duplex)	11 14 99 10 05
6	2	Elastic washer Ø16	-
7	1	Washer Ø16 wide series	-
		Filter kit:	**11 14 99 95 00**
-	1	Filter (St st)	11 12 03 20 00
-	2	Lock Ring (St st)	-
		1,5" Cap kit:	
13	1	1,5" std BSP Cap + Seal + Chain	11 14 01 31 00
-	1	Cap seal (PTFE)	11 12 03 40 02
		Pressure gauge and protection:	
-	1	Plug 1/4"	70 12 39 37 07
16	1	Gauge	70 18 07 00 34
14	1	Gauge protection	11 14 99 00 01
17	2	Screw HC M5-5	70 13 08 37 10
15*	1	Union Fitting 1/4" BSP-M/F	70 12 13 37 41
		Tools kit:	**11 12 69 94 00**
A	1	Spindle Assembly tool	11 12 69 92 01
B	1	O'Ring Assembly tool	11 12 69 92 02

When the ATCO is installed on recess flange

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed
-Valve should be in closed position during installation

A

B



AIR LINE

2" BSP Threaded ATCO™
ND50 Ball Valve
AD5X7T

The Perolo ND50 ATCO FULL BORE threaded ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. With a super compact design, the DN50 ATCO is light & economic.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	5.6 Kg
Inlet Connection	4 holes x Ø17 x 125 PCD
Outlet Connection	2" BSP*
Working Pressure	7 Bar
Test Pressure	0,5 Bar
Working Temperature	- 40° to +150°C
Design Temperature	- 40° to +200°C

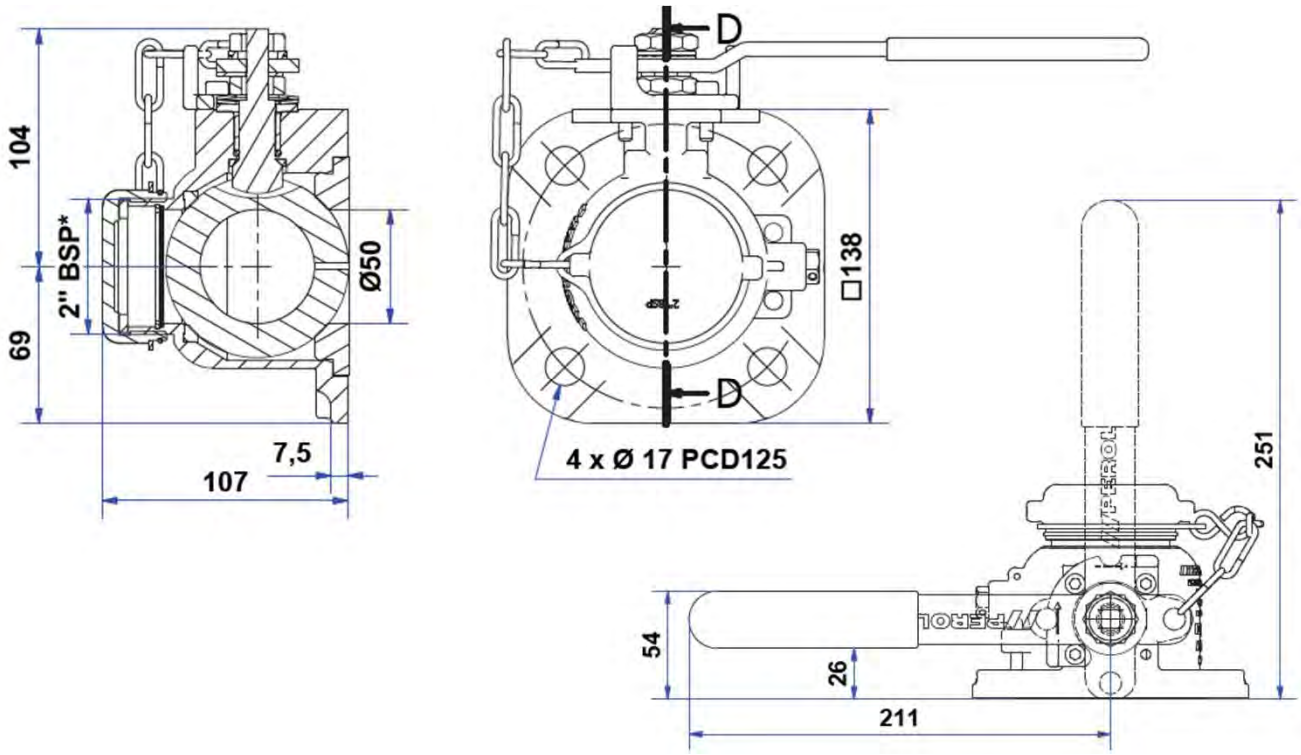
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

ACCESSORIES	
Weld in flange	11 90 25 00 00
Recess flange	11 92 44 00 00
Stud kit	11 91 13 91 00
Remote Control	RMT 53
Remote control Handle	11 14 55 91 00
Pressure gauge	70 18 06 00 95
2" BSP Cap	11 12 03 40 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face to face 107 mm with Caps
Operation	Handle 1/4 Turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

2" BSP Threaded ATCO™
ND50 Ball Valve
AD5X7T

CODIFICATION

STANDARD PART NUMBER	AD5	X	7	T	1	BC	S	0	0	0	0
YOUR PART NUMBER	AD5		7	T	1						

MATERIAL - Other on request	
316L / 1.4409 stainless steel	X

MAWP	
7 Bar	7

WORKING TEMPERATURE	
Ball Seals PTFE +150°C	T

INLET FLANGE DRILLING	
4 x Ø17 x 125 PCD	1

CONNECTION & CAP	
2" BSP with cap	BC
2" NPT with cap	NC
2" BSP without cap	B0
2" NPT without cap	N0

HANDLE	
Standard handle	S
Reverse handle	L
Short handle	M
Reverse short handle	N
Remote contrôle handle	W

GAUGE CONNECTION	
Without	0
With	1

GAUGE PROTECTION	
Without	0
With	1

FILTER	
Without	0
With	1

REMOTE CONTROL	
Without	0
With	1

AIR LINE

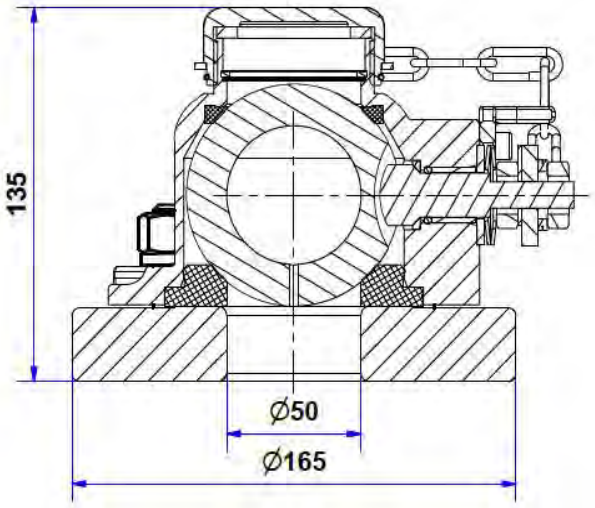
Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 75 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

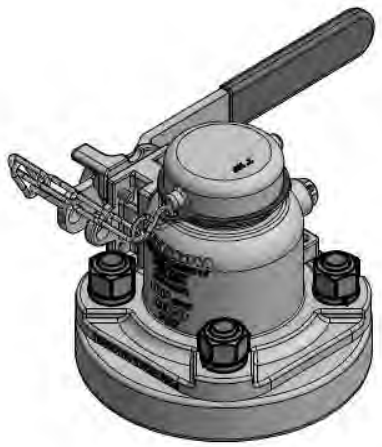
2" BSP Threaded ATCO™
ND50 Ball Valve
AD5X7T

AIR LINE

INSTALLATION

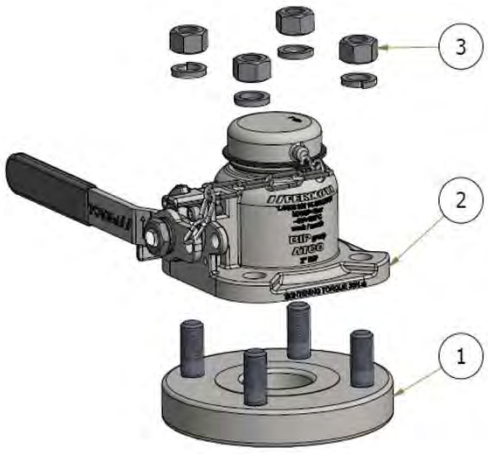
1. check Inlet (tank side) flange is clean, free from dirt and grit.
2. Screw the studs into the weldin flange, lubricate with PTFE added grease before installing the studs
3. Install the valve on the flange, secure with washer and nuts *(refer to PWAGEN001 fort bolt tightening sequence)*
4. Tighten the nuts torque : 30 N.m +/- 5
5. Open and close the valve 3 times to check the operational torque.





ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	Bolting Kit	4 x M16 33 studs 4 x nuts and 4 x washers	11 91 13 91 00
2	ND 50 ATCO "Light"	PN 16 Full Bore, 316 L stainless steel Inlet flange 4xø17x125PCD 2" BSP	AD5X7T1BCS0000
1	Weld-in flange	ext ø 165mm, int ø 50mm thickness 26mm 4xM16x110PCD 316L stainless steel	11 90 25 00 00

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

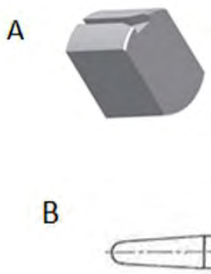


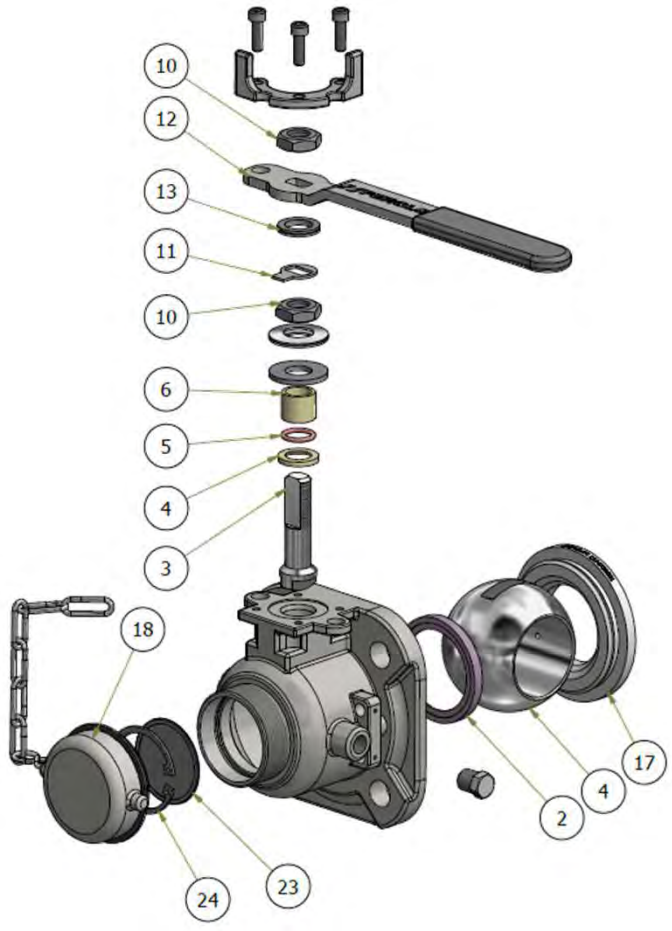
2" BSP Threaded ATCO™
ND50 Ball Valve
AD5X7T

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 14 75 90 00**
2	1	Ball seal (PTFE)	11 12 39 00 09
17	1	Base gasket (PTFE)	11 14 73 00 01
		Ball kit:	**11 12 42 91 00**
4	1	Ball (316)	11 12 42 00 11
		Drilled Ball kit:	**11 12 39 94 00**
4	1	Drilled Ball (316)	11 12 39 00 11
		Handle kit:	**11 14 16 92 00**
12	1	Handle (st st)	-
	1	Handle cap, nitrile	-
10	2	M16 Nut	-
11	1	Stop washer	-
13	1	Washer Nordlock Ø 16	-
		Spindle seals kit:	**11 12 42 93 00**
6	1	Bearing Bush (PTFE)	11 12 39 00 14
5	1	O'Ring (FEP/silicone)	70 10 15 00 12
4	1	Spindle Gasket	11 12 39 00 21
		Spindle kit:	**on request**
3	1	Spindle (St st)	11 12 39 00 16
9	2	Elastic washer (St st)	-
8	1	Washer (St st)	-
		Filter kit:	**11 14 75 92 00**
23	1	Filter (St st)	11 12 03 20 00
24	1	Lock Ring (St st)	70 13 14 37 08
		2" Cap kit:	
18	1	2" BSP Cap + Seal + Chain	11 14 64 20 00
18	1	2" NPT Cap + Seal + Chain	11 14 75 20 00
-	1	Cap seal (PTFE)	11 12 03 40 02
		Tools kit:	**11 12 69 94 00**
A	1	Spindle Assembly tool	11 12 69 92 01
B	1	O'Ring Assembly tool	11 12 69 92 02

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed
-Valve should be in closed position during installation





AIR LINE

2" BSP Threaded ATCO™
Heavy Duty ND50 Ball Valve
AD5X16

The Perolo 2" ATCO™ self-contained ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. With a super compact design, the 2" ATCO™ is light & economic.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	5.9 Kg
Inlet Connection	4 holes Ø 17 x 125 PCD
Outlet Connection	2" BSP or 2" NPT
Working Pressure	16 Bar
Test Pressure	24 Bar
Working Temperature	- 40° to +150°C (+250°C for HT)
Design Temperature	- 40° to +190°C

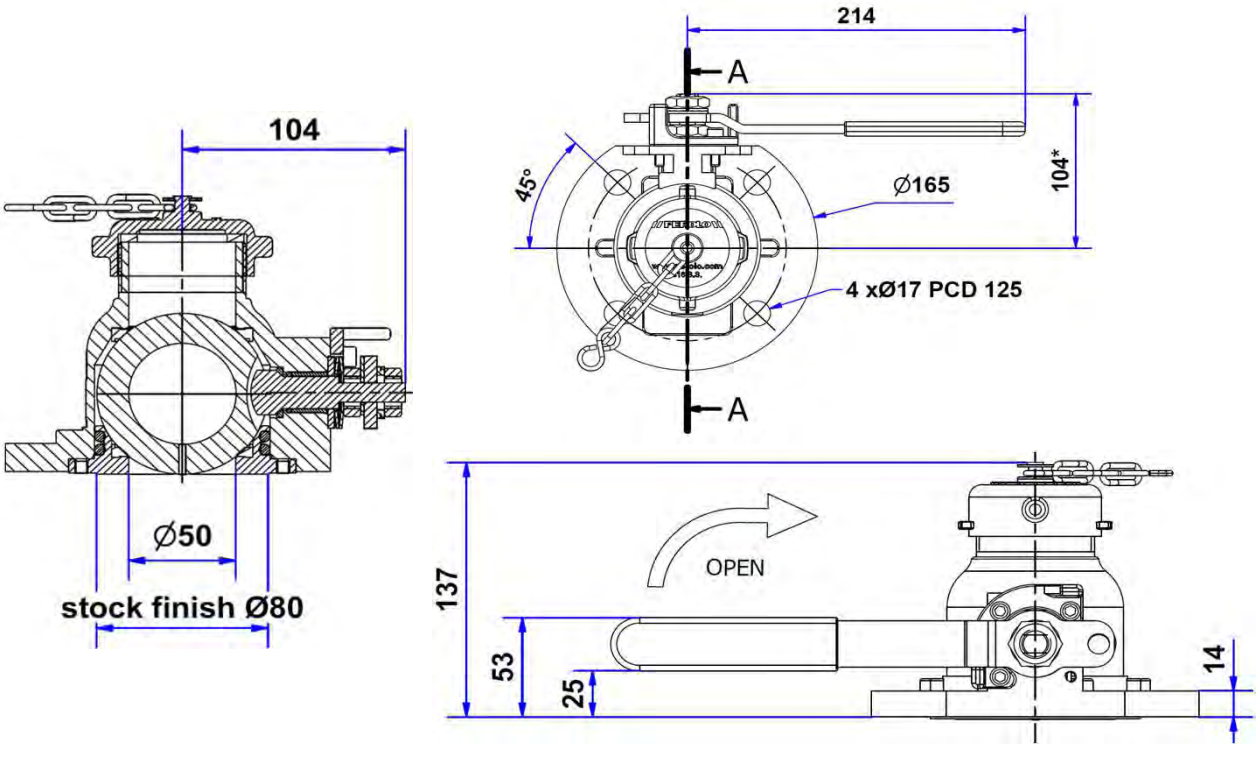
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas - A.C.I

ACCESSORIES	
Weld-in flange	11 90 25 00 00
Bolting kit	17 12 12 96 00
Remote control Handle	11 14 55 91 00
Remote control	RMT 53
Pressure gauge	70 18 06 00 95
2" BSP Cap	11 12 03 40 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face To Face 137mm with Cap
Operation	Handle - 1/4 Turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

2" BSP Threaded ATCO™
Heavy Duty ND50 Ball Valve
AD5X16

CODIFICATION

STANDARD PART NUMBER	AD5	X	16	T	1	BC	W	1	0
YOUR PART NUMBER	AD5	__	16	__	1	BC	__	__	__

MATERIAL - Other on request	
316L / 1.4409 stainless steel	X

MAWP	
16 Bar	16

WORKING TEMPERATURE	
Ball Seals PTFE +150°C	T
High temperature	H

INLET FLANGE DRILLING	
4 x Ø17 x 125 PCD	1

CONNECTION & CAP	
2" BSP with Cap	BC

HANDLE	
With standard handle	S
With remote control handle	W

GAUGE CONNECTION	
Without	0
With	1

GAUGE PROTECTION	
Without	0
With	1

AIR LINE

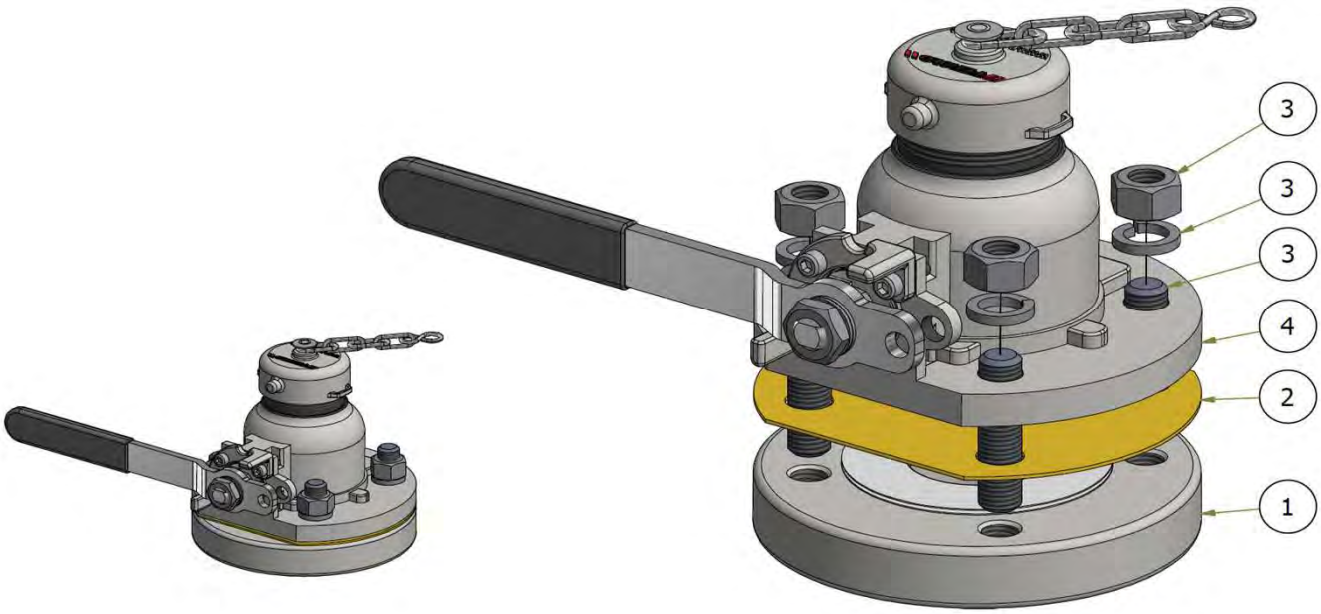
Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 61 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" BSP Threaded ATCO™
Heavy Duty ND50 Ball Valve
AD5X16

INSTALLATION

To install the ND50 ATCO™, the weldin flange must be machined with stock finish (serrated surface).

1. Ensure that the Inlet flange and gasket are clean; free from dirt and grit.
2. Screw the studs from kit 11 91 13 91 00 on the weld in flange (1).
3. Place the gasket (2) on top of the weld in flange.
4. Fit ball valve and secure with washers and nuts.
   The required tightening torque is 50 N.m +/-5 for A2/A4 - 70 bolting class
5. Cross type serration
6- Check torque after 24 hours & tighten at 50 Nm if required
7 - Lubrilate the threads (PTFE grease or MOS2)
8. Refer to PWAGENE001 for bolt tightening sequence.



ITEM	DESCRIPTION	SPECIFICATION	PART N°
4	ND50 ATCO Ball Valve	PN 16 Full Bore, 316 L stainless steel Inlet flange 4xø17x125PCD Outlet flange 2" BSP	AD5X16T1BCS00
3	Bolting Kit	4 x M16 50 studs, 4 x M16 nuts and 4 x grower washers class 70	17 12 12 96 00
2	Gasket	ext ø 165mm, int ø 50mm 4xø18x minimum 125PCD CNAF/PTFE - thickness 2mm	11 90 29 00 00
1	Weld-in flange	ext ø 165mm, int ø 50mm, thickness 21mm 4xM16x125PCD 316L stainless steel	11 90 25 00 00

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

AIR LINE

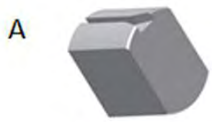
2" BSP Threaded ATCO™
Heavy Duty ND50 Ball Valve
AD5X16

MAINTENANCE

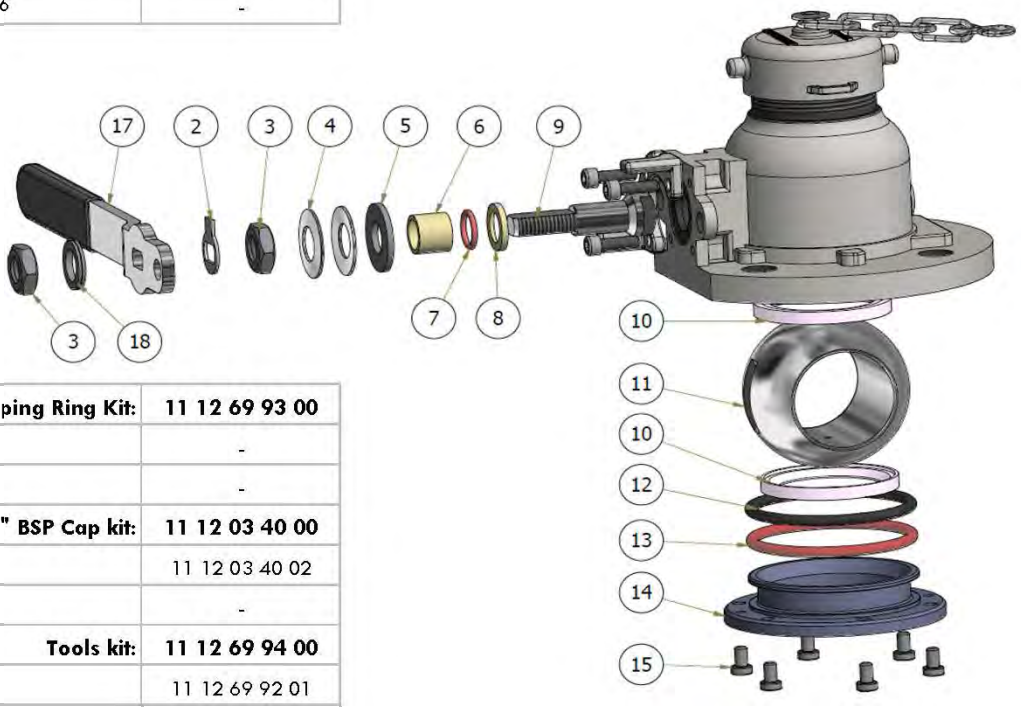
ITEM	QTY	DESIGNATION	PART N°	
			Standard	High Temp
		Ball seals kit:	**11 12 42 92 00**	**on request**
10	2	Ball Seals (PTFE or PEEK)	11 12 39 00 09	11 12 39 00 28
12	1	O'Ring (FEP (or PFA for HT))	70 10 15 00 10	70 10 17 00 40
13	1	O'Ring (Viton (or PFA for HT))	70 10 01 02 39	70 10 17 00 40
		Ball kit:	**11 12 42 91 00**	
11	1	Ball (316)	11 12 42 00 11	
		Handle kit:	**11 14 16 92 00**	
17	1	Handle (st st)	-	
	1	Handle cap, nitrile	-	
3	2	M16 Nut	-	
2	1	Stop washer	-	
18	1	Washer Nordlock Ø 16	-	
		Spindle seals kit:	**11 12 42 93 00**	
6	1	Bearing Bush (PTFE)	11 12 39 00 14	
7	1	O'Ring (FEP/silicone)	70 10 15 00 12	
8	1	Spindle Gasket	11 12 39 00 21	
		Spindle kit:	**11 14 16 91 00**	
9	1	Spindle (St st)	11 12 39 00 16	
4	2	Elastic washer (St st)	-	
5	1	Washer (St st)	-	
3	2	M16 Nut	-	
2	1	Stop washer	-	
18	1	Washer Nordlock Ø 16	-	

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed
-Valve should be in closed position during installation

A



B



		Clamping Ring Kit:	**11 12 69 93 00**
15	6	TBHC Screw	-
14	1	Clamping Ring (St st)	-
		2" BSP Cap kit:	**11 12 03 40 00**
-	1	Cap Seal	11 12 03 40 02
-	1	BSP Cap & Chain	-
		Tools kit:	**11 12 69 94 00**
A	1	Spindle Assembly tool	11 12 69 92 01
B	1	O'Ring Assembly tool	11 12 69 92 02

AIR LINE

2" Flanged TANKFLY™
ND50 Butterfly Valve
TF5X

The 2" TANKFLY™ MKII flanged butterfly valve is designed to be used as a fluid inlet/outlet valve. With a super compact design, the 2" TANKFLY™ is light and economic.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	3.0 Kg
Inlet Connection	4 holes x Ø18 x 125 PCD
Outlet Connection	4 holes slotted x Ø18 x min 114 PCD
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	- 40° to +150°C
Design Temperature	- 40° to +190°C

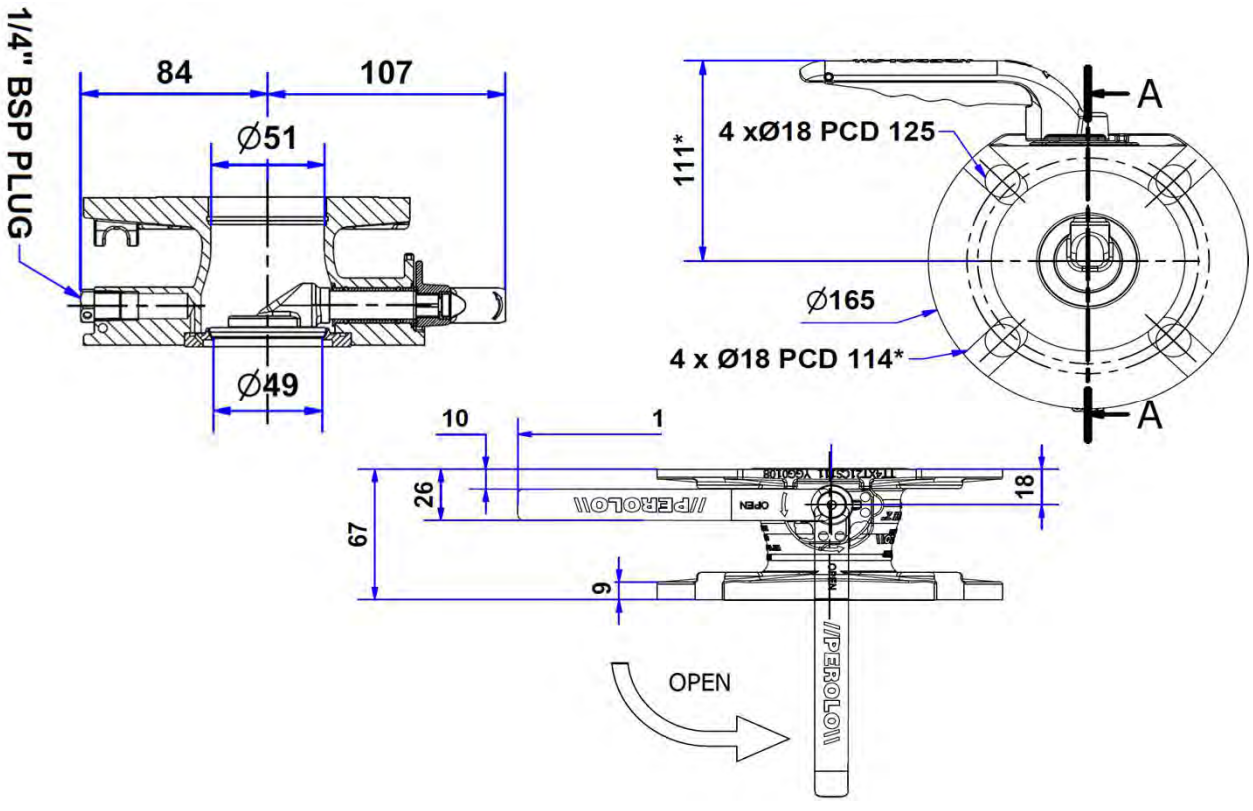
Temperature range vary according to the seal material

ACCESSORIES	
Weld-in flange	11 90 25 00 00
Recess flange	11 92 44 00 00
Blind flange	11 95 88 00 00
Bolting Kit	11 91 13 91 00
Gasket	11 90 29 00 00
Filter	11 12 03 94 00
Remote control	RMT53
Remote control handle	11 17 20 11 00
Pressure gauge	70 18 06 00 95

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas - RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face to face 67 mm
Operation	Handle - 1/4 Turn or Remote Control
Butterfly seal material	PTFE

DIMENSIONAL DRAWING



AIR LINE

2" Flanged TANKFLY™
ND50 Butterfly Valve
TF5X

CODIFICATION

STANDARD PART NUMBER	TF5	X	7	T	1	1	S	1	1	.
YOUR PART NUMBER	TF5	X	7	T	1	__	__	1	__	__

MATERIAL - Other on request	
316L / 1.4409 stainless steel	X
MAWP	
7 Bar	7
WORKING TEMPERATURE	
Butterfly seal PTFE +150°C	T
INLET FLANGE	
4 X Ø18 x 125 PCD	1
OUTLET FLANGE	
4 x open slots Ø18 x 114 PCD	1
4 x M16 x 125 PCD	2
4 x Ø18 x 125 PCD	3
FILTER	
Without	S
With	P
GAUGE CONNECTION & PLUG	
with 1/4" BSP connection & plug	1
GAUGE PROTECTION	
Without	1
With	2
OPERATING SYSTEM	
Standard Handle	.
Pneumatic Actuator	P
With Remote Control Handle	W

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 13 20 00 J
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" Flanged TANKFLY™
ND50 Butterfly Valve
TF5X

INSTALLATION

CAUTION : Closure plate must be closed when valve is fitted onto the flange.

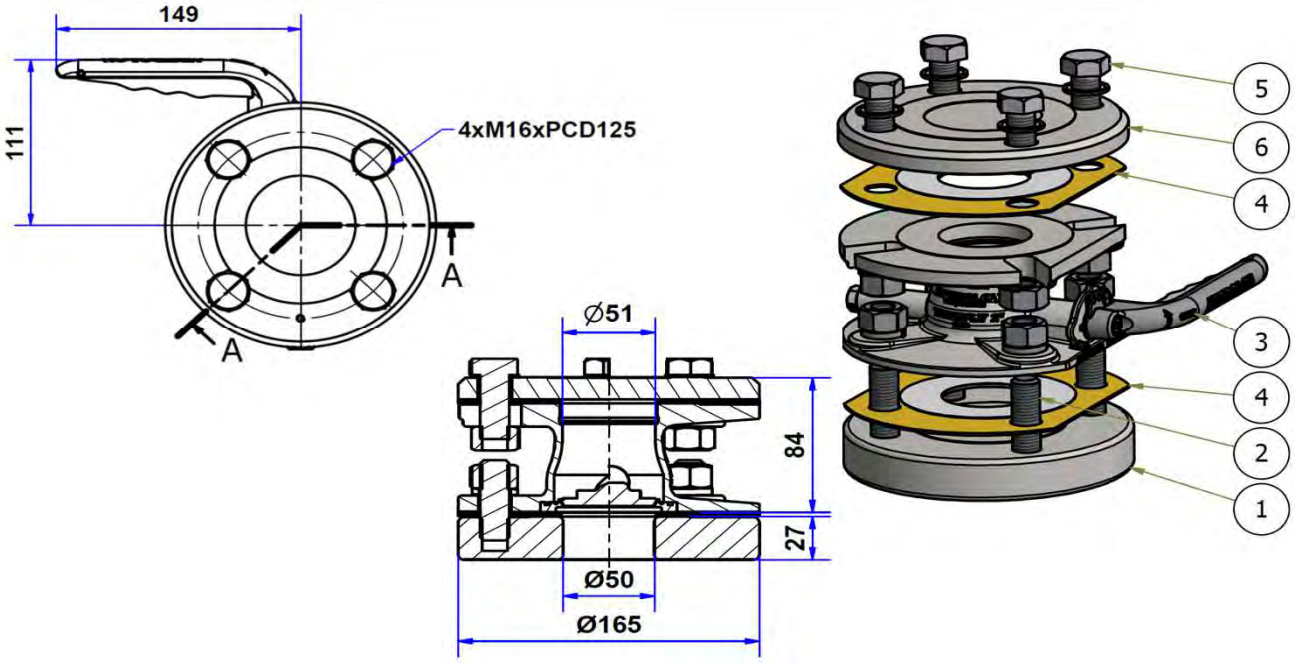
1. Ensure that the Inlet flange and gasket are clean; free from dirt and grit.
2. Align holes of TANKFLY™ (3) to weld in flange (1).
3. Fit Tankfly onto weld in flange and secure with bolting kit (2).

The required tightening torque is 50 N.m +/- 5
The valve flange should be in contact with weld-in flange when tightened.

4. Align holes of gasket (4) and blind flange (6) to TANKFLY™.
5. Fit blind flange onto TANKFLY™ and secure with screw kit (5).
6. Refer to PWAGENE001 for bolt tightening sequence.

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

ITEM	DESCRIPTION	SPECIFICATION	PART N°
6	Blind flange	ext Ø 165mm, thickness 13,5mm, 4x18xPCD125, 316L stainless steel	11 95 88 00 00
5	Screw kit	4xM16-40 screws, 4x lock washers, 4xM16 nuts	12 91 61 92 00
4	Seal	intØ50mm, extØ165mm 2mm thick, 4x18xPCD125,CNAF PTFE	11 90 29 00 00
3	TANKFLY butterfly valve	Nominal diameter: 2", 316L Stainless steel Inlet : 4 x Ø18 on a Ø125PCD Outlet : 4 x slots 18 x min 125PCD	TF5X7T11S11
2	Bolting kit	4xM16-33 studs, 4xH in M16 nuts, 4x grower washer	11 91 13 91 00
1	Weld-in flange	int Ø50mm, ext Ø165mm, thickness 21mm, 4xM16xPCD125, 316L stainless steel	11 90 25 00 00

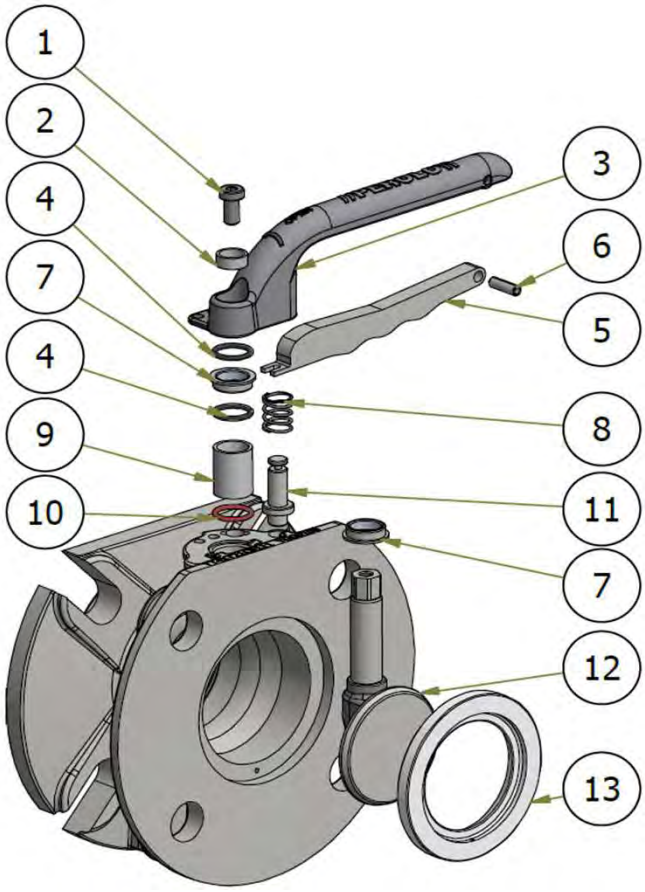


2" Flanged TANKFLY™
ND50 Butterfly Valve
TF5X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit:	**11 17 11 92 00**
1	1	Screw M6-12 (Stainless Steel)	-
2	1	Washer (Stainless Steel)	-
-	1	Handle (St st) Nr 3, 5, 6, 8, 11	11 17 11 41 00
4	2	O'ring (Viton)	70 10 01 02 90
		Spindle & closure plate seals kit:	**11 17 11 91 00**
4	2	O'ring (Viton)	70 10 01 02 90
7	2	Bearing (Stainless Steel)	11 17 11 00 01
9	1	Spacer (PTFE)	11 17 11 00 02
10	1	O'ring (FEP encapsulated Silicone)	70 10 15 00 83
13	1	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate seal kit:	**11 17 11 90 01**
13	1	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate seal kit (10 pcs):	**11 17 11 90 10**
13	10	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate kit:	**11 17 11 90 00**
1	1	Screw M6-12 (Stainless Steel)	-
2	1	Washer (Stainless Steel)	-
4	2	O'ring (Viton)	70 10 01 02 90
7	2	Bearing (Stainless Steel)	11 17 11 00 01
9	1	Spacer (PTFE)	11 17 11 00 02
10	1	O'ring (FEP encapsulated Silicone)	70 10 15 00 83
12	1	Closure plate (Stainless Steel)	11 17 11 30 00
13	1	Closure plate seal (PTFE)	11 17 11 00 03

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.



AIR LINE

2" BSP Threaded TANKFLY™
ND50 Butterfly Valve
TT5X

The 2" TANKFLY™ MKII threaded butterfly valve is designed to be used as a fluid inlet/outlet valve or air inlet valve. With a super compact design, the 2" TANKFLY™ is light and economic.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	1.4 Kg
Inlet Connection	4 slotted holes x Ø11 x 100 to 103.5 PCD
Outlet Connection	2" BSP
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	- 40° to +150°C
Design Temperature	- 40° to +190°C

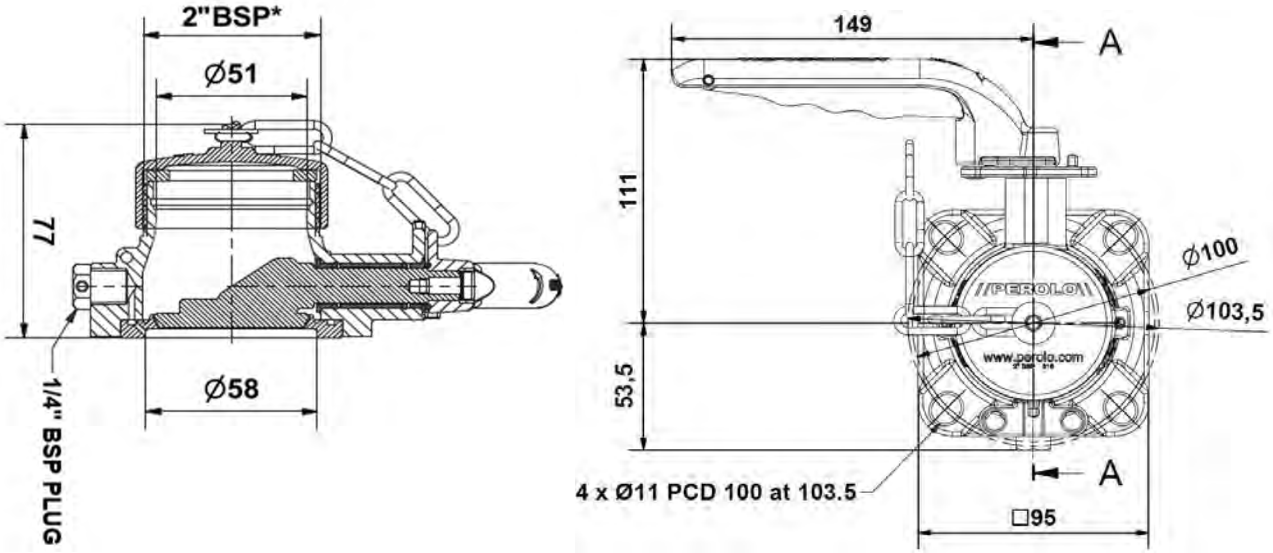
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

ACCESSORIES	
Weld-in flange	11 90 20 00 00
Screw kit	12 91 99 00 00
Filter kit	11 12 03 94 00
Remote control	RMT53 (3 meters)
Remote control handle	11 17 20 11 00
Pressure gauge	70 18 06 00 95
1 1/2" BSP Cap	11 17 11 20 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face to face 77 mm with Cap
Operation	Handle - 1/4 Turn or Remote Control
Butterfly seal material	PTFE

DIMENSIONAL DRAWING



AIR LINE

2" BSP Threaded TANKFLY™
ND50 Butterfly Valve
TT5X

CODIFICATION

STANDARD PART NUMBER	TT5	X	7	T	1	C	S	1	1	.
YOUR PART NUMBER	TT5		7							

MATERIAL - Other on request

316L / 1.4409 stainless steel	X

MAWP

7 Bar	7

WORKING TEMPERATURE

Butterfly seal PTFE +150°C	T

INLET CONNECTION

4 x Slots Ø11 x 100-103.5 PCD	1

OUTLET CONNECTION & CAP

BSP without Cap	1
BSP with Standard Cap	C
BSP with Reinforced Cap	K
BSP with Camlock cap	Q

FILTER

Without	S
With	P

GAUGE CONNECTION & PLUG

with 1/4" BSP connection & plug	1

GAUGE PROTECTION

Without	1
With	2

OPERATING SYSTEM

Standard Handle	.
Pneumatic Actuator	P
With Remote Control Handle	W

AIR LINE

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It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" BSP Threaded TANKFLY™
ND50 Butterfly Valve
TT5X

INSTALLATION

CAUTION: closure plate must be closed when valve is fitted on the flange.

1. Ensure that the Inlet flange and gasket are clean; free from dirt and grit.
2. Align holes in Tankfly to weld in flange. NO GASKET NEEDED
3. Fit Tankfly onto weld in flange and tight the M10 screws with washers.

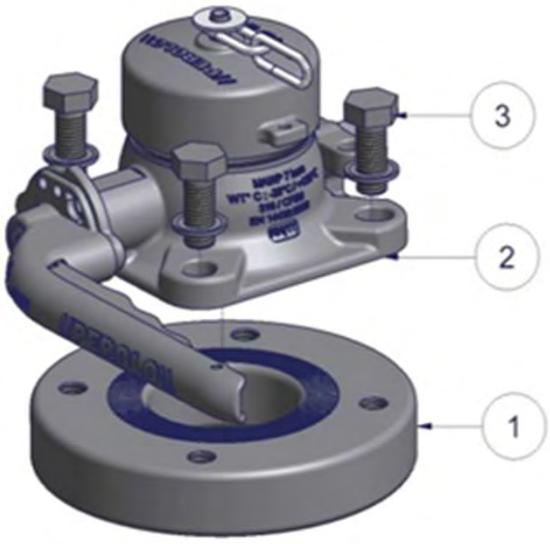
Tightening torque required : 50 N.m +/- 5

- Lubricate threads with PTFE added grease
- cross thight the 4 screws 1-2-3-4
- check torque after 24H

4. Refer to PWAGENE001 for bolt tightening sequence.

The valve flange should be in contact with tank flange when tightened.

ITEM	DESCRIPTION	SPECIFICATION	PART N°
2	TANKFLY Butterfly Valve	2" TANKLFY Airline threaded butterfly valve 4 x Ø11 x 100 PCD	TT5X7T1CS11
3	Screw Kit	4 x M10-25 screws, 4 x grower washers	12 91 99 00 00
1	Weld-in Flange	int Ø50mm, ext Ø165mm, thickness 27mm 4 x M10 x 100 PCD 316L Stainless Steel	11 90 20 00 00



if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

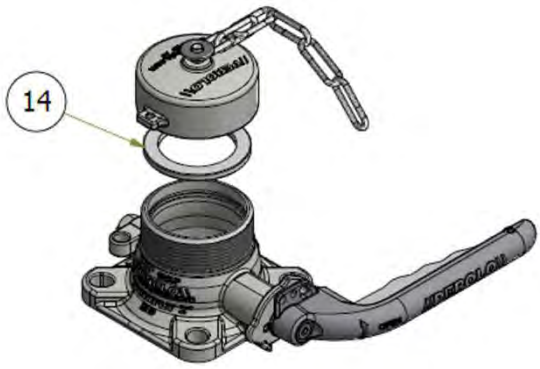
AIR LINE

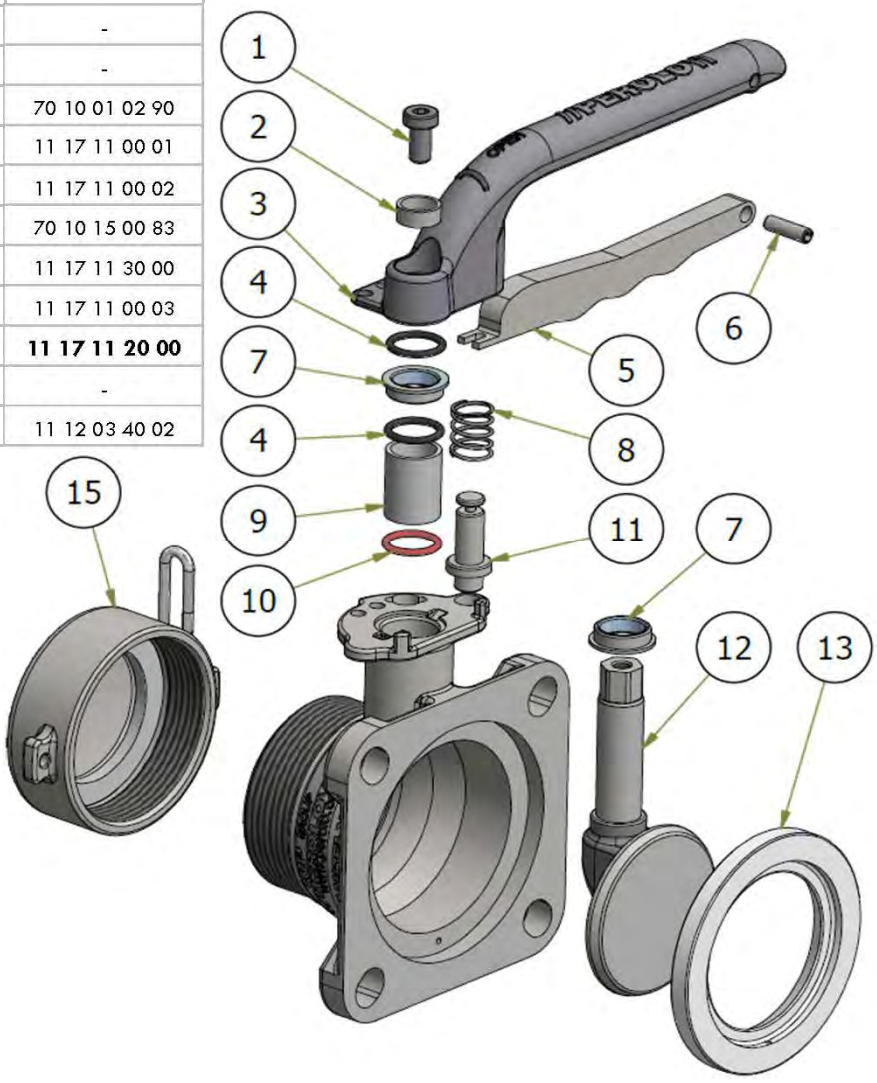
2" BSP Threaded TANKFLY™
ND50 Butterfly Valve
TT5X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit:	**11 17 11 92 00**
1	1	Screw M6-12 (Stainless Steel)	-
2	1	Washer (Stainless Steel)	-
-	1	Handle (St st) Nr 3, 5, 6, 8, 11	11 17 11 41 00
4	2	O'ring (Viton)	70 10 01 02 90
		Spindle & closure plate seals kit:	**11 17 11 91 00**
4	2	O'ring (Viton)	70 10 01 02 90
7	2	Bearing (Stainless Steel)	11 17 11 00 01
9	1	Spacer (PTFE)	11 17 11 00 02
10	1	O'ring (FEP encapsulated Silicone)	70 10 15 00 83
13	1	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate seal kit:	**11 17 11 90 01**
13	1	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate seal kit (10 pcs):	**11 17 11 90 10**
13	10	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate kit:	**11 17 11 90 00**
1	1	Screw M6-12 (Stainless Steel)	-
2	1	Washer (Stainless Steel)	-
4	2	O'ring (Viton)	70 10 01 02 90
7	2	Bearing (Stainless Steel)	11 17 11 00 01
9	1	Spacer (PTFE)	11 17 11 00 02
10	1	O'ring (FEP encapsulated Silicone)	70 10 15 00 83
12	1	Closure plate (Stainless Steel)	11 17 11 30 00
13	1	Closure plate seal (PTFE)	11 17 11 00 03
		2" BSP Cap kit:	**11 17 11 20 00**
15	1	2" BSP CAP & Chain	-
14	1	Cap seal (PTFE)	11 12 03 40 02

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.





AIR LINE

2" Clamped TANKFLY™
ND50 Butterfly Valve
TP5X

The 2" TANKFLY™ MKII clamped butterfly valve is designed to be used between flanges as an air inlet/outlet valve - one part closure plate seal - light - 1/4" BSP gauge connection.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	1.1 Kg
Inlet Connection	Clamped between DIN or ANSI Flange
Outlet Connection	
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	- 40° to +150°C
Design Temperature	- 40° to +190°C

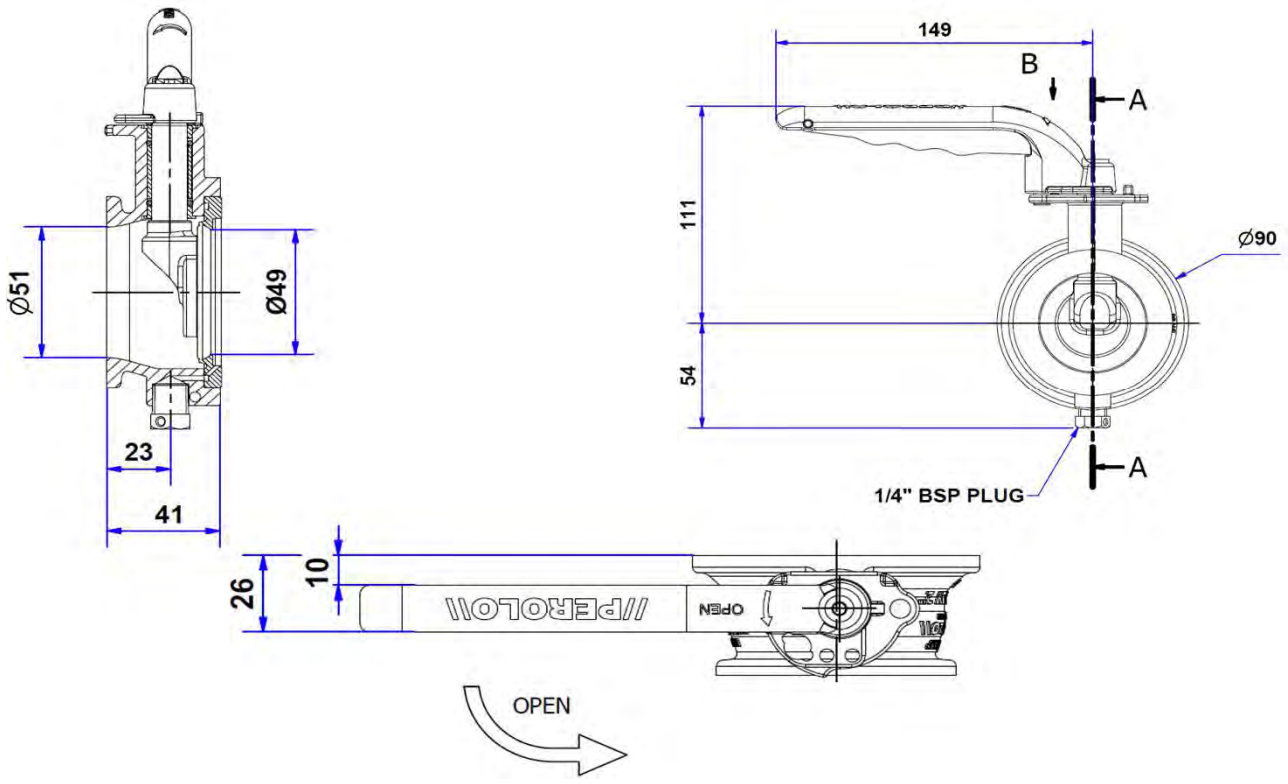
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

ACCESSORIES	
Weld-in flange	11 92 51 00 00
2" BSP nozzle + Cap	11 17 15 20 00
Pressure gauge	70 18 06 00 95

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face To Face 41mm
Operation	Handle - 1/4 Turn lockable
Butterfly seal material	PTFE

DIMENSIONAL DRAWING



AIR LINE

2" Clamped TANKFLY™
ND50 Butterfly Valve
TP5X

CODIFICATION

STANDARD PART NUMBER	TP5	X	7	T	1	1	.
YOUR PART NUMBER	TP5	__	7	T	1	__	__

MATERIAL - Other on request

316L / 1.4409 stainless steel	X

MAWP

7 Bar	7

WORKING TEMPERATURE

Butterfly seal PTFE +150°C	T

GAUGE PLUG

With	1

GAUGE PROTECTION

Without	1
With	2

OPERATING SYSTEM

Standard Handle	.
Pneumatic Actuator	P

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 15 20 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" Clamped TANKFLY™
ND50 Butterfly Valve
TP5X

AIR LINE

INSTALLATION

The leakproof faces of the valve are machined with spiral serrated stock finish

CAUTION: The weld-in flange must be machined in identical or similar fashion.

Closure plate must be closed when the TANKFLY is fitted on the tank.

To assemble the 2" Tankfly clamped butterfly valve:

1. Ensure that the Inlet flange and gasket are clean; free from dirt and grit.
2. Install the valve on the welding flange handle in closed position. NO GASKET NEEDED
3. Install the 4 screws and washers.

In order to avoid any damage caused to the valve seat when doing the equipment assembly, **ensure the butterfly is in the closed** position. Valve must be centered before clamping

Minimum torque of clamp bolts is 50 N.m +/- 5

4. Open and close the valve 3 times before operating to mark the PTFE seals.
5. The valve flange should be in contact with tank flange when tightened.
6. Refer to PWAGENE001 for bolt tightening sequence.

ITEM	DESCRIPTION	SPECIFICATION	PART N°
2	2" Outlet nozzle	BSP thread, 105x105mm, 4 x Ø14 x 110PCD	11 17 15 20 00
4	Stud kit	4 studs M16-75mm, 4 washers & 4 nuts M16	11 17 07 90 00
5	Seal	Flat Gasket	on request
1	2" TANKFLY	ND50 TANKFLY clamped butterfly valve	TP5X7T11
3	Weld-in flange	4 x M16 x 110 PCD, thickness: 30mm 316L/1.4404	11 92 51 00 00

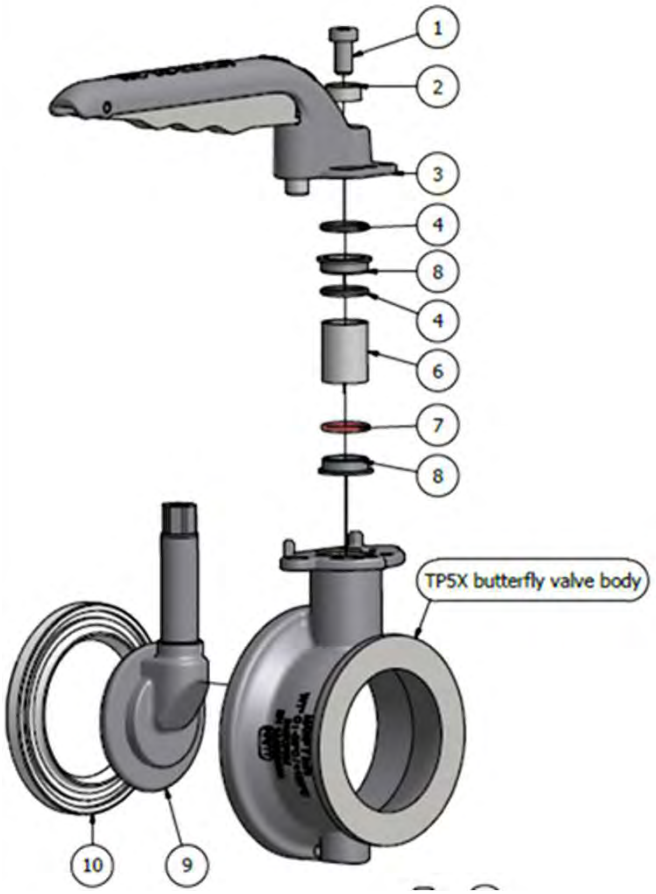
if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

2" Clamped TANKFLY™
ND50 Butterfly Valve
TP5X

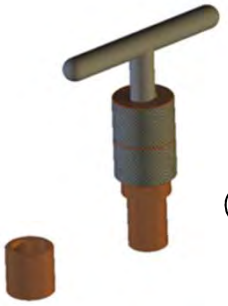
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit:	**11 17 11 92 00**
1	1	Screw M6-12 (Stainless Steel)	-
2	1	Washer (Stainless Steel)	-
3	1	Handle (Stainless Steel)	11 17 11 41 00
4	2	O'ring (Viton)	70 10 01 02 90
		Spindle & closure plate seals kit:	**11 17 11 91 00**
4	2	O'ring (Viton)	70 10 01 02 90
8	2	Bearing (Stainless Steel)	11 17 11 00 01
6	1	Spacer (PTFE)	11 17 11 00 02
7	1	O'ring (FEP encapsulated Silicone)	70 10 15 00 83
10	1	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate seal kit:	**11 17 11 90 01**
10	1	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate seal kit (10 pcs):	**11 17 11 90 10**
10	10	Closure plate seal (PTFE)	11 17 11 00 03
		Closure plate kit:	**11 17 11 90 00**
1	1	Screw M6-12 (Stainless Steel)	-
2	1	Washer (Stainless Steel)	-
4	2	O'ring (Viton)	70 10 01 02 90
8	2	Bearing (Stainless Steel)	11 17 11 00 01
6	1	Spacer (PTFE)	11 17 11 00 02
7	1	O'ring (FEP encapsulated Silicone)	70 10 15 00 83
9	1	Closure plate (Stainless Steel)	11 17 11 30 00
10	1	Closure plate seal (PTFE)	11 17 11 00 03
		Outlet nozzle kit:	**11 17 15 20 00**
13	1	Outlet nozzle with cap and chain	-
		Tools kit:	**AS188**
A	1	Valve Assembly tool	-

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.



13



A

AIR LINE

1 ½" BSP Threaded ATCO™
Pneumatic ND32 Ball Valve
AP3X

The Perolo 1 ½" ATCO™ pneumatic ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	32mm (outlet 1 ½")
Weight	3.95 Kg
Inlet Connection	4 x Ø11 x 103.5 PCD
Outlet Connection	1 1/2" BSP
Working Pressure	6.7 Bar
Test Pressure	10 Bar
Working Temperature	- 40° to +150°C
Design Temperature	- 40° to +190°C
Actuator Air Pressure	4 to 8 Bar
Actuator Working T°	-40°C to +80°C

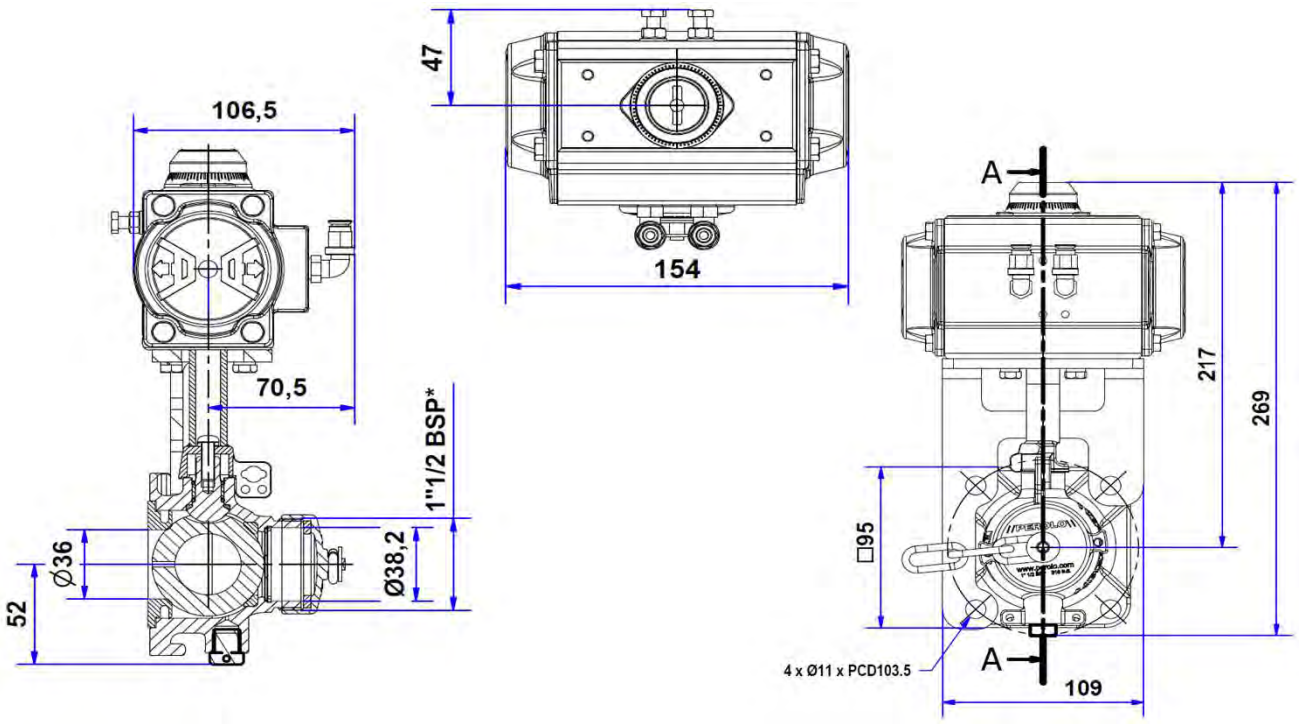
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

ACCESSORIES	
Weld-in flange	11 90 20 00 00
Screw kit	12 91 99 00 00
Filter	11 14 01 95 00
Pressure gauge	70 18 06 00 95
1 1/2" BSP Cap	11 14 01 30 00
Screw kit	11 14 51 90 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face To Face 78mm
Operation	Pneumatic - 1/4 Turn single acting actuator
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

1 ½" BSP Threaded ATCO™
Pneumatic ND32 Ball Valve
AP3X

CODIFICATION

STANDARD PART NUMBER	AP3	X	7	T	2	1C	S	2	1	.
YOUR PART NUMBER	AP3	__	7	T	2	__	__	2	__	__

MATERIAL - Other on request

316L / 1.4409 stainless steel	X

MAWP

6.7 Bar	7

WORKING TEMPERATURE

Ball Seals PTFE +150°C	T

INLET FLANGE

4 x Ø10.5 x 103.5 PCD	2

CONNECTION & CAP

BSP without Cap	1
BSP with Standard Cap/Chain	1C
BSP with Std Cap/Reinforced Chain	1K
NPT with Standard Cap	2C

FILTER

Without	S
With	P

GAUGE CONNECTION & PLUG

1/4" BSP Connection & Plug	2

GAUGE PROTECTION

Without	1
With	2

POSITION SENSOR

Without	.
With	L

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 51 20 00 I
It is the customer's responsability to check the compatibility between valves material and transported product.
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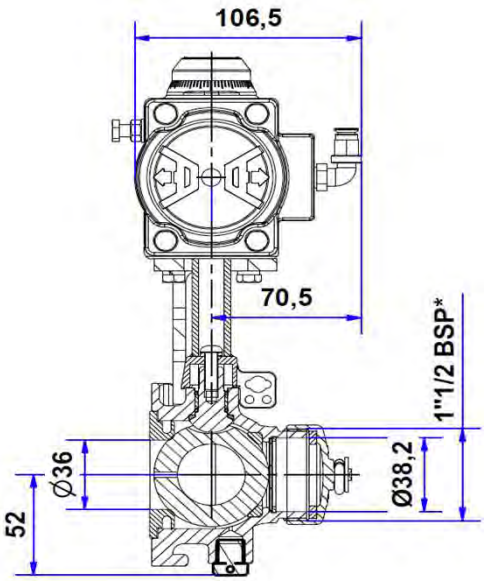
1 ½" BSP Threaded ATCO™
Pneumatic ND32 Ball Valve
AP3X

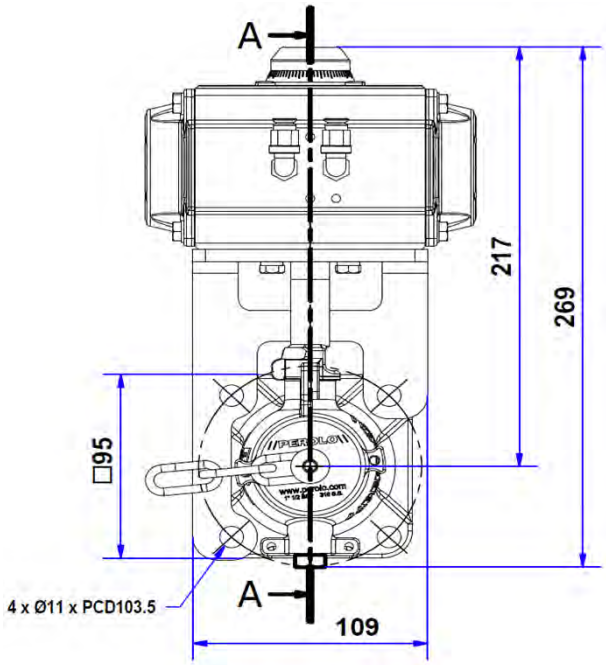
INSTALLATION

Caution: Do not disassemble the actuator from ATCO Ball valve.
For better and long time performances, assembly and settings are realized in PEROLO factory.

CONDITIONS

- Minimum operating pressure: 4 bar (Maximum operating pressure: 8bar)
- Actuator connections: Ø6mm ext
- Actuator Working Temperature: -40°C to 80°C
- For corrosive products application, provide a protection to avoid contact with actuator





INSTALLATION ON THE TANK

- Align holes in gasket to screws (screw kit: 11 14 51 90 00).
- Fit ball valve onto tank pad and secure with washers and nuts. The required tightening torque is 3 daN.m. ± 1
- Lubricate threads with PTFE added grease
- Cross thight the 4 screws 1-2-3-4
- Check torque after 24H
- Adjust at 3 diaNm if required

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

SETTING UP THE ACTUATOR

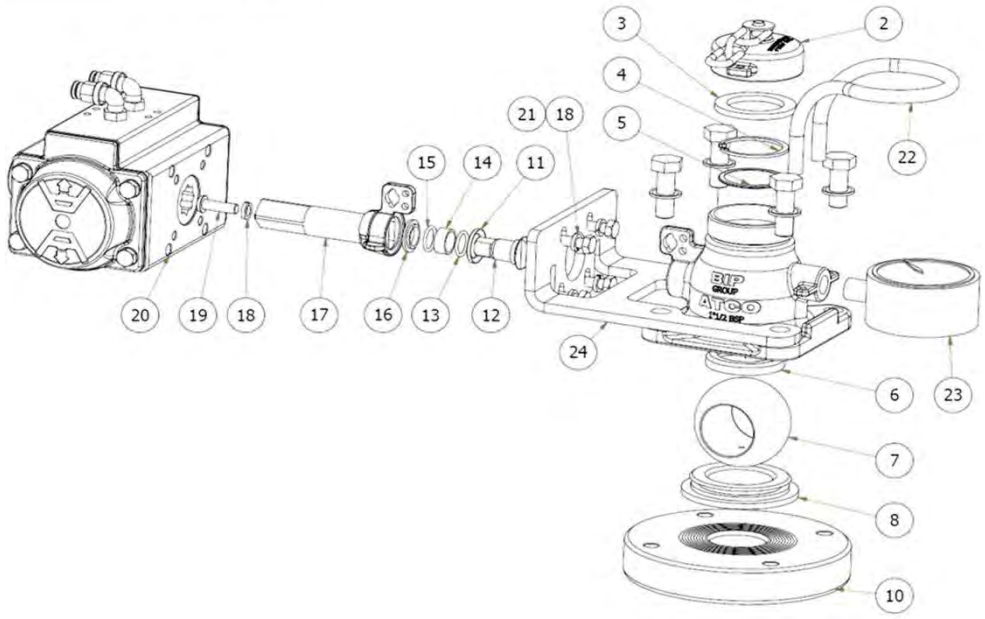
- Connect inlet to connection 2 and outlet to connection 4.
- It is a simple effect actuator able to work with only one inlet connection. However, we recommend to use two inlet connections. It reduces the time of closure. Use lubricated air to keep the actuator in good working condition. The air feeding should be rated from 5.5 bar up to 6.5 bar (do not exceed 8 bar).

1 ½" BSP Threaded ATCO™
Pneumatic ND32 Ball Valve
AP3X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 14 01 93 00**
6	1	Ball seal (PTFE)	11 14 01 00 01
8	1	Main seal (PTFE)	11 14 01 00 04
		Ball kit:	**11 14 01 94 00**
7	1	Ball (316)	12 96 06 00 09
		Actuator coupling kit:	**11 14 51 92 00**
17	1	Coupling (CF8M)	-
19	1	Screw M6x20 (st st)	-
18	1	Washer W6 (st st)	-
16	1	Bearing (PTFE)	12 96 06 00 13
15	1	O'ring (Viton)	70 10 03 02 08
14	1	Spacer (PTFE)	11 14 02 00 03
13	1	O'ring (FEP/Silicon)	70 10 15 00 66
11	1	Spindle seal (PTFE)	11 14 02 00 01
12	1	Spindle (st st)	11 14 02 00 02
		Spindle seals kit:	**11 14 01 92 00**
11	1	Spindle seal (PTFE)	11 14 02 00 01
15	1	O'ring (Viton)	70 10 03 02 08
14	1	Spacer (PTFE)	11 14 02 00 03
13	1	O'ring (FEP/Silicon)	70 10 15 00 66
16	1	Bearing (PTFE)	12 96 06 00 13
		Filter kit:	**11 14 01 95 00**
5	1	Filter (st st)	-
4	1	Lock Ring (st st)	-
		1 1/2" BSP Cap kit:	**11 14 01 30 00**
2	1	1 1/2" BSP cap & chain	-
3	1	Cap seal	12 85 97 00 10

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed simultaneously.
- Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details



AIR LINE

1 ½" Flanged ATCO™
Pneumatic ND32 Ball Valve
AG3X

The Perolo 1 ½" ATCO™ pneumatic ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	32mm (outlet 1 ½")
Weight	3.5 Kg
Inlet Connection	4 x Ø11 x 103.5 PCD
Outlet Connection	4 x M16 x 98.4 PCD
Working Pressure	6.7 Bar
Test Pressure	10 Bar
Working Temperature	- 40° to +150°C
Design Temperature	- 40° to +190°C
Actuator Air Pressure	4 to 8 Bar
Actuator Working T°	-40°C to +80°C

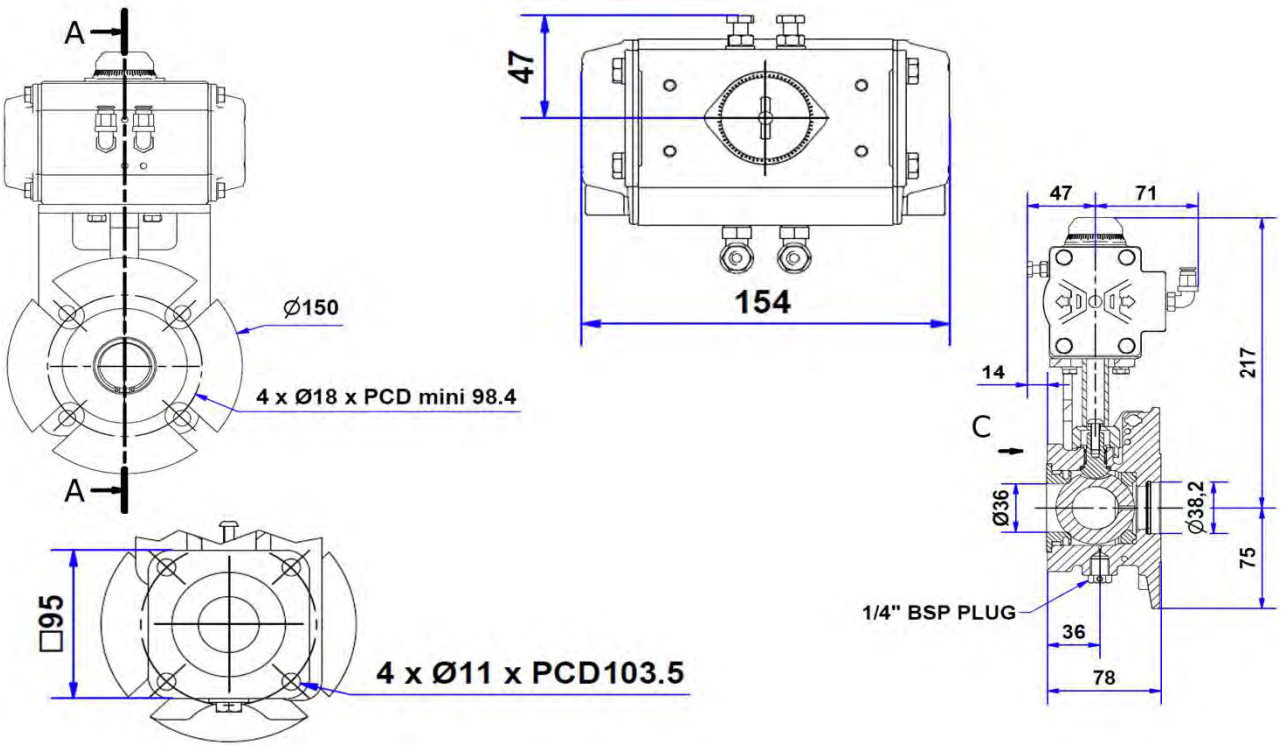
Temperature range vary according to the seal material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

ACCESSORIES	
Weld-in flange	11 90 20 00 00
Blind Flange	11 95 87 00 00
Bolting Kit	17 12 10 95 00
Screw kit	11 14 51 90 00
Filter kit	11 14 01 95 00
Pressure gauge	70 18 06 00 95

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face To Face 94mm with Caps
Operation	Pneumatic - 1/4 Turn single acting actuator
Ball seals material	PTFE

DIMENSIONAL DRAWING



AIR LINE

1 ½" Flanged ATCO™
Pneumatic ND32 Ball Valve
AG3X

CODIFICATION

STANDARD PART NUMBER	AG3	X	7	T	2	4	S	2	1	.
YOUR PART NUMBER	AG3	__	7	T	__	__	__	__	__	__

MATERIAL - Other on request

316L / 1.4409 stainless steel	X

MAWP

6.7 Bar	7

WORKING TEMPERATURE

Ball Seals PTFE +150°C	T

INLET FLANGE

4 x Ø10.5 x 103.5 PCD	2

OUTLET FLANGE - other on request

4 x M16 x 98.4 PCD	4
4 x slots Ø18 x min98.4 PCD	5
4 x M16 x 125 PCD	6
4 x M16 x 110 PCD	8
4 x Ø18 x 110 PCD	9

FILTER

Without	S
With	P

GAUGE CONNECTION & PLUG

1/4" BSP Connection & Plug	2

GAUGE PROTECTION

Without	1
With	2

POSITION SENSOR

Without	.
With	L

AIR LINE

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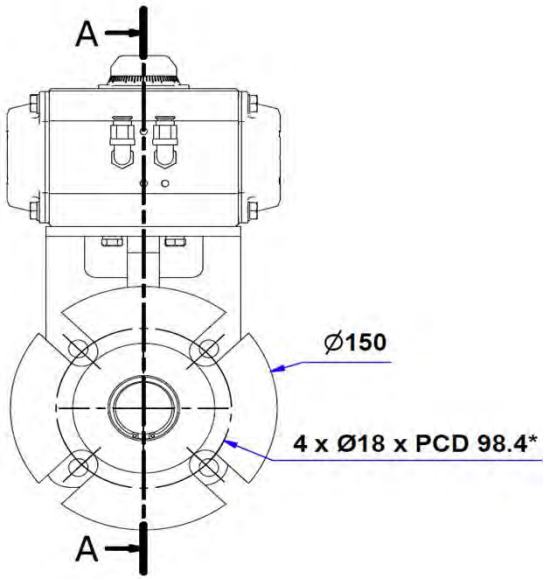
1 ½" Flanged ATCO™
Pneumatic ND32 Ball Valve
AG3X

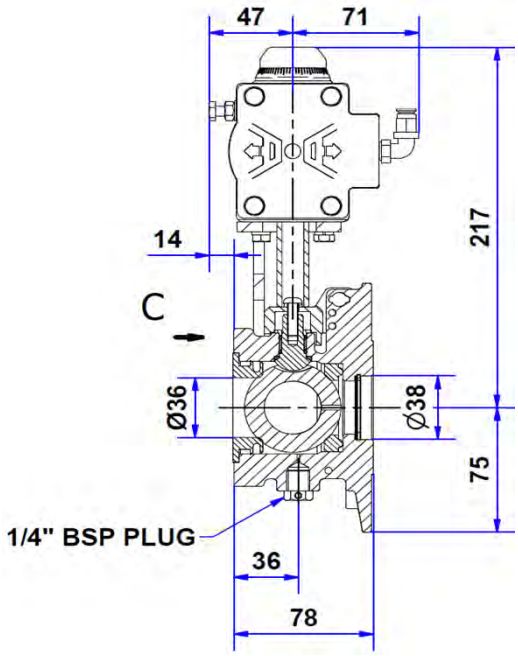
INSTALLATION

Caution: Do not disassemble the actuator from ATCO Ball valve.
For better and long time performances, assembly and settings are realized in PEROLO factory.

CONDITIONS

- Minimum operating pressure: 4 bar (Maximum operating pressure: 8bar)
- Actuator connections: Ø6mm ext
- Actuator Working Temperature: -40°C to 80°C
- For corrosive products application, provide a protection to avoid contact with actuator





INSTALLATION ON THE TANK

- Align holes in gasket to screws (screw kit: 11 14 51 90 00).
- Fit ball valve onto tank pad and secure with washers and nuts. The required tightening torque is 3 daN.m. ± 1
- Lubricate threads with PTFE added grease
- Cross thight the 4 screws 1-2-3-4
- Check torque after 24H
- Adjust at 3 diaNm if required

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

SETTING UP THE ACTUATOR

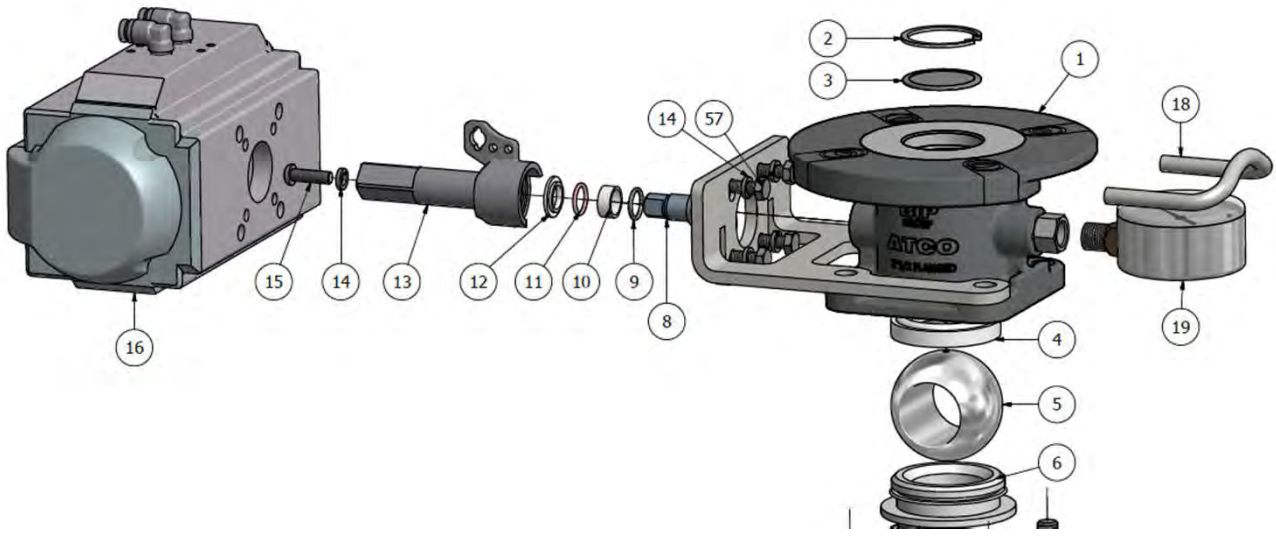
- Connect inlet to connection 2 and outlet to connection 4.
- It is a simple effect actuator able to work with only one inlet connection. However, we recommend to use two inlet connections. It reduces the time of closure. Use lubricated air to keep the actuator in good working condition. The air feeding should be rated from 5.5 bar up to 6.5 bar (do not exceed 8 bar).

1 ½" Flanged ATCO™
Pneumatic ND32 Ball Valve
AG3X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 91 09 00 00**
4	1	Ball seal (PTFE)	12 96 06 00 10
6	1	Main seal (PTFE)	11 14 02 00 06
		Ball kit:	**11 14 01 94 00**
5	1	Ball (316)	12 96 06 00 09
		Actuator coupling kit:	**on request**
13	1	Coupling (CF8M)	-
15	1	Screw M6x20 (st st)	-
14	1	Washer W6 (st st)	-
12	1	Bearing (PTFE)	12 96 06 00 13
11	1	O'ring (Viton)	70 10 03 02 08
10	1	Spacer (PTFE)	11 14 02 00 03
9	1	O'ring (FEP/Silicon)	70 10 15 00 66
7	1	Spindle Gasket (PTFE)	11 14 02 00 01
8	1	Spindle (st st)	11 14 02 00 02
		Spindle seals kit:	**11 14 01 92 00**
12	1	Bearing (PTFE)	12 96 06 00 13
11	1	O'ring (Viton)	70 10 03 02 08
10	1	Spacer (PTFE)	11 14 02 00 03
9	1	O'ring (FEP+Silicon)	70 10 15 00 66
7	1	Spindle Gasket (PTFE)	11 14 02 00 01
		Filter kit:	**11 14 01 95 00**
3	1	Filter (st st)	-
2	1	Lock Ring (st st)	-

Nb Perolo recommends :
- The spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- Both ball gaskets should be changed simultaneously.
- Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details



AIR LINE

SAFETY RELIEF VALVES

FUNCTION

A safety relief valve is designed to protect tanks against accidental overpressure and excessive vacuum. Depending on the model, the relief valve can be single acting or double acting. Perolo offers relief valves with sizes ranging from DN20 until DN80 depending on the application and the requested flow rate.

- Various diameters:
  YAK: 1", 1¼", 1½", 2"
  MEGASUPERVENTIX: 2½", 3"
- Single and/or Double acting
- Inlet connection:
  BSP Threaded or flanged
- Special uses:
  Hygienic DIN11851 connection
  H2O2 breather
- High flow model
- High Pressure model
- Contact Materials:
  316L stainless steel
- Large choice of accessories:
  Gauges, Gauge guard
  Flame proof gauze
  Setting gasket material
- Working Temperature:
  Standard or High (+250°C)
- Surface Finish:
  Acid etched, Halar lined
- ADR, RID, IMDG,
  PED on request

2½" BSP MEGA MX65F

2½" Flanged MEGA MX65

3" Flanged MEGA MX80

3" HIGH FLOW MG80

2½" HIGH PRESSURE MH65F

3" HIGH PRESSURE MH65S

2½" BSP DIN11851 MX65M

2" H2O2 Breather

1" Double acting YAK VXS2

1¼" Single acting YAK VXS3

1½" Double acting YAK VXS4

2" Single acting YAK VXS5

2 1/2" (ND65) SAFETY RELIEF VALVE

BSP MEGA SUPERVENTIX MX65F & MP65F(PED)

2 1/2" MEGA SUPERVENTIX™ threaded safety relief valve is designed to protect tanks against accidental overpressure and excessive vacuum. With a smart compact design, the 2 1/2" BSP MEGA SUPERVENTIX™ SRV has a high flow performance.

SPECIFICATION	
Nominal diameter	65 mm (2 1/2")
Weight	3 kg
Connection	2 1/2" BSP
Pressure settings	From 0.5 Bar to 7.4 Bar
Vacuum settings	From 50 to 410 mBar
Working Temperature	-40° to +190°C (HT +250°C)

*Temperature range may vary according to the seal material

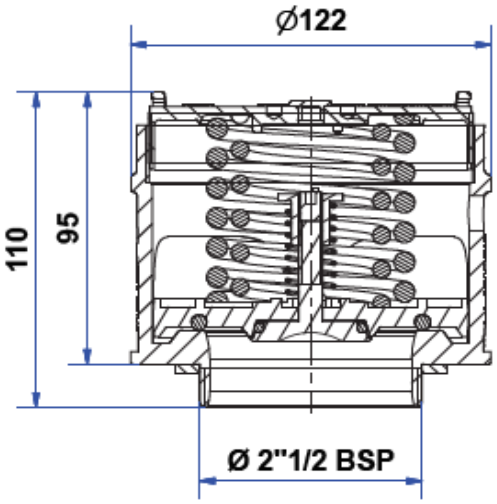
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID / ISO 4126-1
Design Approval: Lloyd's Register - RMR
PED (for MP65 only)
PED 2014/68/EU - Category IV - Module H1 - Fluid gr.1
Lloyd's Register Espana

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone - others on request

ACCESSORIES	
Weld-in flange	12 91 14 00 00
Recess flange	12 92 14 00 00
Seal	11 10 01 60 00
Screw kit	12 12 60 91 00
Bursting Disc Adaptator	11 91 15 00 01
Pressure gauge	70 18 06 00 95

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



Flow measurement conditions: 101.3 kPa at 15°C

Flow reduction with gauze <3%

Set pressure			FLOW RATE AT 120% MX65				FLOW RATE AT 110% MX65			
			Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze		Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze	
Bar	kPa	PSIG	Sm³/S	Scfm	Sm³/S	Scfm	Sm³/S	Scfm	Sm³/S	Scfm
Max (±3% SP; 0,1B/1,45PSI)										
2.18	218	31.6	2.56	5425	2.30	4883	1.98	4205	1.78	3785
3.75	375	54.4	3.92	8306	3.53	7475	3.34	7081	3.01	6373
4.4	440	63.8	4.48	9492	4.03	8543	3.9	7356	3.51	6620
7.3	730	105.9	6.99	14810	6.29	13329	6.4	13584	5.76	12226

SAFETY RELIEF VALVE

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2 1/2" (ND65) SAFETY RELIEF VALVE
BSP MEGA SUPERVENTIX
MX65F & MP65F(PED)

CODIFICATION

STANDARD PART NUMBER	Standard: MX65 PED: MP65	F	44	T	21	G	C	.	.
YOUR PART NUMBER	MX65/MP65	F	__	T	__	__	__	__	.

BODY CONNECTION

2 1/2" BSP Threaded	F

PRESSURE SETTING - Other on request

No pressure	00
. bar (ex: 4.4bar)	44

SETTING GASKET

FEP	T
PFA/Silicone (HT +250°C)	H

VACUUM SETTING - Other on request

No Vacuum	.
0.** bar (ex: 0.21 bar)	21

FLAME PROOF GAUZE

With	G
Without	S

SETTING GASKET

Composite / PTFE	C
Perbunan / PTFE	P

VACUUM TEST BUTTON

With	B
Without	.

HALAR COATING

Without	.

For coating options please see high corrosion catalogue

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 26 20 00 N
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" (ND65) SAFETY RELIEF VALVE
BSP MEGA SUPERVENTIX
MX65F & MP65F(PED)

INSTALLATION

STORAGE

Keep the "MEGA SUPERVENTIX" safety relief valve in original packaging, until ready to use. This prevents ingress of dust and protects against impact damage.

OPERATING

The pressure relief valve (PRV) is designed to be installed on pressurized tanks static or transportation application. The main function of the relief valve is to quickly evacuate the over pressure of the tank specially in case of fire, the high flow capacity of the valve allow this quick flow relief (refer to specification sheet for flow rate values).

Check the compatibility between safety relief valve material and fluid before operating.

The valve operates as following :
The pressure setting is admised to be 1,1 x tank pressure service. This pressure setting is atteined by the spring set compression on the pressure plate. When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure plate closes.
In case of high pressure increase, the pressure plate translation is about 23 mm to open a maximum of surface to allow a high flow evacuation.The set pressure range is 0,5 to 7,4 bar. Service temperature -40°C to 190°C. A vacuum relief function can also be added and set from 0,05 to 0,41 bar.

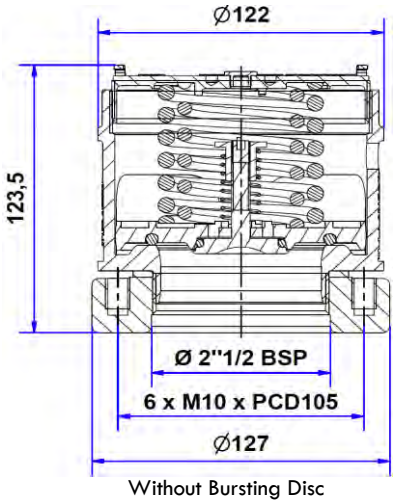
ASSEMBLY

Before assembly check the PRV set pressure is compatible with tank MAWP and the carried cargo with PRV material.
The relief valve should be installed in protected area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.
Make sure the "Mega-SUPERVENTIX" safety relief valve is compatible with the working pressure of the tank.
The valve set pressure is 1.1 x Tank MAWP.

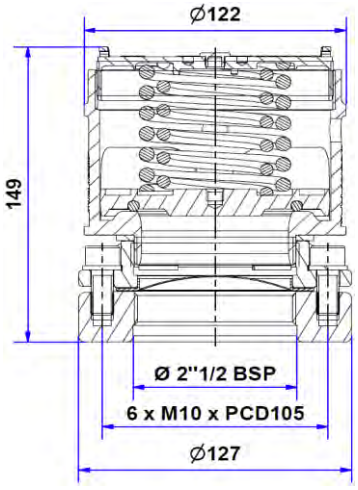
1. Remove the protective cap from the 2 ½ " BSP thread.
2. Remove the gauze temporarily fitted on the valve.
3. Make sure that no dirt or grit is present on the receiving flange & gasket and that the threads are perfectly clean.
4. Manually screw on the valve with gasket (or bursting disc ND65).
Then keep turning until firmly tight using a 'C' Spanner.
Tightening torque settings depend on the setting gasket material:
Perbunan / PTFE gasket : **80 N.m ± 5**
Composite / PTFE gasket: **230 N.m ±5**
PTFE gasket: **240 N.m ±5**

CAUTION : Do not remove the wire weld seal between the body and cover to avoid to lose the warranty before first periodical check.
The valve can be sealed to the tank using the wiring clamps located on the valve cover.

In case of installation of bursting disc and adaptor flange, the 6 screws M10 of adaptor flange should be tightened with **30 N.m ±5**



Without Bursting Disc



Bursting Disc Assembly

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
BSP MEGA SUPERVENTIX
MX65F & MP65F(PED)

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Check no leak exists between the poppet seal and the seat, this check can be done with leakage detection fluid.
If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further check

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the screws that hold the MEGA-SUPERVENTIX to the tank pad, remove the valve and the disc.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.
Further valve disassembly should be done in a workshop.

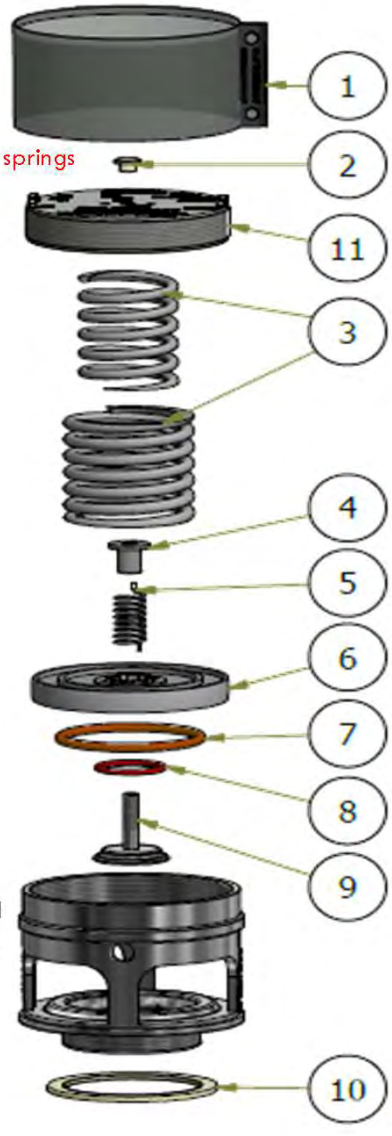
RELIEF VALVE DISASSEMBLY

We recommend this operation every two years.

CAUTION: the springs load is between 100 daN en 400 daN, when cap is removed the springs can violently be ejected.

We recommend to safely disassemble the cap and slowly relief the spring set load.

1. Remove carefully the cap (11) considering the above caution.
2. Remove springs set (3) and pressure plate (6).
3. Remove pressure and vacuum seals (7) & (8)
4. Install a set of new seals
5. Reassemble the valve.



SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5. The leakage is detected with leak test fluid at pressure seal interface.

1. Slowly increase the pressure. **When first bubbles appear at pressure plate seal, note the pressure.**
2. Adjust the cap (screw/unscrew) to get the set pressure.

If set pressure can not be achieved easily, we recommend to change the spring set (3) (see page 3 for item references). Only PEROLO provided spring sets should be installed

3. When setting complete, seal the cap with a stainless steel wire spot welded between cap and body.

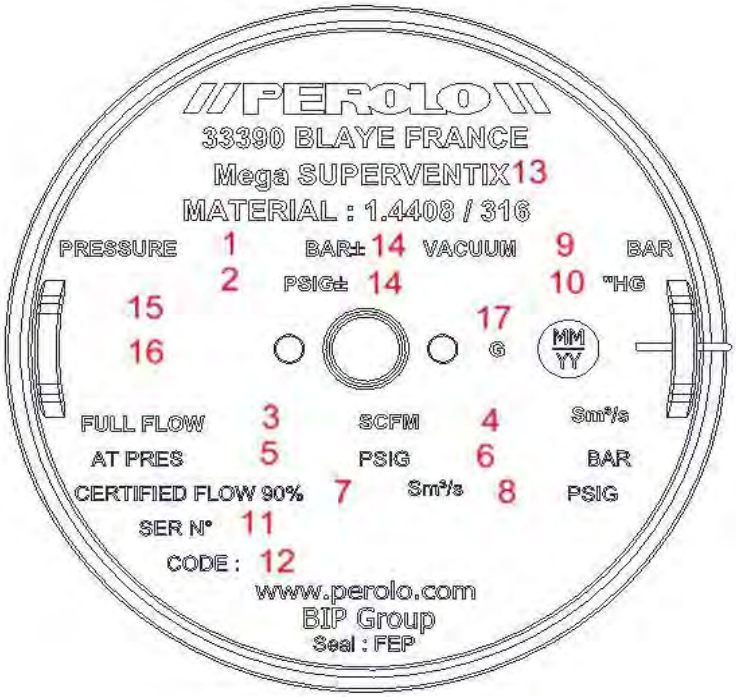
The relief valve is ready to operate on the tank.

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
BSP MEGA SUPERVENTIX
MX65F & MP65F(PED)

SAFETY RELIEF VALVE

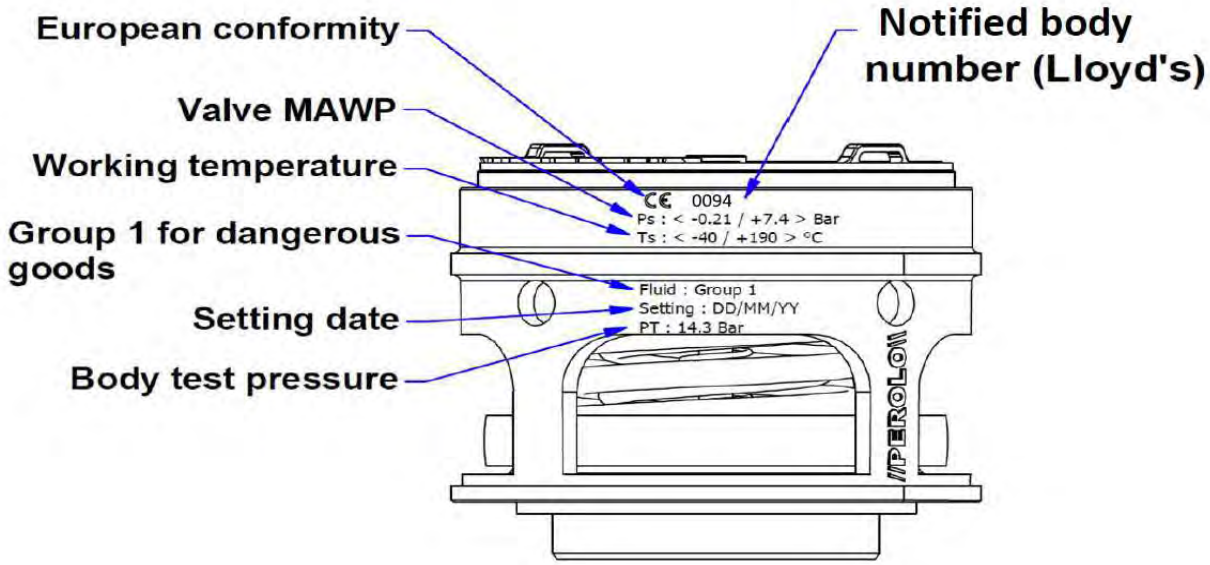
BODY MARKING (For MX65F & MP65F)



1) Setting Pressure (Bar)
2) Setting Pressure (Psi)
3) Full Flow (SCFM)
4) Full Flow (Sm3/S)
5) Full Flow Pressure (Psi)
6) Full Flow Pressure (Bar)
7) Certified Flow 90% (SCFM)
8) Certified Flow 90% (Psi)
9) Vacuum (Bar)
10) Vacuum ("Hg)
11) Serial number
12) P/N (MX65F or MP65F)
13) Nominal diameter (ND65)
14) Pressure tolerance (Bar)
15) Flow section (mm²)
16) Standard : ISO 4126-1
17) "G" indication for gaz

PED ADDITIONNAL BODY MARKING (For MP65F Only)

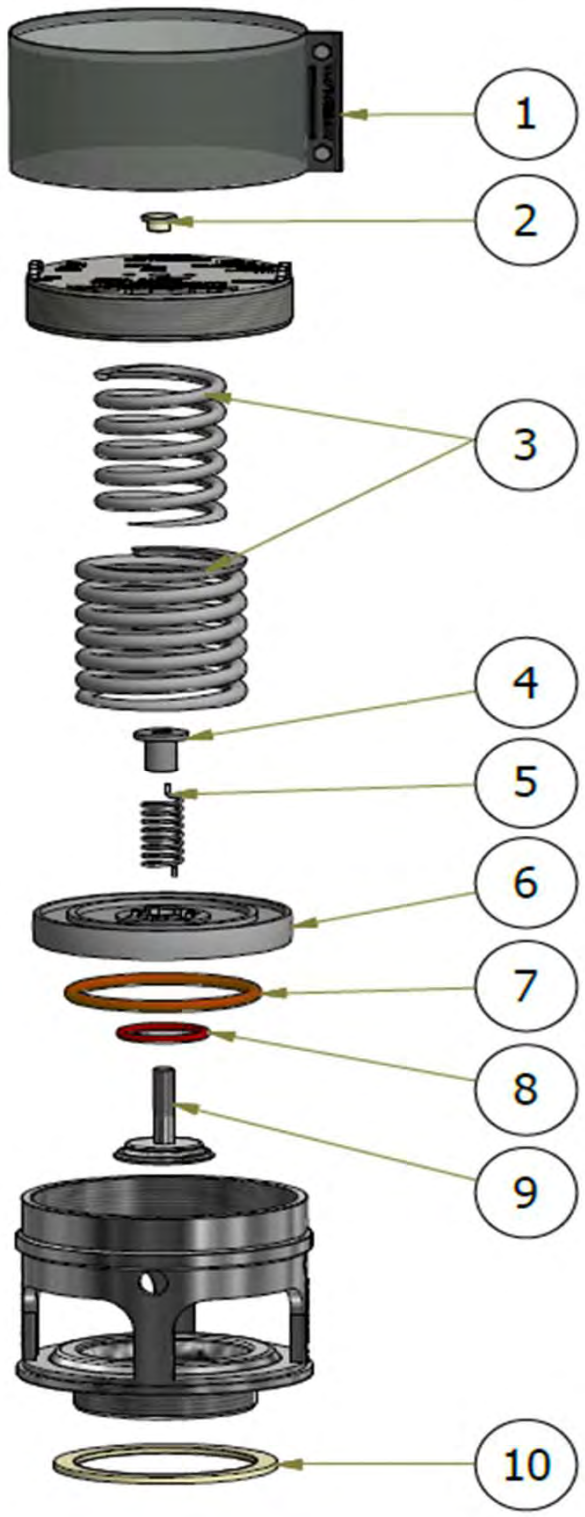
Body marking is only made on PED approved safety relief valve (when reference starts with MP)



2 1/2" (ND65) SAFETY RELIEF VALVE
BSP MEGA SUPERVENTIX
MX65F & MP65F(PED)

ITEM	QTY	DESIGNATION	PART N°
		Pressure & Vacuum O'ring:	
7	1	Pressure O'ring (FEP/Silicone)	11 10 01 99 05
7	10	Pressure O'ring (FEP/Silicone)	11 10 01 99 06
7, 8	1	Pres. & Vac. O'ring (FEP/Si)	11 10 01 91 00
8	10	Vacuum O'ring (FEP/Silicone)	11 10 17 99 00
		Spring sets:	
3	1	Spring set pressure 2.18 Bar	11 10 01 95 00
3	1	Spring set pressure 3.30 Bar	12 12 84 90 00
3	1	Spring set pressure 3.75 Bar	11 10 01 96 00
3	1	Spring set pressure 4.40 Bar	11 10 01 97 00
5	1	Spring set vacuum 0.21 bar	11 10 01 98 00
		Plates, Gauze & Cap:	
6	1	Pressure plate for pressure only	11 10 01 99 01
6	1	Pressure plate, pressure/vacuum	11 10 01 99 00
9	1	Vacuum poppet	11 10 01 99 02
2, 4, 5, 6, 7, 8, 9	1	Vacuum set kit (plug, adjustment nut, pressure plate, o'rings, spring set and vacuum poppet)	11 10 01 99 08
4	1	Adjustement nut	11 10 02 99 06
1	1	Flame proof gauze kit	11 10 01 50 91
1	10	Flame proof gauze kit	11 10 01 50 95
2	1	Plug (LDPE)	70 14 06 90 15
		Setting gasket:	
10	1	Gasket Comp/PTFE	12 91 31 91 01
10	10	Gasket Comp/PTFE	11 10 01 92 00
10	1	Gasket Perbun./PTFE	11 10 01 93 00

DESIGNATION	SPECIFICATIONS	PART N°
Weld-in flange	Ext Ø 127mm, int Ø 76mm thickness 25 mm, 316L st st drilling 6 x M10 x 105 PCD	12 91 14 00 00
Recess flange	Ext Ø 166mm, int Ø 76mm min thick. 22 mm max th. 61 mm drilling 6 x M10 x 105 PCD 316L stainless steel	12 92 14 00 00
Adaptator Flange for Bursting Disc Assembly	Ext Ø 127mm, int Ø 2" BSP thickness 22.5 mm, 316L st st drilling 6 x 11 x 105 PCD 1/4" BSP gauge connection	11 91 15 00 00 + 11 91 15 00 01 (without gauge protection)
Bursting disc	4.84 Bar	DRS65P484T20
Gauge (0-6 bar) bottom entry 1/4"BSP	Full stainless steel	70 18 06 00 95
	Stainless steel case	70 21 02 00 22



SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
DN65 FLANGED MEGA SUPERVENTIX MX65S/P & MP65S/P(PED)

2 1/2" MEGA SUPERVENTIX™ flanged safety relief valve - designed to protect tanks against accidental overpressure or excessive vacuum. With a smart compact design, the 2 1/2" MEGA SUPERVENTIX™ has a high flow performance.

SPECIFICATION	
Nominal diameter	65 mm (2 1/2")
Weight	5 kg
Connection	4 slotted holes x Ø18 on a 145 to 152.4 PCD
Pressure settings	From 0.5 to 7.4 Bar
Vacuum settings	From 50 to 410 mBar
Working Temperature	-40° to +190°C (HT +220°C)

*Temperature range may vary according to the seal material

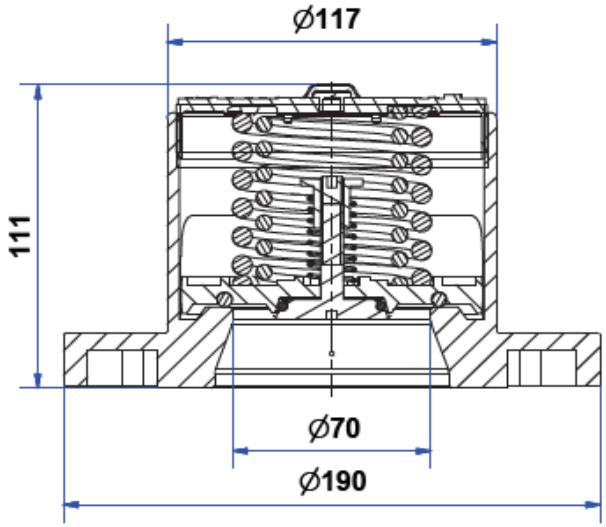
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID / ISO 4126-1
Design Approval: Lloyd's Register - RMR
PED (for MP65 only)
PED 2014/68/EU - Category IV - Module H1 - Fluid gr.1
Lloyd's Register Espana

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone - others on request

ACCESSORIES	
Weld-in flange	12 92 03 00 00
Recess flange	11 90 63 00 00
Seal	12 92 04 00 00
Screw kit	11 10 09 91 00
Stud kit	17 12 12 91 00
Pressure gauge	70 18 06 00 95

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



Flow measurement conditions: 101.3 kPa at 15°C

Flow reduction with gauze <12%

Set pressure			FLOW RATE AT 120% MX65				FLOW RATE AT 110% MX65			
			Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze		Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze	
Bar	kPa	PSIG	Sm³/S	Scfm	Sm³/S	Scfm	Sm³/S	Scfm	Sm³/S	Scfm
Max (±3% SP; 0,1B/1,45PSI)										
2.18	218	31.6	2.56	5425	2.30	4883	1.98	4205	1.78	3785
3.75	375	54.4	3.92	8306	3.53	7475	3.34	7081	3.01	6373
4.4	440	63.8	4.48	9492	4.03	8543	3.9	7356	3.51	6620
7.3	730	105.9	6.99	14810	6.29	13329	6.4	13584	5.76	12226

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 51 20 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" (ND65) SAFETY RELIEF VALVE
DN65 FLANGED MEGA SUPERVENTIX
MX65S/P & MP65S/P(PED)

CODIFICATION

STANDARD PART NUMBER	Standard: MX65 PED: MP65	P	44	T	21	G	.	.	.
YOUR PART NUMBER	MX65/MP65	__	__	__	__	__	__	.	__

BODY DRILLING - Other on request

4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection	S
4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection and plug	P

PRESSURE SETTING - Other on request

No pressure	00
. bar (ex: 4.4bar)	44

O'RING MATERIAL

FEP/Silicone	T
PFA (High Temp 220°C WT at 4,4b)	H

VACUUM SETTING - Other on request

No Vacuum	.
0.** bar (ex: 0.21 bar)	21

FLAME PROOF GAUZE

With	G
Without	S

VACUUM TEST BUTTON

With	B
Without	.

HALAR COATING

Without	.

For coating options please see high corrosion catalogue

GAUGE PROTECTION

With	P
Without	.

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 51 20 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" (ND65) SAFETY RELIEF VALVE
DN65 FLANGED MEGA SUPERVENTIX
MX65S/P & MP65S/P(PED)

SAFETY RELIEF VALVE

INSTALLATION

STORAGE

Keep the "MEGA SUPERVENTIX" safety relief valve in original packaging, until ready to use. This prevents ingress of dust and protects against impact damage.

OPERATING

The pressure relief valve is designed to be installed on pressurized tanks static or transportation application. The main function of the relief valve is to quickly evacuate the over pressure of the tank specially in case of fire, the high flow capacity of the valve allow this quick flow relief (refer to specification sheet for flow rate values).

Check the compatibility between safety relief valve material and fluid before operating.

The valve operates as following :

The pressure setting is 1,1 x tank pressure service. This pressure setting is atteined by the spring set compression on the pressure plate. When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure plate closes.

In case of high pressure increase, the pressure plate translation is about 20 mm to open a maximum of surface to allow a high flow evacuation.The set pressure range is 0,5 to 7,4 bar. Service temperature -40°C to 190°C. A vacuum relief function can also be added and set from 0,05 to 0,41 bar.

ASSEMBLY

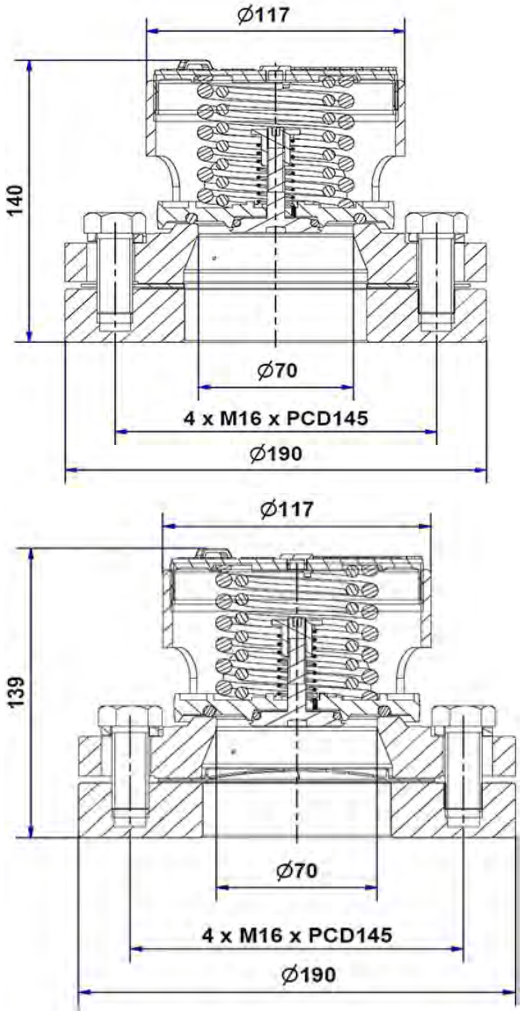
The relief valve should be installed in protected area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.

Make sure the "Mega-SUPERVENTIX" safety relief valve is compatible with the working pressure of the tank.
The valve set pressure should be 1.1 x MAWP(maximum allowable working pressure of the tank)

1. Check there is no dirt or grid on value flange and weld flange.
2. Install the **3" (ND80)** bursting disc (BD) or the gasket
Caution: BD must be 3"
3. Install the valve.
4. Install and tighten the screw set.
   Tighten to a torque of **70N.m. +/-5**
   Cross type Tighten

Check there is no leakage at gasket or bursting disc interface.
The valve is ready to operate.

CAUTION : Do not remove the wire weld seal between the body and cover to avoid losing warranty before first periodical check.



2 1/2" (ND65) SAFETY RELIEF VALVE
DN65 FLANGED MEGA SUPERVENTIX
MX65S/P & MP65S/P(PED)

ISO RAIL ROAD IBC

SAFETY RELIEF VALVE

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Check no leak exists between the poppet seal and the seat, this check can be done with leakage detection fluid.
If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further check

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the screws that hold the MEGA-SUPERVENTIX to the tank pad, remove the valve and the disc.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.

Further valve disassembly should be done in a workshop.

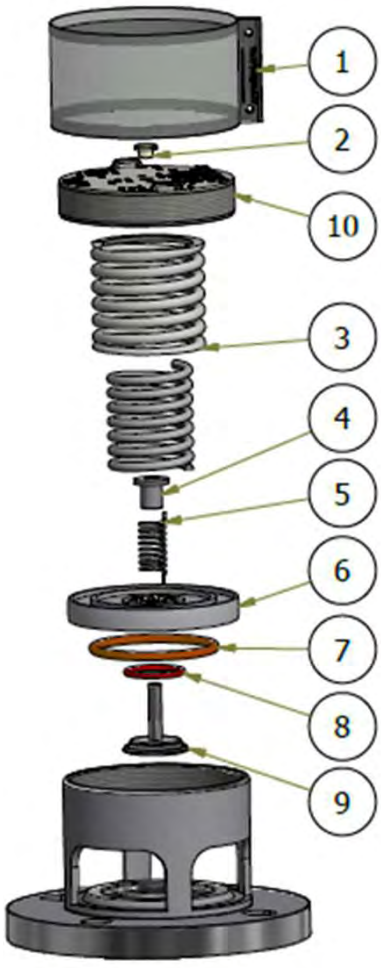
RELIEF VALVE DISASSEMBLY

We recommend this operation every two years.

CAUTION: the springs load is between 100 daN en 400 daN, when cap is removed the springs can violently be ejected.

We recommend to safely disassemble the cap and slowly relief the spring set load.

1. Remove carefully the cap (10) considering the above caution.
2. Remove springs set (3) and pressure plate (6).
3. Remove pressure and vacuum seals (7) & (8)
4. Install a set of new seals
5. Reassemble the valve.



SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5. The leakage is detected with leak test fluid at pressure seal interface.

1. Slowly increase the pressure. **When first bubbles appear at pressure plate seal, note the pressure.**
2. Adjust the cap (screw/unscrew) to get the set pressure.

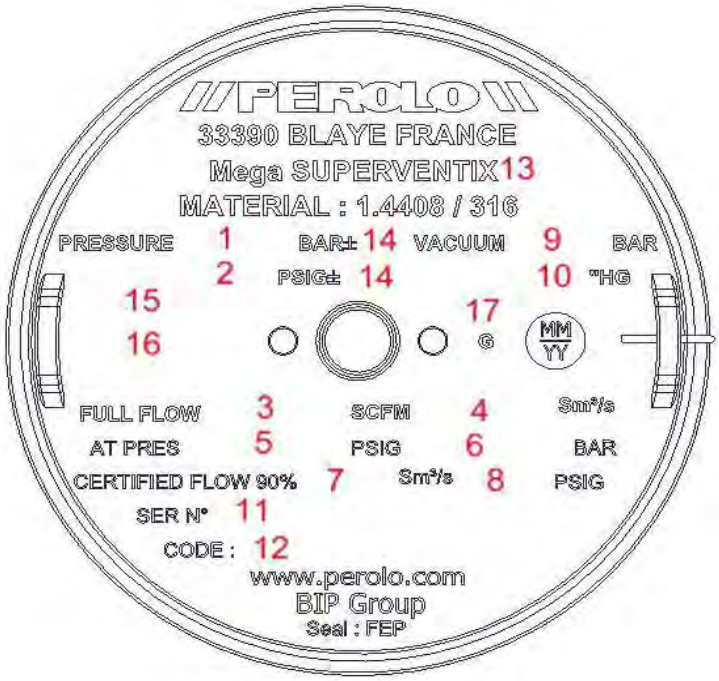
If set pressure can not be achieved easily, we recommended to change the spring set (3) (see page 3 for item references).

3. When setting complete, seal the cap with a stainless steel wire spot welded between cap and body.

The relief valve is ready to operate on the tank.

2 1/2" (ND65) SAFETY RELIEF VALVE
DN65 FLANGED MEGA SUPERVENTIX MX65S/P & MP65S/P(PED)

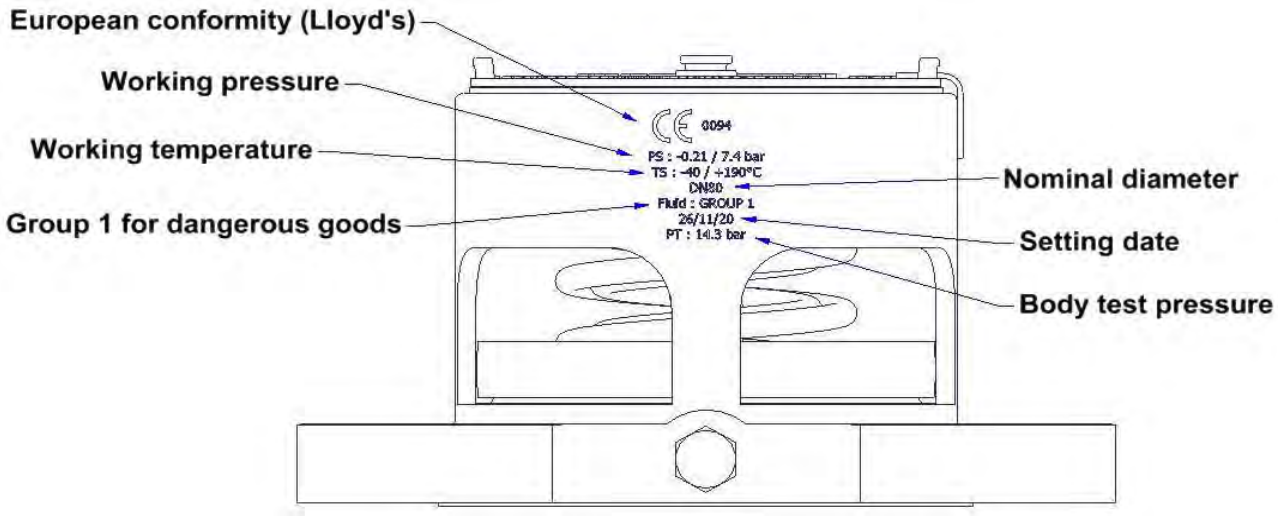
BODY MARKING (For MX65S/P & MP65S/P)



1) Setting Pressure (Bar)
2) Setting Pressure (Psi)
3) Full Flow (SCFM)
4) Full Flow (Sm3/S)
5) Full Flow Pressure (Psi)
6) Full Flow Pressure (Bar)
7) Certified Flow 90% (SCFM)
8) Certified Flow 90% (Psi)
9) Vacuum (Bar)
10) Vacuum ("Hg)
11) Serial number
12) P/N (MX65S/P or MP65S/P)
13) Nominal diameter (ND65)
14) Pressure tolerance (Bar)
15) Flow section (mm²)
16) Standard : ISO 4126-1
17) "G" indication for gaz

PED ADDITIONNAL BODY MARKING (For MP65S/P Only)

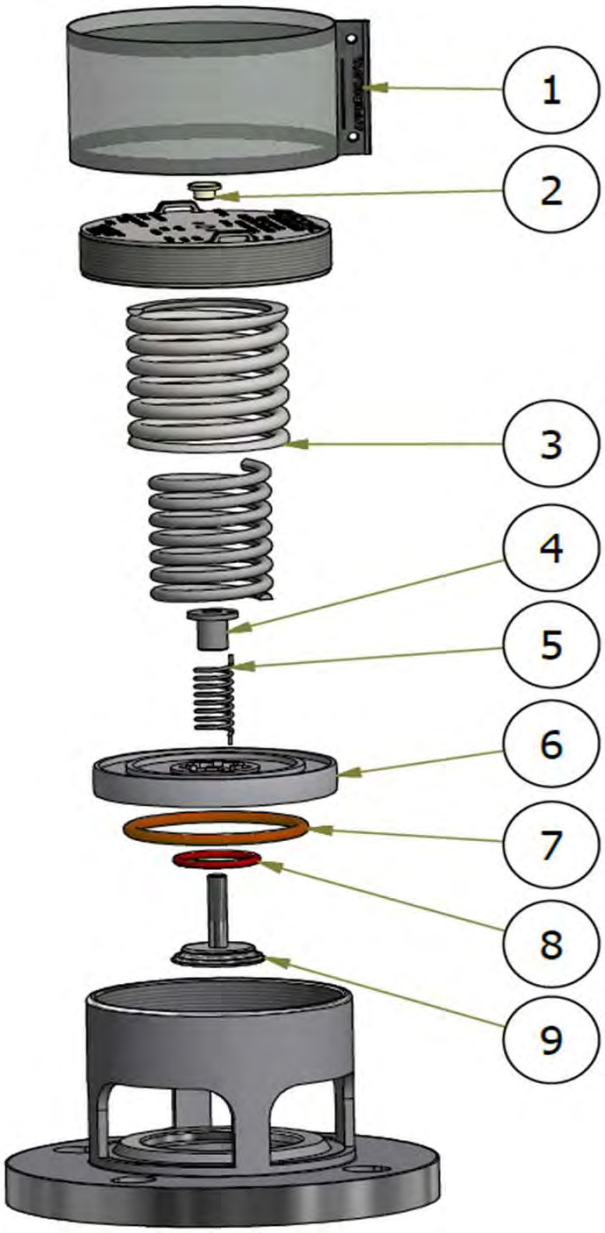
Body marking is only made on PED approved safety relief valve (when reference starts with MP)



2 1/2" (ND65) SAFETY RELIEF VALVE
DN65 FLANGED MEGA SUPERVENTIX
MX65S/P & MP65S/P(PED)

ITEM	QTY	DESIGNATION	PART N°
		Pressure & Vacuum O'ring:	
7	1	Pressure O'ring (FEP/Silicone)	11 10 01 99 05
7	10	Pressure O'ring (FEP/Silicone)	11 10 01 99 06
7, 8	1	Pres. & Vac. O'ring (FEP/Si)	11 10 01 91 00
8	10	Vacuum O'ring (FEP/Silicone)	11 10 17 99 00
		Spring sets:	
3	1	Spring set pressure 2.18 Bar	11 10 01 95 00
3	1	Spring set pressure 3.30 Bar	12 12 84 90 00
3	1	Spring set pressure 3.75 Bar	11 10 01 96 00
3	1	Spring set pressure 4.40 Bar	11 10 01 97 00
5	1	Spring set vacuum 0.21 bar	11 10 01 98 00
		Plates, Gauze & Cap:	
6	1	Pressure plate for pressure only	11 10 01 99 01
6	1	Pressure plate, pressure/vacuum	11 10 01 99 00
9	1	Vacuum poppet	11 10 01 99 02
2, 4, 5, 6, 7, 8, 9	1	Vacuum set kit (plug, adjustment nut, pressure plate, o'rings, spring set and vacuum poppet)	11 10 01 99 08
4	1	Adjustement nut	11 10 02 99 06
1	1	Flame proof gauze kit	11 10 01 94 00
2	1	Plug (LDPE)	70 14 06 90 15

DESIGNATION	SPECIFICATIONS	PART N°
Weld-in flange	Ext Ø 190mm, int Ø 82mm thickness 26 mm drilling 4 x M16 x 145 PCD 316L stainless steel	1292030000
Recess flange	Ext Ø 217mm, int Ø 82,6mm min thick. 31 mm max th. 65 mm 4 x M16 x 145 PCD 316L stainless steel	1190630000
Screw kit	4 x M16-40 screws 4 x grower washers	1110099100
Bursting disc	4.84 Bar	DRS80P484T20
Setting gasket	CNAF/PTFE Ø176mm Ø82mm Round, flat, drilling 4 x 18 x 145 PCD	1292040000
Gauge (0-6 bar) bottom entry 1/4"BSP	Full stainless steel	7018060095
	Stainless steel case brass internals & connection	7021020022



SAFETY RELIEF VALVE

3" (ND80) SAFETY RELIEF VALVE
FLANGED MEGA SUPERVENTIX MX80 & MP80(PED)

3" MEGA SUPERVENTIX™ flanged safety relief valve - designed to protect tanks against accidental overpressure or excessive vacuum. With a smart compact design, the 3" MEGA SUPERVENTIX™ has a high flow performance.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	5 kg
Connection	4 slotted holes x Ø18 on a 145 to 152.4 PCD
Pressure settings	From 0.5 to 7.4 Bar
Vacuum settings	From 50 to 410 mBar
Working Temperature	-40° to +190°C (HT +220°C)

*Temperature range may vary according to the seal material

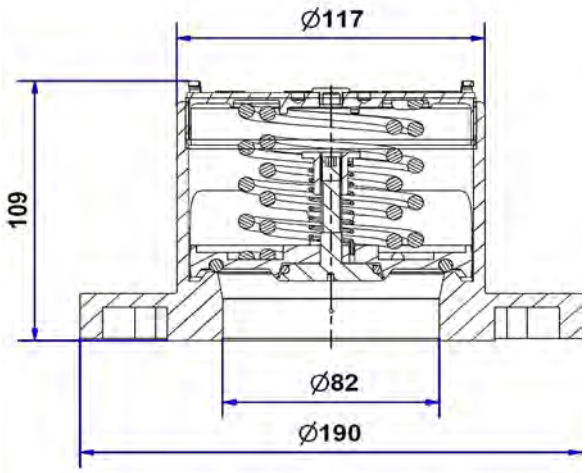
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID / ISO 4126-1
Design Approval: Lloyd's Register - RMR
PED (for MP80 only)
PED 2014/68/EU - Category IV - Module H1 - Fluid gr.1
Lloyd's Register Espana

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone - others on request

ACCESSORIES	
Weld-in flange	12 92 03 00 00
Recess flange	11 90 63 00 00
Seal	12 92 04 00 00
Stud kit	17 12 12 91 00
Screw kit	11 10 09 91 00
Pressure gauge	70 18 06 00 95

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



Flow measurement conditions: 101.3 kPa at 15°C

Flow reduction with gauze <12%

Set pressure			FLOW RATE AT 120% MX80				FLOW RATE AT 110% MX80			
			Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze		Measured flow rate at 110% without gauze		Certified flow rate with the federal register coefficient 90% without gauze	
Bar	kPa	PSIG	Sm³/S	Scfm	Sm³/S	Scfm	Sm³/S	Scfm	Sm³/S	Scfm
Max (±3% SP; 0,1B/1,45PSI)										
2.18	218	31.6	2.78	5888	2.50	5299	1.93	4088	1.74	3679
3.3	330	47.9	4.11	8705	3.70	7834	3.09	6545	2.78	5890
3.75	375	54.4	4.6	9743	4.14	8769	3.56	7540	3.20	6786
4.4	440	63.8	5.35	11331	4.82	10198	4.24	8980	3.82	8082
7.3	730	105.9	7.372	15614	6.63	14053	7.23	15313	6.51	13782

SAFETY RELIEF VALVE

3" (ND80) SAFETY RELIEF VALVE
FLANGED MEGA SUPERVENTIX
MX80 & MP80(PED)

CODIFICATION

STANDARD PART NUMBER	Standard: MX80 PED: MP80	P	44	T	21	G	.	.	.
YOUR PART NUMBER	MX80/MP80						.		

BODY DRILLING - Other on request

4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection	S
4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection and plug	P

PRESSURE SETTING - Other on request

No pressure	00
. bar (ex: 4.4bar)	44

O'RING MATERIAL

FEP/Silicone	T
PFA (High Temp 220°C WT at 4,4b)	H

VACUUM SETTING - Other on request

No Vacuum	.
0.** bar (ex: 0.21 bar)	21

FLAME PROOF GAUZE

With	G
Without	S

VACUUM TEST BUTTON

With	B
Without	.

HALAR COATING

Without	.

For coating options please see high corrosion catalogue

GAUGE PROTECTION

With	P
Without	.

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 24 20 00 L
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) SAFETY RELIEF VALVE
FLANGED MEGA SUPERVENTIX
MX80 & MP80(PED)

ISO RAIL ROAD IBC

INSTALLATION

STORAGE

Keep the “MEGA SUPERVENTIX” safety relief valve in original packaging, until ready to use. This prevents ingress of dust and protects against impact damage.

OPERATING

The pressure relief valve is designed to be installed on pressurized tanks static or transportation application. The main function of the relief valve is to quickly evacuate the over pressure of the tank specially in case of fire,the high flow capacity of the valve allow this quick flow relief (refer to specification sheet for flow rate values).

Check the compatibility between safety relief valve material and fluid before operating.

The valve operates as following :

The pressure setting is admised to be 1,1 x tank pressure service. This pressure setting is got by the spring set compression on the pressure plate. When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure plate closes.

In case of high pressure increase, the pressure plate translation is about 20 mm to open a maximum of surface to allow a high flow evacuation.The set pressure range is 0,5 to 7,4 bar. Service temperature -40°C to 190°C. A vacuum relief function can also be added and set from 0,05 to 0,41 bar.

ASSEMBLY

The relief valve should be installed in protect area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.

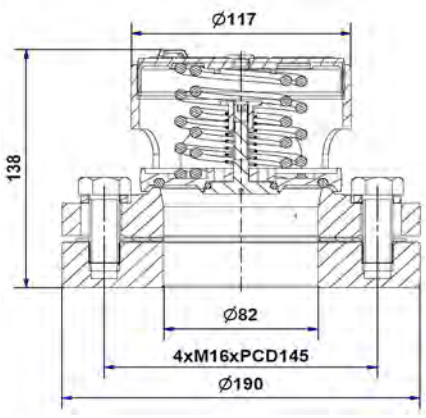
Make sure the “Mega-SUPERVENTIX” safety relief valve is compatible with the working pressure of the tank.
The valve set pressure valve is 1.1 x Tank MAWP.

1. Check there is no dirt or grid on value flange and weld flange.
2. Install the 3" bursting disc (BD) or the gasket (BD must be 3")
3. Install the valve.
4. Install and tighten the screw set.
   Tighten to rhe torque recomanded by disc supplier.
   If no disc installed torque 70 N.m. +/-5 for cnaf/ptfe gasket

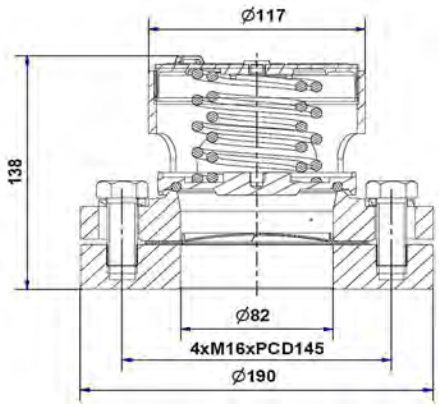
Check there is no leakage at gasket or bursting disc interface.
The valve is ready to operate.

CAUTION : Do not remove the wire weld seal between the body and cover to avoid to lose the warranty before first value periodical check.

The MEGA SUPERVENTIX™ safety relief vent valve is now operational.



without bursting disc



with bursting disc

SAFETY RELIEF VALVE

3" (ND80) SAFETY RELIEF VALVE
FLANGED MEGA SUPERVENTIX
MX80 & MP80(PED)

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Check no leak exists between the poppet seal and the seat, this check can be done with leakage detection fluid. If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further check

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the screws that hold the MEGA-SUPERVENTIX to the tank pad, remove the valve and the disc.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.

Further valve disassembly should be done in a workshop.

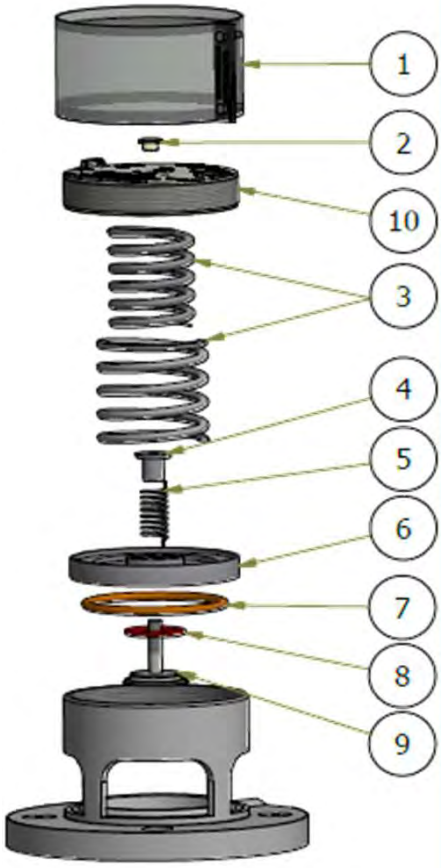
RELIEF VALVE DISASSEMBLY

We recommend this operation every two years.

CAUTION: the springs load is between 100 daN en 400 daN, when cap is removed the springs can violently be ejected.

We recommend to safely disassemble the cap and slowly relief the spring set load.

1. Remove carefully the cap (10) considering the above caution.
2. Remove springs set (3) and pressure plate (6).
3. Remove pressure and vacuum seals (7) & (8)
4. Install a set of new seals
5. Reassemble the valve.



SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5. The leakage is detected with leak test fluid at pressure seal interface.

1. Slowly increase the pressure. **When first bubbles appear at pressure plate seal, note the pressure.**
2. Adjust the cap (screw/unscrew) to get the set pressure

If set pressure can not be achieved easily, we recommended to change the spring set (3) (see page 3 for item references).

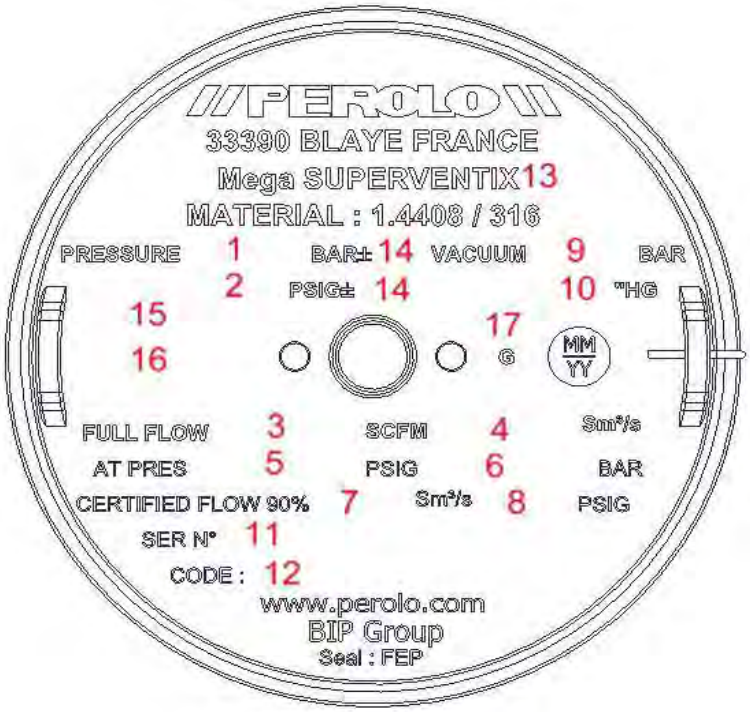
3. When setting complete, seal the cap with a stainless steel wire spot welded between cap and body.

The relief valve is ready to operate on the tank.

SAFETY RELIEF VALVE

3" (ND80) SAFETY RELIEF VALVE
FLANGED MEGA SUPERVENTIX
MX80 & MP80(PED)

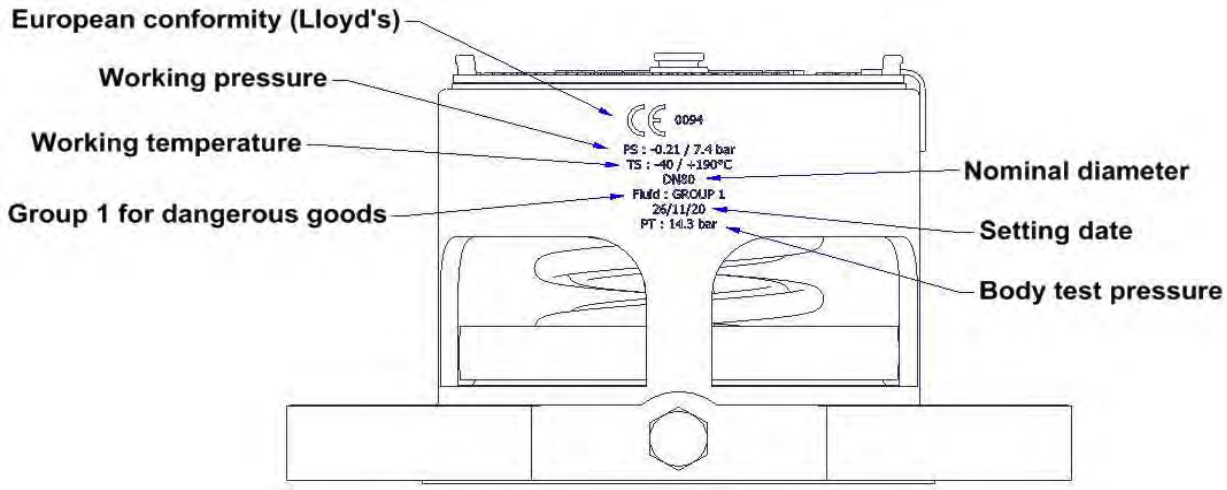
BODY MARKING (For MX80 & MP80)



1) Setting Pressure (Bar)
2) Setting Pressure (Psi)
3) Full Flow (SCFM)
4) Full Flow (Sm3/S)
5) Full Flow Pressure (Psi)
6) Full Flow Pressure (Bar)
7) Certified Flow 90% (SCFM)
8) Certified Flow 90% (Psi)
9) Vacuum (Bar)
10) Vacuum ("Hg)
11) Serial number
12) P/N (MX80 or MP80)
13) Nominal diameter (ND80)
14) Pressure tolerance (Bar)
15) Flow section (mm²)
16) Standard : ISO 4126-1
17) "G" indication for gaz

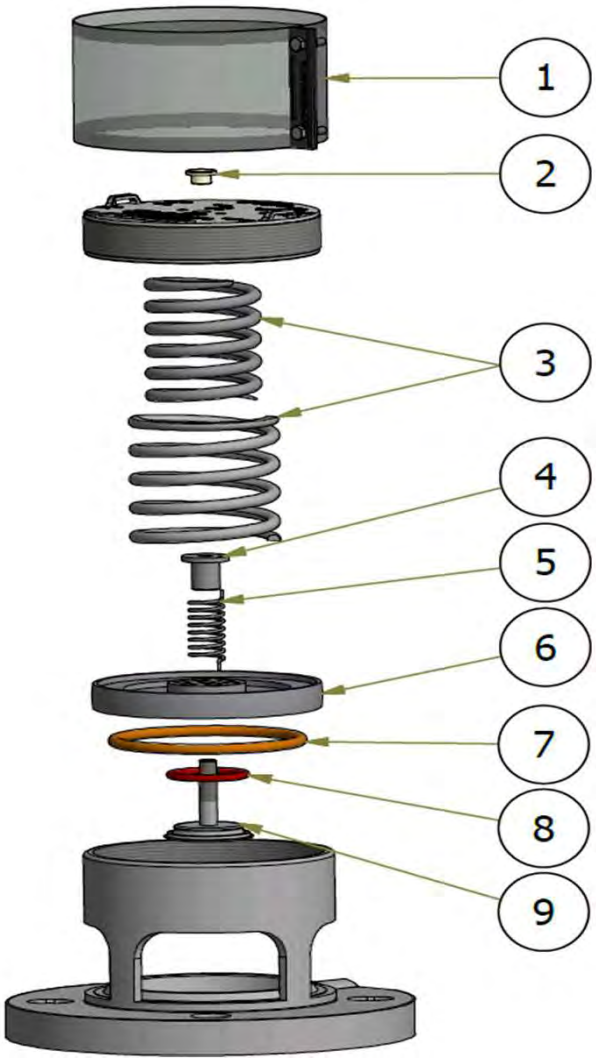
PED ADDITIONNAL BODY MARKING (For MP80 Only)

Body marking is only made on PED approved safety relief valve (when reference starts with MP)



3" (ND80) SAFETY RELIEF VALVE
FLANGED MEGA SUPERVENTIX
MX80 & MP80(PED)

ITEM	QTY	DESIGNATION	PART N°
		Pressure & Vacuum O'ring:	
7	1	Pressure O'ring (FEP/Silicone)	11 10 17 98 00
7	10	Pressure O'ring (FEP/Silicone)	11 10 17 99 06
7, 8	1	Pres. & Vac. O'ring (FEP/Si)	11 10 17 97 00
8	10	Vacuum O'ring (FEP/Silicone)	11 10 17 99 00
		Spring sets:	
3	1	Spring set pressure 2.18 Bar	11 10 17 91 00
3	1	Spring set pressure 3.30 Bar	11 10 17 92 00
3	1	Spring set pressure 3.75 Bar	11 10 17 93 00
3	1	Spring set pressure 4.40 Bar	11 10 17 94 00
5	1	Spring set vacuum 0.21 bar	11 10 01 98 00
		Plates, Gauze & Cap:	
6	1	Pressure plate for pressure only	11 10 17 99 02
6	1	Pressure plate, pressure/vacuum	11 10 17 99 01
9	1	Vacuum poppet	11 10 01 99 02
4	1	Adjustement nut	11 10 02 99 06
1	1	Flame proof gauze kit	11 10 01 94 00
2	1	Plug (LDPE)	70 14 06 90 15



DESIGNATION	SPECIFICATIONS	PART N°
Weld-in flange	Ext Ø 190mm, int Ø 82mm thickness 26 mm drilling 4 x M16 x 145 PCD 316L stainless steel	1292030000
Recess flange	Ext Ø 217mm, int Ø 82,6mm min thick. 31 mm max th. 65 mm 4 x M16 x 145 PCD 316L stainless steel	1190630000
Screw kit	4 x M16-40 screws 4 x grower washers	1110099100
Bursting disc	4.84 Bar	DRS80P484T20
Setting gasket	CNAF/PTFE Ø176mm Ø82mm Round, flat, drilling 4 x 18 x 145 PCD	1292040000
Gauge (0-6 bar) bottom entry 1/4"BSP	Full stainless steel	7018060095
	Stainless steel case brass internals & connection	7021020022

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 10 24 40 00 I
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) SAFETY RELIEF VALVE
HIGH FLOW FLANGED MEGA SUPERVENTIX
MG80

The Perolo 3" MEGA SUPERVENTIX™ safety relief valve is designed to protect tanks against accidental overpressure or excessive vacuum.
This product has a very high flow performance, even more than the standard 3" Mega.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	5 kg
Connection	4 slotted holes x Ø18 on a 145 to 152.4 PCD
Pressure settings	2.18 / 3.30 / 3.75 / 4.40 Bar
Vacuum settings	0.410 Bar
Working Temperature	-40° to +190°C

*Temperature range may vary according to the seal material

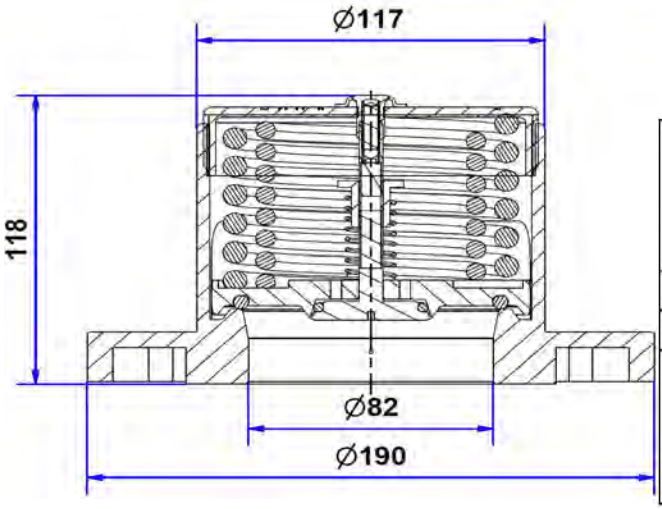
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID / ISO 4126-1
Notified Body : Lloyd's Register Verification - RMR

ACCESSORIES	
Weld-in flange	12 92 03 00 00
Recess flange	11 90 63 00 00
Seal	12 92 04 00 00
Screw kit	11 10 09 91 00
Pressure gauge	70 21 02 00 24

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone - others on request

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



Set Pressure			Measured flow rate at 120 % without gauze		Certified flow rate with the federal Register coefficient 90% without gauze	
Bar	KPa	PSIG	Sm³/s	Scfm	Sm³/s	Scfm
Max (±3% SP; 0,1B/1,45PSI)						
2,18	218	31,6	3,2	6789	2,88	6110
3,3	330	47,8	4,59	9725	4,13	8752
3,75	375	54,4	5,09	10785	4,58	9707
4,4	440	63,8	5,87	12438	5,28	11194

Flow measurement conditions: 101.3 kPa at 15°C
Flow reduction with gauze <12%

SAFETY RELIEF VALVE

3" (ND80) SAFETY RELIEF VALVE
HIGH FLOW FLANGED MEGA SUPERVENTIX
MG80

CODIFICATION

STANDARD PART NUMBER	MG80	S	44	T	21	S	.

YOUR PART NUMBER	MG80			T			

BODY DRILLING - Other on request	
4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection	S
4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection and plug	P

PRESSURE SETTING - Other on request	
No pressure	00
. bar (ex: 4.4bar)	44

SETTING GASKET	
FEP	T

VACUUM SETTING - Other on request	
No Vacuum	.
0.** bar (ex: 0.21 bar)	21

FLAME PROOF GAUZE	
With	G
Without	S

GAUGE PROTECTION	
With	P
Without	.

SAFETY RELIEF VALVE

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It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) SAFETY RELIEF VALVE
HIGH FLOW FLANGED MEGA SUPERVENTIX
MG80

INSTALLATION

STORAGE

Keep the “MEGA SUPERVENTIX” safety relief valve in original packaging, until ready to use.
This prevents ingress of dust and protects against impact damage.

OPERATING

The pressure relief valve is designed to be installed on pressurized tanks static or transportation application. The main function of the relief valve is to quickly evacuate the over pressure of the tank specially in case of fire,the high flow capacity of the valve allow this quick flow relief (refer to specification sheet for flow rate values).

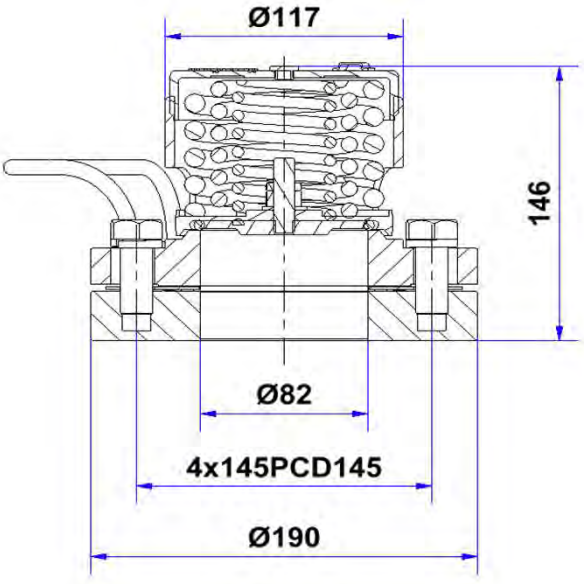
The valve operates as following :
The pressure setting is admised to be 1,1 x tank pressure service. This pressure setting is got by the spring set compression on the pressure plate. When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure plate closes.
In case of high pressure increase, the pressure plate translation is about 20 mm to open a maximum of surface to allow a high flow evacuation.The set pressure range is 0,5 to 4,4 bar. Service temperature -40°C to 190°C. A vacuum relief function can also be added and set from 0,05 to 0,41 bar.

ASSEMBLY

The relief valve should be installed in protected area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.
Make sure the “Mega-SUPERVENTIX” safety relief valve is compatible with the working pressure of the tank. The valve set pre

1. Check there is no dirt or grid on value flange and weld flange.
2. Install the 3" bursting disc (BD) or the gasket (BD must be 3")
3. Install the valve.
4. Install and tighten the screw set. Tighten to a torque of 8 to 10 mdaN.

Check there is no leakage at gasket or bursting disc interface.
The MEGA SUPERVENTIX™ safety relief vent valve is now operational.
CAUTION : Do not remove the wire weld seal between the body and cover to avoid to lose the warranty before first value periodical check.



SAFETY RELIEF VALVE

3" (ND80) SAFETY RELIEF VALVE
HIGH FLOW FLANGED MEGA SUPERVENTIX
MG80

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Check no leak exists between the poppet seal and the seat, this check can be done with leakage detection fluid.
If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further check

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the screws that hold the MEGA-SUPERVENTIX to the tank pad, remove the valve and the disc.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.

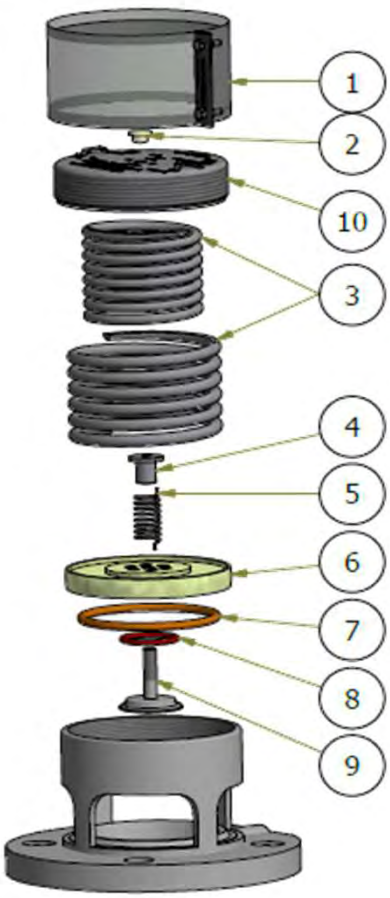
Further valve disassembly should be done in a workshop.

RELIEF VALVE DISASSEMBLY

We recommend this operation every two years.

CAUTION: the springs load is between 100 daN en 400 daN, when cap is removed the springs can violently be ejected.
We recommend to safely disassemble the cap and slowly relief the spring set load.

1. Remove carefully the cap (10) considering the above caution.
2. Remove springs set (3) and pressure plate (6).
3. Remove pressure and vacuum seals (7) & (8)
4. Install a set of new seals
5. Reassemble the valve.



SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5. The leakage is detected with leak test fluid at pressure seal interface.

1. Slowly increase the pressure. **When first bubbles appear at pressure plate seal, note the pressure.**
2. Adjust the cap (screw/unscrew) to get the set pressure.

If set pressure can not be achieved easily, we recommended to change the spring set (3) (see page 3 for item references).

3. When setting complete, seal the cap with a stainless steel wire spot welded between cap and body.

The relief valve is ready to operate on the tank.

SAFETY RELIEF VALVE

3" (ND80) SAFETY RELIEF VALVE
HIGH FLOW FLANGED MEGA SUPERVENTIX MG80

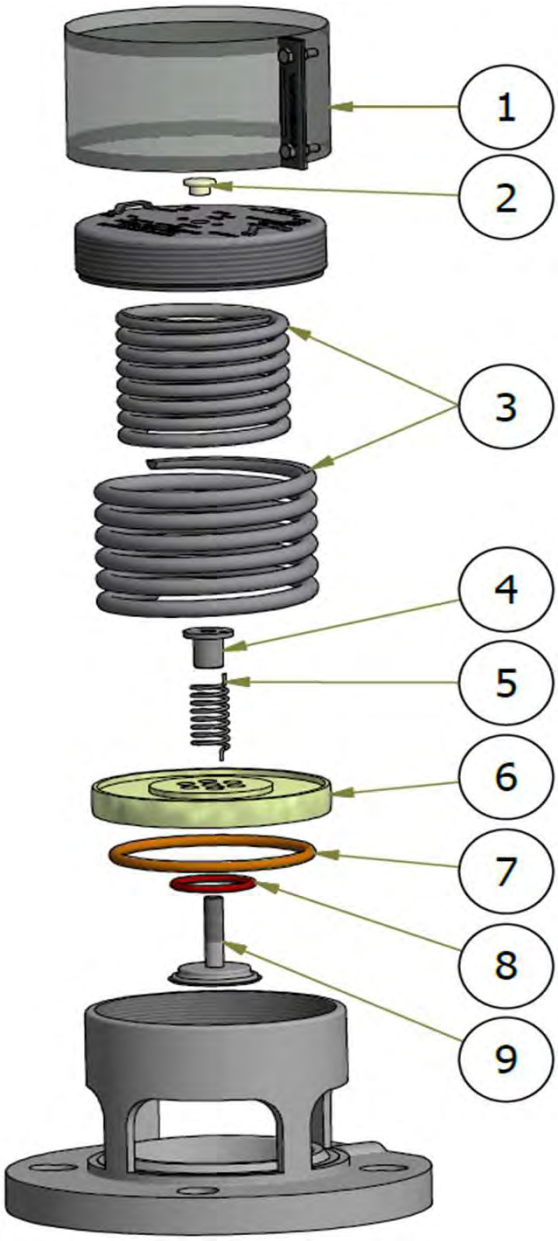
SAFETY RELIEF VALVE

BODY MARKING

1) Setting Pressure (Bar)
2) Setting Pressure (Psi)
3) Full Flow (SCFM)
4) Full Flow (Sm3/S)
5) Full Flow Pressure (Psi)
6) Full Flow Pressure (Bar)
7) Certified Flow 90% (SCFM)
8) Certified Flow 90% (Psi)
9) Vacuum (Bar)
10) Vacuum ("Hg)
11) Serial number
12) P/N (MG80)
13) Nominal diameter (ND80)
14) Pressure tolerance (Bar)
15) Flow section (mm^2)
16) Standard : ISO 4126-1
17) "G" indication fc

3" (ND80) SAFETY RELIEF VALVE
HIGH FLOW FLANGED MEGA SUPERVENTIX
MG80

ITEM	QTY	DESIGNATION	PART N°
		Pressure & Vacuum O'ring:	
7	1	Pressure O'ring (FEP/Silicone)	11 10 17 98 00
7	10	Pressure O'ring (FEP/Silicone)	11 10 17 99 06
7, 8	1	Pres. & Vac. O'ring (FEP/Si)	11 10 17 97 00
8	10	Vacuum O'ring (FEP/Silicone)	11 10 17 99 00
		Spring sets:	
3	1	Spring set pressure 3.30 Bar	11 10 17 50 00
5	1	Spring set vacuum 0.21 bar	11 10 01 98 00
		Plates, Gauze & Cap:	
6	1	Pressure plate for pressure only	11 10 17 99 02
6	1	Pressure plate, pressure/vacuum	11 10 17 99 01
9	1	Vacuum poppet	11 10 01 99 02
4	1	Adjustement nut	11 10 02 99 06
1	1	Flame proof gauze kit	11 10 01 94 00
2	1	Plug (LDPE)	70 14 06 90 15



DESIGNATION	SPECIFICATIONS	PART N°
Weld-in flange	Ext Ø 190mm, int Ø 82mm thickness 26 mm drilling 4 x M16 x 145 PCD 316L stainless steel	1292030000
Recess flange	Ext Ø 217mm, int Ø 82,6mm min thick. 31 mm max th. 65 mm 4 x M16 x 145 PCD 316L stainless steel	1190630000
Screw kit	4 x M16-40 screws 4 x grower washers	1110099100
Bursting disc	pressure setting on request	DRS80P???T20
Setting gasket	CNAF/PTFE Ø176mm Ø82mm Round, flat, drilling 4 x 18 x 145 PCD	1292040000
Gauge (0-6 bar) bottom entry 1/4"BSP	Full stainless steel	7018060095
	Stainless steel case brass internals & connection	7021020022

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE BSP MEGA SUPERVENTIX
MH65F

2 1/2" MEGA SUPERVENTIX™ BSP Threaded High Pressure safety relief valve - designed to protect tanks against accidental overpressure. It is a single acting SRV and suited for pressure ranges going from 8 bar up till 14 bar.

SPECIFICATION	
Nominal diameter	65 mm (2 1/2")
Weight	4,4 kg
Connection	2 1/2" BSP
Pressure settings	8.0 / 11.0 / 14.0 Bar
Vacuum settings	N/A
Working Temperature	-40° to +190°C

*Temperature range may vary according to the seal material

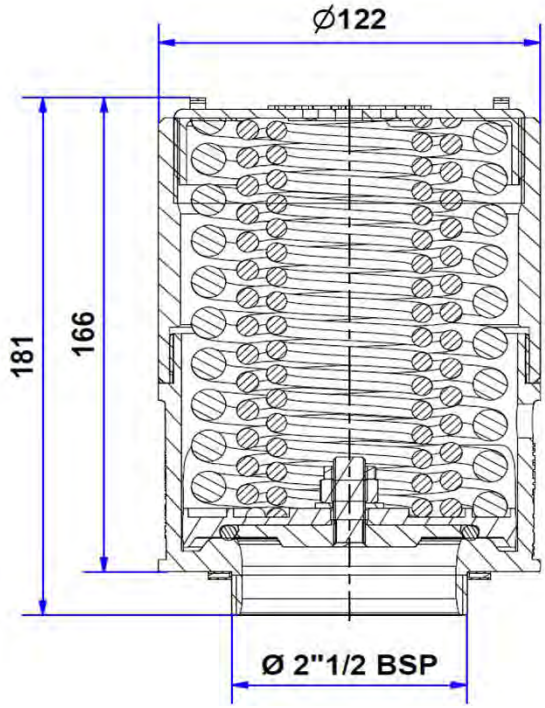
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID / ISO 4126-1
Notified Body : Lloyd's Register Verification

ACCESSORIES	
Weld-in flange	12 91 14 00 00
Recess flange	12 92 14 00 00
Seal	11 10 01 60 00
Bursting disc adaptor flan	11 91 15 00 00
Screw kit	12 12 60 91 00
Pressure gauge 0-10b	70 21 02 00 24

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone - others on request

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



Set Pressure			Measured flow rate at 120 % without gauze		Certified flow rate with the federal Register coefficient	
Bar	KPa	PSIG	Sm³/s	Scfm	Sm³/s	Scfm
Max (±3% SP; 0,1B/1,45PSI)						
8,00	800	116,1	6,76	14323	6,08	12892
11,0	110	159,6	9,16	19409	8,24	17461
14,0	140	203,1	11,56	24494	10,40	22038

Flow measurement conditions: 101.3 kPa at 15°C

Flow reduction with gauze <12%

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE BSP MEGA SUPERVENTIX
MH65F

CODIFICATION

STANDARD PART NUMBER	MH65	F	11	00	T	S	C
YOUR PART NUMBER	MH65	F	__	__	T	__	__

BODY CONNECTION	
2 1/2" BSP	F

PRESSURE SETTING - Other on request	
8 Bar	08
11 Bar	11
14 Bar	14

VACUUM SETTING	
No vacuum	00

SETTING GASKET	
FEP	T

FLAME PROOF GAUZE	
With	G
Without	S

GASKET MATERIAL	
Fluolion	CF
Perbunan/PTFE	P
Composite/PTFE	C

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 49 20 00 E
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE BSP MEGA SUPERVENTIX
MH65F

INSTALLATION

STORAGE

Keep the “MEGA SUPERVENTIX” safety relief valve in original packaging, until ready to use. This prevents ingress of dust and protects against impact damage.

OPERATING

The pressure relief valve is designed to be installed on pressurized tanks static or transportation application. The main function of the relief valve is to quickly evacuate the over pressure of the tank specially in case of fire,the high flow capacity of the valve allow this quick flow relief (refer to specification sheet for flow rate values).

The valve operates as following :
The pressure setting is admised to be 1,1 x tank pressure service. This pressure setting is got by the spring set compression on the pressure plate. When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure plate closes.
In case of high pressure increase, the pressure plate translation is about 20 mm to open a maximum of surface to allow a high flow evacuation.The set pressure range is 8 to 14 bar. Service temperature -40°C to 190°C.

ASSEMBLY

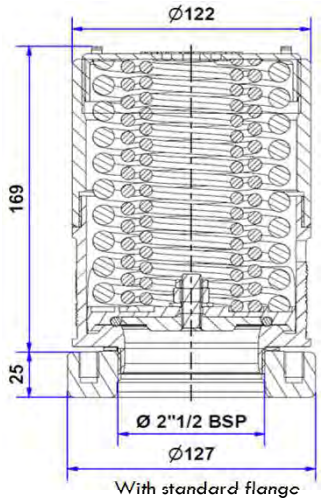
The relief valve should be installed in protected area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.
Make sure the “Mega-SUPERVENTIX” safety relief valve is compatible with the working pressure of the tank.
The valve set pressure is admised to be 1.1 x MAWP (tank pressure)

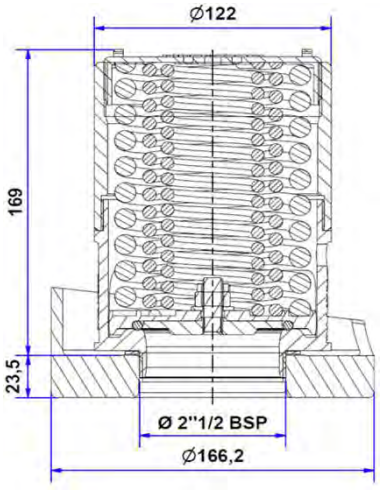
1. Remove the protective cap from the 2.5” BSP thread.
2. Remove the gauze temporarily fitted on the valve.
3. Make sure that no dirt or grit is present on the receiving flange and gasket and the threads are perfectly clean.
4. Manually screw on the valve with gasket. Then keep turning until firmly tight using a 'C' Spanner.
   Tighten to a torque of 20 **daN.m ±5.**
5. Put the gauze back on junction of filter connection (indicated “ JUNCTION GAUZE”). Tighten the filter bolts
(use 7mm open end spanner or socket).

Check there is no leakage at gasket or bursting disc interface.
The valve is ready to operate.

CAUTION : Do not remove the wire weld seal between the body and cover to avoid to lose the warranty before first value periodical check.

The MEGA SUPERVENTIX™ safety relief vent valve is now operational.
The valve can be sealed to the tank using the Tir tabs located on the valve cover.





SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE BSP MEGA SUPERVENTIX
MH65F

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Check no leak exists between the poppet seal and the seat, this check can be done with leakage detection fluid.
If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further check

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the screws that hold the MEGA-SUPERVENTIX to the tank pad, remove the valve and the disc.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.

Further valve disassembly should be done in a workshop.

RELIEF VALVE DISASSEMBLY

We recommend this operation every two years.

CAUTION: the springs load is between 100 daN en 700 daN, when cap is removed the springs can violently be ejected.

We recommend to safely disassemble the cap and slowly relief the spring set load.

1. Remove carefully the cap considering the above caution.
2. Remove springs set (2) and pressure plate (6)
3. Remove pressure seal (7)
4. Install a new seal
5. Reassemble the valve.

RELIEF VALVE ASSEMBLY

1. Fit the new O'ring (7) around the poppet.
2. Fit the poppet (8) into the pressure plate (6).
3. Apply threadlocker (as loctite) and screw the two nuts.
4. Compress the springs (2) and screw the top cap.

1

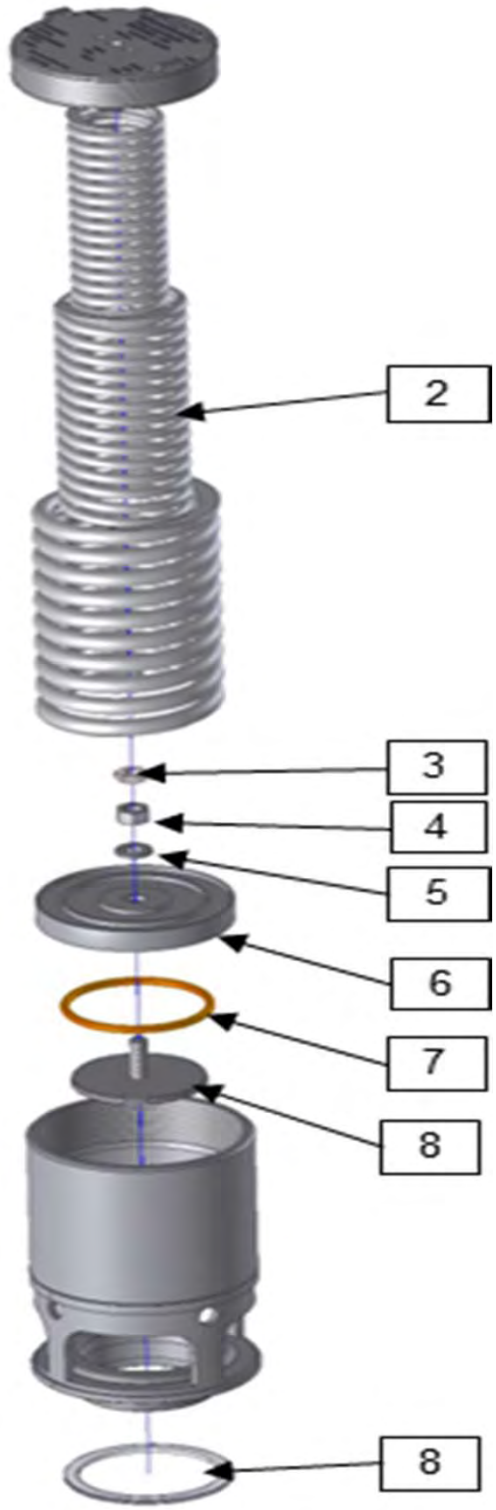
2

3

4

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE BSP MEGA SUPERVENTIX
MH65F



ITEM	QTY	DESIGNATION	PART N°
		Spring sets:	
2	1	Spring set pressure 9 Bar	11 10 49 00 02
2	1	Spring set pressure 11 Bar	11 10 49 00 03
2	1	Spring set pressure 14 Bar	11 10 49 00 08
		O'ring:	
7	1	Pressure O'ring (FEP/Silicone)	11 10 01 99 05

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE FLANGED MEGA SUPERVENTIX
MH65S

2 1/2" MEGA SUPERVENTIX™ flanged High Pressure safety relief valve - designed to protect tanks against accidental overpressure. It is a single acting SRV and suited for pressure ranges going from 8 bar up till 14 bar.

SPECIFICATION	
Nominal diameter	65 mm (2 1/2")
Weight	10 kg
Connection	4 slotted holes x Ø18 on a 145 to 152.4 PCD
Pressure settings	8.0 / 11.0 / 14.0 Bar
Vacuum settings	N/A
Working Temperature	-40° to +190°C

*Temperature range may vary according to the seal material

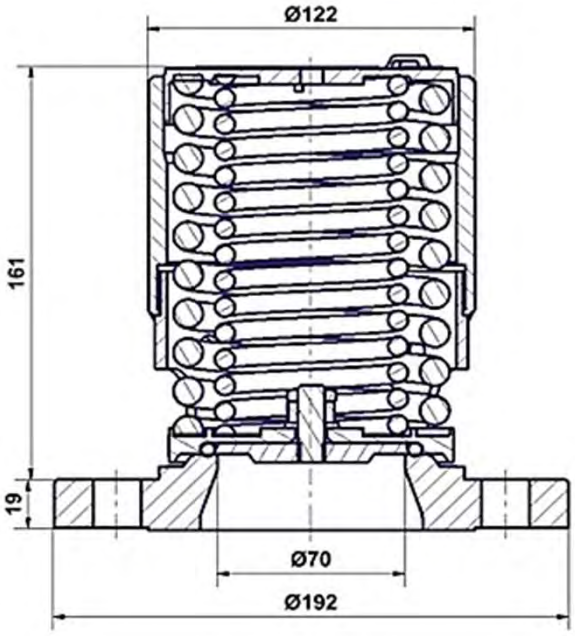
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID / ISO 4126-1
Notified Body : Lloyd's Register Verification

ACCESSORIES	
Weld-in flange	12 92 03 00 00
Recess flange	11 90 63 00 00
Seal	12 92 04 00 00
Pressure gauge 0-10b	70 21 02 00 24

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone - others on request

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



Set Pressure			Measured flow rate at 120 % without gauze		Certified flow rate with the federal Register coefficient	
Bar	KPa	PSIG	Sm³/s	Scfm	Sm³/s	Scfm
Max (±3% SP; 0,1B/1,45PSI)						
8,00	800	116,1	6,76	14323	6,08	12892
11,0	110	159,6	9,16	19409	8,24	17461
14,0	140	203,1	11,56	24494	10,40	22038

Flow measurement conditions: 101.3 kPa at 15°C

Flow reduction with gauze <12%

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE FLANGED MEGA SUPERVENTIX
MH65S

CODIFICATION

STANDARD PART NUMBER	MH65	S	11	00	T	S	S

YOUR PART NUMBER	MH65	S	__	__	T	__	__

BODY DRILLING - Other on request

4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection	S

PRESSURE SETTING - Other on request

8 Bar	08
11 Bar	11
14 Bar	14

VACUUM SETTING

No vacuum	00

SETTING GASKET

FEP	T

FLAME PROOF GAUZE

With	G
Without	S

GAUGE PROTECTION

With	P
Without	S

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 54 20 00 E
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PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE FLANGED MEGA SUPERVENTIX
MH65S

INSTALLATION

STORAGE

Keep the “MEGA SUPERVENTIX” safety relief valve in original packaging, until ready to use. This prevents ingress of dust and protects against impact damage.

OPERATING

The pressure relief valve is designed to be installed on pressurized tanks static or transportation application. The main function of the relief valve is to quickly evacuate the over pressure of the tank specially in case of fire,the high flow capacity of the valve allow this quick flow relief (refer to specification sheet for flow rate values).

The valve operates as following :
The pressure setting is admised to be 1,1 x tank pressure service. This pressure setting is got by the spring set compression on the pressure plate. When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure plate closes.
In case of high pressure increase, the pressure plate translation is about 20 mm to open a maximum of surface to allow a high flow evacuation.The set pressure range is 8 to 14 bar. Service temperature -40°C to 190°C.

ASSEMBLY

The relief valve should be installed in protected area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.

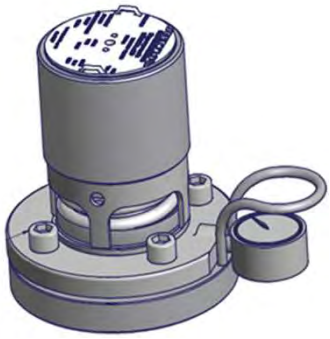
Make sure the “Mega-SUPERVENTIX” safety relief valve is compatible with the working pressure of the tank.
The valve set pressure is admised to be 1.1 x MAWP.

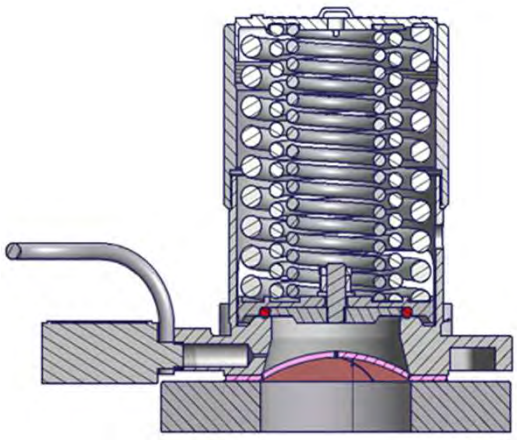
1. Check there is no dirt or grid on value flange and weld flange.
2. Install the 3" bursting disc (BD) or the gasket.
CAUTION: Bursting disc must be 3"
3. Install the valve.
4. Install and tighten the screw set.
   Tighten to a torque of **8 to 10 mdaN.**

Check there is no leakage at gasket or bursting disc interface.
The valve is ready to operate.
CAUTION : Do not remove the wire weld seal between the body and cover to avoid to lose the warranty before first value periodical check.

The MEGA SUPERVENTIX™ safety relief vent valve is now operational.

Description	Part Number
Screw kit	11 10 09 91 00
Weld-in flange (316L/1.4404)	12 92 03 00 00
Bursting disc (316L/1.4404)	DRR80P1210T20





Bursting disc assembly

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE FLANGED MEGA SUPERVENTIX
MH65S

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Check no leak exists between the poppet seal and the seat, this check can be done with leakage detection fluid.
If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further check

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the screws that hold the MEGA-SUPERVENTIX to the tank pad, remove the valve and the disc.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.

Further valve disassembly should be done in a workshop.

RELIEF VALVE DISASSEMBLY

We recommend this operation every two years.

CAUTION: the springs load is between 100 daN en 700 daN, when cap is removed the springs can violently be ejected.

We recommend to safely disassemble the cap and slowly relief the spring set load.

1. Remove carefully the cap (2) considering the above caution.
2. Remove springs set (3) and pressure plate (7)
3. Remove pressure seal (8)
4. Install a new seal
5. Reassemble the valve.

RELIEF VALVE ASSEMBLY

1. Fit the new O'ring (8) around the poppet.
2. Fit the poppet (9) into the pressure plate (7).
3. Apply threadlocker (as loctite) and screw the two nuts.
4. Compress the springs (3) and screw the top cap (2).

1

2

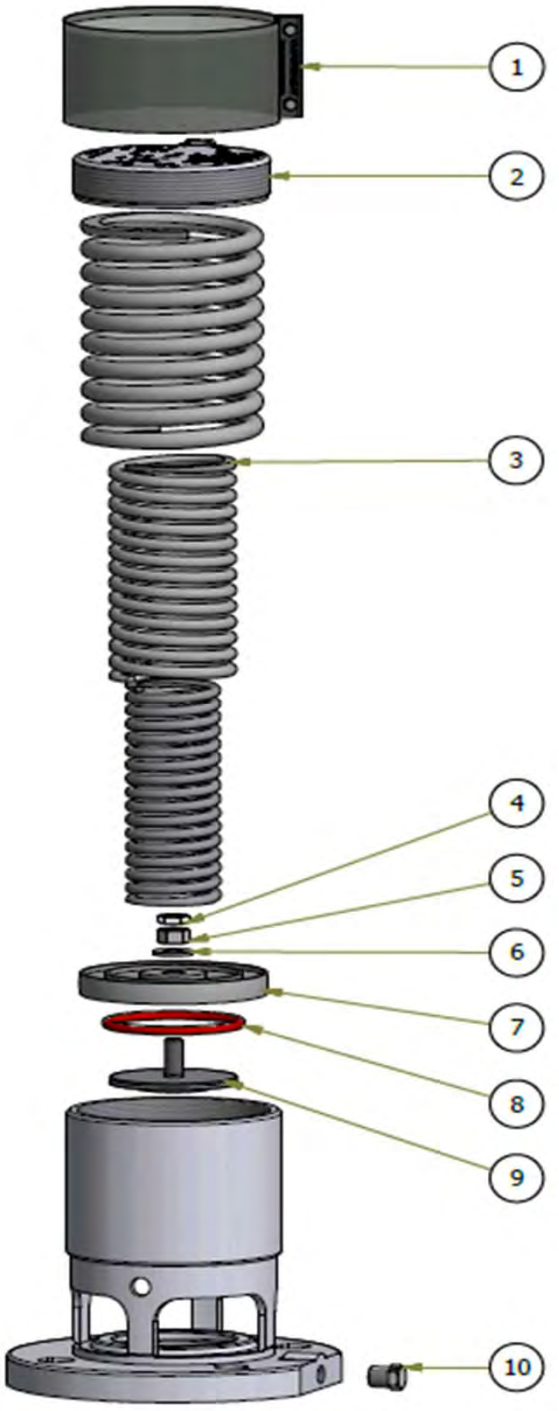
3

4

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
HIGH PRESSURE FLANGED MEGA SUPERVENTIX
MH65S

ITEM	QTY	DESIGNATION	PART N°
		Spring sets:	
3	1	Spring set pressure 9 Bar	11 10 49 91 02
3	1	Spring set pressure 11 Bar	11 10 49 91 03
3	1	Spring set pressure 14 Bar	11 10 49 00 08
		Plate, O'ring & Gauze:	
-	1	Pressure plate kit (Nr 4, 5, 6, 7, 8, 9)	11 10 49 91 06
8	1	Pressure O'ring (FEP/Silicone)	11 10 01 91 00
1	1	Flame proof gauze kit	11 10 01 50 91
1	10	Flame proof gauze kit	11 10 01 50 95



SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
DIN 11851 MEGA SUPERVENTIX MX65M/N & MP65M/N(PED)

The Perolo 2 ½" MEGA SUPERVENTIX™ hygienic connection is designed to protect tanks against accidental overpressure or excessive vacuum. With a compact design, this SRV has a high flow performance. Made of 316/1.4408 stainless steel, this product is suitable for ISO tank containers, IBC's, road tankers and static tanks dedicated to the transport & storage of liquid foodstuff.

SPECIFICATION	
Nominal diameter	65 mm (2 1/2")
Weight	3 kg
Connection	2 1/2" DIN 11851 (ND65 & ND80)
Pressure settings	From 0.5 Bar to 7.4 Bar
Vacuum settings	From 30 to 410 mBar
Working Temperature	-40° to +190°C

*Temperature range may vary according to the seal material

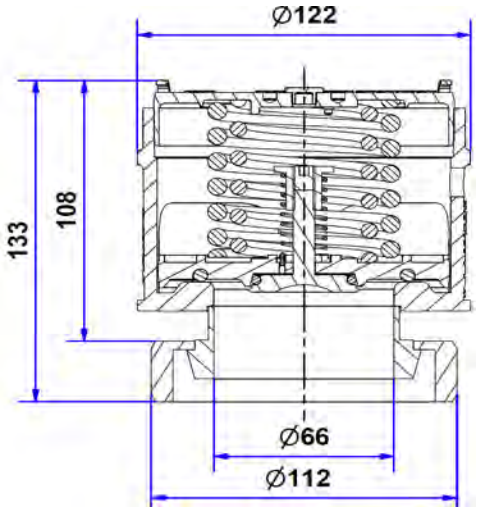
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID / ISO 4126-1
Design Approval: Lloyd's Register - RMR
PED (for MP65 only)
PED 2014/68/EU - Category IV - Module H1 - Fluid gr.1
Lloyd's Register Espana

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone - others on request

ACCESSORIES	
Pressure gauge	70 18 06 00 95

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



DIN 11851 2 1/2"

Flow measurement conditions: 101.3 kPa at 15°C

Flow reduction with gauze <3%

Set pressure			FLOW RATE AT 120% MX65				FLOW RATE AT 110% MX65			
			Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze		Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze	
Bar	kPa	PSIG	Sm³/S	Scfm	Sm³/S	Scfm	Sm³/S	Scfm	Sm³/S	Scfm
Max (±3% SP; 0,1B/1,45PSI)										
2.18	218	31.6	2.56	5425	2.30	4883	1.98	4205	1.78	3785
3.75	375	54.4	3.92	8306	3.53	7475	3.34	7081	3.01	6373
4.4	440	63.8	4.48	9492	4.03	8543	3.9	7356	3.51	6620
7.3	730	105.9	6.99	14810	6.29	13329	6.4	13584	5.76	12226

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE
DIN 11851 MEGA SUPERVENTIX
MX65M/N & MP65M/N(PED)

CODIFICATION

STANDARD PART NUMBER	Standard: MX65 PED: MP65	M	20	T	10	G	.	.
YOUR PART NUMBER	MX65/MP65							

BODY CONNECTION	
DIN 11851 ND65	M
DIN 11851 ND80	N

PRESSURE SETTING - Other on request	
No pressure	00
. bar (ex: 2.bar)	20

SETTING GASKET	
FEP	T
Foodstuff Nitrile	A

VACUUM SETTING - Other on request	
No Vacuum	.
0.** bar (ex: 0.10bar)	10

FLAME PROOF GAUZE	
With	G
Without	S

ELECTROPOLISHED VALVE	
With	E
Without	.

VACUUM TEST BUTTON	
With	B
Without	.

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 44 20 00 G
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" (ND65) SAFETY RELIEF VALVE
DIN 11851 MEGA SUPERVENTIX
MX65M/N & MP65M/N(PED)

SAFETY RELIEF VALVE

INSTALLATION

STORAGE

Keep the “MEGA SUPERVENTIX” safety relief valve in original packaging, until ready to use. This prevents ingress of dust and protects against impact damage.

OPERATING

The pressure relief valve (PRV) is designed to be installed on pressurized tanks static or transportation application. The main function of the relief valve is to quickly evacuate the over pressure of the tank specially in case of fire, the high flow capacity of the valve allow this quick flow relief (refer to specification sheet for flow rate values).

Check the compatibility between safety relief valve material and fluid before operating.

The valve operates as following :

The pressure setting is admised to be 1,1 x tank pressure service. This pressure setting is atteined by the spring set compression on the pressure plate. When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuatedd, then where the pressure balance is reached the pressure plate closes.

In case of high pressure increase, the pressure plate translation is about 23 mm to open a maximum of surface to allow a high flow evacuation.The set pressure range is 0,1 to 7,4 bar. Service temperature -40°C to 190°C. A vacuum relief function can also be added and set from 30 to 410 mbar.

ASSEMBLY

Before assembly check the PRV set pressure is compatible with tank MAWP and the carried cargo compatible with PRV.

The relief valve should be installed in protected area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.

Make sure the “Mega-SUPERVENTIX” safety relief valve is compatible with the working pressure of the tank.

The valve set pressure is admised to be 1.1 x MAWP.

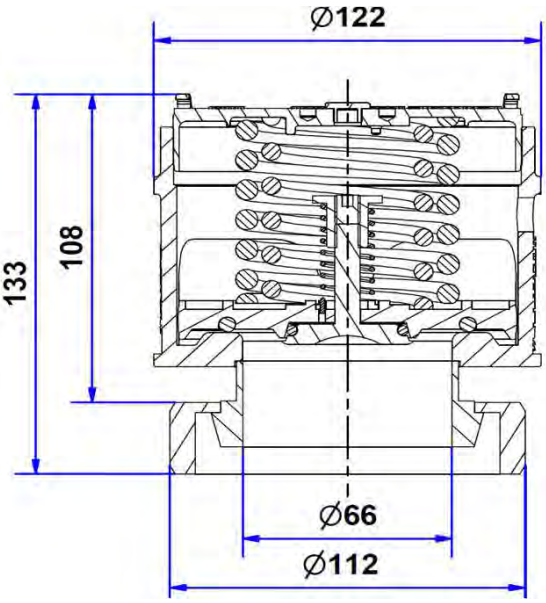
1. Remove the protective cap.
2. Make sure that no dirt or grit is present on the receiving
4. Manually screw on the valve bolt.

Then keep turning until firmly tight using a 'C' Spanner. The tightening torque required is **70 to 130 N.m ±0,5**

CAUTION : Do not remove the wire weld seal between the body and cover to avoid to lose the warranty before first value periodical check.

The MEGA SUPERVENTIX ™ safety relief vent valve is now operational.

The valve can be sealed to the tank using the wiring clamps located on the valve cover.



2 1/2" (ND65) SAFETY RELIEF VALVE
DIN 11851 MEGA SUPERVENTIX
MX65M/N & MP65M/N(PED)

SAFETY RELIEF VALVE

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Check no leak exists between the poppet seal and the seat, this check can be done with leakage detection fluid.
If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further check

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the screws that hold the MEGA-SUPERVENTIX to the tank pad, remove the valve and the disc.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.

Further valve disassembly should be done in a workshop.

RELIEF VALVE DISASSEMBLY

We recommend this operation every two years.

CAUTION: the springs load is between 100 daN en 400 daN, when cap is removed the springs can violently be ejected.

We recommend to safely disassemble the cap and slowly relief the spring set load.

1. Remove carefully the cap (11) considering the above caution.
2. Remove springs set (3) and pressure plate (6).
3. Remove pressure and vacuum seals (7) & (8)
4. Install a set of new seals
5. Reassemble the valve.

SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5. The leakage is detected with leak test fluid at pressure seal interface.

1. Slowly increase the pressure. **When first bubbles appear at pressure plate seal, note the pressure.**
2. Adjust the cap (screw/unscrew) to get the set pressure.

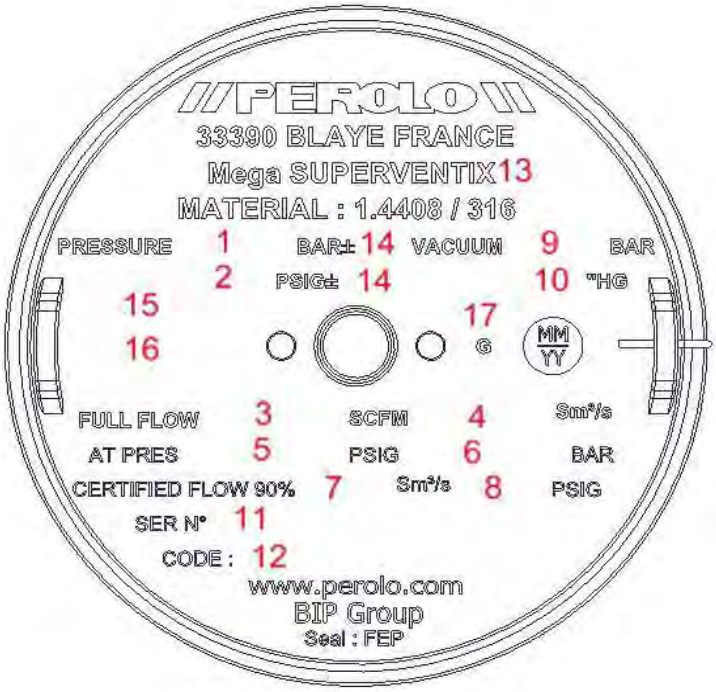
If set pressure can not be achieved easily, we recommend to change the spring set (3) (see page 3 for item references). Only PEROLO provided spring sets should be installed

3. When setting complete, seal the cap with a stainless steel wire spot welded between cap and body.

The relief valve is ready to operate on the tank.

2 1/2" (ND65) SAFETY RELIEF VALVE
DIN 11851 MEGA SUPERVENTIX
MX65M/N & MP65M/N(PED)

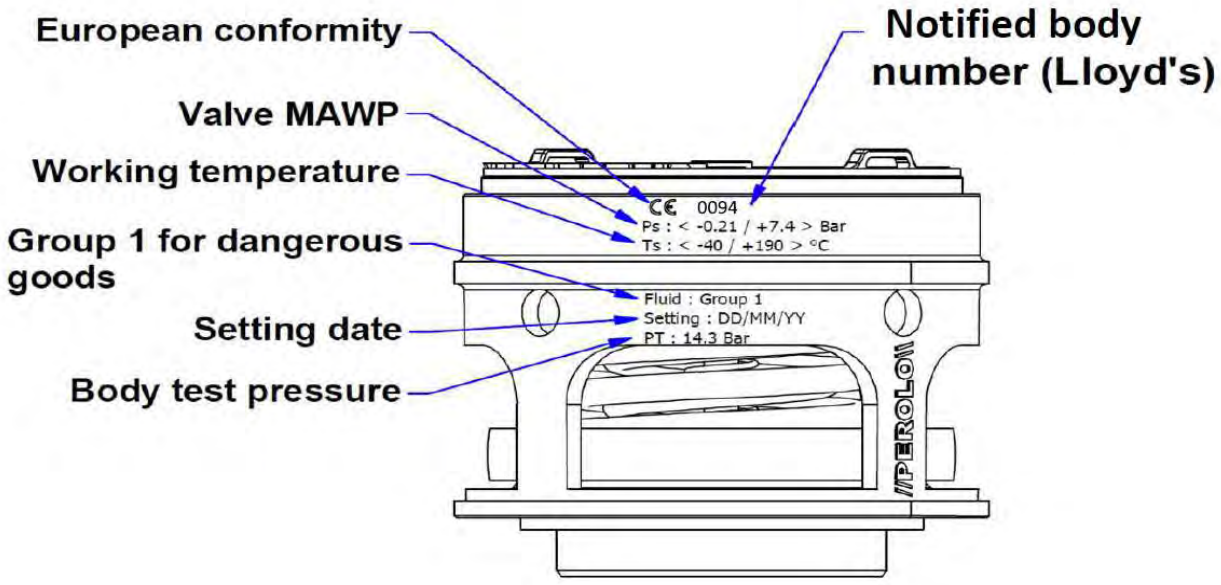
BODY MARKING (For MX65M & MP65M and MX65N & MP65N)



1) Setting Pressure (Bar)
2) Setting Pressure (Psi)
3) Full Flow (SCFM)
4) Full Flow (Sm3/S)
5) Full Flow Pressure (Psi)
6) Full Flow Pressure (Bar)
7) Certified Flow 90% (SCFM)
8) Certified Flow 90% (Psi)
9) Vacuum (Bar)
10) Vacuum ("Hg)
11) Serial number
12) P/N (MX65M or MP65M)
13) Nominal diameter (ND65)
14) Pressure tolerance (Bar)
15) Flow section (mm²)
16) Standard : ISO 4126-1
17) "G" indication for gaz

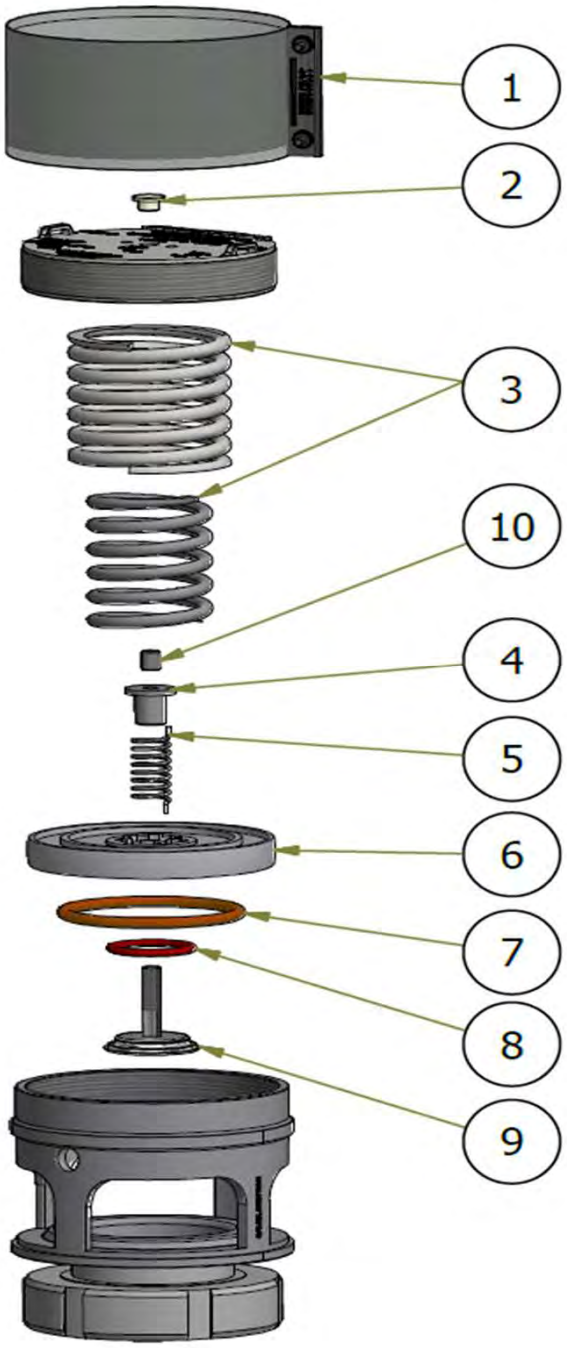
PED ADDITIONNAL BODY MARKING (For MP65M & MP65N Only)

Body marking is only made on PED approved safety relief valve (when reference starts with MP)



2 1/2" (ND65) SAFETY RELIEF VALVE
DIN 11851 MEGA SUPERVENTIX
MX65M/N & MP65M/N(PED)

ITEM	QTY	DESIGNATION	PART N°
		Pressure & Vacuum O'ring:	
7	1	Pressure O'ring (FEP/Silicone)	11 10 01 99 05
7	10	Pressure O'ring (FEP/Silicone)	11 10 01 99 06
7, 8	1	Pres. & Vac. O'ring (FEP/Si)	11 10 01 91 00
8	10	Vacuum O'ring (FEP/Silicone)	11 10 17 99 00
		Spring sets:	
3	1	Spring set pressure 2.18 Bar	11 10 01 95 00
3	1	Spring set pressure 3.30 Bar	12 12 84 90 00
3	1	Spring set pressure 3.75 Bar	11 10 01 96 00
3	1	Spring set pressure 4.40 Bar	11 10 01 97 00
5	1	Spring set vacuum 0.21 bar	11 10 01 98 00
		Plates, Gauze & Cap:	
6	1	Pressure plate for pressure only	11 10 01 99 01
6	1	Pressure plate, pressure/vacuum	11 10 01 99 00
9	1	Vacuum poppet	11 10 01 99 02
2, 4, 5, 6, 7, 8, 9	1	Vacuum set kit (plug, adjustment nut, pressure plate, o'rings, spring set and vacuum poppet)	11 10 01 99 08
4	1	Adjustement nut	11 10 02 99 06
1	1	Flame proof gauze kit	11 10 01 50 91
1	10	Flame proof gauze kit	11 10 01 50 95
2	1	Plug (LDPE)	70 14 06 90 15



SAFETY RELIEF VALVE

2" (DN50) H2O2 Breather

HYDROGEN PEROXIDE H2O2 BREATHER

2" H2O2 breather - used on Hydrogen Peroxide tanks to vent the tank during transport - This product is composed by 2 sintered discs.
One in the body acts as a filter for the compressed air when pressurizing the tank in unloading operations. In case of tank rollover, the liquid flow rates goes through the second disc. The second disc limits the water flow rate.
The high filtering capability of the discs limit the liquid flow rate at 1 ltr/h max at 0.2 Bar pressure in the tank.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	3 kg
Connection	6 holes x Ø11 x 105PCD 4 holes x Ø18 x 125PCD
Working Pressure	4 Bar
Test Pressure	6 Bar
Working Temperature	-40° to +190°C

Temperature range may vary according to the seal material

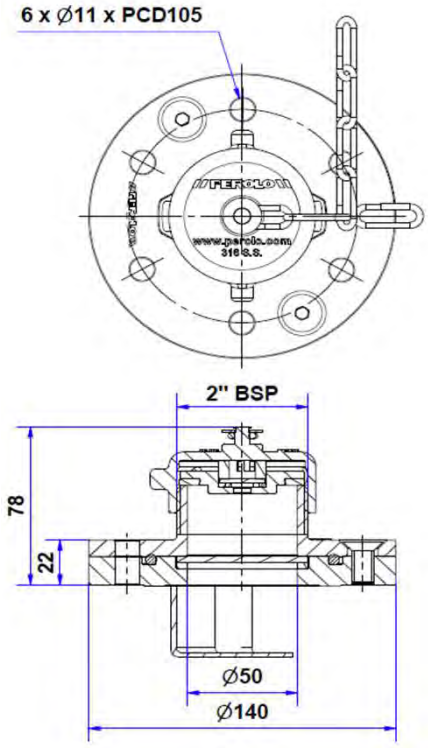
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID
Notified Body : Lloyd's Register Verification - TUV

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Setting Gasket	PTFE & FEP/Silicone

ACCESSORIES	
Weld-in flange 6xM10x105	11 91 35 00 00
Weld-in flange 4xM16x125	11 92 45 00 01
Stud kit 6 x M10 studs&nuts	11 91 35 91 00
Screw kit (4 x M16 screws)	17 12 10 99 03
Seal (6x11x105)	11 10 01 60 00
Seal (4x18x125)	11 90 29 00 00
Low flow filter	12 93 11 60 00
High flow filter	12 93 11 60 00B
2" BSP Cap with chain	12 93 11 21 00
Sintered disc kit	11 15 13 93 00

DIMENSIONAL DRAWING

1115130000



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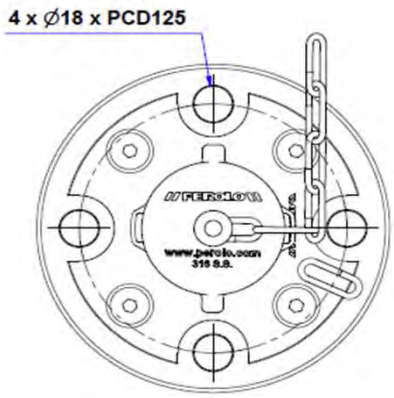


SAFETY RELIEF VALVE

2" (DN50) H2O2 Breather
HYDROGEN PEROXIDE H2O2 BREATHER

INSTALLATION

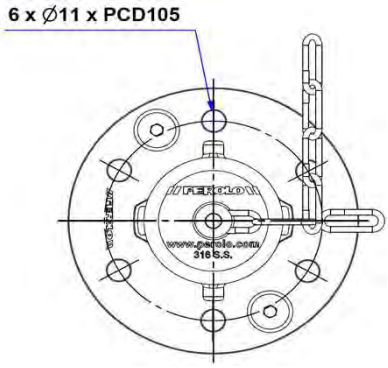
WORKING CONDITIONS

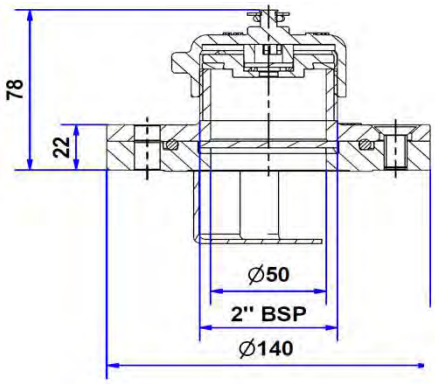
During service, the cap including disc is installed on the body.
The tank breathes through disc (a) and disc (b), external particles are filtered by the 2 discs, consequently can not go into tank. During tank unloading under pressure: the cap including disc is removed from the body.
Compressed air flows through disc is filtered at this stage.In case of tank rollover, the liquid flow rate goes through disc (a) and disc (b). The high filtering capability of thediscs limits the liquid flow rate to 1 liter per hour at 0.2 bar pressure in the tank.

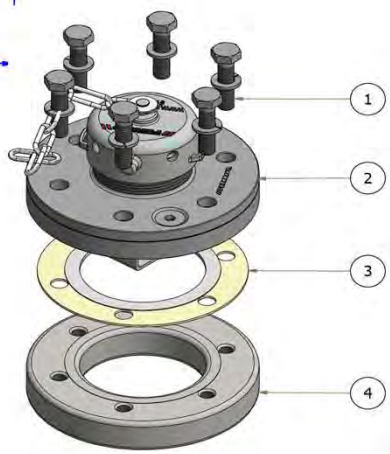
INSTALLATION ON THE TANK

The breather is installed on a weld-in flange previously welded by the customer. This pad should be welded "out of axis" from the tank center line.

1. Check there is no dirt or grit on both weld-in flange and breather pad
2. Screw the 6 studs
3. Install the seal
4. Instal the breather, the "wave breaker" opening perpendicular to tank center line.
5. Install the 6 washers and nuts
6. Tighten to a torque between 2 and 3 m.daN
7. Tighten the 2 screws which get free when the 6 nuts have been tightened.







ITEM	DESCRIPTION	SPECIFICATION	PART NUMBER
2	H2O2 Breather	low flow rate - drilling: 6 holes x Ø11 x 105 PCD	11 15 13 00 00
3	Seal	PTFE lined, Round, flat, drilling 6 x 1 1 x 105PCD	11 10 01 60 00
4	Weld-in flange	int dia 76mm, ext dia 140mm, 316L st st, Thickness 22mm, drilling 6 x M10 / PCD105	11 91 35 00 00
1	Screw kit	6 x M10-40 screws & 6 x washers	11 15 13 92 00
Stud kit - not on drawing		6 x M10-35 studs, 6 x nuts & 6 x washers, 316L/1.4404	11 91 35 91 00

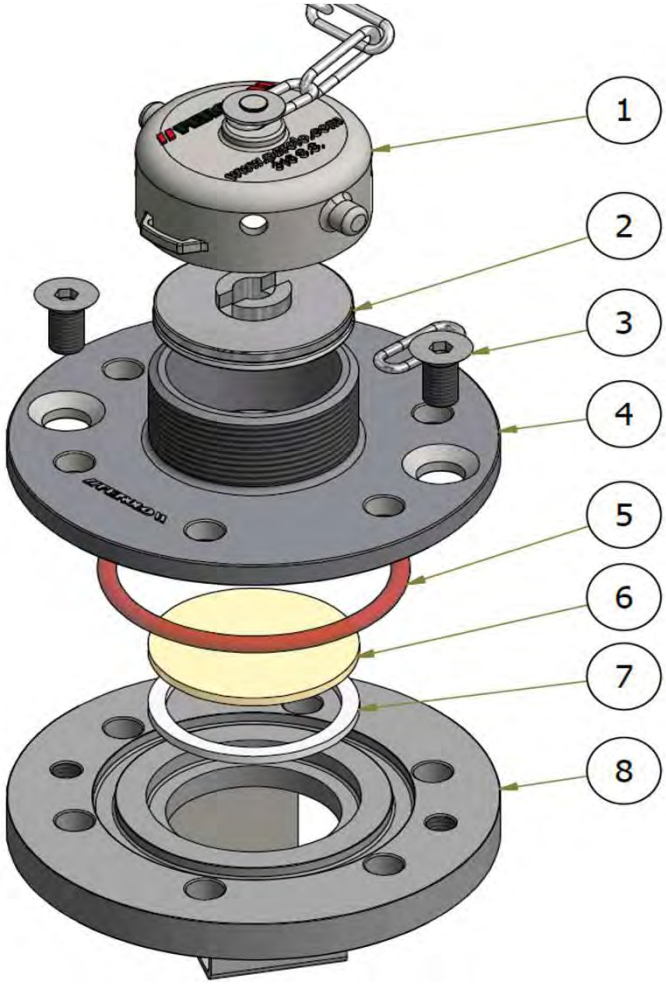
SAFETY RELIEF VALVE

2" (DN50) H2O2 Breather
HYDROGEN PEROXIDE H2O2 BREATHER

MAINTENANCE

H2O2 Breather: 11 15 13 00 00			
ITEM	QTY	DESIGNATION	PART N°
		Spring sets:	
1	1	Cap	12 93 11 21 00
2	1	High flow filter kit, st st + PTFE	12 93 11 60 00 B
2	1	Low flow filter kit, st st + PTFE	12 93 11 60 00
6	1	Sintered disc, 316L/1.4404	11 15 13 93 00
7	1	Sintered disc seal, PTFE	
3	1	2 x M10-20 screws, 304/1.4301	
5	1	O'ring, FEP/Silicon	

H2O2 Breather: 11 18 37 00 00			
ITEM	QTY	DESIGNATION	PART N°
		Spring sets:	
1	1	Cap	12 93 11 21 00
2	1	High flow filter kit, st st + PTFE	12 93 11 60 00 B
2	1	Low flow filter kit, st st + PTFE	12 93 11 60 00
6	1	Sintered disc, 316L/1.4404	11 15 13 93 00
7	1	Sintered disc seal, PTFE	
3	1	2 x M10-20 screws, 304/1.4301	
5	1	O'ring, FEP/Silicon	



SAFETY RELIEF VALVE

1" (ND25) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND/OR VACUUM SRV
YAK VXS2M

1" YAK VX safety relief valve - designed to protect the tank from any over-pressure or any vacuum excess. Made of 316 / 1.4408 / 1.4404 stainless steel, this small product is suitable for IBC tankers dedicated to the transport and storage of chemicals and liquid foodstuff.

SPECIFICATION	
Nominal diameter	25 mm (1")
Weight	250 g
Connection	1" BSP or NPT
Pressure settings	from 0.07 to 2.0 Bar
Vacuum settings	from 15 to 300 mBar
Working Temperature	-40° to +200°C*

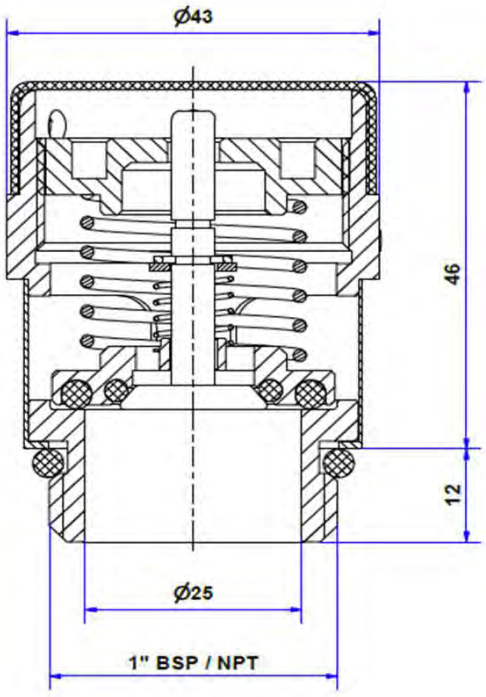
*Temperature range may vary according to the seal material

CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID

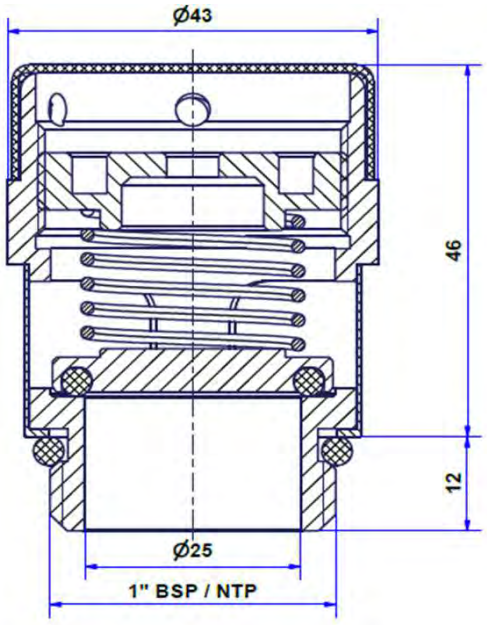
ACCESSORIES	
Weld-in flange	On request
Gauze	On request

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
O'rings material	FEP silicone / Viton / FFKM / EPDM
Push button	Always installed with vacuum option

DIMENSIONAL DRAWING



Pressure & vacuum



Pressure only

Set pressure tolerance: Max (±3%; 0,1b/1,45 PSI)

SAFETY RELIEF VALVE

1" (ND25) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND/OR VACUUM SRV
YAK VXS2M

CODIFICATION

STANDARD PART NUMBER	VXS2	M	B	D	03	30	F	G
YOUR PART NUMBER	VXS2	M						

OUTLET CONNECTION

Male threaded outlet	M

TYPE OF THREAD

1" BSP	B
1" NPT	N

PRESSURE & VACUUM SETTING

Pressure & Vacuum (with test button)	D
Pressure Only (without test button)	P
Vacuum Only (with test button)	V

PRESSURE SETTING - Other on request

No Pressure	00
. bar (ex: 0.3bar)	03

VACUUM SETTING - Other on request

No Vacuum	00
0.** bar (ex: 0,30bar)	30

O'RING & SETTING GASKET MATERIAL

Setting gasket: FKM/Viton Pressure O'ring: FKM/Viton Tcoat Vacuum O'ring: FKM/Viton Tcoat WT -20°C to +200°C	V
Setting gasket: FEP/Silicone Pressure O'ring: FEP/Silicone Vacuum O'ring: FFKM/Kalrez WT -15°C (P&V) or -40°C (P only) to +200°C	K
Setting gasket: FEP/Silicone Pressure O'ring: FEP/Silicone WT -40°C to +200°C (P only)	F
Setting gasket: EPDM Pressure O'ring: EPDM Vacuum O'ring: EPDM WT -40°C to +150°C	E

FLAME PROOF GAUZE

With	G
Without	0

SAFETY RELIEF VALVE

1" (ND25) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND/OR VACUUM SRV
YAK VXS2M

INSTALLATION

STORAGE

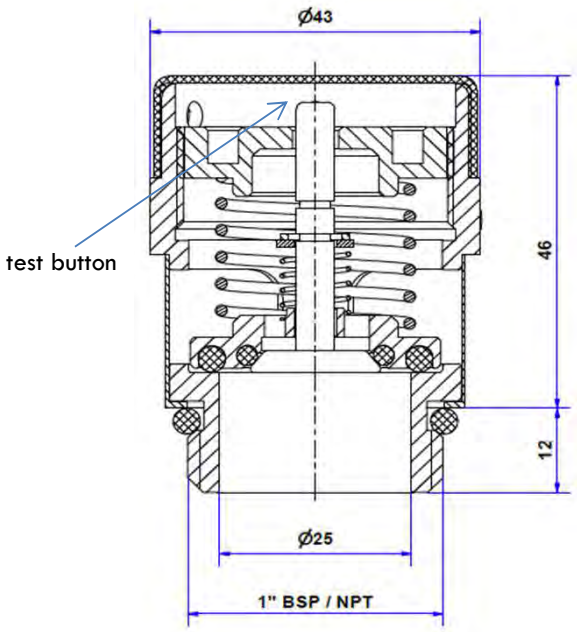
Keep the "YAK VXS2" safety relief valve in original packaging, until ready to use. It will prevent the valve from dust and will protect it againts potential impacts. The weldin flange must be machined BSP or NPT as per Safety Relief Valve.

OPERATING

The pressure relief valve is designed to be installed on pressurized tanks up to maximum allowable working pressure (MAWP) of 4 Bar. The material used are 1.4409 & 1.4404 316L stainless steel. Before installing the valve, always check the fluid compatibility. The YAK VX™ 1" safety relief valve is designed to protect tanks against accidental over-pressure and/or excessive vacuum depending on the chosen options. The vacuum setting range is from -0,015 Bar to -0,3 Bar and the pressure setting range is from 0,07 Bar to 2 Bar.

This pressure or vacuum setting is obtained by compression of the spring set on the pressure plate. When the set pressure is reached, the pressure poppet opens and allows the over pressure to be evacuated, then when the pressure balance is reached the pressure poppet closes. In case of high pressure increase, the pressure poppet fully opens to allow a high flow evacuation.
When a vacuum is created into the tank, the vacuum poppet slides down, opens and allows air to come into the tank avoiding tank implosion.

On valve with vacuum, press the test button top part of the vacuum poppet on the figure to evacuate air pressure or air vacuum inside the tank.



ASSEMBLY

The relief valve should be installed in a protected area to avoid operators to be in contact with the fluid when overpressure and opening of the valve.
Before installing the relief valve, Ensure that the Inlet flange and valve are clean; free from dirt and grit.
Place the valve in 1" BSP tank pad and screw it manually until seal and flange are in contact (connection depending on the valve model). Tighten to a torque equal to 40 N.m with a spanner of 40mm (remove the protection cap if necessary and do not forget to replace it afterwards)
Check that the o'ring is correctly compressed, no leak should occur between flange and gasket when tank is pressurized.
Seal the YAK to the flange by welding a steel wire. The cable goes through the holes of the yak and the proctection cap.

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 17 10 42 30 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1" (ND25) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND/OR VACUUM SRV
YAK VXS2M

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Make sure tank pressure is lower than service pressure.
2. Check if there is any leak between the poppet seal and the seat using leak test fluid at pressure seal interface

If a leak exists, follow next points to remove dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

Make sure the tank is without pressure

1. Unscrew the YAK VX from the flange using a spanner of 40mm, and remove the valve.
2. Plug the tank pad with a blind cap to protect the tank from dust.

Further valve disassemble should be done in a workshop.

RELIEF VALVE DISASSEMBLY

We recommend this operation every two years it must be done in a workshop

CAUTION : the springs load can achieve 15 daN, so that when cap is removed the springs can violently be ejected.

1. Remove protection cap (5)
2. Remove the welded sealing on top of the coverlid with pliers
3. Unscrew the coverlid (4) with a pliers and remove the pressure spring (13)
4. Remove the pressure/ vacuum assembly (7+8+9+10+11+12+3)
5. Remove the clips (3) from vacuum assembly
6. Remove the washer (9) and the vacuum spring (12) and vacuum poppet (8)
7. Remove the pressure poppet seal (10) and vacuum poppet seal (11)

(use a sharp pointed instrument. Be careful not to damage the sealing face)

CAUTION : cap must be installed when valve is fitted on the tank.

SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5.

The leakage is detected with leak test fluid at pressure seal interface.

1. Make sure the valve is properly reassembled with the coverlid correctly screwed into the body
2. For the first pressure setting test, screw the coverlid just below the 4 holes on the body of the YAK. Then screw or unscrew the cap until the suitable setting pressure. (NB make sure the holes remain available for sealing - check 6)
3. Slowly increase the pressure. When first bubbles appear at pressure plate seal, note the pressure. If this is not the case, screw or unscrew the coverlid
4. If set pressure can not be achieved, we recommended to change the spring set (12) or (13).
5. When pressure setting is complete, seal the cap with a stainless steel wire spot welded between cap and body.
6. Put the protection cap on top of the YAK and seal the YAK to the flange by welding a cable. The cable goes through the holes of the yak and the proctection cap.
7. The valve is now ready to operate on the tank

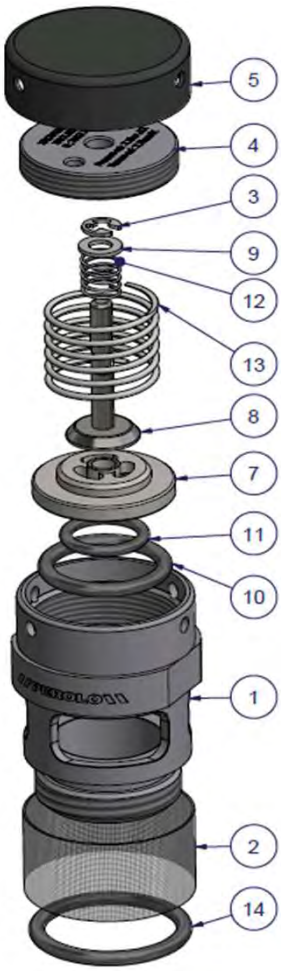
SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 17 10 42 40 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1" (ND25) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND/OR VACUUM SRV
YAK VXS2M

ITEM	QTY	DESIGNATION	PART N°
		Pressure O'ring*:	
10	1	O'ring (FKM/Viton Tcoat)	17 10 35 00 11
10	1	O'ring (EPDM)	70 10 01 05 16
10	1	O'ring (FEP/Silicone)	70 10 17 00 71
		Vacuum O'ring*:	
11	1	O'ring (FKM/Viton Tcoat)	17 10 35 00 10
11	1	O'ring (EPDM)	70 10 01 05 12
11	1	O'ring (FFKM/Kalrez)	70 10 01 93 18
		Setting gasket*:	
14	1	O'ring (FKM/Viton)	70 10 01 02 22
14	1	O'ring (EPDM)	70 10 01 05 22
14	1	O'ring (FEP/Silicone)	70 10 15 00 22
		Poppet, Gauze & Cap:	
7	1	Pressure poppet (P&V)	17 10 35 00 01
7	1	Pressure poppet (P only)	17 10 35 01 01
8	1	Vacuum poppet	17 10 42 00 03
2	1	Flame proof gauze	17 10 35 00 06
8	1	Protection cap	17 10 35 00 07

* seal material is marked on the coverlid



BODY MARKING (Exemple)

body material
seal material
serial number
pressure setting
vacuum setting

WORKING TEMPERATURE DEPENDING ON O'RING		
Material	**T° min (°C)**	**T° max (°C)**
FEP/Silicon	-40	+200
FKM/Viton Tcoat	-15	+200
FFKM	-30	+200
EPDM	-40	+150
Other on request		

SAFETY RELIEF VALVE

1 1/4" (ND32) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS3

1 1/4" YAK VX safety relief valve - designed to protect the tank from any over-pressure or any vacuum excess - Made of 316L/1.4409 stainless steel, this small product is suitable for road tankers dedicated to the transport and storage of chemicals and liquid foodstuff.

SPECIFICATION	
Nominal diameter	32 mm (1 1/4")
Weight	560 g
Connection	1 1/4" BSP or NPT
Pressure settings	from 0.2 to 3.5 Bar
Vacuum settings	from 20 to 800 mBar
Working Temperature	-40° to +190°C*

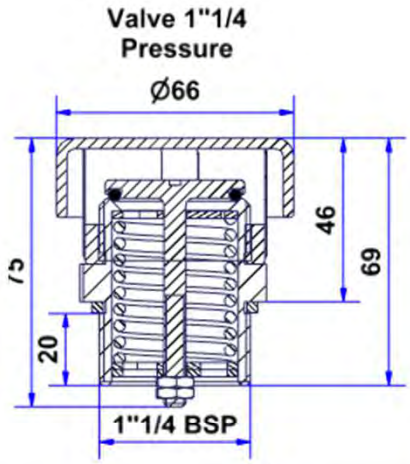
*Temperature range may vary according to the seal material

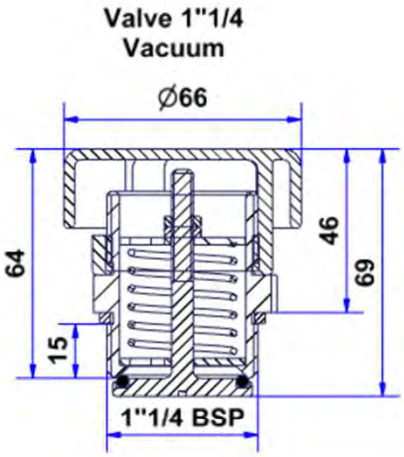
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID

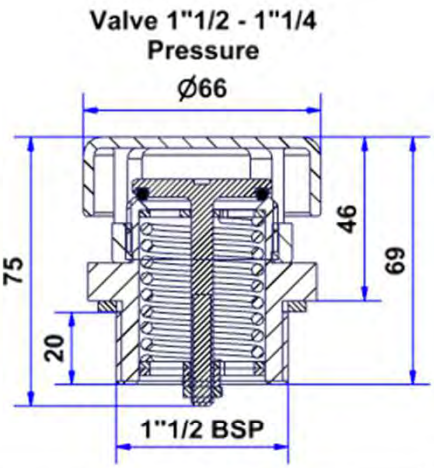
ACCESSORIES	
Weld-in flange	On request
1 1/4" Seal PTFE	17 10 06 00 06
1 1/2" Seal PTFE	17 10 10 00 14

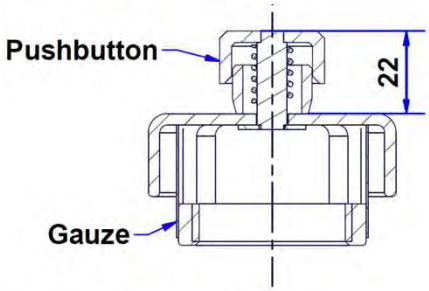
STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Setting Gasket	PTFE / Composite PTFE / Perbunan PTFE
O'rings material	FEP silicone / PFA silicone / Viton / Foodstuff Nitrile / Viton silicone

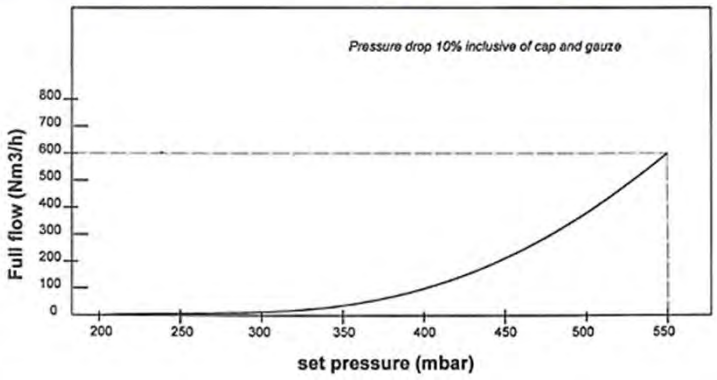
DIMENSIONAL DRAWING











Flow measurement conditions:

1 1/4" vacuum valve flow performance for 210 mBar

Set pressure tolerance: Max (±3%; 0,1b/1,45 PSI)

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 17 10 06 20 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 1/4" (ND32) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS3

CODIFICATION

STANDARD PART NUMBER	VXS3	B	F	P	22	F	Z	S	0
YOUR PART NUMBER	VXS3								

TYPE OF THREAD

BSP	B
NPT	N

INLET & OUTLET CONNECTION

Inlet 1"1/4 - outlet 1"1/4	F
Inlet 1"1/2 - outlet 1"1/4	W
Inlet 1"1/2 - outlet 1"1/2 (no cover)	A

PRESSURE OR VACUUM

Pressure	P
Vacuum	V

PRESSURE OR VACUUM SETTING

For pressure: *.* bar (ex: 2.2bar)	22
For vacuum: 0.** bar (ex: 0.21bar)	21

O'RING

FEP/Silicone (-40° to +200°C)	F
FKM Viton (-15° to +200°C)	V
FEP/Viton (-20° to +200°C)	B
Foodstuff nitrile (-30° to +120°C)	A
PFA/Silicone (-40° to +250°C)	H

SETTING GASKET MATERIAL

Perbunan PTFE	P
Composite PTFE	C
PTFE	Z

COVERLID

With lid (only for 1 1/4" BSP outlet)	C
With lid and vacuum test button	P
Without lid	S

FLAME PROOF GAUZE

With gauze	G
Without gauze	0

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 17 10 06 20 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 1/4" (ND32) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS3

INSTALLATION

STORAGE

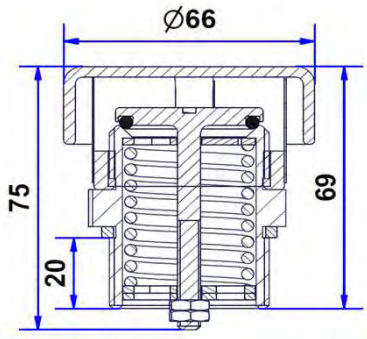
Keep the "YAK VXS3" safety relief valve in original packaging, until ready to use. It will prevent the valve from dust and will protect it againts potential impacts. The weldin flange must be machined BSP or NPT as per Safety Relief Valve.

OPERATING

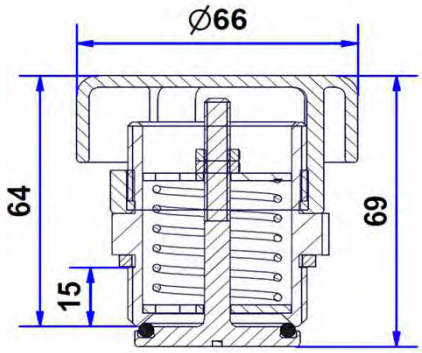
The pressure relief valve is designed to be installed on pressurized tanks up to maximum available working pressure 4 Bar. The material used is 1.4404 316L stainless steel, please check the fluid compatibility.The YAK VX™ 1 1/4" safety relief valve is designed to protect tanks against accidental over-pressure OR excessive vacuum. The vacuum setting range is from -0,02 Bar to -0,8 Bar and the pressure setting range is from 0,2 Bar to 3,5 bar.

This pressure or vacuum setting is done by the spring set compression on the pressure plate.For Pressure valve (drawing on the left) : When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure plate closes.In case of high pressure increase, the pressure plate translates to open a maximum of surface to allow a high flow evacuation. For Vacuum valve (drawing on the right) : When a vacuum is created into the tank,the poppet slides down, opens and allows air to come into the tank avoiding tank implosion.

Valve 1"1/4 Pressure



Valve 1"1/4 Vacuum



ASSEMBLY

The coverlid protects the operator in case of valve opening. Nevertheless the relief valve should be installed in protect area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.
Before installing, the relief valve check there is no dirt or dust on both valve and flange threads. Tighten to a torque equal to 40Nm with a 46 mm spanner on a 1 1/4" BSP tank pad 46mm(or 1 1/4" NPT, or 1 1/2" BSP or 1 1/2" NPT depending on your YAK VX model). Check the PTFE gasket is correctly compressed, no leak should occur between flange and gasket area when tank is pressurized.

SAFETY RELIEF VALVE

1 1/4" (ND32) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS3

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Make sure tank pressure is lower than service pressure.
2. Go to the area where the valve is installed.
3. Remove the cap to have a visual check on the poppet seal.
4. Check no leak exists between the poppet seal ant the seat, this check can be done with leakage detection fluid. If a leak exists, follow next points to dismantle the relief valve for further maintenance.

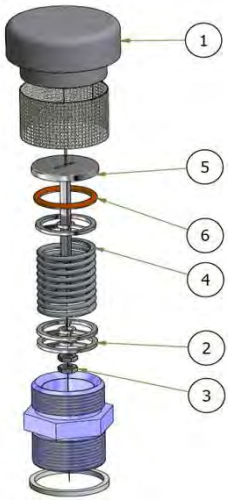
REMOVE THE VALVE FROM THE TANK

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the YAK VX from the flange using a 46mm spanner, remove the valve.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.
   Further valve disassemble should be done in a workshop.

RELIEF VALVE DISASSEMBLY

We recommend this operation **every two years.**

CAUTION: the springs load can achieve 40 daN, when tightening nut on the poppet is removed, the springs can violently relief this preload.



For Pressure valve :

1. Remove screwed cap (1)
2. Install the valve under a shop press, push with a tube on the spring retainer plate (2) to maintain spring compressed.
3. Remove the nut and the counter nut (3)
4. Release slowly pressure on the press to relieve the spring (4).
5. Remove spring (4), plate (2) and poppet (5).
6. Replace poppet seal (6) by a new one. Heat new o'ring in hot water before installing it around poppet groove.
7. Reassemble the valve.

CAUTION: cap must be installed when valve is fitted on the tank.

For Vacuum valve :

1. Remove the cap (1)
2. Push on plate (2) to relief spring (4).
3. Remove the nut and the counter nut (3)
4. Remove spring (4), plate (2) and poppet (5).
5. Replace poppet seal (6) by a new one. Heat new o'ring in hot water before installing it around poppet groove.
6. Reassemble the valve.

CAUTION: cap must be installed when valve is fitted on the tank.

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - PWA 17 10 06 40 00 C
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 1/4" (ND32) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS3

SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5.

The leakage is detected with leak test fluid at pressure seal interface.

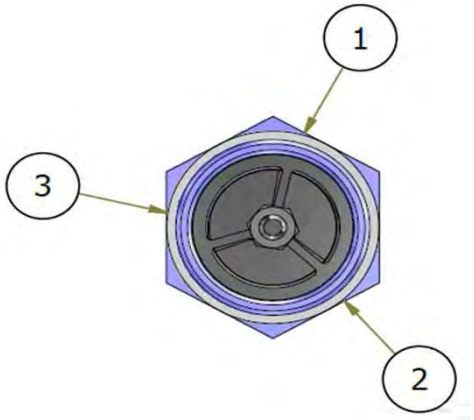
1. Slowly increase the pressure. When first bubbles appear at pressure plate seal, note the pressure.

2. Adjust the counter nut to get the set pressure.

If set pressure can not be achieved easily, we recommended to change the spring set (4).

The relief valve is ready to operate on the tank.

BODY MARKING (Example)



	MARKING	EXAMPLE
Face 1	Product	YAK VX
	Connection	1 1/4"
	Poppet seal material	Silicone
	Codification	VXS3BFP03FZCG
Face 2	Type of valve	Pressure valve
Face 3	Setting (bar)	0.032 bar
	Serial number	343.12

WORKING TEMPERATURE DEPENDS ON O'RING MATERIAL		
MATERIAL	T° min (°C)	T° max (°C)
FEP/Silicone	-40°C	+200°C
FKM Viton	-15°C	+200°C
FEP/Viton	-20°C	+200°C
Foodstuff nitrile	-30°C	+120°C
PFA/Silicone	-40°C	+250°C

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - PWA 17 10 06 40 00 C
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 1/2" (ND40) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND VACUUM SRV
YAK VXS4 & VXP4(PED)

1 1/2" YAK VX safety relief valve - designed to protect the tank from any over-pressure or any vacuum excess
Made of 316 / 1.4408 stainless steel, this small product is suitable for road tankers dedicated to the transport and storage of chemicals and liquid foodstuff.

SPECIFICATION	
Nominal diameter	40 mm (1 1/2") OR 35 mm (1 1/4")
Weight	850 g
Connection	1 1/2" BSP (or NPT) or 1 1/4" BSP (or NPT)
Pressure settings	from 0.09 to 4.4 Bar
Vacuum settings	from 20 to 800 mBar
Working Temperature	-40° to +190°C

*Temperature range may vary according to the seal material

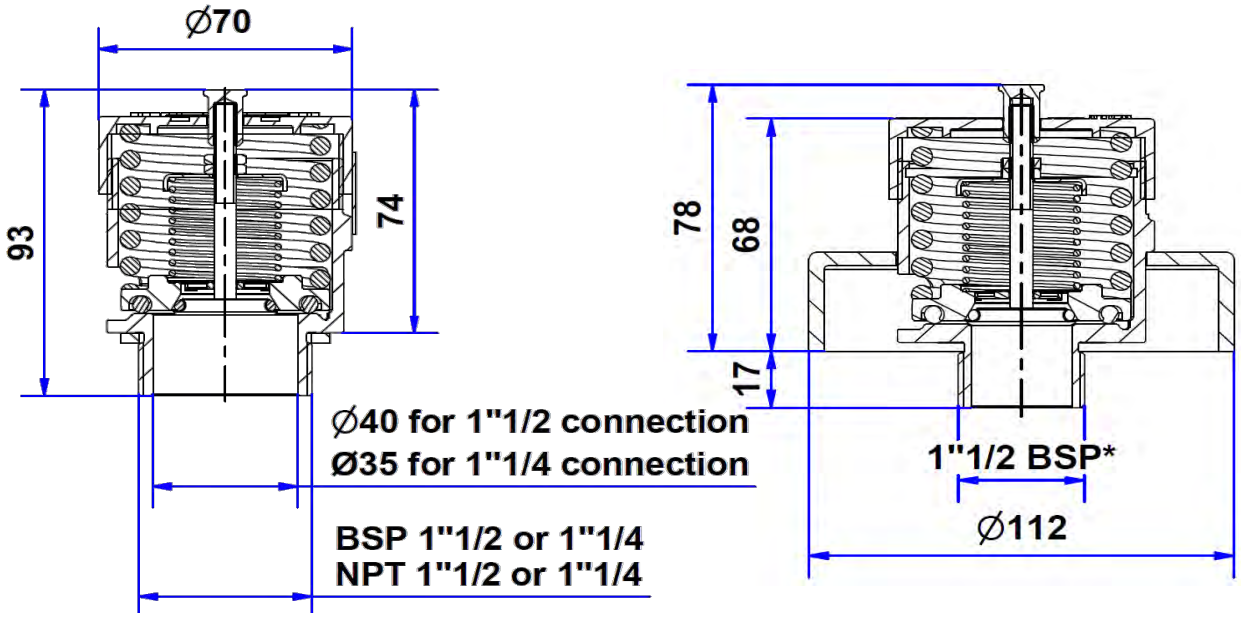
CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID
Design Approval: Lloyd's Register
PED (for VXP4)
PED 2014/68/EU - Category IV - Module H1 - Fluid gr.1
Lloyd's Register Espana

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone / PFA silicone/ Viton / Nitrile / Foodstuff Nitrile

ACCESSORIES	
1 1/2" BSP Weldin flang	11 92 16 00 00
1 1/2" PTFE Gasket	17 10 10 00 14
1 1/4" PTFE Gasket	17 10 06 00 06

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



Set pressure tolerance: Max (±3%; 0,1b/1,45 PSI)

With Protection (option)

SAFETY RELIEF VALVE

1 1/2" (ND40) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND VACUUM SRV
YAK VXS4 & VXP4(PED)

CODIFICATION

STANDARD PART NUMBER	STD: VXS4 / PED: VXP4	BF	D	26	05	F	Z	P	G
YOUR PART NUMBER	VXS4/VXP4								

OUTLET CONNECTION	
1.5" BSP male	BF
1.25" BSP male	BR
1.5" NPT male	NF
1.25" NPT male	NR

PRESSURE & VACUUM SETTING	
Pressure & Vacuum valve	D
Pressure	P

PRESSURE SETTING - Other on request	
No pressure	00
. bar (ex: 2.6bar)	26

VACUUM SETTING - Other on request	
No Vacuum	00
0.** bar (ex: 0.05 bar)	05

POPPET O'RING	
Pressure & vacuum ring in FEP/SIL	F
Pressure & vacuum ring in VITON	V
Press. in FEP/SIL & vacuum in Viton	B
Press. in FEP/SIL & vacuum in Nitrile	N
Pressure & vacuum ring in EPDM	E

SETTING GASKET - Other on request	
PTFE	Z

TEST BUTTON	
With (obligatory when vacuum)	P
Without	0

PROTECTION	
With flame proof gauze	G
With protection on exhaust	E
Without	0

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 17 10 10 20 00 K
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 1/2" (ND40) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND VACUUM SRV
YAK VXS4 & VXP4(PED)

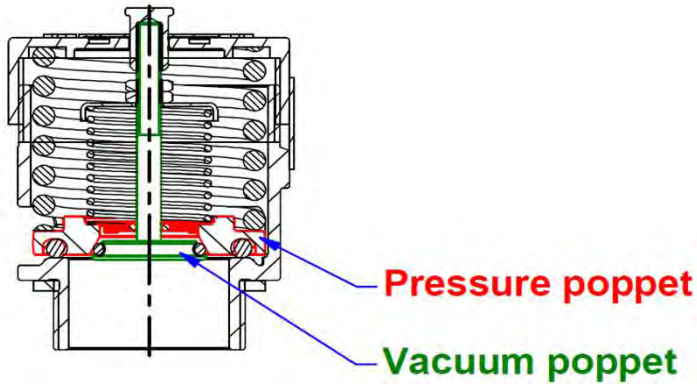
INSTALLATION

STORAGE

Keep the “YAK” safety relief valve in original packaging, until ready to use. It will prevent the valve from dust and will protect it againts potential impacts. The weldin flange must be machined BSP or NPT as per Safety Relief Valve.

OPERATING

The pressure relief valve (PRV) is designed to be installed on pressurized tanks up to maximum available working pressure 4 Bar. The material used are 1.4408 & 1.4404 stainless steel, please check the fluid compatibility. The YAK VX™ 1 1/2" safety relief valve is designed to protect tanks against accidental over-pressure and/or excessive vacuum depending on the chosen options.
The vacuum setting range is from -0,02 Bar to -0,8 Bar and the pressure setting range is from 0,09 Bar to 4.4 bar.
Check the compatibility between safety relief valve material and fluid before operating.
This pressure or vacuum setting is got by the spring set compression on the pressure plate.When the set pressure is reached, the pressure poppet opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure poppet closes. In case of high pressure increase, the pressure poppet translates and open a maximum of surface to allow a high flow evacuation.When a vacuum is created into the tank, the vacuum poppet slides down, opens and allows air to come into the tank avoiding tank implosion.



ASSEMBLY

Before assembly check the PRV set pressure is compatible with tank MAWP and the carried cargo compatible with PRV material.
The coverlid protects the operator in case of valve opening. Nevertheless the relief valve should be installed in protect area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank.
Before installing, the relief valve check there is no dirt or dust on both valve and flange threads. Tighten to a torque equal to 40Nm with 46 mm spanner on a 1 1/2" BSP tank pad 46mm (or 1 1/2" NPT depending on your YAK VX model). Check the PTFE gasket is correctly compressed, no leak should occur between flange and gasket area when tank is pressurized.

ITEM	DESCRIPTION	SPECIFICATION	PART N°
1	Weld-in flange	External Ø 85mm, internal 1.5 "BSP, thickness 17mm, 316L/1,4404	11 92 16 00 00

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 17 10 10 30 00 K
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 1/2" (ND40) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND VACUUM SRV
YAK VXS4 & VXP4(PED)

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Check no leak exists between the poppet seal ant the seat, this check can be done with leakage detection fluid. If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further check

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the YAK VX from the flange using a 46mm spanner, remove the valve.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.
   Further valve disassemble should be done in a workshop.

RELIEF VALVE DISASSEMBLY

We recommend this operation **every two years.**

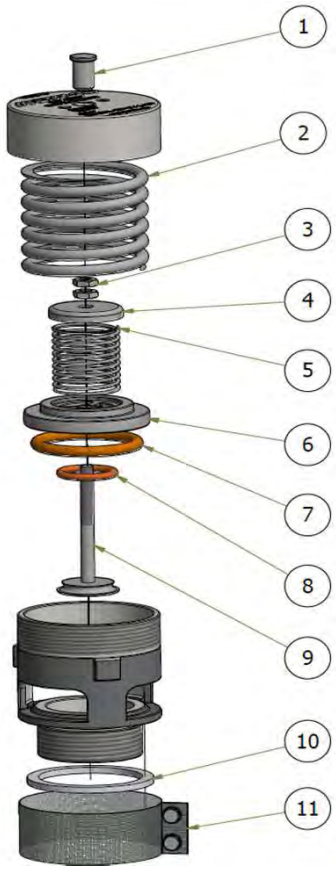
CAUTION : the springs load can achieve 37 daN, when tightening nut on the poppet is removed the springs can violently relief this preload.

1. Remove screwed cap (use a special tool to avoid any risk)
2. Remove the pressure spring (2)
3. Remove the vacuum assembly
4. Unscrew and remove the small nuts (3) from vacuum assembly
5. Remove the vacuum spring (5)
6. Remove the pressure poppet seal (7) and vacuum poppet seal (8)

CAUTION : cap must be installed when valve is fitted on the tank.

ITEM	QTY	DESIGNATION	PART N°
1	1	Vacuum test button	17 10 10 91 00
2	1	Spring set press. 1,95b	17 10 10 98 00
6, 7	1	Pressure poppet kit for Pres&Vac or Vac SRV	17 10 10 92 00
7	1	Pressure O'ring (FEP/Si)	17 10 10 93 00
7	5	Pressure O'ring (FEP/Si)	17 10 11 93 00
8	1	Vacuum O'ring (FEP/Si)	17 10 10 94 00
-	1	Vacuum kit (spring at -0,21b) Nr 3, 4, 5, 8, 9	17 10 10 95 00
10	1	Setting gasket Composite/PTFE	17 10 10 96 00
10	5	Setting gasket Composite/PTFE	17 10 10 99 00
11	1	Flame proof gauze kit	17 10 10 97 00

For other settings, please contact our sales team:
Email: sales@perolo.com
Tel: +33 557 426 700



SAFETY RELIEF VALVE

1 1/2" (ND40) SAFETY RELIEF VALVE
DOUBLE ACTING PRESSURE AND VACUUM SRV
YAK VXS4 & VXP4(PED)

SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5.

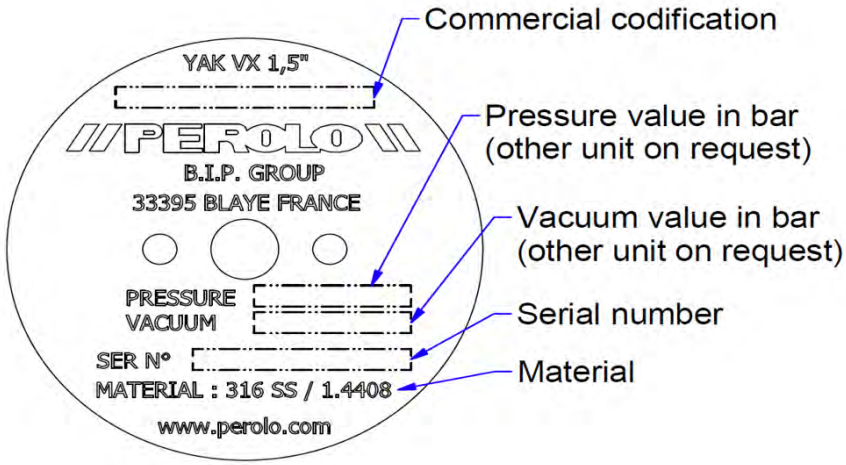
The leakage is detected with leak test fluid at pressure seal interface.

1. Slowly increase the pressure. When first bubbles appear at pressure plate seal, note the pressure.
2. Adjust the counter nut to get the set pressure.

If set pressure can not be achieved easily, we recommended to change the spring set (2). (See page 1 for item references). Only PEROLO provided spring sets should be installed.

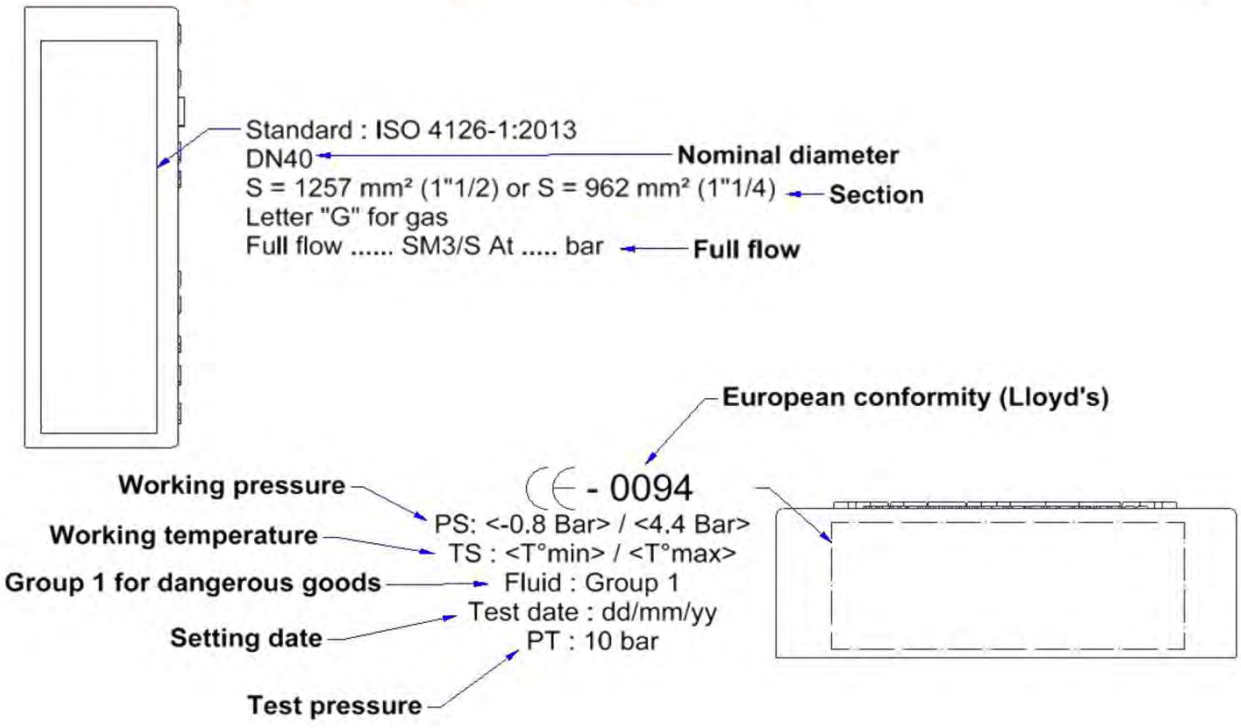
Once pressure and vacuum have been tested and set, the relief valve is ready to operate on the tank.

BODY MARKING (For VXS4 & VXP4)



PED ADDITIONNAL BODY MARKING (For VXP4 Only)

Body marking is only made on PED approved safety relief valve (when reference starts with VXP)



2" (ND50) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS5 & VXP5(PED)

2" YAK VX safety relief valve - designed to protect the tank from any over-pressure or any vacuum excess - Made of 316 / 1.4408 stainless steel, this small product is suitable for road tankers dedicated to the transport and storage of chemicals and liquid foodstuff.

SPECIFICATION	
Nominal diameter	50 mm (2")
Weight	1,5 kg
Connection	2" BSP or NPT
Pressure settings	from 0.02 to 3 Bar
Vacuum settings	from 20 to 800 mBar
Working Temperature	-40° to +190°C (HT +250°C)

*Temperature range may vary according to the seal material

CONFORMITY & CERTIFICATION
Complies with ADR / IMDG / RID
Design Approval: Lloyd's Register
PED (for VXP5)
PED 2014/68/EU - Category IV - Module H1 - Fluid gr.1
Lloyd's Register Espana

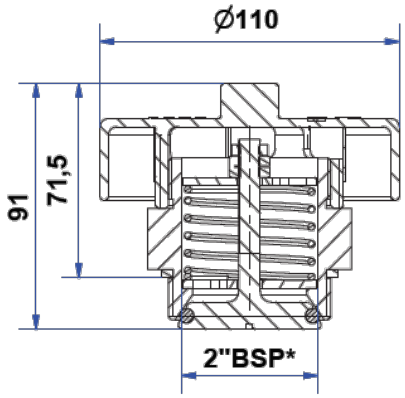
ACCESSORIES	
BSP Weldin flange	11 90 44 00 00
PTFE Flat Gasket	17 10 01 00 04
Burst Disc Adapt Flange	12 92 25 00 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Setting Gasket	PTFE
O'rings material	FEP silicone / PFA silicone/ Viton / Nitrile / Foodstuff Nitrile

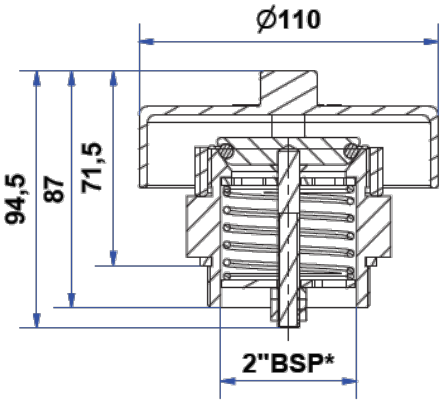
It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING

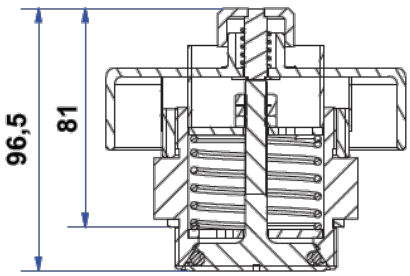
VACUUM



PRESSURE



WITH TEST BUTTON



Set pressure tolerance: Max (±3%; 0,1b/1,45 PSI)

SAFETY RELIEF VALVE

2" (ND50) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS5 & VXP5(PED)

CODIFICATION

STANDARD PART NUMBER	STD: VXS5	BF	V	21	F	Z	P	G
	PED: VXP5							
YOUR PART NUMBER	VXS5/VXP5	__	__	__	__	__	__	__

OUTLET CONNECTION	
2" BSP male	BF
2" NPT male	NF

PRESSURE OR VACUUM	
Pressure	P
Vacuum	V

PRESSURE OR VACUUM SETTING	
For pressure: *.* bar (ex: 2.6bar)	26
For vacuum: 0.** bar (ex: 0.21bar)	21

POPPET O'RING	
FEP/SILICONE	F
VITON	V
NITRILE (black)	N
PFA/SILICONE (HT +250°C)	H
FOOD (white nitrile FDA & CE1935)	A
SILICONE	S

SETTING GASKET	
PTFE	Z

COVERLID	
Without coverlid	S
With coverlid	C
With coverlid and test button	P

GAUZE FILTER	
Without	0
With	G

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 17 10 17 20 00 J
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" (ND50) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS5 & VXP5(PED)

INSTALLATION

STORAGE

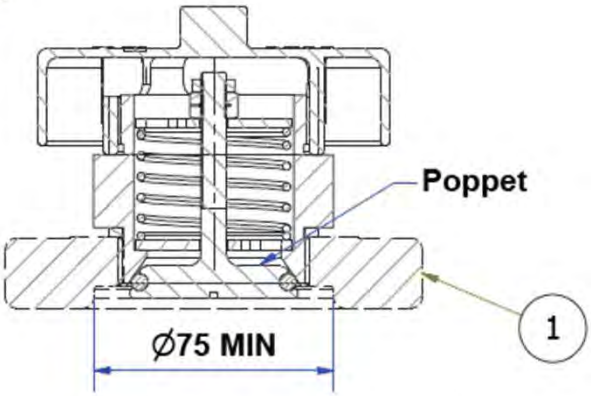
Keep the "YAK VX" safety relief valve in original packaging, until ready to use. This prevents ingress of dust and protects against impact damage. The leak proof face of the valve has a serrated surface. The mounting flange must be machined in identical or similar fashion.

OPERATING

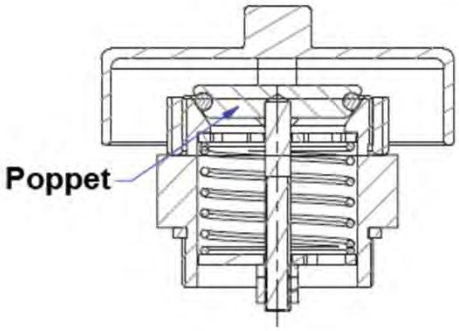
The pressure relief valve (PRV) is designed to be installed on pressurized tanks up to maximum available working pressure 4 Bar. The material used are 1.4408 & 1.4404 316 stainless steel, please check the fluid compatibility. The YAK VX™ 2" safety relief valve is designed to protect tanks against accidental over-pressure or excessive vacuum. The vacuum setting range is from -0.02 Bar to -0.8 Bar and the pressure setting range is from 0.02 Bar to 3.0 bar.

Check the compatibility between safety relief valve material and fluid before operating.

This pressure or vacuum setting is got by the spring set compression on the pressure plate. For Pressure valve (drawing on the left) : When the set pressure is reached, the pressure plate opens and allows the over pressure to be evacuated, then where the pressure balance is reached the pressure plate closes.In case of high pressure increase, the pressure plate translates to open a maximum of surface to allow a high flow evacuation. For Vacuum valve (drawing on the right) : When a vacuum is created into the tank,the poppet slides down, opens and allows air to come into the tank avoiding tank implosion.



Vacuum settings (poppet is at the bottom)



Pressure settings (poppet is on)

ASSEMBLY

Before assembly check the PRV set pressure is compatible with tank MAWP and the carried cargo compatible with PRV material. The coverlid protects the operator in case of valve opening. Nevertheless the relief valve should be installed in protected area where there is no risk for an operator to be in contact with the gas or liquid when valve opens in case of over pressure in the tank. Before installing, the relief valve check there is no dirt or dust on both valve and flange threads (to avoid anysizing during installation). Tighten to a torque equal to 50 N.m with a 65 mm spanner on a 2" BSP tank pad (or 2" NPT depending on your YAK VX model). Check the PTFE gasket is correctly compressed, no leak should occur between flange and gasket area when tank is pressurized.

ITEM	DESCRIPTION	SPECIFICATION	PART N°
1	Weld-in flange	External Ø 130mm, internal 2 "BSP, thickness 21.5mm, 316L/1.4404	11 90 44 00 00

SAFETY RELIEF VALVE

2" (ND50) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS5 & VXP5(PED)

MAINTENANCE

VALVE INSTALLED ON THE TANK

1. Check the tank pressure is lower than the service pressure.
2. Go to the area where the valve is installed.
3. Remove the cap to have a visual check on the poppet seal.
4. Check no leak exists between the poppet seal and the seat, this check can be done with leakage detection fluid. If a leak exists, follow next points to dismantle the relief valve for further maintenance.

REMOVE THE VALVE FROM THE TANK

This operation is done to change the valve for further checking

1. Before any operation, check there is no pressure in the tank.
2. Unscrew the YAK VX from the flange using a 65mm spanner, remove the valve.
3. Plug the hole in the flange with a blind cap to protect the tank from dust.

Further valve disassemble should be done in a workshop.

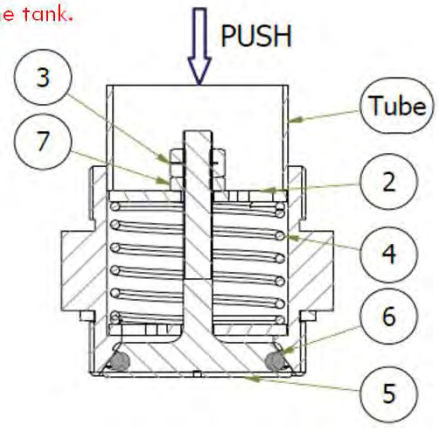
RELIEF VALVE DISASSEMBLY

We recommend this operation every **two years.**

CAUTION : the springs load can achieve 70 daN, when tightening nut on the poppet is removed the springs can violently relief this preload.

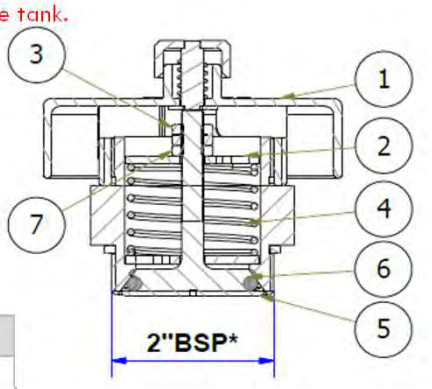
For Pressure valve : CAUTION : cap must be installed when valve is fitted on the tank.

1. Remove screwed cap
2. Install the valve under a shop press, push with a tube on the spring retainer plate (2) to maintain spring compressed.
3. Remove the nut (7) and the counter nut (3)
4. Release slowly pressure on the press to relieve the spring (4).
5. Remove spring (4), plate (2) and poppet (5).
6. Replace poppet seal (6) by a new one. Heat new o'ring in hot water before installing it around poppet groove.
7. Reassemble the valve.



For Vacuum valve : CAUTION : cap must be installed when valve is fitted on the tank.

1. Remove the cap (1)
2. Push on plate (2) to relief spring (4).
3. Remove the nut (7) and the counter nut (3)
4. Remove spring (4), plate (2) and poppet (5).
5. Replace poppet seal (6) by a new one. Heat new o'ring in hot water before installing it around poppet groove.
6. Reassemble the valve.



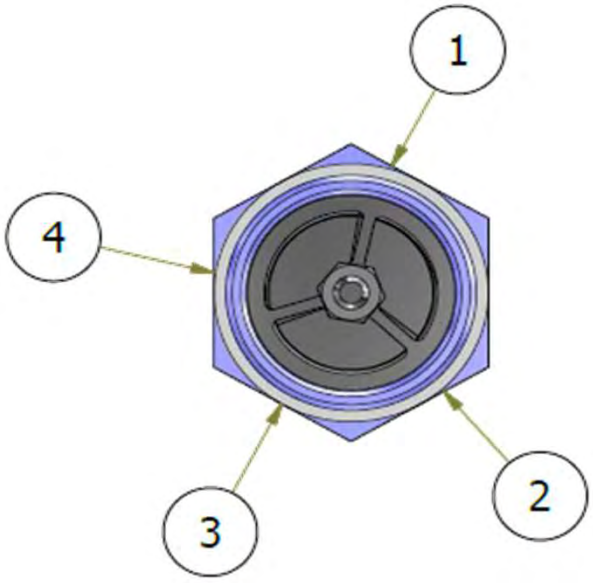
Designation	Part Number
5 x FEP/Silicone poppet seal	17 10 11 93 00
5 x Viton poppet seal	17 10 11 91 00

2" (ND50) SAFETY RELIEF VALVE
SINGLE ACTING PRESSURE OR VACUUM SRV
YAK VXS5 & VXP5(PED)

SET PRESSURE RIGGING

The valve is installed on a test rig. The pressure is measured with a manometer precision class 0,5.
The leakage is detected with leak test fluid at pressure seal interface.
1. Slowly increase the pressure. When first bubbles appear at pressure plate seal, note the pressure.
2. Adjust the counter nut to get the set pressure.
If set pressure can not be achieved easily, we recommended to change the spring set (4).
Once pressure and vacuum have been tested and set, the relief valve is ready to operate on the tank.

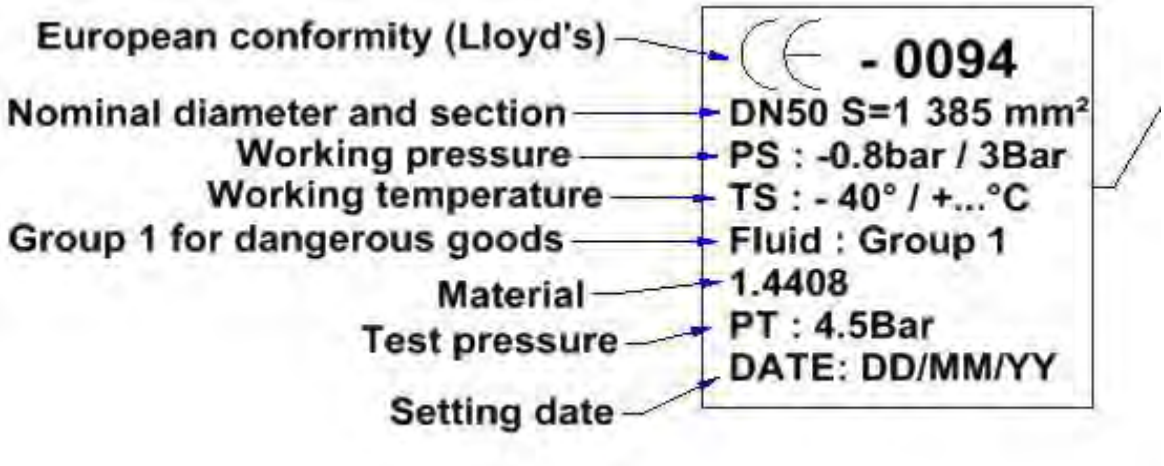
BODY MARKING (For VXS5 & VXP5)



	MARKING	EXAMPLE
Face 1	Product:	YAK VX
	Connection:	2" BSP (or 2" NPT)
	Seal material:	Silicon
	Reference:	VXS5BFV21FZCG
Face 2	Type of valve:	Vacuum valve
Face 3	Setting:	0.21 bar
	Serial number:	253211

PED ADDITIONNAL BODY MARKING (For VXP5 Only)

Body marking is only made on PED approved safety relief valve (when reference starts with VXP)



BALL VALVES

FUNCTION

A Ball valve is designed as a fluid inlet/outlet. It can be installed on top of a tank or tanker as top discharge provision, or at the bottom as part of a bottom outlet. Different sizes are available ranging from ND40 to ND100. Ball valves can have a straight or a flanged outlet depending on the application. The Perolo range comprises regular ball valves and self -contained ball valves that exceed the requirements of a T22 tank and offer a very high protection.

- Various internal diameters: 1½" (40mm), 2" (50mm) 3" (80mm), 4" (100mm)
- Self-contained or compact
- Inlet connection: Various drilling patterns Open or closed slots
- Outlet connection: Flanged, Threaded (BSP or NPT)
- Operating mechanism: Manual: handle or remote lever Pneumatic actuator
- Contact Materials: 316L stainless steel
- Steam heating
- Working Temperature: Standard or High (+250°C)
- Surface Finish: Acid etched, Halar lined
- Large choice of accessories: Gauges, Gauge guard
- ADR, RID, IMDG

1½" Ball Valve AFN4

2" Ball Valve AFN5

3"-2" Ball Valve AHN5

3" Ball Valve AFN8

3" Ball Valve Compact AFC8

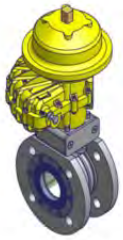
2" Ball Valve ALN5/AKN5

4' Ball Valve AF10

1½" Pneumatic BV AGN4

2" Pneumatic BV AGN5

3" Pneumatic BV AGN8



2" Pneumatic BV AP5

Remote Control
Section Accessories

1 1/2" ATCO AFN4X
ND40 BALL VALVE
AFN4X

The Perolo ND40 ATCO™ FULL BORE self-contained ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	40 mm (1 1/2")
Weight	4.7 Kg
Inlet flange drilling	4 holes x Ø18 x 110 PCD*
Outlet flange drilling	4 slotted Ø18 x 110 to 125 PCD*
Working Pressure	16 Bar / 4 bar for HT°
Test Pressure	24 Bar / 6 bar for HT°
Working Temperature	STD:-40°C to +150°C/HT°:-40°C to +250°C
Design Temperature	190°C / 250°C for HT°

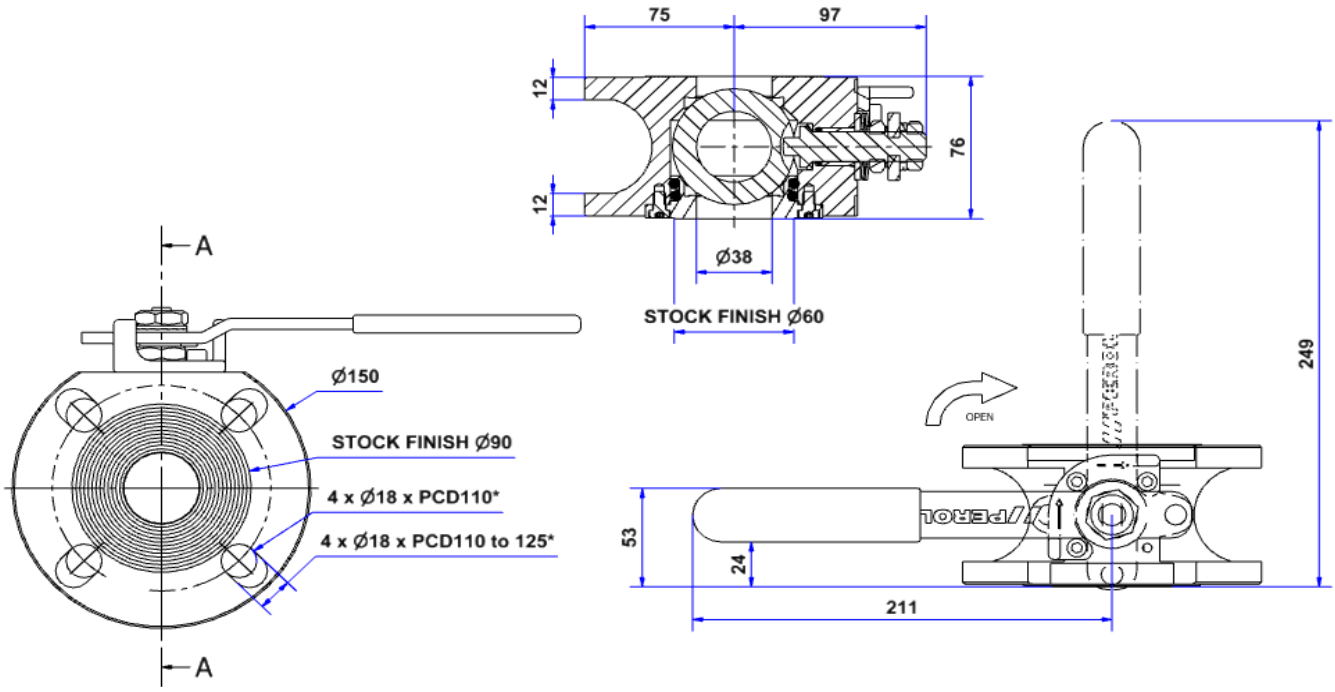
NB: temperature range vary according to o'ring material

ACCESSORIES	
Weld-in flange	11 96 63 00 00
Recess flange	11 97 18 00 00
Blind flange EN14025	11 95 87 00 00
Bolting Kit	11 12 69 91 00
Gaskets	12 91 90 00 00
Remote Control	RMT 33
Remote Control Handle	11 14 55 91 00

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I, RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4408 Stainless steel
Compact design	Face to face 76 mm
Operation	Handle 1/4 Turn or Remote Control
Ball gaskets material	PTFE or PTFE/Graphite for HT°

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

1 1/2" ATCO AFN4X
ND40 BALL VALVE
AFN4X

CODIFICATION

STANDARD PART NUMBER	AFN4	X	16T	1	2	0	0	S	0	0	.
YOUR PART NUMBER	AFN4							S	0	0	

MATERIAL - Other on request	
316L / 1.4408 stainless steel	X

SEAT MATERIAL & MAWP	
Standard PTFE seal 16 Bar	16T
HT 250°C PTFE/Graphite seal 4 Bar	4H

INLET FLANGE	
4 x Ø18 x 110 PCD	1
4 x Ø18 x 110 PCD 53°	2
4 slotted 18 x 110 to 125 PCD	3

OUTLET FLANGE	
4 slotted Ø18 x 110 to 125 PCD	1
4 x M16 x 110 PCD	2
4 slotted Ø18 x 98,5 to 125 PCD	3
4 x M16 x 110 PCD 55°	4
4 x slotted Ø18 x 110 to 150 PCD	5
4 x slotted Ø17 x 98,5 to 110 PCD	6
4 x Ø18 x 110 PCD	7

GAUGE CONNECTION	
Without	0
1/4" BSP gauge connection at 180°	1
1/4" BSP gauge connection at 90°	2

GAUGE PROTECTION	
Without	0
With at 180°	1
With at 90°	2

SEALING	
With	S

STEAM HEATING	
Without	0

HALAR COATING	
Without	0

For coated option, please refer to HIGH CORROSION Catalogue sheet.

HANDLE			
With standard handle	.	Reverse handle	L
Off set handle Ht=100	M	Short handle	S
Off set handle Ht=110	N	Remote control handle	W

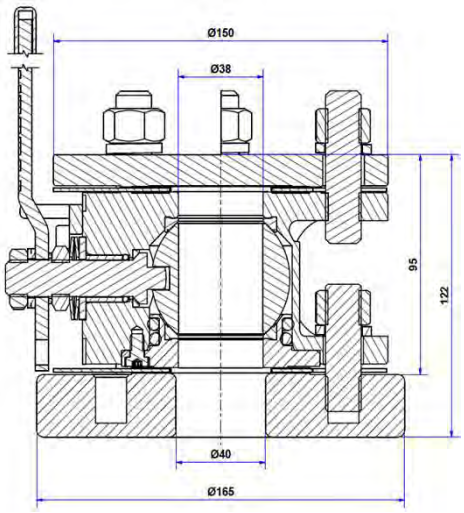
BALL VALVE

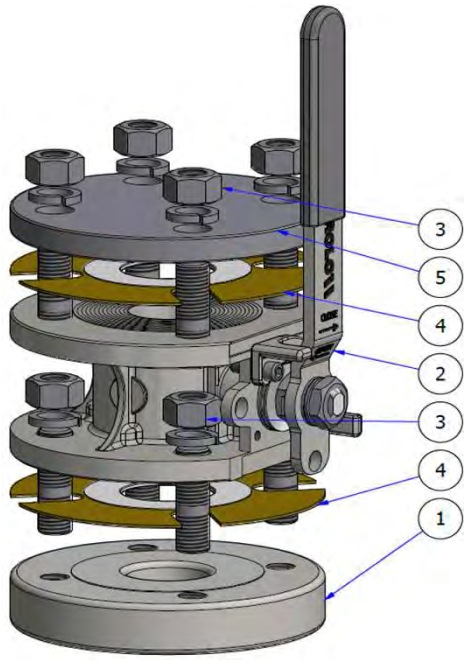
Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 16 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 1/2" ATCO AFN4X
ND40 BALL VALVE
AFN4X

INSTALLATION

1. Check Inlet (tank side) flange is clean, free from dirt and grit.
2. Screw the studs into the weldin flange, lubricate with PTFE added grease before installing the studs
3. Place the gasket
4. Install the valve on the flange, secure with washer and nuts *(refer to PWAGEN001 fort bolt tightening sequence)*
5. Tighten the nuts torque : 40 N.m +/- 5
6. Install gasket, bolting kit and blind flange on the valve outlet flange
(refer to PWAGEN001 fort bolt tightening sequence)
7. Open and close the valve 3 times to check the operationnal torque.





ITEM	DESCRIPTION	SPECIFICATION	PART N°
5	Blind flange	ext ø 150mm, thickness 13.5mm 4 x Ø18x110 PCD 316L stainless steel	11 95 87 00 00
3	Bolting Kit	4 x M16 36/28 studs4 x nuts and 4 x washers	11 12 69 91 00
4	Gasket	ext Ø 150mm, int Ø 45mm, 4x slotted Ø19x minimum 101,4 PCD, CNAF/PTFE	12 91 90 00 00
2	ND 40 ATCO	PN 16 Full Bore, 316 L stainless steel Inlet flange 4xø17x125PCD Outlet flange 4xM16x110PCD	AFN4X 16T1200000
3	Bolting Kit	4 x M16 36/28 studs4 x nuts and 4 x washers	11 12 69 91 00
4	Gasket	ext Ø 150mm, int Ø 45mm, 4x slotted Ø19x minimum 101,4 PCD, CNAF/PTFE	12 91 90 00 00
1	Weld-in flange	ext ø 150mm, int Ø 40mm thickness 27mm 4 x M16 x 110PCD 316L stainless steel	11 96 63 00 00

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

1 1/2" ATCO AFN4X
ND40 BALL VALVE
AFN4X

MAINTENANCE

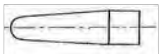
ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 12 82 91 00**
12	2	Ball seals (PTFE)	11 12 82 00 03
14	1	O'ring (FEP encapsulated)	70 10 15 00 95
17	1	O'ring (Viton)	70 10 01 02 32
		Ball kit:	**11 12 82 92 00**
15	1	Ball (stainless steel)	11 12 82 00 05
		Drilled Ball kit:	**11 12 39 97 00**
15	1	Drilled Ball (stainless steel)	11 12 82 00 17
		Spindle seals kit:	**11 12 42 93 00**
11	1	Bearing bush (PTFE)	11 12 39 00 14
9	1	O'ring (FEP encapsulated)	70 10 15 00 12
10	1	Spindle seal (PTFE)	11 12 39 00 21
		Spindle:	
2	1	Spindle (stainless steel)	11 12 39 00 16
		Handle kit:	**11 14 16 92 00**
3	1	Handle (stainless steel)	11 12 39 00 22
	1	Handle cap, nitrile	-
5	1	Stop washer (stainless steel)	-
6	2	Nuts	-
7	1	Washer (stainless steel)	-
		Clamping ring kit:	**11 12 82 97 00**
13	1	Clamping ring (stainless steel)	-
20	4	Screws M6 (stainless st.)	-
		Tools kit:	**11 12 82 96 00**
A	1	Spindle assembly tool	11 12 82 95 01
B	1	O'ring assembly tool	11 12 82 95 02

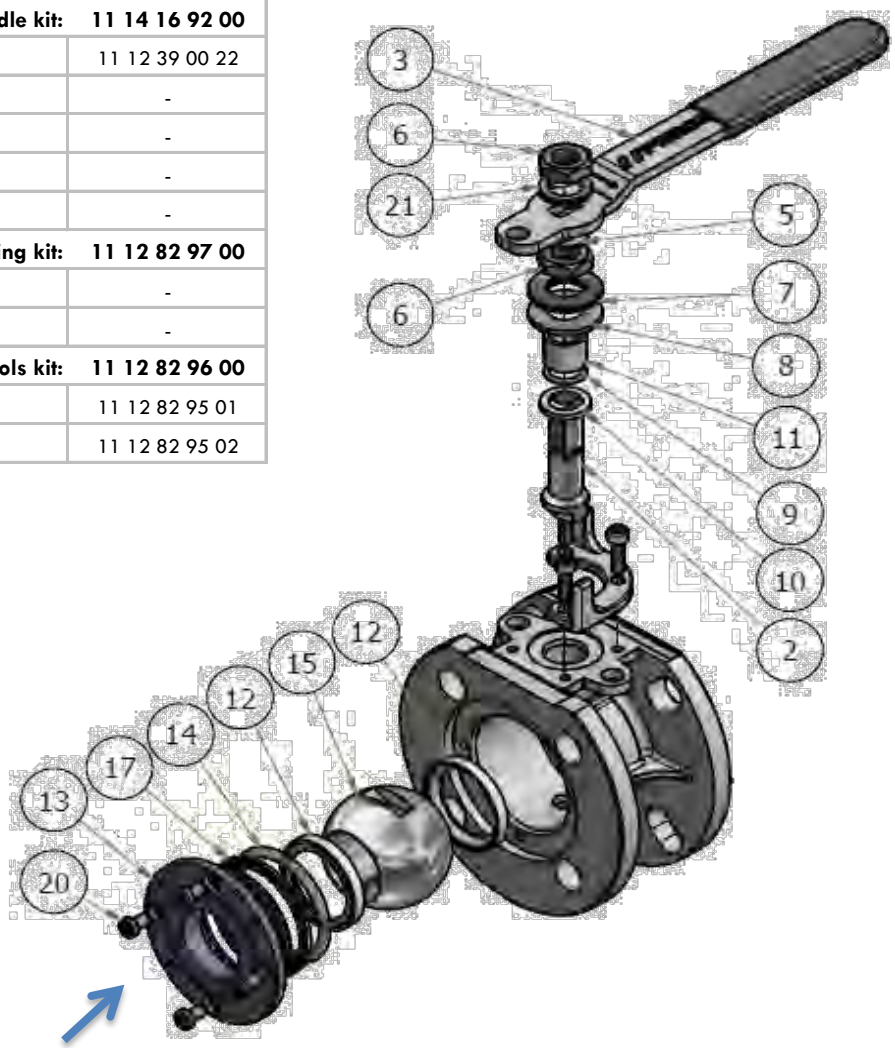
Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details

Nb: Perolo recommends
- the spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- both ball gaskets should be changed simultaneously
- when clamping ring is changed we recommend to also order ball seals kit

A

B





tightening torque 10 N.m.+/-5

BALL VALVE

2" ATCO AFN5X
ND50 BALL VALVE
AFN5X

The Perolo ND50 ATCO™ FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	6.0 Kg
Inlet flange drilling	4 holes x Ø17 x 125 PCD
Outlet flange drilling	4 holes x Ø17 x 125 PCD*
Working Pressure	16 Bar / 4 bar for HT°
Test Pressure	24 Bar / 6 bar for HT°
Working Temperature	STD:-40°C to +150°C/HT°:-40°C to +250°C
Design Temperature	190°C / 250°C for HT°

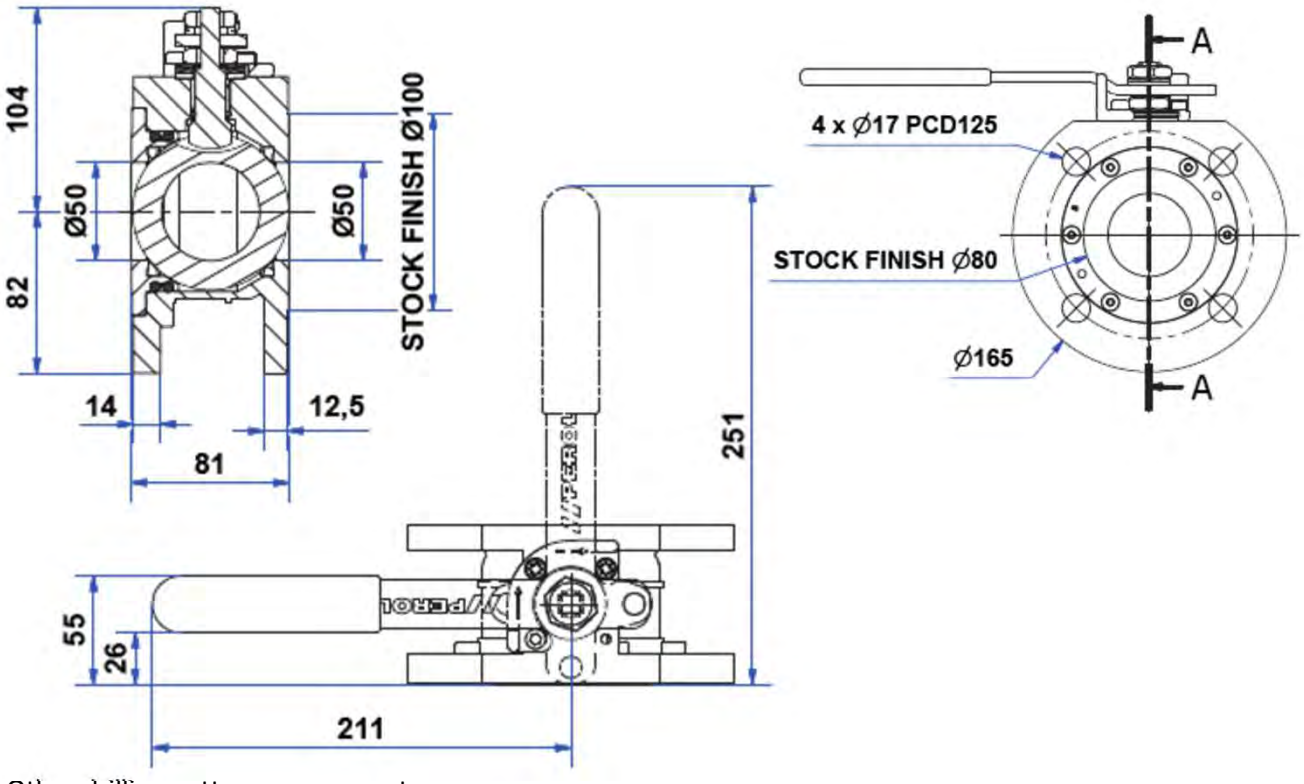
NB: temperature range vary according to o'ring material

ACCESSORIES	
Weld-in flange	11 90 25 00 00
Recess flange	11 92 44 00 00
Blind flange	11 95 88 00 00
Bolting Kit	11 12 69 91 00
Gaskets	11 90 29 00 00
Stud kit	17 12 12 91 00
Remote Control	RMT 53
Remote Control Handle	11 14 55 91 00

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I, RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact design	Face to face 81 mm
Operation	Handle 1/4 Turn or Remote Control
Ball gaskets material	PTFE or PTFE/Graphite for HT°

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

2" ATCO AFN5X
ND50 BALL VALVE
AFN5X

CODIFICATION

STANDARD PART NUMBER	AFN5	X	16T	1	2	0	0	S	0	0	.	.
YOUR PART NUMBER	AFN5	X	__	1	__	__	__	S	__	0	__	__

MATERIAL - Other on request	
316L / 1.4409 stainless steel	X

SEAT MATERIAL & MAWP	
Standard PTFE seal 16 Bar	16T
HT 250°C PTFE/Graphite seal 4 Bar	4H

INLET FLANGE	
4 x Ø17 x 125 PCD	1

OUTLET FLANGE - other on request	
4 x Ø17 x 125 PCD	1
4 x M16 x 125 PCD	2
4 slotted x Ø18 x min 114 PCD	3

GAUGE CONNECTION	
Without	0
1/4" NPT gauge connection & drilled ball	1
1/4" BSP gauge connection & drilled ball	2
With drilled ball only (H2O2)	3

GAUGE PROTECTION	
Without	0
With	1

SEALING	
With	S

STEAM HEATING	
Without	0
With	S

HALAR COATING	
Without	0

For coated option, please refer to HIGH CORROSION Catalogue sheet

SPECIAL OPTIONS	
Without	.
With cut-side flanges	C
With C276 ball	D

HANDLE	
Standard handle	.
Remote control handle	W
Reverse handle	L

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 12 69 20 00 H
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" ATCO AFN5X
ND50 BALL VALVE
AFN5X

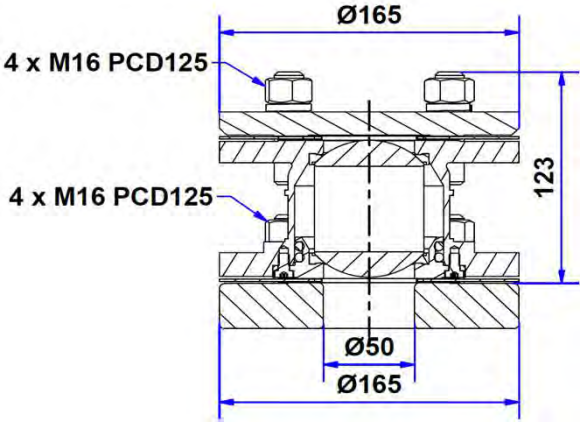
INSTALLATION

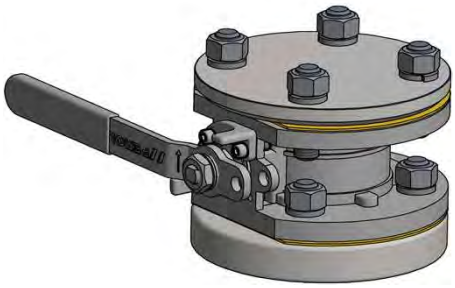
To use the ND50 ATCO™ full bore as an air inlet/outlet:

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Screw the studs into the weld in flange. Lubricate the threads with PTFE added grease.
3. Place the gasket.
4. Fit ball valve on the weld in flange and secure with washers and nuts.
   Tighten the nuts torque : 50 N.m +/-5 (use our bolting kit A2-70 or A4-70 class)
5. Assemble gasket and blind flange on the top same bolting procedure them above
6. check torque after 24 hours

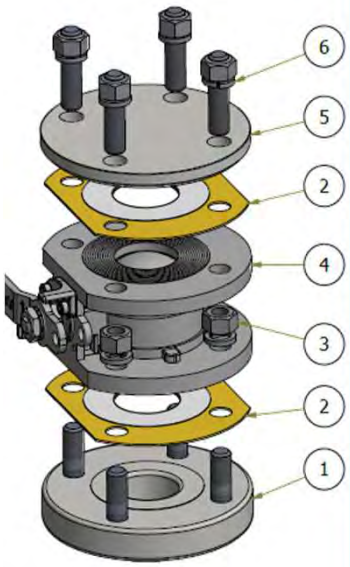
Refer to our PWAGEN001 sheet for flange tightening details

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs





ITEM	DESCRIPTION	SPECIFICATION	PART N°
5	Blind flange	ext ø 165mm, thickness 13.5mm 4 xø18x125 PCD 316L stainless steel	11 95 88 00 00
6	Bolting Kit	4 x studs M16-50 studs 4 x M16 nuts and 4 x grower washers	17 12 12 91 00
2	Gasket	ext ø 165mm, int ø 50mm 4xø18x125PCD CNAF/PTFE	11 90 29 00 00
4	ND 50 ATCO	PN 16 Full Bore, 316 L stainless steel Inlet flange 4xø17x125PCD Outlet flange 4xM16x110PCD	AFN5X16T1200 S00
3	Bolting Kit	4 x M16 35 studs 4 x nuts and 4 x washers	11 12 69 91 00
2	Gasket	ext ø 165mm, int ø 50mm 4xø18x125PCD CNAF/PTFE	11 90 29 00 00
1	Weld-in flange	ext ø 165mm, int ø 50mm thickness 26,5 mm 4xM16x125PCD 316L stainless steel	11 90 25 00 00



BALL VALVE

2" ATCO AFN5X
ND50 BALL VALVE
AFN5X

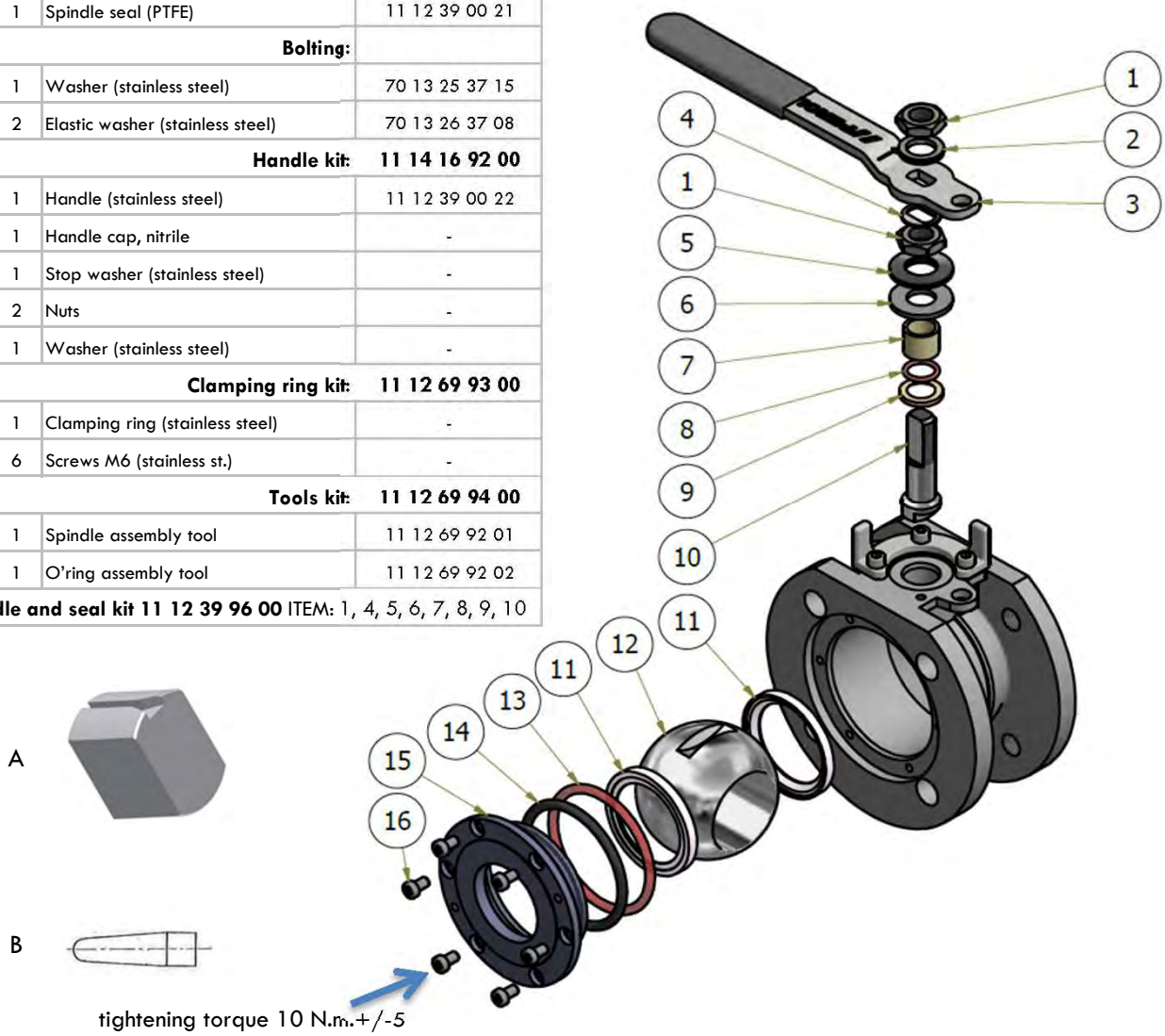
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N° Standard	PART N° High Temp
		Ball seals kit:	**11 12 42 92 00**	**11 12 42 96 00**
11	2	Ball seals (PTFE / PTFE/Graphite)	11 12 39 00 09	11 12 39 00 17
13	1 / 2	O'ring (FEP / PFA)	70 10 15 00 10	70 10 17 00 40
14	1	O'ring (Viton)	70 10 01 02 39	/
		Ball kit: 11 12 42 91 00		
12	1	Ball (stainless steel)	11 12 42 00 11	
		Drilled ball kit: 11 12 39 94 00		
12	1	Drilled ball (stainless steel)	11 12 39 00 11	
		Spindle:		
10	1	Spindle (stainless steel)	11 12 39 00 16	
		Spindle seals kit: 11 12 42 93 00		
7	1	Bearing bush (PTFE)	11 12 39 00 14	
8	1	O'ring (FEP encapsulated)	70 10 15 00 12	
9	1	Spindle seal (PTFE)	11 12 39 00 21	
		Bolting:		
5	1	Washer (stainless steel)	70 13 25 37 15	
6	2	Elastic washer (stainless steel)	70 13 26 37 08	
		Handle kit:	**11 14 16 92 00**	
3	1	Handle (stainless steel)	11 12 39 00 22	
	1	Handle cap, nitrile	-	
4	1	Stop washer (stainless steel)	-	
1	2	Nuts	-	
2	1	Washer (stainless steel)	-	
		Clamping ring kit:	**11 12 69 93 00**	
15	1	Clamping ring (stainless steel)	-	
16	6	Screws M6 (stainless st.)	-	
		Tools kit:	**11 12 69 94 00**	
A	1	Spindle assembly tool	11 12 69 92 01	
B	1	O'ring assembly tool	11 12 69 92 02	

Spindle and seal kit 11 12 39 96 00 ITEM: 1, 4, 5, 6, 7, 8, 9, 10

Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details

Nb: Perolo recommends
- the spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- both ball gaskets should be changed simultaneously
- when clamping ring is changed we recommend to also order ball seals kit



BALL VALVE

3"/2" ATCO AHN5X
ND80/ND50 BALL VALVE
AHN5X

The Perolo ND50/80 ATCO™ FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve. A perfect solution to fit a 2" ball valve onto a 3" existing flange

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	6.85 Kg
Inlet flange drilling	4 x Ø18 x 160PCD & 6 x Ø14 x 168 PCD*
Outlet flange drilling	4 holes x M16 x 125 PCD*
Working Pressure	16 Bar / 4 bar for HT°
Test Pressure	24 Bar / 6 bar for HT°
Working Temperature	STD:-40°C to +150°C/HT°:-40°C to +250°C
Design Temperature	190°C / 250°C for HT°

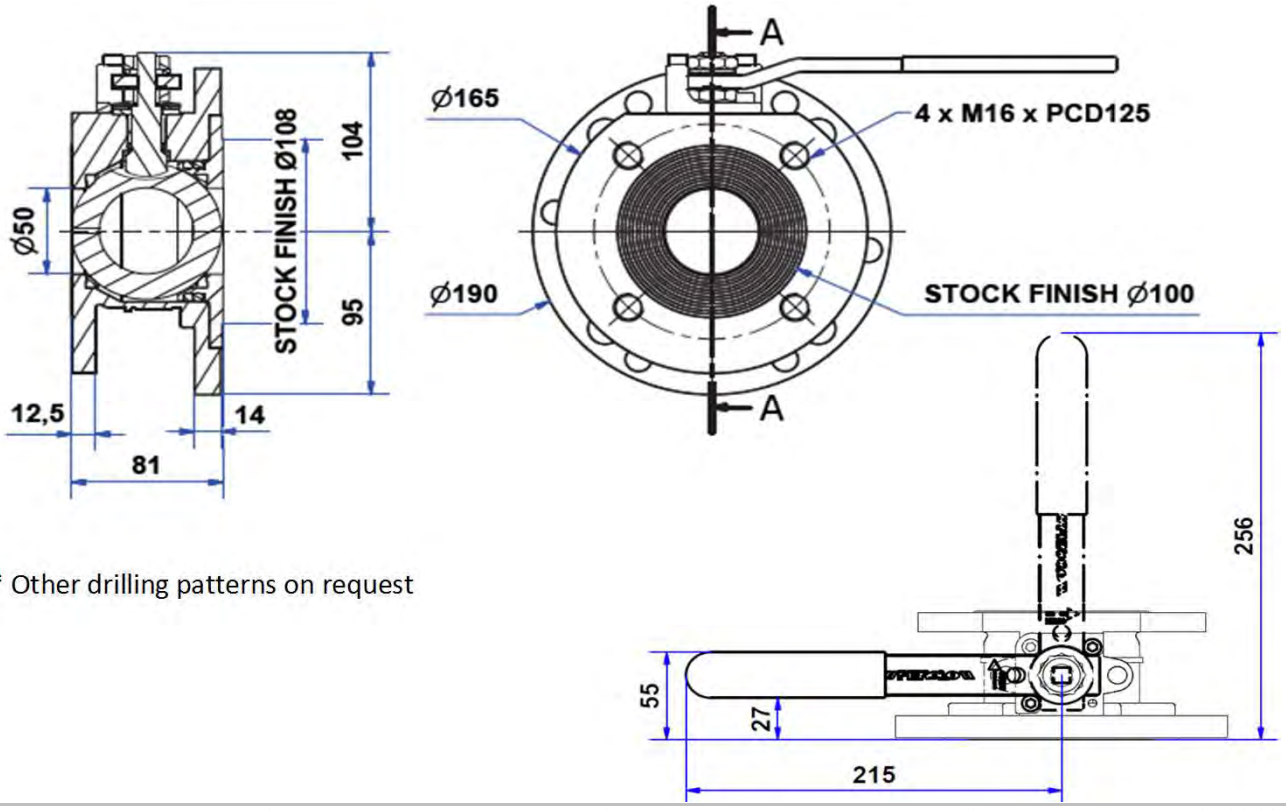
NB: temperature range vary according to o'ring material

ACCESSORIES	
Weld-in flange	11 90 36 00 00
Recess flange	11 90 98 00 00
Blind flange	11 95 88 00 00
Bolting Kit	17 12 12 91 00
Outlet gasket	11 90 29 00 00
Inlet gasket	12 91 40 00 00
Remote Control	RMT 53
Stud kit	11 12 82 93 00

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact design	Face to face 81 mm
Operation	Handle 1/4 Turn or Remote Control
Ball gaskets material	PTFE or PTFE/Graphite for HT°

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

3"/2" ATCO AHN5X
ND80/ND50 BALL VALVE
AHN5X

CODIFICATION

STANDARD PART NUMBER	AHN5	X	16T	1	2	0	0	S	0	0	.
YOUR PART NUMBER	AHN5	X	__	__	__	__	__	S	__	0	__

MATERIAL - Other on request	
316L / 1.4409 stainless steel	X

SEAT MATERIAL & MAWP	
Standard PTFE seal 16 Bar	16T
HT 250°C PTFE/Graphite seal 4 Bar	4H

INLET FLANGE - other on request	
4xØ18x160PCD & 6xØ14x168PCD	1

OUTLET FLANGE - other on request	
4 x Ø17 x 125 PCD	1
4 x M16 x 125 PCD	2
4 slotted x Ø18 x min 114 PCD	3

GAUGE CONNECTION	
Without	0
1/4" NPT gauge connection & drilled ball	1
1/4" BSP gauge connection & drilled ball	2
With drilled ball only (H2O2)	3

GAUGE PROTECTION	
Without	0
With	1

SEALING	
With	S

STEAM HEATING	
Without	0
With	S

HALAR COATING	
Without	0

For coated option, please refer to HIGH CORROSION Catalogue sheet

HANDLE	
Standard handle	.
Reverse handle	L

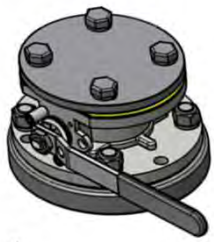
BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 07 20 00 E
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3"/2" ATCO AHN5X
ND80/ND50 BALL VALVE
AHN5X

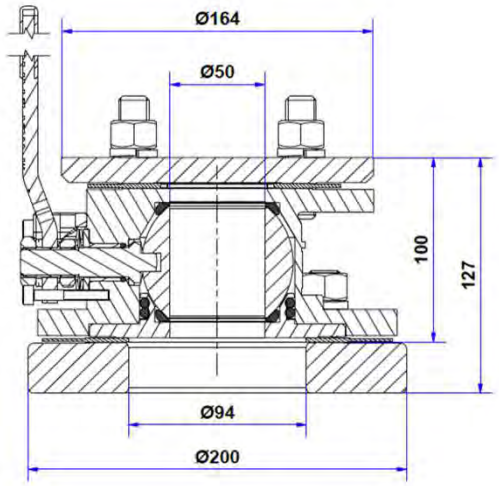
INSTALLATION

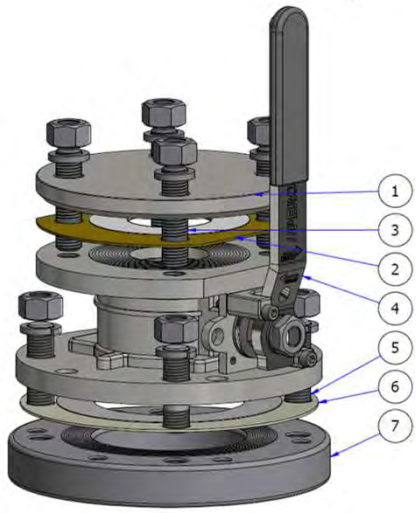
To use the ND50 ATCO™ full bore as an air inlet/outlet:



1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Screw the studs into the weld in flange (1).
3. Place the gasket (2) in the middle of the studs (3).
4. Fit ball valve on the weld in flange and secure with washers and nuts.
   The required tightening torque is 50 N.m +/-5. Please check PWAGEN001 for more details.
5. Assemble gasket and blind flange on the top and secure using bolting kit.

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs





ITEM	DESCRIPTION	SPECIFICATION	PART N°
1	Blind flange	Ext Ø 165mm, thickness 13.5mm, drilling: 4 x Ø18.5 x 125 PCD, 316L/1.4404 st st	11 95 88 00 00
3	Bolting Kit	4 x M16-50 studs, 4 x nuts, 4 x washers	17 12 12 91 00
2	Gasket	Ext Ø 165mm, int Ø 50mm, thickness 2mm, drilling: 4 x 18 x 125 PCD, PTFE	11 90 29 00 00
4	ND 50/80 ATCO	Inlet 3" Outlet 2" Ball valve	AHN5X16T120 0000
5	Stud Kit	4 x M16 36/28 studs, 4 x nuts, 4 x washers	11 12 82 93 00
6	Gasket	Ext Ø 185mm, int Ø 94mm, thickness 2.5mm, drilling: 4 x holes Ø18 x 160 PCD, CNAF/PTFE	12 91 40 00 00
7	Weld-in flange	Ext Ø 200mm, int Ø 94mm, thickness 27mm, drilling: 4 x M16 x 160 PCD & 6 x M12 x 168 PCD, 316L stainless steel	11 90 36 00 00

3"/2" ATCO AHN5X
ND80/ND50 BALL VALVE
AHN5X

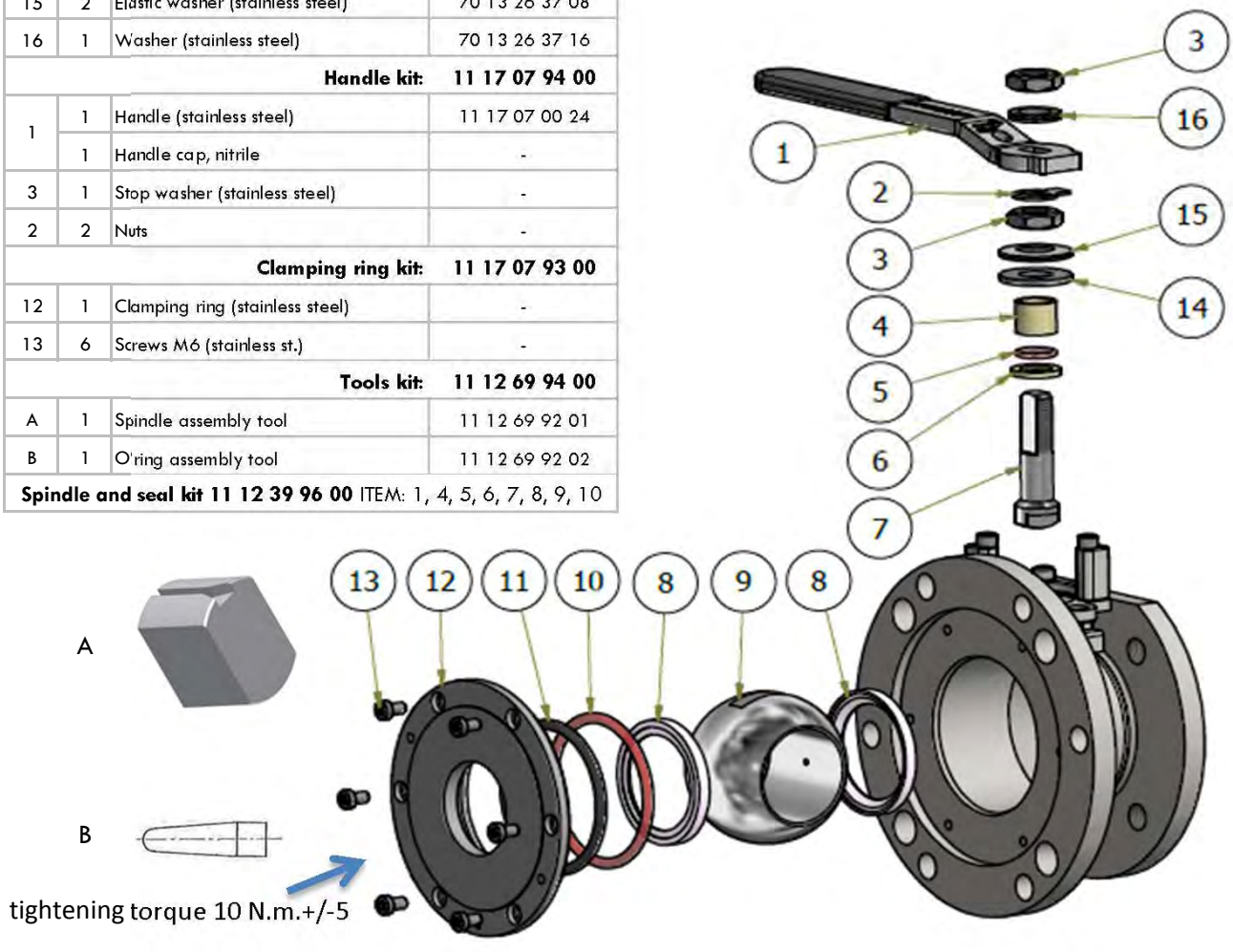
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N° Standard	PART N° High Temp
		Ball seals kit:	**11 12 42 92 00**	**11 12 42 96 00**
8	2	Ball seals (PTFE / PTFE/Graphite)	11 12 39 00 09	11 12 39 00 17
10	1 / 2	O'ring (FEP / PFA)	70 10 15 00 10	70 10 17 00 40
11	1	O'ring (Viton)	70 10 01 02 39	/
		Ball kit: 11 12 42 91 00		
9	1	Ball (stainless steel)	11 12 42 00 11	
		Drilled ball kit: 11 12 39 94 00		
9	1	Drilled ball (stainless steel)	11 12 39 00 11	
		Spindle:		
7	1	Spindle (stainless steel)	11 12 39 00 16	
		Spindle seals kit: 11 12 42 93 00		
4	1	Bearing bush (PTFE)	11 12 39 00 14	
5	1	O'ring (FEP encapsulated)	70 10 15 00 12	
6	1	Spindle seal (PTFE)	11 12 39 00 21	
		Bolting:		
14	1	Washer (stainless steel)	70 13 25 37 15	
15	2	Elastic washer (stainless steel)	70 13 26 37 08	
16	1	Washer (stainless steel)	70 13 26 37 16	
		Handle kit:	**11 17 07 94 00**	
1	1	Handle (stainless steel)	11 17 07 00 24	
	1	Handle cap, nitrile	-	
3	1	Stop washer (stainless steel)	-	
2	2	Nuts	-	
		Clamping ring kit:	**11 17 07 93 00**	
12	1	Clamping ring (stainless steel)	-	
13	6	Screws M6 (stainless st.)	-	
		Tools kit:	**11 12 69 94 00**	
A	1	Spindle assembly tool	11 12 69 92 01	
B	1	O'ring assembly tool	11 12 69 92 02	

Spindle and seal kit 11 12 39 96 00 ITEM: 1, 4, 5, 6, 7, 8, 9, 10

Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details

Nb: Perolo recommends
- the spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- both ball gaskets should be changed simultaneously
- when clamping ring is changed we recommend to also order ball seals kit



BALL VALVE

3" ATCO AFN8X
ND80 BALL VALVES
AFN8X

The Perolo ND80 ATCO™ FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	80mm (3")
Weight	15 Kg
Inlet flange drilling	8 holes x Ø17 x 160 PCD*
Outlet flange drilling	8 holes x Ø18 x 160 PCD*
Working Pressure	16 Bar / 4 bar for HT°
Test Pressure	24 Bar / 6 bar for HT°
Working Temperature	STD:-40°C to +150°C/HT°:-40°C to +250°C
Design Temperature	190°C / 250°C for HT°

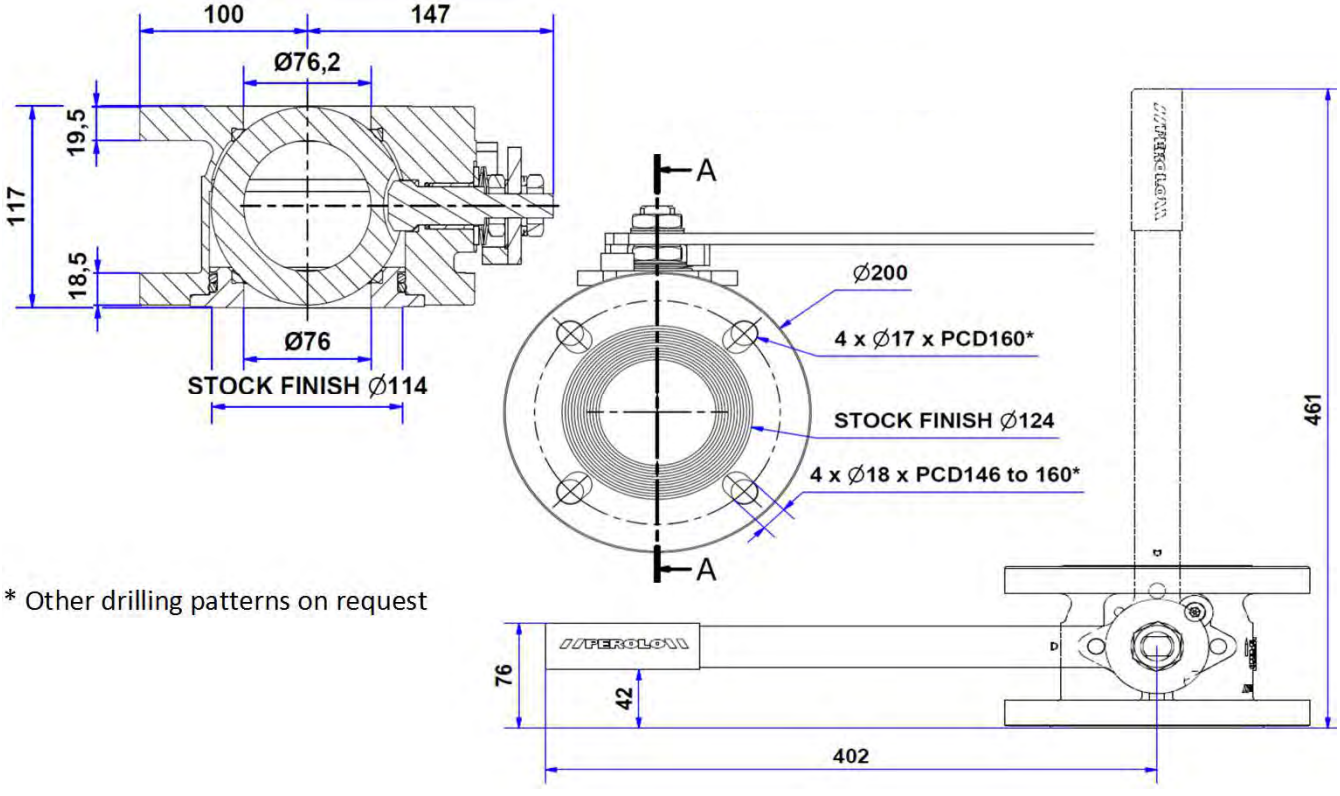
NB: temperature range vary according to o'ring material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, RMR, A-C-I

ACCESSORIES	
Weld-in flange	11 90 36 00 00
Recess flange	11 90 98 00 00
Blind flange	11 95 90 00 00
Bolting Kit	11 17 07 95 00
Gaskets	12 91 40 00 00
Remote Control	on request
Screw kit	19 11 08 90 02

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact design	Face to face 117 mm
Operation	Handle 1/4 Turn (or Remote Control)
Ball gaskets material	PTFE or PTFE/Graphite for HT°

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

3" ATCO AFN8X
ND80 BALL VALVES
AFN8X

CODIFICATION

STANDARD PART NUMBER	AFN8	X	16T	2	4	S	0	0	0

YOUR PART NUMBER AFN8 ___ ___ ___ ___ ___ ___ ___ ___ ___

BODY MATERIAL other on request	
316L / 1.4409 stainless steel	X

SEAT MATERIAL & MAWP	
Standard PTFE: 16b MAWP, 150°C	16T
High temp.: 4b MAWP, 250°C	4H

INLET FLANGE - other on request	
6 x Ø14 x 168.3 PCD	1
8 x Ø17 x 160 PCD	2
4 x Ø17 x 160 PCD	3
8 slots x Ø18 x 160 to 168 PCD	4
4 x Ø17 x 160 PCD & 6 x Ø14 x 168 PCD	5

OUTLET FLANGE - other on request	
4 open slots x Ø18 x min 146.1 PCD	1
8 x Ø18 x 160 PCD (ISO PN16)	2
8 slots x Ø18 x 152.4 to 160 PCD	3
4 x M16 & 4 x Ø18 x 160 PCD	4
4 slots x Ø18 x 146 to 160 PCD	5
4 x M16 & 4 x Ø18 x 160 PCD shifted 22°30'	6
4 xØ19x152,4 PCD + 8xØ18 x 160 PCD	7
8 x M16 x 160 PCD	8
4 slots x Ø18 x 146 to 160 PCD (with cut side)	9

SEALING	
with	S
Sealing & pin	N

PIN BRACKET ON OUTLET FLANGE	
with	P
without	0

STEAM HEATING	
without	0
with	S

HALAR COATING	
without	0

For coated option, please refer to HIGH CORROSION Catalogue sheet

HANDLE other on request	
standard	.
left	L

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 12 71 20 00 G
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" ATCO AFN8X
ND80 BALL VALVES
AFN8X

BALL VALVE

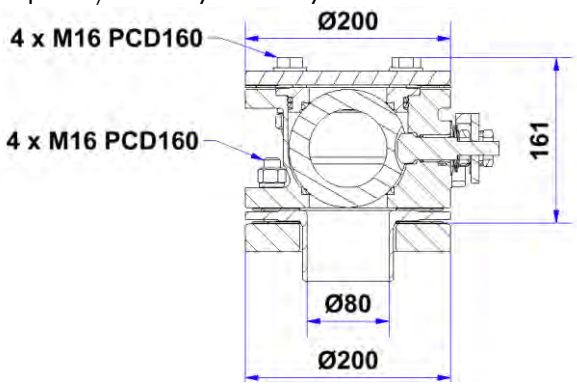
INSTALLATION

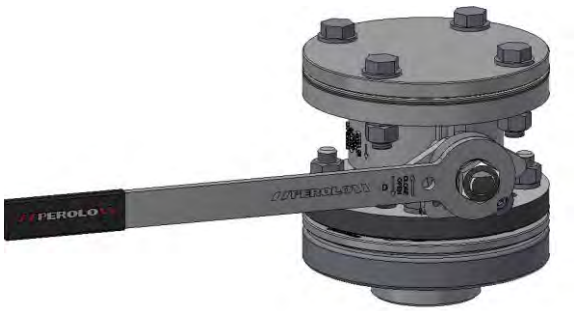
To use the ND80 ATCO™ full bore as an air inlet/outlet:

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Screw the studs into the weld in flange.
3. Place the gasket and siphon tube if any
4. Fit ball valve on the weld in flange and secure with washers and nuts.
   The required tightening torque is 70 N.m +/-5
5. Assemble gasket and blind flange on the top and secure using bolting kit.

Refer to our PWAGEN001 sheet for flange tightening details

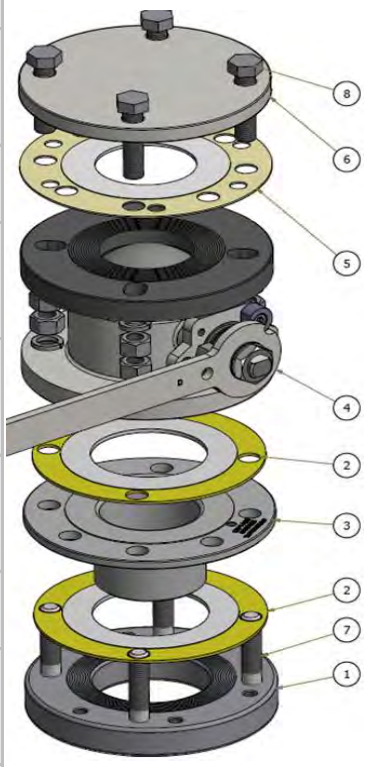
To be opened/closed by hand only





ITEM	DESCRIPTION	SPECIFICATION	PART N°
8	Bolting Kit	4 x M16-65 Screw 4 x M16 nuts and 4 x grower washers and 4 washers	19 11 08 90 02
7	Bolting Kit	4 x M16-64 studs 4 x M16 nuts and 4 x grower washers	11 17 07 95 00
6	Blind flange	ext Ø 200mm, thickness 13.5mm 4x18,5x160PCD, 316L stainless steel	11 95 90 00 00
5	Gasket	ext Ø 200mm, int Ø 81mm - CNAF/PTFE 8xØ18x160PCD and 6xØ14x168PCD	17 11 23 00 00
4	ND 80 ATCO	PN 16 Full Bore, 316 L stainless steel Inlet flange 8xØ17x160PCD Outlet flange 8xØ18x160PCD	AFN8X16T22S000
3	Syphon Tube	ext Ø 200mm, int tube Ø 80mm ext tube guide Ø 93mm, thickness 8mm 6x14xPCD168 & 4x18xPCD160 - 316L	12 92 18 10 00
2	Gasket	ext Ø 185mm, int Ø 94mm 4xØ18x 160PCD, CNAF/PTFE	12 91 40 00 00
1	Weld-in flange	ext Ø 200mm, int Ø 94mm thickness 26.5 mm 6xM12xPCD168 & 4xM16xPCD160 316L stainless steel	11 90 36 00 00

ASSEMBLY EXAMPLE
WITH SYPHON TUBE



if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 12 71 30 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" ATCO AFN8X
ND80 BALL VALVES
AFN8X

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N° Standard	PART N° High Temp
		Ball seals kit:	**11 12 52 92 00**	**11 14 09 91 50**
14	2	Ball seals (PTFE / PTFE/Graphite)	11 12 71 00 07	11 12 71 00 09
17	1 / 2	O'ring (FEP / PFA)	70 10 15 00 92	70 10 17 00 50
18	1	O'ring (Viton)	70 10 01 02 49	/
		Spindle seals kit:	**11 12 52 93 00**	**11 14 09 92 50**
5	1	Bearing bush (PTFE)	11 12 52 00 09	11 12 71 00 10
4	1	O'ring (FEP encapsulated)	70 10 15 00 91	70 10 17 00 60
3	1	Spindle seal (PTFE)	11 12 52 00 10	11 12 71 00 11
		Ball kit:	**11 12 52 91 00**	
15	1	Ball (stainless steel)	11 12 52 00 051	
		Spindle:		
2	1	Spindle (stainless steel)	11 12 52 00 24	
		Bolting:		
6	1	Washer (stainless steel)	70 13 25 37 24	
7	2	Elastic washer (stainless steel)	70 13 26 37 09	
20	1	Washer (stainless steel)	70 13 26 37 18	
		Handle kit:	**11 12 71 98 00**	
10	1	Handle (stainless steel)	11 14 05 00 28	
	1	Handle cap, Nitrile	-	
8	1	Nuts (stainless steel)	-	
9	2	Stop washer (stainless steel)	-	
		Clamping ring kit:	**11 12 71 90 00**	
16	1	Clamping ring (stainless steel)	-	
19	6	Screws M6 (stainless st.)	-	
		Tools kit:	**11 12 71 96 00**	
A	1	Spindle assembly tool	11 12 71 93 01	
B	1	O'ring assembly tool	11 12 71 93 02	

Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details

Nb: Perolo recommends
- the spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- both ball gaskets should be changed simultaneously
- when clamping ring is changed we recommend to also order ball seals kit

BALL VALVE

3" ATCO COMPACT AFC8X
ND80 BALL VALVES
AFC8X

The Perolo 3" ATCO™ is a new range of PEROLO ball valve designed to be used as a fluid inlet/outlet valve. With a compact design, the 3" ATCO™ is a lighter & economical solution. It is equipped with a self-retaining base gasket.

SPECIFICATION	
Nominal diameter	80mm (3")
Weight	12.4 Kg
Inlet flange drilling	8 holes x Ø17 x 160 PCD*
Outlet flange drilling	8 holes x Ø18 x 160 PCD*
Working Pressure	7 Bar
Test Pressure	10,5 Bar
Working Temperature	-40° to +150°C
Design Temperature	-40°C to +190°C

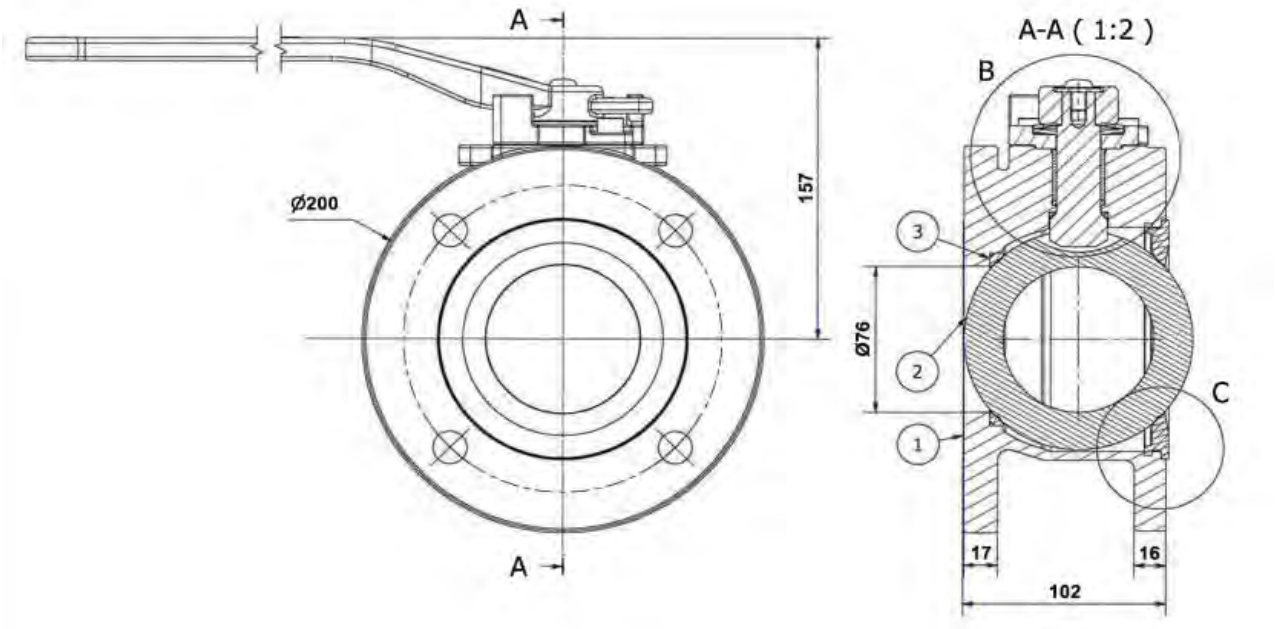
NB: temperature range vary according to o'ring material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

ACCESSORIES	
Weld-in flange	11 90 36 00 00
Recess flange	11 90 98 00 00
Blind flange	11 95 90 00 00
Bolting Kit	11 17 07 95 00
Gaskets	12 91 40 00 00
Remote Control	on request
Screw kit	19 11 08 90 02

STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1.4408 stainless steel
Compact design	Face to face 102mm
Operation	Handle 1/4 Turn (or Remote Control)
Ball gaskets material	PTFE

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 12 11 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

3" ATCO COMPACT AFC8X
ND80 BALL VALVES
AFC8X

CODIFICATION

STANDARD PART NUMBER	AFC8	X	7T	2	4	S	0	0	.

YOUR PART NUMBER AFC8 _ _ _ _ _ _ _ _ _

BODY MATERIAL other on request

316 / 1.4408 stainless steel	X

SEAT MATERIAL & MAWP

Standard PTFE & 7b MAWP	7T

INLET FLANGE - other on request

6 x Ø14 x 168.3 PCD	1
8 x Ø17 x 160 PCD	2
4 x Ø17 x 160 PCD	3
8 slots x Ø18 x 159 to 168 PCD	4
4 x Ø17 x 160 PCD & 6 x Ø14 x 168 PCD	5
4x Ø17 x 160 PCD & cut side	6

OUTLET FLANGE - other on request

4 open slots x Ø18 x min 146.1 PCD	1
8 x Ø18 x 160 PCD (ISO PN16)	2
8 slots x Ø18 x 152.4 to 160 PCD	3
4 x M16 & 4 x Ø18 x 160 PCD	4
4 slots x Ø18 x 146 to 160 PCD	5
4 x M16 & 4 x Ø18 x 160 PCD shifted 22°30'	6
4 xØ19x152,4 PCD + 8xØ18 x 160 PCD	7
8 x M16 x 160 PCD	8
4 slots x Ø18 x 146 to 160 PCD (with cut side)	9

SEALING

with	S
Sealing & pin	N

PIN BRACKET ON OUTLET FLANGE

with	P
without	0

HALAR COATING

without	0

For coated option, please refer to HIGH CORROSION Catalogue sheet

HANDLE other on request

standard	.
left	L

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 12 11 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" ATCO 90° ALN5X or AKN5X
ND50 BALL VALVE
ALN5X or AKN5X

The Perolo 2" 90° angled ATCO™ FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve and offers 2" or 3" inlet flange.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	5.2 Kg
Inlet flange drilling	ALN5: 4 holes Ø17 x PCD 125mm AKN5: 4 holes Ø18 x PCD 160mm
Outlet flange drilling	2" BSP*
Working Pressure	6 Bar
Test Pressure	9 Bar
Working Temperature	-40° to +150°C
Design Temperature	190°C

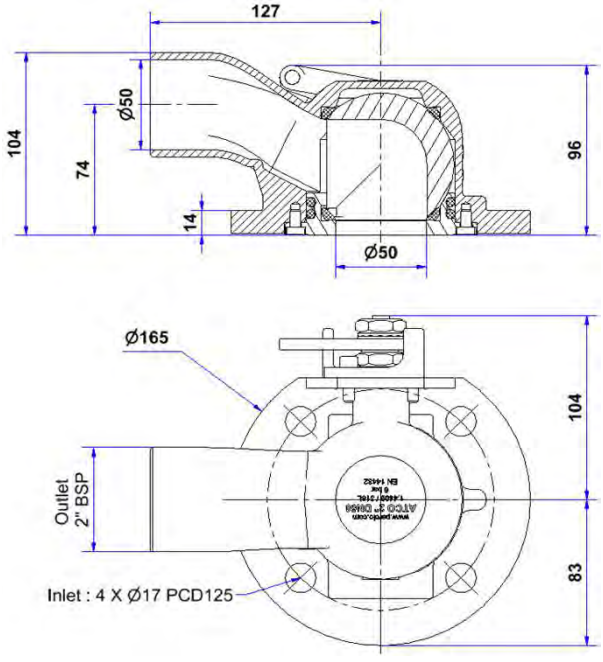
ACCESSORIES	
Weld-in flange (ALN5)	11 90 25 00 00
Weld-in flange (AKN5)	11 90 36 00 00
Recess flange	on request
Cap 2" BSP	11 17 11 20 00
Stud Kit	11 12 82 93 00
Inlet Gasket (ALN5)	11 90 29 00 00
Inlet Gasket (AKN5)	12 91 40 00 00
Remote Control	RMT 53
Remote Control Handle	11 14 75 91 00

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I

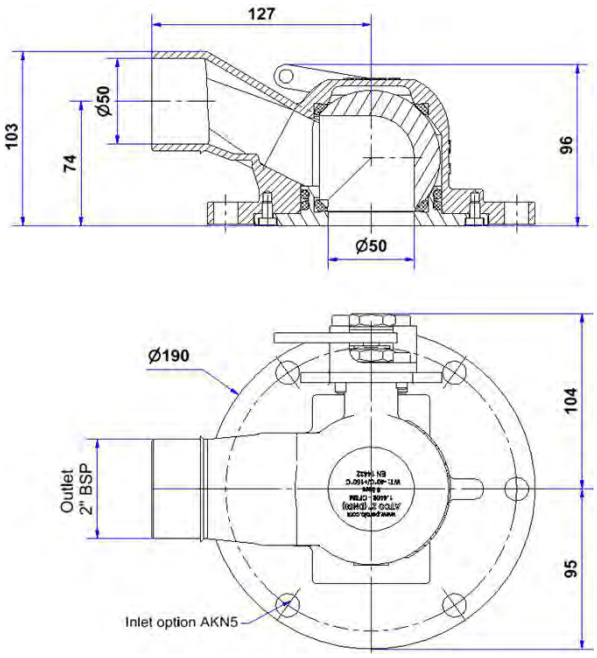
STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact design	Face to face 103 mm
Operation	Handle 1/4 Turn or Remote Control
Ball gaskets material	PTFE

DIMENSIONAL DRAWING

ALN5: ND50 inlet flange



AKN5: ND80 inlet flange



* Other drilling patterns on request

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 60 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" ATCO 90° ALN5X or AKN5X
ND50 BALL VALVE
ALN5X or AKN5X

CODIFICATION

STANDARD PART NUMBER	ALN5/AKN5	X	6T	1	1	W

YOUR PART NUMBER	AL	5		6T			

INLET FLANGE SIZE	
2" inlet flange	N
3" inlet flange	K

MATERIAL	
316L / 1.4409 stainless steel	X

MAWP & SEAT MATERIAL	
6 bar & PTFE	6T

INLET FLANGE DRILLING	
ALN5: 4 holes Ø17 x PCD 125	1
AKN5: 4 holes Ø18 x PCD 160	1
AKN5: 6 holes Ø14 x PCD 168,5	2
AKN5: 4 holes Ø18 x PCD 160 & 6 holes Ø14 x PCD 168,5	3

OUTLET CONNECTION	
2" BSP	1
2" NPT	2
90° elbow	3

HANDLE	
Manually operated	M
Remote Control	W

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 60 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

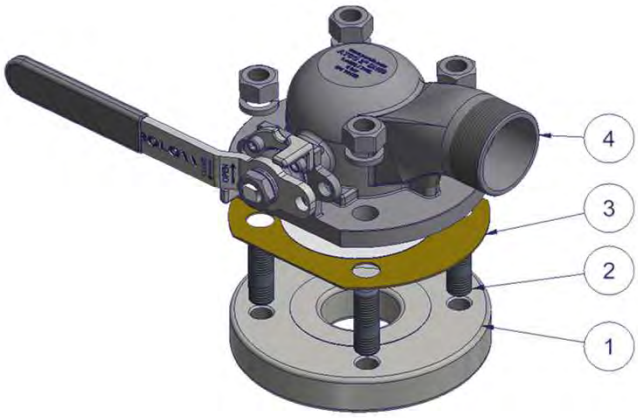
2" ATCO 90° ALN5X or AKN5X
ND50 BALL VALVE
ALN5X or AKN5X

INSTALLATION

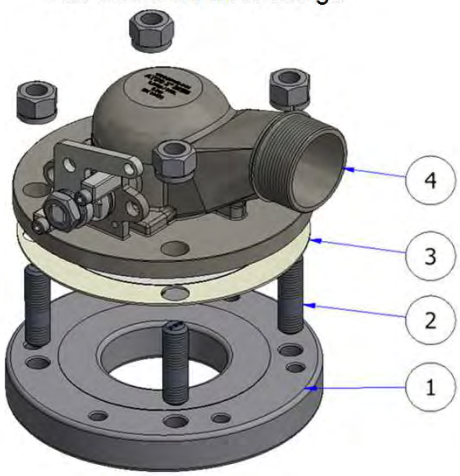
To use the ND50 ATCO™ full bore as an air inlet/outlet:

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Screw the studs into the weld in flange (1).
3. Place the gasket (2) in the middle of the studs (3).
4. Fit ball valve on the weld in flange and secure with washers and nuts.
   The required tightening torque is 50 N.m +/-5. Please check PWAGEN001 for more details.

ALN5: ND50 inlet flange



AKN5: ND80 inlet flange



ITEM	DESCRIPTION	SPECIFICATION ALN5	PART N°
4	ND50 ATCO Ball Valve	PN 16 Full Bore, 316 L stainless steel Inlet flange 4xø17x125PCD, Outlet flange 2" BSP	ALN5X6T11M
3	Bolting Kit	4 x M16 40/30 studs 4 x M16 nuts and 4 x grower washers	11 12 82 93 00
2	Gasket	ext ø 165mm, int ø 50mm 4xø18x 125PCD, CNAF/PTFE, thickness 2mm	11 90 29 00 00
1	Weld-in flange	ext ø 165mm, int ø 50mm, thickness 26,5mm 4xM16x125PCD, 316L stainless steel	11 90 25 00 00

ITEM	DESCRIPTION	SPECIFICATION AKN5	PART N°
4	ND50 ATCO Ball Valve	PN 16 Full Bore, 316 L stainless steel Inlet flange 4xø18x160PCD, Outlet flange 2" BSP	AKN5X6T11W
3	Bolting Kit	4 x M16 40/30 studs 4 x M16 nuts and 4 x grower washers	11 12 82 93 00
2	Gasket	ext ø 185mm, int ø 94mm 4xø18x 160PCD, CNAF/PTFE, thickness 2,5mm	12 91 40 00 00
1	Weld-in flange	ext ø 200mm, int ø 94mm, thickness 27mm, 316L st. st. 6x M12 x PCD Ø168 & 4 x M16 x PCD Ø160	11 90 36 00 00

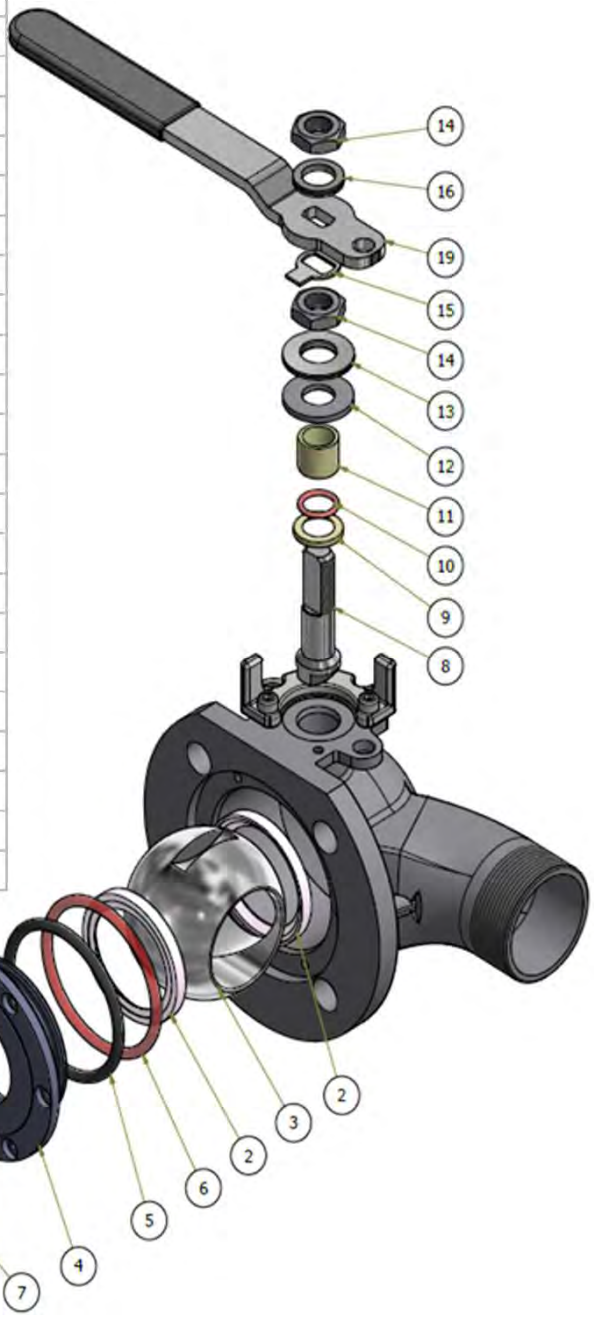
if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

2" ATCO 90° ALN5X or AKN5X
ND50 BALL VALVE
ALN5X or AKN5X

MAINTENANCE

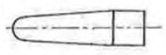
ITEM	QTY	DESIGNATION	PART N°
		Ball seals kit:	**11 12 42 92 00**
2	2	Ball seals (PTFE)	11 12 39 00 09
6	1	O'ring (FEP encapsulated)	70 10 15 00 10
5	1	O'ring (Viton)	70 10 01 02 39
		Ball kit:	
3	1	Ball (stainless steel)	11 14 59 00 11
		Spindle kit:	
8	1	Spindle (stainless steel)	11 12 39 00 16
		Spindle seals kit:	**11 12 42 93 00**
11	1	Bearing bush (PTFE)	11 12 39 00 14
10	1	O'ring (FEP encapsulated)	70 10 15 00 12
9	1	Spindle seal (PTFE)	11 12 39 00 21
		Washers:	**11 12 42 93 00**
12	1	Washer (stainless steel)	70 13 25 37 15
13	2	Elastic washer (stainless steel)	70 13 26 37 08
		Handle kit:	**11 14 16 92 00**
19	1	Handle	11 12 39 00 22
	1	Handle cap, nitrile	-
14	2	Nuts (stainless steel)	-
15	1	Stop washer (stainless steel)	-
16	1	Washer (stainless steel)	-
		Remote control lever kit:	**11 14 75 91 00**
19	1	Remote control lever (stainless steel)	11 14 70 00 01
14	2	Nuts (stainless steel)	-
15	1	Stop washer (stainless steel)	-
16	1	Washer (stainless steel)	-
		Clamping ring kit:	**11 12 69 93 00**
4	1	Clamping ring (stainless steel)	-
7	6	Screws M6 (stainless st.)	-
		Tools kit:	**11 12 69 94 00**
A	1	Spindle assembly tool	11 12 69 92 01
B	1	O'ring assembly tool	11 12 69 92 02

Nb: Perolo recommends
- the spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- both ball gaskets should be changed simultaneously
- when clamping ring is changed we recommend to also order ball seals kit



A

B



tightening torque 10 N.m.+/-5

BALL VALVE

4" ATCO AF10
ND100 BALL VALVE
AF10

The Perolo ND100 ATCO™ FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	100mm (4")
Weight	27 Kg
Inlet flange drilling	8 holes x Ø18 x 180 PCD*
Outlet flange drilling	8 holes x M16 x 180 PCD (for studs only)*
Working Pressure	16 Bar (4 Bar for HT version)
Test Pressure	24 Bar (6 Bar for HT version)
Working Temperature	Standard: -40°C to +150°C HT & Steam heated: -40°C to +250°C

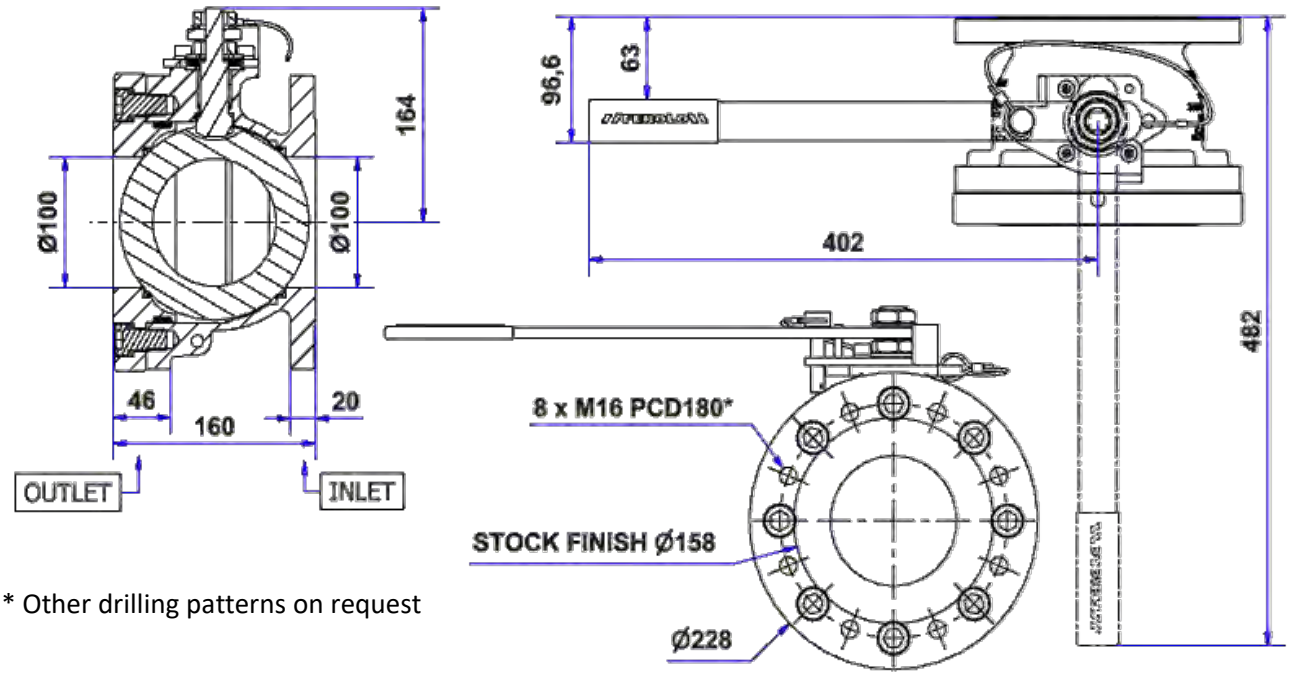
NB: temperature range vary according to o'ring material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I

ACCESSORIES	
Recess flange	On request
Blind flange	11 95 37 00 00
Bolting kit St. Steel	17 12 10 96 00
Gaskets PTFE	11 15 26 00 01
Outlet bolt. kit Carbon st.	19 14 01 91 00
Gaskets TOPCHEM	19 90 05 30 01

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L/1.4409 stainless steel - other on request
Compact design	Face to face 160 mm
Operation	Handle 1/4 Turn
Ball gaskets material	PTFE

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

4" ATCO AF10
ND100 BALL VALVE
AF10

CODIFICATION

STANDARD PART NUMBER	AF10	X	16T	1	1	0	0	.

YOUR PART NUMBER AF10 ____ ____ ____ ____ ____ ____ ____ ____

BODY MATERIAL

Body: 316L/1.4409 stainless steel Ball: 316L stainless steel Ring clamp: 316 stainless steel	X
Body: GP240GH carbon steel Ball: 304L stainless steel Ring clamp: P235GH carbon steel	C
Body: 316L/1.4409 stainless steel Ball: 304L stainless steel Ring clamp: P235GH carbon steel	B
Body: 304/1.4308 stainless steel Ball: 304L stainless steel Ring clamp: 304 stainless steel	Y

SEAL MATERIAL & MAWP

Standard PTFE: 150°C, 16b MAWP	16T
High temp: WT 250°C, 4b MAWP	4H

INLET FLANGE

8 x M16 x 180 PCD (ISO PN16 ND100)	1
8 x Ø18 x 180 PCD (ISO PN16 ND100)	2
8 x Ø19 x 190.5 PCD (ANSI B16.5 S150)	3

OUTLET FLANGE

8 x M16 x 180 PCD (ISO PN16 ND100)	1
8 x M16 x 190.5 PCD (ANSI B16.5 S150)	2

BALL TYPE - other on request

Solid ball	0

HANDLE TYPE

Standard right handle	0
Left handle	L

STEAM HEATING

Without	.
With	V

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 21 20 00 H
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

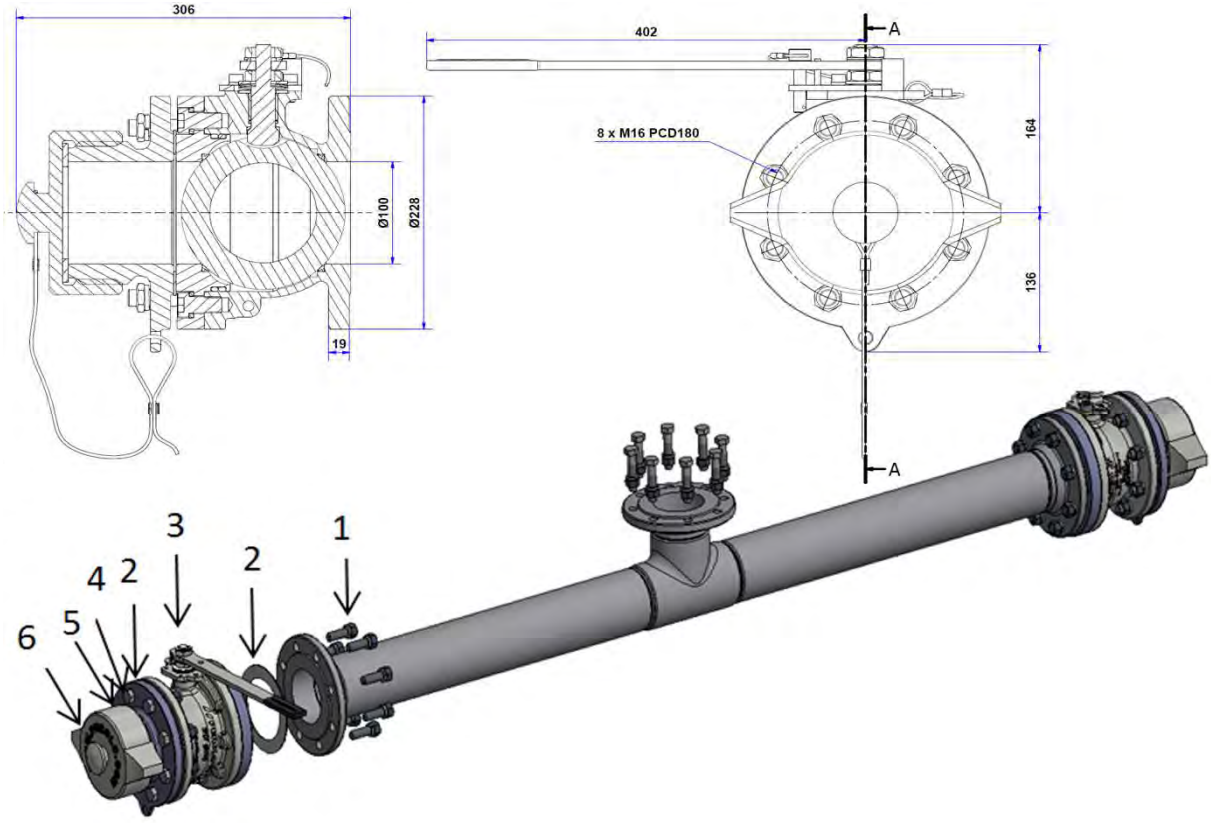
4" ATCO AF10
ND100 BALL VALVE
AF10

INSTALLATION

To use the ND100 ATCO™ full bore as an air inlet/outlet:

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Screw the studs into the weld in flange.
3. Place the gasket in the middle of the studs.
4. Fit ball valve on the flange and secure with washers and nuts.
   The required tightening torque is between 100 mN +/-5 Please check PWAGEN001 for more details.
5. Assemble gasket and blind flange on the top and secure using bolting kit.

NB: only studs can be installed on the thick flange



ITEM	DESCRIPTION	SPECIFICATION	PART N°
6	Cap	Cap kit including 1 cap (PA6 black), 2 sleeves, 1 cable, 1 seal (white nitrile)	19 90 55 00 00
5	Outlet nozzle	8xø18x180PCD, ext ø 228mm, int ø 100mm, carbon steel, with 5 1/2" BSW thread	19 14 01 20 03
3	ND100 ATCO Ball Valve	PN 16 Full Bore, 316 L carbon steel, Inlet flange 8xø18x180PCD, Outlet flange 8xM16x180PCD	AF10C16T110000
4&2	Bolting Kit + flat gasket	8 x M16 46 bm 24 studs 8 x M16 nuts and 8 x grower washers + flat gasket	19 14 01 91 00
2	Gasket	ext ø 220mm, int ø 117mm 8xø18x 180PCD CNAF / PTFE	19 90 05 30 01
1&2	Screw Kit + flat gasket	8 x M16-70 screws 8 x M16 nuts and 16 x grower washers + flat gasket	19 14 01 92 00

BALL VALVE

VALVES SOLUTIONS FOR YOUR

4" ATCO AF10
ND100 BALL VALVE
AF10

RAIL

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°	
			Standard	High Temp
		Ball seals kit:	**11 14 21 90 00**	**on request**
8	2	Ball seals (PTFE / PTFE/Graphite)	11 14 21 00 02	11 14 21 00 05
6	1 / 2	O'ring (FEP / PFA/Silicone)	70 10 17 00 42	70 10 17 00 66
5	1	O'ring (Viton)	70 10 01 02 92	/
		Spindle seals kit:	**11 12 52 93 00**	**on request**
12	1	Bearing bush (PTFE)	11 12 52 00 09	11 12 52 00 27
11	1	O'ring (FEP encapsulated)	70 10 15 00 91	70 10 17 00 60
10	1	Spindle seal (PTFE)	11 12 52 00 10	11 12 52 00 10
		Ball kit:		
7	1	Solid ball (stainless steel)	11 14 21 00 01	
7	1	Hollow ball (stainless steel)	11 14 21 00 04	
		Spindle kit:	**11 12 71 95 00**	
9	1	Spindle (stainless steel)	11 12 52 00 24	
13	2	Washer (stainless steel)	-	
14	2	Elastic washer (stainless steel)	-	
15	2	Nuts (stainless steel)	-	
16	1	Stop washer (stainless steel)	-	
20	1	Washer (stainless steel)	-	
		Bolting:		
19	3	Screw, (stainless steel)	70 13 01 37 15	
22	3	Washer, (stainless steel)	70 13 26 37 18	
23	1	Stop pin sealing (stainless steel)	11 14 21 93 00	
		Handle kit:	**11 12 71 97 00**	
17	1	Handle (stainless steel)	11 12 52 00 20	
	1	Handle cap, nitrile	-	
15	2	Nuts (stainless steel)	-	
16	1	Stop washer (stainless steel)	-	
		Clamping ring kit (stainless steel):	**11 14 21 91 00**	
2	1	Clamping ring (stainless steel)	11 14 21 00 03	
2	N/A	Clamping ring (carbon steel)*	19 14 01 00 03	
4	8	Washers (stainless steel)	70 13 26 37 02	
3	8	Screws M16 (stainless st.)	70 13 01 37 34	
		Tools kit:		
A	1	Spindle assembly tool	11 14 21 97 01	
B	1	O'ring assembly tool	11 12 71 93 02	

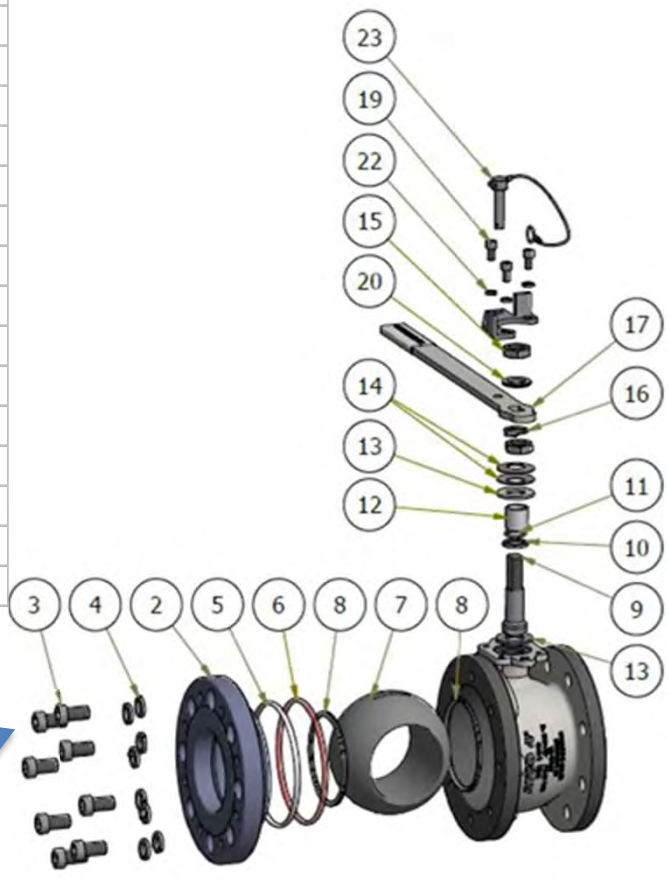
*Carbon st clamping ring kit on request (not included in st st kit)

Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details

Nb: Perolo recommends
- the spindle seals should be changed simultaneously to maintain the ball valve spindle perfectly tight.
- both ball gaskets should be changed simultaneously
- when clamping ring is changed we recommend to also order ball seals kit

A

B



tightening torque 120 N.m.+/-5

BALL VALVE

1 1/2" ATCO Pneumatic AGN4X

ND40 BALL VALVE
AGN4X

The Perolo ND40 ATCO™ pneumatic self-contained ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	40mm (1 1/2")
Weight	12.5 Kg
Inlet flange drilling	4 holes x Ø18 x 110 PCD*
Outlet flange drilling	please refer to spec sheet of AFN4X
Pneumatic Air Pressure	4 to 8 bar
Working Pressure	16 Bar
Test Pressure	24 Bar
Working Temperature	Valve : -40°C to +150°C
	Actuator : -20°C to +80°C

NB: temperature range vary according to o'ring material

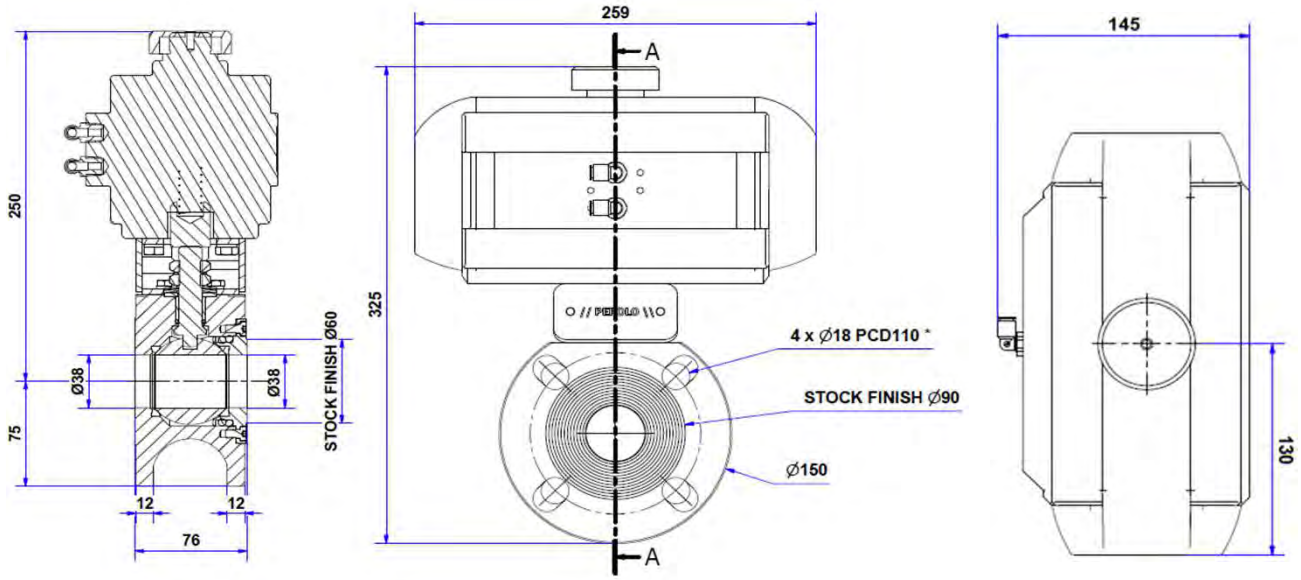
CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I

ACCESSORIES	
Weld-in flange	11 96 63 00 00
Blind flange EN14025	11 95 87 00 00
Stud kit	17 12 12 91 00
Gaskets	12 91 90 00 00
Bolting kit	12 91 90 00 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact design	Face to face 76 mm
Operation	Pneumatic - 1/4 Turn single acting actuator
Ball gaskets material	PTFE

CODIFICATION	
Pneumatic Ball Valve - Inlet flange: 4 x 18 / PCD110mm, outlet flange: 4 x M16 / PCD110mm, 316 L stainless steel	AGN4X16T1200000

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 19 20 00 E
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 1/2" ATCO Pneumatic AGN4X
ND40 BALL VALVE
AGN4X

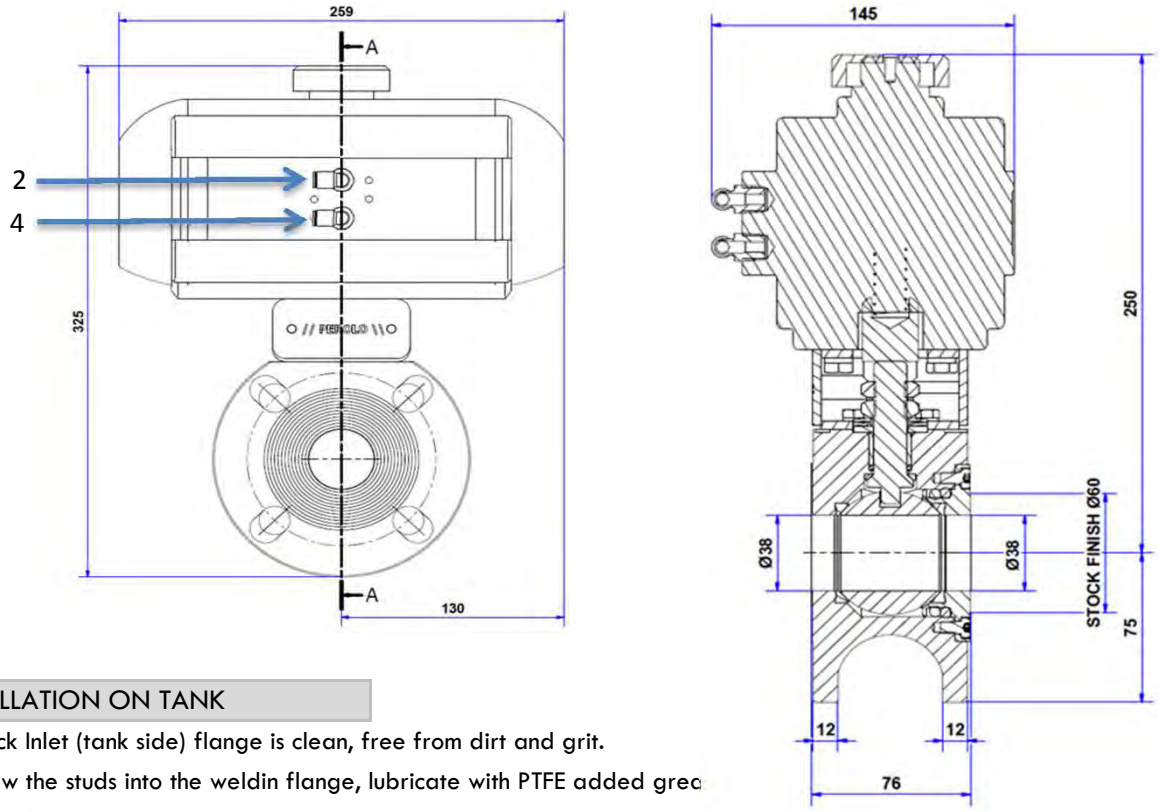
INSTALLATION

CONDITIONS

Caution: Do not disassemble the actuator from ATCO™ Ball valve.
For better and long time performances, assembly and settings are realized in PEROLO factory.

PNEUMATIC OPERATOR

- Minimum operating pressure: 4 bar (Maximum operating pressure: 8bar)
- Actuator connections: Ø6mm ext
- Actuator Working Temperature: -40°C to 80°C
- For corrosive products application, provide a protection to avoid contact with actuator



INSTALLATION ON TANK

1. Check Inlet (tank side) flange is clean, free from dirt and grit.
2. Screw the studs into the weldin flange, lubricate with PTFE added grea
3. Place the gasket
4. Install the valve on the flange, secure with washer and nuts *(refer to PWAGEN001 fort bolt tightening sequence)*
5. Tighten the nuts torque : 40 N.m +/- 5
6. Install gasket, bolting kit and blind flange on the valve outlet flange
(refer to PWAGEN001 fort bolt tightening sequence)
7. Open and close the valve 3 times to check the operationnal torque.

SETTING UP THE ACTUATOR

4. Connect inlet to connection "2" and outlet to connection "4" written on the

It is a simple effect actuator able to work with only one inlet connection. However, we recommend to use two inlet connections. It reduces the time of closure. Use lubricated air to keep the actuator in good working condition.

The air feeding should be rated from 5.5 bar up to 6.5 bar (do not exceed 8 bar).

NB: To get information about maintenance, please refer to the AFN4X maintenance sheet PWA1114614000

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

2" ATCO Pneumatic AGN5X

ND50 BALL VALVE AGN5X

The Perolo ND50 ATCO™ pneumatic FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	12.5 Kg
Inlet flange drilling	4 holes x Ø17 x 125 PCD
Outlet flange drilling	please refer to spec sheet of AFN5X
Pneumatic Air Pressure	4 to 8 bar
Working Pressure	16 Bar
Test Pressure	24 Bar
Working Temperature	Valve : -40°C to +150°C
	Actuator : -20°C to +80°C

NB: temperature range vary according to o'ring material

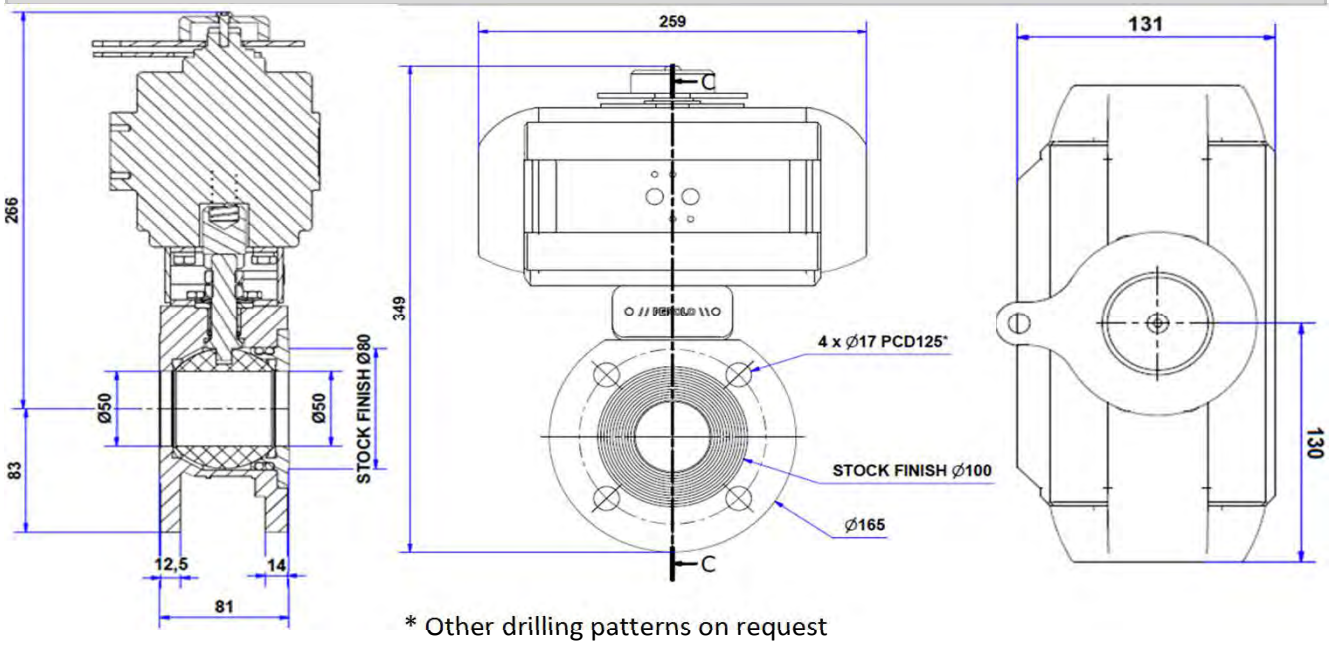
CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I

ACCESSORIES	
Weld-in flange	11 90 25 00 00
Blind flange EN14025	11 95 88 00 00
Stud kit	17 12 12 91 00
Gaskets	11 90 29 00 00
Bolting kit	11 90 29 00 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact design	Face to face 81 mm
Operation	Pneumatic - 1/4 Turn single acting actuator
Ball gaskets material	PTFE or PTFE/Graphite for HT°

CODIFICATION	
Pneumatic Ball Valve - Inlet flange: 4 x 17 / PCD125, outlet flange: 4 x 17 / PCD125, 316 L stainless steel	AGN5X16T1100S00

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

2" ATCO Pneumatic AGN5X
ND50 BALL VALVE
AGN5X

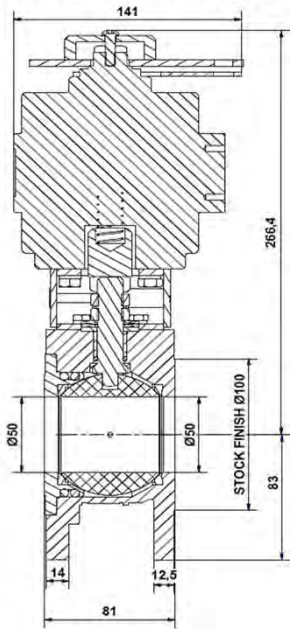
INSTALLATION

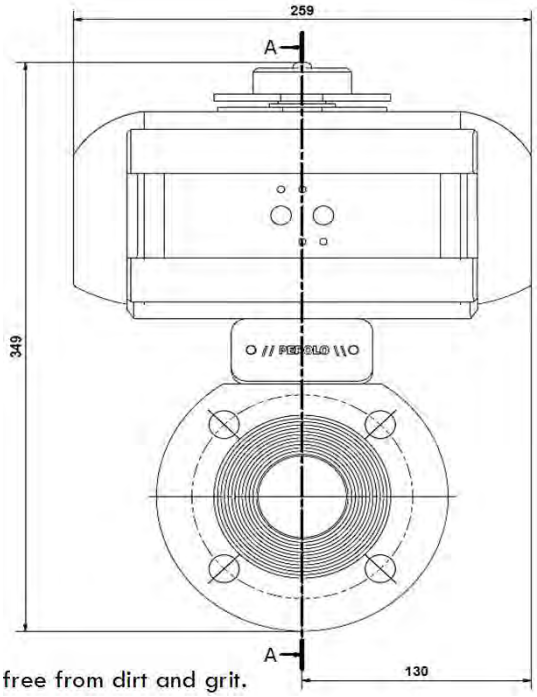
CONDITIONS

Caution: Do not disassemble the actuator from ATCO™ Ball valve.
For better and long time performances, assembly and settings are realized in PEROLO factory.

PNEUMATIC OPERATOR

- Minimum operating pressure: 4 bar (Maximum operating pressure: 8bar)
- Actuator connections: Ø6mm ext
- Actuator Working Temperature: -40°C to 80°C
- For corrosive products application, provide a protection to avoid contact with actuator





INSTALLATION ON TANK

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Screw the studs into the weld in flange. Lubricate the threads with PTFE added grease.
3. Place the gasket.
4. Fit ball valve on the weld in flange and secure with washers and nuts.
   Tighten the nuts torque : 50 N.m +/-5 (use our bolting kit A2-70 or A4-70 class)
5. Assemble gasket and blind flange on the top same bolting procedure them above
6. check torque after 24 hours

Refer to our PWAGEN001 sheet for flange tightening details

SETTING UP THE ACTUATOR

4. Connect inlet to connection "2" and outlet to connection "4" written on the actuator

It is a simple effect actuator able to work with only one inlet connection. However, we recommend to use two inlet connections. It reduces the time of closure. Use lubricated air to keep the actuator in good working condition. The air feeding should be rated from 5.5 bar up to 6.5 bar **(do not exceed 8 bar).**

NB: To get information about maintenance, please refer to the AFN5X maintenance sheet PWA1112694000

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

3" ATCO Pneumatic AGN8X

ND80 BALL VALVE AGN8X

The Perolo ND80 ATCO™ pneumatic self-contained ball valve is designed to be used as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	80mm (3")
Weight	12.5 Kg
Inlet flange drilling	8 holes x Ø17 x 160 PCD*
Outlet flange drilling	8 holes x Ø18 x 160 PCD*
Pneumatic Air Pressure	4 to 8 bar
Working Pressure	16 Bar
Test Pressure	24 Bar
Working Temperature	Valve : -40°C to +150°C
	Actuator : -20°C to +80°C

NB: temperature range vary according to o'ring material

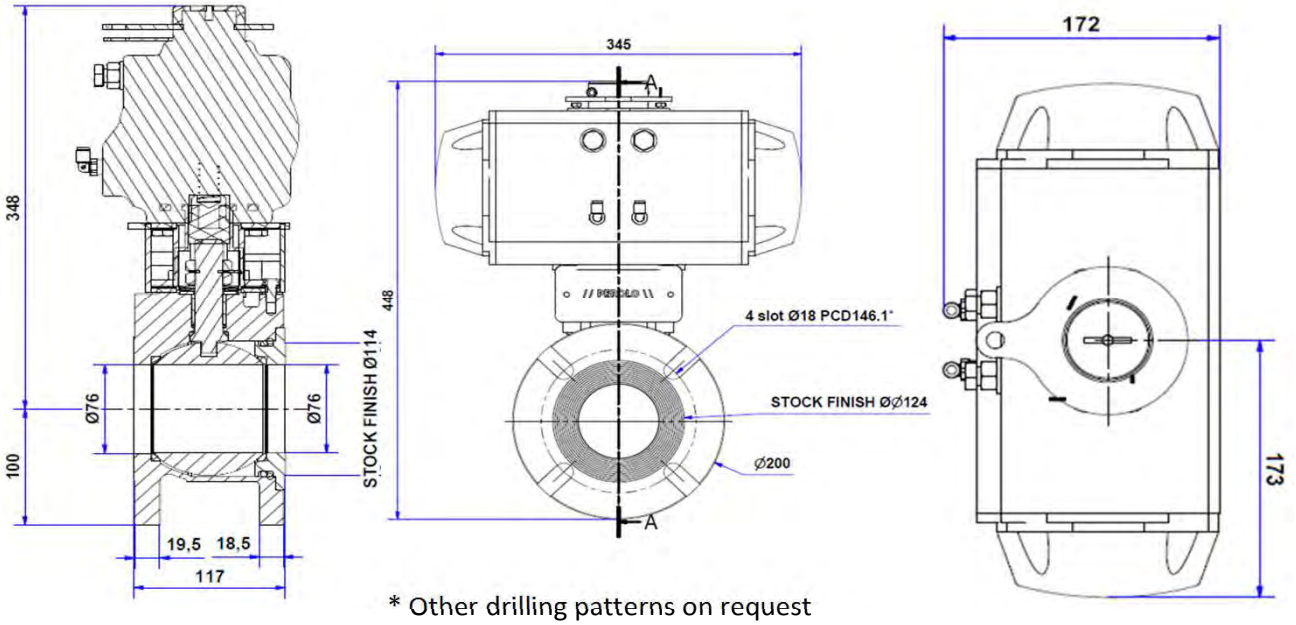
CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I

ACCESSORIES	
Weld-in flange	11 90 36 00 00
Blind flange	11 95 90 00 00
Bolting Kit	11 17 07 95 00
Gaskets	12 91 40 00 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact design	Face to face 117 mm
Operation	Pneumatic - 1/4 Turn single acting actuator
Ball gaskets material	PTFE

CODIFICATION	
Pneumatic Ball Valve - Inlet flange: 8 x Ø17 x PCD Ø160, outlet flange: 8 x Ø18 x PCD Ø160, 316 L stainless steel	AGN8X16T22U000

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

3" ATCO Pneumatic AGN8X
ND80 BALL VALVE
AGN8X

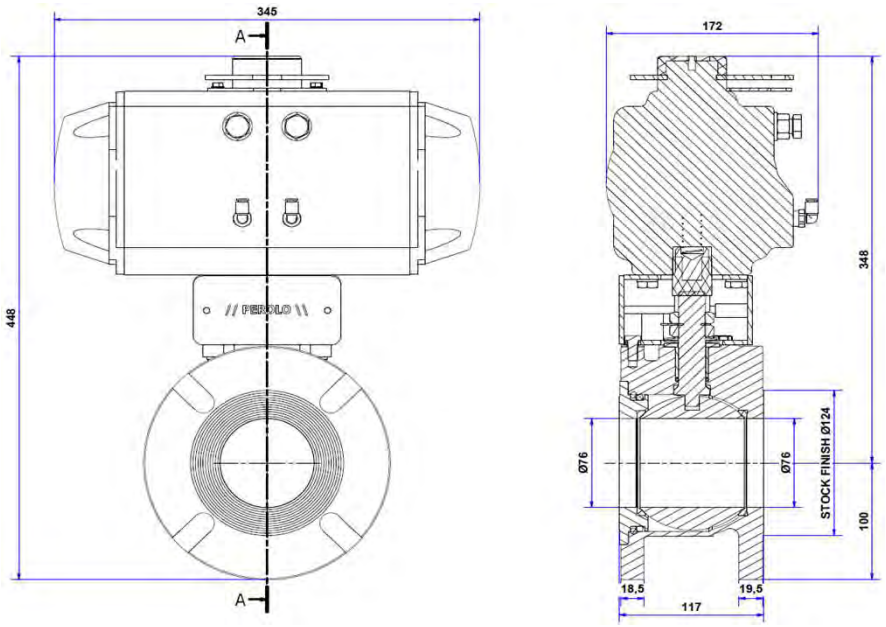
INSTALLATION

CONDITIONS

Caution: Do not disassemble the actuator from ATCO™ Ball valve.
For better and long time performances, assembly and settings are realized in PEROLO factory.

PNEUMATIC OPERATOR

- Minimum operating pressure: 4 bar (Maximum operating pressure: 8bar)
- Actuator connections: Ø6mm ext
- Working Temperature: -40°C to 80°C
- For corrosive products application, provide a protection to avoid contact with actuator



INSTALLATION ON TANK

To use the ND80 ATCO™ full bore as an air inlet/outlet:

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Screw the studs into the weld in flange.
3. Place the gasket and siphon tube if any
4. Fit ball valve on the weld in flange and secure with washers and nuts.
   The required tightening torque is 50 N.m +/-5
5. Assemble gasket and blind flange on the top and secure using bolting kit.

Refer to our PWAGEN001 sheet for flange tightening details

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

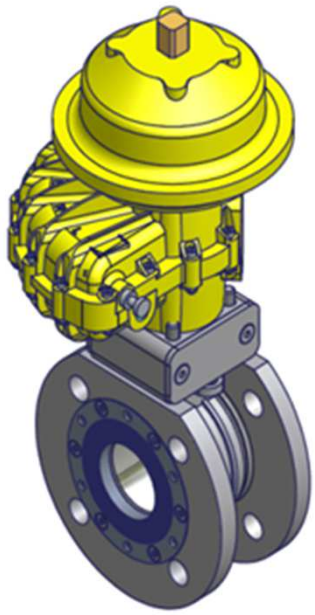
SETTING UP THE ACTUATOR

4. Connect inlet to connection "2" and outlet to connection "4" written on the actuator

It is a simple effect actuator able to work with only one inlet connection. However, we recommend to use two inlet connections. It reduces the time of closure. Use lubricated air to keep the actuator in good working condition. The air feeding should be rated from 5.5 bar up to 6.5 bar **(do not exceed 8 bar).**

NB: To get information about maintenance, please refer to the AFN8X maintenance sheet PWA1114614000.

2" ATCO Pneumatic AP5X
ND50 BALL VALVE
AP5X



The Perolo ND50 ATCO™ pneumatic FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve.

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	10.0 Kg
Inlet flange drilling	4 holes x Ø17 x 125 PCD
Outlet flange drilling	please refer to spec sheet of AFN5X
Pneumatic Air Pressure	4 to 8 bar
Working Pressure	16 Bar
Test Pressure	24 Bar
Working Temperature	Valve : -40°C to +150°C

NB: temperature range vary according to o'ring material

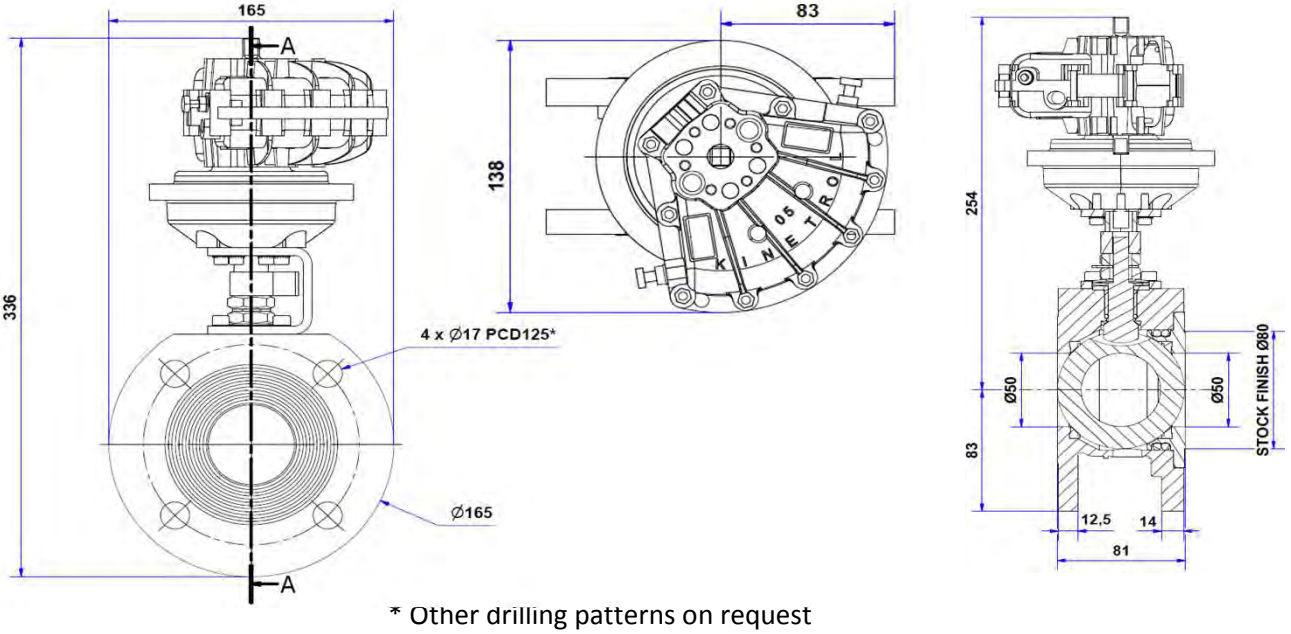
CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, A-C-I

ACCESSORIES	
Weld-in flange	11 90 25 00 00
Blind flange EN14025	11 95 88 00 00
Stud kit	17 12 12 91 00
Gaskets	11 90 29 00 00
Bolting kit	11 90 29 00 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact design	Face to face 81 mm
Operation	Pneumatic - 1/4 Turn single acting actuator
Ball gaskets material	PTFE

CODIFICATION	
Pneumatic Ball Valve - Inlet flange: 4 x 17 / PCD125, outlet flange: 4 x 18 slotted / min PCD114, 316 L stainless steel	AP5X16T1300S0

DIMENSIONAL DRAWING



* Other drilling patterns on request

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 08 20 00 E
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" ATCO Pneumatic AP5X
ND50 BALL VALVE
AP5X

INSTALLATION

CONDITIONS

Caution: Do not disassemble the actuator from ATCO™ Ball valve.
For better and long time performances, assembly and settings are realized in PEROLO factory.

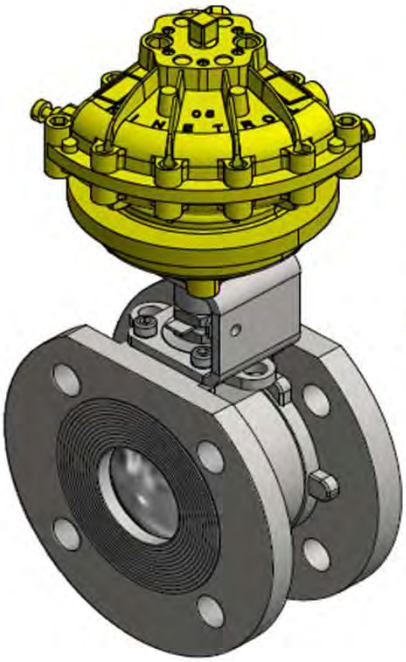
INSTALLATION ON TANK

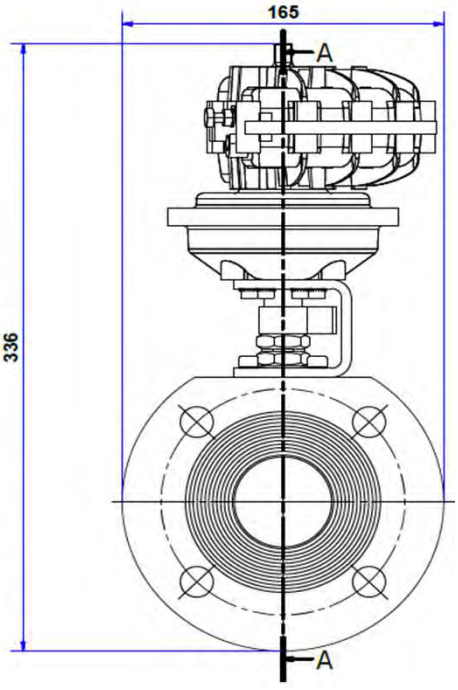
1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Screw the studs into the weld in flange. Lubricate the threads with PTFE added grease.
3. Place the gasket.
4. Fit ball valve on the weld in flange and secure with washers and nuts.
   Tighten the nuts torque : 50 N.m +/-5 (use our bolting kit A2-70 or A4-70 class)
5. Assemble gasket and blind flange on the top same bolting procedure them above
6. check torque after 24 hours

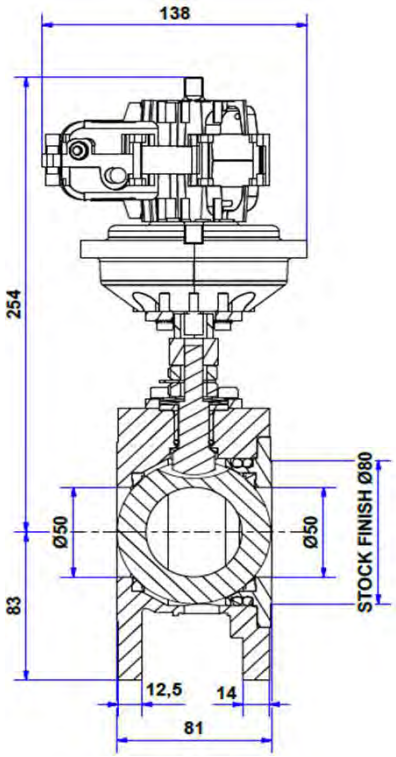
Refer to our PWAGEN001 sheet for flange tightening details

Caution: If over tightened, the valve may be difficult to be opened.

It works with a pneumatic single-acting actuator.
The minimum air supply is 5.5 bar.







NB: To get information about maintenance, please refer to the AFN5X maintenance sheet PWA1112694000

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

BUTTERFLY VALVES

FUNCTION

A butterfly valve is designed as a fluid inlet/outlet. It can be installed on top of a tank or tanker as top discharge provision, or at the bottom as part of a bottom outlet. Different sizes are available ranging from ND50 to ND100. Butterfly valves are either clamped, flanged or have a BSP threaded outlet. For ND50 size, please check airline chapter

- Various internal diameters: 3" (80mm) or 4" (100mm)
- Clamped or Flanged
- Inlet & outlet connection: Various drilling patterns Open or closed slots
- Locking key for Road Tanker range
- Operating mechanism: Manual: handle or remote lever Pneumatic actuator
- Contact Materials: 316L stainless steel
- Working Temperature: Standard or High (+250°C)
- Surface Finish: Acid etched, Halar lined
- CE1935/2004
- ADR, RID, IMDG

3" Clamped Butterfly Valve	3" Flanged Butterfly Valve	3" Clamped BV for Road	3" Flanged BV for Road
4" Clamped Butterfly Valve	4" Clamped BV for ROAD	Remote Control *Section Accessories*	

3" (ND80) TANKFLY
Clamped Butterfly Valve
TPS8

The 3" TANKFLY clamped butterfly valve is designed to be clamped between 2 flanges or directly onto the pipe work. NEW model with splined shaft for even better handling.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	3,15 KG
Inlet flange	Clamped between DIN or ANSI Flange
Outlet flange	Clamped between DIN or ANSI Flange
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	STD:-40° to +150°C/ HT°:-20° to 250°C
Design Temperature	STD:-40° to +250°C/ HT°:-20° to 250°C

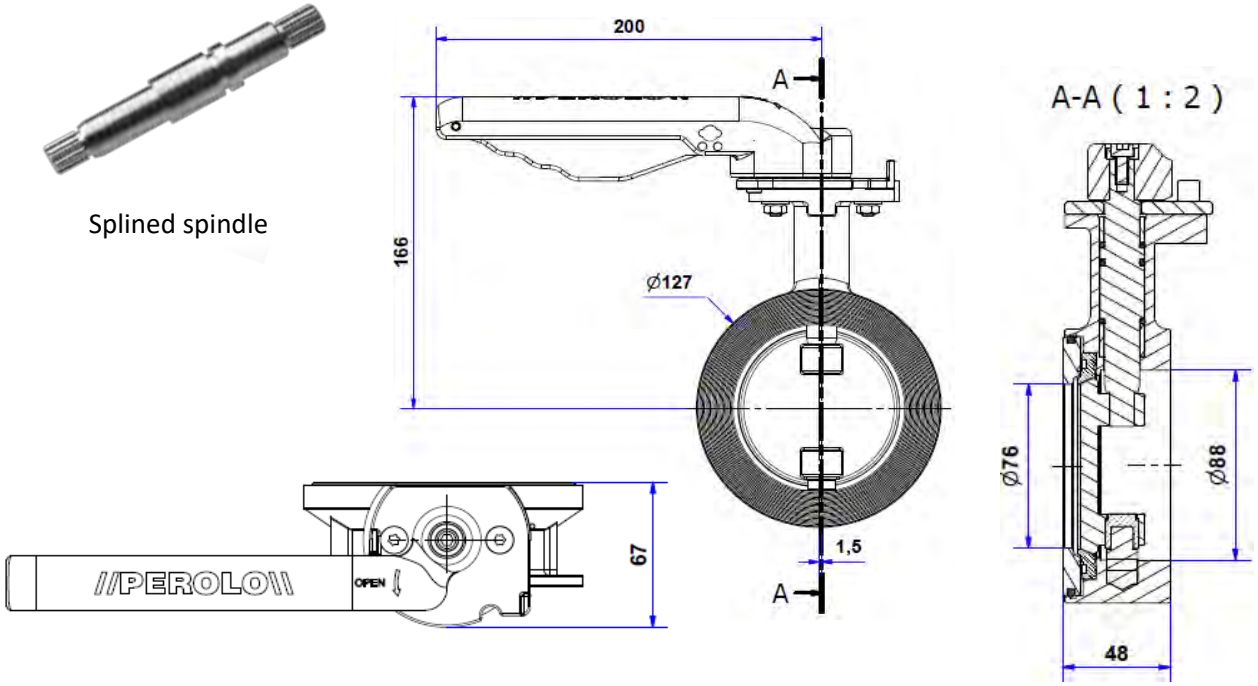
ACCESSORIES	
Weld in flange	11 90 36 00 00
Seal	12 91 40 00 00
Adaptor flange	12 91 41 00 03
Blind flange	11 95 90 00 00
Screw kit M16-90	12 85 57 00 24
Screw kit M16-40	11 10 09 91 00
Remote control	RMT53

CONFORMITY, CERTIFICATION & APPROVAL
Comply with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas - RMR

STANDARD FEATURES	
Made of (material)	316 / 1.4408 stainless steel
Compact Design	Face to face 48 mm
Operation	Handle - 1/4 turn with Locking
Closure plate seal	PTFE

DIMENSIONAL DRAWING

Splined spindle



BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 58 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) TANKFLY
Clamped Butterfly Valve
TPS8

CODIFICATION

STANDARD PART NUMBER	TPS8	X	7	T	0	0

YOUR PART NUMBER	TPS8	__	7	__	__	__

MATERIAL	
316 / 1.4408 stainless steel	X
904L stainless steel	U

MAWP	
7 bar	7

SEAL MATERIAL & WORKING TEMP.	
PTFE seal - Standard WT	T
HT seal - WT: -20°C to 250°C	H
PTFE seal - CE1935/2004	A

CLOSURE PLATE	
Standard closure plate	0
Spheric closure plate	S

OPENING MECHANISM	
Standard manual opening	0
Manual with remote control handle	W
Pneumatic with double rack pneumatic actuator	P1
Pneumatic with rotating actuator	P2

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 58 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) TANKFLY
Clamped Butterfly Valve
TPS8

INSTALLATION

CAUTION : Tankfly must be in closed position when the valve is fitted on the tank.

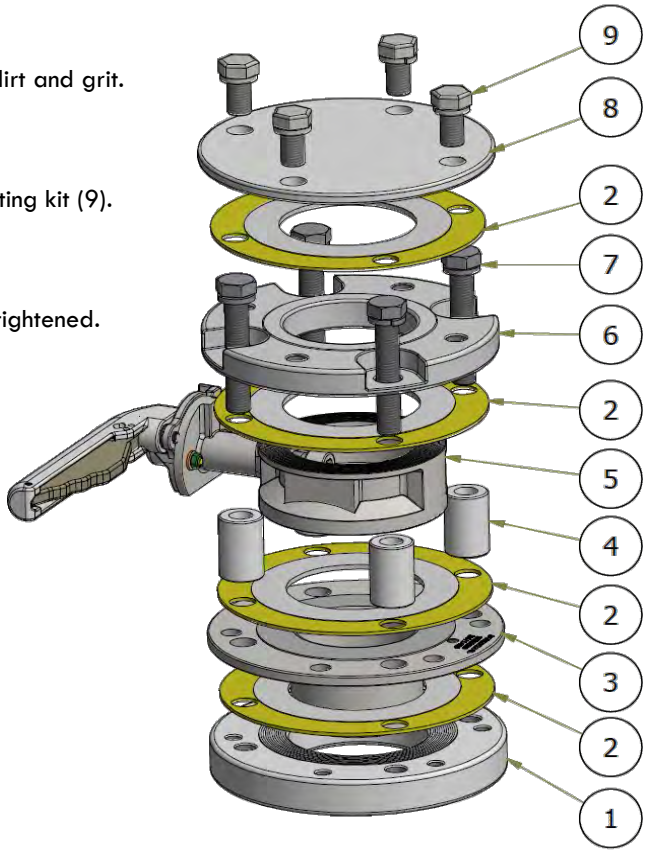
To assembly the 3" Tankfly clamped butterfly valve:

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Align drilling pattern of gaskets with that of tank pad.
3. Center the TANKFLY with weld in flange hole
4. Mount outlet gasket and blind flange and secure using bolting kit (9).

The required tightening torque is 70 N.m +/- 5

5. Refer to PWAGENE001 for bolt tightening sequence.

The valve flange should be in contact with tank flange when tightened.



ITEM	DESCRIPTION	SPECIFICATION	PART N°
8	Blind flange	Ext Ø 200mm, drilling 4 x 18 x 160 PCD, thickness: 13,5mm , 316L/1.4404	11 95 90 00 00
9	Screw kit M16-40	4 x M16-40 screws & 4 x grower washers, 304L/1.4301	11 10 09 91 00
6	Adaptor flange	Ext Ø 200mm, ND80, drillings: 4 x 18 x 160 PCD + 4 x M16 x 160PCD, thickness: 24mm, 316M/1.4408	12 91 41 00 03
7	Screw kit M16-90	4 x M16-90 screws & 4 x washers 316L/1.4404	12 85 57 00 24
4	Spacer	4 x spacers , plastic	12 85 57 91 00
5	3" TANKFLY clamped	With standard PTFE seals, 7 bar MAWP, Working temperature -40°C +150°C, 316L stainless steel	TPS8X7T00
3	Syphon tube	Ext Ø 200 mm, Tube Ø 80 mm, Tube lenght 69mm, Drilling 6 x 14 x 168 PCD & 4 x 18 160 PCD, 316L/1.4404	12 92 18 10 00
2	Seal	Ext Ø 185mm, Int Ø 94mm, drilling 4 x 18 x 160 PCD thickness 2.5mm, CNAF/PTFE	12 91 40 00 00
1	Weld in flange	Int Ø 94mm, ext Ø 200mm, drillings: 6 x M12 x 168PCD & 4 x M16 x 160 PCD, thickness 27mm, 316L/1.4404	11 90 36 00 00

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 17 58 30 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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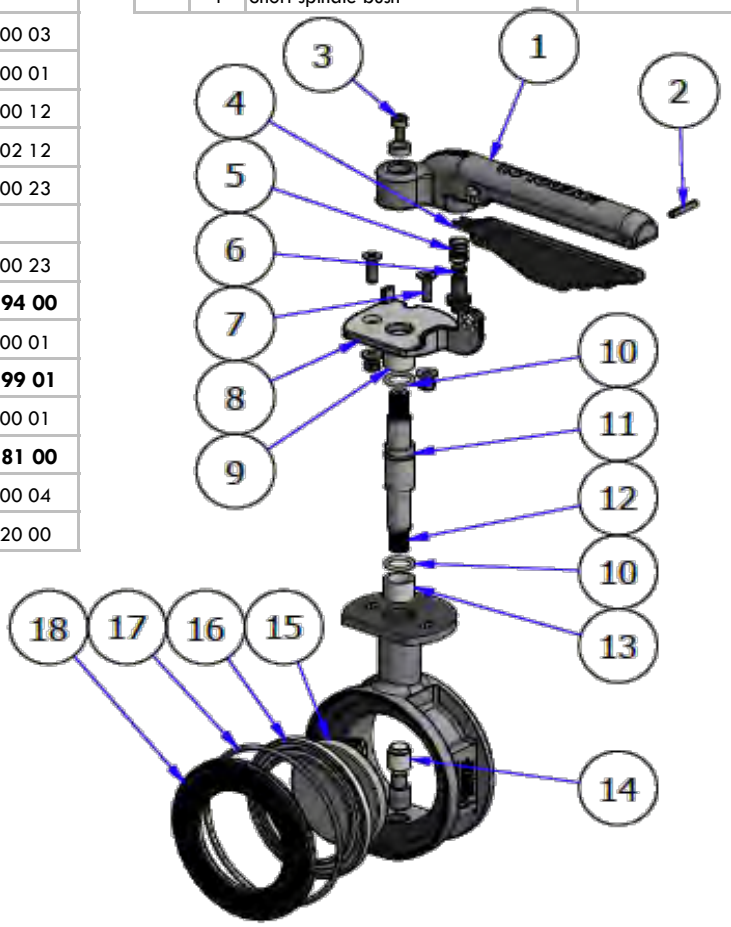
3" (ND80) TANKFLY
Clamped Butterfly Valve
TPS8

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit:	**11 14 85 90 00**
1	1	Handle, stainless steel	11 14 85 40 01
2	1	Split pin, stainless steel	-
3	1	Screw M6-12	-
	1	Washer	-
4	1	Handle lever, stainless steel	11 11 83 40 03
5	1	Handle Spring, stainless steel	-
7	1	Locking Pin, stainless steel	-
		Spindle & Spindle Seals kit:	**11 12 62 97 00**
6	2	Screws M6-20	-
	2	Washers	-
	2	Nuts	-
9& 13	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
	1	Short spindle bush	11 12 62 00 01
10	2	O'ring, FEP & silicone	70 10 15 00 12
11	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
12	1	Spindle, 1.4462	12 85 57 00 23
		Spindle:	
12	1	Spindle, 1.4462	12 85 57 00 23
		Closure plate seal kit:	**12 85 57 94 00**
16	1	Closure plate seal, PTFE	12 85 57 00 01
		Closure plate seal kit (10 pack):	**12 85 57 99 01**
16	10	Closure plate seal, PTFE (10 pack)	12 85 57 00 01
		Closure plate kit:	**12 85 57 81 00**
14	1	Closure plate bush, PTFE & glass 25%	12 85 57 00 04
15	1	Closure plate, stainless steel	11 14 85 20 00

ITEM	QTY	DESIGNATION	PART N°
		Clamping ring kit:	**12 85 57 95 00**
17	1	O'Ring, nitril	70 10 03 07 05
18	1	Clamping ring, stainless steel	12 85 57 00 02
		Spindle Seals kit:	**11 12 62 91 00**
3	1	Screw M6-12,	-
11	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
10	2	O'ring, FEP & silicone	70 10 15 00 12
6	2	Screws M6-20	-
	2	Washers	-
	2	Nuts	-
9& 13	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
	1	Short spindle bush	11 12 62 00 01



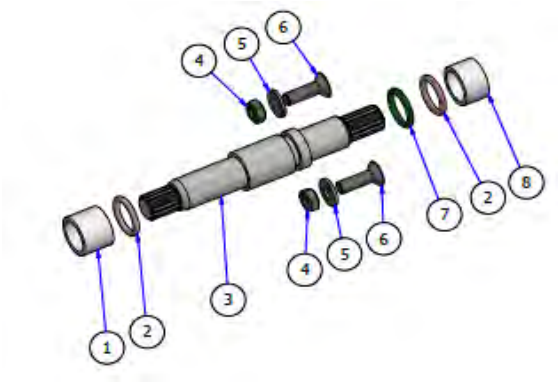
If you want to upgrade a hexagonal spindle to a splined spindle,
please order spindle & spindle seals kit 1112629700,
handle kit 1114859000 and closure plate kit 1285578100

BUTTERFLY VALVE

3" (ND80) TANKFLY
Clamped Butterfly Valve
TPS8

SPINDLE & SEALS KIT 1112629700 for TANKFLY™ butterfly valve Flanged & Clamped

ITEM	QTY	DESIGNATION	PART N°
		Spindle & Spindle Seals kit:	**11 12 62 97 00**
6	2	Screws M6-20	-
5	2	Washers	-
4	2	Nuts	-
1	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
8	1	Short spindle bush	11 12 62 00 01
2	2	O'ring, FEP & silicone	70 10 15 00 12
7	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
3	1	Spindle, 1.4462	12 85 57 00 23



Tankfly Spindle Seals assembly according PEROLO Serial Number

PREVIOUS ASSEMBLY

Serial Number	
From :	Until :
Beginning	0354 14

O'ring, FEP & silicone

O'ring, FEP & silicone

O'ring, FEP & silicone

O'ring, FEP & silicone or FFKM

Serial Number	
From :	Until :
0010 15	0173 16
&	
R 0385 15	R 4676 16

11 12 62 94 00 can replace the 2 previous assemblies with hexagonal spindle

CURRENT ASSEMBLY WITH HEXAGONAL SPINDLE

O'ring, PTFE & silicone

O'ring, FKM/Viton (green)

O'ring, FEP & silicone or FFKM

Serial Number	
From :	Until :
0247 16	now
&	
R 4677 16	now

CURRENT ASSEMBLY WITH SPLINED SPINDLE

O'ring, PTFE & silicone

O'ring, FKM/Viton (green)

O'ring, FEP & silicone or FFKM

Serial Number	
From :	Until :
0247 16	/
&	
R 4677 16	/

If you want to upgrade a hexagonal spindle to a splined spindle, please order spindle & spindle seals kit 1112629700, handle kit 1114859000 and closure plate kit 1285578100

BUTTERFLY VALVE

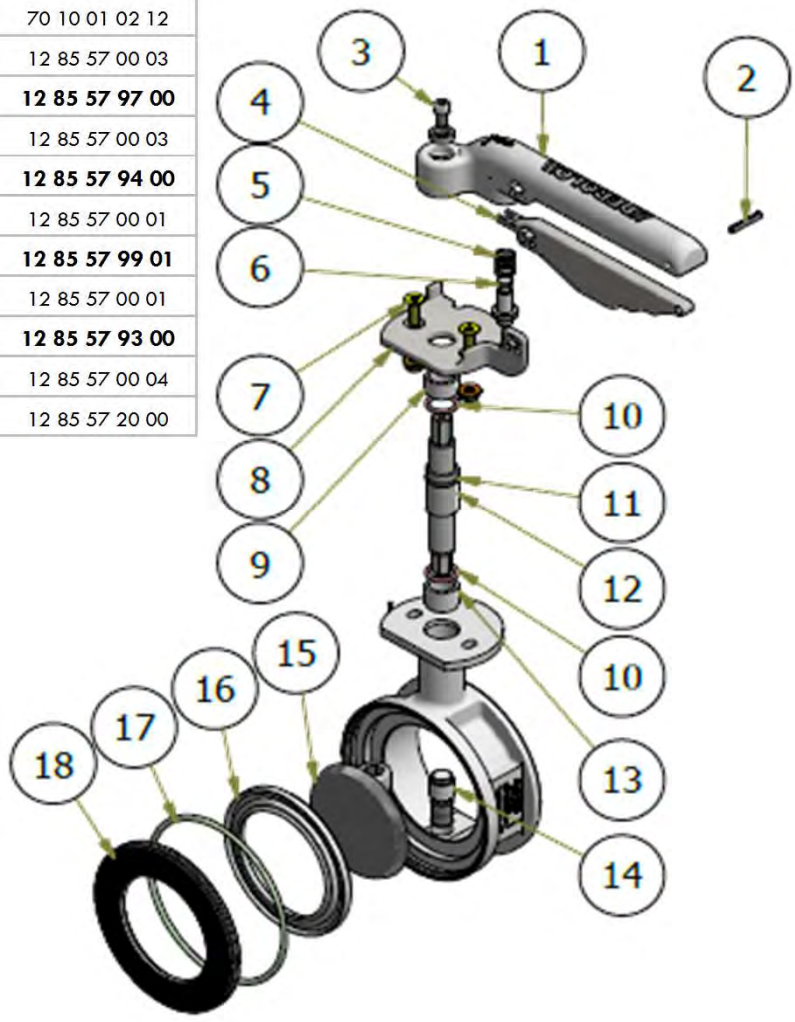
Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 17 58 40 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) TANKFLY
Clamped Butterfly Valve
11 12 62 00 00 D

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit:	**11 12 62 95 00**
1	1	Handle, stainless steel	11 11 83 40 01
2	1	Split pin, stainless steel	-
3	1	Screw M6-12	-
	1	Washer	-
4	1	Handle lever, stainless steel	11 11 83 40 03
5	1	Handle Spring, stainless steel	-
7	1	Locking Pin, stainless steel	-
		Spindle & Spindle Seals kit:	**11 12 62 94 00**
6	2	Screws M6-20	-
	2	Washers	-
	2	Nuts	-
9& 13	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
	1	Short spindle bush	11 12 62 00 01
10	2	O'ring, FEP & silicone	70 10 15 00 12
11	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
12	1	Spindle, 1.4462	12 85 57 00 03
		Spindle:	**12 85 57 97 00**
12	1	Spindle, 1.4462	12 85 57 00 03
		Closure plate seal kit:	**12 85 57 94 00**
16	1	Closure plate seal, PTFE	12 85 57 00 01
		Closure plate seal kit (10 pack):	**12 85 57 99 01**
16	10	Closure plate seal, PTFE (10 pack)	12 85 57 00 01
		Closure plate kit:	**12 85 57 93 00**
14	1	Closure plate bush, PTFE & glass 25%	12 85 57 00 04
15	1	Closure plate, stainless steel	12 85 57 20 00

ITEM	QTY	DESIGNATION	PART N°
		Clamping ring kit:	**12 85 57 95 00**
17	1	O'Ring, nitril	70 10 03 07 05
18	1	Clamping ring, stainless steel	12 85 57 00 02
		Spindle Seals kit:	**11 12 62 91 00**
3	1	Screw M6-12,	-
11	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
10	2	O'ring, FEP & silicone	70 10 15 00 12
6	2	Screws M6-20	-
	2	Washers	-
	2	Nuts	-
9& 13	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
	1	Short spindle bush	11 12 62 00 01

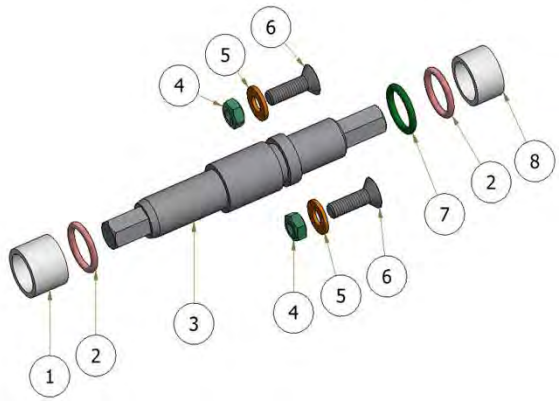


BUTTERFLY VALVE

3" (ND80) TANKFLY
Clamped Butterfly Valve
11 12 62 00 00 D

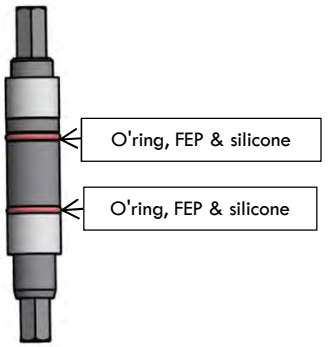
SPINDLE & SEALS KIT 1112629400 for TANKFLY™ butterfly valve Flanged & Clamped

ITEM	QTY	DESIGNATION	PART N°
		Spindle & Spindle Seals kit:	**11 12 62 94 00**
6	2	Screws M6-20	-
5	2	Washers	-
4	2	Nuts	-
1	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
8	1	Short spindle bush	11 12 62 00 01
7	2	O'ring, FEP & silicone	70 10 15 00 12
2	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
3	1	Spindle, 1.4462	12 85 57 00 03

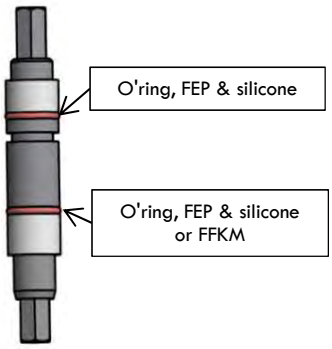


Tankfly Spindle Seals assembly according PEROLO Serial Number

PREVIOUS ASSEMBLY



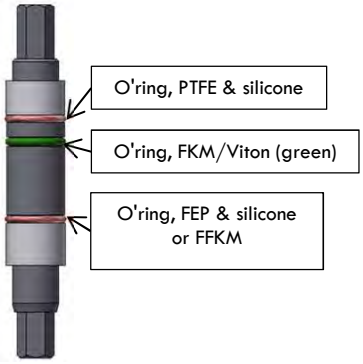
Serial Number	
From :	Until :
Beginning	0354 14



Serial Number	
From :	Until :
0010 15	0173 16
&	
R 0385 15	R 4676 16

11 12 62 94 00 can replace the 2 previous assemblies

CURRENT ASSEMBLY



Serial Number	
From :	Until :
0247 16	now
&	
R 4677 16	now

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 12 62 40 00 D B
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) TANKFLY
Flanged Butterfly Valve
TFS8

The 3" TANKFLY flanged butterfly valve is designed to be used as a fluid inlet/outlet valve - with a compact design this product offers an easy opening/closing with a 1/4 turn lockable handling operation. NEW model with splined shaft for even better handling.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	7.0 Kg
Inlet flange	6 x Ø14 x 168 PCD / 4 x Ø18 x 160 PCD
Outlet flange	4 slots x Ø18 on a 146 PCD*
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	STD:-40° to +150°C/ HT°:-20° to 250°C
Design Temperature	STD:-40° to +250°C/ HT°:-20° to 250°C

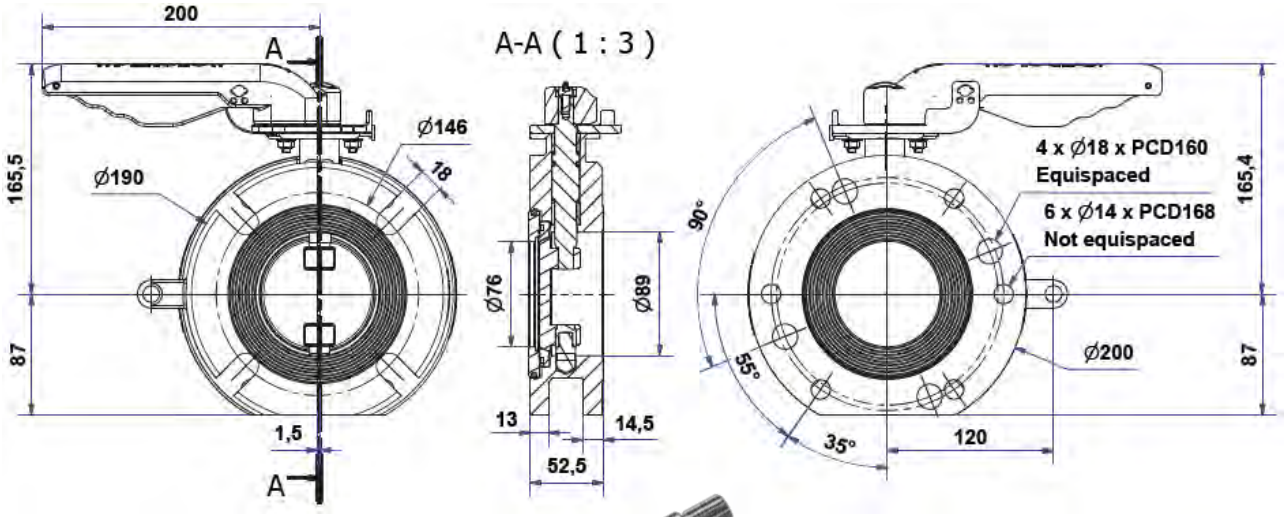
CONFORMITY, CERTIFICATION & APPROVAL
Comply with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas - TÜV - RMR

ACCESSORIES	
Blind Flange	11 90 41 00 00
Weld-in flange	11 90 36 00 00
Recess flange	11 90 98 00 00
Gasket	12 91 40 00 00
Screw kit	12 91 61 92 00
Pivot pin	12 91 52 00 00
Remote control	RMT53

STANDARD FEATURES	
Made of (material)	316 / 1.4408 stainless steel
Compact Design	Face to face 54mm
Operation	Handle - 1/4 turn with Locking
Closure plate seal	PTFE

*for more drilling options please check codification

DIMENSIONAL DRAWING



Splined spindle

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 57 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

3" (ND80) TANKFLY
Flanged Butterfly Valve
TFS8

CODIFICATION

STANDARD PART NUMBER	TFS8	X	7	T	1	1	0	0
YOUR PART NUMBER	TFS8		7					

MATERIAL	
316 / 1.4408 stainless steel	X
904L stainless steel	U

MAWP	
7 bar	7

SEAL MATERIAL & WORKING TEMP.	
PTFE seal - Standard WT	T
HT seal - WT: -20°C to 250°C	H
PTFE seal - CE1935/2004	A

INLET FLANGE DRILLING	
6 x Ø14 x 168 PCD / 4 x Ø18 x 160 PCD	1
8 x Ø18 x 160 PCD	2
4 x M16 x 160 PCD	3
4 x 18 x 160 PCD	4
6 x Ø14 x 168 PCD / 8 x Ø18 x 160 PCD	5
6 x 14 x 168 PCD	6

OUTLET FLANGE DRILLING	
4 x slotsØ18 x 146 PCD	1
4 x slotsØ18 x 146 PCD / 4 x M16 x 160PCD	2

CLOSURE PLATE	
Standard closure plate	0
Spheric closure plate	S

OPENING MECHANISM	
Standard manual opening	0
Manual with remote control handle	W
Pneumatic with double rack pneumatic actuator	P1
Pneumatic with rotating actuator	P2

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 57 20 00 A
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PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

3" (ND80) TANKFLY
Flanged Butterfly Valve
TFS8

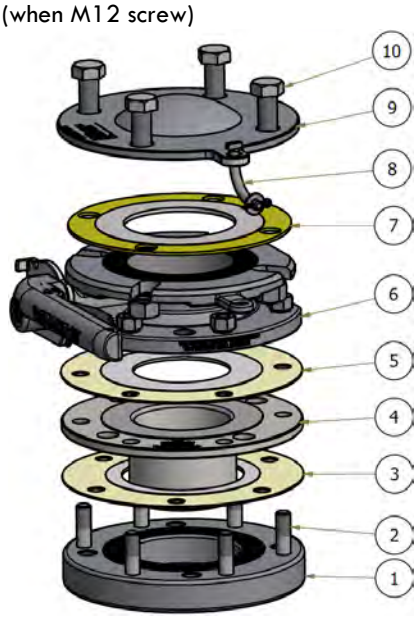
ISO ROAD

BUTTERFLY VALVE

INSTALLATION

CAUTION : Tankfly must be in closed position when the valve is fitted on the tank.

1. Check Inlet (tank side) flange is clean, free from dirt and grit.
2. Screw the studs into the weldin flange, lubricate with PTFE added grease before installing the studs
3. Place the gasket
4. Install the valve on the flange, secure with washer and nuts *(refer to PWAGEN001 fort bolt tightening sequence)*
5. Tighten the nuts torque : 70 N.m +/- 5 (when M16 screw), 50 N.m +/- 5 (when M12 screw)
6. Install gasket, bolting kit and blind flange on the valve outlet flange *(refer to PWAGEN001 fort bolt tightening sequence)*
7. Open and close the valve 3 times to check the operationnal torque.



ITEM	DESCRIPTION	SPECIFICATION	PART N°
9	Dished blind flange	Ext Ø 200 mm, thickness 8 mm, 4 x 18 x 160PCD, 316L stainless steel	11 90 41 00 00
10	Screw kit	4xM16-40 screws + 4 x lock washers and 4xM16 nuts	12 91 61 92 00
8	Pivot pin	Pivot Pin	12 91 52 00 00
7	Gasket	Ext Ø 200mm, Int Ø 94mm, thickness 2mm drilling 4x18x160PCD, CNAF/PTFE gasket	12 91 40 00 00
6	3" TANKFLY butterfly valve	inlet drilling 6 x Ø14 x 168 PCD / 4 x Ø18 x 160 PCD , outlet 4 x slotsØ18 x 146 PCD, 316L st st	TFS8X7T1100
2	Stud kit	6xM12-40 studs + 6 x lock washers and 6 x M12 nuts	12 91 61 91 00
5	Gasket	Ext Ø 200mm, Int Ø 81mm, thickness 2mm drilling 6x14x168PCD, CNAF/PTFE gasket	12 91 48 00 00
4	Siphon tube	Ext Ø 200 mm, Tube length 89 mm 6x14x168PCD non equispaced & 4x18x160PCD equispaced CF8M, EN 10213-4	12 92 18 10 00
3	Gasket	Ext Ø 200mm, Int Ø 94mm, thickness 2mm drilling 6x14x168PCD, CNAF/PTFE gasket	11 90 92 00 00
1	Weld in flange	Int Ø94mm, ext Ø200mm, thickness 26.5mm, 6xM12x168PCD & 4xM16x160PCD, 316L Stainless Steel	11 90 36 00 00

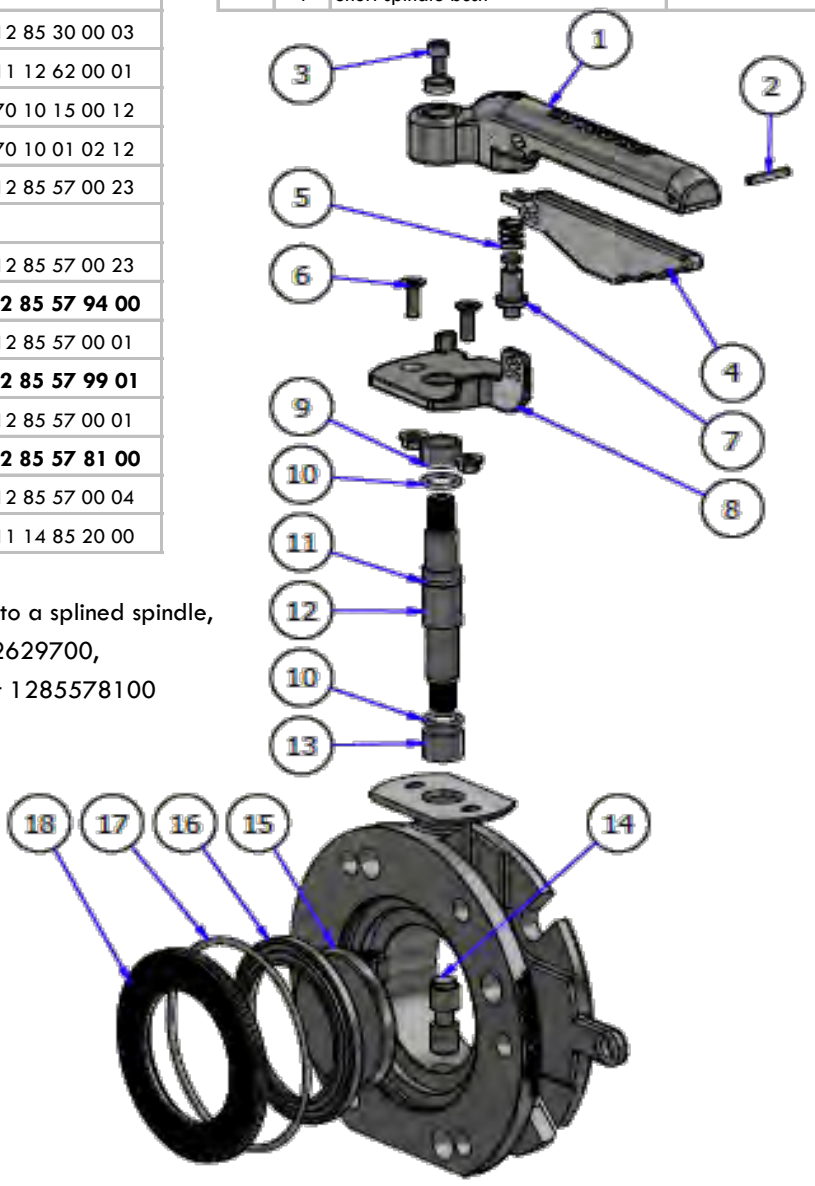
if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

3" (ND80) TANKFLY
Flanged Butterfly Valve
TFS8

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit:	**11 14 85 90 00**
1	1	Handle, stainless steel	11 14 85 40 01
2	1	Split pin, stainless steel	-
3	1	Screw M6-12	-
	1	Washer	-
4	1	Handle lever, stainless steel	11 11 83 40 03
5	1	Handle Spring, stainless steel	-
7	1	Locking Pin, stainless steel	-
		Spindle & Spindle Seals kit:	**11 12 62 97 00**
6	2	Screws M6-20	-
	2	Washers	-
	2	Nuts	-
9	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
	1	Short spindle bush	11 12 62 00 01
10	2	O'ring, FEP & silicone	70 10 15 00 12
11	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
12	1	Spindle, 1.4462	12 85 57 00 23
		Spindle:	
12	1	Spindle, 1.4462	12 85 57 00 23
		Closure plate seal kit:	**12 85 57 94 00**
16	1	Closure plate seal, PTFE	12 85 57 00 01
		Closure plate seal kit (10 pack):	**12 85 57 99 01**
16	10	Closure plate seal, PTFE (10 pack)	12 85 57 00 01
		Closure plate kit:	**12 85 57 81 00**
14	1	Closure plate bush, PTFE & glass 25%	12 85 57 00 04
15	1	Closure plate, stainless steel	11 14 85 20 00

ITEM	QTY	DESIGNATION	PART N°
		Clamping ring kit:	**12 85 57 95 00**
14	1	O'Ring, nitril	70 10 03 07 05
17	1	Clamping ring, stainless steel	12 85 57 00 02
		Spindle Seals kit:	**11 12 62 91 00**
3	1	Screw M6-12,	-
11	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
10	2	O'ring, FEP & silicone	70 10 15 00 12
6	2	Screws M6-20	-
	2	Washers	-
	2	Nuts	-
9	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
	1	Short spindle bush	11 12 62 00 01



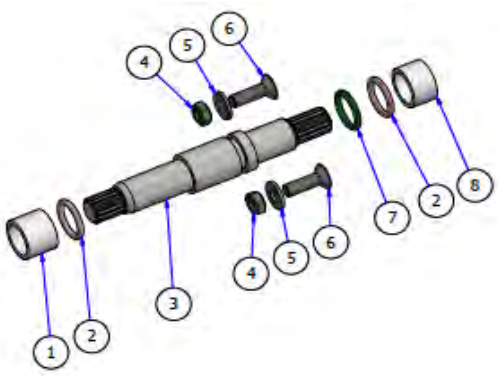
If you want to upgrade a hexagonal spindle to a splined spindle, please order spindle & spindle seals kit 1112629700, handle kit 1114859000 and closure plate kit 1285578100

BUTTERFLY VALVE

3" (ND80) TANKFLY
Flanged Butterfly Valve
TFS8

SPINDLE & SEALS KIT 1112629700 for TANKFLY™ butterfly valve Flanged & Clamped

ITEM	QTY	DESIGNATION	PART N°
		Spindle & Spindle Seals kit:	**11 12 62 97 00**
6	2	Screws M6-20	-
5	2	Washers	-
4	2	Nuts	-
1	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
8	1	Short spindle bush	11 12 62 00 01
2	2	O'ring, FEP & silicone	70 10 15 00 12
7	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
3	1	Spindle, 1.4462	12 85 57 00 23



Tankfly Spindle Seals assembly according PEROLO Serial Number

PREVIOUS ASSEMBLY

Serial Number	
From :	Until :
Beginning	0354 14

O'ring, FEP & silicone

O'ring, FEP & silicone

O'ring, FEP & silicone

O'ring, FEP & silicone or FFKM

Serial Number	
From :	Until :
0010 15	0173 16
&	
R 0385 15	R 4676 16

11 12 62 94 00 can replace the 2 previous assemblies with hexagonal spindle

CURRENT ASSEMBLY WITH HEXAGONAL SPINDLE

O'ring, PTFE & silicone

O'ring, FKM/Viton (green)

O'ring, FEP & silicone or FFKM

Serial Number	
From :	Until :
0247 16	now
&	
R 4677 16	now

CURRENT ASSEMBLY WITH SPLINED SPINDLE

O'ring, PTFE & silicone

O'ring, FKM/Viton (green)

O'ring, FEP & silicone or FFKM

Serial Number	
From :	Until :
TBD	/
&	
TBD	/

If you want to upgrade a hexagonal spindle to a splined spindle, please order spindle & spindle seals kit 1112629700, handle kit 1114859000 and closure plate kit 1285578100

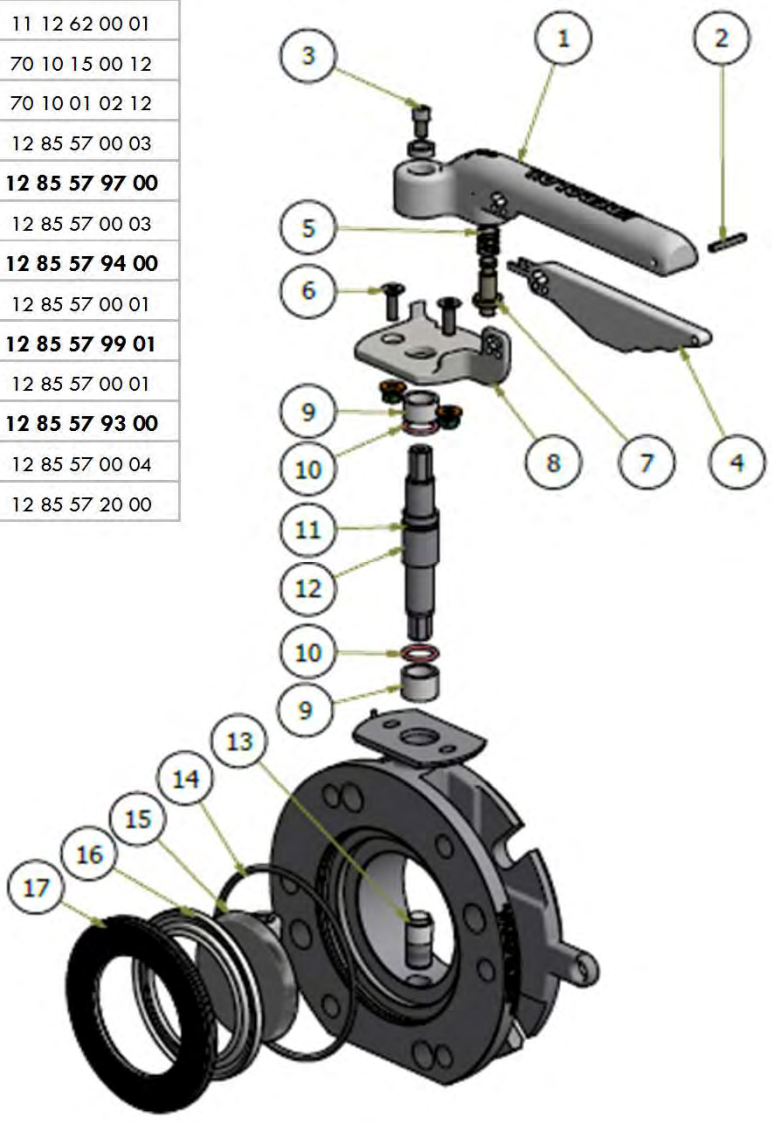
BUTTERFLY VALVE

3" (ND80) TANKFLY
Flanged Butterfly Valve
11 12 31 00 00 D

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit:	**11 12 62 95 00**
1	1	Handle, stainless steel	11 11 83 40 01
2	1	Split pin, stainless steel	-
3	1	Screw M6-12	-
	1	Washer	-
4	1	Handle lever, stainless steel	11 11 83 40 03
5	1	Handle Spring, stainless steel	-
7	1	Locking Pin, stainless steel	-
		Spindle & Spindle Seals kit:	**11 12 62 94 00**
6	2	Screws M6-20	-
	2	Washers	-
	2	Nuts	-
9	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
	1	Short spindle bush	11 12 62 00 01
10	2	O'ring, FEP & silicone	70 10 15 00 12
11	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
12	1	Spindle, 1.4462	12 85 57 00 03
		Spindle:	**12 85 57 97 00**
12	1	Spindle, 1.4462	12 85 57 00 03
		Closure plate seal kit:	**12 85 57 94 00**
16	1	Closure plate seal, PTFE	12 85 57 00 01
		Closure plate seal kit (10 pack):	**12 85 57 99 01**
16	10	Closure plate seal, PTFE (10 pack)	12 85 57 00 01
		Closure plate kit:	**12 85 57 93 00**
13	1	Closure plate bush, PTFE & glass 25%	12 85 57 00 04
15	1	Closure plate, stainless steel	12 85 57 20 00

ITEM	QTY	DESIGNATION	PART N°
		Clamping ring kit:	**12 85 57 95 00**
14	1	O'Ring, nitril	70 10 03 07 05
17	1	Clamping ring, stainless steel	12 85 57 00 02
		Spindle Seals kit:	**11 12 62 91 00**
3	1	Screw M6-12,	-
11	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
10	2	O'ring, FEP & silicone	70 10 15 00 12
6	2	Screws M6-20	-
	2	Washers	-
	2	Nuts	-
9	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
	1	Short spindle bush	11 12 62 00 01

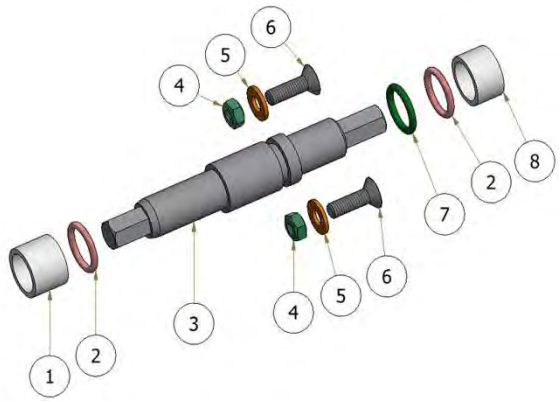


BUTTERFLY VALVE

3" (ND80) TANKFLY
Flanged Butterfly Valve
11 12 31 00 00 D

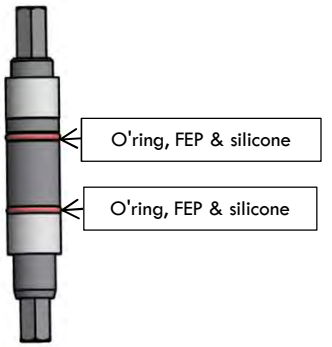
SPINDLE & SEALS KIT 1112629400 for TANKFLY™ butterfly valve Flanged & Clamped

ITEM	QTY	DESIGNATION	PART N°
		Spindle & Spindle Seals kit:	**11 12 62 94 00**
6	2	Screws M6-20	-
5	2	Washers	-
4	2	Nuts	-
1	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
8	1	Short spindle bush	11 12 62 00 01
7	2	O'ring, FEP & silicone	70 10 15 00 12
2	1	O'ring, FKM/Viton (Green)	70 10 01 02 12
3	1	Spindle, 1.4462	12 85 57 00 03

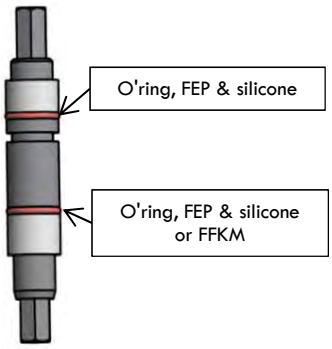


Tankfly Spindle Seals assembly according PEROLO Serial Number

PREVIOUS ASSEMBLY



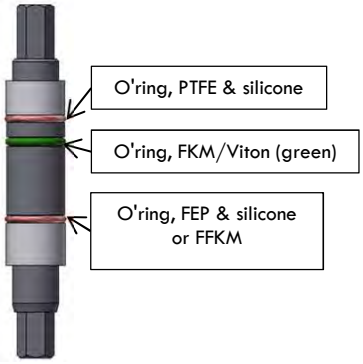
Serial Number	
From :	Until :
Beginning	0354 14



Serial Number	
From :	Until :
0010 15	0173 16
&	
R 0385 15	R 4676 16

11 12 62 94 00 can replace the 2 previous assemblies

CURRENT ASSEMBLY



Serial Number	
From :	Until :
0247 16	now
&	
R 4677 16	now

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 12 31 40 00 D B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) CHEMFLY
Clamped Butterfly Valve
17 12 01 00 00

The 3" CHEMFLY clamped butterfly valve is designed to be clamped between 2 flanges or directly onto the pipe work.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	3.0 Kg
Inlet flange	Clamped between DIN or ANSI Flange
Outlet flange	Clamped between DIN or ANSI Flange
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	-40° to +150°C (HT +250°C on request*)

*HT version only for standard version, not for CE1935/2004 version

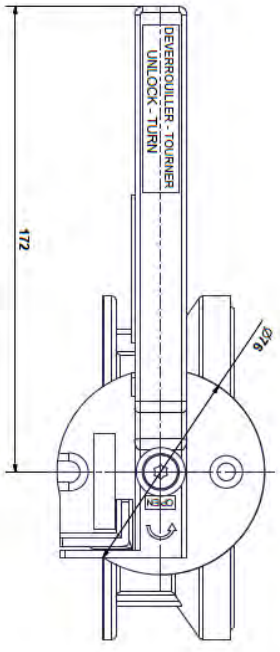
CONFORMITY, CERTIFICATION & APPROVAL
Comply with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

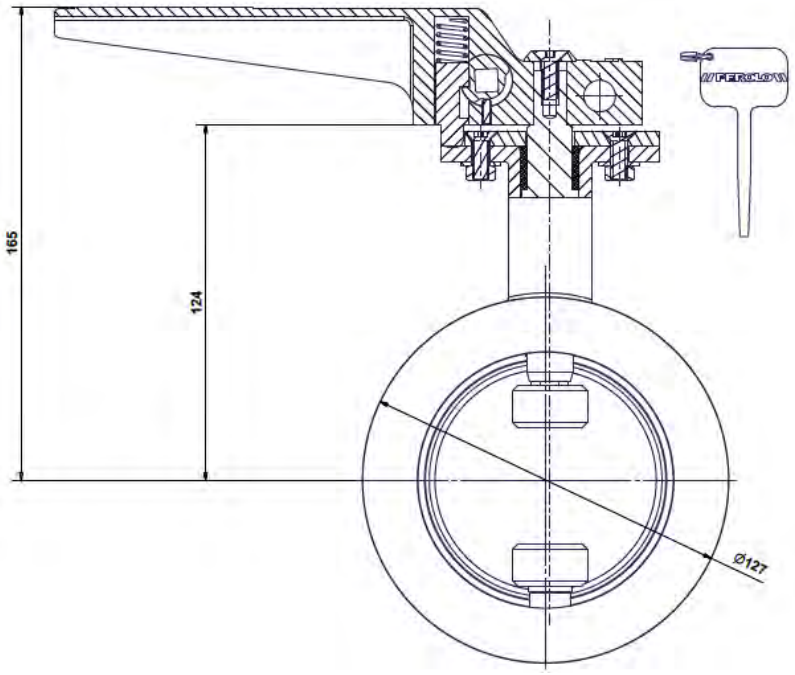
ACCESSORIES	
Handle locking key	17 12 01 30 00
3" BSP Outlet Nozzle	12 85 69 10 00
Bolting kit	12 85 84 91 00
Flat Gasket	12 91 42 00 00
Flat Gasket	12 91 39 00 00
Spacer set	12 85 57 91 00

STANDARD FEATURES	
Made of (material)	316 / 1.4408 stainless steel
Compact Design	Face to face 47 mm
Operation	Locking Handle - 1/4 turn
Closure plate seal	PTFE

CODIFICATION	
3'' CHEMFLY clamped butterfly valve - standard	17 12 01 00 00
3'' CHEMFLY clamped butterfly valve - CE1935/2004	17 12 01 00 00 A

DIMENSIONAL DRAWING





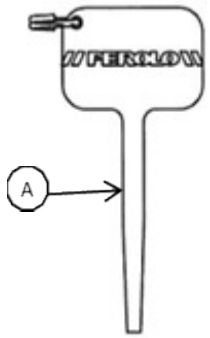
BUTTERFLY VALVE

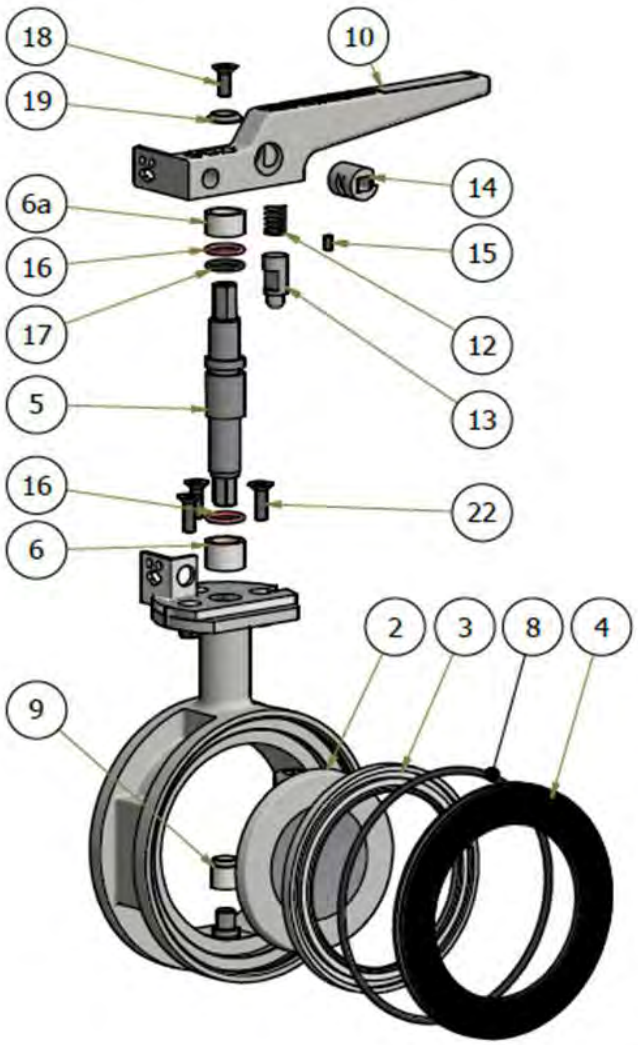
3" (ND80) CHEMFLY
Clamped Butterfly Valve
17 12 01 00 00

ROAD

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit	
18	1	Screw M6-16, Stainless steel	17 12 01 90 00
10	1	Handle, Stainless steel	
19	1	Washer	
12	1	Spring, Stainless steel	
13	1	Handle spindle, Stainless steel	
14	1	Center boss, Stainless steel	
15	1	M4-8 screw	
		Spindle Seals kit	
18	1	Screw M6-16, Stainless steel	17 12 01 91 00
19	1	Washer	
22	3	Screws M6-20, Stainless steel	
	3	Washers, Stainless steel	
	3	Nuts, Stainless steel	
6a	1	Spindle bush, PTFE & glass 25%	
6	1	Spindle bush, PTFE & glass 25%	
17	1	O'ring, Viton	
16	2	O'ring, PTFE & silicon	
9	1	Closure plate bush, PTFE & glass 25%	
		Spindle	
5	1	Spindle (stainless steel)	12 85 57 97 00
		Closure plate seal kit	
3	1	Closure plate seal, PTFE	12 85 57 94 00
		Closure plate kit	
9	1	Closure plate bush, PTFE & glass 25%	12 85 57 93 00
2	1	Closure plate, Stainless steel	
		Clamping ring kit	
8	1	O'Ring, nitril	12 85 57 95 00
4	1	Clamping ring, Stainless steel	
		Handle Locking Key	
A	1	Key	17 12 01 30 00





BUTTERFLY VALVE

3" (ND80) CHEMFLY "Light"

Flanged Butterfly Valve

LY3

The 3" CHEMFLY light butterfly valve is designed to discharge liquid chemicals and foodstuffs carried in road tankers, rail tank cars or stored in static tanks.

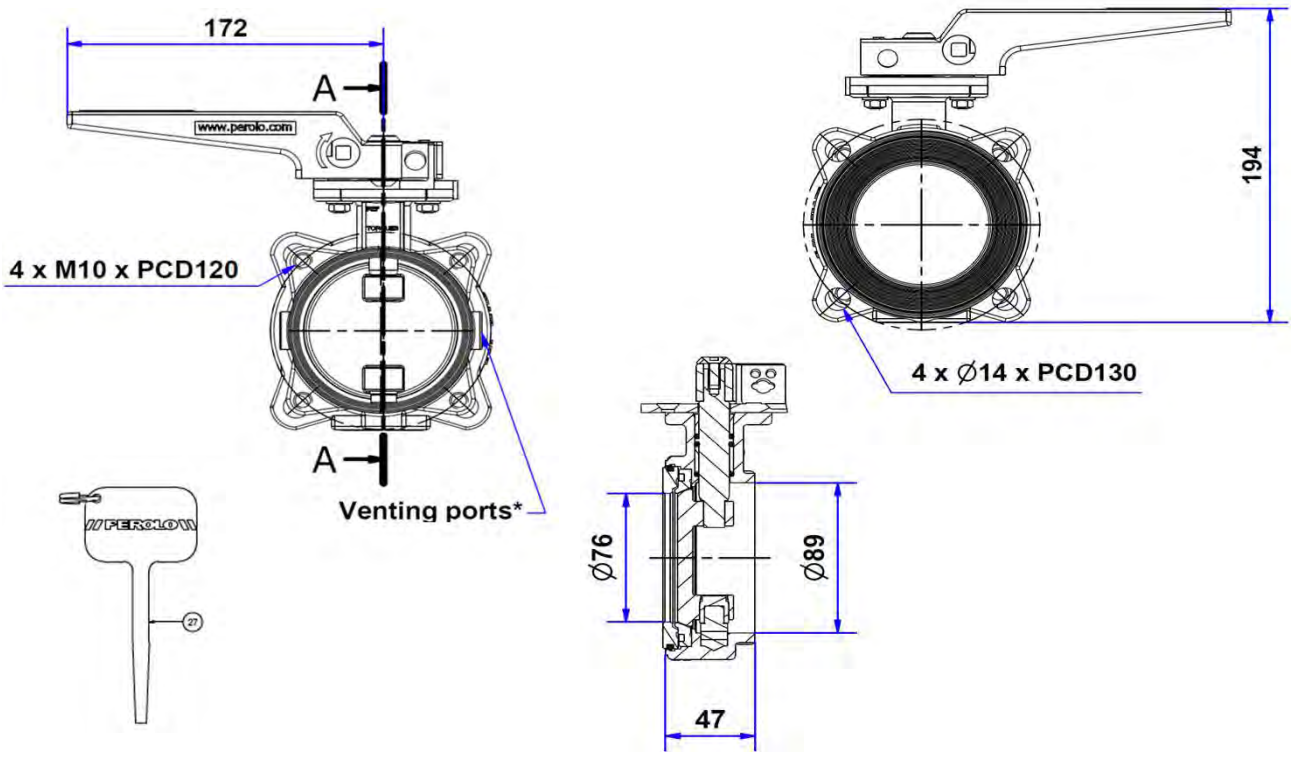
SPECIFICATION	
Nominal diameter	76 mm (3")
Weight	2.8 Kg
Inlet flange	4 holes x Ø14 x 130 PCD
Outlet flange	4 holes x M10 x 120 PCD
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

ACCESSORIES	
Inlet Nozzle	17 12 09 80 00
Inlet gasket	17 12 21 10 00
Inlet screw kit	17 12 11 92 00
3" BSP short outlet nozzle	17 12 09 70 00
3" BSP long outlet nozzle	17 12 09 71 00
Outlet gasket	17 12 21 00 01
Outlet screw kit	17 12 11 91 00
3" BSP cap kit	12 85 59 91 00

CONFORMITY, CERTIFICATION & APPROVAL
Comply with ADR / IMDG / RID / EN14432
Approved by Bureau Verita

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face to face 47 mm
Operation	Locking Handle - 1/4 turn
Closure plate seal	PTFE

DIMENSIONAL DRAWING



BUTTERFLY VALVE

3" (ND80) CHEMFLY "Light"
Flanged Butterfly Valve
LY3

ROAD

BUTTERFLY VALVE

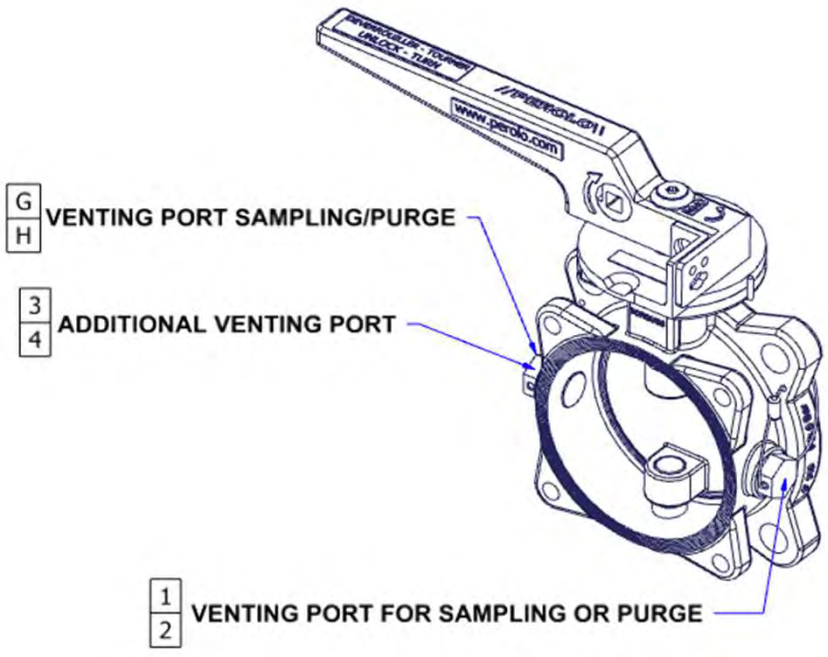
CODIFICATION

STANDARD PART NUMBER	LY3	1	1	0
YOUR PART NUMBER	LY3	1	1	

INLET FLANGE	
4 x Ø14 x 130 PCD	1

OUTLET FLANGE	
4 x M10 x 120 PCD	1

VENTING PORTS	
None	0
1 x 3/8" NPT	1
1 x 3/8" BSP	2
2 x 3/8" NPT	3
2 x 3/8" BSP	4
1 x 3/8" BSP	G
1 x 1/2" BSP	H



3" (ND80) CHEMFLY "Light"
Flanged Butterfly Valve
LY3

ROAD

INSTALLATION

CAUTION : Butterfly valve must be in closed position when the valve is installed.

To assemble the 3" CHEMFLY:

1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. In order to avoid any damage caused to the valve seat when doing the equipment assembly, ensure the butterfly is in closed position.
3. Fit the valve in position and begin to screw the 4 bolts Inlet side.
4. Cross-tighten the bolts symmetrically torque of 30 N.m +/- 5, for more details please check PWAGEN001

OPERATION

To open the valve, unlock with the special key then rotate handle to the opening position

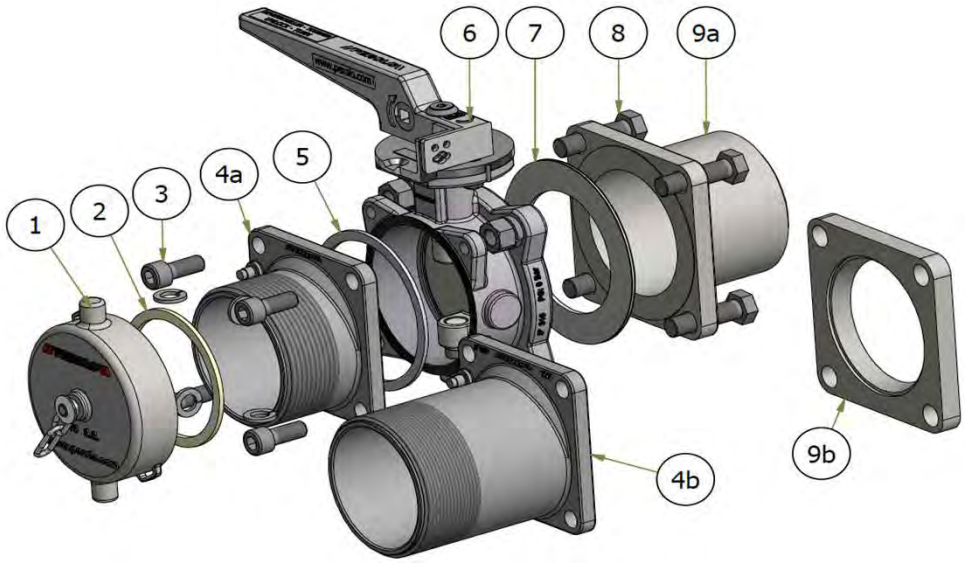
To close the valve, rotate handle to the close position until it locks.

When locked, the valve cannot operate

CAUTION : Do not open the valve under pressure when it is not connected to any downstream pipe

Please check the chemical compatibility of the cargo with the valve material

Item	Designation	Part number
1, 2	3" BSP cap kit	12 85 59 91 00
4a	3" BSP short outlet	17 12 09 70 00
4b	3" BSP long outlet	17 12 09 71 00
3	Screw kit: 4 x M10 CHC screws, 4 washers	17 12 11 91 00
5	Outlet gasket PTFE	17 12 21 00 01
6	3" CHEMFLY butterfly valve	LY311
8	Screw kit: 4 x M12 CHC screws, 4 washers, 4 nuts	17 12 11 92 00
7	Inlet gasket (synthetic fibers)	17 12 21 10 00
9a	Inlet welding	17 12 09 80 00
9b	Square inlet flange	17 12 09 81 00

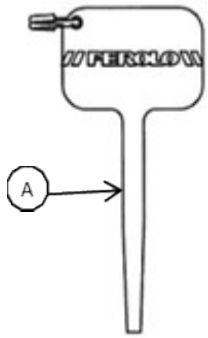


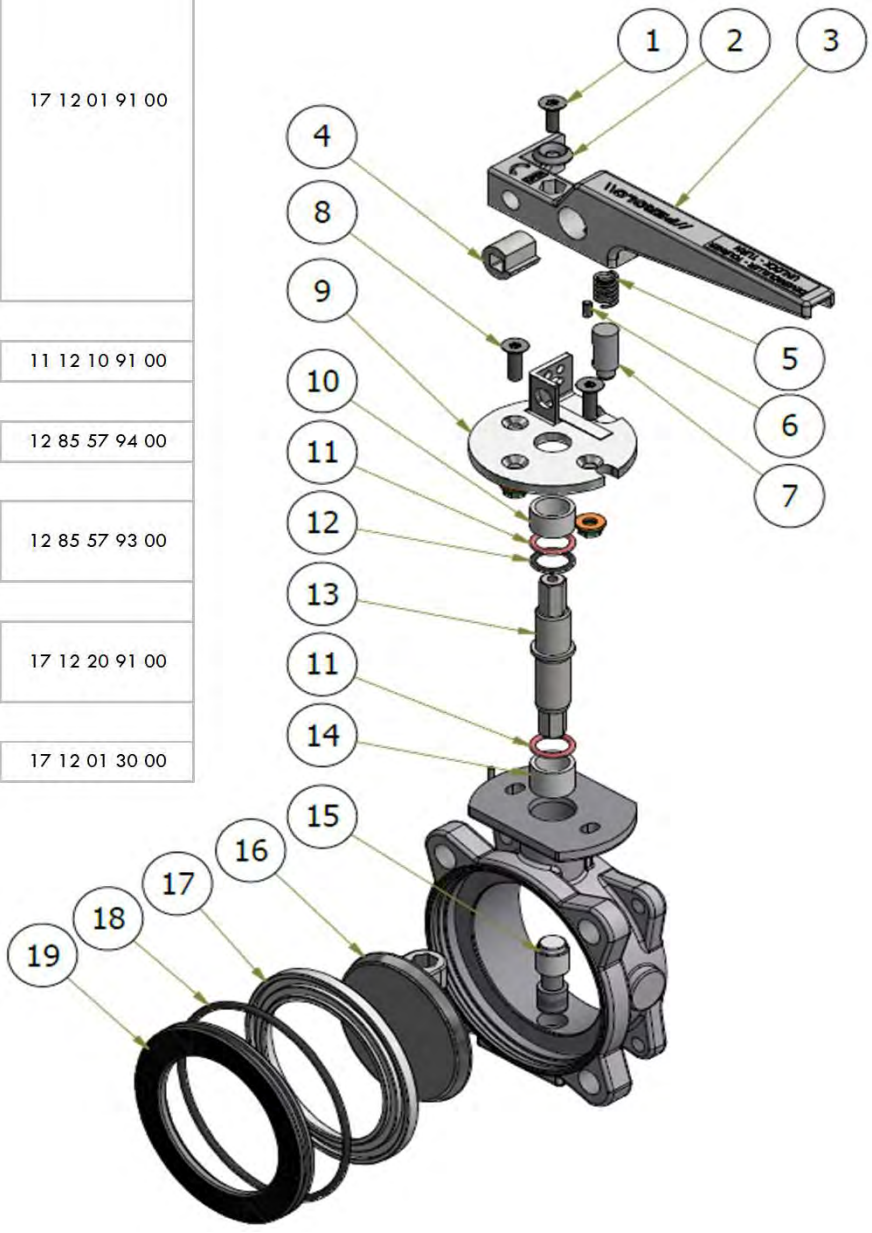
BUTTERFLY VALVE

3" (ND80) CHEMFLY "Light"
Flanged Butterfly Valve
LY3

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit	
1	1	Screw M6-16, Stainless steel	17 12 01 90 00
3	1	Handle, Stainless steel	
2	1	Washer	
5	1	Spring, Stainless steel	
7	1	Handle spindle, Stainless steel	
4	1	Center boss, Stainless steel	
6	1	M4-8 screw	
		Spindle Seals kit	
1	1	Screw M6-16, Stainless steel	17 12 01 91 00
2	1	Washer	
8	3	Screws M6-20, Stainless steel	
9	3	Washers, Stainless steel	
	3	Nuts, Stainless steel	
10	1	Spindle bush, PTFE & glass 25%	
14	1	Spindle bush, PTFE & glass 25%	
12	1	O'ring, Viton	
11	2	O'ring, PTFE & silicon	
15	1	Closure plate bush, PTFE & glass 25%	
		Spindle	
13	1	Spindle (stainless steel)	11 12 10 91 00
		Closure plate seal kit	
17	1	Closure plate seal, PTFE	12 85 57 94 00
		Closure plate kit	
15	1	Closure plate bush, PTFE & glass 25%	12 85 57 93 00
16	1	Closure plate, Stainless steel	
		Clamping ring kit	
18	1	O'Ring, nitril	17 12 20 91 00
19	1	Clamping ring, Stainless steel	
		Handle Locking Key	
A	1	Key	17 12 01 30 00





BUTTERFLY VALVE

4" (ND100) TANKFLY
Clamped Butterfly Valve
TP10X

The 4" TANKFLY clamped butterfly valve is designed to be clamped between 2 flanges or directly onto the pipe work.

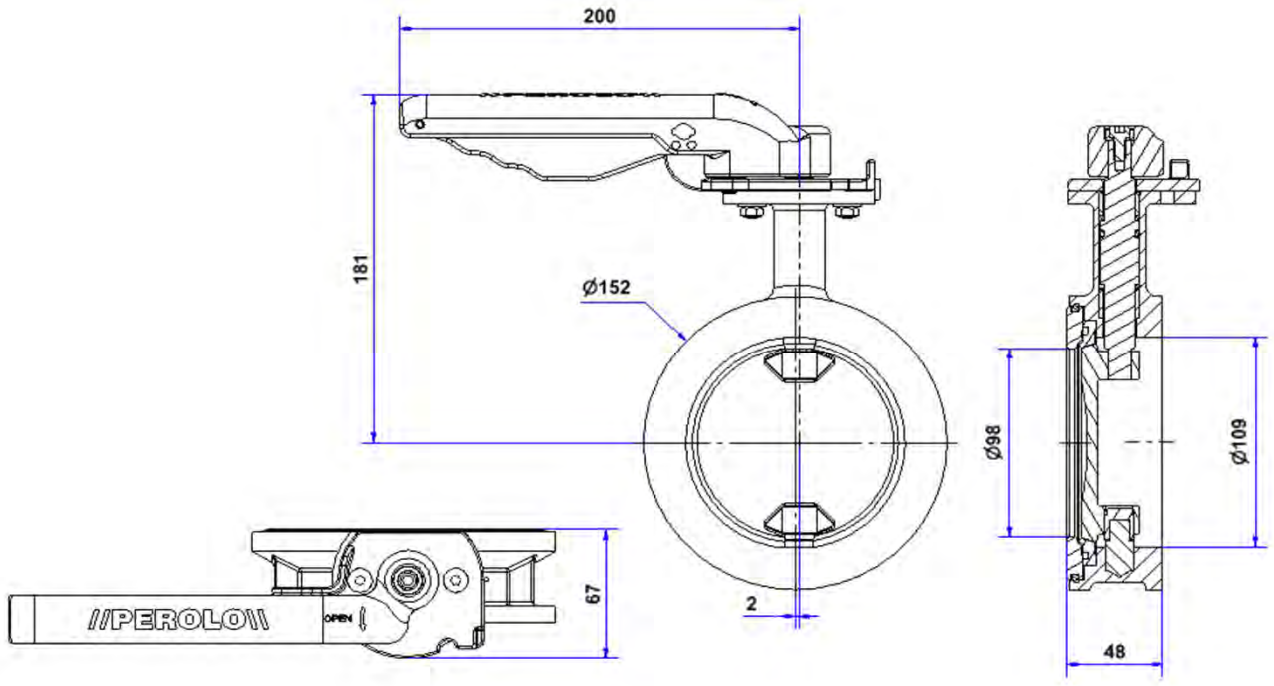
SPECIFICATION	
Nominal diameter	100mm (4")
Weight	4.3 Kg
Inlet flange	Clamped between DIN or ANSI Flange
Outlet flange	Clamped between DIN or ANSI Flange
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

CONFORMITY, CERTIFICATION & APPROVAL
Comply with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Compact Design	Face to face 48 mm
Operation	Handle - 1/4 turn lockable
Closure plate seal	PTFE

ACCESSORIES	
Weld-in flange	11 91 37 00 00
Seal	11 15 26 00 01
Adaptor flange	11 94 99 00 00

DIMENSIONAL DRAWING



BUTTERFLY VALVE

4" (ND100) TANKFLY
Clamped Butterfly Valve
TP10X

CODIFICATION

STANDARD PART NUMBER	TP10	X	7	T	.
YOUR PART NUMBER	TP10	__	7	T	__

MATERIAL	
316L / 1.4409 stainless steel	X
904L stainless steel	U

MAWP	
7 bar	7

SEAL MATERIAL	
PTFE seal	T

HALAR COATING	
Without	.

*for coated options, please see high corrosion catalogue

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 01 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
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4" (ND100) TANKFLY
Clamped Butterfly Valve
TP10X

INSTALLATION (TOP DISCHARGE)

CAUTION : Tankfly must be in closed position when the valve is fitted on the tank.

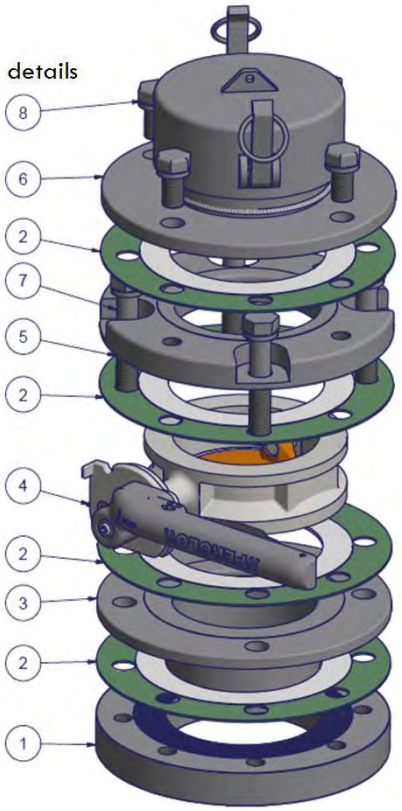
To assembly the 4" Tankfly clamped butterfly valve:

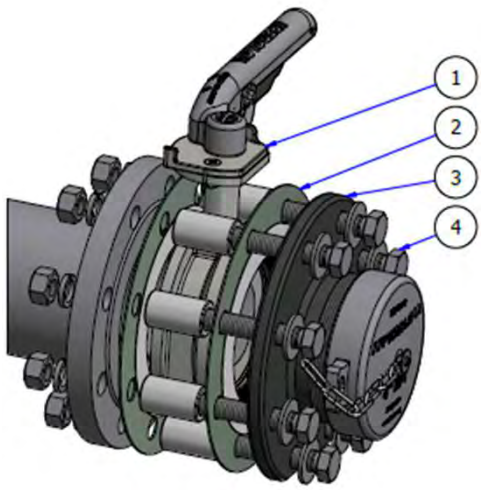
1. Ensure that the Inlet (tank side) flange is clean; free from dirt and grit.
2. Align holes in gasket to studs.
3. Center the TANKFLY with weld in flange hole
4. Mount outlet gasket and blind flange and secure using bolting kit (9).

The required tightening torque is 40 N.m +/- 5. Please refer to PWAGEN001 for more details

The valve BODY flange should be in contact with weld-in flange when tightened.

ITEM	DESCRIPTION	SPECIFICATION	PART N°
6	Camlock nozzle	Ext Ø 200mm, drilling 4 x 18 x 180 PCD , 316L/1.4404	on request
8	Screw kit M16-30	4 x M16-30 screws & 4 x grower washers, 304L/1.4301	11 90 01 92 00
7	Screw kit M16-100	4 x M16-100 screws & 4 x washers & 4 x nuts, 316L	12 85 57 92 00
5	Adaptor flange	Ext Ø200mm, 4 x 18mm x 180 PCD & 4 x M16 x 180 PCD, 316L	11 94 99 00 00
4	4" TANKFLY clamped	4" clamped butterfly valve, 316L/1.4404	TP10X7T
3	Syphon tube	4" syphon tube, 4 x Ø18 x 180PCD, 316L/1.4404	11 94 98 00 00
2	Seal	Ext Ø 220mm, int Ø 117mm, 8 x 18 x 180 PCD, PTFE	11 15 26 00 01
1	Weld in flange	Ext Ø220mm, int Ø116mm, 8x M16x180PCD, th. 26.5mm, 316L	11 91 37 00 00





ITEM	DESCRIPTION	SPECIFICATION	PART N°
4	Screw kit M16-100	8 x M16-100 screws, 8 x washers, 8 x grower washers & 4 x nuts, 316L, 8 spacers	11 16 31 91 00
3	4" NPT nozzle	Round outlet nozzle, 4" NPT, including cap, gasket and chain, 8 x 18 x 180 PCD	1116315000
2	Seal	Ext Ø 220mm, int Ø 117mm, 8 x 18 x 180 PCD, PTFE	11 15 26 00 01
1	4" TANKFLY clamped	4" clamped butterfly valve, 316L/1.4404	TP10X7T
2	Seal	Ext Ø 220mm, int Ø 117mm, 8 x 18 x 180 PCD, PTFE	11 15 26 00 01

if a piping system linked to the tank is installed on the process side of the valve, we recommend to have a flexible hose in-between the valve and the piping system to avoid any additional stress on the fastening studs

4" (ND100) CHEMFLY
Clamped Butterfly Valve
17 12 05 00 00

The 4" CHEMFLY clamped butterfly valve is designed to be clamped between 2 flanges or directly onto the pipe work.

SPECIFICATION	
Nominal diameter	98mm (4")
Weight	3.8 Kg
Inlet flange	Clamped between DIN or ANSI Flange
Outlet flange	Clamped between DIN or ANSI Flange
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

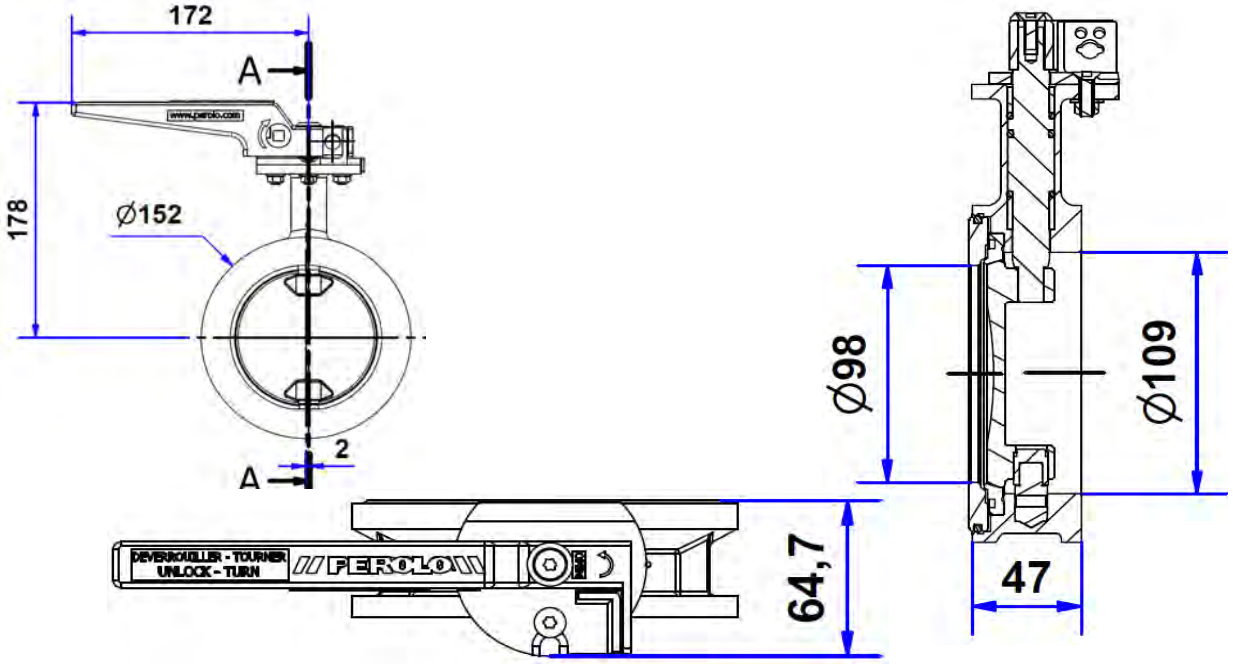
CONFORMITY, CERTIFICATION & APPROVAL
Comply with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas

STANDARD FEATURES	
Made of (material)	316 / 1.4408 stainless steel
Compact Design	Face to face 47 mm
Operation	Locking Handle - 1/4 turn
Closure plate seal	PTFE

ACCESSORIES	
Handle Locking Key	17 12 01 30 00

CODIFICATION	
4" CHEMFLY clamped butterfly valve - standard - PTFE gasket	17 12 05 00 00
4" CHEMFLY clamped butterfly valve - CE1935/2004 - PTFE gasket	17 12 05 00 00 A

DIMENSIONAL DRAWING



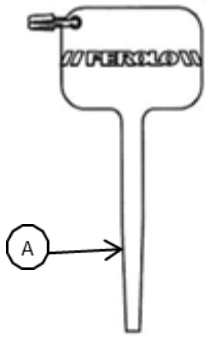
BUTTERFLY VALVE

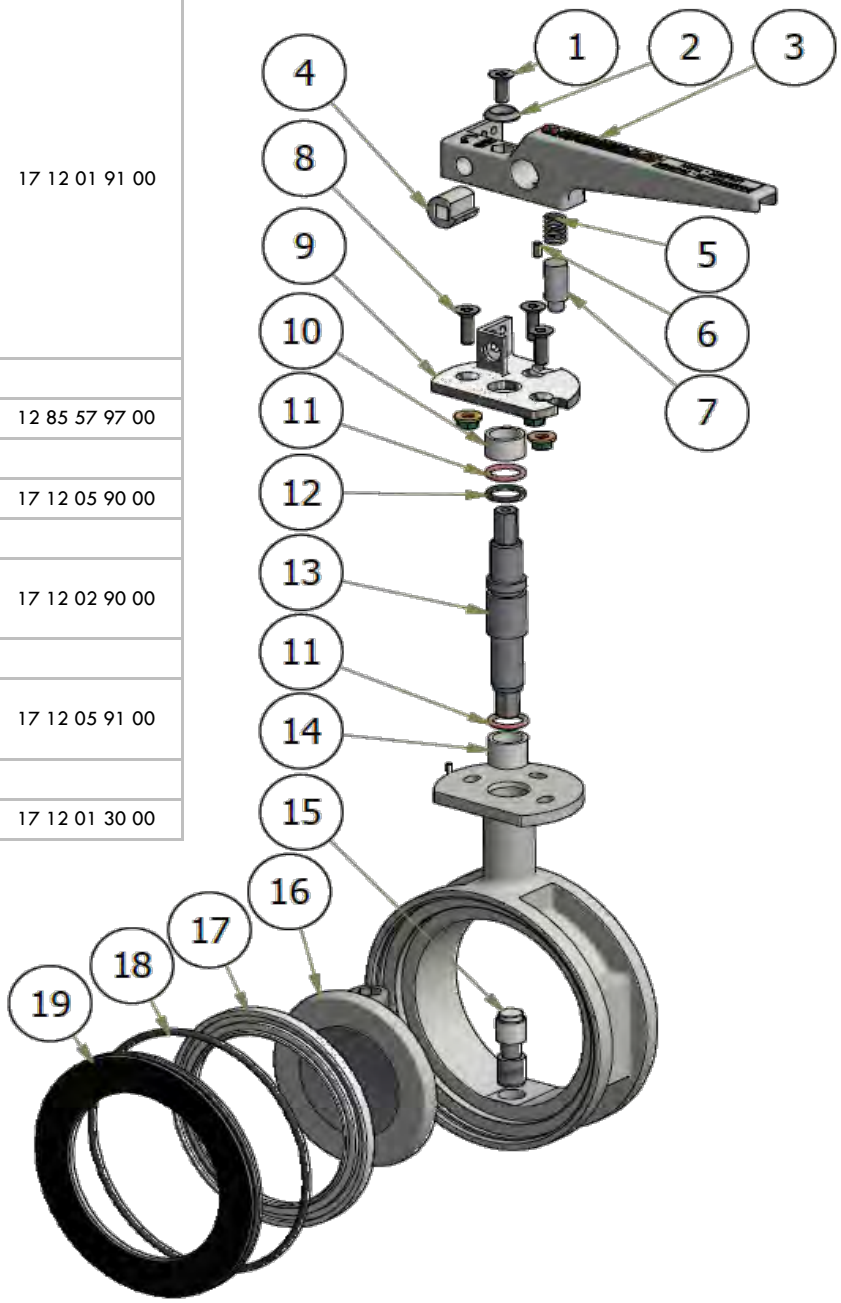
4" (ND100) CHEMFLY
Clamped Butterfly Valve
17 12 05 00 00

ROAD

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Handle kit	
1	1	Screw M6-16, Stainless steel	17 12 01 90 00
3	1	Handle, Stainless steel	
2	1	Washer	
5	1	Spring, Stainless steel	
7	1	Handle spindle, Stainless steel	
4	1	Center boss, Stainless steel	
6	1	M4-8 screw	
		Spindle Seals kit	
1	1	Screw M6-16, Stainless steel	17 12 01 91 00
2	1	Washer	
8	3	Screws M6-20, Stainless steel	
9	3	Washers, Stainless steel	
	3	Nuts, Stainless steel	
10-14	2	Spindle bush, PTFE &glass 25%	
12	1	O'ring, Viton	
11	2	O'ring, PTFE & silicon	
15	1	Closure plate bush, PTFE & glass 25%	
		Spindle	
13	1	Spindle (stainless steel)	12 85 57 97 00
		Closure plate seal kit	
17	1	Closure plate seal, PTFE	17 12 05 90 00
		Closure plate kit	
15	1	Closure plate bush, PTFE & glass 25%	17 12 02 90 00
16	1	Closure plate, Stainless steel	
		Clamping ring kit	
18	1	O'Ring, nitril	17 12 05 91 00
19	1	Clamping ring, Stainless steel	
		Handle Locking Key	
A	1	Key	17 12 01 30 00





BUTTERFLY VALVE

FOOTVALVES & BOTTOM DISCHARGES

FUNCTION

A bottom discharge is designed to be fitted at the bottom of a tank or tanker to ensure loading and unloading. It generally consists of a footvalve, followed by a second closure valve (a ball valve or butterfly valve), and lastly an outlet nozzle with cap or a coupling. The footvalve can have an internal or external spring. Perolo offers both mechanic and pneumatic actuated footvalves. For ISO tank containers, footvalves have an angle of 45° or 30°, whereas with Road Tankers or Offshore tanks an angle of 90° is more common.

- Various internal diameters: 3" (80mm) or 4" (100mm)
- Internal or external spring
- Body angle: 30° or 45° (ISO) 90° (ROAD)
- Single or combined body
- Inlet connection: Various drilling patterns
- Outlet connection: 3' BSP nozzle, Camlock, flange,....
- Operating mechanism: Manual: handle or remote lever Pneumatic actuator
- Contact Materials: 316L stainless steel
- Working Temperature: Standard or High (+250°C)
- Steam heating
- Surface Finish: Acid etched, Halar lined
- Special Uses: Bitumen
- ADR, RID, IMDG

3" 30° or 45° NEATCO FV	3" 30° or 45°NEATFLOW FV	3" 30° or 45° NEATCO BD	3" 30° or 45°NEATFLOW BD
3" 30° or 45° TWINCO BD	3" 30° or 45° TWINFLOW BD	4" 45° NEATFLOW FV	3" CHEMFLY BD
3" 90° CHEMFLOW FV	4" 90° CHEMFLOW FV	4" 90° CHEMFLOW BITUMEN	

BOTTOM DISCHARGE

3" (ND80) NEATCO MKII
Footvalve
NT3

3" NEATCO MKII 45° or 30° footvalve - designed to be fitted at the bottom of tank containers - no retention, better flow performance - more robust but no heavier - new stuffing box fastening - conical poppet - overall external dimensions reduced allowing more rear cabinet clearances.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	7.2 Kg
Inlet flange	8 holes x Ø14 on a PCD178
Outlet flange	4 holes x Ø18 on a PCD160*
Working pressure	4 Bar
Test pressure	6 Bar
Working temperature	-40° to +190°C
Design temperature	-40° to +200°C

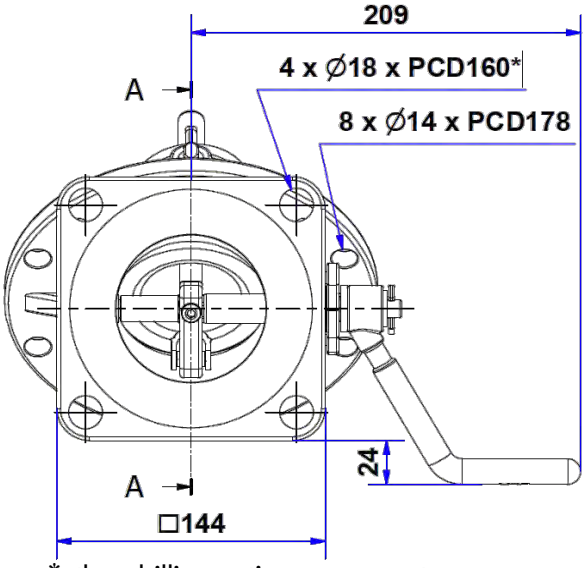
NB: temperature range vary according to o'ring material

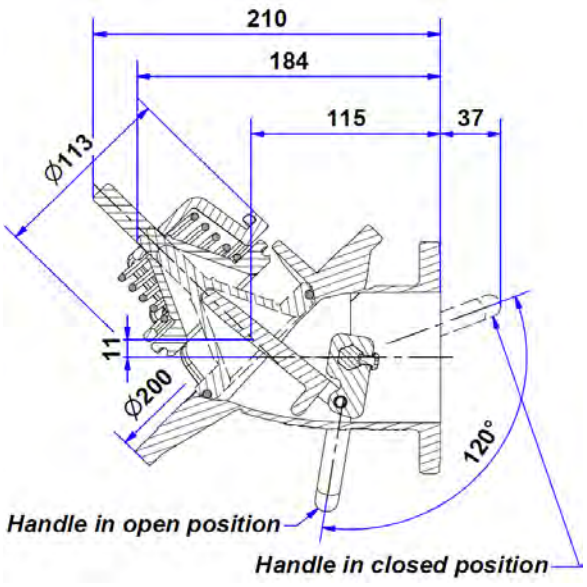
CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433
Approved by : Bureau Veritas - RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Body length	115mm (or 97mm but only for 30°)
Body length	115mm (or 97mm but only for 30°)
O'rings material	FEP/Silicone
Seals material	PTFE
Operating	Side operated handle
Footvalve spring	Inside the tank

ACCESSORIES	
45° weld-in flange	11 90 59 00 00
30° weld-in flange	11 91 16 00 00
Flange gasket	12 91 20 00 00
Screw kit	12 91 61 94 00

DIMENSIONAL DRAWING





*other drilling options on request

drawing of 45° footvalve, for drawing of 30° footvalve please contact our sales department

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 16 01 20 00 H
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) NEATCO MKII
Footvalve
NT3

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	NT3	4	115	A	1	.	.	.

YOUR PART NUMBER NT3 ____ ____ ____ ____ ____ ____ ____ ____

OUTLET ANGLE	
45°	4
30°	3

BODY LENGTH	
115 mm	115
97mm (only for 30° model)	97

INLET FLANGE	
8 x Ø14 x 178 PCD (only 45°)	A
8 x Ø14 & 4 x Ø18 x 178 PCD (30°	B

OUTLET FLANGE - others on request	
Square, equidist. 4 x Ø18 x160PCD	1
Round, equidist. 4 x Ø18 x160PCD	2
Square, equidist. 4 x M16 x160PCD	3
Round, non-equid. 6 x Ø18 x168PCD	5
Round, non-equid. 6 x Ø14 x168PCD	6

SEAT OPTIONS	
Stainless st 316L / 1.4409 seat	.
Hastelloy C276 coated seat	Q

STEAM HEATING	
Without	.
With steam heating	V

HANDLE	
Standard handle	.
Short handle	C

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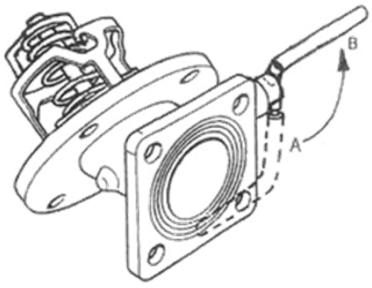
3" (ND80) NEATCO MKII
Footvalve
NT3

BOTTOM DISCHARGE

INSTALLATION



Ensure that the Inlet flange and gasket are clean; free from dirt and grit.

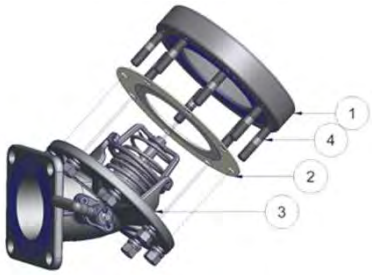
1. Check Inlet (tank side) flange is clean, free from dirt and grit.
   The operating lever can be turned to B position.
2. Screw the studs into the weldin flange, lubricate with PTFE added grease before installing the studs
3. Place the gasket
4. Install the valve on the flange, secure with washer and nuts
5. Tighten the nuts torque : 50 N.m +/- 10 (M12); for 2nd closure valve 70 N.m +/- 5 (M16) behind neatco *(refer to PWAGEN001 fort bolt tightening sequence)*
6. Install gasket, bolting kit and blind flange on the valve outlet flange
7. Open and close the valve 3 times to check the operationnal torque.

Picture 1

Picture 2

Picture 3

Picture 4



45° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	NEATCO MK2 3" 45°	3" 45° footvalve, inlet flange 8x14xPCD178, Outlet flange 4x18xPCD160	NT34115A1
4	Stud kit	6x M12 34/25 studs, 2x M12 40/30 studs, 8x grower washers, 8xM12 standard nuts	12 91 61 94 00
2	Inlet Gasket	ext ø 200 mm, int ø114mm, 8 x 14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 90 59 00 00

30° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	NEATCO MK2 3" 30°	3" 30° footvalve, inlet flange 8x14xPCD178 & 4x18xPCD178, Outlet flange 4x18xPCD160	NT33115B1
4	Stud kit	6x M12 34/25 studs, 2x M12 40/30 studs, 8x grower washers, 8xM12 standard nuts	12 91 61 94 00
2	Inlet Gasket	ext ø 200 mm, int ø 114 mm8 x 14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm thickness 35 mm 8 x M12 x PCD 178 316L stainless steel	11 91 16 00 00

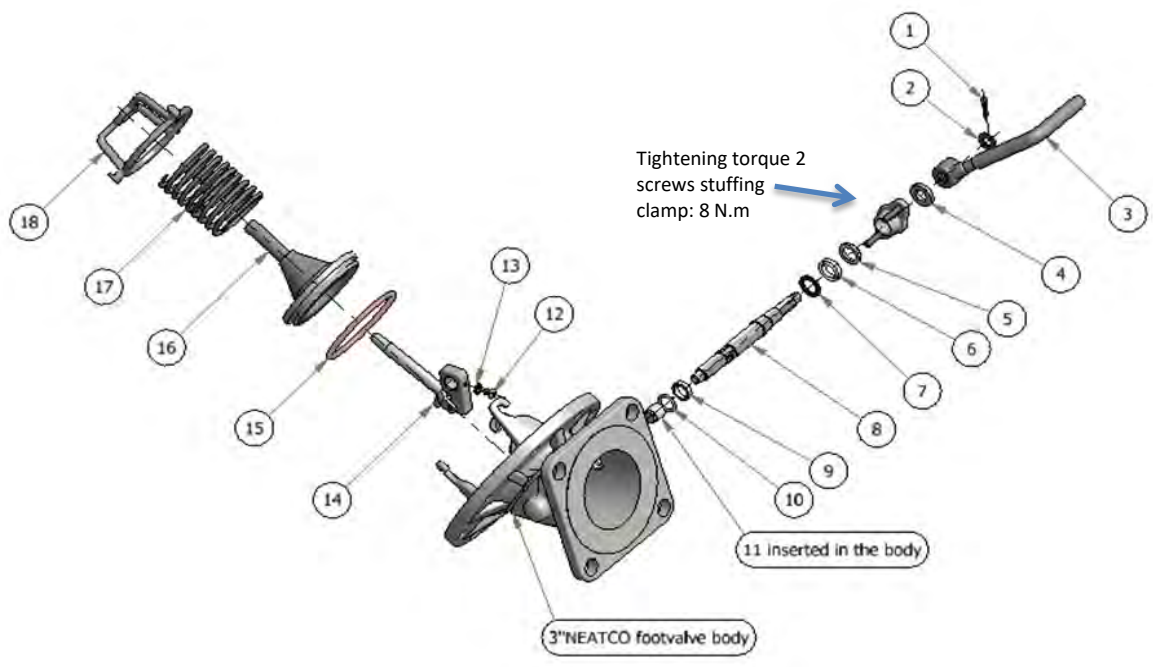
3" (ND80) NEATCO MKII
Footvalve
NT3

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Spindle kit:	
8	1	Spindle, stainless steel	11 16 01 00 01
		Spindle seal kit:	**11 16 01 93 00**
1	1	Split pin, stainless steel	-
2	1	Washer, stainless steel	-
4	1	Spacer, stainless steel	-
6 , 5	1	Packing gland, PTFE	12 85 30 30 00
7	1	Packing ring, PTFE	12 85 30 00 02
9	1	Spindle bush, PTFE	11 16 01 00 03
10	1	O'ring, FEP/silicone	70 10 15 00 12
11	1	End spindle bush, PTFE & 25% glass	17 16 23 10 01
		Pushrod kit:	**11 16 01 92 00**
12	1	M6-8 screw, stainless steel	-
13	1	M6 washer, stainless steel	-
14	1	Crank & pushrod, stainless steel	11 16 01 30 00
		Bayonet cap & spring:	
18	1	Bayonet cap	11 16 01 00 04
17	1	Spring, stainless steel	12 85 38 00 02

ITEM	QTY	DESIGNATION	PART N°
		Poppet kit:	**1111149700mk2**
16	1	Poppet, stainless steel	11 16 01 20 00
15	1	O'ring, FEP/silicone	70 10 15 00 53
		Poppet seal kit:	**11 11 24 91 00**
15	1	O'ring, FEP/silicone	70 10 15 00 53
		Handle kit:	**12 85 24 93 00**
1	1	Split pin, stainless steel	-
2	1	Washer, stainless steel	-
3	1	Handle, stainless steel	11 11 69 0A 00

Nb: Perolo recommends the spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.



BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 16 01 40 00 H
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) NEATFLOW MKIV

Footvalve

NX3

The Perolo 3" NEATFLOW™ MKIV external spring footvalve is designed to be used as a bottom discharge for chemicals and liquid foodstuff products carried in tank containers and IBC's. The external spring design is perfectly suited to tanks containing viscous fluids and allows easy cleaning and maintenance - Discharge time improved - compact design

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	9.5 Kg
Inlet flange	8 holes x Ø14 on a 178 PCD
Outlet flange	4 holes x Ø18 on a 160 PCD
Working pressure	4 Bar
Test pressure	6 Bar
Working temperature	-40° to +190°C (HT +220°C)
Design temperature	-40° to +200°C (HT +250°C)

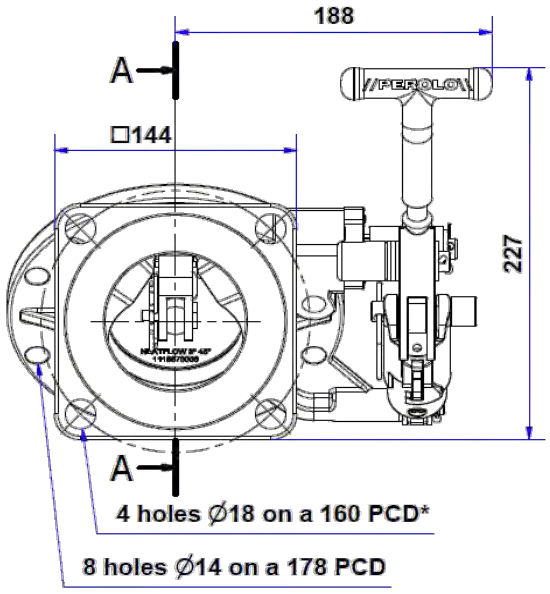
NB: temperature range vary according to o'ring material

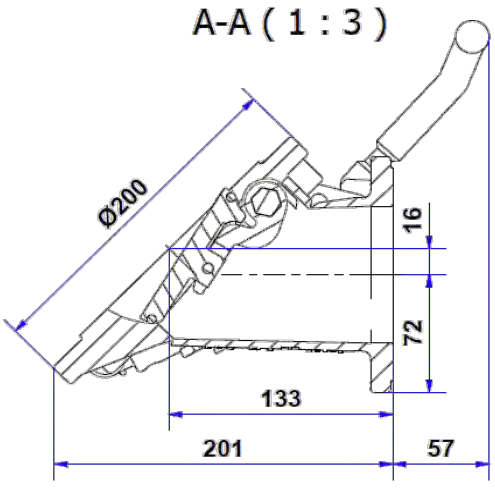
CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433
Approved by : Bureau Veritas - RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 (Hast. C22 on request)
Angle	45°
Body length	133 mm from center to outlet
O'rings material	FEP/Silicone
Seals material	PTFE
Operating	Side operated handle or Pneumatic option
Footvalve spring	Spring outside the tank allows easier cleaning

ACCESSORIES	
45° weld-in flange	11 90 59 00 00
30° weld-in flange	11 91 16 00 00
Flange gasket	12 91 20 00 00
Screw kit	12 91 61 94 00

DIMENSIONAL DRAWING





*other drilling options on request

drawing of 45° footvalve, for drawing of 30° footvalve please contact our sales department

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 16 87 20 00 C
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) NEATFLOW MKIV
Footvalve
NX3

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	45°: NX3	4	A	X	1	.	.	W	.	.	.
	30°: NF3	3	A	X	1	.	.	W	.	.	.
YOUR PART NUMBER	45°: NX3	4	A	X							
	30°: NF3	3	A	.							

OUTLET ANGLE	
45°	4
30°: only in MKII NF33 version	3

INLET FLANGE	
8 x Ø14 x 178 PCD	A

MATERIAL	
316L stainless steel (45° version)	X
316L & Hastelloy C276 coated seat	Q
316L stainless steel (30° version)	.

OUTLET FLANGE	
Square flange 4 x Ø18 x 160 PCD	1
Square flange 4 x M16 x 160 PCD	2

STEAM HEATING	
Without	.
With steam heating	V

CONICAL POPPET	
Without	.
With conical poppet	K

REMOTE CONTROL	
Without	.
With remote control lever	W

HIGH TEMPERATURE	
Without (WT +190°C)	.
With High temperature (WT +220°C)	HTA

PNEUMATIC ACTUATOR	
Without	.
With	P

HANDLE	
Standard Handle	.
Without	Z
L Handle	L
Short Handle	C
Left L Handle	G

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 16 87 20 00 C
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3" (ND80) NEATFLOW MKIV
Footvalve
NX3

BOTTOM DISCHARGE

INSTALLATION

Ensure that the Inlet (tank side) flange and gasket are clean; free from dirt and grit.

1. Screw the studs into the tank container flange. Take note of the position of the studs (pic 1)
2. Fit the gasket (picture 2).
3. Fit the NEATFLOW MKIV Footvalve in position (picture 3).
4. Tighten the nuts torque : 50 N.m +/- 10 (M12); for 2nd closure valve 70 N.m +/- 5 (M16) behind neatflow.

For more details please refer to PWAGEN001

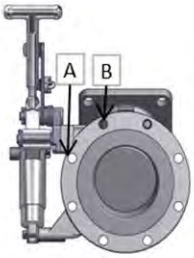
Pneumatic version: Compressed air pressure for actuator should be in-between 4 bar and 6 bar to ensure correct opening & closing of the poppet

A Stud M12 63/33 with standard nuts

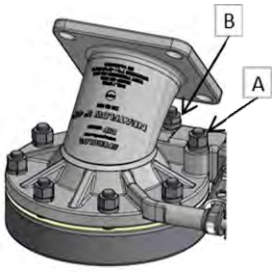
B Stud M12 48/30 with thin nut

Picture 1

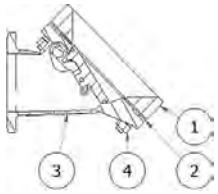
Picture 2



Picture 3



Picture 4



45° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	3" NEATFLOW 45° MK IV	3" 45° footvalve, inlet flange 8x14xPCD178 Square Outlet flange 4xØ 18xPCD160	NX34AX1
4	Stud kit	6x M12 34/25 studs, 1x M12 40/30 stud(B)1x M12 63/33 stud(A), 8x grower washers, 7xM12 standard nuts, 1x thin nut (B)	11 11 27 93 00
2	Inlet Gasket	ext ø 200 mm, int ø 114 mm, 8 x 14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 90 59 00 00

30° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	3" NEATFLOW 30° MK II	3" 30° footvalve, inlet flange 8x14xPCD178 Square Outlet flange 4xØ 18xPCD160	NF33A1
4	Stud kit	6x M12 34/25 studs, 1x M12 40/30 stud(B)1x M12 63/33 stud(A), 8x grower washers, 7xM12 standard nuts, 1x thin nut (B)	11 11 27 93 00
2	Inlet Gasket	ext ø 200 mm, int ø 114 mm, 8 x 14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 91 16 00 00

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 16 87 30 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

3" (ND80) NEATFLOW MKIV

Footvalve

NX3

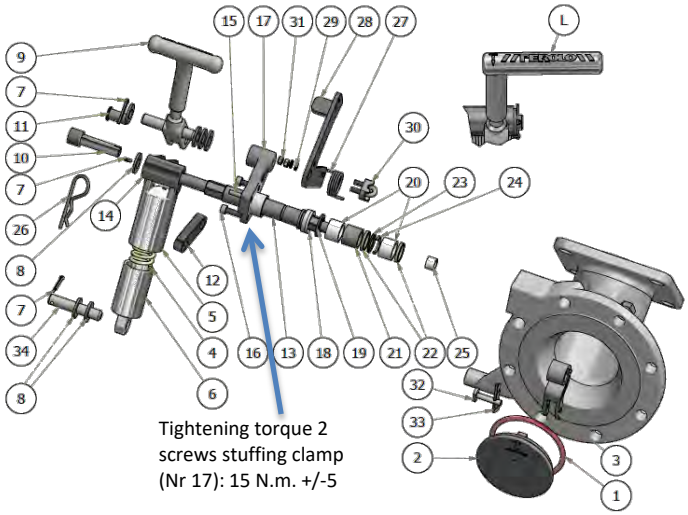
ISO ROAD IBC

BOTTOM DISCHARGE

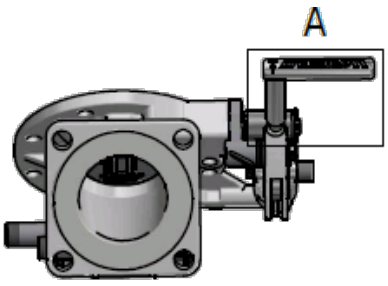
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Spindle kit:	**on request**
13	1	Spindle, Stainless steel	17 11 25 00 01
15	2	Grooved pin, Stainless steel	-
17	1	Stuffing clamp, Stainless steel	-
		Spindle seals kit:	**11 11 24 95 00**
25	1	Spindle bush, PTFE	12 85 30 00 05
22	2	Gasket Ekonol, PTFE	70 10 90 00 93
20	2	Spindle bush, PTFE	11 11 24 00 06
24	1	Elastic ring, Stainless steel	-
23	1	Stop ring, Stainless steel	-
21	1	Spacer, Stainless steel	-
19	1	Packing gland, PTFE	11 11 24 00 15
18	3	Packing ring, Stainless steel	-
8	1	Washer, S.S.	-
7	1	V-pin, S.S.	-
		Pusher Mk4:	
3	1	Pusher, Stainless steel	11 16 87 00 02
		Poppet seal kit:	**11 11 24 91 00**
1	1	O'ring, FEP/Silicon	70 10 15 00 53
		Poppet kit:	**11 16 87 90 00**
1	1	O'ring, FEP/Silicon	70 10 15 00 53
2	1	Poppet, Stainless steel	11 16 87 00 06
32	1	Poppet Spindle, Stainless steel	11 16 87 00 05
33	1	Beta Pin	11 16 87 00 04
		Remote control kit:	**11 16 25 95 00**
28	1	Remote control lever, Stainless steel	11 16 25 00 02
27	1	Spring, Stainless steel	11 16 25 00 03
31	2	M5 nut, Stainless steel	-
29	2	M5 washer, Stainless steel	-
30	1	Rope clamp, Stainless steel	-

ITEM	QTY	DESIGNATION	PART N°
		Spring system kit:	**11 11 14 99 19**
6	1	Female MK2 case, Stainless steel	11 11 70 0A 03
5	1	Male MK2 case, Stainless steel	11 11 70 0A 04
26	1	Beta pin, Stainless steel	70 13 20 37 14
8	2	Washer, Stainless steel	-
7	1	V-pin, Stainless steel	-
4	1	Spring, Stainless steel	11 11 70 00 16
-	1	Chain	-
		Handle kit:	**11 16 25 94 00**
9	1	Handle, Stainless steel	11 11 83 1A 10
12	1	Connecting rod, Stainless steel	-
11	1	Spindle roller, Stainless steel	-
8	2	Z12 washer, Stainless steel	-
7	1	Split pin, Stainless steel	-
		L Handle kit:	**11 12 99 10 00**
L	1	L handle, Stainless steel	11 11 70 0A 08
12	1	Connecting rod, Stainless steel	-
11	1	Spindle roller, Stainless steel	-
8	2	Z12 washer, Stainless steel	-
7	1	Split pin, Stainless steel	-
		High Temperature	
1	1	O'ring, PFA/Silicon	70 10 17 00 32
19	1	Packing gland, PTFE Graphite	17 11 71 00 03
20	2	Spindle bush, PTFE Graphite	17 11 71 00 02
22a	1	Gasket, KALREZ	80 10 01 93 20
22b	1	Gasket, PTFE	70 10 90 00 93
25	1	Spindle bush, PTFE Graphite	17 11 71 00 01



Tightening torque 2 screws stuffing clamp (Nr 17): 15 N.m. +/-5



Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details.

Nb: Perolo recommends the spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 16 87 40 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) NEATCO MKII
Bottom Discharge
VNT3

3" NEATCO MKII 45° or 30° bottom discharge assembly combines one 3" NEATCO™ MKII footvalve, one 3" TANKFLY™ butterfly valve, and one 3" BSP threaded outlet nozzle with cap, all fully assembled together with gaskets and screws - This version improves the flow performance avoiding retention and also improves solidness due to FEA Design. Each bottom discharge is individually tested on a dedicated test bench and has the Perolo guarantee for leak-tightness.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	16.0 Kg
Inlet flange	8 holes x Ø14 on a 178 PCD
Outlet flange	3" BSP male threaded
Working pressure	4 Bar
Test pressure	6 Bar
Working temperature	-40° to +150°C (+190°C for Footvalve)
Design temperature	-40° to +190°C

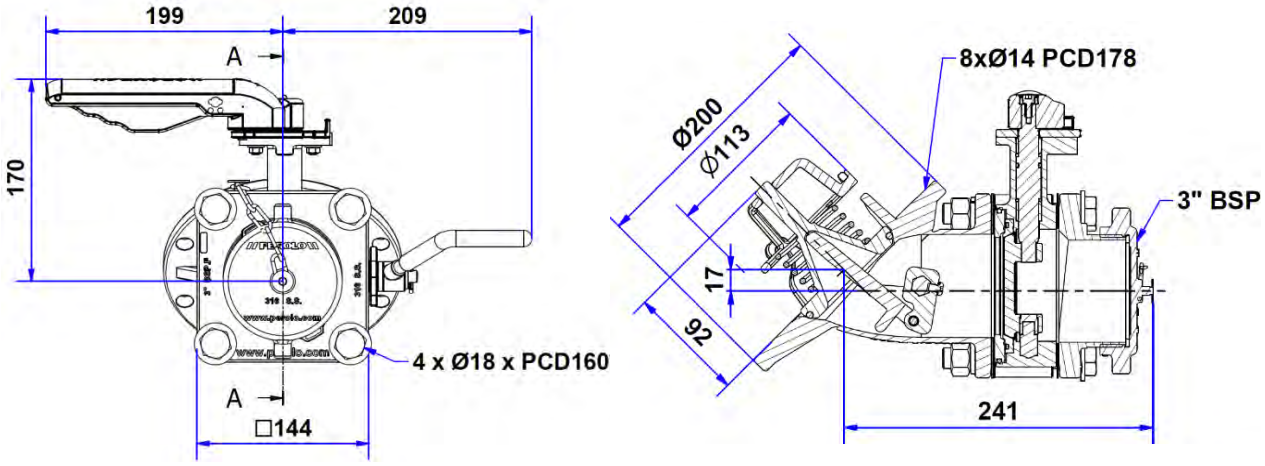
NB: temperature range vary according to o'ring material

CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433 / EN14432
Approved by : Bureau Veritas - RMR

ACCESSORIES	
45° weld-in flange	11 90 59 00 00
30° weld-in flange	11 91 16 00 00
Seal flange-footvalve	12 91 20 00 00
Screw kit	12 91 61 94 00
Seal footvalve-butt. valv	12 91 39 00 00
Seal butt. valve-outlet fl.	12 91 42 00 00
3" BSP outlet nozzle kit	12 85 96 00 00

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Angle	45° or 30°
Body length	depends on options
O'rings material	FEP/Silicone
Seals material	PTFE
Operating	Side operated handle
Footvalve spring	Inside the tank

DIMENSIONAL DRAWING



*other drilling options on request

drawing of 45° footvalve, for drawing of 30° footvalve please contact our sales department

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 16 28 20 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) NEATCO MKII
Bottom Discharge
VNT3

BIP GROUP – VALVES SOLUTIONS FOR YOUR – PEROLO – ISO RAIL IBC ROAD

CODIFICATION

STANDARD PART NUMBER	VNT 3	4	115	A	1	.	.	.	S	B	R

YOUR PART NUMBER: VNT3 ____ ____ ____ ____ ____ ____ ____ ____ ____ ____ ____

OUTLET ANGLE	
45°	4
30°	3
BODY LENGTH	
115 mm	115
97mm (only for 30° model)	97
INLET FLANGE	
8 x Ø14 x 178 PCD (only 45°)	A
8 x Ø14 & 4 x Ø18 x 178 PCD (30°	B
OUTLET FLANGE - others on request	
Square, equidist. 4 x Ø18 x160PCD	1
Round, equidist. 4 x Ø18 x160PCD	2
Square, equidist. 4 x M16 x160PCD	3
Round, non-equid. 6 x Ø18 x168PCD	5
Round, non-equid. 6 x Ø14 x168PCD	6
SEAT OPTIONS	
Stainless st 316L / 1.4409 seat	.
Hastelloy C276 coated seat	Q
STEAM HEATING	
Without	.
With steam heating	V
HANDLE	
Standard handle	.
Short handle	C

SECOND CLOSURE SYSTEM	
Clamped butterfly valve ND 80 with hexagonal spindle	S
Flanged butterfly valve ND 80 with hexagonal spindle	F
3" ATCO full bore PN16 ball valve	A
Clamped butterfly valve ND 80 with splined spindle	N
Flanged butterfly valve ND 80 with splined spindle	G

NOZZLE & CAP - other on request	
3" standard BSP Nozzle & cap	B
3" Camlock nozzle and cap	C
3" blind flange	P
REMOTE CONTROL	
Wire control device without cable	R
Complete wire control device	W

For installation, please refer to installation sheet of neatco footvalve

For maintenance, please refer to maintenance sheet of neatco footvalve & maintenance sheet of 2nd closure valve & accessories sheet for outlet (e.g. nozzle, cap)

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 16 28 20 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) NEATFLOW MKIV

Bottom Discharge

VNX3

The Perolo 3" NEATFLOW™ MKIV external spring footvalve is designed to be used as a bottom discharge for chemicals and liquid foodstuff products carried in tank containers and IBC's. The external spring design is perfectly suited to tanks containing viscous fluids and allows easy cleaning and maintenance - Discharge time improved - compact design

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	20.7 Kg
Inlet flange	8 holes x Ø14 on a 178 PCD
Outlet flange	3" BSP or 3" CAMLOCK
Working pressure	4 Bar
Test pressure	6 Bar
Working temperature	-40° to +150°C (HT +220°C)
Design temperature	-40° to +200°C (HT +250°C)

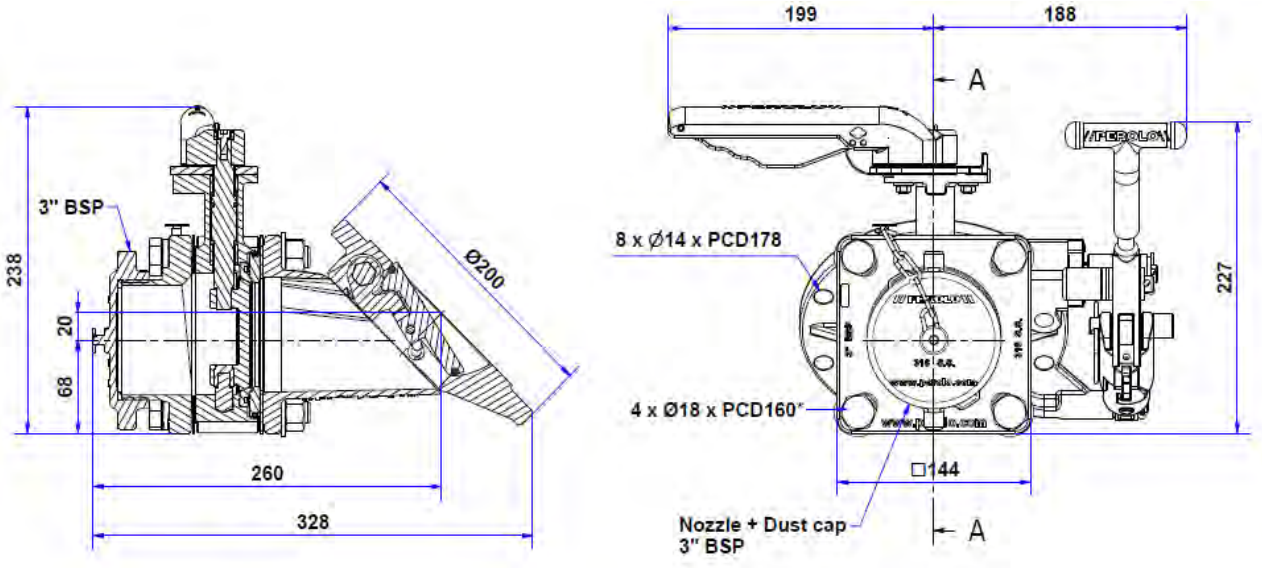
NB: temperature range vary according to o'ring material

CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433 / EN14432
Approved by : Bureau Veritas - RMR (footvalve & butterfly valve)

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Angle	45°
Body length	-
O'rings material	FEP/Silicone
Seals material	PTFE
Operating	Side operated handle
Footvalve spring	Spring outside the tank allows easier cleaning

ACCESSORIES	
45° weld-in flange	11 90 59 00 00
30° weld-in flange	11 91 16 00 00
Flange gasket	12 91 20 00 00
Screw kit	12 91 61 94 00

DIMENSIONAL DRAWING



*other drilling options on request

drawing of 45° footvalve, for drawing of 30° footvalve please contact our sales department

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 16 93 20 00 C
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

3" (ND80) NEATFLOW MKIV

Bottom Discharge

VNX3

CODIFICATION

STANDARD PART NUMBER	45°: VNX34	A	X	1	.	.	S	B	W	.	.	.
	30°: VNF33	A	X	1	.	.	S	B	W	.	.	.
YOUR PART NUMBER	45°: VNX34	A	X									
	30°: VNF33	A	.									

INLET FLANGE	
8 x Ø14 x 178 PCD	A
30°: only in MKII NF33 version	3
MATERIAL	
316L stainless steel (45° version)	X
316L & Hastelloy C276 seat (45°)	Q
316L stainless steel (30° version)	.
OUTLET FLANGE	
Square flange 4 x Ø18 x 160 PCD	1
Square flange 4 x M16 x 160 PCD	2
STEAM HEATING	
Without	.
With steam heating	V
CONICAL POPPET	
Without	.
With conical poppet	K
2ND CLOSURE VALVE - others on request	
Clamped butterfly valve ND 80 with hexagonal spindle	S
Clamped butterfly valve ND 80 with splined spindle	N
Flanged butterfly valve ND 80 with splined spindle	G
NOZZLE & CAP	
BSP Nozzle with cap	B
BSP Nozzle without cap	N
Camlock Nozzle	C
REMOTE CONTROL & CABLE	
Without	.
With remote control lever, no cable	W
With remote control lever & cable	WR
HIGH TEMPERATURE	
Without (WT +190°C)	.
With High temperature (WT +220°C)	HTA
PNEUMATIC ACTUATOR	
Without	.
With	P
HANDLE	
Standard Handle	.
Without	Z
L Handle	L
Short Handle	C
Left L Handle	G

For installation, please refer to installation sheet of neatflow footvalve

For maintenance, please refer to maintenance sheet of neatflow footvalve & maintenance sheet of 2nd closure valve & accessories sheet for outlet (e.g. nozzle, cap)

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 16 93 20 00 C
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) TWINCO MKII
Bottom Discharge
VTSCX3

Perolo 3" TWINCO bottom discharge assembly combines one 3" TWINCO™ footvalve with splined butterfly shaft, and one 3" BSP threaded outlet nozzle with cap, all fully assembled together with gaskets and screws. It is designed to be fitted at the bottom of tank containers. Made of stainless steel 316L/1.4409, this product is suitable for ISO tank containers dedicated to the transport and storage of chemicals and liquid foodstuff.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	15.0 Kg
Inlet flange	8 x Ø14 on a 178 PCD
Outlet flange	3" BSP*
Working pressure	4 Bar
Test pressure	6 Bar
Working temperature	-40° to +150° (+190°C for Footvalve)
Design temperature	-40° to +200°C

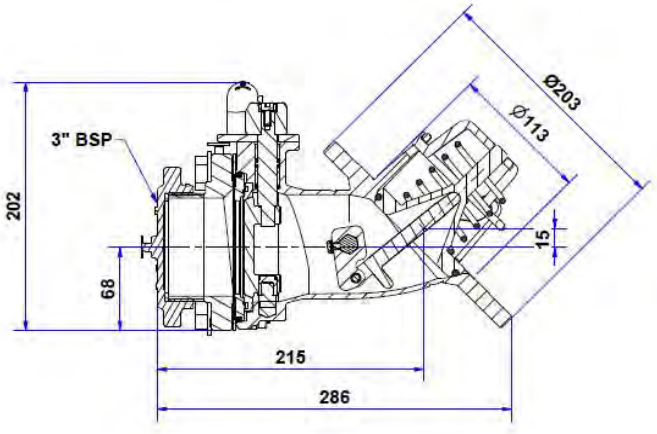
NB: temperature range vary according to o'ring material

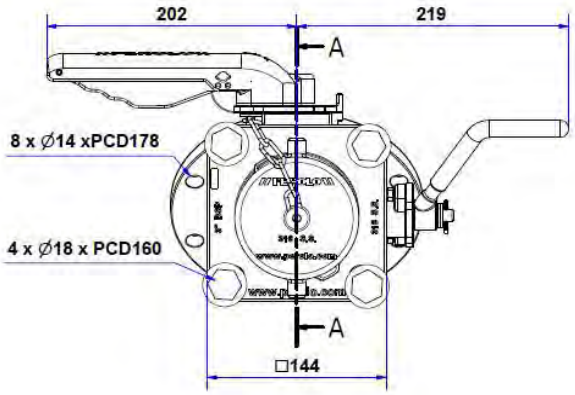
CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433 / EN14432
Approved by : Bureau Veritas - RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Angle	45° or 30°
Body length	216 mm from center to cap
O'rings material	FEP/Silicone
Seals material	PTFE
Operating	Side operated handle
Footvalve spring	Inside the tank

ACCESSORIES	
45° weld-in flange	11 90 59 00 00
30° weld-in flange	11 91 16 00 00
Seal	12 91 20 00 00
Stud kit	12 91 61 94 00

DIMENSIONAL DRAWING





drawing of 45° footvalve, for drawing of 30° footvalve please contact our sales department

* other outlet options on request (please see codification page)

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 22 48 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) TWINCO MKII
Bottom Discharge
VTSCX3

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	45° VTSCX3	4	S	.	.	.	.	C	R	.
	30° VTCS3	3S	S	.	.	.	.	C	R	.
YOUR PART NUMBER	45° VTSCX3									
	30° VTCS3		S							

OUTLET ANGLE	
45°	4
30° (MKI version only)	3S

OUTLET FLANGE DRILLING	
4 x Ø18 x 160 PCD	S

STEAM HEATING	
Without	.
With steam heating	V

SEAT MATERIAL	
Stainless st 316L / 1.4409 seat	.
Hastelloy C276 coated seat	Q

VENTING PORT	
Without	.
1 port 1/4" BSP	1
2 ports 1/4" BSP	2
1 port 1/4" NPT	3
2 ports 1/4" NPT	4

SAMPLING PORT - other on request	
Without	.

OUTLET SQUARE NOZZLE	
Without cap	N
With cap	C
With female cap	F
With 3" camlock nozzle & cap	B

REMOTE CONTROL	
With remote control system	R
With remote control system + cable	W
With remote control system + kick remote	K
With remote control system + handle set	D

HANDLE	
Standard	.
Short handle (45° footvalve)	C
Short handle (30° footvalve)	L

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 22 48 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) TWINCO MKII
Bottom Discharge
VTSCX3

INSTALLATION

Ensure that the Inlet flange and gasket are clean; free from dirt and grit.

1. Pull out the split pin and remove the washer on the control spindle.
   The operating lever can be turned to B position.
   Replace the washer and the split pin.
2. Screw the studs into the tank container flange.
3. Fit the PTFE gasket (picture 2).
4. Fit the TWINCO Footvalve in position.
5. Tighten the studs and torque of 50 N.m +/-5. For more details please refer to PWAGEN001

during loading/unloading operations the weight of the piping system fixed onto the footvalve should be lower than 700KG

1

2

45° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	3" TWINCO 45°	3" 45° footvalve, inlet flange 8xØ14xPCD178, Outlet flange 4xØ18xPCD160	VTSCX34SCR
4	Stud kit	6x M12 34/25 studs, 2x M12 40/30 stud(B), 8x grower washers, 8xM12 standard nuts	12 91 61 94 00
2	Inlet Gasket	ext ø 200 mm, int ø 109 mm8 x Ø14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 90 59 00 00

30° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	3" TWINCO 30°	3" 30° footvalve, inlet flange 8xØ14xPCD178, Outlet flange 4xØ18xPCD160	VTCS33SSCR
4	Stud kit	6x M12 34/25 studs, 2x M12 40/30 stud(B), 8x grower washers, 8xM12 standard nuts	12 91 61 94 00
2	Inlet Gasket	ext ø 200 mm, int ø 109 mm8 x Ø14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.9 mm thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 91 16 00 00

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 22 48 30 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

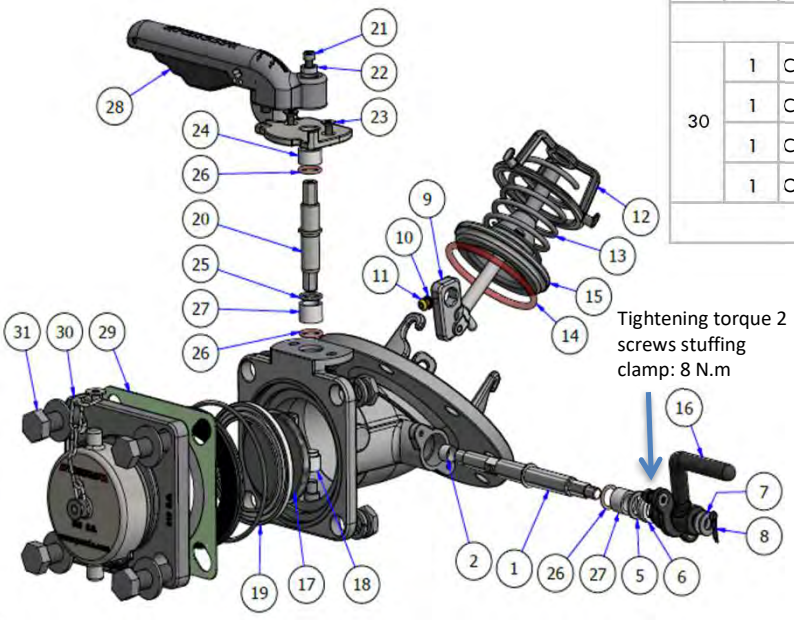
3" (ND80) TWINCO MKII
Bottom Discharge
VTCX3

BOTTOM DISCHARGE

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Spindle:	
1	1	Spindle, stainless steel	11 11 14 00 05
		Spindle seals kit:	**11 11 73 93 00**
8	1	Split pin, stainless steel	-
7	2	Washer, stainless steel	-
6	1	Packing gland, PTFE	12 85 30 30 00
5	1	Packing ring, stainless steel	-
27	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
26	1	O'ring, FEP/Silicon	70 10 15 00 12
2	1	Spindle bush, PTFE	12 85 30 00 05
-	2	Screw M6-25, stainless steel	-
		Pushrod kit:	**11 16 01 92 00**
9	1	M6-8 screw, stainless steel	-
10	1	M6 washer, stainless steel	-
11	1	Crank & pushrod, stainless steel	11 16 01 30 00
		Bayonet cap & spring:	
18	1	Bayonet cap, stainless steel	11 16 01 00 04
17	1	Spring, stainless steel	12 85 38 00 02
		Poppet kit:	**1111149700mk2**
15	1	Poppet, stainless steel	11 16 01 70 00
14	1	O'ring, FEP/Silicon	70 10 15 00 53
		Closure plate kit:	**12 85 57 93 00**
17	1	Closure plate, stainless steel	12 85 57 20 00
18	1	Closure plate bush, PTFE & glass	12 85 57 00 04
		Closure plate seal kit:	**12 85 57 94 00**
19	1	Closure plate seal, PTFE	12 85 57 00 01

ITEM	QTY	DESIGNATION	PART N°
		Butterfly valve Spindle:	**11 12 10 91 00**
20	1	Short spindle, Duplex / 1.4462	11 12 10 00 01
		Butterfly valve spindle seals kit:	**11 16 92 40 00**
27	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
26	2	O'rings, FEP & silicon	70 10 15 00 02
23	2	Screw, Nuts & Washer stainless steel	-
21	2	Screw M6-12, Stainless steel	-
22	1	Ring washer, Stainless steel	-
18	1	Closure plate bush, PTFE & glass 25%	12 85 57 00 04
24	1	Short Spindle bush, PTFE & glass 25%	11 12 62 00 01
25	1	O'ring, FKM/Viton	70 10 01 02 12
		Ring clamp kit:	**12 85 57 95 00**
39	1	O'ring, Nitrile	70 10 03 07 05
32	1	Ring clamp, stainless steel	12 85 57 00 02
		Butterfly valve handle kit:	**11 12 62 95 00**
28	1	Handle, stainless steel	11 11 83 40 00
21	1	Screw M6-12, stainless steel	-
22	1	Ring washer, stainless steel	-
		Footvalve handle kit:	**12 85 24 93 00**
16	1	Handle, stainless steel	11 11 69 0A 00
8	1	Split pin	-
7	1	Washer	-
		Bolting kit:	**11 16 92 30 00**
31	4	M16 screw, stainless steel	-
	4	M16 washer, stainless steel	-
	4	W16 grower washer, stainless steel	-
	4	Nut, stainless steel	-
		Outlet nozzle kit:	**12 85 69 00 00**
30	1	Chain, stainless steel	-
	1	Cap seal, PTFE	12 85 59 00 01
	1	Cap, stainless steel	12 85 59 30 01
	1	Outlet nozzle square, stainless steel	12 85 69 60 00
		CNAF/PTFE Gasket (item 29):	**12 91 42 00 00**



Tightening torque 2 screws stuffing clamp: 8 N.m

Nb: Perolo recommends the spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 22 06 40 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) TWINFLOW MKIV

Bottom Discharge

VTSXS3

Perolo 3" Twinflow MKIV is designed to be used as a bottom discharge. This combined body comprises both Footvalve and butterfly valve with splined shaft, together with cap, to provide the 3 closures required to comply with Transport Regulations. Made of stainless steel AISI 316L/1.4409, this product is suitable for ISO tank containers, IBC's, road tankers and static tanks dedicated to the transport and storage of chemicals and liquid foodstuff.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	15 Kg
Inlet flange	8 holes x Ø14 x 178 PCD
Outlet flange	3" BSP
Working pressure	4 Bar
Test pressure	6 Bar
Working temperature	-40° to +150°C (+190°C for Footvalve)
Design temperature	-40° to +200°C

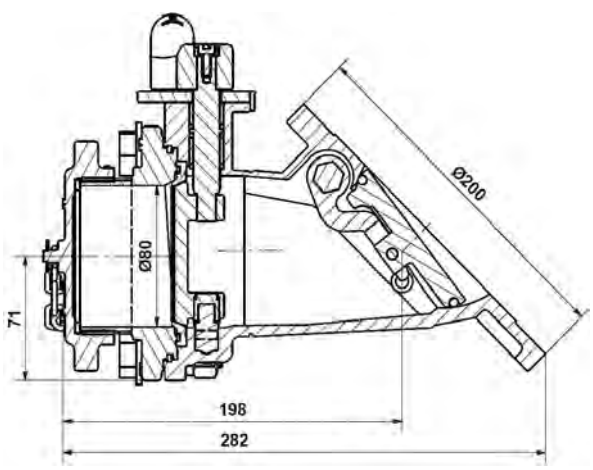
NB: temperature range vary according to o'ring material

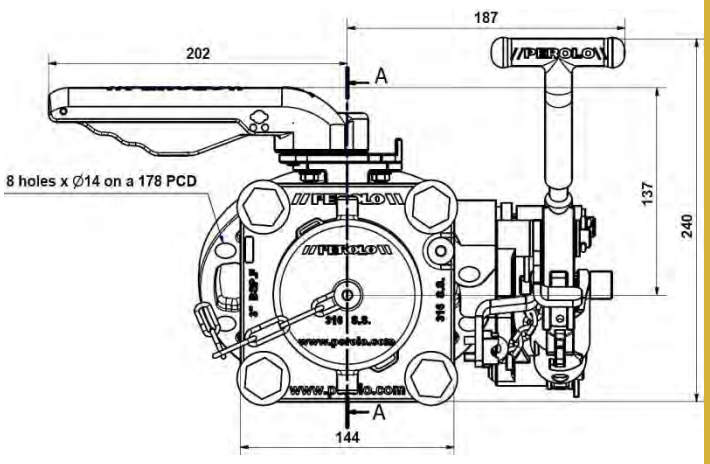
CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433 / EN14432
Approved by : Bureau Veritas - RMR

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Body length	45°
Body length	198 mm from center to outlet
O'rings material	FEP/Silicone
Seals material	PTFE
Operating	Side operated handle
Footvalve spring	Spring outside the tank allows easier cleaning

ACCESSORIES	
45° weld-in flange	11 90 59 00 00
Seal	12 91 20 00 00
Screw kit	11 11 27 93 00

DIMENSIONAL DRAWING





*other drilling options on request

drawing of 45° footvalve, for drawing of 30° footvalve please contact our sales department

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 22 47 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) TWINFLOW MKIV
Bottom Discharge
VTSXS3

CODIFICATION

STANDARD PART NUMBER	VTSXS3	4	C	W	.	.	.
YOUR PART NUMBER	VTSXS3	4	__	W	__	__	__

OUTLET ANGLE	
45°	4
30°	3

OUTLET NOZZLE - other on request	
With nozzle and cap	C

REMOTE CONTROL	
With remote control system	W

HASTELLOY SEAT	
Without	.
With Hastelloy C276 coated seat	Q

HANDLE	
Standard handle	.
L Handle	L
Reversed L handle	RL
Short handle	C

OUTLET NOZZLE	
Standard 3" BSP outlet nozzle	.
Outlet nozzle with shortened chain	N

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 22 47 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" (ND80) TWINFLOW MKIV
Bottom Discharge
VTSXS3

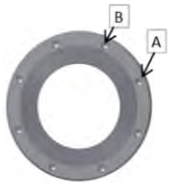
INSTALLATION

Ensure that the Inlet (tank side) flange and gasket are clean; free from dirt and grit.

1. Screw the studs into the tank container flange. Take careful note of the position of the studs (picture 1).
2. Fit the PTFE gasket (picture 2).
3. Fit the TWINFLOW MKIV Footvalve in position (picture 3).
4. Tighten the studs and nuts torque of 50 N.m. +/-5. Fore more details please refer to PWAGEN001

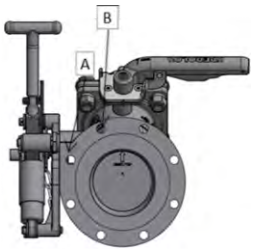
A Stud M12 63/33 with standard nuts

B Stud M12 48/30 with thin nut

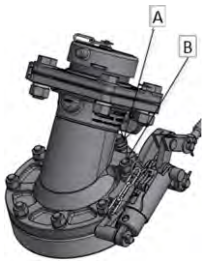


Picture 1

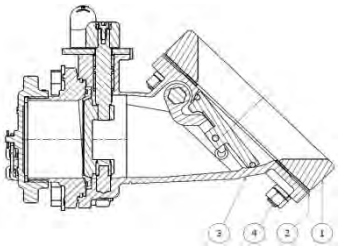
Picture 2



Picture 3



Picture 4



45° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	3" TWINFLOW 45° MK IV	3" 45° footvalve, inlet flange 8x14xPCD178 3" BSP Outlet	VTSXS34CW
4	Stud kit	6x M12 - 52 studs, 1x M12 - 66 stud 1x M12 80 stud, 8x grower washers, 7xM12 standard nuts, 1x thin nut	11 11 27 93 00
2	Inlet Gasket	ext ø 200 mm, int ø 114 mm, 8 x Ø14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel 316/316L	11 90 59 00 00

30° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	3" TWINFLOW 30° MK IV	3" 30° footvalve, inlet flange 8x14xPCD178 3" BSP Outlet	VTSXS33CW
4	Stud kit	6x M12 - 52 studs, 1x M12 - 66 stud 1x M12 80 stud, 8x grower washers, 7xM12 standard nuts, 1x thin nut	11 11 27 93 00
2	Inlet Gasket	ext ø 200 mm, int ø 114 mm, 8 x Ø14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 91 16 00 00

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 22 47 30 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

3" (ND80) TWINFLOW MKIV

Bottom Discharge

VTXS3

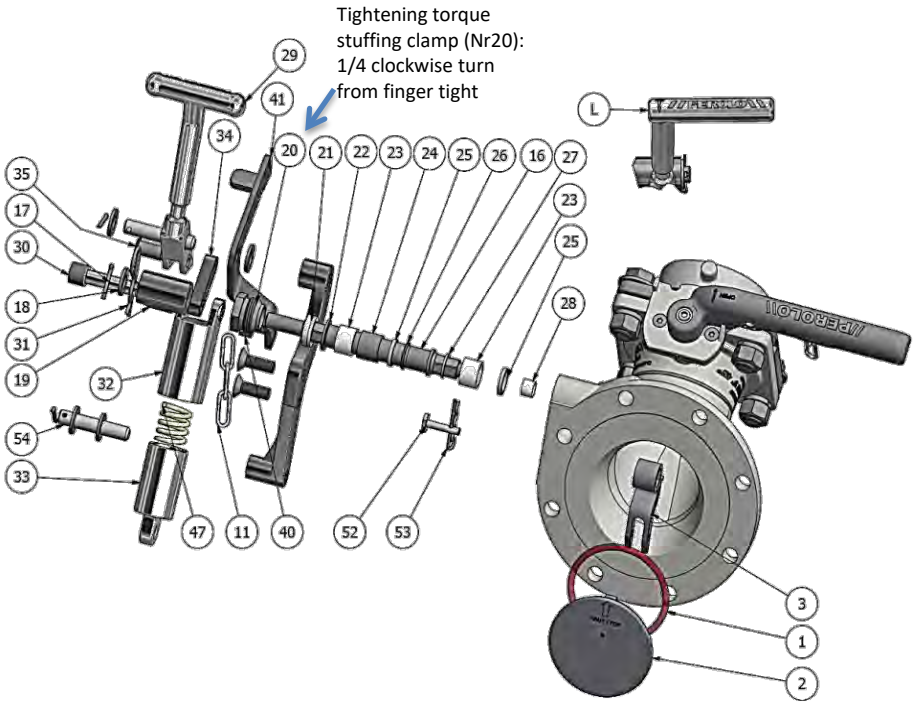
ISO ROAD IBC

BOTTOM DISCHARGE

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Spindle kit:	**on request**
16	1	Spindle, Stainless steel	17 11 25 00 01
19	1	Drive yoke, Stainless steel	-
20	1	Stuffing clamp, Stainless steel	-
		Spindle seals kit:	**11 11 24 95 00**
17	1	V-pin, Stainless steel	-
18	1	Washer, Stainless steel	-
21	3	Packing ring, Stainless steel	-
22	1	Packing gland, PTFE	11 11 24 00 15
23	2	Spindle bush, PTFE & glass 25%	11 11 24 00 06
24	1	Spacer, Stainless steel	-
25	2	Gasket, Ekonol	70 10 90 00 93
26	1	Stop ring, Stainless steel	-
27	1	Elastic ring, Stainless steel	-
28	1	Spindle bush, PTFE	12 85 30 00 05
		Pusher MK4:	
3	1	Pusher, Stainless steel	11 16 92 00 02
		Poppet kit:	**11 16 92 90 00**
1	1	O'ring, FEP/Silicon	70 10 15 00 53
2	1	Poppet, Stainless steel	11 16 92 00 06
52	1	Poppet Spindle, Stainless steel	11 16 87 00 05
53	1	Beta Pin	11 16 87 00 04
		Poppet seal kit:	**11 11 24 91 00**
1	1	O'ring, FEP/Silicon	70 10 15 00 53

ITEM	QTY	DESIGNATION	PART N°
		Spring system kit:	**11 11 14 99 19**
33	1	Female MK2 cage, Stainless steel	11 11 70 0A 03
32	1	Male MK2 cage, Stainless steel	11 11 70 0A 04
31	1	Beta pin, Stainless steel	70 13 20 37 14
18	2	Washer, Stainless steel	-
17	1	V-pin, Stainless steel	-
47	1	Spring, Stainless steel	11 11 70 00 16
11	1	Chain, Stainless steel	-
		Handle kit:	**11 16 92 20 00**
29	1	Handle, Stainless steel	11 11 83 1A 10
34	1	Connecting rod, Stainless steel	-
35	1	Spindle roller, Stainless steel	-
18	2	Z12 washer, Stainless steel	-
17	1	Split pin, Stainless steel	-
		L Handle kit:	**11 12 99 10 00**
L	1	Reverse handle, Stainless steel	11 11 70 0A 08
34	1	Connecting rod, Stainless steel	-
35	1	Spindle roller, Stainless steel	-
18	2	Z12 washer, Stainless steel	-
17	1	Split pin, Stainless steel	-
		Remote control kit:	**11 11 98 90 00**
41	1	Remote control lever, Stainless steel	11 11 70 00 17
40	1	Rope clamp	-
-	1	Spring, Stainless steel	11 11 57 00 02



Nb: Perolo recommends the spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 16 92 40 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

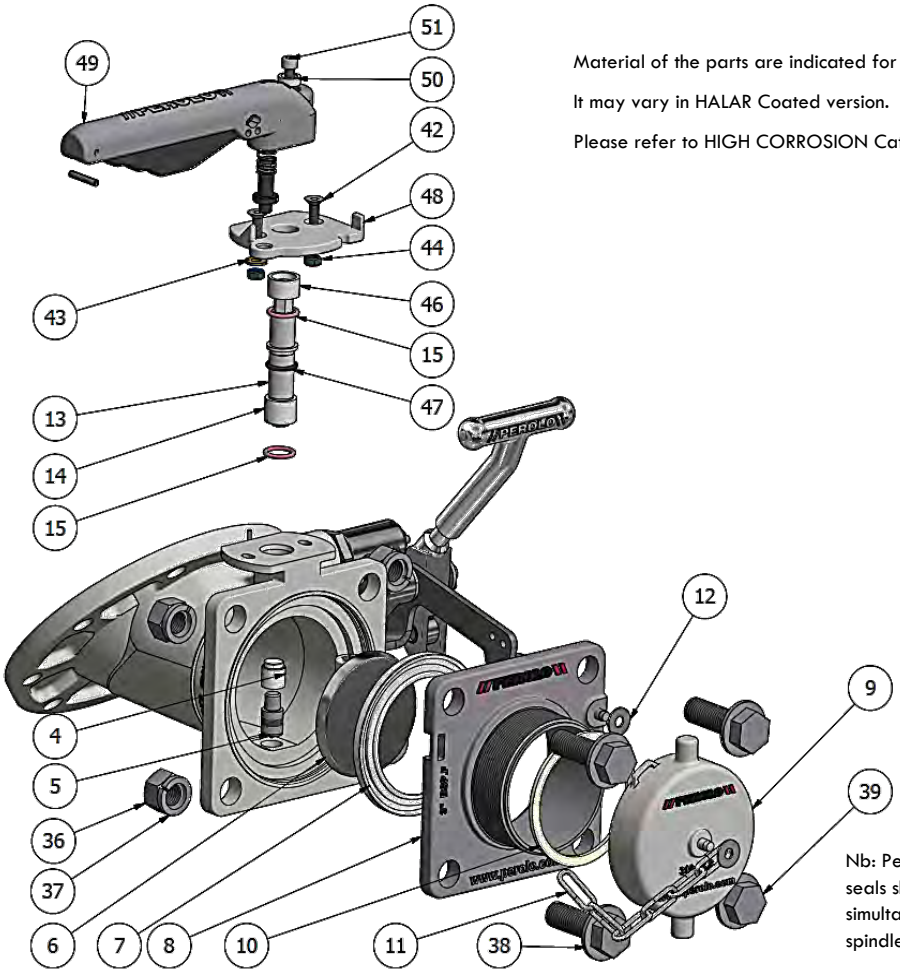
3" (ND80) TWINFLOW MKIV
Bottom Discharge
VTXS3

BOTTOM DISCHARGE

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Closure plate kit:	**12 85 57 93 00**
6	1	Closure plate, Stainless steel	12 85 57 20 00
4	1	Closure plate bush, PTFE & glass 25%	12 85 57 00 04
		Closure plate seal:	**12 85 57 94 00**
7	1	Closure plate seal, PTFE	12 85 57 00 01
		Butterfly valve spindle kit:	**11 12 10 91 00**
13	1	Spindle	11 12 10 00 01
		Butterfly valve spindle seals kit:	**11 16 92 40 00**
4	1	Closure plate bush, PTFE & glass 25%	12 85 57 00 04
51	1	Screw M6-12, Stainless steel	-
50	1	Ring washer, Stainless steel	-
42	2	Screws M6-20, Stainless steel	-
44	2	Nuts, Stainless steel	-
43	2	Washers, Stainless steel	-
14	1	Spindle bush, PTFE & glass 25%	12 85 30 00 03
15	2	O'rings, FEP & silicon	70 10 15 00 12
47	1	O'rings, FKM & viton	70 10 01 02 12
46	1	Short Spindle bush, PTFE & glass 25%	11 12 62 00 01

ITEM	QTY	DESIGNATION	PART N°
		Butterfly valve handle kit:	**11 12 62 95 00**
51	1	M6-12 screw, Stainless steel	-
50	1	Washer, Stainless steel	-
49	1	Handle Stainless steel	11 11 83 40 00
		Outlet nozzle kit:	**11 11 83 93 00**
7	1	Closure plate seal, PTFE	12 85 57 00 01
12	2	Washer, Stainless steel	-
11	1	Chain, Stainless steel	-
10	1	Cap seal, PTFE	12 85 59 00 01
9	1	Cap, Stainless steel	12 85 59 30 01
8	1	Outlet nozzle square, Stainless steel	11 11 83 20 01
		Bolting kit:	**11 16 92 30 00**
39	4	M16 screw, Stainless steel	-
38	4	M16 washer, Stainless steel	-
37	4	W16 grower washer, Stainless steel	-
36	4	Nut, Stainless steel	-



Material of the parts are indicated for Stainless Steel Version.

It may vary in HALAR Coated version.

Please refer to HIGH CORROSION Catalogue for more details.

Nb: Perolo recommends the spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.

4" (ND100) NEATFLOW
Footvalve
NX4

The Perolo 4" NEATFLOW™ MKIV external spring footvalve is designed to be used as a bottom discharge for chemicals and liquid foodstuff products carried in tank containers and IBC's. The external spring design is perfectly suited to tanks containing viscous fluids and allows easy cleaning and maintenance - Discharge time improved - compact design.

SPECIFICATION	
Nominal diameter	100 mm (4")
Weight	16.0 Kg
Inlet flange	8 holes x Ø14 x 190 PCD
Outlet flange	4 holes x Ø18 x 170 PCD*
Working pressure	4 Bar
Test pressure	6 Bar
Working temperature	-40° to +190°C
Design temperature	-40° to +200°C

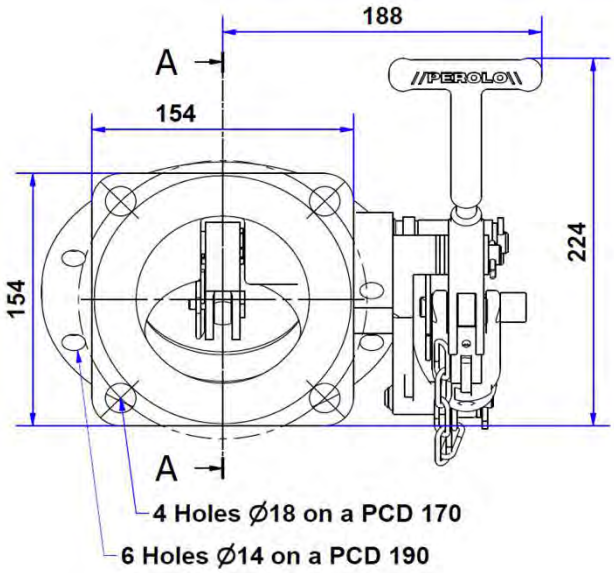
NB: temperature range vary according to o'ring material

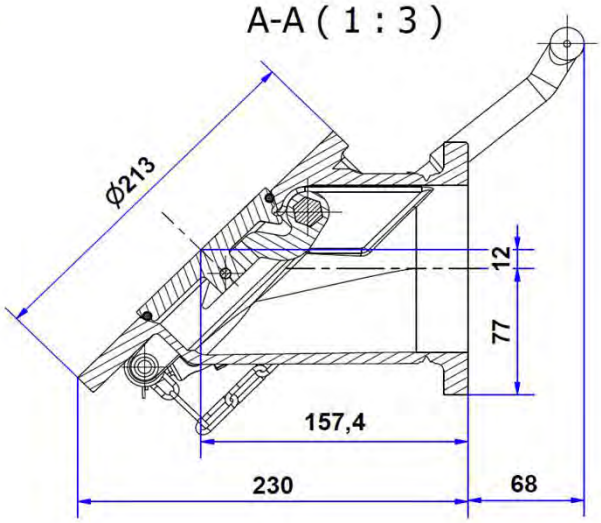
CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433
Approved by : Bureau Veritas

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409
Angle	45°
Body length	155 mm from center to outlet
O'rings material	FEP/Silicone
Seals material	PTFE
Operating	Side operated handle
Footvalve spring	Spring outside the tank allows easier cleaning

ACCESSORIES	
45° weld-in flange	11 11 59 50 01
PTFE	11 11 59 90 01
Screw kit	11 11 59 99 01

DIMENSIONAL DRAWING





*other drilling options on request

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 22 08 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

4" (ND100) NEATFLOW
Footvalve
NX4

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	NX4	4	A	1	W
YOUR PART NUMBER	NX4	4	A	__	W

OUTLET ANGLE	
45°	4

INLET FLANGE	
8 x Ø14 x 190 PCD	A

OUTLET FLANGE	
Square flange 4 x Ø18 x 170 PCD	1
Round flange 8 x Ø18 x 180 PCD	2

REMOTE CONTROL	
With remote control lever	W
Without	.

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 22 08 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

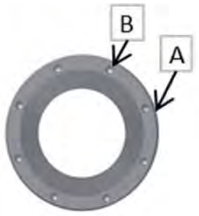
4" (ND100) NEATFLOW
Footvalve
NX4

BOTTOM DISCHARGE

INSTALLATION

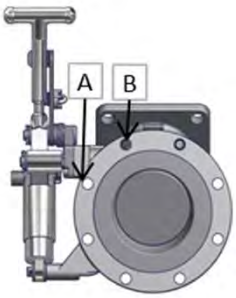
Ensure that the Inlet (tank side) flange and gasket are clean; free from dirt and grit.

1. Screw the studs into the tank container flange. Take careful note of the position of the studs (pic 1)
2. Fit the gasket (picture 2).
3. Fit the NEATFLOW MKIV Footvalve in position (picture 3).
4. Tighten the studs and nuts at a maximum torque of 50 N.m. +/-5. For more details please refer to PWAGEN001

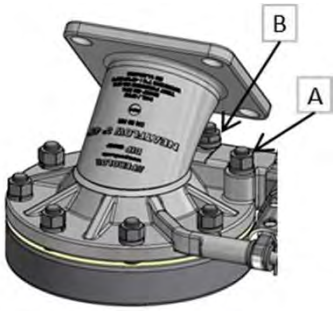


Picture 1

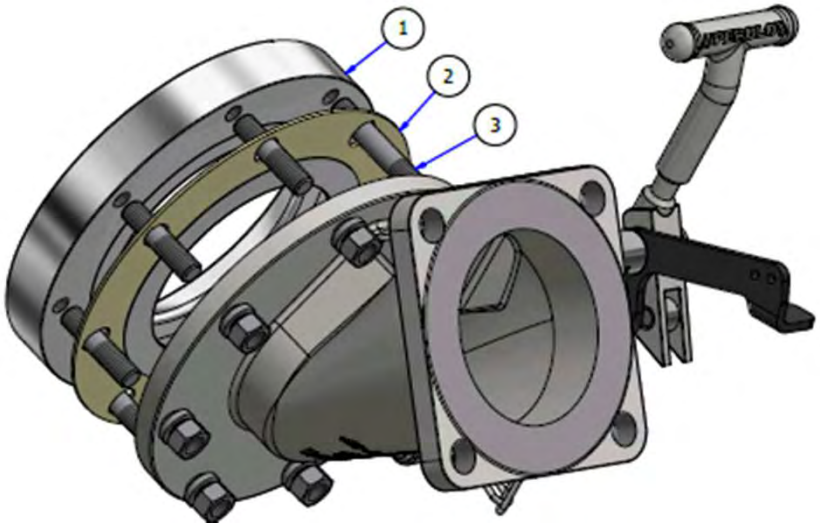
Picture 2



Picture 3



Picture 4



ITEM	DESCRIPTION	SPECIFICATION	PART N°
	4" NEATFLOW 45° MK IV	4" 45° footvalve, inlet flange 8x14xPCD190 Square Outlet flange 4xØ18xPCD170	NX44A1W
3	Stud kit	6x M12 34/18 studs, 2x M12 57/18 studs, 8x grower washers, 8xM12 standard nuts	11 11 59 99 01
2	Inlet Gasket	ext ø 213 mm, int ø 114 mm, 8 slots x14 x PCD184.2-190, CNAF / PTFE	11 11 59 90 01
1	Weld-in flange	ext ø 200 mm, int ø 120 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 11 59 50 01

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 22 08 30 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

4" (ND100) NEATFLOW
Footvalve
NX4

BOTTOM DISCHARGE

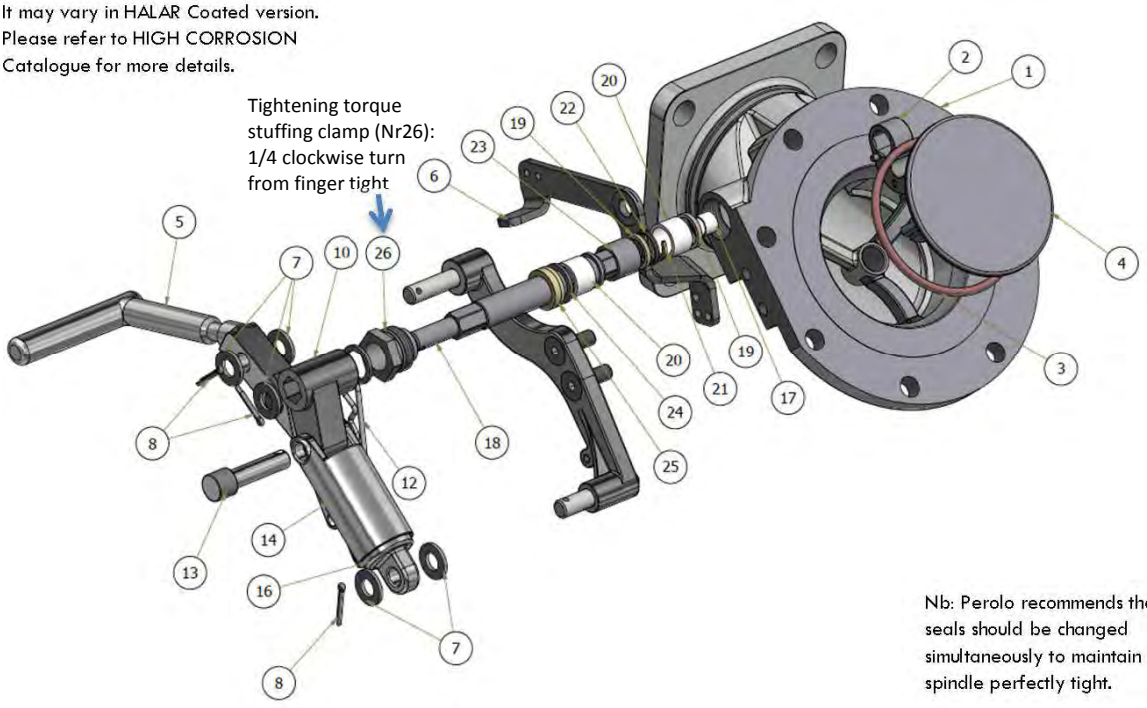
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Spindle:	
18	1	Spindle, Stainless steel	17 11 25 00 01
		Spindle seals kit:	**11 11 24 95 00**
20	2	Spindle bush, PTFE	11 11 24 00 06
19	2	Gasket Ekonol, PTFE	70 10 90 00 93
17	1	Spindle bush, PTFE	12 85 30 00 05
21	1	Elastic ring, Stainless steel	-
22	1	Stop ring, Stainless steel	-
23	1	Spacer, Stainless steel	-
25	1	Packing gland, PTFE	11 11 24 00 15
24	2	Packing ring, Stainless steel	-
7	1	Washer, Stainless steel	-
8	1	V-pin, Stainless steel	-
		Pusher kit:	**on request**
2	1	Lift fork, Stainless steel	11 22 08 00 02
		Poppet seal:	**17 11 02 93 00**
3	1	O'ring, FEP/Silicon	70 10 15 00 85
		Poppet kit:	
3	1	O'ring, FEP/Silicon	70 10 15 00 85
4	1	Poppet, Stainless steel	11 22 08 00 06
-	1	Poppet Spindle, Stainless steel	11 16 87 00 05
-	1	Beta Pin	11 16 87 00 04

ITEM	QTY	DESIGNATION	PART N°
		Spring system kit:	**11 11 14 99 19**
14	1	Female MK2 case, Stainless steel	11 11 70 0A 03
16	1	Male MK2 case, Stainless steel	11 11 70 0A 04
8	1	V-pin, Stainless steel	-
7	2	Washer, Stainless steel	-
12	1	Beta pin, Stainless steel	70 13 20 37 14
15	1	Spring, Stainless steel	11 11 70 00 16
28	1	Chain, Stainless steel	-
		Handle kit:	**11 16 25 94 00**
5	1	T handle, Stainless steel	11 11 83 1A 10
-	1	Connecting rod, Stainless steel	-
9	1	Spindle roller, Stainless steel	-
7	2	Washer, Stainless steel	-
8	1	V-pin, Stainless steel	-

Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION Catalogue for more details.

Tightening torque stuffing clamp (Nr26): 1/4 clockwise turn from finger tight



Nb: Perolo recommends the spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.

3" (ND80) CHEMFLY
Bottom discharge
VLY

Perolo 3" CHEMFLY bottom discharge valve is designed to suit chemical or foodstuff road tankers - includes a 3" CHEMFLY butterfly valve, a 3" BSP threaded outlet and a 3" welding neck.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	8 Kg
Inlet flange	4 holes x Ø14 on a 130 PCD
Outlet flange	3" BSP
Working pressure	7 bar
Test pressure	10.5 bar
Working temperature	-40° to +150°C
Design temperature	-40° to +190°C

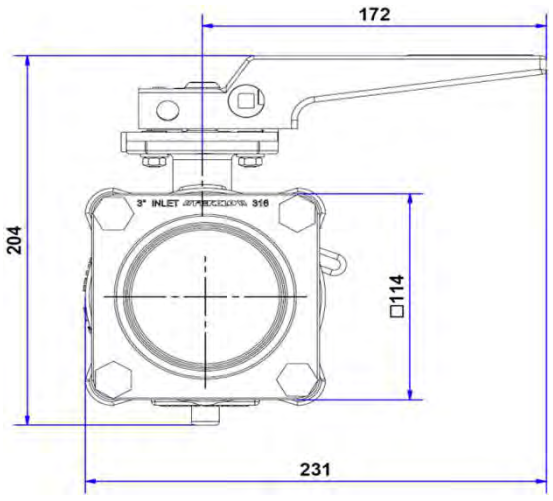
NB: temperature range vary according to o'ring material

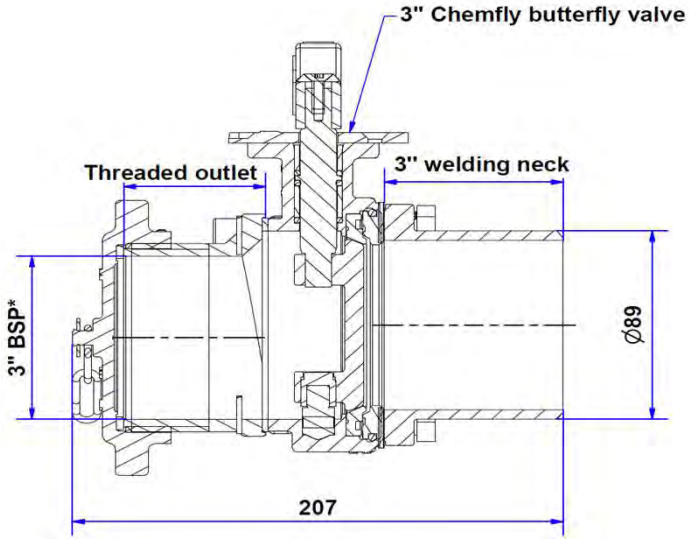
CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433
Approved by : Bureau veritas

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Angle	180°
Body length	depends on options
O'rings material	Not applicable
Seals material	PTFE
Operating	Locking handle - 1/4 turn
Footvalve spring	Without spring

ACCESSORIES	
45° weld-in flange	11 11 59 50 01
PTFE	11 11 59 90 01
Screw kit	11 11 59 99 01

DIMENSIONAL DRAWING





BOTTOM DISCHARGE

3" (ND80) CHEMFLY
Bottom discharge
VLY

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	VLY3	1	1	0	.	1	2	1	0

YOUR PART NUMBER	VLY3	1	1						

INLET FLANGE	
4 x Ø14 x 130 PCD	1

OUTLET FLANGE	
4 x M10 x 120 PCD	1

VENTING PORTS *	
None	0
1 x 3/8" NPT	1
1 x 3/8" BSP	2
2 x 3/8" NPT	3
2 x 3/8" BSP	4
1 x 3/8" BSP	G
1 x 1/2" BSP	H

MATERIAL	
316L / 1.4409 stainless steel	.
904l / 1.4539 stainless Steel	U

WELDING INLET - Others on request	
3"	1

OUTLET NOZZLE LENGTH	
Short outlet length (60mm)	1
Long outlet length (110mm)	2

OUTLET THREAD	
Without thread	0
3" BSP	1
3" NPT	2

DUST CAP	
None	0
3" BSP cap	1
3" NPT cap	2

* Venting ports position are described on specification sheet of chemfly light butterfly valve

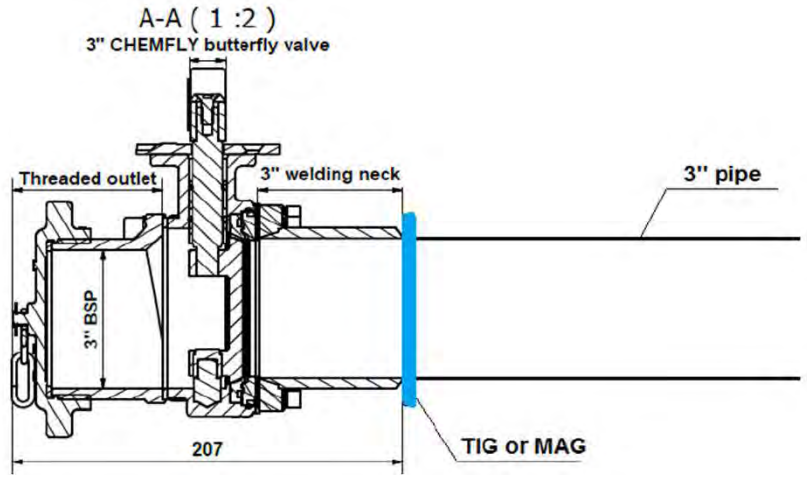
Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 17 12 21 20 00 F
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) CHEMFLY
Bottom discharge
VLY

BOTTOM DISCHARGE

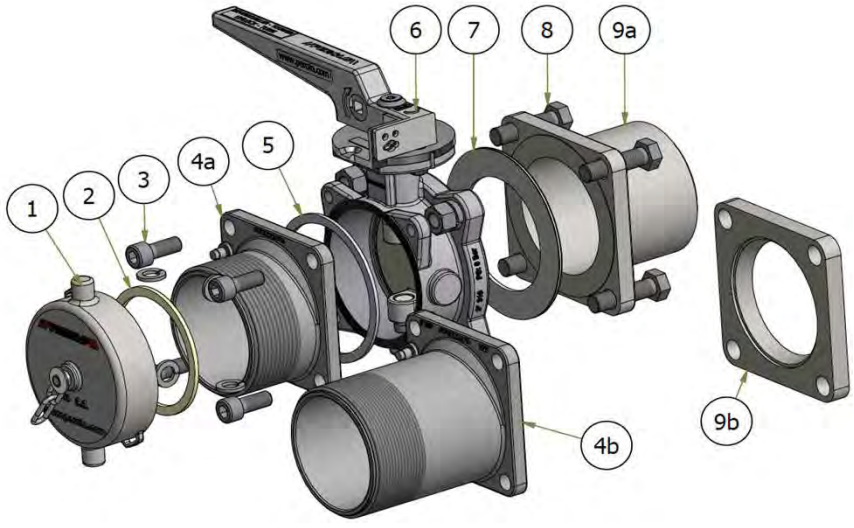
INSTALLATION

Weld the nozzle (9) on a 3" pipe (Ø89 x Ø80). Use MAG or TIG welding process with a fully interpenetrated welding process. Before welding the welding neck, please disassemble the valve from the welding neck. Reassemble once welding completed.



MAINTENANCE

ITEM	DESCRIPTION	PART N°
1, 2	3" BSP cap kit	12 85 59 91 00
4a	3" BSP short outlet	17 12 09 70 00
4b	3" BSP long outlet	17 12 09 71 00
3	Screw kit: 4 x M10 CHC screws, 4 washers	17 12 11 91 00
5	Outlet gasket PTFE	17 12 21 00 01
6	3" CHEMFLY butterfly valve	LY3110
8	Screw kit: 4 x M12 CHC screws, 4 washers, 4 nuts	17 12 11 92 00
7	Inlet gasket (synthetic fibers)	17 12 21 10 00
9a	Inlet welding	17 12 09 80 00
9b	Square inlet flange	17 12 09 81 00



for maintenance of chemfly butterfly valve, please check maintenance sheet of chemfly butterfly valve

3" CHEMFLOW 90° MKIV

Footvalve

CX39

3" CHEMFLOW™ 90° MK4 pneumatic or manual footvalve - designed to discharge liquid chemicals and foodstuffs carried in road tankers. The external spring design is perfectly suited to tanks containing viscous fluids and allows easy cleaning and maintenance. With removable seat (optional): when the seat of your Chemflow is damaged by pitting, you only change the removable seat and not your complete footvalve.

SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	12 Kg
Inlet flange	8 holes x Ø14 on a 178 PCD
Outlet flange	4 holes x Ø18 on a 160 PCD
Working pressure	4 Bar (pneumatic air pressure 5 Bar)
Test pressure	6 Bar
Working temperature	-40° to +190°C
Design temperature	-40° to +200°C

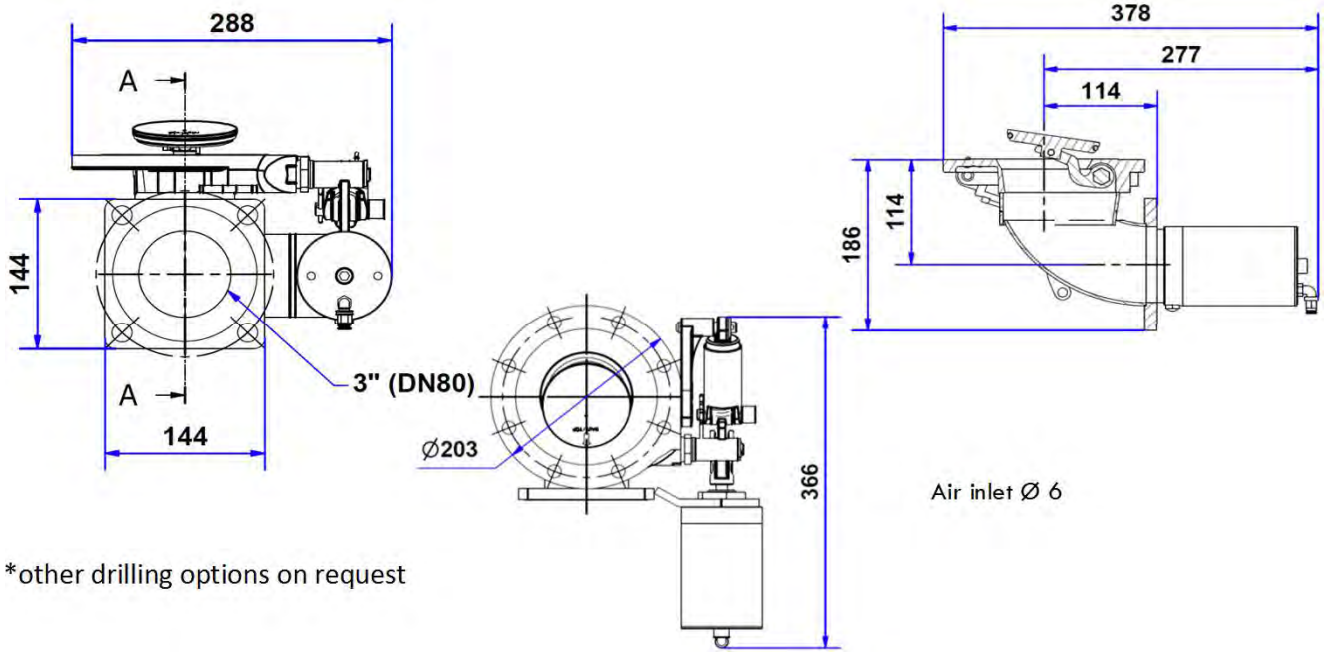
NB: temperature range vary according to o'ring material

CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / RID / EN14433
Approved by : Bureau Veritas

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel
Angle	90°
Body length	-
O'rings material	FEP/Silicone
Seals material	FEP / PTFE
Operating	St. St. spring return actuator or hand operated
Footvalve spring	Spring outside the tank allows easier cleaning

ACCESSORIES	
Weld-in flange	11 90 59 00 00
Inlet seal	12 91 20 00 00
Outlet seal	12 91 39 00 00
Bolting kit	11 11 27 93 00
Removable seat kit	17 11 89 90 00

DIMENSIONAL DRAWING



*other drilling options on request

BOTTOM DISCHARGE

3" CHEMFLOW 90° MKIV

Footvalve
CX39

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	CX39	1	R	C	.	.	.
YOUR PART NUMBER	CX39	__	R	__	__	__	__

OUTLET FLANGE	
Square outlet flange 4 x18x160PCD	1
Round outlet flange 4 x18x160PCD	2
Removable seat square outlet flange 4 x 18x160 PCD	3

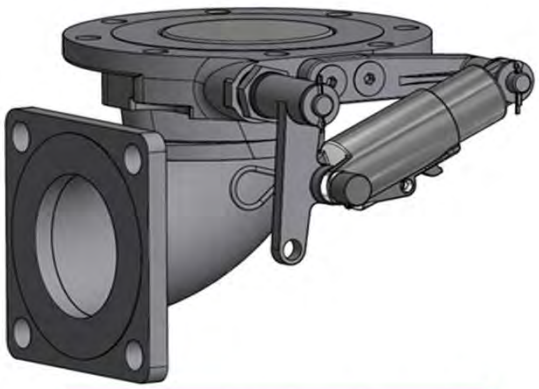
POPPET	
Flat poppet	R

ACTUATOR	
Side actuator	C
Mechanical lever (remote operated)	M
Mechanical handle	H

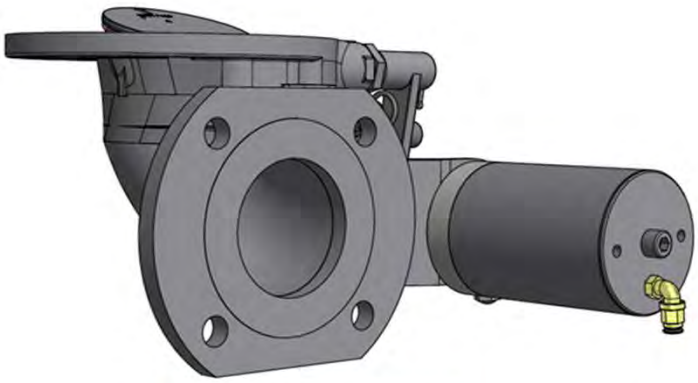
REMOVABLE SEAT	
Without	.
With removable seat	X

BODY MATERIAL	
316L	.
904L	U

INTERNAL COMPONENT	
316L Stainless steel	.
C276 Hastelloy	H



Mechanical (Without Actuator)



Round Outlet Flange

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 17 16 18 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" (ND80) NEATFLOW MKIV
Footvalve
NX3

BOTTOM DISCHARGE

INSTALLATION

Ensure that the Inlet (tank side) flange and gasket are clean; free from dirt and grit.

1. Screw the studs into the tank container flange. Take note of the position of the studs (pic 1)
2. Fit the gasket (picture 2).
3. Fit the NEATFLOW MKIV Footvalve in position (picture 3).
4. Tighten the nuts torque : 50 N.m +/- 10 (M12); for 2nd closure valve 70 N.m +/- 5 (M16) behind neatflow.

For more details please refer to PWAGEN001

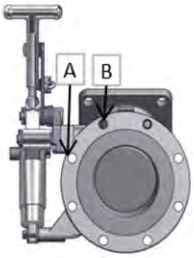
Pneumatic version: Compressed air pressure for actuator should be in-between 4 bar and 6 bar to ensure correct opening & closing of the poppet

A Stud M12 63/33 with standard nuts

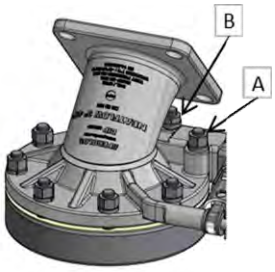
B Stud M12 48/30 with thin nut

Picture 1

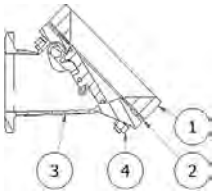
Picture 2



Picture 3



Picture 4



45° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	3" NEATFLOW 45° MK IV	3" 45° footvalve, inlet flange 8x14xPCD178 Square Outlet flange 4xØ 18xPCD160	NX34AX1
4	Stud kit	6x M12 34/25 studs, 1x M12 40/30 stud(B)1x M12 63/33 stud(A), 8x grower washers, 7xM12 standard nuts, 1x thin nut (B)	11 11 27 93 00
2	Inlet Gasket	ext ø 200 mm, int ø 114 mm, 8 x 14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 90 59 00 00

30° INSTALLATION			
ITEM	DESCRIPTION	SPECIFICATION	PART N°
3	3" NEATFLOW 30° MK II	3" 30° footvalve, inlet flange 8x14xPCD178 Square Outlet flange 4xØ 18xPCD160	NF33A1
4	Stud kit	6x M12 34/25 studs, 1x M12 40/30 stud(B)1x M12 63/33 stud(A), 8x grower washers, 7xM12 standard nuts, 1x thin nut (B)	11 11 27 93 00
2	Inlet Gasket	ext ø 200 mm, int ø 114 mm, 8 x 14 x PCD 178, CNAF / PTFE	12 91 20 00 00
1	Weld-in flange	ext ø 200 mm, int ø 114.3 mm, thickness 35 mm, 8 x M12 x PCD 178, 316L stainless steel	11 91 16 00 00

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 16 87 30 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
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3" CHEMFLOW 90° MKIV
Footvalve
CX39

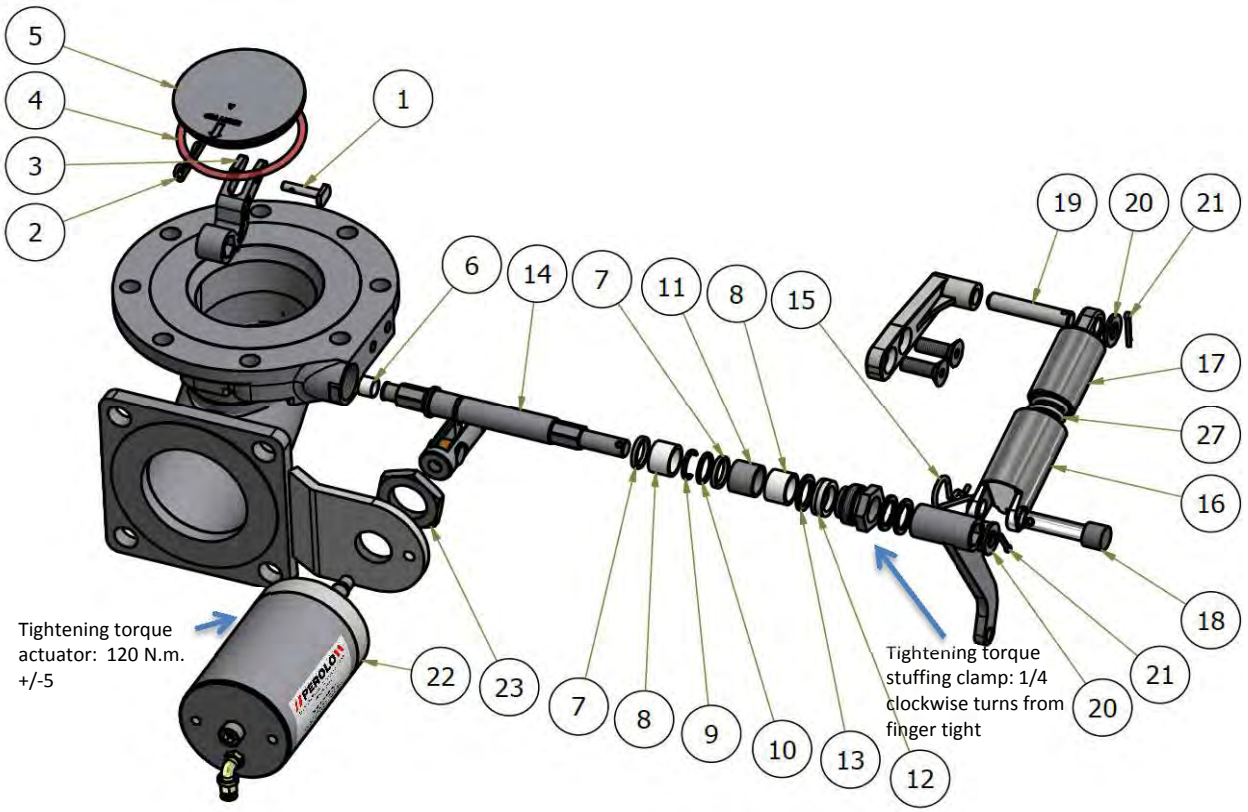
ROAD

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Spindle kit:	**on request**
14	1	Spindle, Stainless steel	11 11 14 99 11
		Spindle seals kit:	**11 11 24 95 00**
6	1	Spindle bush, PTFE	12 85 30 00 05
7	2	Gasket Ekonol, PTFE	70 10 90 00 93
8	2	Spindle bush, PTFE	11 11 24 00 06
9	1	Elastic ring, Stainless steel	-
10	1	Stop ring, Stainless steel	-
11	1	Spacer, Stainless steel	-
12	1	Packing gland, PTFE	11 11 24 00 15
13	3	Packing ring, Stainless steel	-
20	1	Washer, S.S.	-
21	1	V-pin, S.S.	-
		Lift fork kit:	**on request**
3	1	Lift fork kit, Stainless steel	11 16 92 00 02
		Poppet seal kit:	**11 11 24 91 00**
4	1	O'ring, FEP/Silicon	70 10 15 00 53
		Poppet kit:	**17 16 17 90 00**
5	1	Flat poppet, 316L/1.4404	17 16 17 00 01
4	1	O'ring, FEP/Silicon	70 10 15 00 53
2	1	Beta Pin, Stainless steel	11 16 87 00 04
1	1	Poppet Spindle, Stainless steel	11 16 87 00 05

ITEM	QTY	DESIGNATION	PART N°
		Spring system kit:	**11 11 14 99 19**
16	1	Female MK2 case, Stainless steel	11 11 70 0A 03
17	1	Male MK2 case, Stainless steel	11 11 70 0A 04
15	1	V-pin, Stainless steel	-
20	2	Washer, Stainless steel	-
21	1	Beta pin, Stainless steel	70 13 20 37 14
27	1	Spring, Stainless steel	11 11 70 00 16
-	1	Chain, Stainless steel	-
		Actuator kit:	**17 11 01 95 00**
22	1	Actuator, Stainless steel	-
23	1	Nut, Stainless steel	-

Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION
Nb: Perolo recommends the spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.



BOTTOM DISCHARGE

4" CHEMFLOW
Footvalve
CF49

BOTTOM DISCHARGE

4" CHEMFLOW™ MK2 90° pneumatic or manual footvalve - designed to discharge liquid chemicals and foodstuffs carried in road tankers - The external spring design is perfectly suited to tanks containing viscous fluids and allows easy cleaning and maintenance.

SPECIFICATION	
Nominal diameter	100 mm (4")
Weight	15.0 Kg
Inlet flange	8 x Ø14 x 184.2 to 190 PCD
Outlet flange	8 x Ø12 x 150 PCD
Working pressure	4 Bar (pneumatic air pressure 5 Bar)
Test pressure	6 Bar
Working temperature	-40° to +190°C
Design temperature	-40° to +200°C

NB: temperature range vary according to o'ring material

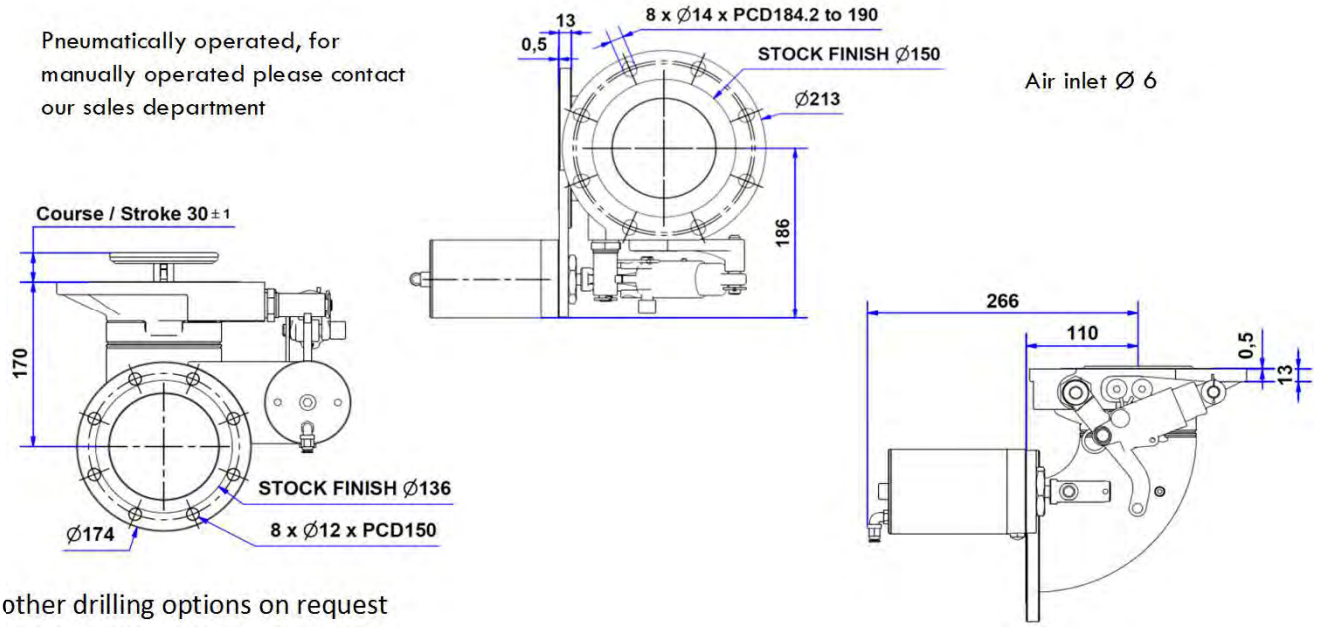
CONFORMITY, CERTIFICATION & APPROVAL
ADR / IMDG / EN14433
Approved by : Bureau Veritas

STANDARD FEATURES (See Codification for option)	
Made of (material)	316L / 1.4409 stainless steel (C22 Hastelloy)
Angle	90°
Body length	-
O'rings material	FEP/Silicone
Seals material	FEP / PTFE
Operating	St. St. spring return actuator or hand operated
Footvalve spring	Spring outside the tank allows easier cleaning

ACCESSORIES	
Weld-in flange	17 11 03 90 01
Seal	17 11 03 90 02
Bolting kit	17 11 03 92 00

DIMENSIONAL DRAWING

Pneumatically operated, for manually operated please contact our sales department



*other drilling options on request

4" CHEMFLOW
Footvalve
CF49

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	CF492	1A	R	C	.

YOUR PART NUMBER CF492 ___ ___ ___ ___

OUTLET FLANGE	
Round, 8 x Ø12x150PCD (pneumatic)	1A
without outlet flange (pneumatic)	1B
Round, 8 x Ø12 x 150 PCD (manual)	1
Square 4 x Ø18 x 150PCD (manual)	2
without outlet flange (manual)	3

POPPET	
Conical Poppet	S
Flat Poppet	R
Blade Poppet	B

ACTUATOR	
Side Stainless Steel Actuator	C
Manually operated with lever	M

SPECIAL OPTIONS	
Without special options	.
Body in 904L stainless steel	U
Body in Hastelloy C22	H
With removable seat in 316L	X

removable seat marked in red

4" CHEMFLOW
Footvalve
CF49

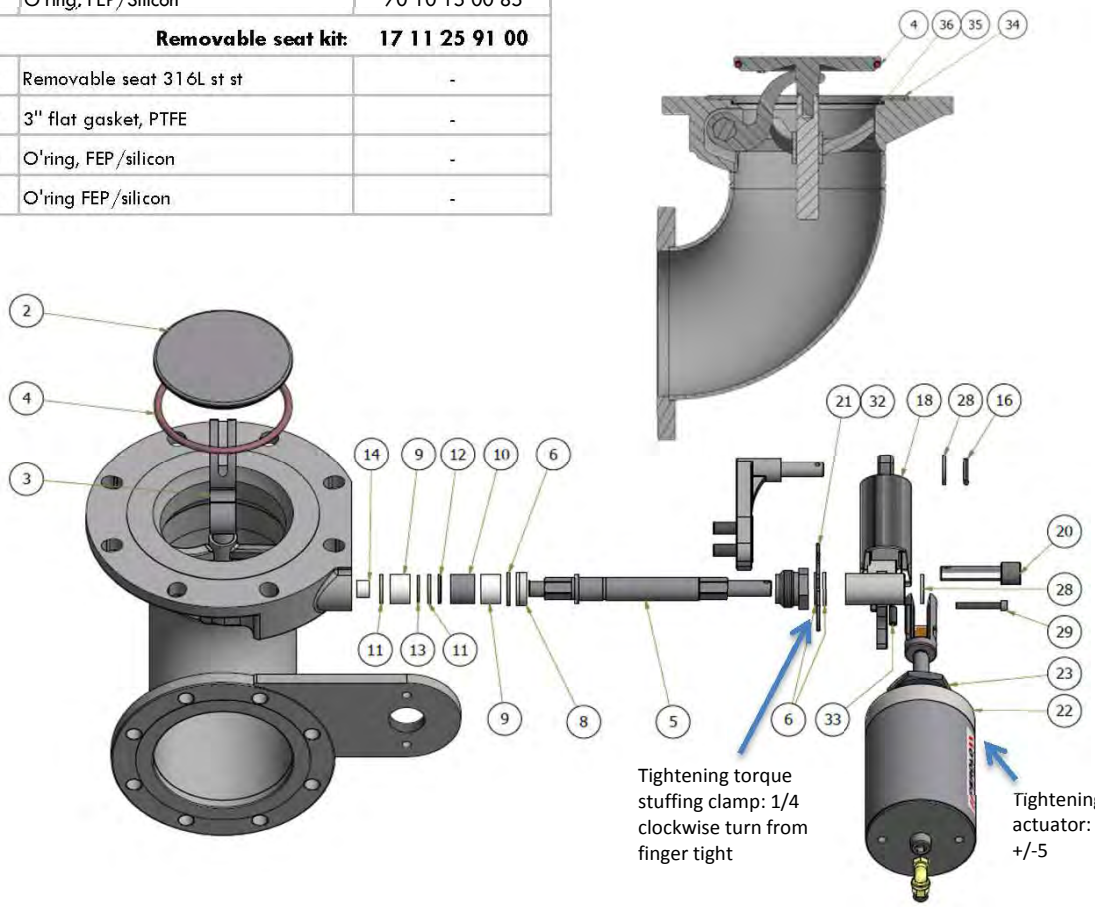
BOTTOM DISCHARGE

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Spindle kit:	**on request**
5	1	Spindle, Stainless steel	17 11 25 00 01
		Spindle seals kit:	**11 11 24 95 00**
14	1	Spindle bush, PTFE	12 85 30 00 05
11	2	Gasket Ekonol, PTFE	70 10 90 00 93
9	2	Spindle bush, PTFE	11 11 24 00 06
13	1	Elastic ring, Stainless steel	-
12	1	Stop ring, Stainless steel	-
10	1	Spacer, Stainless steel	-
8	1	Packing gland, PTFE	11 11 24 00 15
6	3	Packing ring, Stainless steel	-
28	1	Washer, S.S.	-
16	1	V-pin, S.S.	-
		Lift fork kit:	**on request**
3	1	Lift fork kit, Stainless steel	17 16 04 00 09
		Poppet seal kit:	**17 11 02 93 00**
4	1	O'ring, FEP/Silicon	70 10 15 00 85
		Poppet kit:	**17 11 77 91 00**
2	1	Flat poppet, 316L/1.4404	17 16 04 02 00
4	1	O'ring, FEP/Silicon	70 10 15 00 85
		Removable seat kit:	**17 11 25 91 00**
34	1	Removable seat 316L st st	-
35	1	3" flat gasket, PTFE	-
36	1	O'ring, FEP/silicon	-
4	1	O'ring FEP/silicon	-

ITEM	QTY	DESIGNATION	PART N°
		Spring system kit:	**11 11 14 99 19**
18	1	Female MK2 case, Stainless steel	11 11 70 0A 03
-	1	Male MK2 case, Stainless steel	11 11 70 0A 04
16	1	V-pin, Stainless steel	-
28	2	Washer, Stainless steel	-
21	1	Beta pin, Stainless steel	70 13 20 37 14
-	1	Spring, Stainless steel	11 11 70 00 16
-	1	Chain, Stainless steel	-
		Actuator kit:	**17 11 01 95 00**
22	1	Actuator, Stainless steel	-
23	1	Nut, Stainless steel	-

Material of the parts are indicated for Stainless Steel Version.
It may vary in HALAR Coated version.
Please refer to HIGH CORROSION
Nb: Perolo recommends the spindle seals should be changed simultaneously to maintain the valve spindle perfectly tight.



4" CHEMFLOW BITUMEN

Footvalve

17 16 01 00 00

4" CHEMFLOW™ BITUMEN 90° pneumatic footvalve is designed to discharge hot bitumen from road tankers. Made of 304L 1.4307 stainless Steel, this product is designed with appropriate seals to work in a high temperature environment (250°C). The conical poppet with blades facilitates the opening in case of solidification of the bitumen. Time of discharge is improved due to the 40mm stroke. The footalve can be manually operated in case of pneumatic failure.

SPECIFICATION	
Nominal diameter	100 mm (4")
Weight	16.4 Kg
Inlet flange	6x Ø18 + 2x Ø17 on a 180 PCD
Outlet flange	8x Ø18 on a 180 PCD
Working pressure	4 Bar (pneumatic air pressure 4 to 10 Bar)
Test pressure	6 Bar
Working temperature	-40° to +250°C
Design temperature	-40° to +250°C

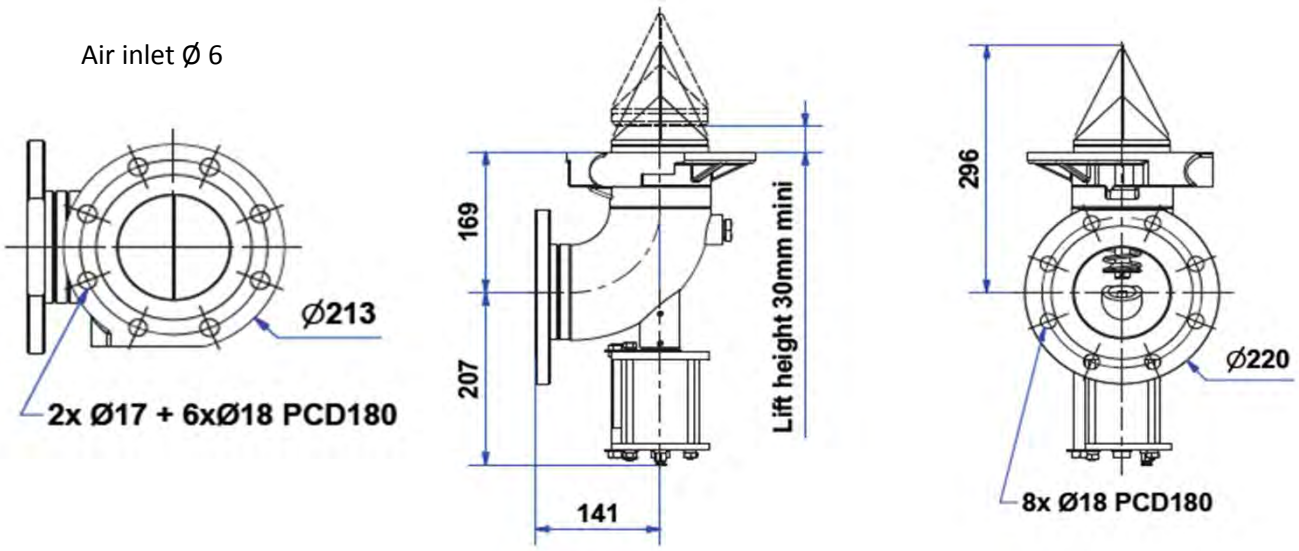
NB: temperature range vary according to o'ring material

CONFORMITY, CERTIFICATION & APPROVAL
ADR / RID / EN14433
Approved by : Bureau Veritas

STANDARD FEATURES (See Codification for option)	
Made of (material)	304L / 1.4307 stainless steel
Angle	90°
Body length	-
O'rings material	PFA/Silicone
Seals material	PFA/Silicone and graphite loaded PTFE
Operating	Simple effect actuator (manually operated in case of pneumatic failure)
Footvalve spring	Inside the body

ACCESSORIES	
Weld-in flange	17 16 01 70 00
Weld-in fl. steam. heated	17 16 01 70 10
Outlet square flange	17 11 38 10 02
Seal	11 15 26 00 01
Bolting kit inlet flange	17 16 01 93 00
Bolting kit outlet flange	17 16 01 94 00

DIMENSIONAL DRAWING



*other drilling options on request

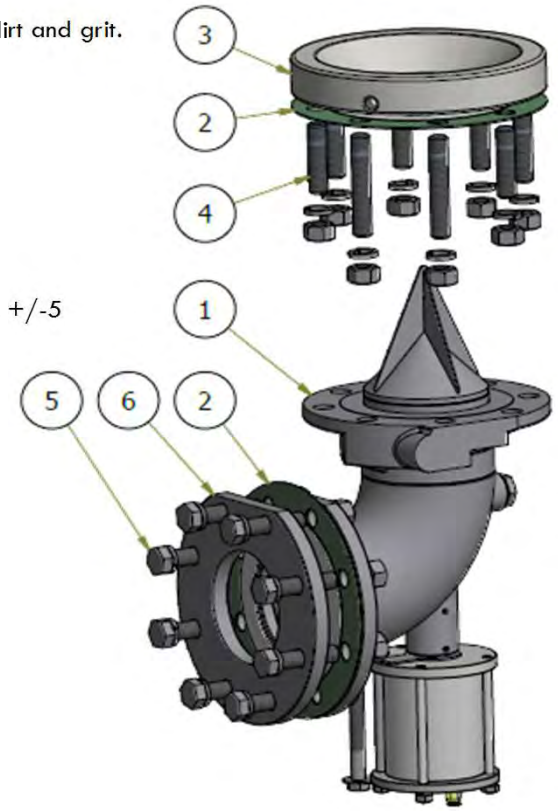
BOTTOM DISCHARGE

4" CHEMFLOW BITUMEN
Footvalve
17 16 01 00 00

BOTTOM DISCHARGE

INSTALLATION

Ensure that the Inlet flange and gasket are clean; free from dirt and grit.



1. Screw the studs into the tank container flange.
   Take careful note of the position of the studs.
2. Fit the PTFE gasket.
3. Fit the CHEMFLOW Footvalve in position.
4. Tighten the studs and nuts at a maximum torque of 50 N.m. +/-5
5. Refer to PWAGENE001 for bolt tightening sequence.

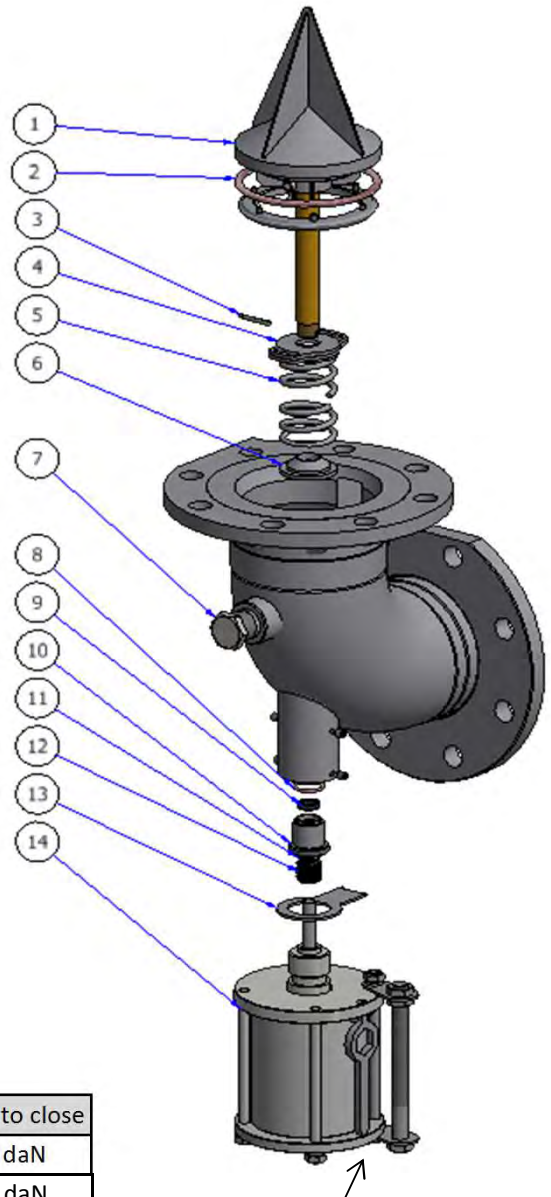
ITEM	DESCRIPTION	SPECIFICATION	PART N°
6	Outlet flange	ext Ø220 mm, int Ø106mm, thickness 16mm, 8 x Ø18 x PCD 180, 304L stainless steel	17 11 38 10 02
5	Outlet screw kit	Outlet Flange Screw Kit 8x M16-65 containing 8 screws, 8 washers and 8 nuts	17 16 01 94 00
4	Inlet studs kit	Inlet Flange Kit 6 Studs M16-50 + 2 studs M16-80, 8 washers and 8 nuts	17 16 01 93 00
3	Weld in flange steam heated	ext Ø220 mm, int Ø114.3 mm, thickness 36 mm, 8 x M16 x PCD 180, 316L stainless steel	17 16 01 70 10
3	Weld in flange	ext Ø220 mm, int Ø114.3 mm, thickness 34 mm, 8 x M16 x PCD 180, 316L stainless steel	17 16 01 70 00
2	Gasket	ext Ø220 mm, int Ø117 mm, 8 x Ø18 x PCD 180, CNAF / PTFE	11 15 26 00 01
1	CHEMFLOW MK2 4" BITUMEN	4" footvalve, inlet flange 6xØ18+2xØ17xPCD180, outlet flange 8xØ18xPCD180	17 16 01 00 00

4" CHEMFLOW BITUMEN
Footvalve
17 16 01 00 00

BOTTOM DISCHARGE

MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
		Poppet Seal:	**17 11 71 95 00**
2	1	O'ring, PFA Silicon	70 10 17 00 30
		Poppet Kit	**17 16 01 90 00**
1,2	1	Poppet, Stainless steel & O'ring	17 16 01 60 00
3	1	Split pin, Stainless steel	-
		Spring Kit	**17 16 01 91 00**
4	1	Spring plate, Stainless steel	-
5	1	Spring, Stainless steel	17 16 01 00 10
6	1	Disc, Stainless steel	-
3	1	Split pin, Stainless steel	-
		Cylinder Seals Kit	**17 16 01 92 00**
8	1	O'ring, PFA Silicon	70 10 17 00 31
9	1	Wiper spring seal, PTFE Graphite	70 10 91 00 41
10	1	Cartridge, Stainless steel	-
11	1	V Washer, Stainless steel	-
12	4	Chevron Packing, PTFE Graphite	17 11 71 00 06
13	1	Cartridge Tool, Stainless steel	-
		Actuator kit*:	**17 16 01 50 00**
14	1	Actuator, Stainless steel	-
-	1	Manual emergency opening key	-
		Plug kit:	**on request**
7	1	3/4" NPT Plug, Stainless steel	70 12 16 37 19



*Including manual emergency opening " key "

Pressure	Break to open	End to close	Break to close
4 bar	201 daN	175 daN	51 daN
5 bar	251 daN	225 daN	52 daN
6 bar	301 daN	275 daN	53 daN
10 bar	502 daN	476 daN	54 daN

The tightening torque of the cylinder is 20 Nm.do not exceed torque it can damage the packing
It must be checked every 18 months or in case of leakage.

HIGH CORROSION & HIGH PURITY

FUNCTION

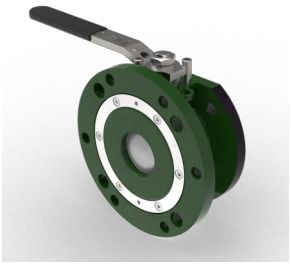
Perolo offers a wide range of high corrosion or high purity solutions. We can offer metallic solutions in high resistant materials as Hastelloy or 904L stainless steel. Next, we offer coated solutions with Halar coated valve bodies and wetted parts in resistant material (Hastelloy, Titanium, Ceramic, PVDF) depending on the transported cargo. Lastly, we offer customized solutions as manlids specifically designed for coating for even better resistance than a standard coated manlid.

Various materials for wetted parts:
High corrosion: Hastelloy, Titanium
High purity (no metal contact): PVDF, Ceramic

Surface Finish:
Halar lined

ADR, RID, IMDG

Large choice of accessories:
Lined gauge guard
Swingbolts with protective washer
Protective cap for SRV

1½" Ball Valve AF3	2" Butterfly Valve TF5	1½" Ball Valve AFN4	2" Ball Valve AFN5
 3"-2" Ball Valve AHN5	3" Ball Valve AFN8	3" Butterfly Valve TP8	3" Butterfly Valve TF8
2½" SRV MX65S	3" SRV MX80S	2½" SRV MX65F	Halar lined Coverlid
3" 45°NEATFLOW NF3	3" 90° CHEMFLOW CF3		

1 ½" Flanged ATCO™ AIRLINE
HIGH CORROSION ND32 Ball Valve
AF3X

The Perolo 1 ½" ATCO™ ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. With a super compact design, the 1 ½" ATCO™ is light & economic.
This product consists of a HALAR & PTFE/PFA coated body and Hastelloy ball, PFA coated ball, Ceramic ball or PVDF ball.

Caution: Please check the cargo compatibility with proposed materials

SPECIFICATION	
Nominal diameter	32mm (outlet 1 ½")
Weight	3.1 Kg
Inlet flange drilling	4 holes x Ø10,5on a 103.5 PCD
Outlet flange drilling	4 open slots x Ø18 on a min 98,4 PCD*
Working Pressure	6,7 Bar
Test Pressure	10 Bar
Working Temperature	-40°C to 150°C
Design Temperature	-40°C to 190°C

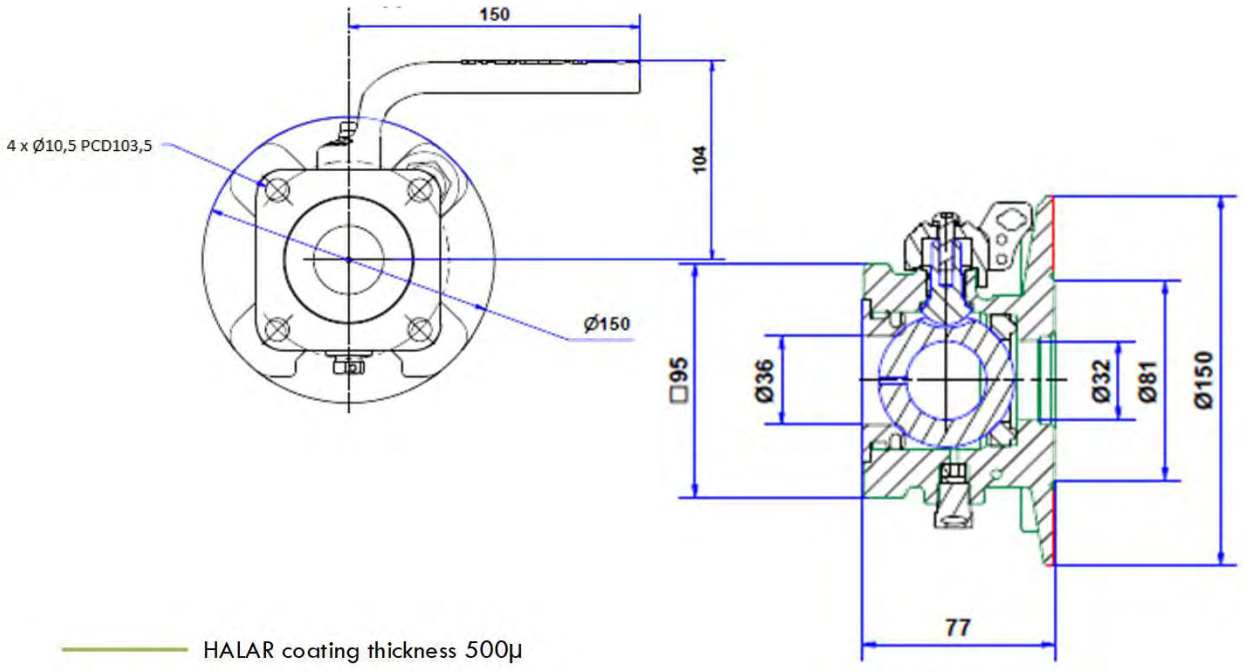
NB: temperature range vary according to o'ring material & ball material

ACCESSORIES	
Weld-in flange	11 90 20 00 00
Recess flange	11 91 12 00 00
Blind flange	11 95 87 10 00
Bolting Kit	on request
Screw kit	on request
Gasket	12 91 90 00 00
Filter Kit	11 14 01 95 00
Remote control	RMT33
Remote control handle	11 14 02 21 00
Pressure gauge	70 18 06 00 95

CONFORMITY & CERTIFICATION	
ADR / IMDG / RID / EN14432	
Bureau Veritas, RMR	
STANDARD FEATURES (See Codification for option)	
Body: 316/1.4408 & HALAR and PFA/PTFE coated Ball & Spindle: Hastelloy C276, Ceramic, 316L/PFA, PVDF	
Compact design	Face to face 76 mm
Handling operation	Handle - 1/4 turn or Remote Control
Ball seals material	PTFE

*Other drilling on request

DIMENSIONAL DRAWING



HALAR coating thickness 500µ
High corrosion material, please see p2 for ball & spindle options
PFA or PTFE coating

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 24 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

1 ½" Flanged ATCO™ AIRLINE
HIGH CORROSION ND32 Ball Valve
AF3X

CODIFICATION

STANDARD PART NUMBER	AF3	X	7	T	2	5	S	2	1	R	.	.
YOUR PART NUMBER	AF3		7		2			2				

MATERIAL - Other on request	
316 / 1.4408 stainless steel	X
MAWP	
6.7 Bar	7
BALL SEALS MATERIAL	
Ball Seals PTFE	T
INLET FLANGE	
4 x Ø10,5 x 103.5 PCD	2
OUTLET FLANGE - other on request	
4 x M16 x 98.4 PCD	4
4 x open slots Ø18 x min98.4 PCD	5
4 x M16 x 110 PCD	8
FILTER	
Without	S
With	P
GAUGE CONNECTION & PLUG	
1/4" BSP Connection & Plug	2
GAUGE PROTECTION	
Without	1
With	2
HIGH CORROSION - Coated Body	
Solid ball in Hastelloy C276	R
Ceramic ball	C
Hastelloy C276 FEP coated ball	F
PFA coated 316L ball	J
PVDF Solid Ball (max WT 80°C)	D
REMOTE CONTROL HANDLE	
Without remote control handle	.
With remote control handle	W
HIGH CORROSION - Spindle	
Hastelloy C276	.
Titanium	T
Hastelloy C276 FEP coated	F

ISO ROAD IBC

AIR LINE

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It is the customer's responsability to check the compatibility between valves material and transported product.
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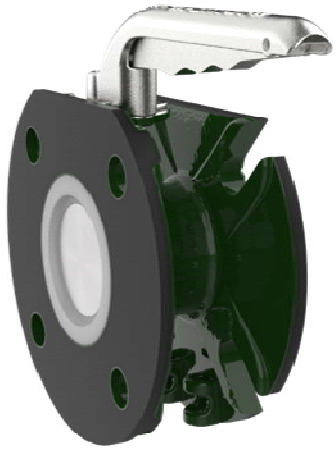
2" TANKFLY AIRLINE
HIGH CORROSION ND50 Butterfly Valve
TF5R



The 2" TANKFLY™ MKII flanged butterfly valve is designed to be used as a fluid inlet/outlet valve. With a super compact design, the 2" TANKFLY™ is light and economic.
This product consists of a HALAR & PTFE/PFA coated body and Hastelloy or titanium closure plate insuring an extreme resistance to corrosion.

Caution: Please check the cargo compatibility with proposed materials

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	3.0 Kg
Inlet flange drilling	4 holes x Ø18 on a 125 PCD
Outlet flange drilling	4 holes x Ø18 on a min 114 PCD*
Working Pressure	7 Bar
Test Pressure	10.5 Bar
Working temperature	-40° to +150°C
Design Temperature	-40° to +190° C

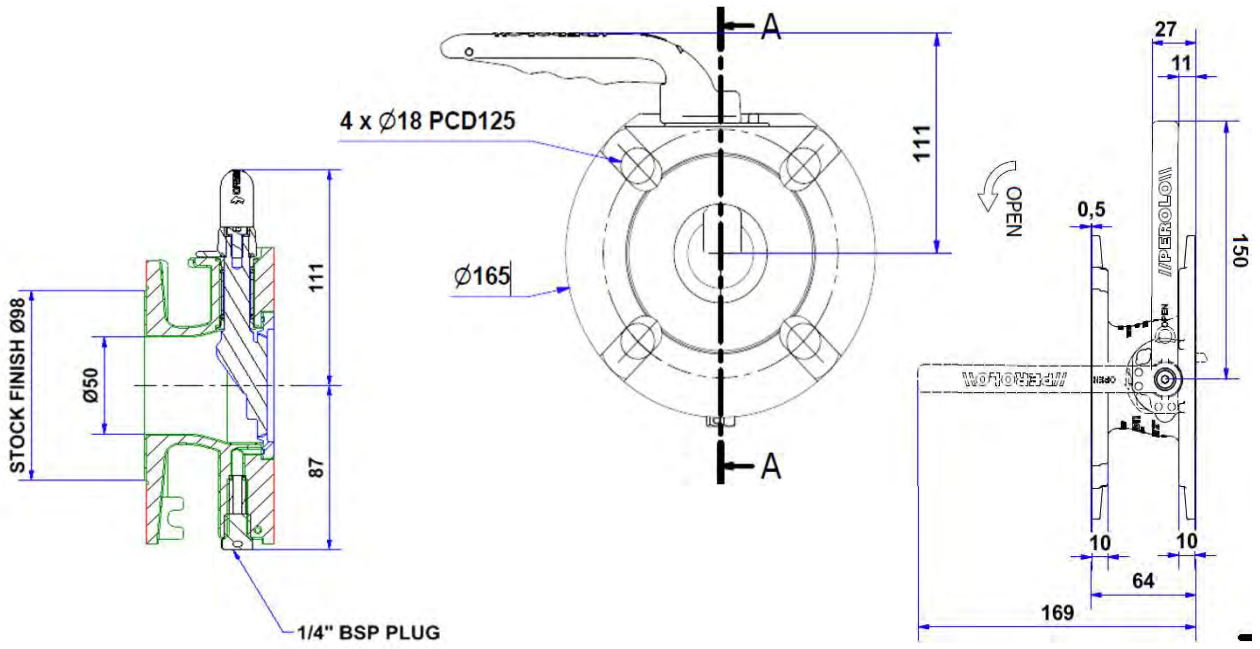
NB: temperature range vary according to o'ring material

ACCESSORIES	
Weld-in flange	11 90 25 00 00
Recess flange	11 92 44 00 00
Blind flange	11 95 88 10 00
Bolting Kit	on request
Gasket	11 90 29 00 00
Remote control	RMT53
Remote control handle	11 17 20 11 00
Pressure gauge	70 18 06 00 95

CONFORMITY & CERTIFICATION	
ADR / IMDG / RID / EN14432	
Bureau Veritas, RMR	
STANDARD FEATURES (See Codification for option)	
Body: 316/1.4408 & HALAR and PFA/PTFE coated Closure plate: Hastelloy C276 or Titanium	
Compact design	Face to face 64 mm
Operation	Handle - 1/4 turn or Remote Control
Gasket material	PTFE

*Other drilling on request

DIMENSIONAL DRAWING



High corrosion material, please see p2 for ball & spindle options

PFA or PTFE coating

HALAR coating thickness 500µ

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 14 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" TANKFLY AIRLINE
HIGH CORROSION ND50 Butterfly Valve
TF5R

CODIFICATION

STANDARD PART NUMBER	TF5	R	7	T	1	1	S	1	1	.	H
YOUR PART NUMBER	TF5	R	7	T	1	__	S	1	__	__	__

HIGH CORROSION - Coated Body

316/1.4408 Halar-PFA/PTFE coating	R

MAWP

7 Bar	7

WORKING TEMPERATURE

150 °C (PTFE seal)	T

INLET FLANGE

4 slots x Ø18 x 125 PCD	1

OUTLET FLANGE

4 open slots x Ø18 x min 114 PCD	1
4 x M16 x 125 PCD	2
4 slots x Ø18 x 125 PCD	3

FILTER

Without filter	S

GAUGE CONNECTION & PLUG

with 1/4" BSP connection & plug	1

GAUGE PROTECTION

without	1
with	2

OPERATING SYSTEM

Standard Handle	.
Pneumatic Actuator	P
With Remote Control Handle	W

HIGH CORROSION - Butterfly Material

Hastelloy C276	H
Titanium	T

AIR LINE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 14 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

1 ½" ATCO BALL VALVE
HIGH CORROSION ND40 Ball Valve
AFN4X

The Perolo ND40 ATCO™ FULL BORE self-contained ball valve is designed to be used as an air inlet or as a fluid inlet/outlet valve. This product consists of a HALAR & PTFE/PFA coated body and Hastelloy ball, 316L / PFA or halar coated ball or Ceramic Ball.

SPECIFICATION	
Nominal diameter	40 mm (1"1/2)
Weight	5.8 Kg
Inlet flange drilling	4 holes x Ø18 on a 110 PCD*
Outlet flange drilling	4 slotted holes Ø18 on a 110 to 125 PCD*
Working Pressure	16 Bar
Test Pressure	24 Bar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +250°C

NB: temperature range vary according to o'ring material

CONFORMITY, CERTIFICATION & APPROVAL
Complies with ADR / IMDG / RID / EN14432
Approved by Bureau Veritas, RMR, A-C-I

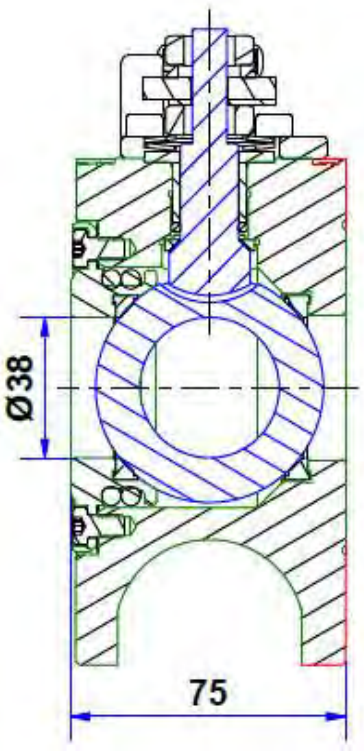
ACCESSORIES	
Recess flange	11 97 18 00 00
Blind flange PN16	11 95 87 10 00
Gasket	12 91 90 00 00
Stud kit	11 12 69 91 00
Remote Control	RMT 33

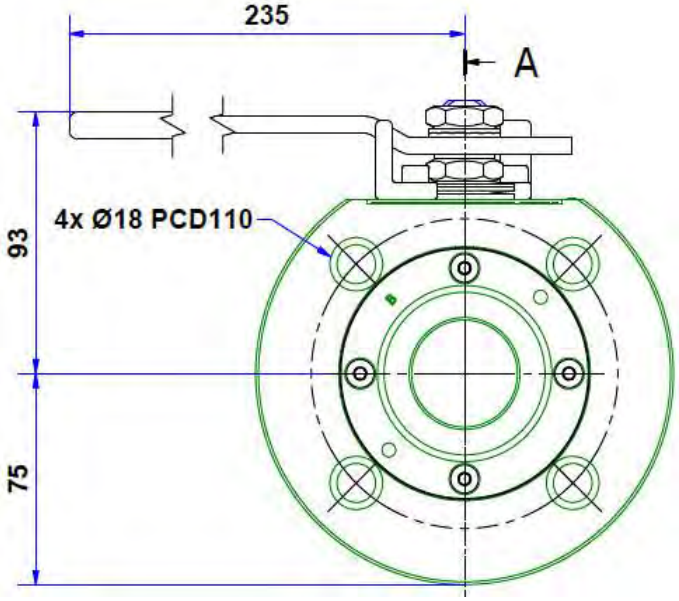
* other drilling patterns on request

STANDARD FEATURES (See Codification for option)	
Body: 316L/1.4409 & HALAR PFA/PTFE coated Ball: Hastelloy, Ceramic, 316L/PFA or Halar coated	
Compact design	Face to face 74 mm
Handling operation	1/4 turn
Ball seals material	PTFE

Caution: Please check the cargo compatibility with proposed materials

DIMENSIONAL DRAWING





Halar coating thickness 500 µm
Hastelloy spindle, for ball check options on next page
PFA or PTFE coating

BALL VALVE

1 ½" ATCO BALL VALVE
HIGH CORROSION ND40 Ball Valve
AFN4X

BALL VALVE

CODIFICATION

STANDARD PART NUMBER	AFN4	X	16T	1	1	0	0	S	0	R	.
YOUR PART NUMBER	AFN4	X	__	__	__	0	0	S	0	__	__

BODY MATERIAL	
316 / 1.4408 stainless steel	X

SEAL MATERIAL & MAWP	
PTFE seal & 16 Bar	16T
PTFE seal & 4 Bar (PVDF ball only)	4T

INLET FLANGE - others on request	
4 x Ø18 x 110 PCD	1

OUTLET FLANGE	
4 slots x Ø18 x 110 to 125 PCD	1
4 x M16 x 110 PCD	2
4 slots x Ø18 x 98,5 to 125 PCD	3
4 x M16 spaced 55° x 110 PCD	4
4 slots x Ø18 x 110 to 150 PCD	5

GAUGE CONNECTION	
without	0

GAUGE PROTECTION	
without	0

SEALING	
with	S

STEAM HEATING	
without	0

HIGH CORROSION - Coated Body	
Hastelloy C276 solid ball & spindle	R
Halar coated 316L ball & Hastelloy C276 spindle	E
PFA coated 316L ball & Hastelloy C276 spindle	I
Ceramic solid ball & Hastelloy C276 spindle	C
Hastelloy C22 solid ball & spindle	Z
PVDF solid ball & FEP coated Hastelloy C276 spindle. MAWP 4 bar. Max WT +80°C	D

HANDLE - other on request	
Standard handle	.
Reverse handle	L

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 06 20 00 E
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2" ATCO BALL VALVE
HIGH CORROSION ND50 Ball Valve
AFN5X

The Perolo ND50 ATCO™ FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve.
This product is composed with HALAR & PTFE/PFA coated body and Hastelloy ball, 316L / PFA coated ball, Ceramic ball or PVDF ball.

Caution: Please check the cargo compatibility with proposed materials

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	6.0 Kg
Inlet flange drilling	4 holes x Ø17 on a 125 PCD
Outlet flange drilling	4 holes x M16 on a 125 PCD*
Working Pressure	16 Bar
Test Pressure	24 Bar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

NB: temperature range vary according to o'ring material & ball options

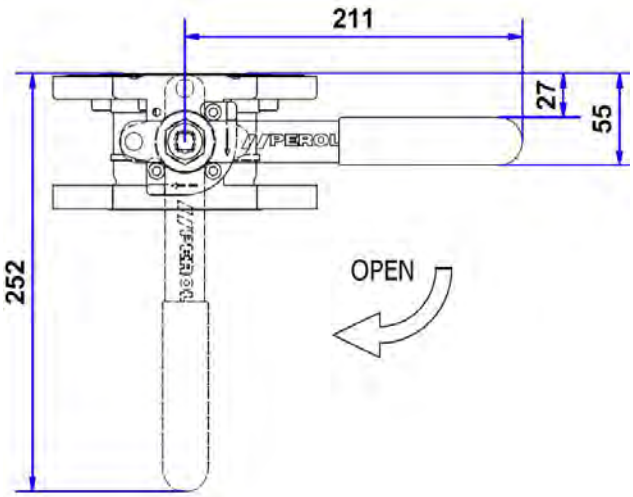
CONFORMITY & CERTIFICATION
ADR / IMDG / EN14432
Bureau Veritas, RMR, A-C-I

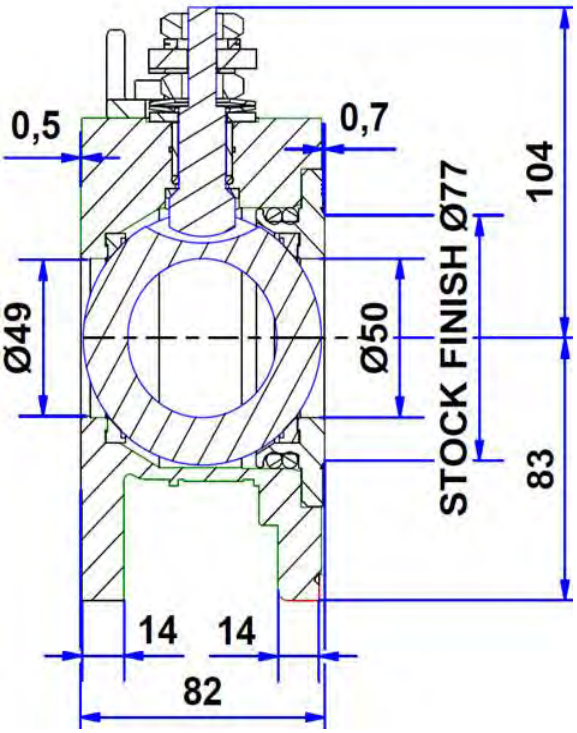
ACCESSORIES	
Weld-in flange	11 90 25 00 00
Recess flange	11 92 44 00 00
Blind flange	11 95 88 10 00
Bolting Kit	on request
Gaskets	11 90 29 00 00
Stud kit	on request
Remote Control	RMT 53
Remote Control Handle	11 14 55 91 00

STANDARD FEATURES (See Codification for option)	
Body: 316/1.4408 & HALAR and PFA/PTFE coated Ball: Hastelloy C276, Ceramic, 316L/PFA, PVDF	
Compact design	Face to face 81 mm
Operation	Handle - 1/4 turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING

*Other drilling on request





High corrosion material, please see p2 for ball & spindle options

PFA/PTFE coating or Topcoat

HALAR coating thickness 500µ

BALL VALVE

2" ATCO BALL VALVE
HIGH CORROSION ND50 Ball Valve
AFN5X

CODIFICATION

STANDARD PART NUMBER	AFN5	X	16T	1	2	0	0	S	0	R	R	.
YOUR PART NUMBER	AFN5	X	__	1	__	__	__	S	__	__	__	__

BODY MATERIAL

316 / 1.4408 stainless steel	X

SEAL MATERIAL & MAWP

PTFE seal & 16 Bar	16T
PTFE seal & 4 Bar (PVDF ball only)	4T

INLET FLANGE DRILLING

4 x Ø17 x 125 PCD	1

OUTLET FLANGE DRILLING*

4 x Ø17 x 125 PCD	1
4 x M16 x 125 PCD	2
4 slotted x Ø18 x min 114 PCD	3

*other drilling on request

GAUGE CONNECTION - on request

without	0

GAUGE PROTECTION

without	0

SEALING

with	S

STEAM HEATING

without	0

HIGH CORROSION - Coated Body

Halar - PFA/PTFE coating	R
Halar -Topcoat coating	T

HIGH CORROSION - Ball* & Spindle

Hastelloy C276 solid ball & spindle	R
Ceramic ball & Hastelloy C276 spindle	C
PFA coated 316L ball & Hastelloy C276 spindle	I
PVDF solid ball with HastC276 inserts & FEP coated Hastelloy C276 spindle. MAWP 4 bar. Max WT +80°C	D

*drilled ball on request

HANDLE

Standard handle	.
Remote control handle	W
Reverse handle	L

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 12 95 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
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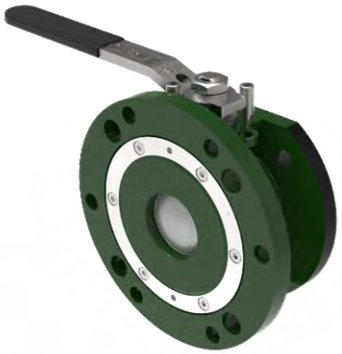
3"/2" ATCO AHN5X
HIGH CORROSION ND80/ND80 Ball Valve
AHN5X



The Perolo ND80/ND50 ATCO™ FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve.
This product is composed with HALAR & PTFE/PFA coated body and Hastelloy ball, Ceramic ball or PVDF ball.

Caution: Please check the cargo compatibility with proposed materials

SPECIFICATION	
Nominal diameter	50mm (2")
Weight	5,84 Kg
Inlet flange drilling	4 x Ø18 x 160PCD & 6 x Ø14 x 168 PCD*
Outlet flange drilling	4 holes x M16 x 125 PCD*
Working Pressure	16 Bar (4 Bar for PVDF Ball)
Test Pressure	24 Bar
Working Temperature	-40° to +150°C (80°C for PVDF Ball)
Design Temperature	-40° to +190°C

NB: temperature range vary according to o'ring material & ball options

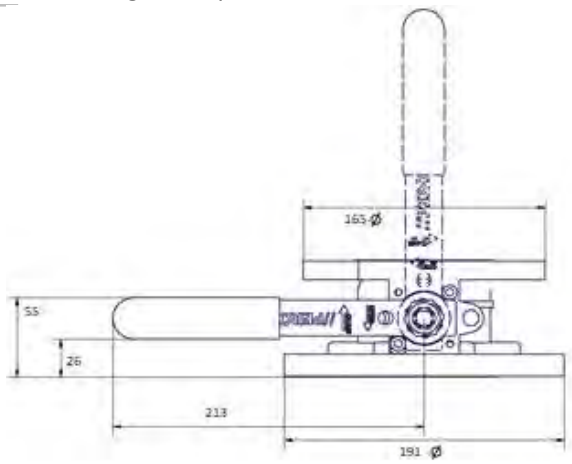
CONFORMITY & CERTIFICATION
ADR / IMDG / EN14432
Bureau Veritas, RMR, A-C-I

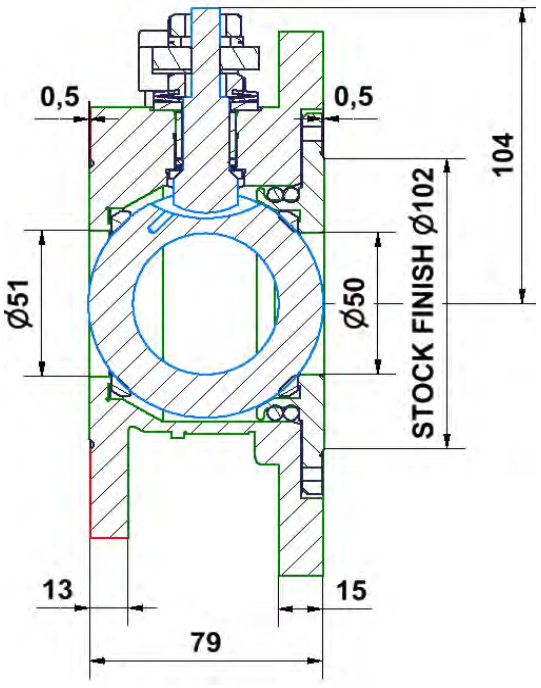
ACCESSORIES	
Weld-in flange	11 90 36 00 00
Recess flange	11 90 98 00 00
Blind flange	11 95 88 10 00
Bolting Kit	on request
Outlet Gasket	11 90 29 00 00
Inlet Gasket	12 91 40 00 00
Stud kit	on request
Remote Control	RMT 53

STANDARD FEATURES (See Codification for option)	
Body: 316/1.4408 & HALAR and PFA/PTFE coated Ball: Hastelloy C276, Ceramic, PVDF	
Compact design	Face to Face 79mm
Operation	Handle - 1/4 turn or Remote Control
Ball seals material	PTFE

DIMENSIONAL DRAWING

*Other drilling on request





High corrosion material, please see p2 for ball & spindle options

PFA/PTFE coating

HALAR coating thickness 500µ

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 17 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3"/2" ATCO AHN5X
HIGH CORROSION ND80/ND80 Ball Valve
AHN5X

CODIFICATION

STANDARD PART NUMBER	AHN5	X	16T	1	2	0	0	S	0	R	
YOUR PART NUMBER	AHN5	X		1				S			

BODY MATERIAL	
316 / 1.4408 stainless steel	X

SEAL MATERIAL & MAWP	
PTFE seal & 16 Bar	16T
PTFE seal & 4 Bar (PVDF ball only)	4T

INLET FLANGE DRILLING	
4xØ18x160PCD & 6xØ14x168PCD	1

OUTLET FLANGE DRILLING*	
4 x Ø17 x 125 PCD	1
4 x M16 x 125 PCD	2
4 slotted x Ø18 x min 114 PCD	3

*other drilling on request

GAUGE CONNECTION - on request	
without	0

GAUGE PROTECTION	
without	0

SEALING	
with	S

STEAM HEATING	
without	0

HIGH CORROSION - Coated body	
Hastelloy C276 solid ball & spindle	R
Ceramic ball & Hastelloy C276 spindle	C
PFA coated 316L ball & Hastelloy C276 spindle	I
PVDF solid ball & FEP coated Hastelloy C276 spindle. MAWP 4 bar. Max WT +80°C	D

HANDLE	
Standard handle	.
Remote control handle	W
Reverse handle	L

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 17 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" ATCO BALL VALVE
HIGH CORROSION ND80 Ball Valve
AFN8X

The Perolo ND80 ATCO™ FULL BORE self-contained ball valve is designed to be used as a fluid inlet/outlet valve.This product is composed with HALAR & PTFE/PFA coated body and Hastelloy ball, 316L / PFA coated ball, or Ceramic Ball

Caution: Please check the cargo compatibility with proposed materials

SPECIFICATION	
Nominal diameter	80mm (3")
Weight	12.5 Kg
Inlet flange drilling	8 holes x Ø17 on a 160 PCD*
Outlet flange drilling	8 holes x Ø18 on a 160 PCD*
Working Pressure	16 bar
Test Pressure	24 Bar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

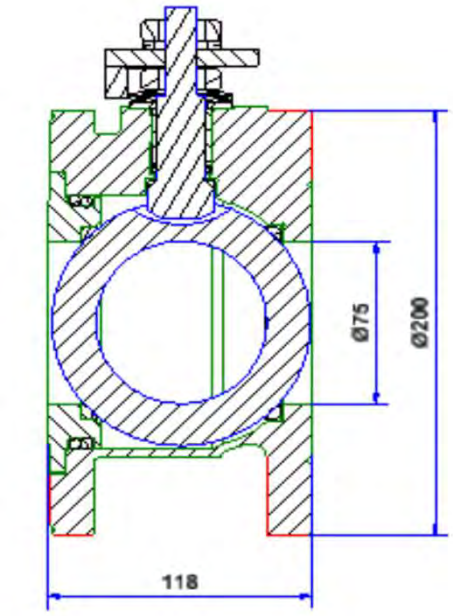
NB: temperature range vary according to o'ring material & ball material

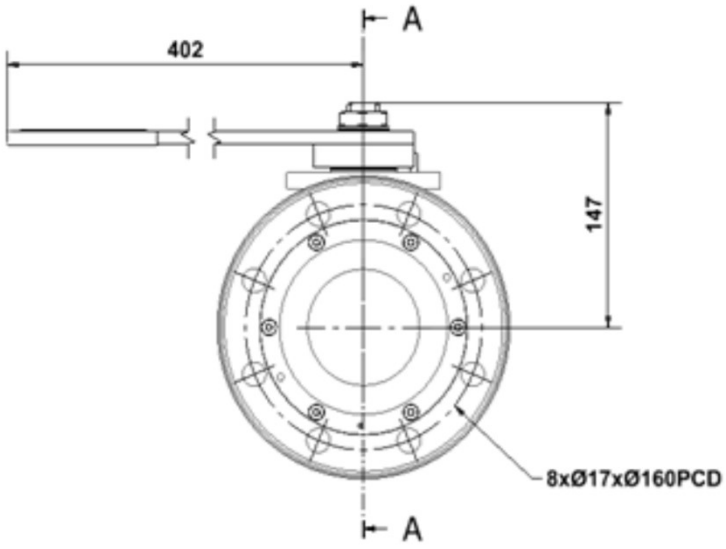
ACCESSORIES	
Recess flange	11 90 98 00 00
Blind flange	11 95 90 00 00
Bolting Kit	on request
Gaskets	12 91 40 00 00
Tools kit	11 12 71 96 00
Remote Control	on request
Screw kit	on request

CONFORMITY & CERTIFICATION	
ADR / IMDG / RID / EN14432	
Bureau Veritas, RMR, A-C-I	
STANDARD FEATURES (See Codification for option)	
Body: 316/1.4408 & HALAR and PFA/PTFE coated Ball: Hastelloy C276, Ceramic, 316L/PFA	
Compact design	Face to face 115 mm
Operation	Handle - 1/4 turn or Remote Control
Ball seals material	PTFE

*other drilling on request

DIMENSIONAL DRAWING





High corrosion material, please see p2 for ball & spindle options

PFA or PTFE coating

HALAR coating thickness 500µ

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 05 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

3" ATCO BALL VALVE
HIGH CORROSION ND80 Ball Valve
AFN8X

CODIFICATION

STANDARD PART NUMBER	AFN8	X	16T	2	4	S	0	0	R	.
YOUR PART NUMBER	AFN8							0		

BODY MATERIAL other on request	
316 / 1.4408 stainless steel	X

SEAL MATERIAL & MAWP	
PTFE seal & 16 Bar	16T

INLET FLANGE DRILLING other on request	
6 x Ø14 x 168.3 PCD	1
8 x Ø17 x 160 PCD	2
4 x Ø17 x 160 PCD	3
8 slots x Ø18 x 160 to 168 PCD	4
4 x Ø17 x 160 PCD & 6 x Ø14 x 168 PCD	5

OUTLET FLANGE DRILLING others on request	
4 open slots x Ø18 x min 146.1 PCD	1
8 x Ø18 x 160 PCD (ISO PN16)	2
8 slots x Ø18 x 152.4 to 160 PCD	3
4 x M16 & 4 x Ø18 x 160 PCD	4
4 slots x Ø18 x 146 to 160 PCD	5
4 x M16 & 4 x Ø18 x 160 PCD shifted 22°30'	6
4 xØ19x152,4 PCD + 8xØ18 x 160 PCD	7
8 x M16 x 160 PCD	8
4 open slots x Ø18 x 146 (with cut side)	9

SEALING	
Sealing	S
Sealing & pin	N

PIN BRACKET ON OUTLET FLANGE	
with	P
without	0

STEAM HEATING	
without	0

HIGH CORROSION - Body Halar/PFA/PTFE coated	
Hastelloy C276 solid ball & spindle	R
Hastelloy C22 solid ball & spindle	Z
Ceramic ball & Hastelloy C276 spindle	C
PFA coated 316L ball & Hastelloy C276 spindle	I

HANDLE other on request	
standard	.
left	L

BALL VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 14 05 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" TANKFLY CLAMPED
HIGH CORROSION ND80 Butterfly Valve
TP8R

The 3" HALAR TANKFLY™ clamped butterfly valve is designed to be clamped between two flanges or directly onto the pipe work.
This product consists of an internal HALAR coated body, Hastelloy or Titanium or PVDF closure plate.

Caution: Please check the cargo compatibility with proposed materials

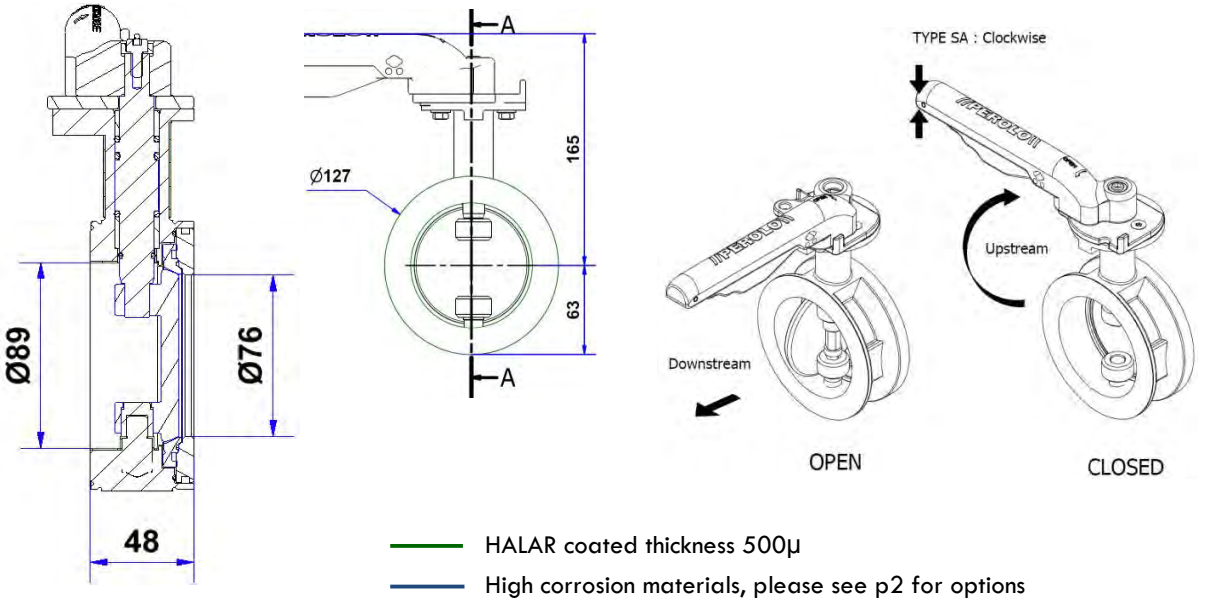
SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	3.4 Kg
Inlet flange drilling	Clamped between DIN or ANSI flanges
Working Pressure	7 Bar
Test Pressure	10,5 Bar
Working Temperature	-20° to +150°C (PVDF max +80°C)
Design Temperature	-40° to +190°C

NB: temperature range vary according to o'ring material

CONFORMITY & CERTIFICATION	
ADR / IMDG / EN 14332 / A-C-I	
Bureau Veritas	
STANDARD FEATURES (See Codification for option)	
Made of (material)	Body: 316/1.4408 & HALAR coated Closure plate: Hastelloy C22, Titanium, PVDF
Maintenance	Face to face 48 mm
Handling operation	Handle - 1/4 turn with locking
Closure plate seal	PTFE

ACCESSORIES	
Weld in flange	11 90 36 00 00
Seal	12 91 40 00 00
Adaptor flange	12 91 41 00 03
Blind flange	11 95 90 10 00
Screw kit M16-90	on request
Screw kit M16-40	on request
Remote control	RMT83

DIMENSIONAL DRAWING



HALAR coated thickness 500µ

High corrosion materials, please see p2 for options

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 48 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" TANKFLY CLAMPED
HIGH CORROSION ND80 Butterfly Valve
TP8R

CODIFICATION

STANDARD PART NUMBER	TP8	R	7	H
YOUR PART NUMBER	TP8	R	__	__

HIGH CORROSION - Coated Body	
316/1.4408 Halar	R

MAWP	
7 Bar	7
4 Bar (PVDF closure plate)	4

HIGH CORROSION - closure plate & spindle & clamping ring	
Hastelloy C22 closure plate, Hastelloy C276 spindle, Hastelloy C276 clamping ring	H
Titanium closure plate, Titanium spindle, Titanium clamping ring	T
PVDF closure plate, Halar FDA coated Hastelloy C276 spindle, PVDF clamping ring, max WT +80°C	P

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 48 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" TANKFLY FLANGED
HIGH CORROSION ND80 Butterfly Valve
TF8R

The PEROLO HALAR 3" flanged TANKFLY™ butterfly valve is designed to be used as a fluid inlet/outlet valve.
This product consists of an internal HALAR coated body, Hastelloy or Titanium or PVDF closure plate.

Caution: Please check the cargo compatibility with proposed materials

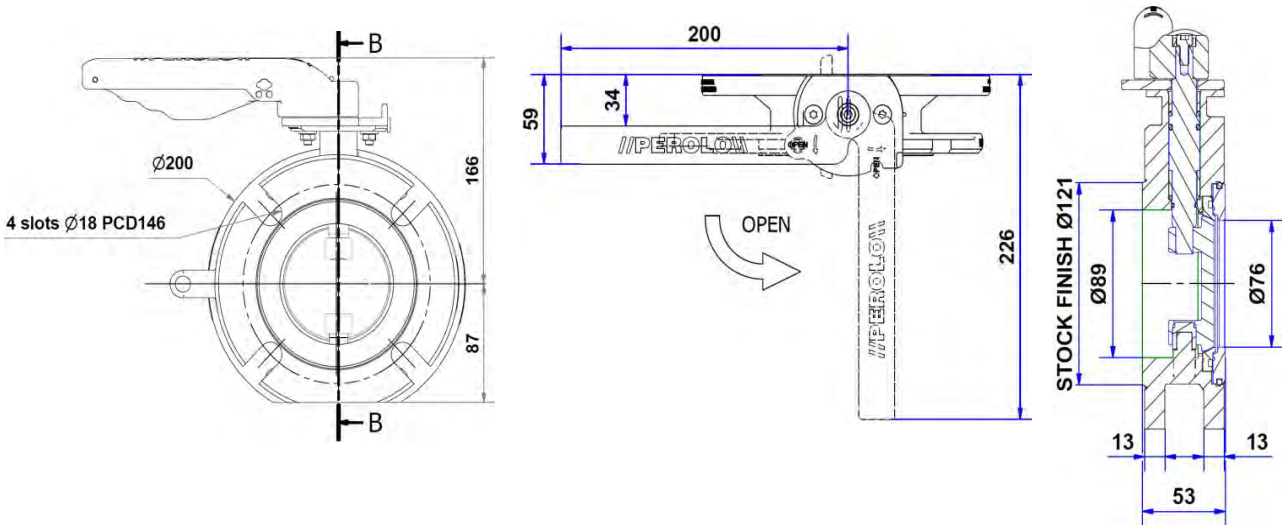
SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	6,8 Kg
Inlet flange drilling	6 or 4 holes
Outlet flange drilling	4 x slots 18 on PCD 146 to 160
Working Pressure	7 Bar
Test Pressure	10,5 Bar
Working Temperature	-20° to +150°C (PVDF max +80°C)
Design Temperature	-40° to +190°C

NB: temperature range vary according to o'ring material & poppet material

CONFORMITY & CERTIFICATION	
ADR / IMDG / RMR / EN 14332 / A-C-I	
Bureau Veritas	
STANDARD FEATURES (See Codification for option)	
Made of (material)	Body: 316/1.4408 & HALAR coated Closure plate: Hastelloy C22, Titanium, PVDF
Maintenance	Face to face 55mm
Handling operation	Handle - 1/4 turn with locking
Closure plate seal	PTFE

ACCESSORIES	
Blind Flange	11 97 48 00 00
Weld-in flange	11 90 36 00 00
Recess flange	11 90 98 00 00
Gasket	12 91 40 00 00
Screw kit	12 91 61 92 00
Pivot pin	12 91 52 00 00
Remote control	RMT83

DIMENSIONAL DRAWING



HALAR coated thickness 500µ

High corrosion materials, please see p2 for options

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 47 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

3" TANKFLY FLANGED
HIGH CORROSION ND80 Butterfly Valve
TF8R

CODIFICATION

STANDARD PART NUMBER	TF8	R	7	4	1	H
YOUR PART NUMBER	TF8	R				

HIGH CORROSION - Coated Body	
316/1.4408 Halar coating	R

MAWP	
7 Bar	7
4 Bar (PVDF closure plate)	4

INLET FLANGE - other on request	
6 x Ø14 x 168 PCD	1
4 x Ø18 x 160 PCD	2
4 x M16 x 160 PCD	3
6 x Ø14 x 168 PCD & 4 x Ø18 x 160 PCD	4
6 x Ø14 x 168 PCD & 8 x Ø18 x 160 PCD	5

OUTLET FLANGE	
4 open slots x Ø18 x min 146 PCD	1
4 open slots x Ø18 x min 146 PCD & 4 x M16 x 160 PCD	2

HIGH CORROSION - closure plate & spindle & clamping ring	
Hastelloy C22 closure plate, Hastelloy C276 spindle, Hastelloy C276 clamping ring	H
Titanium closure plate, Titanium spindle, Titanium clamping ring	T
PVDF closure plate, Halar FDA coated Hastelloy C276 spindle, PVDF clamping ring, max WT +80°C	P

BUTTERFLY VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 17 47 20 00 B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" MEGA SUPERVENTIX SRV

HIGH CORROSION FLANGED ND65 SRV

MX65S/P

2 1/2" MEGA SUPERVENTIX™ flanged safety relief valve - designed to protect tanks against accidental overpressure or excessive vacuum. With a smart compact design, the 2 1/2" MEGA SUPERVENTIX™ has a high flow performance. This product consists of a HALAR coated body and Hastelloy, Titanium & PVDF poppet insuring an extreme resistance to corrosion. Its plastic protection cap placed on top of the coverlid maximizes corrosion resistance.

Caution: Please check the cargo compatibility with proposed materials

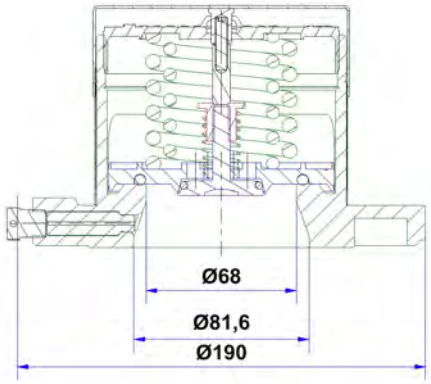
SPECIFICATION	
Bore Diameter	68 mm
Weight	5.63 Kg
Drilling	4 slotted holes x Ø18 on a 145 to 152.4 PCD
Pressure Setting	From 0.5 to 7.4 Bar
Vacuum Setting	From 50 to 410 mBar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

NB: temperature range vary according to o'ring & poppet material

ACCESSORIES	
Weld-in flange	12 92 03 00 00
Recess flange	11 90 63 00 00
Seal	12 92 04 00 00
Screw kit	11 10 09 91 00
Stud kit	17 12 12 91 00
Gauge protection Kit	11 10 24 00 01
HALAR gauge prot. Kit	11 10 69 10 00
Pressure gauge	70 18 06 00 95

CONFORMITY & CERTIFICATION	
ADR / IMDG / RID / ISO4126-1	
Lloyd's	
STANDARD FEATURES (See Codification for option)	
Made of (material)	Body: 316/1.4408 & HALAR coated, Poppet: C276 Hastelloy, PVDF, Titanium
Seals Material	PTFE
O Ring Material	FEP/Silicone - others on request

DIMENSIONAL DRAWING



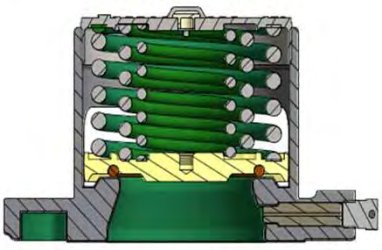
Set pressure			Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze	
Bar	kPa	PSIG	Sm³/S	Scfm	Sm³/S	Scfm
Max (±3% SP; 0,1B/1,45PSI)						
2,18	218	31,6	2,56	5425	2,30	4883
3,75	375	54,4	3,92	8306	3,53	7475
4,4	440	63,8	4,48	9492	4,03	8543
7,3	730	105,9	6,99	14810	6,29	13329

Halar coating thickness 500 µm

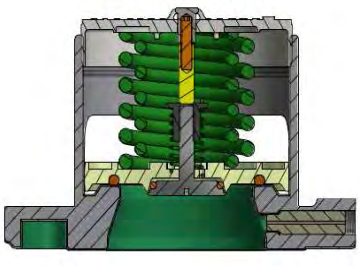
C276 Hastelloy, PVDF, Titanium: see p2 for poppet material options

Hastelloy C276 adjustment nut

without vacuum option



with vacuum option



SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 50 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" MEGA SUPERVENTIX SRV

HIGH CORROSION FLANGED ND65 SRV

MX65S/P

CODIFICATION

STANDARD PART NUMBER	MX65	P	44	T	21	G	.	H	J
YOUR PART NUMBER	MX65	__	__	T	__	__	__	__	J

BODY DRILLING

4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection	S
4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection and plug	P
4 slots x Ø18 x 145-152.4 PCD without connection	A

SET PRESSURE - Other on request

No pressure	00
. bar (ex: 4.4bar)	44

SEAL MATERIAL

FEP	T

VACUUM - other on request

No Vacuum	.
0.** bar (ex: 0.21 bar)	21

FLAME PROOF GAUZE

With	G
Without	S

VACUUM TEST BUTTON

With	B
Without	.

HIGH CORROSION - poppet material

Pressure Plate: Hastelloy C276 - Vacuum Plate: Hastelloy C276 Max pressure setting: 7,3b - Max working temperature: +150°C	H
Pressure Plate: Titanium - Vacuum Plate: PVDF Max pressure setting: 7,3b - Max working temperature: +80°C	T
Pressure Plate: PVDF - Vacuum Plate: PVDF Max pressure setting: 4,4b - Max working temperature: +80°C	R

HIGH CORROSION halar coated body

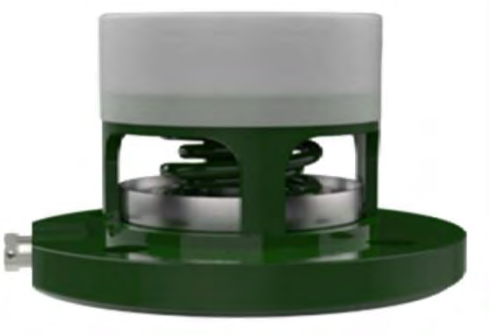
Internal & external HALAR coating	J

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 50 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

3" MEGA SUPERVENTIX SRV

HIGH CORROSION ND80 SRV MX80



3" MEGA SUPERVENTIX™ flanged safety relief valve - designed to protect tanks against accidental overpressure or excessive vacuum. With a smart compact design, the 3" MEGA SUPERVENTIX™ has a high flow performance. This product consists of a HALAR coated body and Hastelloy, Titanium & PVDF poppet insuring an extreme resistance to corrosion. Its plastic protection cap placed on top of the coverlid maximizes corrosion resistance.

Caution: Please check the cargo compatibility with proposed materials

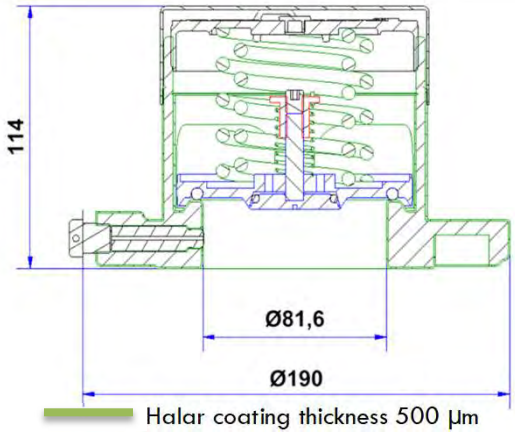
SPECIFICATION	
Bore Diameter	82 mm
Weight	5.2 Kg
Drilling	4 slotted holes x Ø18 on a 145 to 152.4 PCD
Pressure Setting	1 bar to 7,3 bar
Vaccum Setting	0,05 to 0,41 bar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

NB: temperature range vary according to o'ring material

ACCESSORIES	
Weld-in flange	12 92 03 00 00
Recess flange	11 90 63 00 00
Seal	12 92 04 00 00
Screw kit	11 10 09 91 00
Stud kit	17 12 12 91 00
Gauge protection Kit	11 10 24 00 01
HALAR gauge prot. Kit	11 10 69 10 00
Pressure gauge	70 18 06 00 95

CONFORMITY & CERTIFICATION	
ADR / IMDG / RID / ISO4126-1	
Lloyd's	
STANDARD FEATURES (See Codification for option)	
Made of (material)	Body: 316/1.4408 & HALAR coated, Poppet: C276 Hastelloy, PVDF, Titanium
Seals Material	PTFE
O Ring Material	FEP/Silicone - others on request

DIMENSIONAL DRAWING



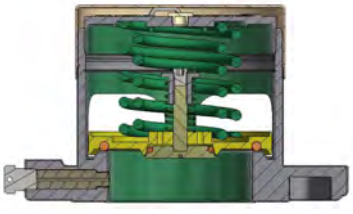
Halar coating thickness 500 µm

C276 Hastelloy, PVDF, Titanium: see p2 for poppet material options

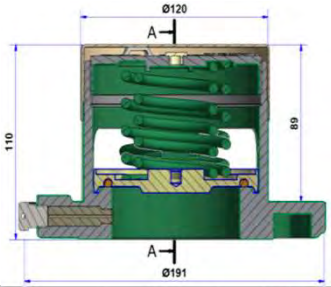
Hastelloy C276 adjustment nut

Set pressure			Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze	
Bar	kPa	PSIG	Sm³/S	Scfm	Sm³/S	Scfm
Max (±3% SP; 0,1B/1,45PSI)						
2,18	218	31,6	2,78	5888	2,50	5299
3,3	330	47,9	4,11	8705	3,70	7834
3,75	375	54,4	4,6	9743	4,14	8769
4,4	440	63,8	5,35	11331	4,82	10198
7,3	730	105,9	7,372	15614	6,63	14053

With vacuum option



Without vacuum option



SAFETY RELIEF VALVE

VALVES SOLUTIONS FOR YOUR

3" MEGA SUPERVENTIX SRV

HIGH CORROSION ND80 SRV MX80

CODIFICATION

STANDARD PART NUMBER	MX80	P	44	T	21	G	.	H	J
YOUR PART NUMBER	MX80	__	__	T	__	__	__	__	J

BODY DRILLING	
4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection	S
4 slots x Ø18 x 145-152.4 PCD with 1/4" BSP connection and plug	P
4 slots x Ø18 x 145-152.4 PCD without connection	E

SET PRESSURE - Other on request	
No pressure	00
. bar (ex: 4.4bar)	44

SEAL MATERIAL	
FEP	T

VACUUM - other on request	
No Vacuum	.
0.** bar (ex: 0.21 bar)	21

FLAME PROOF GAUZE	
With	G
Without	S

VACUUM TEST BUTTON	
With	B
Without	.

HIGH CORROSION - poppet material	
Pressure Plate: Hastelloy C276 - Vacuum Plate: Hastelloy C276 Max pressure setting: 7,3b - Max working temperature: +150°C	H
Pressure Plate: Titanium - Vacuum Plate: PVDF Max pressure setting: 7,3b - Max working temperature: +80°C	T
Pressure Plate: PVDF - Vacuum Plate: PVDF Max pressure setting: 4,4b - Max working temperature: +80°C	R

HIGH CORROSION halar coated body	
Internal & external HALAR coating	J

ISO RAIL ROAD IBC

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 69 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

2 1/2" (ND65) SAFETY RELIEF VALVE

BSP MEGA SUPERVENTIX MX65F

2 1/2" MEGA SUPERVENTIX™ threaded safety relief valve is designed to protect tanks against accidental overpressure and excessive vacuum. With a smart compact design, the 2 1/2" BSP MEGA SUPERVENTIX™ SRV has a high flow performance. Its halar coated springs offer a better protection against corrosion

SPECIFICATION	
Nominal Diameter	65 mm (2 1/2")
Weight	3 kg
Connection	2 1/2" BSP
Pressure settings	From 0,5 Bar to 7,4 Bar
Vacuum settings	From 50 to 410 mbar
Working Temperature	-40°C +150°C

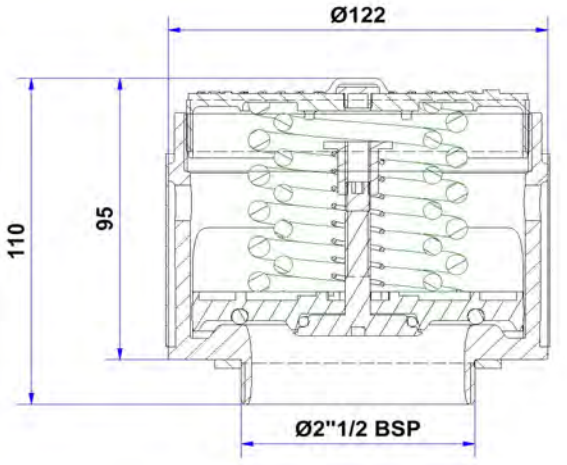
*Temperature range may vary according to the seal material

ACCESSORIES	
Weld-in flange	12 91 14 00 00
Recess flange	12 92 14 00 00
Seal	11 10 01 60 00
Screw kit	12 12 60 91 00
Bursting Disc Adaptator	11 91 15 00 01
Pressure gauge	70 18 06 00 95

CONFORMITY & CERTIFICATION	
Complies with ADR / IMDG / RID / ISO 4126-1	
Notified Body: Lloyd's Reigster Verification - RMR	
STANDARD FEATURES (See Codification for option)	
Made of (material)	316 / 1,4408 stainless steel
Setting Gasket	PTFE
O'rings material	Fep Silicone - others on request

It is the responsibility of the user to ensure that the fluids in contact with the equipment are compatible with the materials listed in the product sheet

DIMENSIONAL DRAWING



Set pressure			Measured flow rate at 120% without gauze		Certified flow rate with the federal register coefficient 90% without gauze	
Bar	kPa	PSIG	Sm³/S	Scfm	Sm³/S	Scfm
Max (±3% SP; 0,1B/1,45PSI)						
2,18	218	31,6	2,56	5425	2,30	4883
3,75	375	54,4	3,92	8306	3,53	7475
4,4	440	63,8	4,48	9492	4,03	8543
7,3	730	105,9	6,99	14810	6,29	13329

Flow measurement conditions: 101.3 kPa at 15°C

Flow reduction with gauze <3%

SAFETY RELIEF VALVE

2 1/2" (ND65) SAFETY RELIEF VALVE

BSP MEGA SUPERVENTIX
MX65F

CODIFICATION

STANDARD PART NUMBER	MX65	F	44	T	21	G	C	I	.
YOUR PART NUMBER	MX65	F	__	T	__	__	__	I	__

BODY CONNECTION

2 1/2" BSP Threaded	F

PRESSURE SETTING - Other on request

No pressure	00
. bar (ex: 4.4bar)	44

SETTING GASKET

FEP	T

VACUUM SETTING - Other on request

No Vacuum	.
0.** bar (ex: 0.21 bar)	21

FLAME PROOF GAUZE

With	G
Without	S

SETTING GASKET

Composite / PTFE	C
Perbunan / PTFE	P

HIGH CORROSION halar coating

Halar coated springs	I

VACUUM TEST BUTTON

With	B
Without	.

SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 43 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND500 MULTIDO COVERLID
HIGH CORROSION Manlid

The PEROLO DN500 HALAR MULTIDO™ Coverlid is designed to be installed on PEROLO neckrings welded on tank containers and road tankers intended for high corrosive products.

Caution: Please check the cargo compatibility with proposed materials

SPECIFICATION	
Nominal diameter	500 mm (20")
Weight	18 Kg
Fastening Point	6 or 8 (others on request)
Working Pressure	4 Bar
Test Pressure	6 Bar
Working Temperature	-40° to 130 °C
Design Temperature	-40° to 200 °C

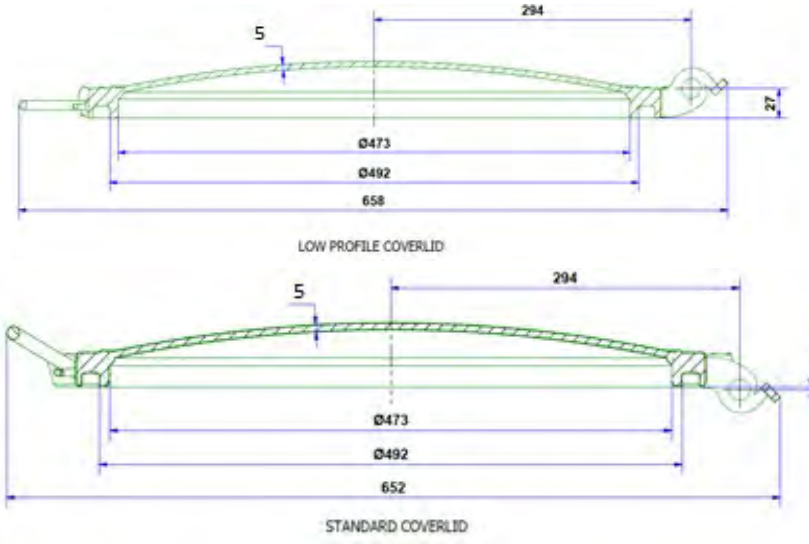
NB: temperature range vary according to o'ring material

CONFORMITY & CERTIFICATION	
ADR / IMDG / RID	
Lloyd's Register	
STANDARD FEATURES	
Made of (material)	316L stainless steel and HALAR TM coated
HALAR Thickness	600µ to 800µ
Opening angle	130° 0/-10°

ACCESSORIES	
Long eye bolt kit (93mm)	11 20 13 90 10
Short eye bolt kit (73mm)	11 20 13 90 11
PTFE/EPDM Seal(14x14)	11 13 75 00 19
Hinge kit (stainless steel)	12 13 68 91 00

CODIFICATION	
ND500, 8 points, low profile coverlid, HALAR coated	11 15 18 60 00
ND500, 8 points, standard coverlid, HALAR coated	11 15 20 60 00
ND500, 6 points, low profile coverlid, HALAR coated	11 20 32 10 00

DIMENSIONAL DRAWING



HALAR coated thickness 600µ to 800µ

MANLIDS

3" NEATFLOW MKIV
HIGH CORROSION ND80 Footvalve
NX34

The Perolo 3" NEATFLOW™ MKIV HALAR external spring footvalve is designed to be used as a bottom discharge for corrosive chemicals carried in tank containers and IBC's. Wetted parts in Hastelloy C276, Hastelloy C22, Titanium or PVDF. Available in 45° (standard version) or 30°.

Caution: Please check the cargo compatibility with proposed materials

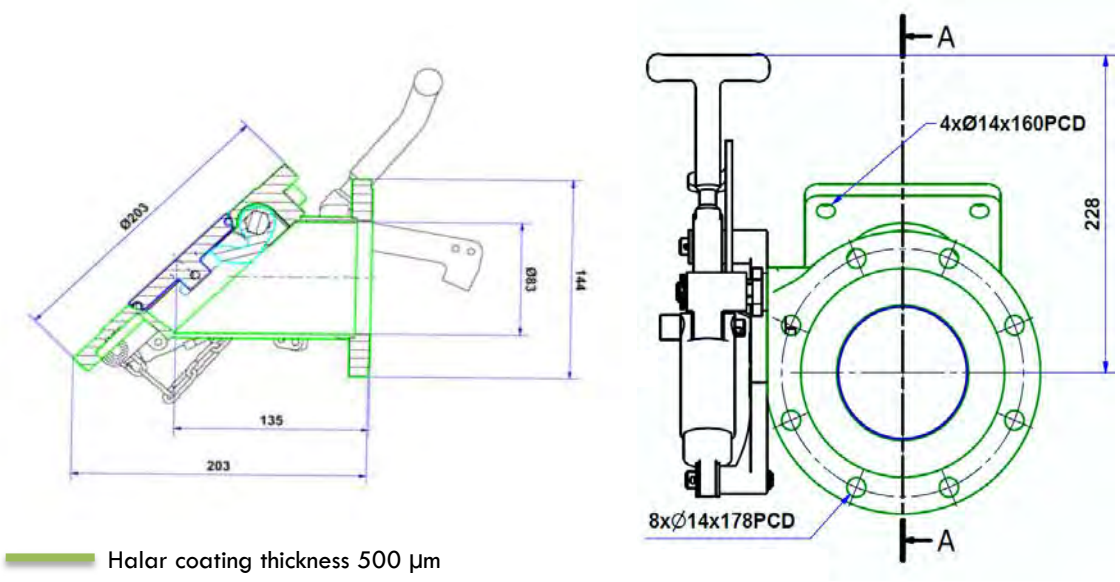
SPECIFICATION	
Nominal diameter	80 mm (3")
Weight	9,5 kg
Inlet flange drilling	8 x Ø14 x 178 PCD
Outlet flange drilling	4 x Ø18 x 160 PCD
Working Pressure	4 Bar
Test Pressure	6 Bar
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

NB: temperature range vary according to o'ring material & material internal parts

CONFORMITY & CERTIFICATION	
ADR / IMDG / RID / EN14433	
Bureau Veritas	
STANDARD FEATURES (See Codification for option)	
Made of (material)	Body: 316L/1.4409 & HALAR coated Poppet, pusher & spindle: Hastelloy, Titanium, FDA coated 316, PVDF (see p2)
Angle	45° or 30° (MKII version)
Body length	135mm from center to oulet
O'rings material	FEP/Silicone
Seals material	PTFE
Operating	Side operated handle or Pneumatic option
Footvalve spring	Spring outside the tank allows easier cleaning

ACCESSORIES	
45° Weld in flange	11 90 59 00 00
30° Weld in flange	11 91 16 00 00
Seal	12 91 20 00 00
Screw kit	11 11 27 93 00

DIMENSIONAL DRAWING (45° version)



Halar coating thickness 500 µm

High corrosion material, please see p2 for options

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 22 18 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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VALVES SOLUTIONS FOR YOUR

3" NEATFLOW MKIV
HIGH CORROSION ND80 Footvalve
NX34

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	NX3	4	A	X	1	R	H	WR	P
YOUR PART NUMBER	NX3	__	A	__	1	R	__	__	__

OUTLET ANGLE

45°	4

30°: contact our sales department

INLET FLANGE

8 x Ø14 x 178 PCD	A

MATERIAL

316L stainless steel	X

OUTLET FLANGE - other on request

Square flange 4 x Ø18 x 160 PCD	1

HIGH CORROSION - halar coated body

Halar coated body	R

HIGH CORROSION - Poppet, spindle & pusher material (45° model)

Hast. C276 poppet, Hast. C276 spindle, Hast. C276 pusher WT -40°C + 150°C	H
FDA coated Hast. C276 poppet, Hast. C276 spindle, Hast. C276 pusher WT -20°C + 150°C	K
Titanium poppet, Titanium spindle, Titanium pusher WT -40°C + 150°C	T
Hast. C22 poppet, Hast. C22 spindle, Hast. C22 pusher WT -40°C + 150°C	Z
FDA coated Hast. C22 poppet, Hast. C22 spindle, Hast. C22 pusher WT -20°C + 150°C	F
PVDF poppet, Hast. C276 spindle, Hast. C22 pusher WT -20°C + 80°C	P

for 30° model options, please contact our sales department

REMOTE CONTROL

Without	00
With remote control lever	W0
With remote control lever & cable	WR

HANDLE

Standard Handle	P
Without	Z
L Handle	L
Short Handle	C
Left L Handle	G

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 22 18 20 00 A
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3" 90° CHEMFLOW
HIGH CORROSION

PEROLO offers a CHEMFLOW 90° footvalves dedicated to the bottom discharge of corrosive chemicals.
These CHEMFLOW are composed with HALAR coated body and C276 hastelloy or Titanium poppet, push rod and spindle insuring an extreme resistance to corrosion. Available in pneumatic or manual version.

Caution: Please check the cargo compatibility with proposed materials

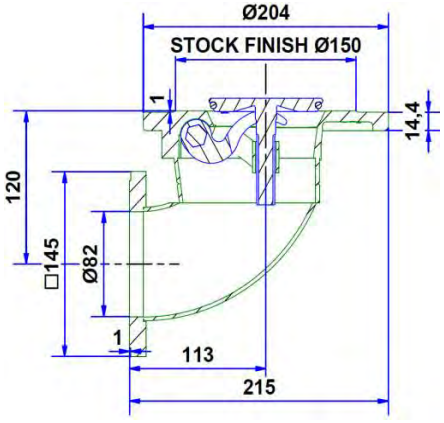
SPECIFICATION	
Nominal diameter	83 mm (3")
Weight	9.1 Kg
Inlet flange drilling	8 x Ø14 x 178 PCD
Outlet flange drilling	4 x Ø18 x 160 PCD
Working Pressure	4 bar
Test Pressure	6 Bar
Pneumatic Air Pressure	5 Bar
Air Inlet	Ø6
Working Temperature	-40° to +150°C
Design Temperature	-40° to +190°C

NB: temperature range vary according to o'ring material

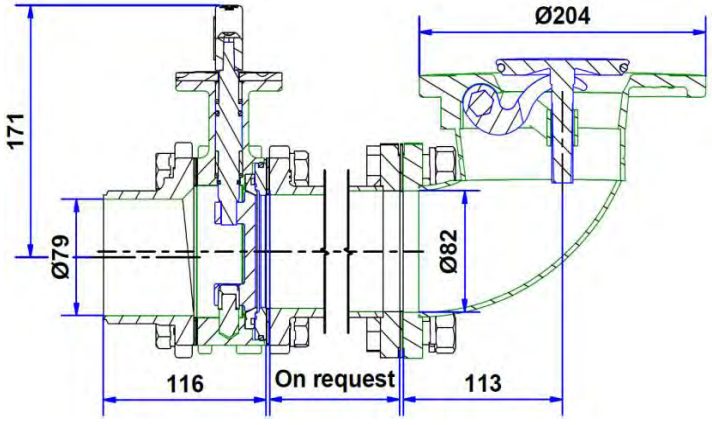
CONFORMITY & CERTIFICATION	
ADR / IMDG / RID / EN14433	
Bureau Veritas	
STANDARD FEATURES (See Codification for option)	
Made of (material)	Body: 316/1.4408/CF8M & HALAR coated or 904L without halar coating Poppet, pushrod & spindle: Hastelloy C276 or Titanium
Outlet Angle	90°
Operation	St. St. spring return actuator or hand operate
Footvalve spring	Spring outside the tank allows easier cleaning

ACCESSORIES	
Weld-in flange	11 90 59 00 00
Inlet seal	12 91 20 00 00
Outlet seal	12 91 39 00 00
Stud kit	17 11 03 92 00

DIMENSIONAL DRAWING



Footvalve



Complete bottom discharge

HALAR coated thickness 500µ

Hastelloy or Titanium, please see p2

BOTTOM DISCHARGE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 17 11 63 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
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VALVES SOLUTIONS FOR YOUR

3" 90° CHEMFLOW
HIGH CORROSION

BOTTOM DISCHARGE

CODIFICATION

STANDARD PART NUMBER	CF3	9	1	R	C	R	.
YOUR PART NUMBER	CF3	9	1	__	C	__	__

OUTLET ANGLE

90°	9

OUTLET FLANGE - other on request

Square outlet flange 4 x18x160PCD	1

POPPET

Flat Poppet	R

OPERATING MECHANISM

Side Actuator	C
Mechanical (without Actuator)	M

HIGH CORROSION - poppet, spindle & pusher

Hastelloy C22 poppet, Hastelloy C276 spindle & pusher	R
Titanium poppet, spindle & pusher	T

HIGH CORROSION - halar coated body

Halar coated body in 316L	.
904L body (no halar coating)	U

mechanical version

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 17 11 63 20 00 D
It is the customer's responsability to check the compatibility between valves material and transported product.
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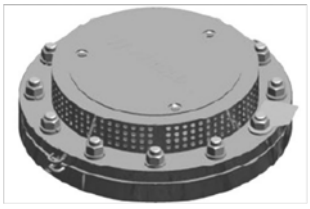
FUNCTION

Perolo offer a wide range of accessories for your tank or tanker:

- Steam Valves
- Manlid gaskets in different sizes and materials
- Flat gaskets
- Caps & Nozzles & Spigot flanges
- Remote control for ball valve or butterfly valve
- Flanges
- Bursting discs
- Pressure gauges
- Thermometers

¾" Steam Valve SVM	Manlid Gaskets	Flat Gaskets	Caps
Outlet Nozzles & Spigots	Remote Control	Flanges	Bursting Disc
 10" Bursting Disc Assembly	Pressure Gauges	Thermometer	

3/4" (ND20) STEAM VALVE

Steam Valve
SVM

3/4" steam valve - protects steam circuits by opening when excess pressure is reached - compact, individually tested and requiring minimal maintenance, these valves have been specially designed for the steam heating circuits used in tank containers.

SPECIFICATION	
Nominal diameter	20 mm (3/4")
Weight	120 g
Connection	3/4" BSP
Pressure settings	from 1 to 10 Bar
Working Temperature	-40° to +180°C

Temperature range may vary according to the seal material

STANDARD FEATURES	
Made of (material)	316L / 1.4409 stainless steel
O'rings material	Viton

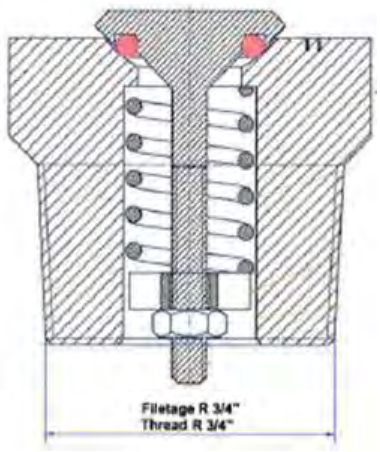
CODIFICATION

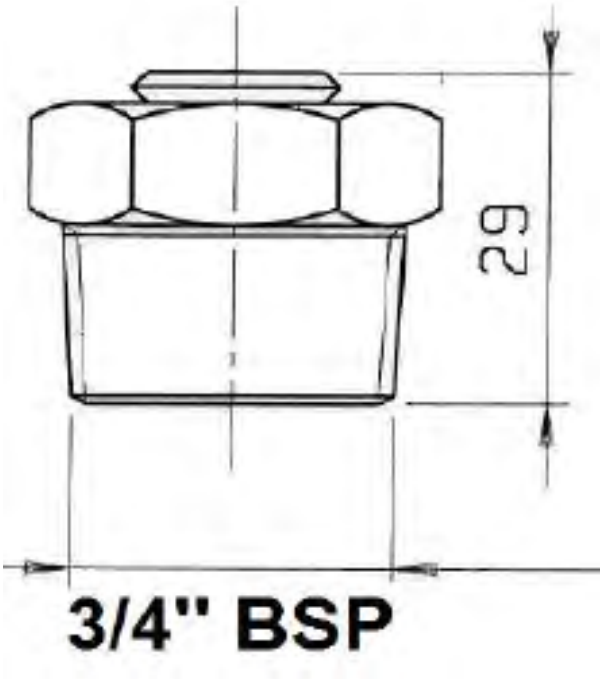
STANDARD PART NUMBER	SV	44	M
YOUR PART NUMBER	SV	__	M

PRESSURE SETTING - Other on request	
. bar (ex: 4,4 bar)	44

THREAD TYPE	
3/4" BSP Male Thread	M

DIMENSIONAL DRAWING





SAFETY RELIEF VALVE

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA 11 10 12 20 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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Manlid Gasket
Accessories
Gasket

Perolo offers a comprehensive range of manlid gaskets to suit all applications and a large range of temperatures. Many dimensions are available on request as well as special gaskets to suit specific manlids.

Best sellers marked in gold

ND 150		
CHEMICAL	Black Nitrile - 14 x 10 - 70Sh	11 13 91 10 02
ND 300		
FOOD	White Nitrile - 14 x 14 - 60Sh	11 13 75 00 26
FOOD	FEP/ Silicone - 14 x 14 - FDA certified	70 10 17 00 15
FOOD	FEP/ Viton - 14 x 14 - FDA certified	70 10 17 00 22
BASIC	Braided PTFE - 15 x 10	11 13 75 00 17
BASIC	Braided PTFE - 15 x 14	11 13 75 00 12
BASIC	Braided PTFE - 14 x 14	11 13 75 00 41
BASIC	Neoprene - 14 x 14	11 13 75 00 35
CHEMICAL	Black Nitrile - 14 x 14	11 13 75 00 11
CHEMICAL	Nitrile (inner)/ PTFE (envelope) - 14 x 14	11 13 75 00 13
CHEMICAL	EPDM (inner)/ PTFE (envelope) - 14 x 14	11 13 75 00 20
CHEMICAL	EPDM (inner)/ PTFE (envelope) - 14 x 12	11 13 75 00 42
HIGH TEMPERATURE	PTFE Graphite Added - 14 x 14	11 13 75 00 30
ND 400		
CHEMICAL	EPDM (inner)/ PTFE (envelope) - 12 x 12	11 13 75 00 52
CHEMICAL	Black Nitrile - 10 x 10 - 70Sh	11 13 75 00 68
ND 450		
POWDER	White Nitrile - 10 x 14 - 50Sh	13 15 01 00 04
POWDER	White EPDM - 10 x 14	13 15 24 00 04
POWDER	White EPDM - 10 x 14 -FDA certified	13 15 24 00 04 A
ND460		
CHEMICAL	Rubber - 15 x 15 - 40/45Sh	11 13 75 00 16
CHEMICAL	Black Nitrile - 14 x 14	11 13 75 00 24
FOOD	White Nitrile - 14 x 14 - 70Sh	11 18 84 00 01
CHEMICAL	EPDM (inner)/ PTFE (envelope) - 14 x 14	11 13 75 00 44
CHEMICAL	EPDM (inner)/ PTFE (envelope) - 15 x 15	11 13 75 00 50
FOOD	FEP/ Silicone - 14 x 14	70 10 17 00 18
FOOD	FEP/ Silicone - 14 x 14 - FDA certified	70 10 17 00 18 A
FOOD	FEP/ Viton - 14 x 14	70 10 17 00 25
FOOD	FEP/ Viton - 14 x 14	70 10 17 00 25

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA MANLID GASKET B
It is the customer's responsability to check the compatibility between valves material and transported product.
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ND 500		
FOOD	White Nitrile - 15 x 10	11 13 75 00 01
FOOD	White Nitrile - 14 x 14 - 70Sh	11 13 75 00 18
FOOD	White Nitrile - 14 x 14 - 50Sh	11 13 75 00 39
FOOD	White Nitrile - 14 x 10 - 70Sh	11 13 75 00 58
FOOD	White EPDM - 14 x 14 - 40Sh	11 13 75 00 75
FOOD	White EPDM - 14 x 14 - 60/70Sh	11 13 75 00 77
HIGH CORROSION	Virgin PTFE - 14.5 x 11.7	11 13 75 00 03
HIGH CORROSION	Virgin PTFE - 12 x 4	11 13 75 00 43
BASIC	Braided PTFE - 14 x 14	11 13 75 00 15
BASIC	Braided PTFE - 14 x 12	11 13 75 00 36
CHEMICAL	Viton - 14 x 10 - 80Sh	11 13 75 00 47
CHEMICAL	Viton - 14 x 14 - 70Sh	11 13 75 00 21
CHEMICAL	Viton - 14 x 14 - 50Sh	11 13 75 00 49
CHEMICAL	Black Nitrile - 14 x 14 - 70Sh	11 13 75 00 23
CHEMICAL	Neoprene - 14 x 14 - 60Sh	11 13 75 00 29
CHEMICAL	EPDM - 14 x 14 - 70Sh	11 13 75 00 31
CHEMICAL	EPDM - 14 x 14 - 50Sh	11 13 75 00 79
CHEMICAL	Nitrile (inner) /PTFE (envelope) - 14 x 10	11 13 75 00 04
CHEMICAL	Nitrile (inner) /PTFE (envelope) - 14 x 14	11 13 75 00 14
CHEMICAL	EPDM (inner)/ PTFE (envelope) - 14 x 14	11 13 75 00 19
CHEMICAL	EPDM (inner)/ PTFE (envelope) - 14 x 12	11 13 75 00 40
CHEMICAL	EPDM (inner)/ PTFE (envelope) - 14 x 10	11 13 75 00 76
CHEMICAL	EPDM / PTFE Lined - 13 x 13	11 13 75 00 61
HIGH TEMPERATURE	PTFE Graphite Added - 14 x 14	11 13 75 00 22
FOOD	PFA (inner) / Silicone (encapsulated) - 14 x 12	11 13 75 00 72
BASIC	Acrylic - 14 x 10	11 13 75 00 56
BASIC	Acrylic / PTFE - 14 x 14	11 13 75 00 37
FOOD	Silicone (inner)/ FEP (encaps) - 14x14	70 10 17 00 19
FOOD	Silicone (inner)/ FEP (encaps) - 14x14 - FDA certified	70 10 17 00 19 A
FOOD	Silicone (inner)/ FEP (encapsulated) - 14 x 14	70 10 17 00 64
ND 600		
CHEMICAL	Neoprene - 14 x 14 - 60Sh	11 13 75 00 28
HIGH TEMPERATURE	PTFE Graphite Added - 14 x 14	11 13 75 00 55
ND 900		
CHEMICAL	Neoprene - 14 x 10	17 13 01 00 01
ND 500 RAIL MANLID		
CHEMICAL	Black nitrile - 25 x 22 - 70Sh	19 13 01 00 05
CHEMICAL	Viton - 25 x 22 - 60Sh	19 13 01 00 06
CHEMICAL	Hypalon - 25 x 22 - 70Sh	19 13 01 00 07
POWDER	Parablond - 25 x 22 - 70Sh	19 13 20 00 01
CHEMICAL	Neoprene - 25 x 22 - 60Sh	19 13 20 00 07

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA MANLID GASKET B
It is the customer's responsability to check the compatibility between valves material and transported product.
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Flange Gasket
Accessories
Gasket

PEROLO flat CNAF/PTFE Gaskets are to be placed in between two flanges and ensure leak tightness.

SPECIFICATION

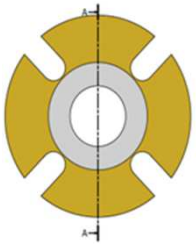
- CNAF Gasket with PTFE product contact area
- U shaped PTFE profile
- Drillings depending on application
- Dimensions and shape depending on application

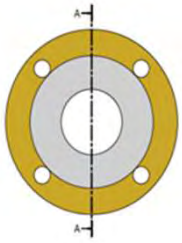
OPTIONS

- Other materials (Fluolion, Gylon,…)
- Full PTFE seals
- Other drillings
- Other diameters

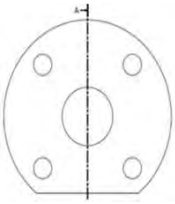
If you require a gasket for a specific valve, please refer to the installation sheet of that valve

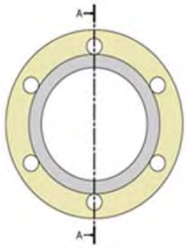
ND 40 1 1/2" Gaskets

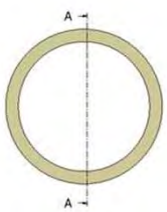
Part number: 12 91 90 00 00	for ATCO ND40 ball valve & ATCO ND32 flanged ball valve
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **45 mm**
	External diameter: **150 mm**
	Drilling: **4 slots x 19 xmin101.4 PCD**

Part number: 11 92 58 00 00	for provision ATCO ND40 ball valve & ATCO ND32 ball valve
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **45 mm**
	External diameter: **127 mm**
	Drilling: **4 x Ø12 x 103.5 PCD**

ND 50 2" & ND 65 2 1/2" Gaskets

Part number: 11 90 29 00 00	for ND50 ATCO ball valves and TANKFLY butterfly valves
	Shape: Round cut
	Material: CNAF/PTFE
	Thickness: 3 mm
	Internal diameter: **50 mm**
	External diameter: **165 mm**
	Drilling: **4 x Ø18 x 125 PCD**

Part number: 11 10 01 60 00	for ND65 MEGASUPERVENTIX Safety relief valve
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **76 mm**
	External diameter: **127 mm**
	Drilling: **6 x Ø11 x 105 PCD**

Part number: 12 91 31 00 00	for ND65 MEGASUPERVENTIX Safety relief valve: setting gasket
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **75,5 mm**
	External diameter: **90 mm**
	Drilling: **no drilling**

ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA FLANGE GASKET A
It is the customer's responsability to check the compatibility between valves material and transported product.
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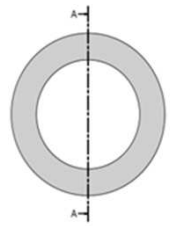
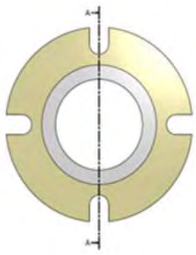
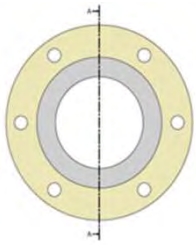
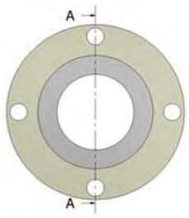
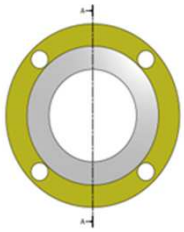
ND 80 3" Gaskets

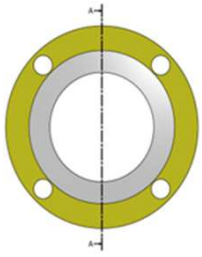
Part number: 17 12 21 10 00	for ND80 CHEMFLY light bottom discharge
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **78 mm**
	External diameter: **116 mm**
	Drilling: **no drilling**

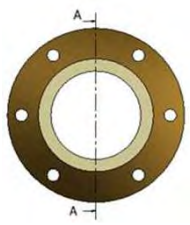
Part number: 12 92 04 00 00	for ND80 MEGASUPERVENTIX Safety relief valve
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **82 mm**
	External diameter: **176 mm**
	Drilling: **4 slots x Ø18 xmin145 PCD**

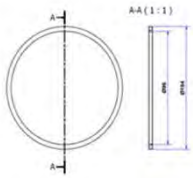
Part number: 11 90 92 00 00	for 3" butterfly valve inlet flange
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **94 mm**
	External diameter: **200 mm**
	Drilling: **6 x Ø14 x 168 PCD**

Part number: 11 98 83 00 00	for 3" top provision
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **80 mm**
	External diameter: **185 mm**
	Drilling: **4 x Ø18 x 160 PCD**

Part number: 12 91 40 00 00	for 3" butterfly valve & 3" ball valves inlet and outlet flanges
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **94 mm**
	External diameter: **185 mm**
	Drilling: **4 x Ø18 x 160 PCD**

Part number: 11 93 98 00 00	for 3" butterfly valve & 3" ball valves inlet and outlet flanges
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 3.5 mm
	Internal diameter: **100 mm**
	External diameter: **185 mm**
	Drilling: **4 x Ø18 x 160 PCD**

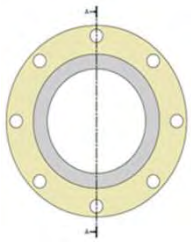
Part number: 11 97 21 00 00	for 3" butterfly valve & 3" ball valves inlet and outlet flanges
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **101.6 mm**
	External diameter: **200 mm**
	Drilling: **6 x Ø14 x 168 PCD**

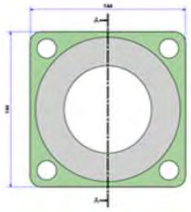
Part number: 12 92 08 00 03	to be placed between syphon tube guide & ND80 weld-in flange
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 3 mm
	Internal diameter: **95 mm**
	External diameter: **105 mm**
	Drilling: **no drilling**

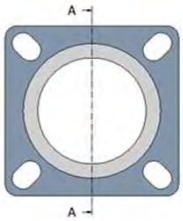
ACCESSORIES

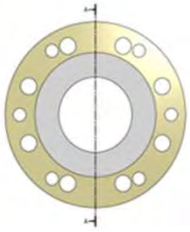
Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA FLANGE GASKET A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

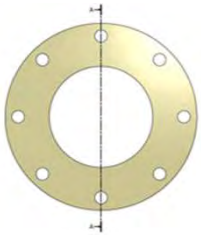
ND 80 3" & ND 100 4" Gaskets for bottom discharge

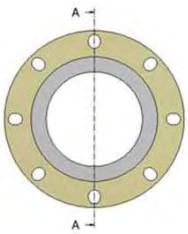
Part number: 12 91 20 00 00	3" bottom discharge: fit between 30° or 45 or 90° footvalve and tank
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **114 mm**
	External diameter: **200 mm**
	Drilling: **8 x Ø14 x 178 PCD**

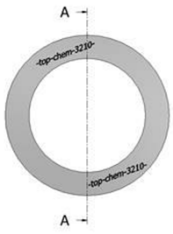
Part number: 12 91 39 00 00	3" bottom discharge: fit between footvalve and butterfly valve
	Shape: Square
	Material: CNAF/PTFE
	Thickness: 2.5 mm
	Internal diameter: **81mm**
	External dia: **144 x 144 mm**
	Drilling: **4 x Ø18 x 160 PCD**

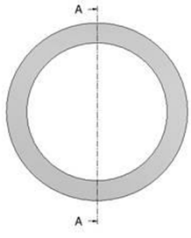
Part number: 12 91 42 00 00	3 "bottom discharge: fit between butterfly valve and outlet nozzle
	Shape: Square
	Material: CNAF/PTFE
	Thickness: 2 mm
	Internal diameter: **90 mm**
	External dia: **144 x 144 mm**
	Drilling: **4 slots x Ø18 x145-162PCD**

Part number: 17 11 23 00 00	for 3" CHIM footvalve
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 2,5 mm
	Internal diameter: **81 mm**
	External diameter: **200 mm**
	Drilling: **8xØ18x160PCD & 6xØ14x168PCD**

Part number: 17 11 03 90 02	for 4" CHEMFLOW bottom discharge
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 3 mm
	Internal diameter: **114 mm**
	External diameter: **213 mm**
	Drilling: **8 x Ø14 x 184.2 PCD**

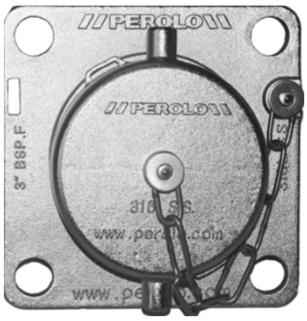
Part number: 11 11 59 90 01	for 4" NEATFLOW bottom discharge
	Shape: Round
	Material: CNAF/PTFE
	Thickness: 3 mm
	Internal diameter: **114 mm**
	External diameter: **213 mm**
	Drilling: **8 slots x Ø14 x 184.2-190PCD**

Part number: 19 90 05 30 01	for 4" rail tank car bottom discharge with 2 cables
	Shape: Round
	Material: TOPCHEM 3210
	Thickness: 2 mm
	Internal diameter: **115 mm**
	External diameter: **162 mm**
	Drilling: **no drilling**

Part number: 19 90 05 40 01	for rail tank car aeration valve
	Shape: Round
	Material: PTFE
	Thickness: 2 mm
	Internal diameter: **169 mm**
	External diameter: **218 mm**
	Drilling: **no drilling**

Outlet Nozzle - Blank caps

Accessories
Nozzles & caps



Perolo offers a comprehensive range of outlet nozzles and blank caps to suit all applications. Many dimensions are available on request as well as special gaskets options as camlock, halar coating or cap gasket material.

Best sellers marked in gold

316L/1.4409 - Stainless steel caps		Part Number
BSP	3/4" BSP with gasket and chain	70 12 16 37 10
	1" BSP with gasket and chain	70 12 16 37 08
	1 1/2" BSP with gasket and chain	11 14 01 30 00
	2" BSP with gasket and chain	11 12 03 40 00
	3" BSP with gasket and chain L200mm	12 85 59 91 00
	3" BSP with gasket and chain L400mm	12 85 59 92 00
NPT	3/4" NPT with gasket and chain	17 11 75 40 00
	1 1/2" NPT with gasket and chain	12 96 03 00 00
	2" NPT with gasket and chain	17 11 75 30 00
CAMLOCK	ND20 Camlock with nitrile gasket	70 12 52 01 19
	ND32 Camlock with nitrile gasket	70 12 52 01 20
	ND50 Camlock with nitrile gasket	70 12 52 01 04
	ND80 Camlock with nitrile gasket	70 12 52 01 06

316L/1.4404 - Stainless steel spigot flanges (eccentric)		Part Number
BSP	2" round BSP spigot flange	17 11 22 31 00
	3" square BSP spigot flange	12 85 69 60 00
	3" square BSP spigot fl. TWINFLOW	11 11 83 20 01
	3" square BSP halar coated spigot flange	12 86 06 00 01
NPT	2" round NPT spigot flange	17 11 22 32 00
CAMLOCK	2" round Camlock spigot flange	11 93 72 00 01
	3" square Camlock spigot flange	11 96 14 10 00

316L/1.4404 - Stainless steel outlet nozzle (cap&spigot)		Part Number
BSP	2" round BSP outlet nozzle	17 11 22 30 00
	3" square BSP outlet nozzle	12 85 69 00 00
	3" square BSP outlet nozzle TWINFLOW with PTFE inlet gasket	11 11 83 93 00
	3" square BSP halar coated outlet nozzle	12 86 06 00 00
NPT	2" round NPT outlet nozzle	17 11 22 40 00
CAMLOCK	2" round Camlock outlet nozzle	11 93 72 00 00
	3" square Camlock outlet nozzle	11 96 14 00 00

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Remote Control
Accessories
RMT

The remote control is designed to open from ground level valves which are installed on top of a container. It guarantees the safety of operators that no longer need to go on top of a tank container. It's available in left or right lever.

Caution: valve, valve lever and bolts are not included

SPECIFICATION

Made of (material)	Stainless steel
Fixation on tank frame	By bolting or welding
Operating	Manually with remote handle
Working temperature	-40° to +190°C

	Cable Ø	Rcable mini	Working load	Maximum Load	Valve lever center distance
RMT3	6 mm	125 mm	340N	500N	66 mm
RMT5	8 mm	125 mm	600N	900N	69 mm

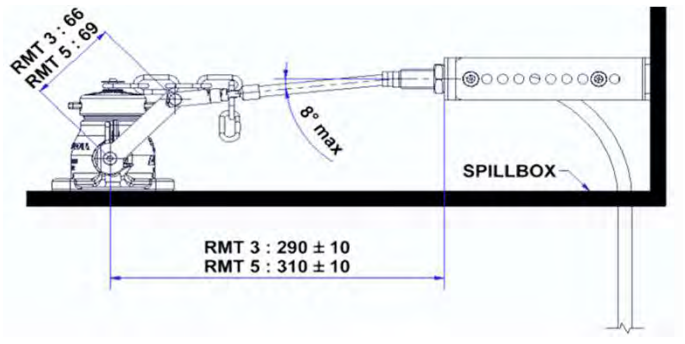
CODIFICATION REMOTE LEVERS

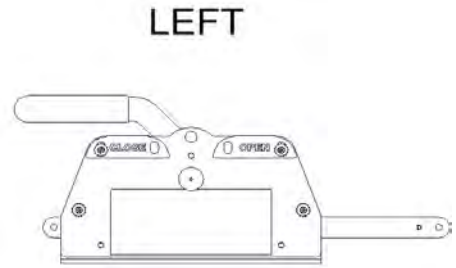
AD3 ND32 ball valve	11 14 15 60 00
AF3 ND32 ball valve	11 14 02 21 00
AT4 ND32 dual ball valve	12 96 07 16 00
TT5 ND50 butterfly valve	11 17 20 11 00
TF5 ND50 butterfly valve	
ATC5 ND50 dual ball valve	11 24 06 00 01
ADN4 ND40 ball valve	11 14 99 30 01
ADC5 ND50 ball valve	

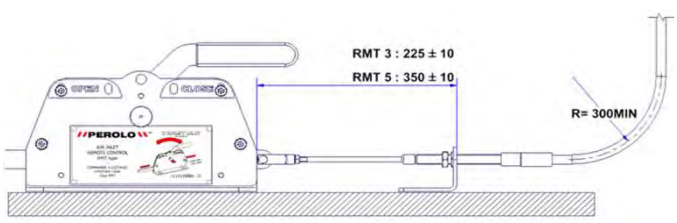
AFN4 ND40 ball valve	11 14 55 91 00
AD5X7 ND50 ball valve	
AD5X16 ND50 ball valve	
AFN5 ND50 ball valve	
ALN5 ND50 ball valve	
AKN5 ND50 ball valve	
AHN5 ND80-ND50 ball valve	on request

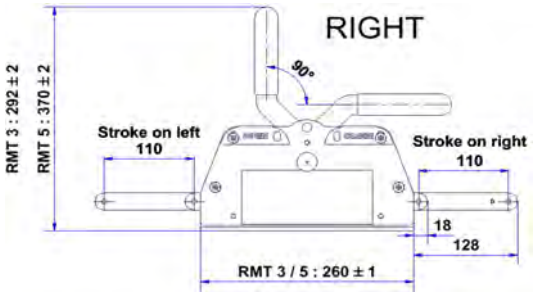
please see p2 for pictures of valves

DIMENSIONAL DRAWING









ACCESSORIES

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Remote Control
Accessories
RMT

CODIFICATION

STANDARD PART NUMBER	RMT	3	3	.

YOUR PART NUMBER RMT _ _ _

MODEL	
ND32 & ND40 ball valve	3
ND50 or ND80 butterfly valve & ND50 ball valve	5

CABLE LENGTH - other on request	
3 meters	3
4 meters	4
5 meters	5
7 meters	7
8 meters	8

POSITION OF THE LEVER	
Left lever	.
Right lever	R

Please contact us for specific demands

RMT3

AD3 Ball Valve

AF3 Ball Valve

AT4 Ball Valve

AFN4 Ball Valve

ADN4 Ball Valve

RMT5

ADC5 Ball Valve

AD5X7 Ball Valve

AD5X16 Ball Valve

TF5 Butterfly Valve

TT5 Butterfly Valve

ATC5 Ball Valve

AFN5 Ball Valve

AHN5 Ball Valve

ALN5 Ball Valve

AKN5 Ball Valve

ACCESSORIES

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INSTALLATION - RMT3

Our recommendations before installing the remote control RMT3

-To open the valve : the cable must push the valve lever
-To close the valve : the cable must pull the valve lever
-For RIGHT RMT :
The side of the RMT with the label must be facing the same direction as the valve lever. (*Fig. 2*)
-For LEFT RMT :
The side of the RMT with the label must be facing the opposite direction as the valve lever. (*Fig. 1*)
-Check p7&8 of this document to reverse the opening sense of the remote control.

Figure 1 : Left version

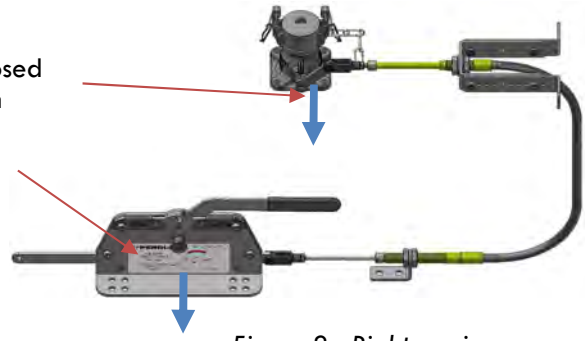

Figure 2 : Right version

The remote control should be installed in 2 areas :
-**the valve area (1)**
-**the remote control area (2)**
following the **right** (**R**) or the **left** (**L**) version,detail of *Fig. 3*.

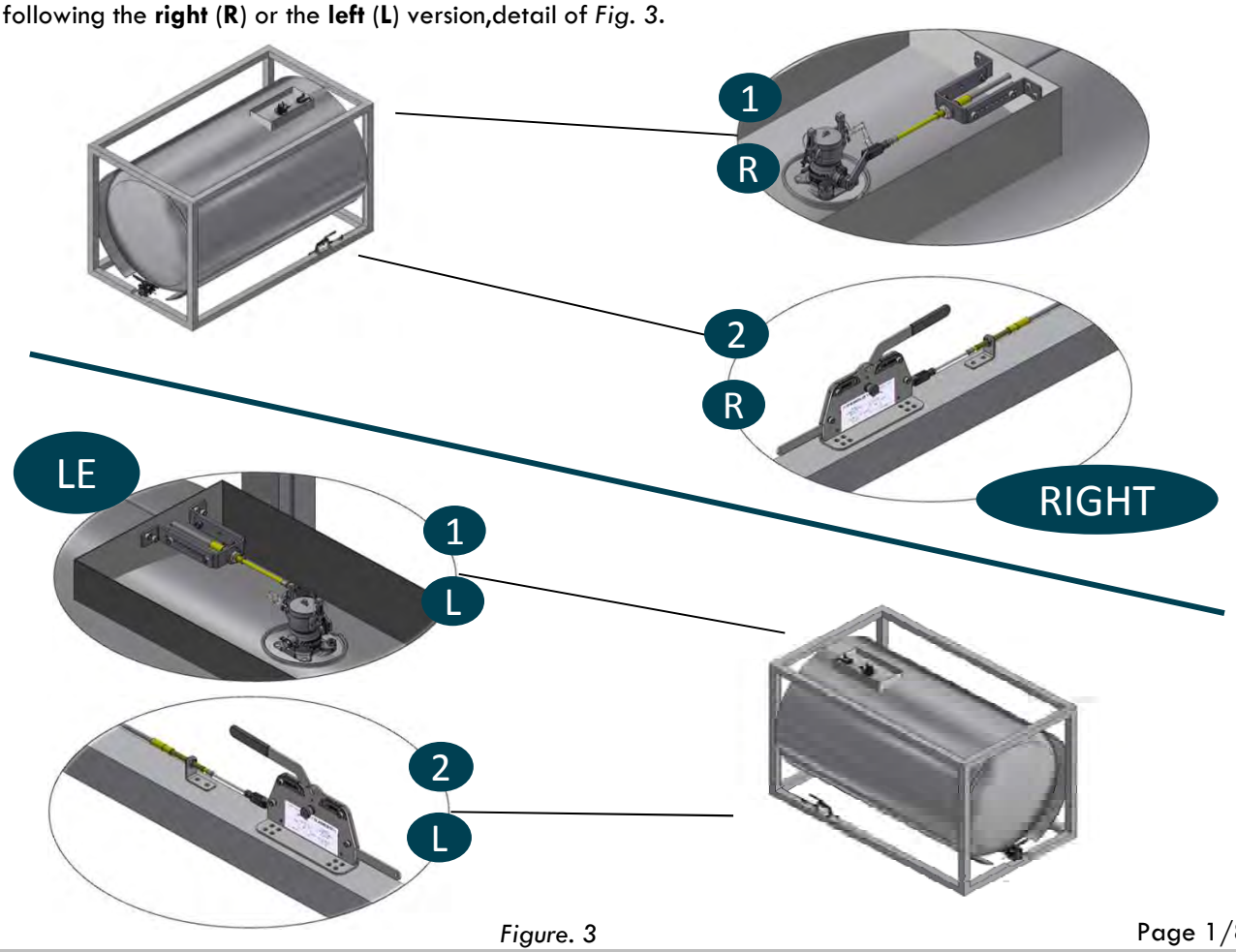

Figure. 3

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1. Preparing the bracket of the valve

-Take the end of the cable with the articulation and shorter thread for the valve side Fig.4.

End with longer thread
"Remote control side"

End with articulation and
shorter thread
'Valve side"

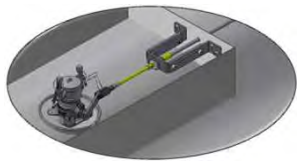

Figure 4

-Remove all the components at the end of the cable and place the bracket (B1) Fig.5 according to the dimension against the valve *Fig. 5*. The bracket can be set to attain the below conditions.

The value X :
-Closed position of the valve
-Cable pulled back in its sleeve
-Must be in the middle of the threaded section
The value Y :
-Must be aligned with the valve lever
The value Z :
-Must be in the middle height of the upper and lower position of the lever axle and the yoke (2)

For welding assembly :
-Mark all the holes (red and blue holes) and drill only the red hole for the cable (Ø24)

For bolted assembly :
-Mark and drill all the holes (red (Ø24) and blue (Ø10,5) holes) for the cable and the bolts

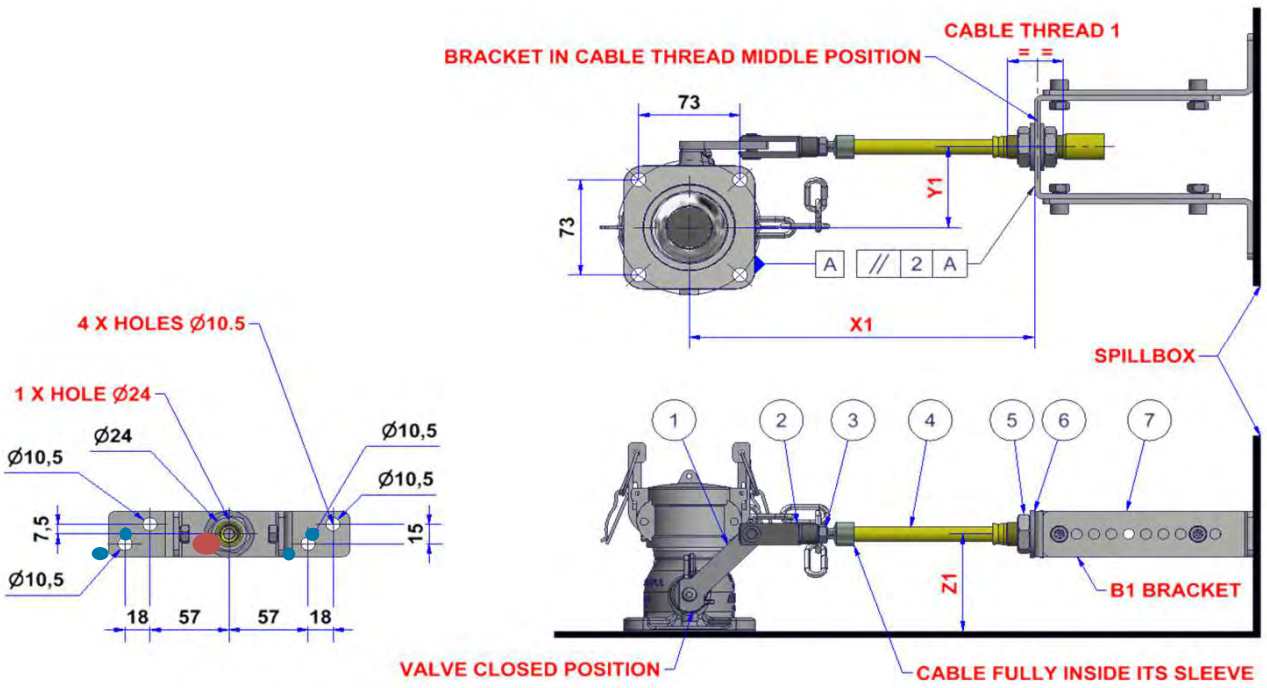

Figure 5

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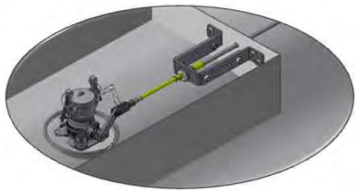

2. Fix and assemble valve on top of container

For welding assembly :
-With the marking, place and weld the bracket legs to the spillbox according the *value X, Y, Z.*
For bolted assembly :
-Assemble the bracket with a suitable nut, washers and bolts (M8) (not included in the RMT kit) *Fig.6.*

-Ensure the inner cable is pulled back so that the valve remains in closed position Fig. *7.*
-Assemble the components according the *Fig. 7* and Lock the yoke (3) to the valve lever (1) with its axis (2) *Fig. 7.*
-Keep a gap between both nuts (7A & 7B) and the bracket (8) before the next step.

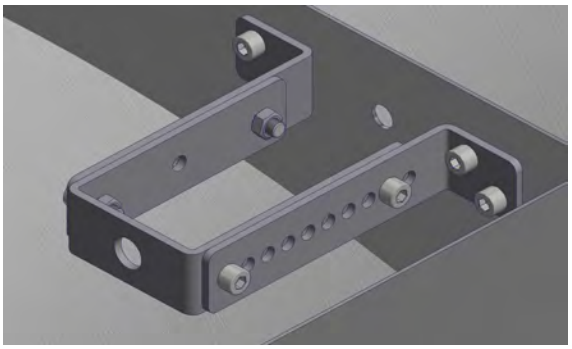

Figure 6

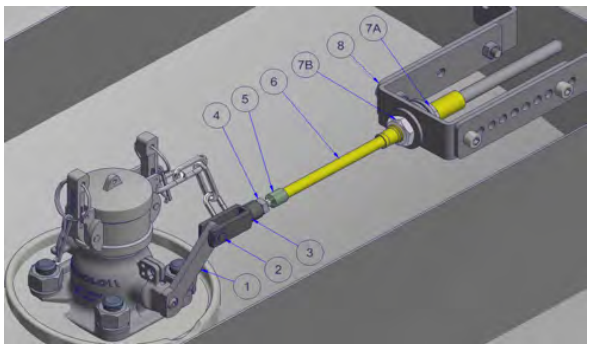

Figure 7

3. Preparing the remote control area on the tank frame
-Take the second end of cable, it must be the one with longer thread *Fig. 8.*

End with longer thread
"Remote control side"

End with articulation and
shorter thread
'Valve side"

Figure 8

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-Remove if necessary the components at the end before inserting the cable trough the sleeve (if there is one) along the frame work until the suitable position of the remote control *Fig. 9*. Check the distance of the cable before.

Caution : the number of 90° angles of the cable must be between 3 and 5.
The minimum radius at the angle is : 100mm

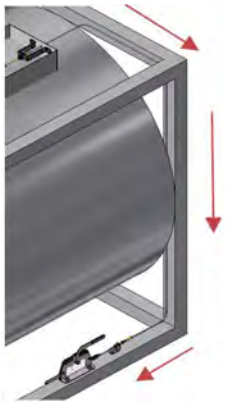


Figure 9

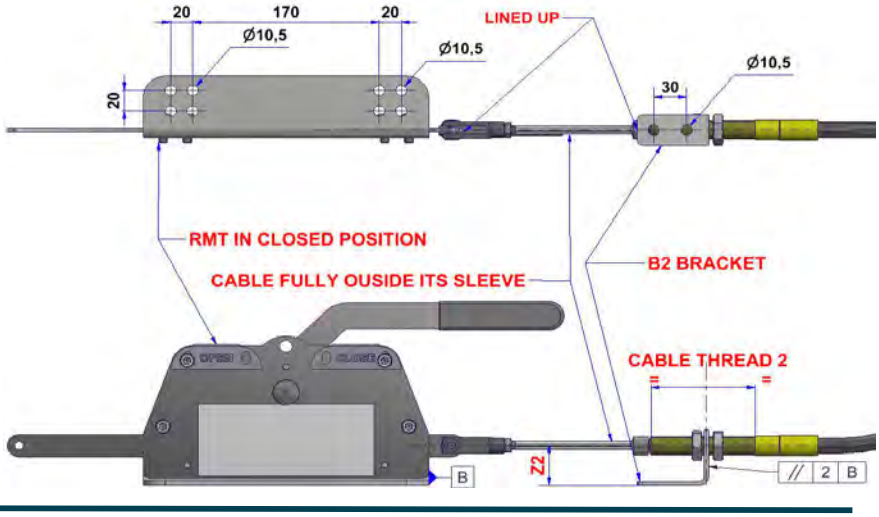
-Pull the cable until a mechanical stop. Place the bracket (B2) and the remote control along the horizontal frame work.. The bracket must be placed in the middle of the threaded section of the cable *Fig. 10 & 11*.
-Once the bracket is approximately placed, the RMT must be aligned with the bracket according the Fig. 10 & 11.

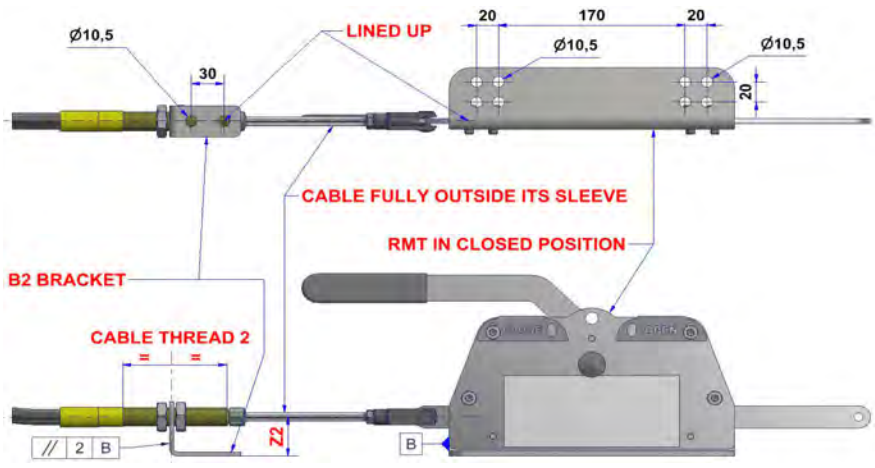
-Below the following assembly instructon:
For welding assembly :
-Mark all the holes and weld the bracket and the remote control to the frametank according the L or R version.
For bolted assembly :
-Mark all the holes and drill a tap M8 of the bracket and the remote control to the frametank according the Fig. 10 & 11.

Bottom view

Right version
Figure 10

Top view





Bottom view

Left version
Figure 11

Top view

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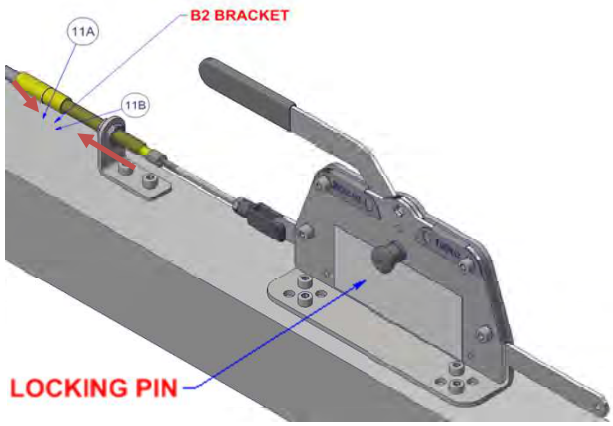
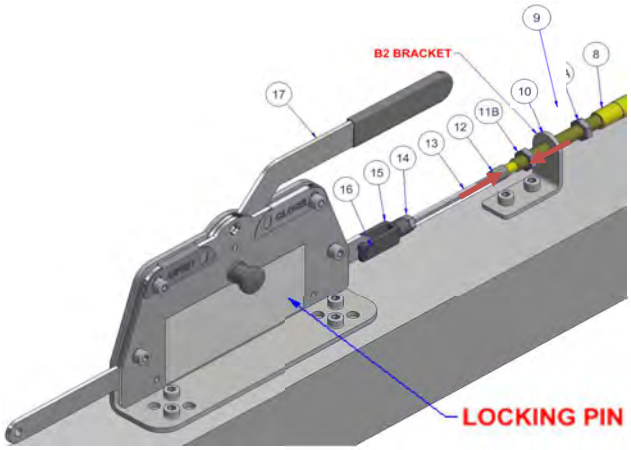
4. Fix and assemble the remote control area
-Once the bracket and the remote control mechanism are aligned correctly and in accordance with the dimensions and in the closed position *Fig.11*, follow the assembly instructon:
For welding assembly :
-Weld the bracket (B2) and the remote control to the framework according to the marking Fig. 11.
For bolted assembly :
-Bolt the bracket (B2) and the remote control to the framework according to the marking with a screw and washers (M8) (non-included in the RMT kit) Fig.10 & 11.

-Assemble and lock the yoke (15) to the valve lever with its axis (16) according the *Fig. 12 & 13* and keep a gap between the nuts (11A and 11B) and the bracket (10), detail on *Fig. 12 & 13* .

-Screw the nut (11A) until resistance and screw again (about 3 times) to have a tension in the cable.
-Check the condition below for a proper assembly :
In order to open the index, you must apply light prerssure to the handle of the remote control while pulling the locking pin towards you at the same time. *
* If not, screw again the nut (11A) until the previous condition.
-Lastly, screw the nut (11B) to tighten the cable to the bracket (10).



Left version - Figure 12



Right version - Figure 13

5. Setting the RMT in the valve area on top of container
-Go back on top of the tank to set the mechanism and ensure the RMT is in closed position and locked.
-Screw the nut (7A) until resistance and screw again (about 3 times) to have a tension in the cable.
-Check the condition for a proper assembly :
The lever must be in mechanical stop with the valve stop Fig.15.*
*If not, screw again the nut (7A) until the previous condition.
-Lastly, screw the nut (7B) to tighten the cable to the bracket (8).

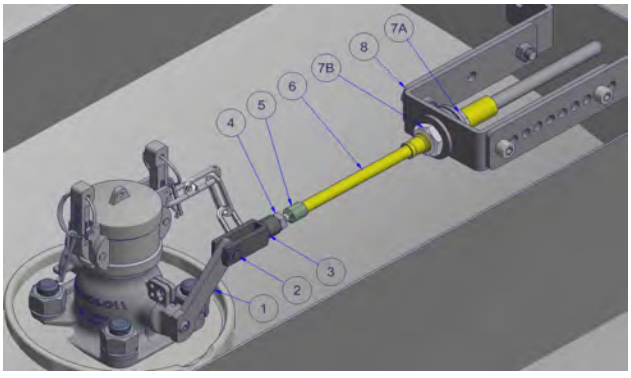


Figure 14

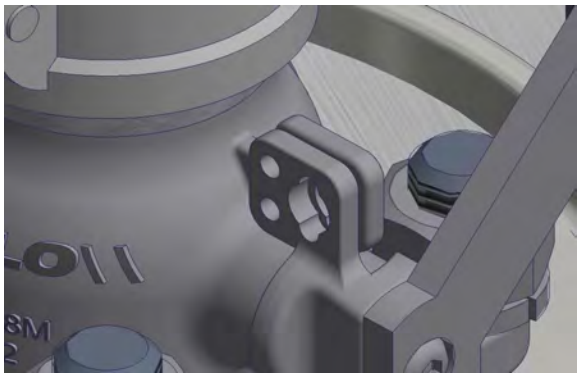


Figure 15

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6. Check the open and close system.
To be sure the system is correctly installed, the below conditions must be fulfilled.
1 -when the RMT is in closed position, the lever of the valve must be in mechanical stop by the valve body.
2 -when the RMT is in open position and locked, the valve is correctly opened even if it is not fully open with a tolerance from 2° to 5°.

Figure 16
RMT IN CLOSED POSITION

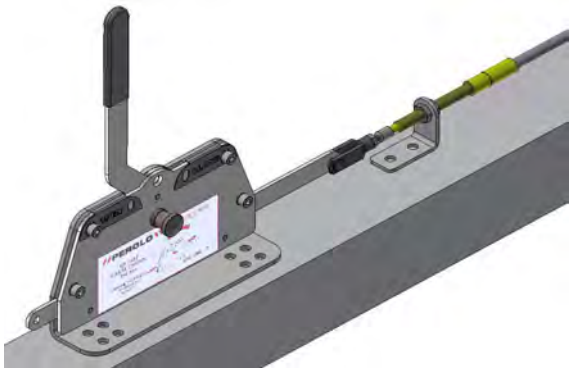


Figure 17
RMT IN OPEN POSITION

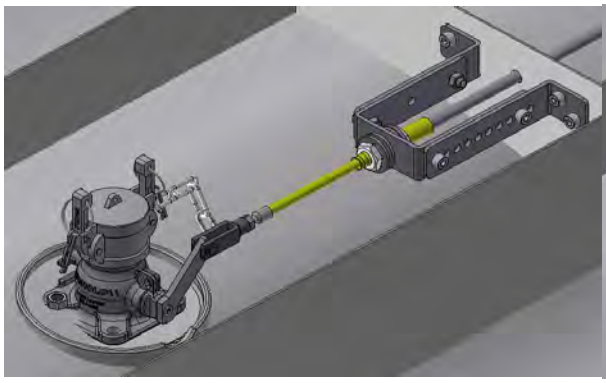


Figure 18
VALVE IN CLOSED POSITION

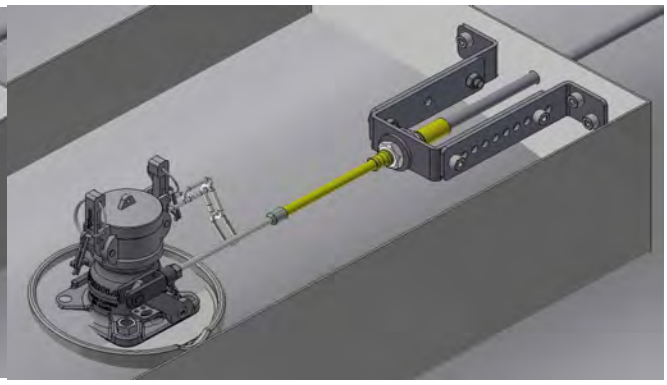


Figure 19
VALVE IN OPEN POSITION

The basic installation conditions :

FROM CLOSED TO OPEN POSITION (from Fig. 16 & 18 to Fig 17 & 19)
-To open : You must apply light pressure to the handle of the remote control while pulling the locking pin towards you at the same time and pull the handle until the mechanical stop.
-During the operation: The cable must pull the valve lever for opening.
-The open position must be locked automatically by the locking pin and the valve must be open with a tolerance of 2° to 5°.

FROM OPEN TO CLOSED POSITION (from Fig. 17 & 19 to Flg. 16 & 18)
-To close : You must pull the locking pin towards you, and push handle until a mechanical stop.
-During the operation: The cable must pull the valve lever for closing
-The closed position must be locked automatically by the locking pin system.

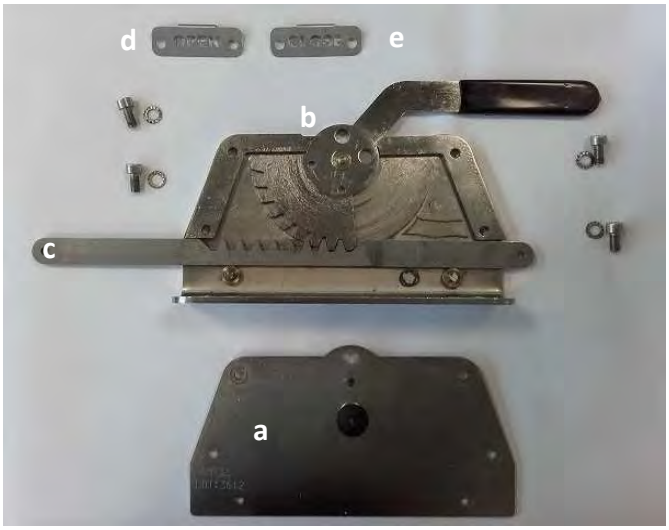
CONTROL TORQUE :
Working torque : **22N.m**
Maximum working torque **: 33N.m**

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 14 15 30 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 57 42 67 00 - Email : sales@perolo.com

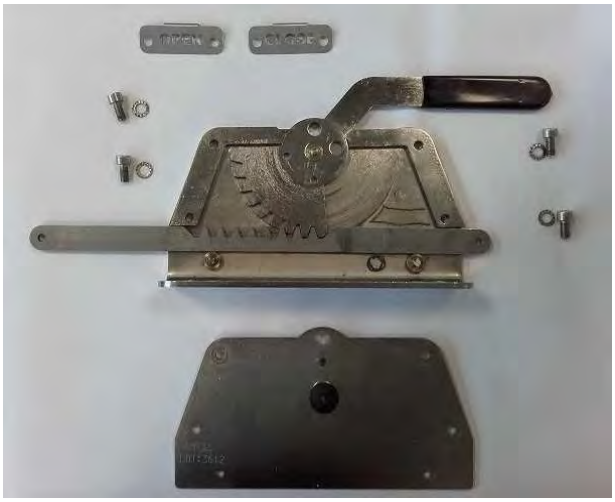
REVERSE OPENING SENSE OF REMOTE CONTROL BOX

Passage de la position fermeture à droite à la position fermeture à gauche.
Transition from the closed position on the right to the closed position on the left.

a : Flasque avant / Front frame
b : Levier à dents / Toothed lever
c : Crémaillére / Rack
d : Plaque open / Open plate
e : Plaque close / Close plate



1 - Enlever les vis et les plaques.
1 - Remove the screws and plates.
2 - Enlever la flasque avant.
2 - Remove the front frame.



3 - Pousser la crémaillére sur la droite.
3 - Push the rack on the right.
4 - Retourner le levier et le placer à gauche, la premiére dent dans le premier creux.
4 - Flip the lever and place it on the left, the first tooth in the first hollow.

ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 14 15 30 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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4 - Repositionner la flasque avant

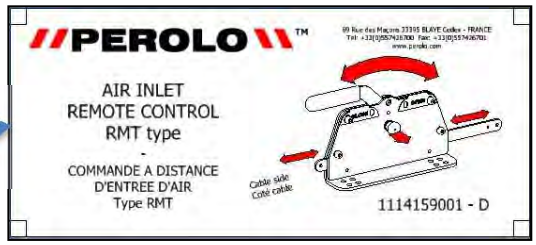
4 - Reposition the front frame

5 - Placer les plaques : close à droite (**e**).
open à gauche (**d**).

5 - Place the plates: close to the right (**e**).
open to the left (**d**).

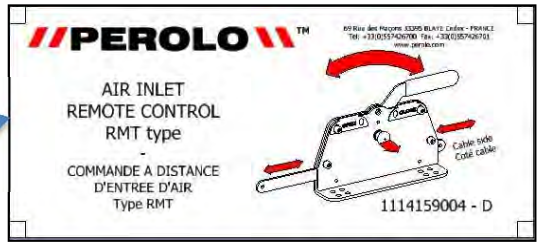
6 - Remettre en place les 4 vis.

6 - Replace the 4 screws



Autocollant levier à gauche : référence 11 14 15 90 01

Left lever sticker: reference 11 14 15 90 01



Autocollant levier à droite : référence 11 14 15 90 04

Right lever sticker: reference 11 14 15 90 04

Due to our policy of continuous product improvement, specifications may change without notice - Installation sheet PWA 11 14 15 30 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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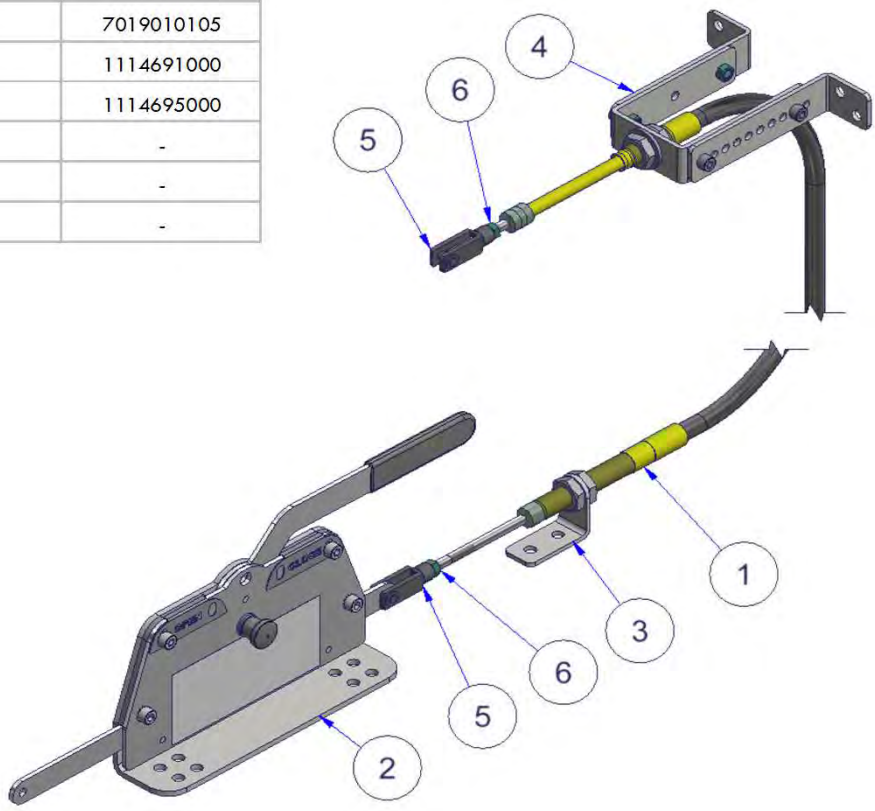
MAINTENANCE

ITEM	QTY	DESIGNATION	PART N°
RMT3 - CABLE Ø6			
1	1	Cable 3m	7019090014
		Cable 4m	7019090015
		Cable 5m	7019090016
		Cable 7m	7019090017
		Cable 8m	7019090018
2	2	Right control box	1114151000
		Left control box	1114158000
3	1	Adaptator bracket	-
4	2	Fix bracket	-
5	2	Yoke	-
RMT5 - CABLE Ø8			
1	1	Cable 1m	7019010103
		Cable 3m	7019010101
		Cable 4m	7019010102
		Cable 5m	7019010104
		Cable 5,6m	7019090028
		Cable 7m	7019010105
2	2	Right control box	1114691000
		Left control box	1114695000
3	1	Adaptator bracket	-
4	2	Fix bracket	-
5	2	Yoke	-



ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 14 15 40 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

TROUBLE SHOOTING

1 - Mechanical slack

After repeated service it is possible that the cable may stretch and require adjustment to achieve full closure.
Adjustment procedure: Readjust the nut against the bracket (at the control box area or/and on the top valve area) in order to take up the mechanical slack, if necessary, check details on the instruction on the "INSTALLATION-PWA-RMT".
Screw the other nut to tighten the cable to the bracket.
Lastly, test the system following the below handling conditions.

2 - Handling difficulty

Because of the wrong handlings or an adverse condition, the remote control may need additional lubrication.
We recommend to apply lubricant to the mechanism (control box) and open/close 10 times.

3 - System Seized

A - If it is completely seized, the fault might come from the remote control area.
Check the remote control area, inpect in case of a foreign body in the mechanism, if so remove it.
If the problem is comes from the control box itself, you can disassemble the control box, clean all components, apply lubricant and reassemble. Use the " installation document" P.7 if needed. .

B - If there is blockage with a slack cable, it may come from the top valve area.
Check the top valve area, if there is a foreign body in the mechanism remove it.

C - Check all the system by disconnecting the cable at the remote control box and operate the control lever independently. If the lever operates freely repeat the same procedure at the valve. If the valve operates freely (when operated independantly from the cable) then the fault points to the cable. Check routing of the cable to ensure soft curves in line with previous set up instructions. Reroute as necessary. In the event that any individual components cannot be repaired then call the PEROLO after-sales service to change the component.

The conditions of manipulation:

FROM CLOSED TO OPEN POSITION
-To open : Apply light pressure to the handle of the remote control while pulling the locking pin towards you at the same time and pull the handle until the mechanical stop.
-During the manipulation : The cable must pull the valve lever for opening.
-The open position must be locked automatically by the locking pin and the valve must be open with a tolerance of 2° to 5°.

FROM OPEN TO CLOSED POSITION
-To close : You must pull the locking pin towards you, and push handle until a mechanical stop.
-During the manipulation : The cable must pull the valve lever for closing
-The closed position must be locked automatically by the locking pin system.

Due to our policy of continuous product improvement, specifications may change without notice - Maintenance sheet PWA 11 14 15 40 00 A
It is the customer's responsability to check the compatibility between valves material and transported product.
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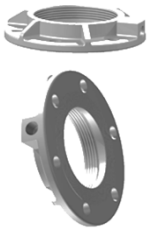
Adaptor Flanges
Accessories
Flanges

Perolo 316L stainless steel adaptor flanges are designed to be fitted on tank containers, IBCs, road tankers, rail tank cars and storage tanks in order to fit valves and accessories for tank filling and discharge purposes.

SPECIFICATION	OPTIONS
• 316L stainless steel construction	• Other diameterd
• Radiused angles	• Other material (carbon steel, 304L, 904L…)
• Serrated gasket contact area	• Other drilling

If you require a gasket or bolting kit for a specific valve, please refer to the installation sheet of that valve

Part Number: 11 91 15 00 00	with 1/4" BSP gauge connection & gauge protection
	Internal - external diameter
	2 1/2" BSP thread. -127mm
	Thickness
	23 mm
	Drilling
	6 x Ø11 x 105 PCD
Seal (CNAF/PTFE)	11 10 01 60 00

Part Number: 11 91 15 00 01	with 1/4" BSP gauge connection
	Internal - external diameter
	2 1/2" BSP thread. - 127mm
	Thickness
	23 mm
	Drilling
	6 x Ø11 x 105 PCD
Seal (CNAF/PTFE)	11 10 01 60 00

Part Number: 12 96 07 17 00	1 1/2" BSP PN10 flange with internal thread
	Internal - external diameter
	1 1/2" BSP thread. -160mm
	Thickness
	9 mm
	Drilling
	4 x Ø18 x 125 PCD
Seal (CNAF/PTFE)	12 91 90 00 00

Part Number: 12 92 25 00 00	2" BSP ND50 PN10 flange with internal thread
	Internal - external diameter
	2" BSP threaded - 165mm
	Thickness
	16 mm
	Drilling
	4 x Ø18 x 125 PCD
Seal (CNAF/PTFE)	11 90 29 00 00

Part Number: 11 93 04 00 00	1 1/2" BSP PN10/PN16 flange with internal thread
	Internal - external diameter
	1 1/2" BSP thread. -150mm
	Thickness
	12 mm
	Drilling
	4 x Ø18 x 110 PCD
Seal (CNAF/PTFE)	12 91 90 00 00

Part Number: 11 93 11 00 00	ND50 Adaptor flange
	Internal - external diameter
	50 mm - 165 mm
	Thickness
	8,5 mm
	Drilling
	4 x Ø18 x 125 PCD
Seal (CNAF/PTFE)	11 90 29 00 00

ACCESSORIES

Adaptor Flanges
Accessories
Flanges

Part Number: 11 95 25 00 00	ND50 Adaptor flange
	Internal - external diameter **50 mm - 165 mm** Thickness **20 mm** Drilling **4 x Ø18 x 125 PCD** **4 x M16 x 125 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00

Part Number: 12 91 41 00 03	ND80 Adaptor flange for clamped 3" TANKFLY
	Internal - external diameter **90 mm - 200 mm** Thickness 21 mm Drilling **4 x Ø18 x 160 PCD** **4 x M16 x 160 PCD**
Seal (CNAF/PTFE)	12 91 40 00 00

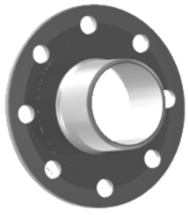
Part Number: 11 94 50 00 00	ND80 Adaptor flange
	Internal - external diameter **80 mm - 200 mm** Thickness **21 mm** Drilling **4 x M16 x 152,4 PCD** **4 x Ø18 x 160 PCD**
Seal (CNAF/PTFE)	12 92 04 00 00 (ND80)
Seal (CNAF/PTFE)	12 91 40 00 00 (ND80)

Part Number: 11 97 96 00 00	ND50 to ND80 2" siphon tube guide
	Internal - external diameter **54 mm - 200 mm** Thickness - Tube length **28 mm - 50 mm** Drilling **4 x M16 x 120,6 PCD** **4 x Ø18,5 x 152,4 PCD**
Seal (CNAF/PTFE)	11 90 29 00 60 (ND50)
Seal (CNAF/PTFE)	12 91 40 00 30 (ND80)

Part Number: 11 96 41 30 00	2" siphon tube guide
	Internal - external diameter **50 mm - 200 mm** Thickness - Tube length **31 mm - 41 mm** Drilling **4 x Ø18 x 160 PCD** **4 x M16 x 125 PCD**
Seal (CNAF/PTFE)	11 90 29 00 60 (ND50)
Seal (CNAF/PTFE)	11 93 98 00 00 (ND80)

Part Number: 11 12 67 20 00	2" siphon tube guide
	Internal - external diameter **52 mm - 104,5 mm** Thickness - Tube length **17 mm - 46 mm** Drilling **no drilling**
Seal (CNAF/PTFE)	12 92 08 00 03

Part Number: 12 92 18 10 00	3" siphon tube guide
	Internal - external diameter **80 mm - 200 mm** Thickness - Tube length **10 mm - 70 mm** Drilling **4 x Ø18 x 160 PCD** **6 x Ø14 x 168 PCD**
Seal (CNAF/PTFE)	17 11 23 00 00

Part Number: 11 93 82 00 00	3" siphon tube guide
	Internal - external diameter **80 mm - 200 mm** Thickness - Tube length **10 mm - 70 mm** Drilling **8 x Ø18 x 160 PCD**
Seal (CNAF/PTFE)	11 92 85 00 00

ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA ADAPTOR FLANGE A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

Part Number: 11 95 57 00 00	ND50 to ND32 Adaptor Flange
	Internal - external diameter **32-50 mm - 165 mm** Thickness **20 mm** Drilling **4 x M10 x 103,5 PCD** **4 x Ø18 x 125 PCD**
Seal (CNAF/PTFE)	11 92 58 00 00 (ND32)
Seal (CNAF/PTFE)	11 90 29 00 00 (ND50)

Part Number: 11 95 95 00 00	ISO PN16 ND80 to ND40
	Internal - external diameter **40-80 mm - 200 mm** Thickness **23 mm** Drilling **4 x M16 x 110 PCD** **4 x Ø18 x 160 PCD**
Seal (CNAF/PTFE)	12 91 90 00 00 (ND40)
Seal (CNAF/PTFE)	12 91 40 00 00 (ND80)

Part Number: 11 92 86 00 00	ISO PN16 ND80 to ND50
	Internal - external diameter **50-88 mm - 200 mm** Thickness **21 mm** Drilling **4 x M16 x 125 PCD** **4 x Ø18 x 160 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00 (ND50)
Seal (CNAF/PTFE)	12 91 40 00 00 (ND80)

Part Number: 11 92 95 00 00	ISO PN16 ND80 to ND62 Adaptor flange
	Internal - external diameter **62 mm - 200 mm** Thickness **21 mm** Drilling **4 x M16 x 125 PCD** **4 x Ø18 x 160 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00 (ND50)
Seal (CNAF/PTFE)	12 91 40 00 00 (ND80)

Blind Flanges

Accessories
Flanges

Perolo 316L stainless steel blind flanges are designed to close valves or flanges welded on tank containers, IBCs, road tankers, rail tank cars and storage tanks.

SPECIFICATION

- 316L stainless steel construction
- Radiused angles
- Serrated gasket contact area

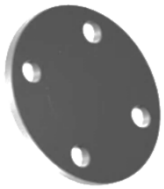
OPTIONS

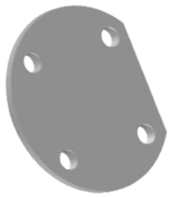
- Other diameterd
- Other material (carbon steel, 304L, 904L…)
- Other drilling

If you require a gasket or bolting kit for a specific valve, please refer to the installation sheet of that valve

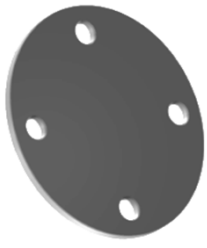
Part Number:	for ND65 BSP Safety relief valve provision
11 90 01 00 00	
	Tank Type - Thickness **T1 to T20 - 8,5mm EN14025** External diameter **127 mm** Drilling **6 x Ø11 x 105 PCD**
Seal (CNAF/PTFE)	11 10 01 60 00

11 92 57 00 00 11 95 86 00 00	for ND32 ball valve provision
	Tank Type - Thickness **T1 to T14 - 8,5mm** **T1 to T22 - 13,5mm EN14025** External diameter **127 mm** Drilling **4 x Ø12 x 103.5 PCD**
Seal (CNAF/PTFE)	11 92 58 00 00

11 92 99 00 00 11 95 87 00 00	For ND40" flanged ball valve ISO PN16 ND40
	Tank Type - Thickness T1 to T14 - 8,5mm **T1 to T22 - 13,5mm EN14025** External diameter **150 mm** Drilling **4 x Ø18 x 110 PCD**
Seal (CNAF/PTFE)	12 91 90 00 00

17 12 12 16 01 11 95 88 00 00	For 2" ball or butterfly valve ISO PN16 ND50
	Tank Type - Thickness **T1 to T14 - 8,5mm** **T1 to T22 - 13,5mm EN14025** External diameter **165 mm** Drilling **4 x Ø18 x 125 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00

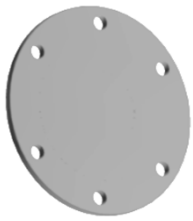
11 90 48 00 00 11 95 89 00 00	for ND65 or ND80 flanged Safety relief valve provision ISO PN16 ND65
	Tank Type - Thickness T1 to T14 - 8,5mm **T1 to T22 - 13,5mm EN14025** External diameter **190 mm** Drilling **4 x Ø18 x 145 PCD**
Seal (CNAF/PTFE)	12 92 04 00 00

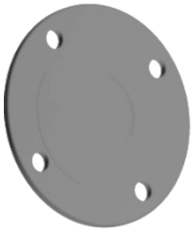
11 90 48 00 00 11 95 89 00 00	for ND65 or ND80 flanged Safety relief valve provision ISO PN16 ND65
	Tank Type - Thickness **T1 to T14 - 8,5mm** **T1 to T22 - 13,5mm EN14025** External diameter **190 mm** Drilling **4 x Ø18 x 145 PCD**
Seal (CNAF/PTFE)	12 92 04 00 00

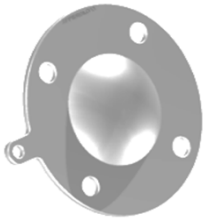
ACCESSORIES

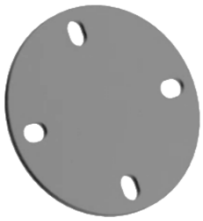
Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA BLIND FLANGE A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

Blind Flanges
Accessories
Flanges

11 90 13 00 00 11 95 94 00 00	For 3" ball or butterfly valve
	Tank Type - Thickness T1 to T14 - 8,5mm **T1 to T22 - 13,5mm EN14025** External diameter **200 mm** Drilling **6 x Ø14 x 168 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

12 85 49 00 01 11 95 90 00 00	For 3" ball or butterfly valve ISO PN16 ND80
	Tank Type - Thickness **T1 to T14 - 8,5mm** **T1 to T22 - 13,5mm EN14025** External diameter **200 mm** Drilling **4 x Ø18 x 160 PCD**
Seal (CNAF/PTFE)	12 91 40 00 00

Part Number: 11 90 41 00 00	Dished for ND80 Butterfly Valve
	Tank Type - Thickness **T1 to T14 - 8,5mm** External diameter **200 mm** Drilling **4 x Ø18 x 160 PCD**
Seal (CNAF/PTFE)	12 91 40 00 00

Part Number: 11 93 09 00 00	Slotted holes, for 3" ball or butterfly valve
	Tank Type - Thickness **T1 to T14 - 8,5mm** External diameter **200 mm** Drilling **4 x Ø18 x 160 PCD**
Seal (CNAF/PTFE)	12 91 40 00 00

ISO
RAIL
ROAD
IBC

ACCESSORIES

ND32 Tank top weld-in flanges

Accessories
Weld-in Flanges - Top Tank

Perolo 316L stainless steel weld-in flanges are designed to be welded on top of tank containers, IBCs, road tankers, rail tank cars and storage tanks in order to fit valves and accessories for tank filling and discharge purposes.

SPECIFICATION

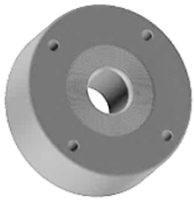
- 316L stainless steel construction
- Nominal diameter 1 1/4 "
- Radiused angles
- Serrated gasket contact area

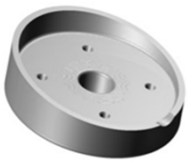
OPTIONS

- Other diameter weld-in flanges
- Other material (carbon steel, 304L, 904L…)
- Other drilling

If you require a gasket or bolting kit for a specific valve, please refer to the installation sheet of that valve

Part Number: 11 90 20 00 00	for ATCO ND32 ball valve
	Tank type T1 to T19
	Internal - external diameter **32 mm - 127 mm**
	Thickness **22 mm**
	Drilling **4 x M10 x 103.5 PCD**
Seal (CNAF/PTFE)	12 91 90 00 00

Part Number: 11 90 38 00 00	for ATCO ND32 ball valve
	Tank type T1 to T19
	Internal - external diameter **32 mm - 127 mm**
	Thickness **50 mm**
	Drilling **4 x M10 x 103.5 PCD**
Seal (CNAF/PTFE)	12 91 90 00 00

Part Number: 11 91 12 00 00	for ATCO ND32 ball valve - Recessed
	Tank type T1 to T19
	Internal - external diameter **32mm - 166 mm**
	Thickness **min 23.5 mm, max 46 mm**
	Drilling **4 x M10 x 103.5 PCD**
Seal (CNAF/PTFE)	12 91 90 00 00

ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA ND32 WELD-IN FLANGE A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND40 & ND50 Tank top weld-in flanges

Accessories

Weld-in Flanges - Top Tank

Perolo 316L stainless steel weld-in flanges are designed to be welded on top of tank containers, IBCs, road tankers, rail tank cars and storage tanks in order to fit valves and accessories for tank filling and discharge purposes.

SPECIFICATION

- 316L stainless steel construction
- Nominal diameter 1 1/2 " & 2"
- Radiused angles
- Serrated gasket contact area

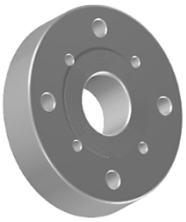
OPTIONS

- Other diameter weld-in flanges
- Other material (carbon steel, 304L, 904L…)
- Other drilling

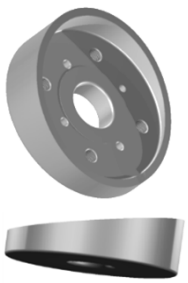
If you require a gasket or bolting kit for a specific valve, please refer to the installation sheet of that valve

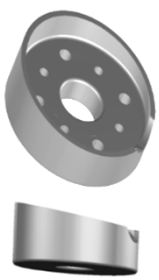
Part Number: 11 90 25 00 00	for Tankfly ND50 butterfly valve - PN16 ND50
	Tank type T1 to T19
	Internal - external diameter **50 mm - 165 mm**
	Thickness **26,5 mm**
	Drilling **4 x M16 x 125 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00

Part Number: 11 92 51 00 00	PN16 ND40
	Tank type T1 to T19
	Internal - external diameter **49 mm - 150 mm**
	Thickness **30 mm**
	Drilling **4 x M16 x 110 PCD**
Seal (CNAF/PTFE)	on request

Part Number: 11 91 19 00 00	PN16 ND50 and other
	Tank type T1 to T19
	Internal - external diameter **50 mm - 165 mm**
	Thickness **50 mm**
	Drilling **4 x M10 x 100 PCD and 4 x M16 x 125 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00

Part Number: 11 91 20 00 00	PN16 ND50 and other
	Tank type T1 to T19
	Internal - external diameter **50 mm - 165 mm**
	Thickness **59 mm**
	Drilling **4 x M10 x 100 PCD and 4 x M16 x 125 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00

Part Number: 11 92 44 00 00	PN16 ND50 and other - Recessed
	Tank type T1 to T19
	Internal - external diameter **50 mm - 190 mm**
	Thickness **min. 27 mm - max 63 mm**
	Drilling **4 x M10 x 100 PCD and 4 x M16 x 125 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00

Part Number: 12 92 21 00 00	1 1/2" ANSI 150 & PN16 ND40 - Recessed
	Tank type T1 to T19
	Internal - external diameter **49 mm - 166 mm**
	Thickness **min. 27 mm - max 58 mm**
	Drilling **4 x M12 x 98.4 PCD and 4 x M16 x 110 PCD**
Seal (CNAF/PTFE)	11 90 29 00 00

ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA ND40 WELD-IN FLANGE A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND80 30° weld-in flanges

Accessories

Weld-in Flanges - Bottom Tank

Perolo 3" 30° 316L stainless steel weld-in flanges are designed to be welded at the bottom of tank containers in order to fit valves for tank filling and discharge pusposes. To be installed on 30° Bottom Discharge. Please note that according to ADR bottom discharge is prohibited on tank types T5, T8, T9, T10, T13, T14, T19.

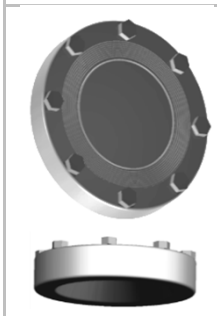
SPECIFICATION

- 316L stainless steel construction
- Nominal diameter 3"
- Radiused angles
- Serrated gasket contact area

OPTIONS

- Other diameter weld-in flanges
- Other material (Aluminium, 304L, 904L…)
- Other drilling

If you require a gasket or bolting kit for a specific valve, please refer to the installation sheet of that valve

Part Number:	Bottom outlet provision - to convert tank without bottom outlet to one with bottom outlet
11 91 94 00 00	
With cover: Tank type T5-8 9-10-13-14-19	
Without cover: Tank type T1-2-3-4-6-7-11-12-15-16-17-18	
	External diameter **200 mm**
	Thickness **35 mm**
	Including **8xH-M12-20 welded screws**
	Drilling **8 x M12 x 178 PCD**
Seal (CNAF/PTFE)	12 91 20 00 00

Part Number:	ND80 Weld In Flange Bottom Discharge 30°
11 91 16 00 00	
	Tank type T1 to T18
	Internal - external diameter **114.9 - 200 mm**
	Thickness **35 mm**
	Drilling **8 x M12 x 178 PCD**
Seal (CNAF/PTFE)	12 91 20 00 00

ACCESSORIES

ND80 45° weld-in flanges

Accessories

Weld-in Flanges - Bottom Tank

Perolo 3" 30° 316L stainless steel weld-in flanges are designed to be welded at the bottom of tank containers in order to fit valves for tank filling and discharge pusposes. To be installed on 30° Bottom Discharge. Please note that according to ADR bottom discharge is prohibited on tank types T5, T8, T9, T10, T13, T14, T19.

SPECIFICATION

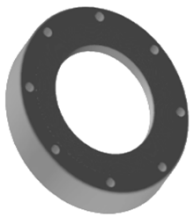
- 316L stainless steel construction
- Nominal diameter 3"
- Radiused angles
- Serrated gasket contact area

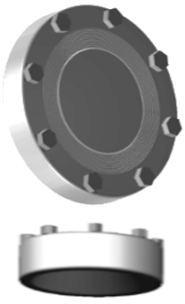
OPTIONS

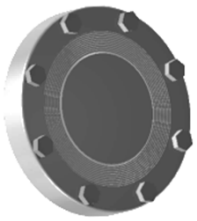
- Other diameter weld-in flanges
- Other material (Aluminium, 304L, 904L…)
- Other drilling

If you require a gasket or bolting kit for a specific valve, please refer to the installation sheet of that valve

Part Number: 11 90 59 00 00	ND80 Weld In Flange Bottom Discharge 45°
	Tank type T1 to T18
	Internal - external diameter **114.3 mm - 200 mm**
	Thickness **35 mm**
	Drilling **8 x M12 x 178 PCD**
Seal (CNAF/PTFE)	12 91 20 00 00

Part Number: 11 90 94 00 00	ND80 Weld In Flange Bottom Discharge 45°
	Tank type T1 to T18
	Internal - external diameter **114.3 mm - 200 mm**
	Thickness **45 mm**
	Drilling **8 x M12 x 178 PCD**
Seal (CNAF/PTFE)	12 91 20 00 00

Part Number: 11 92 46 00 00	Bottom outlet provision - to convert tank without bottom outlet to one with bottom outlet
With cover: Tank type T5-8 9-10-13-14-19 Without cover: Tank type T1-2-3-4-6-7-11-12-15-16-17-18	
	External diameter **200 mm**
	Thickness **35 mm**
	Including **8xH-M12-20 welded screws**
	Drilling **8 x M12 x 178 PCD**
Seal (CNAF/PTFE)	12 91 20 00 00

Part Number: 11 95 76 00 00	Bottom outlet provision - to convert tank without bottom outlet to one with bottom outlet
With cover: Tank type T5-8 9-10-13-14-19 Without cover: Tank type T1-2-3-4-6-7-11-12-15-16-17-18	
	External diameter **200 mm**
	Thickness **35 mm**
	Including **8xH-M12-20 welded screws**
	Drilling **8 x M12 x 178 PCD**
Seal (CNAF/PTFE)	12 91 20 00 00

ACCESSORIES

ND80 45° weld-in flanges

Accessories

Weld-in Flanges - Bottom Tank

Part Number: 11 95 82 00 00	ND80 Weld in Flange
	Tank type T1 to T19
	Internal - external diameter **100 mm - 200 mm**
	Thickness **27 mm**
	Drilling **4 x M16 x 152.4 PCD and 8 x M16 x 160 PCD**
Seal (CNAF/PTFE)	

Part Number: 11 95 65 00 00	ND80 Weld in Flange Recessed
	Tank type T1 to T19
	Internal - external diameter **94 mm - 217 mm**
	Thickness **min. 27 mm - max 57 mm**
	Drilling **4 x M16 x 152.4 PCD and 8 x M16 x 160 PCD**
Seal (CNAF/PTFE)	

Part Number: 11 90 64 00 00	ND80 Weld In Flange Bottom Discharge 45° - **STEAM HEATED**
	Tank type T1 to T14
	Internal - external diameter **114.3 mm - 200 mm**
	Thickness **36 mm**
	Drilling **8 x M12 x 178 PCD**
Seal (CNAF/PTFE)	12 91 20 00 00

Part Number: 11 94 28 00 00	For 45° bottom discharges
	Tank type T1 to T19
	Internal - external diameter **114.3 mm - 200 mm**
	Thickness **24 mm**
	Drilling **6 x M12 x 160 PCD**
Seal (CNAF/PTFE)	on request

ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA ND 80 FLANGE 45° A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND 80 Tank top weld-in flanges

Accessories
Weld-in Flanges - Top Tank

Perolo 316L stainless steel weld-in flanges are designed to be welded on top of tank containers, IBCs, road tankers, rail tank cars and storage tanks in order to fit valves and accessories for tank filling and discharge purposes.

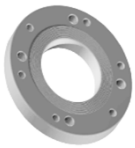
SPECIFICATION

- 316L stainless steel construction
- Nominal diameter 3 "
- Radiused angles
- Serrated gasket contact area

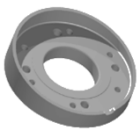
OPTIONS

- Other diameter weld-in flanges
- Other material (carbon steel, 304L, 904L…)
- Other drilling

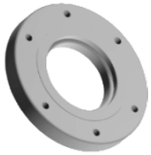
If you require a gasket or bolting kit for a specific valve, please refer to the installation sheet of that valve

Part Number: 11 90 36 00 00	PN16 ND80 - Siphon tube provision
	Tank type T1 to T19
	Internal - external diameter **94 mm - 200 mm**
	Thickness **27 mm**
Siphon tube 12 92 18 10 00 11 90 92 00 00	Drilling **6 x M12 x 168 PCD and 4 x M16 x 160 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

Part Number: 11 90 37 00 00	3" ANSI serie 150 & PN16 ND80 - Siphon tube provision
	Tank type T1 to T19
	Internal - external diameter **94 mm - 200 mm**
	Thickness **27 mm**
Siphon tube & seal 12 92 18 10 00 12 91 40 00 00	Drilling **8 x M16 x 160 PCD and 4 x M16 x 152.4 PCD**
Seal (CNAF/PTFE)	12 92 85 00 00

Part Number: 11 90 98 00 00	PN 16 ND80 Recessed
	Tank type T1 to T19
	Internal - external diameter **94 mm - 217 mm**
	Thickness **min. 31 mm - max 57 mm**
Siphon tube & seal 12 92 18 10 00 11 90 92 00 00	Drilling **6 x M12 x 168 PCD and 4 x M16 x 160 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

Part Number: 11 91 98 00 00	PN16 ND80 - Siphon tube provision
	Tank type T1 to T19
	Internal - external diameter **94 mm - 200 mm**
	Thickness **27 mm**
Siphon tube & seal 17 12 16 00 15 12 92 08 00 03	Drilling **6 x M12 x 168 PCD and 8 x M16 x 160 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

Part Number: 12 92 08 00 01	Siphon tube provision low profile
	Tank type T1 to T19
	Internal - external diameter **95 mm - 200 mm**
	Thickness **27 mm**
Siphon tube & seal 17 12 16 00 15 12 92 08 00 03	Drilling **6 x M12 x 168 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

Part Number: 17 12 16 00 01	PN 16 Siphon tube provision
	Tank type T1 to T19
	Internal - external diameter **94 mm - 200 mm**
	Thickness **27 mm**
Siphon tube & seal 17 12 16 00 15 12 92 08 00 03	Drilling **6 x M12 x 168 PCD and 4 x M16 x 125 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

ACCESSORIES

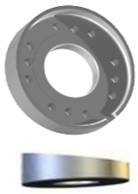
Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA ND80 WELD-IN FLANGE A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

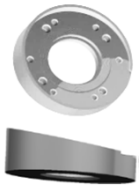
ND 80 Tank top weld-in flanges

Accessories

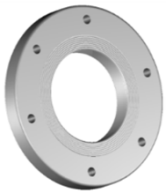
Weld-in Flanges - Top Tank

Part Number: 11 91 87 00 00	PN 16 Siphon tube provision
	Tank type T1 to T19
	Internal - external diameter **95 mm - 200 mm**
	Thickness **27 mm**
Siphon tube 17 12 16 00 15 12 92 08 00 03	Drilling **6 x M12 x 168 PCD and 4 x M16 x 160 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

Part Number: 11 90 99 00 00	PN16 3" ANSI serie 150 Recessed & Siphon tube provision
	Tank type T1 to T19
	Internal - external diameter **94 mm - 216 mm**
	Thickness **min. 31 mm - max 56 mm**
Siphon tube & seal 17 12 16 00 15 12 92 08 00 03	Drilling **4 x M16 x 152.4 PCD and 8 x M16 x 160 PCD**
Seal (CNAF/PTFE)	12 91 40 00 00

Part Number: 11 90 76 00 00	PN 16 Recessed & Siphon tube provision
	Tank type T1 to T19
	Internal - external diameter **94 mm - 217 mm**
	Thickness **min. 31 mm - max 56 mm**
Siphon tube & seal 17 12 16 00 15 12 92 08 00 03	Drilling **6 x M12 x 168 PCD and 4 x M16 x 160 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

Part Number: 11 94 07 00 00	PN16 ND80
	Tank type T1 to T19
	Internal - external diameter **101,6 mm - 200 mm**
	Thickness **27 mm**
	Drilling **6 x M12 x 168 PCD & 4 x M16 x 160 PCD**
Seal (CNAF/PTFE)	11 97 21 00 00

Part Number: 11 90 60 00 00	ND100 Int diameter
	Tank type T1 to T19
	Internal - external diameter **101,6 mm - 200 mm**
	Thickness **27 mm**
	Drilling **6 x M12 x 168 PCD**
Seal (CNAF/PTFE)	11 97 21 00 00

Part Number: 11 94 21 00 00	PN16 ND80
	Tank type T1 to T19
	Internal - external diameter **100 mm - 217 mm**
	Thickness **min. 27 mm - max 58 mm**
	Drilling **4 x M16 x 160 PCD & 6 x M12 x 168 PCD**
Seal (CNAF/PTFE)	11 97 21 00 00

Part Number: 11 92 56 00 00	PN16 ND80 & ND50 - Siphon tube provision
	Tank type T1 to T19
	Internal - external diameter **94 mm - 200 mm**
	Thickness **27 mm**
Siphon tube & seal 17 12 16 00 15 12 92 08 00 03	Drilling: **6 x M12 x 168 PCD 4 x M16 x 160 PCD 4 x M16 x 125 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA ND80 WELD-IN FLANGE A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

ND65 & ND80 Safety relief valve weld-in flanges
Accessories
Weld-in Flanges - Top Tank

Perolo 316L stainless steel weld-in flanges are designed to be welded on top of tank containers, IBCs, road tankers, rail tank cars and storage tanks in order to fit valves and accessories for tank filling and discharge purposes.

SPECIFICATION

- 316L stainless steel construction
- Nominal diameter 3 "
- Radiused angles
- Serrated gasket contact area

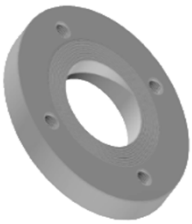
OPTIONS

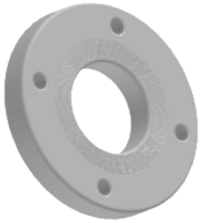
- Other diameter weld-in flanges
- Other material (carbon steel, 304L, 904L…)
- Other drilling

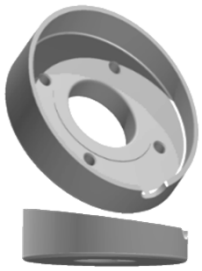
If you require a gasket or bolting kit for a specific valve, please refer to the installation sheet of that valve

Part Number: 12 91 14 00 00	2 1/2" BSP internal thread
	Tank type T1 to T20
	Internal - external diameter **2 1/2" BSP thread -127 mm**
	Thickness **25 mm**
	Drilling **6 x M10 x 105 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

Part Number: 12 92 14 00 00	Recessed
	Tank type T1 to T19
	Internal - external diameter **2 1/2" BSP thread -167 mm**
	Thickness **min. 24 mm - max 61 mm**
	Drilling **6 x M10 x 105 PCD**
Seal (CNAF/PTFE)	11 90 92 00 00

Part Number: 12 92 03 00 00	ISO PN16 ND65
	Tank type T1 to T19
	Internal - external diameter **82 mm-190 mm**
	Thickness **27 mm**
	Drilling **4 x M16 x 145 PCD**
Seal (CNAF/PTFE)	12 92 04 00 00

Part Number: 11 90 15 00 00	3" ANSI Serie 150
	Tank type T1 to T19
	Internal - external diameter **82 mm-190 mm**
	Thickness **27 mm**
	Drilling **4 x M16 x 152,4 PCD**
Seal (CNAF/PTFE)	12 92 04 00 00

Part Number: 11 90 63 00 00	ISO PN16 ND65 - Recessed
	Tank type T1 to T19
	Internal - external diameter **82 mm - 217 mm**
	Thickness **min. 31 mm - max 65 mm**
	Drilling **4 x M16 x 145 PCD**
Seal (CNAF/PTFE)	12 92 04 00 00

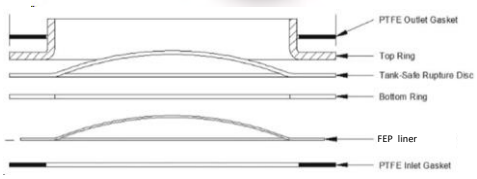
Part Number: 11 93 28 00 00	3" ANSI serie 150
	Tank type T1 to T19
	Internal - external diameter **82 mm-190 mm**
	Thickness **55 mm**
	Drilling **4 x M16 x 152,4 PCD**
Seal (CNAF/PTFE)	12 92 04 00 00

ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA ND 80 WELD-IN FLANGE SRV A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

VALVES SOLUTIONS FOR YOUR

Bursting Discs
Accessories
DRR & DRS



The Perolo bursting discs are designed to burst open and relieve an overpressure or vacuum condition at a predetermined differential pressure, and specific temperature.

PEROLO is an official elfab Bursting Disc Distributor

SPECIFICATION

- 316 stainless steel & Nickel Alloy
- Partially or fully FEP lined product contact area
- Bursting pressure, dimensions and shape depending on application
- A ND65 flanged SRV requires a ND80 (and not a ND65) bursting disc

CODIFICATION

STANDARD PART NUMBER	DR	R	80	P	484	T	20	.	.
YOUR PART NUMBER	DR	__	__	P	_____	T	20	__	__

LINING OPTIONS	
partially lined	S
fully lined (incl. outer ring)	R

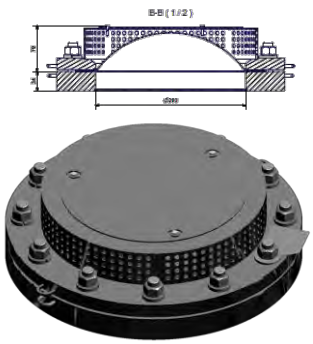
DIMENSIONS - others on request	
ND65	65
ND80	80
ND250	250

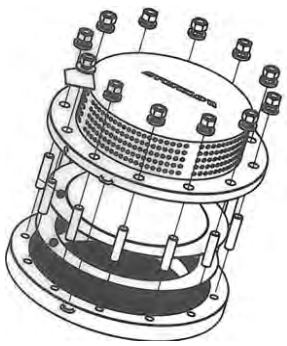
PRESSURE SETTING - others on request	
3,67 Bar +/- 5% Tolerance at 20°C	367
4,84 Bar +/- 5% Tolerance at 20°C	484

LINING MATERIAL	
Standard FEP lining	.
PTFE lining	PTFE
PFA lining	PFA

QUANTITY	
Single bursting disc	.
Pack of 10 bursting discs	K

Assembly 10" bursting disc for H2O2 application
Bursting pressure: 4,5 or 6 Bar
Height: 103 mm
Including weld-in flange (316L st st), cover (316L st st), screw kit and gasket (PTFE)
Part Number:
1119760045 (4,5 bar)
1119760060 (6 bar)



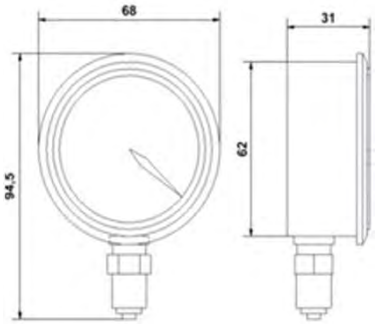


ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA BURSTING DISC B
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

Pressure Gauges
Accessories
Gauges

Perolo ND63 glycerin filled dial pressure gauges are designed to be fitted on tank containers, IBC's, road tankers or ralk tank cars in order to measure and indicate existing pressure.



SPECIFICATION

- Stainless steel body
- Dual pressure scale, bar & psi
- Glycerin filled unless specified otherwise
- Aluminum pointer
- Nominal external diameter 63mm / 2.5"
- Accuracy class 2.5
- Pressure connection ¼" BSP

bottom connection

back connection

OPTIONS

- ¼" NPT pressure connection
- Other pressure scales
- Stainless steel or brass internal parts depending on model
- Bottom or back pressure connection
- Tell tale
- Best Seller: 70 18 06 00 95

Pressure Bar	Pressure PSI	Internals & case	Connection	Glycerin	Tell tale	Part Number
0 to 4	0 to 60	brass with st st case	bottom	✓	-	70 21 02 00 66
0 to 6	0 to 90	brass with st st case	bottom	✓	-	70 21 02 00 22
0 to 6	0 to 90	full stainless steel	bottom	✓	-	70 18 06 00 95
0 to 6	0 to 90	full stainless steel	bottom	✓	✓	70 21 02 00 30
0 to 6	0 to 90	full stainless steel	bottom	-	✓	70 21 02 00 51
0 to 7	0 to 100	brass with st st case	bottom	✓	-	70 21 02 00 40
0 to 7	0 to 100	full stainless steel	bottom	✓	-	70 21 02 00 58
0 to 7	0 to 100	brass with st st case	bottom	✓	✓	70 21 02 00 35
0 to 9	0 to 130	full stainless steel	bottom	✓	-	70 21 02 00 44
-1 to 9	-14 to 130	brass with st st case	bottom	✓	-	70 21 02 00 23
-1 to 9	-14 to 130	full stainless steel	bottom	✓	✓	70 21 02 00 25
-1 to 9	-14 to 130	full stainless steel	bottom	-	✓	70 21 02 00 49
0 to 10	0 to 145	brass with st st case	bottom	✓	-	70 21 02 00 29
0 to 10	0 to 145	full stainless steel	bottom	✓	-	70 21 02 00 24
0 to 10	0 to 145	full stainless steel	bottom	✓	✓	70 21 02 00 47
0 to 10	0 to 145	full stainless steel	bottom	-	✓	70 21 02 00 32
0 to 16	0 to 232	full stainless steel	bottom	✓	-	70 21 02 00 34
0 to 40	0 to 580	full stainless steel	back	✓	-	70 21 02 00 28
0 to 40	0 to 580	full stainless steel	bottom	✓	-	70 21 02 00 61
0 to 40	0 to 580	full stainless steel	bottom	✓	-	70 21 02 00 37

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA PRESSURE GAUGE A
It is the customer's responsability to check the compatibility between valves material and transported product.
PEROLO SAS, 69 Rue des Maçons, 33390 Blaye, FRANCE -Phone : + 33 (0)5 5742 67 00 - Email : sales@perolo.com

4" Thermometer

Accessories

70 21 01 00 02

Perolo ND100 glycerin thermometer is designed to be fitted on tank containers, IBC's , road tankers or rail tank cars in order to measure Temperature.

SPECIFICATION

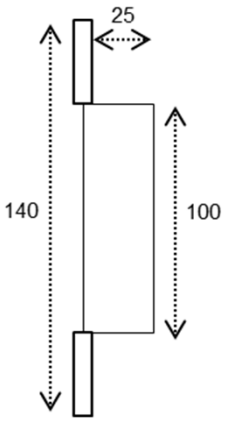
- Temperature sensor
- -40°C to +160°C or -40°F + 320°F
- Brass movement
- Wire length : 2m
- Front flange

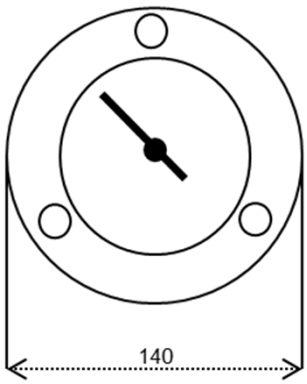
FEATURE

- 304 Stainless steel case
- Stainless steel front flange
- Dual Temperature Celsius & Fahrenheit.
- Glycerin filled
- 4", ND 100
- Back connection

CODIFICATION	
4" Thermometer	70 21 01 00 02

DRAWING





ACCESSORIES

Due to our policy of continuous product improvement, specifications may change without notice - Specification sheet PWA THERMOMETER A
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